

Numerical Linear Algebra and Matrix Factorizations (Tom Lyche) (Z-Library) — formula report

4638 inline formulas, 1056 display equations, 7 tables, 58 unrecovered image regions.

8 rows show a REFINEMENT below, not the OCR reading: 8 verified against the inferred. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex’ holds; this projection chose the refinement for them. Rows: Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0326, Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0369, Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0579, Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0630, Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0670, Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0706 ...

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0706 [conf 0.001] [refined: inferred]	253	■ 0.001 ● W -1	\begin{array}{rl} \boldsymbol{TV}+\boldsymbol{VT} & \\ \& = h^2\boldsymbol{F} \\ \Longleftarrow \boldsymbol{TSXS}+\boldsymbol{SXST} & \\ \& = h^2\boldsymbol{F} \\ \Longleftarrow \boldsymbol{STSXS^2}+ & \boldsymbol{S^2XSTS} \\ \& = h^2\boldsymbol{SFS}=h^2\boldsymbol{G} \\ \Longleftarrow \boldsymbol{S^2DXS^2}+ & \boldsymbol{S^2XS^2D} \\ \& = h^2\boldsymbol{G} \\ \Longleftarrow \boldsymbol{S^2=I/(2h)}\boldsymbol{DX} & +\boldsymbol{XD} & \& = 4h^4\boldsymbol{G}. \end{array}	$\begin{array}{rcl} TV + VT & = & h^2 F \\ \stackrel{V=XS}{\Longleftarrow} TSXS + SXST & = & h^2 F \\ \stackrel{S()S}{\Longleftarrow} STSXS^2 + S^2XSTS & = & h^2 SFS = h^2 G \\ \stackrel{TS=SD}{\Longleftarrow} S^2DXS^2 + S^2XS^2D & = & h^2 G \\ \stackrel{S^2=I/(2h)}{\Longleftarrow} DX + XD & = & 4h^4 G. \end{array}$	$\begin{array}{rcl} TV + VT & = & h^2 F \\ \stackrel{V=XS}{\Longleftarrow} TSXS + SXST & = & h^2 F \\ \stackrel{S()S}{\Longleftarrow} STSXS^2 + S^2XSTS & = & h^2 SFS = h^2 G \\ \stackrel{TS=SD}{\Longleftarrow} S^2DXS^2 + S^2XS^2D & = & h^2 G \\ \stackrel{S^2=I/(2h)}{\Longleftarrow} DX + XD & = & 4h^4 G. \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library_EQ0729 [conf 0.002] [refined: inferred]	258	■ 0.002 ● C +107	(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k} \quad \text{for all } j, k	$(DX + XD)_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k} \quad \text{for all } j, k$	$\begin{array}{rcl} (F_{2m} P_{2m})_{p,q} & = & \omega_{2m}^{j(2k)} = \omega_m^{jk}, \\ (F_{2m} P_{2m})_{p+m,q} & = & \omega_{2m}^{(j+m)(2k)} = \omega_m^{(j+m)k} = \omega_m^{jk}, \\ (F_{2m} P_{2m})_{p,q+m} & = & \omega_{2m}^{j(2k+1)} = \omega_{2m}^j \omega_m^{jk}, \\ (F_{2m} P_{2m})_{p+m,q+m} & = & \omega_{2m}^{(j+m)(2k+1)} = \omega_{2m}^{j+m} \omega_m^{(j+m)k} = \omega_{2m}^j \omega_m^{jk}. \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0326 (5.18) [refined: inferred]	133	■ 0.118 ● W +2	$\begin{aligned} & \boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and} \\ & \boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix}. \end{aligned}$	$\boldsymbol{Q}_1 := [\boldsymbol{q}_1, \dots, \boldsymbol{q}_n], \quad \boldsymbol{q}_j := \frac{\boldsymbol{v}_j}{\ \boldsymbol{v}_j\ _2}, \quad j = 1, \dots, n \text{ and}$ $\boldsymbol{R}_1 := \begin{bmatrix} \ \boldsymbol{v}_1\ _2 & \boldsymbol{a}_2^T \boldsymbol{q}_1 & \boldsymbol{a}_3^T \boldsymbol{q}_1 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_1 & \boldsymbol{a}_n^T \boldsymbol{q}_1 \\ 0 & \ \boldsymbol{v}_2\ _2 & \boldsymbol{a}_3^T \boldsymbol{q}_2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_2 & \boldsymbol{a}_n^T \boldsymbol{q}_2 \\ & 0 & \ \boldsymbol{v}_3\ _2 & \cdots & \boldsymbol{a}_{n-1}^T \boldsymbol{q}_3 & \boldsymbol{a}_n^T \boldsymbol{q}_3 \\ & & \ddots & \ddots & \vdots & \vdots \\ & & & \ddots & \ \boldsymbol{v}_{n-1}\ _2 & \boldsymbol{a}_n^T \boldsymbol{q}_{n-1} \\ & & & & 0 & \ \boldsymbol{v}_n\ _2 \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0029	031	■ 0.138 ● W +0	$\begin{array}{l} \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \\ \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \end{array}$	$\begin{array}{l} \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \\ \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \end{array}$	$\begin{array}{l} \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \\ \begin{array}{c} \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ \vdots \\ \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{array} \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0630 [refined: inferred]	229	■ 0.167 ● C +165	$\begin{aligned} & \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ & \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ & \vdots \\ & \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{aligned}$	$\begin{aligned} & \text{a}_{11} x_1 + \text{a}_{12} x_2 + \cdots + \text{a}_{1n} x_n = b_1 \\ & \text{a}_{21} x_1 + \text{a}_{22} x_2 + \cdots + \text{a}_{2n} x_n = b_2 \\ & \vdots \\ & \text{a}_{m1} x_1 + \text{a}_{m2} x_2 + \cdots + \text{a}_{mn} x_n = b_m \end{aligned}$	$\begin{aligned} & \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} \alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{bmatrix} = \alpha_i \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad i = 1, \dots, r, \\ & \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix} = \begin{bmatrix} -\boldsymbol{A} \boldsymbol{v}_i \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} -\alpha_i \boldsymbol{u}_i \\ \alpha_i \boldsymbol{v}_i \end{bmatrix} = -\alpha_i \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad i = 1, \dots, r, \\ & \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \boldsymbol{A}^* \boldsymbol{u}_i \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \boldsymbol{u}_i \\ \mathbf{0} \end{bmatrix}, \quad i = r + 1, \dots, m, \\ & \begin{bmatrix} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \boldsymbol{v}_i \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} \boldsymbol{v}_i \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} = 0 \begin{bmatrix} \mathbf{0} \\ \boldsymbol{v}_i \end{bmatrix}, \quad i = r + 1, \dots, n. \end{aligned}$

Conf. MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

— no value

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

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K clean

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not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00371	151	■0.194 ● N +1	$\boldsymbol{J}:=\left(\begin{array}{cccccc} 3 & 1 & & & & \\ & 0 & 3 & & & \\ & & & 2 & 1 & \\ & & & 0 & 2 & \\ & & & & & 2 \\ & & & & 0 & 1 & 0 \\ & & & & 0 & 2 & 1 \\ & & & & 0 & 2 & \end{array}\right)$	$\boldsymbol{J}:=\left[\begin{array}{cccc} 3 & 1 & & \\ 0 & 3 & & \\ & & 2 & 1 \\ & & 0 & 2 \\ & & & 2 \\ & & & 0 & 1 & 0 \\ & & & 0 & 2 & 1 \\ & & & 0 & 0 & 2 \end{array}\right]$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00492 (8.20)	193	■0.232 ● W +1	$\begin{aligned} & \boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{I} \text{ and } \boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{I} \\ & \text{if } \boldsymbol{A} \text{ is positive definite, } \boldsymbol{A}^{-1}=\frac{1}{\lambda_n}\boldsymbol{A} \\ & \text{if } \boldsymbol{A} \text{ is normal, } \boldsymbol{A}^{-1}=\frac{1}{ \lambda_n }\boldsymbol{A} \end{aligned}$	$\ \boldsymbol{A}\ _2=\sigma_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{\sigma_n}$ $\ \boldsymbol{A}\ _2=\lambda_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{\lambda_n}, \text{ if } \boldsymbol{A} \text{ is positive definite,}$ $\ \boldsymbol{A}\ _2= \lambda_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{ \lambda_n }, \text{ if } \boldsymbol{A} \text{ is normal.}$	$\ \boldsymbol{A}\ _2=\sigma_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{\sigma_n},$ $\ \boldsymbol{A}\ _2=\lambda_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{\lambda_n}, \text{ if } \boldsymbol{A} \text{ is positive definite,}$ $\ \boldsymbol{A}\ _2= \lambda_1 \text{ and } \ \boldsymbol{A}^{-1}\ _2=\frac{1}{ \lambda_n }, \text{ if } \boldsymbol{A} \text{ is normal.}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00010	024	■0.244 ● C -9	$\begin{aligned} & \text{diag}(d_1, \dots, d_n) := \begin{pmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{pmatrix} \\ & \text{tridiag}(a_1, d_1, c_1, a_2, d_2, c_2, \dots, a_{n-1}, d_{n-1}, c_{n-1}) := \begin{pmatrix} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & d_n & \end{pmatrix} \end{aligned}$	$\text{diag}(d_i) = \text{diag}(d_1, \dots, d_n) := \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix} = \begin{pmatrix} d_1 & & & \\ & \ddots & & \\ & & d_n & \end{pmatrix},$ $\boldsymbol{B} = \text{tridiag}(a_i, d_i, c_i) = \text{tridiag}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{c}) := \begin{pmatrix} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & d_n & \end{pmatrix}.$	$\text{diag}(d_i) = \text{diag}(d_1, \dots, d_n) := \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix} = \begin{pmatrix} d_1 & & & \\ & \ddots & & \\ & & d_n & \end{pmatrix},$ $\boldsymbol{B} = \text{tridiag}(a_i, d_i, c_i) = \text{tridiag}(\boldsymbol{a}, \boldsymbol{d}, \boldsymbol{c}) := \begin{pmatrix} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{pmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00924 (13.53)	314	■0.288 ● W +1	$\begin{aligned} & (L_h v)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m, \\ & v_{j,k} = 0, \quad (j,k) \in \partial I_m, \end{aligned}$	$(L_h v)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m,$ $v_{j,k} = 0, \quad (j,k) \in \partial I_m,$	$(L_h v)_{j,k} := \frac{(dv)_{j,k}}{h^2} = f_{j,k}, \quad (j,k) \in I_m,$ $v_{j,k} = 0, \quad (j,k) \in \partial I_m,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0369 (6.7) [re-fined: inferred]	150	■ 0.298 ● C -93	$d_i \text{ for } i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i \text{ for } i \text{ otherwise. } \in \mathbb{C}^{m \times n} \text{ and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j \text{ matrix } A(i,j) \in \mathbb{C}^{r \times c} \text{ is the submatrix of } A$ c	$d_i \text{ for } i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i \text{ for } i \text{ otherwise. } \in \mathbb{C}^{m \times n} \text{ and } 1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j \text{ matrix } A(i,j) \in \mathbb{C}^{r \times c} \text{ is the submatrix of } A$ c	$J := \text{diag}(U_1, U_2) = \begin{bmatrix} 2 & 1 & 0 & & & & & \\ 0 & 2 & 1 & & & & & \\ 0 & 0 & 2 & & & & & \\ & & & 2 & 1 & & & \\ & & & 0 & 2 & & & \\ & & & & & 2 & & \\ & & & & & & 3 & 1 \\ & & & & & & 0 & 3 \end{bmatrix} \in \mathbb{R}^{8 \times 8}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0375	152	■ 0.305 ● W +0	$\begin{aligned} & \begin{array}{l} \boldsymbol{U}^* \boldsymbol{A} \\ \boldsymbol{U} = \boldsymbol{W}^* \boldsymbol{V}^* \boldsymbol{V}^* \boldsymbol{W} \\ \boldsymbol{A} \boldsymbol{V} \boldsymbol{W} = \\ \left[\begin{array}{l} 1 \\ \mathbf{0} \end{array} \right] \begin{array}{l} 1 \\ \mathbf{0} \end{array} \end{array} \\ & \begin{array}{l} \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{W} = \\ \left[\begin{array}{l} 1 \\ \mathbf{0} \end{array} \right] \begin{array}{l} 1 \\ \mathbf{0} \end{array} \end{array} \\ & \begin{array}{l} \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{W} = \\ \left[\begin{array}{l} 1 \\ \mathbf{0} \end{array} \right] \begin{array}{l} 1 \\ \mathbf{0} \end{array} \end{array} \end{aligned}$	$U^*AU = W^*(V^*AV)W = \begin{bmatrix} \frac{1}{\mathbf{0}} & \frac{\mathbf{0}^*}{W_1^*} \end{bmatrix} \begin{bmatrix} \lambda_1 & x^* \\ \mathbf{0} & M \end{bmatrix} \begin{bmatrix} \frac{1}{\mathbf{0}} & \frac{\mathbf{0}^*}{W_1} \end{bmatrix}$ $= \begin{bmatrix} \lambda_1 & x^*W_1 \\ \mathbf{0} & W_1^*MW_1 \end{bmatrix}$	$U^*AU = W^*(V^*AV)W = \begin{bmatrix} \frac{1}{\mathbf{0}} & \frac{\mathbf{0}^*}{W_1^*} \end{bmatrix} \begin{bmatrix} \lambda_1 & x^* \\ \mathbf{0} & M \end{bmatrix} \begin{bmatrix} \frac{1}{\mathbf{0}} & \frac{\mathbf{0}^*}{W_1} \end{bmatrix}$ $= \begin{bmatrix} \lambda_1 & x^*W_1 \\ \mathbf{0} & W_1^*MW_1 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0366 (6.4)	149	■ 0.315 ● W +0	$\boldsymbol{J}_m(\lambda) := \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & \lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix} = \lambda \boldsymbol{I}_m + \boldsymbol{E}_m, \quad \boldsymbol{E}_m := \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}.$	$J_m(\lambda) := \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{bmatrix} = \lambda \boldsymbol{I}_m + \boldsymbol{E}_m, \quad \boldsymbol{E}_m := \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library__EQ0579 [refined: inferred]	215	■ 0.322 ● C +90	$\text{\text{nal if } } a_{ij} = 0 \text{\text{ for } } i \neq j. \text{\text{ }} \text{\text{triangular or \textbf{right triangular} if } } a_{ij} = 0 \text{\text{ for } } i > j. \text{\text{ }} \text{\text{triangular or \textbf{left triangular} if } } a_{ij} = 0 \text{\text{ for } } i < j. \text{\text{ }} \text{\textbf{Hessenberg if } } a_{ij} = 0 \text{\text{ for } } i > j + 1. \text{\text{ }} \text{\textbf{Hessenberg if } } a_{ij} = 0 \text{\text{ for } } i < j + 1. \text{\text{ }} \text{\textbf{gonal if } } a_{ij} = 0 \text{\text{ for } } i - j > 1. \text{\text{ }} \text{\textbf{ded if } } a_{ij} = 0 \text{\text{ for } } i - j > d. \text{\text{ }} \text{\text{[lex] \text{the following notations for diagonal- and tridiagonal } } n \times n$	(not rendered)	$A^*Ax = \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ \vdots \\ 1 & t_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} m & \sum t_k \\ \sum t_k & \sum t_k^2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$ $= \begin{bmatrix} 1 & \cdots & 1 \\ t_1 & \cdots & t_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} \sum y_k \\ \sum t_k y_k \end{bmatrix} = A^*b,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library__EQ0335 (5.19)	135	■ 0.329 ● N +0	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right] = \left[\begin{array}{rc} c & s \\ -s & c \end{array}\right] \left[\begin{array}{l} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right], [C(:,i)C(:,j)] = [A(:,i)A(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right].$	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right] = \left[\begin{array}{rc} c & s \\ -s & c \end{array}\right] \left[\begin{array}{l} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right], [C(:,i)C(:,j)] = [A(:,i)A(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right].$	$\left[\begin{array}{l} \boldsymbol{B}(i,:) \\ \boldsymbol{B}(j,:) \end{array}\right] = \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right] \left[\begin{array}{l} \boldsymbol{A}(i,:) \\ \boldsymbol{A}(j,:) \end{array}\right], [C(:,i)C(:,j)] = [A(:,i)A(:,j)] \left[\begin{array}{cc} c & s \\ -s & c \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library__EQ0917	312	■ 0.371 ● W +2	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \quad \text{for } k \geq 0$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \quad \text{for } k \geq 0$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library__EQ0547	207	■ 0.373 ● N +0	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}} \right) & \text{otherwise.} \end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}} \right) & \text{otherwise.} \end{cases}$	$c_{i,k+1} = \begin{cases} 0 & \text{if } \sigma_i = \lambda_{k+1}, \\ c_{i,k} \left(1 - \frac{\sigma_i}{\lambda_{k+1}} \right) & \text{otherwise.} \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche__Z-Library__EQ0855	296	■ 0.386 ● C +86	$\begin{array}{ll} \boldsymbol{t}_0 = \left[\begin{array}{c} 3 \\ -3 \end{array} \right] / 2, & \boldsymbol{\alpha}_0 = 1/2, \\ \boldsymbol{x}_1 = \left[\begin{array}{c} -1 \\ 4 \end{array} \right] / 2, & \boldsymbol{r}_1 = \left[\begin{array}{c} 0 \\ 3 \end{array} \right] / 4, \\ \boldsymbol{\beta}_0 = 1/4, & \boldsymbol{p}_1 = \left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], \\ \boldsymbol{t}_1 = \left[\begin{array}{c} 0 \\ 9 \end{array} \right] / 8, & \boldsymbol{\alpha}_1 = 2/3, \\ \boldsymbol{x}_2 = \mathbf{0}, & \boldsymbol{r}_2 = \mathbf{0}. \end{array}$	(not rendered)	$\boldsymbol{t}_0 = \left[\begin{array}{c} 3 \\ -3 \end{array} \right] / 2, \quad \boldsymbol{\alpha}_0 = 1/2, \quad \boldsymbol{x}_1 = \left[\begin{array}{c} -1/4 \\ -1/2 \end{array} \right], \quad \boldsymbol{r}_1 = \left[\begin{array}{c} 0 \\ 3/4 \end{array} \right], \quad \boldsymbol{\beta}_0 = 1/4,$ $\boldsymbol{p}_1 = \left[\begin{array}{c} 3/8 \\ 3/4 \end{array} \right], \quad \boldsymbol{t}_1 = \left[\begin{array}{c} 0 \\ 9/8 \end{array} \right], \quad \boldsymbol{\alpha}_1 = 2/3, \quad \boldsymbol{x}_2 = \mathbf{0}, \quad \boldsymbol{r}_2 = \mathbf{0}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0670 (10.10) [re-fined: inferred]	241	■ 0.390 ● C +97	$\begin{bmatrix} \vdots & \vdots & \cdots & \vdots \\ a_{r_2, c_1} & a_{r_2, c_1+1} & \cdots & a_{r_2, c_2} \end{bmatrix}$	$\begin{aligned} a_{ii} &= 2d, \, i = 1, \dots, n, \\ a_{i,i+1} &= a_{i+1,i} = a, \, \, i = 1, \dots, n-1, \, \, \, i \neq m, 2m, \dots, (m-1)m, \\ a_{i,i+m} &= a_{i+m,i} = a, \, \, i = 1, \dots, n-m, \\ a_{ij} &= 0, \, \, \text{otherwise}, \end{aligned}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0763 (12.14) [re-fined: inferred]	271	■ 0.402 ● C +11	$\langle j_1, j_2 \mid j_c \rangle \begin{bmatrix} \cdot & \cdot & \cdots & \cdot \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{bmatrix}$	$\boldsymbol{G}_\omega = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \omega/2 & 0 \\ \omega^2/4 & \omega/2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1-\omega & \omega/2 \\ \omega(1-\omega)/2 & 1-\omega+\omega^2/4 \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0498	194	■ 0.420 ● C +4	$\begin{aligned} &\texttt{\textbackslash begin{aligned}} \& \texttt{x_{1}}+\texttt{\textbackslash left(1-10^{-16}}\texttt{\textbackslash right)} \texttt{x_{2}}=20 \\ &\texttt{\textbackslash \& x_{1}}+\texttt{\textbackslash left(1-10^{-15}}\texttt{\textbackslash right.} \texttt{\textbackslash end{aligned}} \end{aligned}$	$\begin{aligned} x_1 + (1 - 10^{-16}) x_2 &= 20 \\ x_1 + (1 - 10^{-15} \end{aligned}$	$\begin{aligned} x_1 + x_2 &= 20 \\ x_1 + (1 - 10^{-16})x_2 &= 20 - 10^{-15} \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0262	107	■ 0.421 ● N -1	$\begin{aligned} &\text{\texttt{\textbackslash boldsymbol{x}}^{\texttt{*}}} \text{\texttt{\textbackslash boldsymbol{A}}} \text{\texttt{\textbackslash boldsymbol{x}}=z^{\texttt{*}}} \\ &\text{\texttt{\textbackslash boldsymbol{U}}^{\texttt{*}}} \text{\texttt{\textbackslash boldsymbol{A}}} \text{\texttt{\textbackslash boldsymbol{U}}} \\ &\text{\texttt{\textbackslash boldsymbol{z}}=z^{\texttt{*}}} \text{\texttt{\textbackslash operatorname{diag}}}\texttt{\textbackslash left(} \\ &\texttt{\textbackslash lambda_{1}}, \texttt{\textbackslash ldots}, \texttt{\textbackslash lambda_{n}}\texttt{\textbackslash right)} \text{\texttt{\textbackslash boldsymbol{z}}=} \\ &\texttt{\textbackslash sum_{j=1}^{\texttt{n}}} \texttt{\textbackslash lambda_{j}}\texttt{\textbackslash left z_{j}}\texttt{\textbackslash right }^{\texttt{2}} \texttt{\textbackslash geq} \texttt{0} . \end{aligned}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \text{diag}(\lambda_1, \dots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j z_j ^2 \geq 0.$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{z} = \boldsymbol{z}^* \text{diag}(\lambda_1, \dots, \lambda_n) \boldsymbol{z} = \sum_{j=1}^n \lambda_j z_j ^2 \geq 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0404	161	■ 0.448 ● C +10	$\left[\begin{array}{llllllll} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \end{array}\right] \in \mathbb{R}^{8,8}?$	$\begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & \end{bmatrix} \in \mathbb{R}^{8,8}?$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0914 (13.48)	311	■ 0.453 ● N +0	$\begin{aligned} &\text{\texttt{\textbackslash begin{aligned}}} \& \texttt{\textbackslash boldsymbol{y}}_{\texttt{k+1}}= \\ &\texttt{\textbackslash boldsymbol{y}}_{\texttt{k}}+\texttt{\textbackslash alpha_{k}} \texttt{\textbackslash boldsymbol{q}}_{\texttt{k}}, \texttt{\textbackslash quad} \\ &\texttt{\textbackslash alpha_{k}}=\texttt{z_{k}}^{\texttt{T}} \texttt{z_{k}} / \texttt{\textbackslash boldsymbol{q}}_{\texttt{k}}^{\texttt{T}}\texttt{\textbackslash left(} \\ &\texttt{\textbackslash boldsymbol{C}} \texttt{\textbackslash boldsymbol{A}} \texttt{\textbackslash boldsymbol{C}}^{\texttt{T}}\texttt{\textbackslash right)} \\ &\texttt{\textbackslash boldsymbol{q}}_{\texttt{k}}, \texttt{\textbackslash \& \textbackslash boldsymbol{z}}_{\texttt{k+1}}=\texttt{z_{k}}- \\ &\texttt{\textbackslash alpha_{k}}\texttt{\textbackslash left(\textbackslash boldsymbol{C}} \texttt{\textbackslash boldsymbol{A}} \\ &\texttt{\textbackslash boldsymbol{C}}^{\texttt{T}}\texttt{\textbackslash right)} \texttt{\textbackslash boldsymbol{q}}_{\texttt{k}}, \\ &\texttt{\textbackslash \& \textbackslash boldsymbol{q}}_{\texttt{k+1}}=\texttt{z_{k+1}}+\texttt{\textbackslash beta_{k}} \\ &\texttt{\textbackslash boldsymbol{q}}_{\texttt{k}}, \texttt{\textbackslash quad} \texttt{\textbackslash beta_{k}}=\texttt{z_{k+1}}^{\texttt{T}} \texttt{z_{k+1}} / \\ &\texttt{z_{k}}^{\texttt{T}} \texttt{z_{k}} . \texttt{\textbackslash end{aligned}} \end{aligned}$	$\begin{aligned} \boldsymbol{y}_{k+1} &= \boldsymbol{y}_k + \alpha_k \boldsymbol{q}_k, \quad \alpha_k = \boldsymbol{z}_k^T \boldsymbol{z}_k / \boldsymbol{q}_k^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{q}_k, \\ \boldsymbol{z}_{k+1} &= \boldsymbol{z}_k - \alpha_k \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{q}_k, \\ \boldsymbol{q}_{k+1} &= \boldsymbol{z}_{k+1} + \beta_k \boldsymbol{q}_k, \quad \beta_k = \boldsymbol{z}_{k+1}^T \boldsymbol{z}_{k+1} / \boldsymbol{z}_k^T \boldsymbol{z}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{y}_{k+1} &= \boldsymbol{y}_k + \alpha_k \boldsymbol{q}_k, \quad \alpha_k = \boldsymbol{z}_k^T \boldsymbol{z}_k / \boldsymbol{q}_k^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ \boldsymbol{z}_{k+1} &= \boldsymbol{z}_k - \alpha_k (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{q}_k, \\ \boldsymbol{q}_{k+1} &= \boldsymbol{z}_{k+1} + \beta_k \boldsymbol{q}_k, \quad \beta_k = \boldsymbol{z}_{k+1}^T \boldsymbol{z}_{k+1} / \boldsymbol{z}_k^T \boldsymbol{z}_k. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00245	100	■ 0.458 ● N -1	$\left[\begin{array}{ll} a & \bar{b} \\ b & d \end{array}\right] = \left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array}\right] \left[\begin{array}{ll} d_1 & 0 \\ 0 & d_2 \end{array}\right] \left[\begin{array}{ll} 1 & \overline{l_1} \\ 0 & 1 \end{array}\right] = \left[\begin{array}{ll} d_1 & d_1 \overline{l_1} \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{array}\right]$	$\begin{bmatrix} a & \bar{b} \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_1 & 1 \end{bmatrix} \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \begin{bmatrix} 1 & \overline{l_1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} d_1 & d_1 \overline{l_1} \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{bmatrix}$	$\begin{bmatrix} a & \bar{b} \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l_1 & 1 \end{bmatrix} \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \begin{bmatrix} 1 & \overline{l_1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} d_1 & d_1 \overline{l_1} \\ d_1 l_1 & d_1 l_1 ^2 + d_2 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00370	150	■ 0.466 ● N +1	$\begin{aligned} & \boldsymbol{A} \boldsymbol{s}_1 = 2 \boldsymbol{s}_1, \quad \boldsymbol{A} \boldsymbol{s}_2 = \boldsymbol{s}_1 + 2 \boldsymbol{s}_2, \quad \boldsymbol{A} \boldsymbol{s}_3 = \boldsymbol{s}_2 + 2 \boldsymbol{s}_3, \\ & \boldsymbol{A} \boldsymbol{s}_4 = 2 \boldsymbol{s}_4, \quad \boldsymbol{A} \boldsymbol{s}_5 = \boldsymbol{s}_4 + 2 \boldsymbol{s}_5, \\ & \boldsymbol{A} \boldsymbol{s}_6 = 2 \boldsymbol{s}_6, \\ & \boldsymbol{A} \boldsymbol{s}_7 = 3 \boldsymbol{s}_7, \quad \boldsymbol{A} \boldsymbol{s}_8 = \boldsymbol{s}_7 + 3 \boldsymbol{s}_8. \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{s}_1 &= 2 \boldsymbol{s}_1, & \boldsymbol{A} \boldsymbol{s}_2 &= \boldsymbol{s}_1 + 2 \boldsymbol{s}_2, & \boldsymbol{A} \boldsymbol{s}_3 &= \boldsymbol{s}_2 + 2 \boldsymbol{s}_3, \\ \boldsymbol{A} \boldsymbol{s}_4 &= 2 \boldsymbol{s}_4, & \boldsymbol{A} \boldsymbol{s}_5 &= \boldsymbol{s}_4 + 2 \boldsymbol{s}_5, \\ \boldsymbol{A} \boldsymbol{s}_6 &= 2 \boldsymbol{s}_6, \\ \boldsymbol{A} \boldsymbol{s}_7 &= 3 \boldsymbol{s}_7, & \boldsymbol{A} \boldsymbol{s}_8 &= \boldsymbol{s}_7 + 3 \boldsymbol{s}_8. \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{s}_1 &= 2 \boldsymbol{s}_1, & \boldsymbol{A} \boldsymbol{s}_2 &= \boldsymbol{s}_1 + 2 \boldsymbol{s}_2, & \boldsymbol{A} \boldsymbol{s}_3 &= \boldsymbol{s}_2 + 2 \boldsymbol{s}_3, \\ \boldsymbol{A} \boldsymbol{s}_4 &= 2 \boldsymbol{s}_4, & \boldsymbol{A} \boldsymbol{s}_5 &= \boldsymbol{s}_4 + 2 \boldsymbol{s}_5, \\ \boldsymbol{A} \boldsymbol{s}_6 &= 2 \boldsymbol{s}_6, \\ \boldsymbol{A} \boldsymbol{s}_7 &= 3 \boldsymbol{s}_7, & \boldsymbol{A} \boldsymbol{s}_8 &= \boldsymbol{s}_7 + 3 \boldsymbol{s}_8. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00396	158	■ 0.467 ● C -3	$\begin{aligned} & \beta_k \leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_B(x) \leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_E(x) \\ & \leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in \mathbb{C}^n \\ x \neq \mathbf{0}}} R_E(x) = \alpha_k + \varepsilon_1, \end{aligned}$	$\begin{aligned} \beta_k &\leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_B(x) \leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_E(x) \\ &\leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in \mathbb{C}^n \\ x \neq \mathbf{0}}} R_E(x) = \alpha_k + \varepsilon_1, \end{aligned}$	$\begin{aligned} \beta_k &\leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_B(x) \leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_E(x) \\ &\leq \max_{\substack{x \in S \\ x \neq \mathbf{0}}} R_A(x) + \max_{\substack{x \in \mathbb{C}^n \\ x \neq \mathbf{0}}} R_E(x) = \alpha_k + \varepsilon_1, \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00308	125	■ 0.469 ● N +0	$\boldsymbol{P} \boldsymbol{x} = \boldsymbol{x} - \frac{\boldsymbol{v}^* \boldsymbol{x}}{\boldsymbol{v}^* \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x} - \frac{1}{2} \boldsymbol{v} = \frac{1}{2} (\boldsymbol{x} + \boldsymbol{y}).$	$\boldsymbol{P} \boldsymbol{x} = \boldsymbol{x} - \frac{\boldsymbol{v}^* \boldsymbol{x}}{\boldsymbol{v}^* \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x} - \frac{1}{2} \boldsymbol{v} = \frac{1}{2} (\boldsymbol{x} + \boldsymbol{y}).$	$\boldsymbol{P} \boldsymbol{x} = \boldsymbol{x} - \frac{\boldsymbol{v}^* \boldsymbol{x}}{\boldsymbol{v}^* \boldsymbol{v}} \boldsymbol{v} \stackrel{(5.13)}{=} \boldsymbol{x} - \frac{1}{2} \boldsymbol{v} = \frac{1}{2} (\boldsymbol{x} + \boldsymbol{y}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0600 (0)	221	■ 0.474 ● N +0	$\left[\begin{array}{lll} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right] = \frac{1}{2} \left[\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{array}\right] \times \left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right] = \left[\begin{array}{l} 2 \\ 0 \\ 0 \end{array}\right],$	$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix},$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0628 (1.11)	031	■ 0.479 ● N +0	$\ x\ _2 := \left(\sum_{j=1}^n \left x_j\right ^2\right)^{1/2} = \sqrt{x^*x}.$	$\ x\ _2 := \left(\sum_{j=1}^n x_j ^2\right)^{1/2} = \sqrt{x^*x}.$	$\ x\ _2 := \left(\sum_{j=1}^n x_j ^2\right)^{1/2} = \sqrt{x^*x}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0241 (3.33)	097	■ 0.492 ● C +5	$\boldsymbol{B} := \left[\begin{array}{cc} (1 + \boldsymbol{u}_1^T \boldsymbol{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\boldsymbol{u}_1^T \boldsymbol{B}_{2,2} / u_{1,1} \\ -\boldsymbol{u}_{2,2}^T \boldsymbol{B}_{2,2} / u_{1,1} & \boldsymbol{B}_{2,2} \end{array}\right].$	$\boldsymbol{B} := \left[\begin{array}{cc} (1 + \boldsymbol{u}_1^T \boldsymbol{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\boldsymbol{u}_1^T \boldsymbol{B}_{2,2} / u_{1,1} \\ -\boldsymbol{u}_{2,2}^T \boldsymbol{B}_{2,2} / u_{1,1} & \boldsymbol{B}_{2,2} \end{array}\right].$	$\boldsymbol{B} := \left[\begin{array}{cc} (1 + \boldsymbol{u}_1^T \boldsymbol{B}_{2,2} \boldsymbol{\ell}_1) / u_{1,1} & -\boldsymbol{u}_1^T \boldsymbol{B}_{2,2} / u_{1,1} \\ -\boldsymbol{u}_{2,2}^T \boldsymbol{B}_{2,2} / u_{1,1} & \boldsymbol{B}_{2,2} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0757 (12.11)	270	■ 0.498 ● C +6	$\boldsymbol{A}_{\boldsymbol{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} \end{array}\right], \quad \boldsymbol{A}_{\boldsymbol{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots \\ \ddots & \ddots & \vdots \\ 0 & -a_{n-1,n} & 0 \end{array}\right].$	$\boldsymbol{A}_{\boldsymbol{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} & 0 \end{array}\right], \quad \boldsymbol{A}_{\boldsymbol{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots & -a_{1,n} \\ \ddots & \ddots & \ddots & \vdots \\ & 0 & -a_{n-1,n} & 0 \end{array}\right].$	$\boldsymbol{A}_{\boldsymbol{L}} := \left[\begin{array}{ccc} 0 & -a_{2,1} & 0 \\ \vdots & \ddots & \ddots \\ -a_{n,1} & \cdots & -a_{n,n-1} & 0 \end{array}\right], \quad \boldsymbol{A}_{\boldsymbol{R}} := \left[\begin{array}{ccc} 0 & -a_{1,2} & \cdots & -a_{1,n} \\ \ddots & \ddots & \ddots & \vdots \\ & 0 & -a_{n-1,n} & 0 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0138	064	■ 0.503 ● N +1	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$	$\left[\begin{array}{ll} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array}\right] \left[\begin{array}{ll} B_{11} & B_{12} \\ B_{21} & B_{22} \end{array}\right] = \left[\begin{array}{ll} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0465 (8.1)	186	■ 0.507 ● N +0	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p\right)^{1/p}, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00486	191	■ 0.508 ● W -1	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ x\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, p \geq 1, \quad \ x\ _\infty := \max_{1 \leq j \leq n} x_j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00766	272	■ 0.512 ● N +0	$x_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} x_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} x_k(2) + \frac{3}{4}.$	$x_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} x_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} x_k(2) + \frac{3}{4}.$	$\mathbf{x}_{k+1}(2) = \frac{1}{2} \left(\frac{1}{2} \mathbf{x}_k(2) + \frac{1}{2} \right) + \frac{1}{2} = \frac{1}{4} \mathbf{x}_k(2) + \frac{3}{4}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E01017	345	■ 0.514 ● K +0	$z_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$	$z_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$	$z_0 = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00798	280	■ 0.516 ● W +1	$\begin{aligned} & \left\ \mathbf{x}_k - \mathbf{x} \right\ \leq \left\ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x}) \right\ \leq \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x} \right\ \\ & = \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x} \right\ \leq \left\ \mathbf{G} \right\ \left(\left\ \mathbf{x}_{k-1} - \mathbf{x}_k \right\ + \left\ \mathbf{x}_k - \mathbf{x} \right\ \right). \end{aligned}$	$\begin{aligned} \left\ \mathbf{x}_k - \mathbf{x} \right\ &= \left\ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x}) \right\ \leq \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x} \right\ \\ &= \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x} \right\ \leq \left\ \mathbf{G} \right\ \left(\left\ \mathbf{x}_{k-1} - \mathbf{x}_k \right\ + \left\ \mathbf{x}_k - \mathbf{x} \right\ \right) \end{aligned}$	$\begin{aligned} \left\ \mathbf{x}_k - \mathbf{x} \right\ &= \left\ \mathbf{G}(\mathbf{x}_{k-1} - \mathbf{x}) \right\ \leq \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x} \right\ \\ &= \left\ \mathbf{G} \right\ \left\ \mathbf{x}_{k-1} - \mathbf{x}_k + \mathbf{x}_k - \mathbf{x} \right\ \leq \left\ \mathbf{G} \right\ \left(\left\ \mathbf{x}_{k-1} - \mathbf{x}_k \right\ + \left\ \mathbf{x}_k - \mathbf{x} \right\ \right). \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00837 (13.3)	291	■ 0.516 ● N +0	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$	$Q(\mathbf{y}) := \frac{1}{2} \mathbf{y}^T \mathbf{A} \mathbf{y} - \mathbf{b}^T \mathbf{y} + c.$

Conf. MathPix confidence:

■ ≥0.9
■ 0.5-0.9
■ <0.5
— no value

Residual render vs scan (inkdrill):

● C component
● W weak
● S stable
● N noise
● K clean
● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08854 (13.18)	296	■ 0.521 ● W +0	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k \\ \boldsymbol{p}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \end{aligned}$	$\begin{aligned} \boldsymbol{t}_k &= \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= (\boldsymbol{r}_k^T \boldsymbol{r}_k) / (\boldsymbol{p}_k^T \boldsymbol{t}_k), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k, \\ \beta_k &= (\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}) / (\boldsymbol{r}_k^T \boldsymbol{r}_k), \\ \boldsymbol{p}_{k+1} &:= \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k. \end{aligned}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08329	134	■ 0.523 ● K +0	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08862	300	■ 0.523 ● W -1	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$	$\kappa_P = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\cos^2(\pi h/2)}{\sin^2(\pi h/2)} \text{ and } \sqrt{\kappa_P} = \frac{\cos(\pi h/2)}{\sin(\pi h/2)} \approx \frac{2}{\pi h} \approx \frac{2}{\pi} \sqrt{n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08504 (8.26)	197	■ 0.531 ● W +0	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$	$K_2(\boldsymbol{A}) = \begin{cases} \lambda_1 / \lambda_n, & \text{if } \boldsymbol{A} \text{ is positive definite,} \\ \lambda_1 / \lambda_n , & \text{if } \boldsymbol{A} \text{ is normal,} \\ \sigma_1 / \sigma_n, & \text{in general.} \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08830 (12.46)	287	■ 0.531 ● N +0	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$	$\boldsymbol{A} = \boldsymbol{M} - \boldsymbol{N}, \quad \boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{L})(\boldsymbol{I} - \boldsymbol{L}^T), \quad \boldsymbol{N} = \boldsymbol{L} \boldsymbol{L}^T.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00098	054	■ 0.532 ● C +3	$\begin{aligned} & \& u_{\{1\}}=d_{\{1\}}, \quad \& l_{\{1\}}=a_{\{1\}} / u_{\{1\}}, \\ & \& u_{\{2\}}=d_{\{2\}}-\& l_{\{1\}} c_{\{1\}}, \quad \& l_{\{2\}}=a_{\{2\}} / u_{\{2\}}, \\ & \& u_{\{3\}}=d_{\{3\}}-\& l_{\{2\}} c_{\{2\}} \quad \& \& z_{\{1\}}=b_{\{1\}}, \quad \& z_{\{2\}}=b_{\{2\}}-\& l_{\{1\}} z_{\{1\}}, \quad \& z_{\{3\}}=b_{\{3\}}-\& l_{\{2\}} z_{\{2\}} \\ & \& \& x_{\{3\}}=z_{\{3\}} / u_{\{3\}}, \quad \& x_{\{2\}}=\& \left(z_{\{2\}}-\& c_{\{2\}} x_{\{3\}}\right) / u_{\{2\}}, \quad \& x_{\{1\}}=\& \left(z_{\{1\}}-\& c_{\{1\}} x_{\{2\}}\right) / u_{\{1\}} \end{aligned}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1 / u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2 / u_2, \quad u_3 = d_3 - l_2 c_2 \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2 \\ x_3 &= z_3 / u_3, \quad x_2 = (z_2 - c_2 x_3) / u_2, \quad x_1 = (z_1 - c_1 x_2) / u_1 \end{aligned}$	$\begin{aligned} u_1 &= d_1, \quad l_1 = a_1 / u_1, \quad u_2 = d_2 - l_1 c_1, \quad l_2 = a_2 / u_2, \quad u_3 = d_3 - l_2 c_2, \\ z_1 &= b_1, \quad z_2 = b_2 - l_1 z_1, \quad z_3 = b_3 - l_2 z_2, \\ x_3 &= z_3 / u_3, \quad x_2 = (z_2 - c_2 x_3) / u_2, \quad x_1 = (z_1 - c_1 x_2) / u_1. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00836 (13.2)	291	■ 0.546 ● N +0	$\left \left \boldsymbol{x} \right ^T \boldsymbol{A} \boldsymbol{y} \right ^2 \leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \right) \left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} \right ^2 \leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \right) \left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} \right ^2 \leq \left(\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \right) \left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y} \right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00868	301	■ 0.546 ● N +0	$\left \left \boldsymbol{x} \right - \left \boldsymbol{x}_{\{0\}} \right - \left \boldsymbol{w} \right \right _{\left \left \boldsymbol{A} \right \right } = \left \left \boldsymbol{x} \right - \left \boldsymbol{x}_k \right + \left \boldsymbol{u} \right \right _{\left \left \boldsymbol{A} \right \right } \geq \left \left \boldsymbol{x} \right - \left \boldsymbol{x}_k \right \right _{\left \left \boldsymbol{A} \right \right },$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}} = \left\ \boldsymbol{x} - \boldsymbol{x}_k + \boldsymbol{u} \right\ _{\boldsymbol{A}} \geq \left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _{\boldsymbol{A}},$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _{\boldsymbol{A}} = \left\ \boldsymbol{x} - \boldsymbol{x}_k + \boldsymbol{u} \right\ _{\boldsymbol{A}} \geq \left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _{\boldsymbol{A}},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00918	312	■ 0.547 ● W +0	$\frac{\left\ \boldsymbol{y} - \boldsymbol{y}_k \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\left\ \boldsymbol{y} - \boldsymbol{y}_0 \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$	$\frac{\left\ \boldsymbol{y} - \boldsymbol{y}_k \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\left\ \boldsymbol{y} - \boldsymbol{y}_0 \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$	$\frac{\left\ \boldsymbol{y} - \boldsymbol{y}_k \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}}{\left\ \boldsymbol{y} - \boldsymbol{y}_0 \right\ _{\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T}} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad \text{for } k \geq 0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00386	156	■ 0.548 ● N +0	$\begin{aligned} \boldsymbol{R} &:= \left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \boldsymbol{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{D}_3 \end{array} \right], \boldsymbol{D}_1 := \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right]. \\ \boldsymbol{D}_{\{1\}} &:= \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_{\{2\}} := [1], \boldsymbol{D}_{\{3\}} := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right]. \end{aligned}$	$\boldsymbol{R} := \left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \boldsymbol{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{D}_3 \end{array} \right], \boldsymbol{D}_1 := \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right].$	$\boldsymbol{R} := \left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} \\ \boldsymbol{0} & \boldsymbol{D}_2 & \boldsymbol{R}_{2,3} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{D}_3 \end{array} \right], \boldsymbol{D}_1 := \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array} \right], \boldsymbol{D}_2 := [1], \boldsymbol{D}_3 := \left[\begin{array}{cc} 3 & 2 \\ -1 & 1 \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00185	079	■ 0.548 ● W +0	$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)} \text{ for } k < \min(i, j), \text{ and } l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)} \text{ for } i > j.$	$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)} \text{ for } k < \min(i, j), \text{ and } l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)} \text{ for } i > j.$	$l_{ik}^{(k)} a_{kj}^{(k)} = a_{ij}^{(k)} - a_{ij}^{(k+1)} \text{ for } k < \min(i, j), \text{ and } l_{ij}^{(k)} a_{jj}^{(j)} = a_{ij}^{(j)} \text{ for } i > j.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0313	126	■0.549 ●N +1	\begin{aligned} \boldsymbol{H} \boldsymbol{x} &= \\ \boldsymbol{x}-\left(\boldsymbol{u}^*\boldsymbol{u}\right) \boldsymbol{x}&=\rho\ \boldsymbol{x}\ _2\left(z-\left(\boldsymbol{u}^*\boldsymbol{z}\right)\boldsymbol{u}\right)=\rho\ \boldsymbol{x}\ _2\left(z-\frac{\left(\boldsymbol{z}^*+\boldsymbol{e}_1^*\right)\boldsymbol{z}}{1+z_1}\left(z+\boldsymbol{e}_1\right)\right) \\&=\rho\ \boldsymbol{x}\ _2\left(z-\left(z+\boldsymbol{e}_1\right)\right)=-\rho\ \boldsymbol{x}\ _2\boldsymbol{e}_1=a\boldsymbol{e}_1.\end{aligned}	$\boldsymbol{Hx} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u} = \rho \ \boldsymbol{x} \ _2 (z - (\boldsymbol{u}^* \boldsymbol{z}) \boldsymbol{u}) = \rho \ \boldsymbol{x} \ _2 \left(z - \frac{(z^* + \boldsymbol{e}_1^*) \boldsymbol{z}}{1 + z_1} (z + \boldsymbol{e}_1) \right)$ $= \rho \ \boldsymbol{x} \ _2 (z - (z + \boldsymbol{e}_1)) = -\rho \ \boldsymbol{x} \ _2 \boldsymbol{e}_1 = a \boldsymbol{e}_1$	$\boldsymbol{Hx} = \boldsymbol{x} - (\boldsymbol{u}^* \boldsymbol{x}) \boldsymbol{u} = \rho \ \boldsymbol{x} \ _2 (z - (\boldsymbol{u}^* \boldsymbol{z}) \boldsymbol{u}) = \rho \ \boldsymbol{x} \ _2 (z - \frac{(z^* + \boldsymbol{e}_1^*) \boldsymbol{z}}{1 + z_1} (z + \boldsymbol{e}_1))$ $= \rho \ \boldsymbol{x} \ _2 (z - (z + \boldsymbol{e}_1)) = -\rho \ \boldsymbol{x} \ _2 \boldsymbol{e}_1 = a \boldsymbol{e}_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0129 (89)	063	■0.550 ●N +0	\boldsymbol{A}_{11}=[1], \quad \boldsymbol{A}_{12}=[2,3], \quad \boldsymbol{A}_{21}=\left[\begin{array}{l} 4 \\ 7 \end{array}\right], \quad \text{and } \boldsymbol{A}_{22}=\left[\begin{array}{ll} 5 & 6 \\ 8 & 9 \end{array}\right],	$\boldsymbol{A}_{11} = [1], \quad \boldsymbol{A}_{12} = [2, 3], \quad \boldsymbol{A}_{21} = \left[\begin{array}{c} 4 \\ 7 \end{array} \right], \quad \text{and } \boldsymbol{A}_{22} = \left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array} \right],$	$\boldsymbol{A}_{11} = [1], \quad \boldsymbol{A}_{12} = [2, 3], \quad \boldsymbol{A}_{21} = \left[\begin{array}{c} 4 \\ 7 \end{array} \right], \quad \text{and } \boldsymbol{A}_{22} = \left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0117 (2.27)	060	■0.551 ●C -5	\boldsymbol{T}=\boldsymbol{T}_{\mathrm m}:=\operatorname{tridiag}_{\mathrm m}(-1,2,-1)=\left[\begin{array}{ccccccc} 2&-1&0&&&&\\-1&2&-1&&&&\\0&\ddots&\ddots&\ddots&&&\\&&0&&&&\\&&&-1&2&-1&\\&&&0&-1&&2\end{array}\right]\in\mathbb R^{m\times m}.	$\boldsymbol{T} = \boldsymbol{T}_m := \operatorname{tridiag}_m(-1, 2, -1) = \left[\begin{array}{ccccccc} 2 & -1 & 0 & & & & \\ -1 & 2 & -1 & & & & \\ 0 & \ddots & \ddots & \ddots & & & \\ & & 0 & & & & \\ & & & -1 & 2 & -1 & \\ & & & 0 & -1 & & 2 \end{array} \right] \in \mathbb{R}^{m \times m}.$	$\boldsymbol{T} = \boldsymbol{T}_m := \operatorname{tridiag}_m(-1, 2, -1) = \left[\begin{array}{ccccccc} 2 & -1 & 0 & & & & \\ -1 & 2 & -1 & & & & \\ 0 & \ddots & \ddots & \ddots & & & \\ & & 0 & & & & \\ & & & -1 & 2 & -1 & \\ & & & 0 & -1 & & 2 \end{array} \right] \in \mathbb{R}^{m \times m}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0577	214	■0.552 ●N +0	\boldsymbol{Ax}=\left[\begin{array}{c} p\left(t_1\right) \\ \vdots \\ p\left(t_m\right) \end{array}\right]=\left[\begin{array}{cc} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \end{array}\right]=\left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array}\right]=\boldsymbol{b}.	$\boldsymbol{Ax} = \left[\begin{array}{c} p(t_1) \\ \vdots \\ p(t_m) \end{array} \right] = \left[\begin{array}{cc} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$	$\boldsymbol{Ax} = \left[\begin{array}{c} p(t_1) \\ \vdots \\ p(t_m) \end{array} \right] = \left[\begin{array}{cc} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0569	211	■0.554 ●K +0	\hat{\boldsymbol{s}}=\boldsymbol{b}+\boldsymbol{e}.	$\boldsymbol{A}\hat{\boldsymbol{s}} = \boldsymbol{b} + \boldsymbol{e}.$	$\boldsymbol{A}\hat{\boldsymbol{s}} = \boldsymbol{b} + \boldsymbol{e}.$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00336	136	0.557 C -8	$\begin{aligned} \mathbf{A} = & \left[\begin{array}{cccc} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right] \xrightarrow{P_{12}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array} \right] \xrightarrow{P_{23}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{array} \right] \xrightarrow{P_{34}} \left[\begin{array}{cccc} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{array} \right]. \end{aligned}$	$\mathbf{A} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \xrightarrow{P_{12}} \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ \mathbf{0} & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \xrightarrow{P_{23}} \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & \mathbf{0} & x & x \\ 0 & 0 & x & x \end{bmatrix} \xrightarrow{P_{34}} \begin{bmatrix} r_{11} & r_{12} & r_{13} & r_{14} \\ 0 & r_{22} & r_{23} & r_{24} \\ 0 & 0 & r_{33} & r_{34} \\ 0 & 0 & \mathbf{0} & r_{44} \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01015	344	0.561 N +0	$\mathbf{A} = \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right], \quad \mathbf{z}_0 := \left[\begin{array}{c} 1 \\ 0 \end{array} \right].$	$\mathbf{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{z}_0 := \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00919	312	0.563 N +0	$\begin{aligned} & \left[\begin{array}{c} \mathbf{y} - \mathbf{y}_k \\ \mathbf{C}^T \mathbf{A} \end{array} \right]^2 = \left(\mathbf{y} - \mathbf{y}_k \right)^T \left(\mathbf{C} \mathbf{A} \mathbf{C}^T \right) \left(\mathbf{y} - \mathbf{y}_k \right) \\ & = \left(\mathbf{C}^T \left(\mathbf{y} - \mathbf{y}_k \right) \right)^T \mathbf{A} \left(\mathbf{C}^T \left(\mathbf{y} - \mathbf{y}_k \right) \right) = \left\ \mathbf{x} - \mathbf{x}_k \right\ _{\mathbf{A}}^2, \end{aligned}$	$\begin{aligned} \left\ \mathbf{y} - \mathbf{y}_k \right\ _{\mathbf{C} \mathbf{A} \mathbf{C}^T}^2 &= \left(\mathbf{y} - \mathbf{y}_k \right)^T \left(\mathbf{C} \mathbf{A} \mathbf{C}^T \right) \left(\mathbf{y} - \mathbf{y}_k \right) \\ &= \left(\mathbf{C}^T \left(\mathbf{y} - \mathbf{y}_k \right) \right)^T \mathbf{A} \left(\mathbf{C}^T \left(\mathbf{y} - \mathbf{y}_k \right) \right) = \left\ \mathbf{x} - \mathbf{x}_k \right\ _{\mathbf{A}}^2, \end{aligned}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00132	063	0.564 N +0	$\mathbf{A} \mathbf{B} = \left[\begin{array}{c} a_{1:}^T \\ a_{2:}^T \\ \vdots \\ a_{m:}^T \end{array} \right] \mathbf{B} = \left[\begin{array}{c} a_{1:}^T \mathbf{B} \\ a_{2:}^T \mathbf{B} \\ \vdots \\ a_{m:}^T \mathbf{B} \end{array} \right],$	$\mathbf{A} \mathbf{B} = \begin{bmatrix} a_{1:}^T \\ a_{2:}^T \\ \vdots \\ a_{m:}^T \end{bmatrix} \mathbf{B} = \begin{bmatrix} a_{1:}^T \mathbf{B} \\ a_{2:}^T \mathbf{B} \\ \vdots \\ a_{m:}^T \mathbf{B} \end{bmatrix},$	
Conf. MathPix confidence: 0.9 0.5-0.9 0.5 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0425 (7.4)	171	■ 0.564 ● W -1	$\begin{aligned} & \boldsymbol{U} = [\boldsymbol{U}_1, \boldsymbol{U}_2] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1 := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], \quad \boldsymbol{U}_2 := [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], \\ & \boldsymbol{V} = [\boldsymbol{V}_1, \boldsymbol{V}_2] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1 := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], \quad \boldsymbol{V}_2 := [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], \\ & \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{bmatrix} \in \mathbb{R}^{m \times n}, \text{ where } \boldsymbol{\Sigma}_1 := \text{diag}(\sigma_1, \dots, \sigma_r). \end{aligned}$	$\boldsymbol{U} = [\boldsymbol{U}_1, \boldsymbol{U}_2] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1 := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], \quad \boldsymbol{U}_2 := [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], \\ \boldsymbol{V} = [\boldsymbol{V}_1, \boldsymbol{V}_2] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1 := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], \quad \boldsymbol{V}_2 := [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], \\ \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{bmatrix} \in \mathbb{R}^{m \times n}, \text{ where } \boldsymbol{\Sigma}_1 := \text{diag}(\sigma_1, \dots, \sigma_r).$	$\boldsymbol{U} = [\boldsymbol{U}_1, \boldsymbol{U}_2] \in \mathbb{C}^{m \times m}, \quad \boldsymbol{U}_1 := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r], \quad \boldsymbol{U}_2 := [\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m], \\ \boldsymbol{V} = [\boldsymbol{V}_1, \boldsymbol{V}_2] \in \mathbb{C}^{n \times n}, \quad \boldsymbol{V}_1 := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r], \quad \boldsymbol{V}_2 := [\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n], \\ \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0}_{r, n-r} \\ \boldsymbol{0}_{m-r, r} & \boldsymbol{0}_{m-r, n-r} \end{bmatrix} \in \mathbb{R}^{m \times n}, \text{ where } \boldsymbol{\Sigma}_1 := \text{diag}(\sigma_1, \dots, \sigma_r).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1016	345	■ 0.571 ● N +0	$\begin{aligned} z_1 &= \boldsymbol{A} z_0 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad z_2 = \boldsymbol{A} z_1 = \begin{bmatrix} 5 \\ -4 \end{bmatrix}, \dots, z_k = \frac{1}{2} \begin{bmatrix} 1+3^k \\ 1-3^k \end{bmatrix}, \dots \end{aligned}$	$z_1 = \boldsymbol{A} z_0 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad z_2 = \boldsymbol{A} z_1 = \begin{bmatrix} 5 \\ -4 \end{bmatrix}, \dots, z_k = \frac{1}{2} \begin{bmatrix} 1+3^k \\ 1-3^k \end{bmatrix}, \dots$	$z_1 = \boldsymbol{A} z_0 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad z_2 = \boldsymbol{A} z_1 = \begin{bmatrix} 5 \\ -4 \end{bmatrix}, \dots, z_k = \frac{1}{2} \begin{bmatrix} 1+3^k \\ 1-3^k \end{bmatrix}, \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0834	289	■ 0.578 ● N +1	$\boldsymbol{A} := \begin{bmatrix} \lambda & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & \lambda \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A} := \begin{bmatrix} \lambda & 0 & \cdots & 0 & 0 \\ 0 & \lambda & a & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & \lambda \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A} := \begin{bmatrix} \lambda & a & 0 & \cdots & 0 \\ 0 & \lambda & a & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & \lambda \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{n \times n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0831 (12.47)	287	■ 0.581 ● N +0	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}.$	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}.$	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{x}^T \boldsymbol{N} \boldsymbol{x}}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0420	169	■ 0.585 ● K +0	$\lambda^m \pi_{\boldsymbol{A}^* \boldsymbol{A}}(\lambda) = \lambda^n \pi_{\boldsymbol{A} \boldsymbol{A}^*}(\lambda), \quad \lambda \in \mathbb{C},$	$\lambda^m \pi_{\boldsymbol{A}^* \boldsymbol{A}}(\lambda) = \lambda^n \pi_{\boldsymbol{A} \boldsymbol{A}^*}(\lambda), \quad \lambda \in \mathbb{C},$	$\lambda^m \pi_{\boldsymbol{A}^* \boldsymbol{A}}(\lambda) = \lambda^n \pi_{\boldsymbol{A} \boldsymbol{A}^*}(\lambda), \quad \lambda \in \mathbb{C},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00709	254	0.587 N +0	$\begin{aligned} & \& \boldsymbol{X} = h^4 \boldsymbol{G} / \boldsymbol{M}, \text{ where } \boldsymbol{M} := \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_m \end{array} \right] [1, \dots, 1] + \left[\begin{array}{c} 1 \\ \vdots \\ i \\ \vdots \\ m \end{array} \right] [\sigma_1 \dots \sigma_m], \\ & \boldsymbol{S} = \sin \left(\pi h \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right] [12 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right] \right) \wedge 2. \end{aligned}$	$\boldsymbol{X} = h^4 \boldsymbol{G} / \boldsymbol{M}, \text{ where } \boldsymbol{M} := \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_m \end{array} \right] [1, \dots, 1] + \left[\begin{array}{c} 1 \\ \vdots \\ i \\ \vdots \\ m \end{array} \right] [\sigma_1 \dots \sigma_m],$ $\boldsymbol{S} = \sin \left(\pi h \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right] [12 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \left[\begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} \right] \right) \wedge 2.$	$\boldsymbol{X} = h^4 \boldsymbol{G} / \boldsymbol{M}, \text{ where } \boldsymbol{M} := \left[\begin{array}{c} \sigma_1 \\ \vdots \\ \sigma_m \end{array} \right] [1, \dots, 1] + \left[\begin{array}{c} 1 \\ \vdots \\ i \\ \vdots \\ m \end{array} \right] [\sigma_1 \dots \sigma_m],$ $\boldsymbol{S} = \sin \left(\pi h \left[\begin{array}{c} \frac{1}{2} \\ \vdots \\ m \end{array} \right] [1 \ 2 \dots m] \right), \quad \boldsymbol{\sigma} = \sin \left(\frac{\pi h}{2} \left[\begin{array}{c} \frac{1}{2} \\ \vdots \\ m \end{array} \right] \right) \wedge 2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00644	232	0.587 W +0	$\begin{array}{l} \boldsymbol{A} \boldsymbol{B} \boldsymbol{A} = \boldsymbol{A} \quad (1) \quad \boldsymbol{A} \boldsymbol{C} \boldsymbol{A} = \boldsymbol{A}, \\ \boldsymbol{B} \boldsymbol{A} \boldsymbol{B} = \boldsymbol{B} \quad (2) \quad \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}, \\ (\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{A} \boldsymbol{B} \quad (3) \quad (\boldsymbol{A} \boldsymbol{C})^* = \boldsymbol{A} \boldsymbol{C}, \\ (\boldsymbol{B} \boldsymbol{A})^* = \boldsymbol{B} \boldsymbol{A} \quad (4) \quad (\boldsymbol{C} \boldsymbol{A})^* = \boldsymbol{C} \boldsymbol{A}. \end{array}$	$\begin{array}{l} \boldsymbol{A} \boldsymbol{B} \boldsymbol{A} = \boldsymbol{A} \quad (1) \quad \boldsymbol{A} \boldsymbol{C} \boldsymbol{A} = \boldsymbol{A}, \\ \boldsymbol{B} \boldsymbol{A} \boldsymbol{B} = \boldsymbol{B} \quad (2) \quad \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}, \\ (\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{A} \boldsymbol{B} \quad (3) \quad (\boldsymbol{A} \boldsymbol{C})^* = \boldsymbol{A} \boldsymbol{C}, \\ (\boldsymbol{B} \boldsymbol{A})^* = \boldsymbol{B} \boldsymbol{A} \quad (4) \quad (\boldsymbol{C} \boldsymbol{A})^* = \boldsymbol{C} \boldsymbol{A}. \end{array}$	$\begin{array}{l} \boldsymbol{A} \boldsymbol{B} \boldsymbol{A} = \boldsymbol{A} \quad (1) \quad \boldsymbol{A} \boldsymbol{C} \boldsymbol{A} = \boldsymbol{A}, \\ \boldsymbol{B} \boldsymbol{A} \boldsymbol{B} = \boldsymbol{B} \quad (2) \quad \boldsymbol{C} \boldsymbol{A} \boldsymbol{C} = \boldsymbol{C}, \\ (\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{A} \boldsymbol{B} \quad (3) \quad (\boldsymbol{A} \boldsymbol{C})^* = \boldsymbol{A} \boldsymbol{C}, \\ (\boldsymbol{B} \boldsymbol{A})^* = \boldsymbol{B} \boldsymbol{A} \quad (4) \quad (\boldsymbol{C} \boldsymbol{A})^* = \boldsymbol{C} \boldsymbol{A}. \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00171	076	0.588 W +0	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \quad \text{III}', \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \text{III}', \end{aligned}$	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}' \\ & a_{32}^{(2)} x_2 + a_{33}^{(2)} x_3 = b_3^{(2)}, \quad \text{III}', \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00467 (8.4)	186	0.589 N -1	$\lim_{p \rightarrow \infty} \ \boldsymbol{x}\ _p = \ \boldsymbol{x}\ _\infty \text{ for all } \boldsymbol{x} \in \mathbb{C}^n.$	$\lim_{p \rightarrow \infty} \ \boldsymbol{x}\ _p = \ \boldsymbol{x}\ _\infty \text{ for all } \boldsymbol{x} \in \mathbb{C}^n.$	$\lim_{p \rightarrow \infty} \ \boldsymbol{x}\ _p = \ \boldsymbol{x}\ _\infty \text{ for all } \boldsymbol{x} \in \mathbb{C}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00793	279	0.592 N +0	$k = -\frac{s \log(10)}{\log(1 - \eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$	$k = -\frac{s \log(10)}{\log(1 - \eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$	$k = -\frac{s \log(10)}{\log(1 - \eta)} \approx \frac{s \log(10)}{\eta} = \tilde{k}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01023 (15.6)	346	0.600 W +0	$\begin{array}{l} \text{(i)} \quad \boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\begin{array}{l} \text{(i)} \quad \boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\begin{array}{l} \text{(i)} \quad \boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} \\ \text{(ii)} \quad \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00935	317	0.600 N +0	$\lambda=\frac{x^T}{x^T A_p x} \frac{\boldsymbol{A} \, x}{x}$	$\lambda = \frac{x^T \boldsymbol{A} \boldsymbol{x}}{x^T \boldsymbol{A}_p x}.$	$\lambda = \frac{x^T \boldsymbol{A} \boldsymbol{x}}{x^T \boldsymbol{A}_p x}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00901	308	0.602 W +2	$\begin{aligned} & \left \left \boldsymbol{\epsilon}_k \right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k \right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_k \right \right _{\boldsymbol{A}}^2 - 2 \alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k \\ & \left \left \boldsymbol{\epsilon}_{k+1} \right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k \right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_k \right \right _{\boldsymbol{A}}^2 - 2 \alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k \\ & \left \left \boldsymbol{\epsilon}_{k+1} \right \right _{\boldsymbol{A}}^2 = \left \left \boldsymbol{\epsilon}_k \right \right _{\boldsymbol{A}}^2 - 2 \alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k \end{aligned}$	$\left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k,$ $\left\ \boldsymbol{\epsilon}_{k+1} \right\ _{\boldsymbol{A}}^2 = (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)$ $= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2 \alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}.$	$\left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k = \boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k,$ $\left\ \boldsymbol{\epsilon}_{k+1} \right\ _{\boldsymbol{A}}^2 = (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)^T \boldsymbol{A} (\boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k)$ $= \boldsymbol{\epsilon}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k - 2 \alpha_k \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{\epsilon}_k + \alpha_k^2 \boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k = \left\ \boldsymbol{\epsilon}_k \right\ _{\boldsymbol{A}}^2 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00892 (13.37)	307	0.606 W +0	$a:=\frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b:=\frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$	$a := \frac{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n t_i \lambda_i, \quad b := \frac{\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}}{\boldsymbol{y}^T \boldsymbol{y}} = \sum_{i=1}^n \frac{t_i}{\lambda_i},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01029	348	0.608 K +0	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$	$\boldsymbol{A}_1 := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \boldsymbol{A}_2 := \begin{bmatrix} 1.7 & -0.4 \\ 0.15 & 2.2 \end{bmatrix}, \quad \text{and } \boldsymbol{A}_3 = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00379	154	0.611 W +0	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and } \boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and } \boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$	$\boldsymbol{A} \boldsymbol{A}^* = (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B} \boldsymbol{B}^* \boldsymbol{U}^* \text{ and } \boldsymbol{A}^* \boldsymbol{A} = (\boldsymbol{U} \boldsymbol{B}^* \boldsymbol{U}^*) (\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*) = \boldsymbol{U} \boldsymbol{B}^* \boldsymbol{B} \boldsymbol{U}^*.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1031 (15.7)	349	■ 0.611 ● W +0	$\begin{array}{ll} \text{ (i) } & (\boldsymbol{A} - s\boldsymbol{I})\boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\begin{array}{ll} \text{ (i) } & (\boldsymbol{A} - s\boldsymbol{I})\boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$	$\begin{array}{ll} \text{ (i) } & (\boldsymbol{A} - s\boldsymbol{I})\boldsymbol{y}_k = \boldsymbol{x}_{k-1} \\ \text{ (ii) } & \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ . \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1011	341	■ 0.614 ● K +0	$\boldsymbol{A} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{E} := \left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array} \right].$	$\boldsymbol{A} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{E} := \left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array} \right].$	$\boldsymbol{A} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{E} := \left[\begin{array}{llll} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \epsilon & 0 & 0 & 0 \end{array} \right], \quad \boldsymbol{B} := \left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0011	024	■ 0.618 ● N +0	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \left(\begin{array}{cccc} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{array} \right) = \left[\begin{array}{cccc} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{array} \right].$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \left(\begin{array}{cccc} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{array} \right) = \left[\begin{array}{cccc} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{array} \right].$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) := \boldsymbol{A} \left(\begin{array}{cccc} i_1 & i_2 & \cdots & i_r \\ j_1 & j_2 & \cdots & j_c \end{array} \right) = \left[\begin{array}{cccc} a_{i_1, j_1} & a_{i_1, j_2} & \cdots & a_{i_1, j_c} \\ a_{i_2, j_1} & a_{i_2, j_2} & \cdots & a_{i_2, j_c} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i_r, j_1} & a_{i_r, j_2} & \cdots & a_{i_r, j_c} \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0249 (4.4)	102	■ 0.629 ● W +0	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$	$\boldsymbol{a}_n = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n, \quad \boldsymbol{a}_{nn} = \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00849	295	■ 0.636 ● W +0	$\begin{aligned} & \boldsymbol{r}_{k+1}^T \boldsymbol{r}_j \stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j \\ & \stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right) \\ & \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0 \quad \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k. \end{aligned}$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j \stackrel{(13.13)}{=} (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k)^T \boldsymbol{r}_j$ $\stackrel{(13.11)}{=} \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \left(\boldsymbol{p}_j + \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i \right)$ $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0 \quad \boldsymbol{r}_k^T \boldsymbol{r}_j - \alpha_k \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_j = 0, \quad j = 0, 1, \dots, k.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00592 (9.4)	218	■ 0.637 ● K +0	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^*) .$	$\mathbb{C}^m = \mathcal{R}(\boldsymbol{A}) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A}^*) .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00553 (8.42)	208	■ 0.638 ● N +1	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd} \\ h^{-2} - 1, & m \text{ even} \end{cases}$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_\infty(\boldsymbol{T}) = \frac{1}{2} \begin{cases} h^{-2}, & m \text{ odd}, \\ h^{-2} - 1, & m \text{ even}. \end{cases}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01032	349	■ 0.642 ● W +0	$\begin{aligned} & \text{(i) } \boldsymbol{A} - s_{k-1} \boldsymbol{I} \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \text{(ii) } \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{(iii) } \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{(iv) } \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k. \end{aligned}$	$\begin{aligned} & \text{(i) } (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k = \boldsymbol{x}_{k-1}, \\ & \text{(ii) } \boldsymbol{x}_k = \boldsymbol{y}_k / \ \boldsymbol{y}_k\ , \\ & \text{(iii) } \boldsymbol{s}_k = \boldsymbol{x}_k^* \boldsymbol{A} \boldsymbol{x}_k, \\ & \text{(iv) } \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k. \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00698	250	■0.645 ●W +0	$\begin{aligned} \boldsymbol{A} &= \left[\begin{array}{rrr rrr rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{bmatrix}, \end{aligned}$	$\boldsymbol{A} = \left[\begin{array}{rrr rrr rrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 & 0 \\ \hline -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 & -1 \\ \hline 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right] \\ = \begin{bmatrix} \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} & -\boldsymbol{I} \\ \boldsymbol{0} & -\boldsymbol{I} & \boldsymbol{T} + 2\boldsymbol{I} \end{bmatrix},$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00641	232	■0.645 ●N +0	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$	$\boldsymbol{r} = \boldsymbol{r}(x) = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00601 (9.7)	222	■0.650 ●N -1	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = [\boldsymbol{U}_1, \boldsymbol{U}_2] \begin{bmatrix} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_1^* \\ \boldsymbol{V}_2^* \end{bmatrix} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{\Sigma}_1 = \text{diag}(\sigma_1, \dots, \sigma_r),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00201 (3.11)	085	■0.651 ●N -1	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$	$\boldsymbol{p} := \boldsymbol{p}_n, \text{ where } \boldsymbol{p}_1 := [1, 2, \dots, n]^T, \text{ and } \boldsymbol{p}_{k+1} := \boldsymbol{I}_{r_k, k} \boldsymbol{p}_k \text{ for } k = 1, \dots, n-1.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00947	320	0.654 ● N +0	$\begin{aligned} & \left[\begin{array}{cccc} \mathbf{x}_0, & \mathbf{x}_1, & \mathbf{x}_2, & \mathbf{x}_3 \end{array} \right] = \left[\begin{array}{cccc} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right], \\ & \left[\begin{array}{cccc} \mathbf{r}_0, & \mathbf{r}_1, & \mathbf{r}_2, & \mathbf{r}_3 \end{array} \right] = \left[\begin{array}{cccc} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{array} \right], \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc} \mathbf{x}_0, & \mathbf{x}_1, & \mathbf{x}_2, & \mathbf{x}_3 \end{array} \right] = \left[\begin{array}{cccc} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right], \\ & \left[\begin{array}{cccc} \mathbf{r}_0, & \mathbf{r}_1, & \mathbf{r}_2, & \mathbf{r}_3 \end{array} \right] = \left[\begin{array}{cccc} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{array} \right], \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc} \mathbf{x}_0, & \mathbf{x}_1, & \mathbf{x}_2, & \mathbf{x}_3 \end{array} \right] = \left[\begin{array}{cccc} 0 & 2 & 8/3 & 3 \\ 0 & 0 & 4/3 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right], \\ & \left[\begin{array}{cccc} \mathbf{r}_0, & \mathbf{r}_1, & \mathbf{r}_2, & \mathbf{r}_3 \end{array} \right] = \left[\begin{array}{cccc} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4/3 & 0 \end{array} \right], \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00415	166	0.666 ● N +0	$R(\mathbf{x}-t\mathbf{y})=R(\mathbf{x})-2t(\mathbf{A}\mathbf{x}-R(\mathbf{x})\mathbf{x})^T\mathbf{y}+\mathcal{O}(t^2),$	$R(\mathbf{x}-t\mathbf{y})=R(\mathbf{x})-2t(\mathbf{A}\mathbf{x}-R(\mathbf{x})\mathbf{x})^T\mathbf{y}+\mathcal{O}(t^2),$	$R(\mathbf{x}-t\mathbf{y})=R(\mathbf{x})-2t(\mathbf{A}\mathbf{x}-R(\mathbf{x})\mathbf{x})^T\mathbf{y}+\mathcal{O}(t^2),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00602	222	0.667 ● C +16	$\begin{aligned} & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{U}_1=\mathbf{I}, \quad \mathbf{U}_2^*\mathbf{U}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{U}_1=\mathbf{I}, \quad \mathbf{U}_2^*\mathbf{U}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{U}_1=\mathbf{I}, \quad \mathbf{U}_2^*\mathbf{U}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{U}_1=[\mathbf{u}_1,\dots,\mathbf{u}_r], \quad \mathbf{U}_2=[\mathbf{u}_{r+1},\dots,\mathbf{u}_m], \quad \mathbf{U}_1^*\mathbf{U}_1=\mathbf{I}, \quad \mathbf{U}_2^*\mathbf{U}_2=\mathbf{I}, \\ \mathbf{V}_1=[\mathbf{v}_1,\dots,\mathbf{v}_r], \quad \mathbf{V}_2=[\mathbf{v}_{r+1},\dots,\mathbf{v}_n], \quad \mathbf{V}_1^*\mathbf{V}_1=\mathbf{I}, \quad \mathbf{V}_2^*\mathbf{V}_2=\mathbf{I} \end{array} \right] \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00751	267	0.670 ● W +0	$\begin{aligned} & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \\ & \left[\begin{array}{l} \mathbf{J}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_k(i-1,j)+\mathbf{v}_k(i,j-1)+\mathbf{v}_k(i+1,j)+\mathbf{v}_k(i,j+1)+\mathbf{e}(i,j)\right)/4 \\ \mathbf{GS}: \mathbf{v}_{k+1}(i,j)=\left(\mathbf{v}_{k+1}(i-1,j)+\mathbf{v}_{k+1}(i,j-1)+\mathbf{v}_{k+1}(i+1,j)+\mathbf{v}_{k+1}(i,j+1)+\mathbf{e}(i,j)\right)/4 \end{array} \right] \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00190 (3.8)	080	0.672 ● N +0	$\mathbf{x}_k=\left(\mathbf{b}_k-\sum_{j=l_k}^{k-1}a_{k,j}\mathbf{x}_j\right)/a_{kk}, \quad k=1,2,\dots,n.$	$x_k=\left(b_k-\sum_{j=l_k}^{k-1}a_{k,j}x_j\right)/a_{kk}, \quad k=1,2,\dots,n.$	$x_k=\left(b_k-\sum_{j=l_k}^{k-1}a_{k,j}x_j\right)/a_{kk}, \quad k=1,2,\dots,n.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00719	257	■ 0.674 ● W +0	$\begin{aligned} \left(\boldsymbol{F}_{2m+2}\boldsymbol{z}\right)_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$	$\begin{aligned} (\boldsymbol{F}_{2m+2}\boldsymbol{z})_{k+1} &= \sum_{j=0}^{2m-1} \omega^{jk} z_{j+1} = \sum_{j=1}^m x_j \omega^{jk} - \sum_{j=1}^m x_j \omega^{(2m+2-j)k} \\ &= \sum_{j=1}^m x_j \left(e^{-\pi i j k / (m+1)} - e^{\pi i j k / (m+1)} \right) \\ &= -2i \sum_{j=1}^m x_j \sin \left(\frac{jk\pi}{m+1} \right) = -2i (\boldsymbol{S}_m \boldsymbol{x})_k. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00599	221	■ 0.676 ● N -1	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ccc} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] \text{ and } \boldsymbol{b}=\left[\begin{array}{l} 1 \\ 1 \\ 1 \\ 1 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00726 (11.7)	258	■ 0.678 ● W +0	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right],$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right],$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00513	200	■ 0.679 ● N +1	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1}. \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1} \end{aligned}$	$\begin{aligned} f(x) &= \frac{x_2-x}{x_2-x_1} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2) + (x-x_1)(x-x_2) f''(c)/2 \\ &= (1-\lambda) f(x_1) + \lambda f(x_2) + (x_2-x_1)^2 \lambda (\lambda-1) f''(c)/2, \quad \lambda := \frac{x-x_1}{x_2-x_1}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00607	222	■ 0.679 ● W +1	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger \end{aligned}$	$\begin{aligned} \boldsymbol{A} \boldsymbol{A}^\dagger \boldsymbol{A} &= \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{V}_1^* = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{A} \\ \boldsymbol{A}^\dagger \boldsymbol{A} \boldsymbol{A}^\dagger &= \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{A}^\dagger. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00038	035	■0.682 ● N +1	\begin{aligned} &\&left \begin{array}{lll} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{array} \\ &\& a_{13} \& \& a_{21} \& a_{22} \& a_{23} \& \& a_{31} \& a_{32} \\ &\& a_{33} \end{array} \right = a_{11} a_{22} a_{33} - a_{11} a_{32} a_{23} - a_{21} a_{12} a_{33} \\ &+ a_{21} a_{32} a_{13} + a_{31} a_{12} a_{23} - a_{31} a_{22} a_{13} \end{aligned}	$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00383	155	■0.685 ● K +0	R(x)=R_{\{A\}}(x):=\frac{x^{[*]} \ A \ x}{x^{[*]} \ x}	$R(x) = R_A(x) := \frac{x^*Ax}{x^*x}$	$R(x) = R_A(x) := \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00578	214	■0.686 ● C -6	c_{\{1\}}\left[\begin{array}{c} 1 \\ \vdots \\ t_{\{m\}} \end{array}\right]+c_{\{2\}}\left[\begin{array}{c} t_1 \\ \vdots \\ t_m \end{array}\right]=\left[\begin{array}{c} 0 \\ \vdots \\ 0 \end{array}\right]\Rightarrow \left[\begin{array}{cc} 1 & t_i \\ 1 & t_j \end{array}\right]\left[\begin{array}{c} c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \end{array}\right]\Rightarrow c_1=c_2=0.	$c_1\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + c_2\begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & t_i \\ 1 & t_j \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow c_1 = c_2 = 0.$	$c_1\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + c_2\begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & t_i \\ 1 & t_j \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow c_1 = c_2 = 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00535	203	■0.693 ● N +0	a_{\{n\}}\langle \boldsymbol{x},\boldsymbol{y}\rangle=\frac{1}{4}\left(\left\ a_n\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_n\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).	$a_n\langle \boldsymbol{x},\boldsymbol{y}\rangle = \langle a_n\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\left\ a_n\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_n\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$	$a_n\langle \boldsymbol{x},\boldsymbol{y}\rangle = \langle a_n\boldsymbol{x},\boldsymbol{y}\rangle \stackrel{(8.36)}{=} \frac{1}{4}\left(\left\ a_n\boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ a_n\boldsymbol{x}-\boldsymbol{y}\right\ ^2\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01026	347	■0.696 ● N +0	\left[\begin{array}{c} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{array}\right](\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u})=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{U}^*\boldsymbol{u} \end{array}\right]=\left[\begin{array}{c} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{array}\right].	$\begin{bmatrix} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{bmatrix}(\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}) = \begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{U}^*\boldsymbol{u} \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u} - \lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{bmatrix}.$	$\begin{bmatrix} \boldsymbol{u}^* \\ \boldsymbol{U}^* \end{bmatrix}(\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}) = \begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}^*\boldsymbol{u} \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{U}^*\boldsymbol{u} \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u} - \lambda \\ \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00410 (6.19)	164	■0.702 ● W +0	\mu_{\{\boldsymbol{A}\}}(\lambda):=\prod_{i=1}^k\left(\lambda_i-\lambda\right)^{m_i}\text{ where }\text{\textit{m}}_i:=\max_{1\leq j\leq g_i}\text{\textit{m}}_{i,j},	$\mu_A(\lambda) := \prod_{i=1}^k (\lambda_i - \lambda)^{m_i} \text{ where } m_i := \max_{1 \leq j \leq g_i} m_{i,j},$	$\mu_A(\lambda) := \prod_{i=1}^k (\lambda_i - \lambda)^{m_i} \text{ where } m_i := \max_{1 \leq j \leq g_i} m_{i,j},$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00699 (11.1)	251	■ 0.706 ● C +4	$\left[\begin{array}{cccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & \boldsymbol{C}_2 & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & & \\ & \ddots & \ddots & \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m \end{array}\right] = \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_1 & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right] \left[\begin{array}{cccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & & \\ & \ddots & \ddots & \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_m \end{array}\right].$	$\left[\begin{array}{ccc} \boldsymbol{D}_1 & \boldsymbol{C}_1 & \\ \boldsymbol{A}_1 & \boldsymbol{D}_2 & \boldsymbol{C}_2 \\ & \ddots & \ddots \\ & & \boldsymbol{A}_{m-2} & \boldsymbol{D}_{m-1} & \boldsymbol{C}_{m-1} \\ & & \boldsymbol{A}_{m-1} & \boldsymbol{D}_m & \end{array}\right] = \left[\begin{array}{ccc} \boldsymbol{I} & & \\ \boldsymbol{L}_1 & \boldsymbol{I} & \\ & \ddots & \ddots \\ & & \boldsymbol{L}_{m-1} & \boldsymbol{I} \end{array}\right] \left[\begin{array}{ccc} \boldsymbol{U}_1 & \boldsymbol{C}_1 & \\ & \ddots & \ddots \\ & & \boldsymbol{U}_{m-1} & \boldsymbol{C}_{m-1} \\ & & & \boldsymbol{U}_m \end{array}\right].$	
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00717	256	■ 0.709 ● N +0	$(\boldsymbol{S} \boldsymbol{x})_k = \frac{i}{2} (\boldsymbol{F}_{2m+2} \boldsymbol{z})_{k+1}, \quad k = 1, \dots, m.$	$(\boldsymbol{S} \boldsymbol{x})_k = \frac{i}{2} (\boldsymbol{F}_{2m+2} \boldsymbol{z})_{k+1}, \quad k = 1, \dots, m.$	$(\boldsymbol{S} \boldsymbol{x})_k = \frac{i}{2} (\boldsymbol{F}_{2m+2} \boldsymbol{z})_{k+1}, \quad k = 1, \dots, m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00593	218	■ 0.712 ● N +0	$\ A \boldsymbol{x} - \boldsymbol{b}\ _2^2 = \ (A \boldsymbol{x} - \boldsymbol{b}_1) - \boldsymbol{b}_2\ _2^2 = \ A \boldsymbol{x} - \boldsymbol{b}_1\ _2^2 + \ \boldsymbol{b}_2\ _2^2 \geq \ \boldsymbol{b}_2\ _2^2$	$\ A \boldsymbol{x} - \boldsymbol{b}\ _2^2 = \ (A \boldsymbol{x} - \boldsymbol{b}_1) - \boldsymbol{b}_2\ _2^2 = \ A \boldsymbol{x} - \boldsymbol{b}_1\ _2^2 + \ \boldsymbol{b}_2\ _2^2 \geq \ \boldsymbol{b}_2\ _2^2$	$\ A \boldsymbol{x} - \boldsymbol{b}\ _2^2 = \ (A \boldsymbol{x} - \boldsymbol{b}_1) - \boldsymbol{b}_2\ _2^2 = \ A \boldsymbol{x} - \boldsymbol{b}_1\ _2^2 + \ \boldsymbol{b}_2\ _2^2 \geq \ \boldsymbol{b}_2\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00530 (8.38)	203	■ 0.713 ● W +1	$\begin{aligned} &4\angle x, z \stackrel{(8.36)}{=} \ x+z\ ^2 - \ x-z\ ^2 + \ y+z\ ^2 - \ y-z\ ^2 \\ &= \left\ \left(z + \frac{x+y}{2} \right) + \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) + \frac{y-x}{2} \right\ ^2 \\ &+ \left\ \left(z + \frac{x+y}{2} \right) - \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) - \frac{y-x}{2} \right\ ^2 \\ &\stackrel{(8.35)}{=} 2 \left\ z + \frac{x+y}{2} \right\ ^2 + 2 \left\ \frac{x-y}{2} \right\ ^2 - 2 \left\ z - \frac{x+y}{2} \right\ ^2 - 2 \left\ \frac{y-x}{2} \right\ ^2 \\ &\stackrel{(8.36)}{=} 8 \left\langle \frac{x+y}{2}, z \right\rangle \end{aligned}$	$\begin{aligned} &4\langle x, z \rangle + 4\langle y, z \rangle \stackrel{(8.36)}{=} \ x+z\ ^2 - \ x-z\ ^2 + \ y+z\ ^2 - \ y-z\ ^2 \\ &= \left\ \left(z + \frac{x+y}{2} \right) + \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) + \frac{y-x}{2} \right\ ^2 \\ &+ \left\ \left(z + \frac{x+y}{2} \right) - \frac{x-y}{2} \right\ ^2 - \left\ \left(z - \frac{x+y}{2} \right) - \frac{y-x}{2} \right\ ^2 \\ &\stackrel{(8.35)}{=} 2 \left\ z + \frac{x+y}{2} \right\ ^2 + 2 \left\ \frac{x-y}{2} \right\ ^2 - 2 \left\ z - \frac{x+y}{2} \right\ ^2 - 2 \left\ \frac{y-x}{2} \right\ ^2 \\ &\stackrel{(8.36)}{=} 8 \left\langle \frac{x+y}{2}, z \right\rangle \end{aligned}$	$\begin{aligned} &4\langle \boldsymbol{x}, \boldsymbol{z} \rangle + 4\langle \boldsymbol{y}, \boldsymbol{z} \rangle \stackrel{(8.36)}{=} \ \boldsymbol{x} + \boldsymbol{z}\ ^2 - \ \boldsymbol{x} - \boldsymbol{z}\ ^2 + \ \boldsymbol{y} + \boldsymbol{z}\ ^2 - \ \boldsymbol{y} - \boldsymbol{z}\ ^2 \\ &= \left\ \left(\boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) + \frac{\boldsymbol{x} - \boldsymbol{y}}{2} \right\ ^2 - \left\ \left(\boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) + \frac{\boldsymbol{y} - \boldsymbol{x}}{2} \right\ ^2 \\ &+ \left\ \left(\boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) - \frac{\boldsymbol{x} - \boldsymbol{y}}{2} \right\ ^2 - \left\ \left(\boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2} \right) - \frac{\boldsymbol{y} - \boldsymbol{x}}{2} \right\ ^2 \\ &\stackrel{(8.35)}{=} 2 \ \boldsymbol{z} + \frac{\boldsymbol{x} + \boldsymbol{y}}{2}\ ^2 + 2 \ \frac{\boldsymbol{x} - \boldsymbol{y}}{2}\ ^2 - 2 \ \boldsymbol{z} - \frac{\boldsymbol{x} + \boldsymbol{y}}{2}\ ^2 - 2 \ \frac{\boldsymbol{y} - \boldsymbol{x}}{2}\ ^2 \\ &\stackrel{(8.36)}{=} 8 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00615	226	■0.721 ●N +0	$\boldsymbol{A}^{\ast} \boldsymbol{A} = \left[\begin{array}{ll} 1 & 1 \\ 1 & 2 \end{array} \right], \boldsymbol{A}^{\dagger} = \left(\boldsymbol{A}^{\ast} \boldsymbol{A} \right)^{-1} \boldsymbol{A}^{\ast} = \left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 1 & 0 \end{array} \right].$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00460 (2)	182	■0.722 ●C +10	$\boldsymbol{A} := \left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array} \right] = \left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] \left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \left[\begin{array}{rrr} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{array} \right]$		$\boldsymbol{A} := \left[\begin{array}{rrr} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{array} \right] = \left[\begin{array}{rrrr} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] \left[\begin{array}{rrr} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right] \left[\begin{array}{rrr} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00736	261	■0.724 ●N -1	$\boldsymbol{T} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = \boldsymbol{C}, \text{ where } \boldsymbol{X} = \boldsymbol{V} \boldsymbol{S}, \text{ and } \boldsymbol{C} = \boldsymbol{h}^2 \boldsymbol{F} \boldsymbol{S}.$		$\boldsymbol{T} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = \boldsymbol{C}, \text{ where } \boldsymbol{X} = \boldsymbol{V} \boldsymbol{S}, \text{ and } \boldsymbol{C} = \boldsymbol{h}^2 \boldsymbol{F} \boldsymbol{S}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00668	241	■0.727 ●C +11	$\boldsymbol{A} = \left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right].$		$\boldsymbol{A} = \left[\begin{array}{rrrrrrrr} 4 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 4 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 4 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 4 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 & 4 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 4 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 4 \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00976 (14.1)	330	■0.729 ●N +0	$ \mu - \lambda \leq K \ E\ _2^{1/n}, \quad K = (\ A\ _2 + \ A + E\ _2)^{1-1/n}.$		$ \mu - \lambda \leq K \ E\ _2^{1/n}, \quad K = (\ A\ _2 + \ A + E\ _2)^{1-1/n}.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0676 (10.13)	243	■ 0.737 ● W +2	$\begin{aligned} & (\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B}) \\ & (\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B}, \\ & \boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2 \\ & (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}). \end{aligned}$	$\begin{aligned} & (\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B}) \\ & (\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B}, \\ & \boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2 \\ & (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}). \end{aligned}$	$\begin{aligned} & (\lambda \boldsymbol{A}) \otimes (\mu \boldsymbol{B}) = \lambda \mu (\boldsymbol{A} \otimes \boldsymbol{B}), \\ & (\boldsymbol{A}_1 + \boldsymbol{A}_2) \otimes \boldsymbol{B} = \boldsymbol{A}_1 \otimes \boldsymbol{B} + \boldsymbol{A}_2 \otimes \boldsymbol{B}, \\ & \boldsymbol{A} \otimes (\boldsymbol{B}_1 + \boldsymbol{B}_2) = \boldsymbol{A} \otimes \boldsymbol{B}_1 + \boldsymbol{A} \otimes \boldsymbol{B}_2, \\ & (\boldsymbol{A} \otimes \boldsymbol{B}) \otimes \boldsymbol{C} = \boldsymbol{A} \otimes (\boldsymbol{B} \otimes \boldsymbol{C}). \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0466 (8.3)	186	■ 0.737 ● N +0	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\sum_{j=1}^n x_j y_j \leq \ \boldsymbol{x} \ _p \ \boldsymbol{y} \ _q, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0897	308	■ 0.743 ● W +0	$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}) (\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$	$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}) (\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$	$\frac{(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}) (\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y})}{(\boldsymbol{y}^T \boldsymbol{y})^2} = ab \leq \frac{(M+m)^2}{4Mm},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0582	216	■ 0.744 ● W +0	$\begin{array}{ c c c c } \hline \boldsymbol{u} & \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \\ \hline \boldsymbol{y} & y_1 & y_2 & \cdots & y_m \\ \hline \end{array}$	$\frac{\boldsymbol{u}}{\boldsymbol{y}} = \frac{\boldsymbol{u}_1}{y_1} \frac{\boldsymbol{u}_2}{y_2} \cdots \frac{\boldsymbol{u}_m}{y_m}.$	$\frac{\boldsymbol{u}}{\boldsymbol{y}} = \frac{\boldsymbol{u}_1}{y_1} \frac{\boldsymbol{u}_2}{y_2} \cdots \frac{\boldsymbol{u}_m}{y_m}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0204	085	■ 0.749 ● W +1	$\begin{aligned} & \text{for } i = k+1 : n \\ & \quad a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)} \\ & \quad \text{for } j = k : n \\ & \quad \quad a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} \end{aligned}$	$\begin{aligned} & \text{for } i = k+1 : n \\ & \quad a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)} \\ & \quad \text{for } j = k : n \\ & \quad \quad a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} \end{aligned}$	$\begin{aligned} & \text{for } i = k+1 : n \\ & \quad a_{p_i,k}^{(k)} = a_{p_i,k}^{(k)} / a_{p_k,k}^{(k)} \\ & \quad \text{for } j = k : n \\ & \quad \quad a_{p_i,j}^{(k+1)} = a_{p_i,j}^{(k)} - a_{p_i,k}^{(k)} a_{p_k,j}^{(k)} \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0142	065	■ 0.749 ● K +0	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \mathbf{0} & \boldsymbol{A}_{22} \end{bmatrix}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00707	253	0.749 N +0	$\begin{aligned} \boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} \end{aligned}$ $\boldsymbol{D} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k}$	$(\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D})_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k}$	$(\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D})_{j,k} = \sum_{\ell=1}^m d_{j,\ell} x_{\ell,k} + \sum_{\ell=1}^m x_{j,\ell} d_{\ell,k} = \lambda_j x_{j,k} + \lambda_k x_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0959 (13.68)	322	0.749 N +0	$\omega_k := \begin{cases} 1, & \text{if } k = 0 \\ \left(1 + \omega_{k-1}^{-1} \rho_k / \rho_{k-1}\right)^{-1}, & \text{otherwise.} \end{cases}$	$\omega_k := \begin{cases} 1, & \text{if } k = 0 \\ \left(1 + \omega_{k-1}^{-1} \rho_k / \rho_{k-1}\right)^{-1}, & \text{otherwise.} \end{cases}$	$\omega_k := \begin{cases} 1, & \text{if } k = 0 \\ (1 + \omega_{k-1}^{-1} \rho_k / \rho_{k-1})^{-1}, & \text{otherwise.} \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0575	213	0.751 N +1	$\begin{aligned} & \begin{aligned} & x_1 = 1 \\ & \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x} = [x_1], \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned} \end{aligned}$	$x_1 = 1, \quad \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x} = [x_1], \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$	$\begin{aligned} & x_1 = 1 \\ & x_1 = 1, \quad \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \boldsymbol{x} = [x_1], \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0649	234	0.756 K +0	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V} = \begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V} = \begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V} = \begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0667 (10.8)	240	0.758 W -2	$\begin{aligned} & a_{ii} = 4, i = 1, \dots, n, \\ & a_{i+1,i} = a_{i,i+1} = -1, i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ & a_{i+m,i} = a_{i,i+m} = -1, i = 1, \dots, n-m, \\ & a_{ij} = 0, \text{ otherwise.} \end{aligned}$	$\begin{aligned} & a_{ii} = 4, i = 1, \dots, n, \\ & a_{i+1,i} = a_{i,i+1} = -1, i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ & a_{i+m,i} = a_{i,i+m} = -1, i = 1, \dots, n-m, \\ & a_{ij} = 0, \text{ otherwise.} \end{aligned}$	$\begin{aligned} & a_{ii} = 4, \quad i = 1, \dots, n, \\ & a_{i+1,i} = a_{i,i+1} = -1, \quad i = 1, \dots, n-1, \quad i \neq m, 2m, \dots, (m-1)m, \\ & a_{i+m,i} = a_{i,i+m} = -1, \quad i = 1, \dots, n-m, \\ & a_{ij} = 0, \quad \text{otherwise.} \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1018	345	0.760 K +0	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 = c_1 3^k \boldsymbol{v}_1 + c_2 1^k \boldsymbol{v}_2.$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 = c_1 3^k \boldsymbol{v}_1 + c_2 1^k \boldsymbol{v}_2.$	$\boldsymbol{z}_k = c_1 \lambda_1^k \boldsymbol{v}_1 + c_2 \lambda_2^k \boldsymbol{v}_2 = c_1 3^k \boldsymbol{v}_1 + c_2 1^k \boldsymbol{v}_2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0408	163	0.761 K +0	$\boldsymbol{A} = \frac{1}{6} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$\boldsymbol{A} = \frac{1}{6} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$\boldsymbol{A} = \frac{1}{6} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0651	234	0.763 N +0	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{A}^* \boldsymbol{y} = \mathbf{0}, \boldsymbol{y}^* \boldsymbol{b} \neq 0.$	$(1) \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad (2) \quad \boldsymbol{A}^* \boldsymbol{y} = \mathbf{0}, \boldsymbol{y}^* \boldsymbol{b} \neq 0.$	$(1) \quad \boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \quad (2) \quad \boldsymbol{A}^* \boldsymbol{y} = \mathbf{0}, \boldsymbol{y}^* \boldsymbol{b} \neq 0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00176	077	■ 0.770 ● W +0	$\text{for } i = k+1 : n \quad l_{ik}^{(k)} = a_{ik}^{(k)} / a_{kk}^{(k)}$	$\text{for } i = k+1 : n \quad l_{ik}^{(k)} = a_{ik}^{(k)} / a_{kk}^{(k)}$	for $i = k+1 : n$ $l_{ik}^{(k)} = a_{ik}^{(k)} / a_{kk}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0772 (12.19)	273	■ 0.772 ● N +0	$\rho(\mathbf{I} - \alpha \mathbf{A}) > \rho(\mathbf{I} - \alpha_o \mathbf{A}) = \frac{\kappa - 1}{\kappa + 1}, \quad \alpha \in \mathbb{R} \setminus \{\alpha_o\},$ $\kappa := \frac{\lambda_1}{\lambda_n}, \quad \alpha_o := \frac{2}{\lambda_1 + \lambda_n}.$	$\rho(\mathbf{I} - \alpha \mathbf{A}) > \rho(\mathbf{I} - \alpha_o \mathbf{A}) = \frac{\kappa - 1}{\kappa + 1}, \quad \alpha \in \mathbb{R} \setminus \{\alpha_o\},$ $\kappa := \frac{\lambda_1}{\lambda_n}, \quad \alpha_o := \frac{2}{\lambda_1 + \lambda_n}.$	$\rho(\mathbf{I} - \alpha \mathbf{A}) > \rho(\mathbf{I} - \alpha_o \mathbf{A}) = \frac{\kappa - 1}{\kappa + 1}, \quad \alpha \in \mathbb{R} \setminus \{\alpha_o\},$ $\kappa := \frac{\lambda_1}{\lambda_n}, \quad \alpha_o := \frac{2}{\lambda_1 + \lambda_n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0915 (13.47)	311	■ 0.779 ● K +0	$\mathbf{x}_k := \mathbf{C}^T \mathbf{y}_k, \quad \mathbf{p}_k := \mathbf{C}^T \mathbf{q}_k, \quad \mathbf{s}_k := \mathbf{C}^T \mathbf{z}_k, \quad \mathbf{r}_k := \mathbf{C}^{-1} \mathbf{z}_k$	$\mathbf{x}_k := \mathbf{C}^T \mathbf{y}_k, \quad \mathbf{p}_k := \mathbf{C}^T \mathbf{q}_k, \quad \mathbf{s}_k := \mathbf{C}^T \mathbf{z}_k, \quad \mathbf{r}_k := \mathbf{C}^{-1} \mathbf{z}_k$	$\mathbf{x}_k := \mathbf{C}^T \mathbf{y}_k, \quad \mathbf{p}_k := \mathbf{C}^T \mathbf{q}_k, \quad \mathbf{s}_k := \mathbf{C}^T \mathbf{z}_k, \quad \mathbf{r}_k := \mathbf{C}^{-1} \mathbf{z}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0610	223	■ 0.780 ● K +0	$\mathbf{v} = \mathbf{U} \mathbf{U}^* \mathbf{v} = [\mathbf{U}_1, \mathbf{U}_2 \begin{bmatrix} \mathbf{U}_1^* \\ \mathbf{U}_2^* \end{bmatrix}] \mathbf{v} = \mathbf{s} + \mathbf{t},$	$\mathbf{v} = \mathbf{U} \mathbf{U}^* \mathbf{v} = [\mathbf{U}_1, \mathbf{U}_2 \begin{bmatrix} \mathbf{U}_1^* \\ \mathbf{U}_2^* \end{bmatrix}] \mathbf{v} = \mathbf{s} + \mathbf{t},$	$\mathbf{v} = \mathbf{U} \mathbf{U}^* \mathbf{v} = [\mathbf{U}_1, \mathbf{U}_2 \begin{bmatrix} \mathbf{U}_1^* \\ \mathbf{U}_2^* \end{bmatrix}] \mathbf{v} = \mathbf{s} + \mathbf{t},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0203 (3.13)	085	■ 0.781 ● W +0	$\mathbf{p} = [1, \dots, n]^T;$ for $k = 1 : n-1$ choose $r_k \geq k$ so that $a_{p_{r_k}, k}^{(k)} \neq 0$. $\mathbf{p} = I_{r_k, k} \mathbf{p}$	$\mathbf{p} = [1, \dots, n]^T;$ for $k = 1 : n-1$ choose $r_k \geq k$ so that $a_{p_{r_k}, k}^{(k)} \neq 0$. $\mathbf{p} = I_{r_k, k} \mathbf{p}$	$\mathbf{p} = [1, \dots, n]^T;$ for $k = 1 : n-1$ choose $r_k \geq k$ so that $a_{p_{r_k}, k}^{(k)} \neq 0$. $\mathbf{p} = I_{r_k, k} \mathbf{p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0852 (13.15)	295	■ 0.782 ● W +0	$\mathbf{x}_{k+1} := \mathbf{x}_k + \alpha_k \mathbf{p}_k, \quad \alpha_k := \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k},$ $\mathbf{r}_{k+1} := \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{p}_k,$	For $k = 0, 1, 2, \dots$ $\mathbf{x}_{k+1} := \mathbf{x}_k + \alpha_k \mathbf{p}_k, \quad \alpha_k := \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k},$ $\mathbf{r}_{k+1} := \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{p}_k,$	For $k = 0, 1, 2, \dots$ $\mathbf{x}_{k+1} := \mathbf{x}_k + \alpha_k \mathbf{p}_k, \quad \alpha_k := \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{p}_k^T \mathbf{A} \mathbf{p}_k},$ $\mathbf{r}_{k+1} := \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{p}_k,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00531	203	0.783 N +0	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \langle \boldsymbol{y}, \boldsymbol{z} \rangle = 2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle, \quad \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00099 (1.1)	023	0.784 N +0	$\delta_{ij} := \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$	$\delta_{ij} := \begin{cases} 1 & \text{if } i=j, \\ 0 & \text{otherwise,} \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00108 (2.21)	057	0.788 C -3	$\boldsymbol{T} \boldsymbol{v} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \boldsymbol{b}.$	$\boldsymbol{T} \boldsymbol{v} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \boldsymbol{b}.$	$\boldsymbol{T} \boldsymbol{v} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{m-1} \\ v_m \end{bmatrix} = h^2 \begin{bmatrix} f(h) \\ f(2h) \\ \vdots \\ f((m-1)h) \\ f(mh) \end{bmatrix} =: \boldsymbol{b}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00401	160	0.791 N +0	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda & \boldsymbol{z}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right],$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda & \boldsymbol{z}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right],$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} = \left[\begin{array}{c c} \lambda & \boldsymbol{z}^* \\ \hline \mathbf{0} & \boldsymbol{M} \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00590	218	0.792 W +1	$\begin{aligned} \frac{1}{m} b_{i,j} &= \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ &\approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}. \end{aligned}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}$	$\frac{1}{m} b_{i,j} = \frac{1}{m} \sum_{k=1}^m t_k^{i+j-2} = \frac{1}{m} \sum_{k=1}^m \left(\frac{k-1}{m-1} \right)^{i+j-2} \\ \approx \int_0^1 x^{i+j-2} dx = \frac{1}{i+j-1} = h_{i,j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00096 (0)	054	0.793 W +0	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$	$\begin{bmatrix} d_1 & c_1 & 0 \\ a_1 & d_2 & c_2 \\ 0 & a_2 & d_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{bmatrix} \begin{bmatrix} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} u_1 & c_1 & 0 \\ l_1 u_1 & l_1 c_1 + u_2 & c_2 \\ 0 & l_2 u_2 & l_2 c_2 + u_3 \end{bmatrix},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09563 (8.44)	210	■ 0.822 ● K +0	$\boldsymbol{A} \boldsymbol{s}=\boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{s} = \boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09354	142	■ 0.825 ● N +0	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array}\right],$ $\left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array}\right]=\boldsymbol{Q}^T \left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array}\right],$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array}\right],$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array}\right]=\boldsymbol{Q} \left[\begin{array}{c} \boldsymbol{R} \\ 0 \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09040 (1.14)	036	■ 0.826 ● N +0	$\begin{aligned} \operatorname{det}(\boldsymbol{A}) &= \sum_{j=1}^n(-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } i=1, \ldots, n, \text{ row} \\ \operatorname{det}(\boldsymbol{A}) &= \sum_{i=1}^n(-1)^{i+j} a_{ij} \operatorname{det}(\boldsymbol{A}_{ij}) \text{ for } j=1, \ldots, n, \text{ column.} \end{aligned}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n(-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i=1, \ldots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n(-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j=1, \ldots, n, \text{ column.}$	$\det(\boldsymbol{A}) = \sum_{j=1}^n(-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } i=1, \ldots, n, \text{ row}$ $\det(\boldsymbol{A}) = \sum_{i=1}^n(-1)^{i+j} a_{ij} \det(\boldsymbol{A}_{ij}) \text{ for } j=1, \ldots, n, \text{ column.}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09338	136	■ 0.827 ● N +0	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}\right ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}\right ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$	$\left \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}\right ^2 \leq \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01004	338	■ 0.827 ● N +0	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z}=\sum_{i=m+1}^n \mu_i\left z_i\right ^2 \leq 0,$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z}=\sum_{i=m+1}^n \mu_i\left z_i\right ^2 \leq 0,$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y}=\boldsymbol{y}^* \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E} \boldsymbol{y}=\boldsymbol{z}^* \boldsymbol{B} \boldsymbol{z}=\sum_{i=m+1}^n \mu_i\left z_i\right ^2 \leq 0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09298	123	■ 0.828 ● W +0	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \overline{\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \overline{\boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \overline{\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \overline{\boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$	$\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle} = \overline{\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \overline{\boldsymbol{A} \boldsymbol{x}} = \boldsymbol{y}^* \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09263	108	■ 0.830 ● N +0	$\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8 / 3} \end{array}\right].$	$\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8 / 3} \end{array}\right].$	$\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^*, \quad \boldsymbol{B}=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} \sqrt{3} & 0 \\ 0 & \sqrt{8 / 3} \end{array}\right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0267	109	■ 0.831 ● N +0	$\begin{array}{c} \boldsymbol{L} \boldsymbol{L}^* = \left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{array} \right] \left[\begin{array}{cc} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right] = \left[\begin{array}{cc} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right] = \boldsymbol{A} \\ \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{array} \right] \left[\begin{array}{cc} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right] = \left[\begin{array}{cc} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right] = \boldsymbol{A}$	$\boldsymbol{L} \boldsymbol{L}^* = \left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{array} \right] \left[\begin{array}{cc} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right] = \left[\begin{array}{cc} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right] = \boldsymbol{A}$	$\boldsymbol{L} \boldsymbol{L}^* = \left[\begin{array}{cc} \beta & \mathbf{0} \\ \boldsymbol{v}/\beta & \boldsymbol{L}_1 \end{array} \right] \left[\begin{array}{cc} \beta & \boldsymbol{v}^*/\beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array} \right] = \left[\begin{array}{cc} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array} \right] = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0189 (3.7)	080	■ 0.831 ● N +0	$x_k = \left(b_k - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k = 1, 2, \dots, n.$	$x_k = \left(b_k - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k = 1, 2, \dots, n.$	$x_k = \left(b_k - \sum_{j=1}^{k-1} a_{k,j} x_j \right) / a_{kk}, \quad k = 1, 2, \dots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0122 (2.30)	061	■ 0.833 ● W +0	$\begin{aligned} & \boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ & \lambda_j = d + 2a \cos(j\pi h). \end{aligned}$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j = d + 2a \cos(j\pi h).$	$\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ \lambda_j = d + 2a \cos(j\pi h).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0950 (13.64)	321	■ 0.838 ● N +0	$\begin{aligned} & \langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ & \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2. \end{aligned}$	$\langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2.$	$\langle \boldsymbol{B} \boldsymbol{x}, \boldsymbol{x} \rangle = 0, \quad \boldsymbol{x} \in \mathbb{R}^n, \\ \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _2^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0321 (5.16)	130	■ 0.839 ● N +0	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$	$\int_0^n 4(m-k)(n+r-k) dk = 2m(n+r)^2 - \frac{2}{3}(n+r)^3.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0693 (10.23)	248	■ 0.839 ● C +9	$\begin{array}{c} \Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial\Omega. \end{array}$	$\Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial\Omega.$	$\Delta^2 u(s, t) := \Delta(\Delta u(s, t)) = f(s, t) \quad (s, t) \in \Omega, \\ u(s, t) = 0, \quad \Delta u(s, t) = 0 \quad (s, t) \in \partial\Omega.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0828	287	■ 0.840 ● K +0	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$	$\boldsymbol{x}_k(1) = \boldsymbol{x}_k(2) = 1 - a^k, \quad k \geq 0,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0650	234	0.847 N +1	$\left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & 0 \end{array}\right] y = c.$	$\left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & 0 \end{array}\right] y = c.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0034	033	0.848 K +0	$C=C \ I=C(A \ B)=(C \ A) \ B=I \ B=B$	$C = CI = C(AB) = (CA)B = IB = B$	$C = CI = C(AB) = (CA)B = IB = B$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0084 (2.6)	050	0.850 K +0	$p_{\{i\}}\left(x_{\{i\}}\right)=y_{\{i\}}, \quad p_{\{i\}}\left(x_{\{i+1\}}\right)=y_{\{i+1\}}, \quad p_{\{i\}}^{\prime \prime}\left(x_{\{i\}}\right)=\mu_{\{i\}}, \quad p_{\{i\}}^{\prime \prime}\left(x_{\{i+1\}}\right)=\mu_{\{i+1\}}.$	$p_i\left(x_i\right)=y_i, \quad p_i\left(x_{i+1}\right)=y_{i+1}, \quad p_i^{\prime \prime}\left(x_i\right)=\mu_i, \quad p_i^{\prime \prime}\left(x_{i+1}\right)=\mu_{i+1}.$	$p_i\left(x_i\right)=y_i, \quad p_i\left(x_{i+1}\right)=y_{i+1}, \quad p_i^{\prime \prime}\left(x_i\right)=\mu_i, \quad p_i^{\prime \prime}\left(x_{i+1}\right)=\mu_{i+1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0409	163	0.850 K +0	$\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc} a & 1 \\ 0 & a \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0144	065	0.853 K +0	$\boldsymbol{B} \ \boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array}\right]\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]=\boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array}\right]\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]=\boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A}=\left[\begin{array}{ll} \boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22} \end{array}\right]\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right]=\left[\begin{array}{ll} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0264 (4.8)	109	0.854 N +0	$\boldsymbol{A}=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$	$\boldsymbol{A}=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$	$\boldsymbol{A}=\left[\begin{array}{ll} \alpha & \boldsymbol{v}^* \\ \boldsymbol{v} & \boldsymbol{B} \end{array}\right], \quad \alpha \in \mathbb{C}, \quad \boldsymbol{v} \in \mathbb{C}^{n-1}, \quad \boldsymbol{B} \in \mathbb{C}^{(n-1) \times (n-1)}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0392	157	0.857 N +0	$\max_{\substack{x \in \mathcal{S} \\ x \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$	$\max_{\substack{x \in \mathcal{S} \\ x \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$	$\max_{\substack{x \in \mathcal{S} \\ x \neq \mathbf{0}}} R(\boldsymbol{x}) \geq R(\boldsymbol{y}) = \sum_{j=1}^n \lambda_j c_j ^2 = \sum_{j=1}^k \lambda_j c_j ^2 \geq \sum_{j=1}^k \lambda_k c_j ^2 = \lambda_k,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0032	032	0.859 N +0	$\boldsymbol{B} \boldsymbol{z}=\left[\begin{array}{ll} \boldsymbol{A} & \boldsymbol{b} \end{array}\right]\left[\begin{array}{c} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array}\right]=\boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0}.$	$\boldsymbol{B} \boldsymbol{z}=\left[\begin{array}{ll} \boldsymbol{A} & \boldsymbol{b} \end{array}\right]\left[\begin{array}{c} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array}\right]=\boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0}.$	$\boldsymbol{B} \boldsymbol{z}=\left[\begin{array}{ll} \boldsymbol{A} & \boldsymbol{b} \end{array}\right]\left[\begin{array}{c} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{array}\right]=\boldsymbol{A} \tilde{\boldsymbol{z}}+z_{n+1} \boldsymbol{b}=\mathbf{0}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0891 (13.36)	307	0.859 W +0	$1 \leq \frac{\left(\boldsymbol{y}^T \boldsymbol{y}\right) \boldsymbol{A} \boldsymbol{y}}{\left(\boldsymbol{y}^T \boldsymbol{y}\right)^2} \leq \frac{(M+m)^2}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0}, \boldsymbol{y} \in \mathbb{R}^n,$	$1 \leq \frac{\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}\right) \left(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^T \boldsymbol{y}\right)^2} \leq \frac{(M+m)^2}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0}, \boldsymbol{y} \in \mathbb{R}^n,$	$1 \leq \frac{\left(\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}\right) \left(\boldsymbol{y}^T \boldsymbol{A}^{-1} \boldsymbol{y}\right)}{\left(\boldsymbol{y}^T \boldsymbol{y}\right)^2} \leq \frac{(M+m)^2}{4 M m} \quad \boldsymbol{y} \neq \mathbf{0}, \boldsymbol{y} \in \mathbb{R}^n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0033	033	0.860 W +0	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} \left(-z_{n+1} \boldsymbol{b} \right) = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} \left(-z_{n+1} \boldsymbol{b} \right) = \boldsymbol{b},$	$\boldsymbol{A} \boldsymbol{x} = -\boldsymbol{A} \left(\frac{\tilde{\boldsymbol{z}}}{z_{n+1}} \right) = -\frac{1}{z_{n+1}} \boldsymbol{A} \tilde{\boldsymbol{z}} = -\frac{1}{z_{n+1}} \left(-z_{n+1} \boldsymbol{b} \right) = \boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ00451	180	0.869 W +0	$\boldsymbol{p}_i = \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{q}_i = \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{r}_j = \begin{bmatrix} \boldsymbol{u}_j \\ \mathbf{0} \end{bmatrix}, \text{ for } i = 1, \dots, n, j = n+1, \dots, m.$	$\boldsymbol{p}_i = \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{q}_i = \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{r}_j = \begin{bmatrix} \boldsymbol{u}_j \\ \mathbf{0} \end{bmatrix}, \text{ for } i = 1, \dots, n, j = n+1, \dots, m.$	$\boldsymbol{p}_i = \begin{bmatrix} \boldsymbol{u}_i \\ \boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{q}_i = \begin{bmatrix} \boldsymbol{u}_i \\ -\boldsymbol{v}_i \end{bmatrix}, \quad \boldsymbol{r}_j = \begin{bmatrix} \boldsymbol{u}_j \\ \mathbf{0} \end{bmatrix}, \text{ for } i = 1, \dots, n, j = n+1, \dots, m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0835 (13.1)	290	0.869 W +0	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}, \quad \ \boldsymbol{y}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{y}^T \boldsymbol{A} \boldsymbol{y}}, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00323	132	0.881 C -6	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 3 & 7 & \\ 1 & -1 & -4 \end{array}\right] \boldsymbol{R}=\frac{1}{2}\left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{array}\right] \boldsymbol{Q}=\left[\begin{array}{lll} 2 & 2 & 2 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{rrr} 1 & 3 & 1 \\ 1 & 3 & 7 \\ 1 & -1 & -4 \\ 1 & -1 & 2 \end{array}\right] = \frac{1}{2} \left[\begin{array}{rrrr} 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{array}\right] \times \left[\begin{array}{rrr} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{array}\right] = \boldsymbol{Q}\boldsymbol{R}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00314	127	0.884 N +1	$\boldsymbol{H} \boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{y} \\ \boldsymbol{a} \end{array}\right] \boldsymbol{e}_1 \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{c} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	$\boldsymbol{H}\boldsymbol{x}=\left[\begin{array}{c} \boldsymbol{y} \\ \boldsymbol{a} \end{array}\right] \boldsymbol{e}_1, \text{ where } \boldsymbol{H}:=\boldsymbol{I}-\boldsymbol{u}\boldsymbol{u}^*=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right]-\left[\begin{array}{c} \boldsymbol{0} \\ \hat{\boldsymbol{u}} \end{array}\right]\left[\begin{array}{cc} \boldsymbol{0} & \hat{\boldsymbol{u}}^* \end{array}\right]=\left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \hat{\boldsymbol{H}} \end{array}\right],$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00367 (6.5)	150	0.884 N +0	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	$\boldsymbol{J}:=\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}=\text{diag}(\boldsymbol{U}_1,\dots,\boldsymbol{U}_k), \text{ with } \boldsymbol{U}_i\in\mathbb{C}^{a_i\times a_i},$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00594 (9.5)	219	0.885 N +0	$\left\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\right\}.$	$\{\boldsymbol{x}\in\mathbb{C}^n:\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}_1\}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00064	042	0.887 N +0	$\boldsymbol{A}=\left[\begin{array}{ccc} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	$\boldsymbol{A}=\left[\begin{array}{ccc} 2 & -6 & 3 \\ 3 & -2 & -6 \\ 6 & 3 & 2 \end{array}\right],$	
Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0977	330	0.892 N +0	$\begin{gathered} \left\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\right\ _2 = \left\ (\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{u}_1 - \boldsymbol{A} \boldsymbol{u}_1\right\ _2 = \left\ \boldsymbol{E} \boldsymbol{u}_1\right\ _2 \leq \ \boldsymbol{E}\ _2, \\ \prod_{j=2}^n\left\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\right\ _2 \leq \prod_{j=2}^n(\mu + \ \boldsymbol{A} \boldsymbol{u}_j\ _2) \leq (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A}\ _2)^{n-1} . \end{gathered}$	$\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\ _2 = \ (\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{u}_1 - \boldsymbol{A} \boldsymbol{u}_1\ _2 = \ \boldsymbol{E} \boldsymbol{u}_1\ _2 \leq \ \boldsymbol{E}\ _2,$ $\prod_{j=2}^n\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\ _2 \leq \prod_{j=2}^n(\mu + \ \boldsymbol{A} \boldsymbol{u}_j\ _2) \leq (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A}\ _2)^{n-1} .$	$\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\ _2 = \ (\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{u}_1 - \boldsymbol{A} \boldsymbol{u}_1\ _2 = \ \boldsymbol{E} \boldsymbol{u}_1\ _2 \leq \ \boldsymbol{E}\ _2,$ $\prod_{j=2}^n\ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\ _2 \leq \prod_{j=2}^n(\mu + \ \boldsymbol{A} \boldsymbol{u}_j\ _2) \leq (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A}\ _2)^{n-1} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0147	066	0.896 N +0	$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] = \boldsymbol{I}$	$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] = \boldsymbol{I}$	$\left[\begin{array}{cc} \boldsymbol{A}_{11}^{-1} & -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{array}\right] = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0825	286	0.896 N +0	$\rho(\boldsymbol{G}_J) = \sqrt{ a_{21} a_{12} / a_{11} a_{22} }.$	$\rho(\boldsymbol{G}_J) = \sqrt{ a_{21} a_{12} / a_{11} a_{22} }.$	$\rho(\boldsymbol{G}_J) = \sqrt{ a_{21} a_{12} / a_{11} a_{22} }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0989	333	0.899 N +0	$\boldsymbol{A}_{\boldsymbol{k}} = \left[\begin{array}{cc} \boldsymbol{B}_{\boldsymbol{k}} & \boldsymbol{C}_{\boldsymbol{k}} \\ \boldsymbol{D}_{\boldsymbol{k}} & \boldsymbol{E}_{\boldsymbol{k}} \end{array}\right].$	$\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array}\right].$	$\boldsymbol{A}_k = \left[\begin{array}{cc} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0824 (1.9)	029	0.902 N +0	$\boldsymbol{s}_{\boldsymbol{j}} = \sum_{i=1}^m \boldsymbol{a}_{i \boldsymbol{j}} \text{ for } j=1, \ldots, n .$	$\boldsymbol{s}_j = \sum_{i=1}^m a_{ij} \boldsymbol{v}_i \text{ for } j = 1, \ldots, n.$	$\boldsymbol{s}_j = \sum_{i=1}^m a_{ij} \boldsymbol{v}_i \text{ for } j = 1, \ldots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0614	225	0.902 K +0	$\boldsymbol{A} = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b} = \left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e} = \left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b} = \left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e} = \left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{array}\right], \quad \boldsymbol{b} = \left[\begin{array}{c} 10^{-4} \\ 0 \\ 1 \end{array}\right], \quad \boldsymbol{e} = \left[\begin{array}{c} 10^{-6} \\ 0 \\ 0 \end{array}\right].$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00303	124	0.910 N +0	$\boldsymbol{H}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right] \cdot \left[\begin{array}{l} u_1 \\ u_2 \end{array}\right] \left[\begin{array}{cc} u_1 & u_2 \end{array}\right] = \left[\begin{array}{cc} 1-u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1-u_2^2 \end{array}\right]$	$\boldsymbol{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1-u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1-u_2^2 \end{bmatrix}.$	$\boldsymbol{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1-u_1^2 & -u_1 u_2 \\ -u_2 u_1 & 1-u_2^2 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00252	103	0.911 W +0	$\begin{array}{l} z^* A z = (x-i y)^T A (x+i y) = x^T A x - i y^T A x + i x^T A y - i^2 y^T A y \\ = x^T A x + y^T A y, \end{array}$	$z^* A z = (x - i y)^T A (x + i y) = x^T A x - i y^T A x + i x^T A y - i^2 y^T A y = x^T A x + y^T A y,$	$z^* A z = (x - i y)^T A (x + i y) = x^T A x - i y^T A x + i x^T A y - i^2 y^T A y = x^T A x + y^T A y,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00170	075	0.914 W +0	$\begin{array}{l} a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{array}$	$\begin{array}{l} a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{array}$	$\begin{array}{l} a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ a_{21}^{(1)} x_1 + a_{22}^{(1)} x_2 + a_{23}^{(1)} x_3 = b_2^{(1)}, \quad \text{II} \\ a_{31}^{(1)} x_1 + a_{32}^{(1)} x_2 + a_{33}^{(1)} x_3 = b_3^{(1)}. \quad \text{III.} \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00494	194	0.915 K +0	$\left\ \boldsymbol{A}^* \right\ _F = \left\ \boldsymbol{A} \right\ _F \text{ and } \left\ \boldsymbol{A}^* \right\ _2 = \left\ \boldsymbol{A} \right\ _2.$	$\ \boldsymbol{A}^* \ _F = \ \boldsymbol{A} \ _F \text{ and } \ \boldsymbol{A}^* \ _2 = \ \boldsymbol{A} \ _2.$	$\ \boldsymbol{A}^* \ _F = \ \boldsymbol{A} \ _F \text{ and } \ \boldsymbol{A}^* \ _2 = \ \boldsymbol{A} \ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00485	191	0.917 C +32	$\begin{array}{l} \left\ \boldsymbol{A} \boldsymbol{B} \right\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{B} \boldsymbol{x} \right\ } \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } \\ \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{y} \right\ }{\left\ \boldsymbol{y} \right\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \left\ \boldsymbol{A} \right\ \left\ \boldsymbol{B} \right\ \end{array}$	$\begin{array}{l} \left\ \boldsymbol{A} \boldsymbol{B} \right\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{B} \boldsymbol{x} \right\ } \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } \\ \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{y} \right\ }{\left\ \boldsymbol{y} \right\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \left\ \boldsymbol{A} \right\ \left\ \boldsymbol{B} \right\ . \end{array}$	$\begin{array}{l} \left\ \boldsymbol{A} \boldsymbol{B} \right\ = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \max_{\boldsymbol{B} \boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{B} \boldsymbol{x} \right\ } \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } \\ \leq \max_{\boldsymbol{y} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{y} \right\ }{\left\ \boldsymbol{y} \right\ } \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{B} \boldsymbol{x} \right\ }{\left\ \boldsymbol{x} \right\ } = \left\ \boldsymbol{A} \right\ \left\ \boldsymbol{B} \right\ . \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1022 (15.5)	346	0.918 N +0	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 v_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 v_1,$	$\lim_{k \rightarrow \infty} \frac{z_k}{\lambda_1^k} = c_1 v_1,$

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0580	215	0.924 N +0	$\begin{array}{c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$	$\begin{array}{c c c c} t & 1.0 & 2.0 & 3.0 & 4.0 \\ \hline y & 3.1 & 1.8 & 1.0 & 0.1 \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0912 (13.45)	310	0.931 W +0	$\begin{aligned} A x &= b \\ B A x &= B b \\ (C A C^T) y &= C b, \quad x = C^T y. \end{aligned}$	$\begin{aligned} A x &= b \\ B A x &= B b \\ (C A C^T) y &= C b, \quad x = C^T y. \end{aligned}$	$\begin{aligned} A x &= b \\ B A x &= B b \\ (C A C^T) y &= C b, \quad x = C^T y. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0229 (3.20)	092	0.933 N +0	$\begin{aligned} \boldsymbol{A} &= \boldsymbol{L} \boldsymbol{U} \\ \left[\begin{array}{cccc} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{array} \right] & \left[\begin{array}{cccc} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{mm} \end{array} \right] \end{aligned}$	$A = LU = \begin{bmatrix} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{mm} \end{bmatrix}$	$A = LU = \begin{bmatrix} \boldsymbol{I} & & & \\ \boldsymbol{L}_{21} & \boldsymbol{I} & & \\ \vdots & & \ddots & \\ \boldsymbol{L}_{m1} & \cdots & \boldsymbol{L}_{m,m-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{11} & \cdots & \boldsymbol{U}_{1m} \\ & \boldsymbol{U}_{22} & \cdots & \boldsymbol{U}_{2m} \\ & & \ddots & \vdots \\ & & & \boldsymbol{U}_{mm} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0253	103	0.934 W +1	$\nabla f(\boldsymbol{x}) = \left[\begin{array}{c} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{array} \right] \in \mathbb{R}^n, \quad H f(\boldsymbol{x}) = \left[\begin{array}{ccc} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{array} \right] \in \mathbb{R}^{n \times n}.$	$\nabla f(\boldsymbol{x}) = \left[\begin{array}{c} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{array} \right] \in \mathbb{R}^n, \quad H f(\boldsymbol{x}) = \left[\begin{array}{ccc} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{array} \right] \in \mathbb{R}^{n \times n}.$	$\nabla f(\boldsymbol{x}) = \left[\begin{array}{c} \frac{\partial f(\boldsymbol{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\boldsymbol{x})}{\partial x_n} \end{array} \right] \in \mathbb{R}^n, \quad H f(\boldsymbol{x}) = \left[\begin{array}{ccc} \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\boldsymbol{x})}{\partial x_n \partial x_n} \end{array} \right] \in \mathbb{R}^{n \times n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0659 (10.3)	239	0.934 W +0	$\begin{aligned} -\Delta_h v_{j,k} &= f_{j,k}, \quad (j h, k h) \in \Omega_h, \\ v_{j,k} &= 0, \quad (j h, k h) \in \partial \Omega_h, \end{aligned}$	$\begin{aligned} -\Delta_h v_{j,k} &= f_{j,k}, \quad (j h, k h) \in \Omega_h, \\ v_{j,k} &= 0, \quad (j h, k h) \in \partial \Omega_h, \end{aligned}$	$\begin{aligned} -\Delta_h v_{j,k} &= f_{j,k}, \quad (j h, k h) \in \Omega_h, \\ v_{j,k} &= 0, \quad (j h, k h) \in \partial \Omega_h, \end{aligned}$
Conf. MathPix confidence: <					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00873	303	■ 0.936 ● W -1	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b}\left Q(x)\right ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b}\left Q(x)\right ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b}\left Q(x)\right ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b}\left Q(x)\right ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$	$\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}^2 \leq\left\ \boldsymbol{x}-\boldsymbol{x}_0-\boldsymbol{w}\right\ _{\boldsymbol{A}}^2 \leq \max _{a \leq x \leq b}\left Q(x)\right ^2 \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}=\max _{a \leq x \leq b}\left Q(x)\right ^2\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01052	359	■ 0.936 ● W +0	$\begin{aligned} & \nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^T \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y & -x+2 y \\ 1 & -1 \end{array}\right], \\ & \boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array}\right]=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] . \end{aligned}$	$\nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^T \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y & -x+2 y \\ 1 & -1 \end{array}\right],$ $\boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array}\right]=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$	$\nabla f(x, y)=\left[\begin{array}{c} 2 x-y \\ -x+2 y \end{array}\right], \quad \nabla^T \boldsymbol{g}(x, y)=\left[\begin{array}{cc} 2 x-y & -x+2 y \\ 1 & -1 \end{array}\right],$ $\boldsymbol{H} f(x, y)=\left[\begin{array}{cc} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{array}\right]=\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00782	276	■ 0.938 ● W +0	$\begin{aligned} & \left(\boldsymbol{A} \boldsymbol{x}\right)^* \boldsymbol{y}+\boldsymbol{y}^*\left(\lambda \boldsymbol{A} \boldsymbol{x}\right)=\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_R\right) \boldsymbol{y}+\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_R-\boldsymbol{D}\right) \boldsymbol{y} \\ & =\boldsymbol{y}^*\left(2 \omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega) 1-\lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x} . \end{aligned}$	$\left(\boldsymbol{A} \boldsymbol{x}\right)^* \boldsymbol{y}+\boldsymbol{y}^*\left(\lambda \boldsymbol{A} \boldsymbol{x}\right)=\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_R\right) \boldsymbol{y}+\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_R-\boldsymbol{D}\right) \boldsymbol{y}$ $=\boldsymbol{y}^*\left(2 \omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega) 1-\lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x} .$	$\left(\boldsymbol{A} \boldsymbol{x}\right)^* \boldsymbol{y}+\boldsymbol{y}^*\left(\lambda \boldsymbol{A} \boldsymbol{x}\right)=\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_R\right) \boldsymbol{y}+\boldsymbol{y}^*\left(\omega^{-1} \boldsymbol{D}+\boldsymbol{A}_R-\boldsymbol{D}\right) \boldsymbol{y}$ $=\boldsymbol{y}^*\left(2 \omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{y}=\omega^{-1}(2-\omega) 1-\lambda ^2 \boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00330	134	■ 0.938 ● K +0	$\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^2+s^2=1 .$	$\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^2+s^2=1 .$	$\boldsymbol{P}:=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text { where } c^2+s^2=1 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00770 (12.17)	273	■ 0.939 ● W -1	$\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})} \lambda .$	$\rho(\boldsymbol{A}):=\max _{\lambda \in \sigma(\boldsymbol{A})} \lambda .$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00482	190	■0.940 ● N -1	$\max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{Ay}\ .$	$\max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{Ay}\ .$	$\max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ } = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \left\ \boldsymbol{A} \left(\frac{\boldsymbol{x}}{\ \boldsymbol{x}\ } \right) \right\ = \max_{\ \boldsymbol{y}\ =1} \ \boldsymbol{Ay}\ .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0604 (9.9)	222	■0.940 ● K +0	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$	$\boldsymbol{A}^{\dagger} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0806 (12.35)	282	■0.941 ● K +0	$\left\ (\boldsymbol{I} - \boldsymbol{B})^{-1} \right\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$	$\ (\boldsymbol{I} - \boldsymbol{B})^{-1}\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$	$\ (\boldsymbol{I} - \boldsymbol{B})^{-1}\ \leq \frac{1}{1 - \ \boldsymbol{B}\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ1012	342	■0.942 ● N +0	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$	$\boldsymbol{A}_n = \begin{bmatrix} 10 & 1 & 0 & \cdots & 0 \\ 1 & 10 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 10 & 1 \\ 0 & \cdots & 0 & 1 & 10 \end{bmatrix} \in \mathbb{R}^{n \times n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0174	076	■0.942 ● C -9	$\begin{aligned} \boldsymbol{L} \boldsymbol{U} &:= \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right] \\ &= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array} \right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array} \right] = \boldsymbol{A}. \end{aligned}$	$\boldsymbol{LU} := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right]$ $= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array} \right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array} \right] = \boldsymbol{A}.$	$\boldsymbol{LU} := \left[\begin{array}{ccc} 1 & 0 & 0 \\ l_{21}^{(1)} & 1 & 0 \\ l_{31}^{(1)} & l_{32}^{(2)} & 1 \end{array} \right] \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} \\ 0 & 0 & a_{33}^{(3)} \end{array} \right]$ $= \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ l_{21}^{(1)} a_{11}^{(1)} & l_{21}^{(1)} a_{12}^{(1)} + a_{22}^{(2)} & l_{21}^{(1)} a_{13}^{(1)} + a_{23}^{(2)} \\ l_{31}^{(1)} a_{11}^{(1)} & l_{31}^{(1)} a_{12}^{(1)} + l_{32}^{(2)} a_{22}^{(2)} & l_{31}^{(1)} a_{13}^{(1)} + l_{32}^{(2)} a_{23}^{(2)} + a_{33}^{(3)} \end{array} \right] = \left[\begin{array}{ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & a_{23}^{(1)} \\ a_{31}^{(1)} & a_{32}^{(1)} & a_{33}^{(1)} \end{array} \right] = \boldsymbol{A}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00628	229	■0.943 ● N -1	$\boldsymbol{C}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0}\end{array}\right] \text{ and } \boldsymbol{D}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0}\end{array}\right] \in \mathbb{C}^{(m+n)\times(m+n)}.$	$\boldsymbol{C}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0}\end{array}\right] \text{ and } \boldsymbol{D}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0}\end{array}\right] \in \mathbb{C}^{(m+n)\times(m+n)}.$	$\boldsymbol{C}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0}\end{array}\right] \text{ and } \boldsymbol{D}:=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{B} \\ \boldsymbol{B}^* & \mathbf{0}\end{array}\right] \in \mathbb{C}^{(m+n)\times(m+n)}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01054	359	■0.944 ● W +1	$\begin{aligned} g(0) &= f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}), \quad g'(\boldsymbol{x})=\sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x}+\boldsymbol{h})}{\partial x_i}=\boldsymbol{h}^T \nabla f(\boldsymbol{x}+\boldsymbol{h}), \\ g''(\boldsymbol{x}) &=\sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x}+\boldsymbol{h})}{\partial x_i \partial x_j}=\boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x}+\boldsymbol{h}) \boldsymbol{h}. \end{aligned}$	$g(0)=f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}),$ $g'(t)=\sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_i}=\boldsymbol{h}^T \nabla f(\boldsymbol{x}+t \boldsymbol{h}),$ $g''(t)=\sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_i \partial x_j}=\boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x}+t \boldsymbol{h}) \boldsymbol{h}.$	$g(0)=f(\boldsymbol{x}) \quad g(1)=f(\boldsymbol{x}+\boldsymbol{h}),$ $g'(t)=\sum_{i=1}^n h_i \frac{\partial f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_i}=\boldsymbol{h}^T \nabla f(\boldsymbol{x}+t \boldsymbol{h}),$ $g''(t)=\sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f(\boldsymbol{x}+t \boldsymbol{h})}{\partial x_i \partial x_j}=\boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{x}+t \boldsymbol{h}) \boldsymbol{h}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00373 (6.8)	152	■0.945 ● W -1	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c c}\lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M}\end{array}\right], \text{ for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text{ and } \boldsymbol{x} \in \mathbb{C}^k.$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c c}\lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M}\end{array}\right], \text{ for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text{ and } \boldsymbol{x} \in \mathbb{C}^k.$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V}=\left[\begin{array}{c c}\lambda_1 & \boldsymbol{x}^* \\ \hline \mathbf{0} & \boldsymbol{M}\end{array}\right], \text{ for some } \boldsymbol{M} \in \mathbb{C}^{k \times k} \text{ and } \boldsymbol{x} \in \mathbb{C}^k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00533	203	■0.946 ● W +1	$m^2 \left\langle \frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \right\rangle \stackrel{(8.39)}{=} m \langle n \boldsymbol{x}, \boldsymbol{y} \rangle \stackrel{(8.39)}{=} mn \langle \boldsymbol{x}, \boldsymbol{y} \rangle,$	$m^2 \left\langle \frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \right\rangle \stackrel{(8.39)}{=} m \langle n \boldsymbol{x}, \boldsymbol{y} \rangle \stackrel{(8.39)}{=} mn \langle \boldsymbol{x}, \boldsymbol{y} \rangle,$	$m^2 \left\langle \frac{n}{m} \boldsymbol{x}, \boldsymbol{y} \right\rangle \stackrel{(8.39)}{=} m \langle n \boldsymbol{x}, \boldsymbol{y} \rangle \stackrel{(8.39)}{=} mn \langle \boldsymbol{x}, \boldsymbol{y} \rangle,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00686 (10.19)	246	■0.946 ● N +0	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right)=\frac{1}{4 h^2} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,$	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right)=\frac{1}{4 h^2} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,$	$\left(\boldsymbol{s}_j \otimes \boldsymbol{s}_k\right)^T\left(\boldsymbol{s}_p \otimes \boldsymbol{s}_q\right)=\frac{1}{4 h^2} \delta_{j, p} \delta_{k, q}, \quad j, k, p, q=1, \ldots, m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00324	132	■0.946 ● N +0	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{ccc} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_1 \boldsymbol{R}_1.$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{ccc} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_1 \boldsymbol{R}_1.$	$\boldsymbol{A}=\frac{1}{2}\left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & 1 \end{array}\right] \times\left[\begin{array}{ccc} 2 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]=\boldsymbol{Q}_1 \boldsymbol{R}_1.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00331	135	■ 0.947 ● N +0	$\boldsymbol{x}=\left[\begin{array}{l} x_{1} \\ x_{2} \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_1}{r}, \quad s:=\frac{x_2}{r}, \quad r:=\ \boldsymbol{x}\ _2.$	$\boldsymbol{x}=\left[\begin{array}{l} x_1 \\ x_2 \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_1}{r}, \quad s:=\frac{x_2}{r}, \quad r:=\ \boldsymbol{x}\ _2.$	$\boldsymbol{x}=\left[\begin{array}{l} x_1 \\ x_2 \end{array}\right] \neq \mathbf{0}, \quad c:=\frac{x_1}{r}, \quad s:=\frac{x_2}{r}, \quad r:=\ \boldsymbol{x}\ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01000	337	■ 0.947 ● N +0	$y_{\{0\}}<y_{\{1\}}<z_{\{1\}}<y_{\{2\}}<z_{\{2\}} \\ \cdots <z_{\{k-1\}}<y_{\{k\}}<y_{\{k+1\}} \text{ .}$	$y_0 < y_1 < z_1 < y_2 < z_2 \cdots < z_{k-1} < y_k < y_{k+1}.$	$y_0 < y_1 < z_1 < y_2 < z_2 \cdots < z_{k-1} < y_k < y_{k+1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00999	337	■ 0.950 ● N +0	$p_{\{3\}}(-M)<0, \quad \text{\quad} p_{\{3\}}\left(y_{\{1\}}\right)>0, \quad \text{\quad} p_{\{3\}}\left(y_{\{2\}}\right)<0, \quad \text{\quad} p_{\{3\}}(+M)>0 \text{ .}$	$p_3(-M) < 0, \quad p_3(y_1) > 0, \quad p_3(y_2) < 0, \quad p_3(+M) > 0.$	$p_3(-M) < 0, \quad p_3(y_1) > 0, \quad p_3(y_2) < 0, \quad p_3(+M) > 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00948 (13.61)	321	■ 0.950 ● W +0	$\begin{aligned} & \left[\begin{array}{l} \boldsymbol{A} \boldsymbol{p}_{\{0\}}, \\ \boldsymbol{A} \boldsymbol{p}_{\{1\}}, \boldsymbol{A} \boldsymbol{p}_{\{2\}} \end{array}\right] \&= \left[\begin{array}{rcl} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16/9 \end{array}\right] \\ & \left[\begin{array}{l} \boldsymbol{p}_{\{0\}}, \boldsymbol{p}_{\{1\}}, \boldsymbol{p}_{\{2\}}, \boldsymbol{p}_{\{3\}} \end{array}\right] \&= \left[\begin{array}{rrrr} 4 & 1 & 4/9 & 0 \\ 0 & 2 & 8/9 & 0 \\ 0 & 0 & 12/9 & 0 \end{array}\right] \end{aligned}$	$\begin{aligned} [\boldsymbol{A} \boldsymbol{p}_0, \boldsymbol{A} \boldsymbol{p}_1, \boldsymbol{A} \boldsymbol{p}_2] &= \begin{bmatrix} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16/9 \end{bmatrix}, \\ [\boldsymbol{p}_0, \boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3] &= \begin{bmatrix} 4 & 1 & 4/9 & 0 \\ 0 & 2 & 8/9 & 0 \\ 0 & 0 & 12/9 & 0 \end{bmatrix}, \end{aligned}$	$\begin{aligned} [\boldsymbol{A} \boldsymbol{p}_0, \boldsymbol{A} \boldsymbol{p}_1, \boldsymbol{A} \boldsymbol{p}_2] &= \begin{bmatrix} 8 & 0 & 0 \\ -4 & 3 & 0 \\ 0 & -2 & 16/9 \end{bmatrix}, \\ [\boldsymbol{p}_0, \boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3] &= \begin{bmatrix} 4 & 1 & 4/9 & 0 \\ 0 & 2 & 8/9 & 0 \\ 0 & 0 & 12/9 & 0 \end{bmatrix}, \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00058	041	■ 0.951 ● N +1	$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{a d-b c}$	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{ad-bc}$	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]^{-1}=\alpha\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right], \quad \alpha=\frac{1}{ad-bc},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00220	089	■ 0.951 ● N +0	$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ l_1 & 1 \end{array}\right]\left[\begin{array}{ll} u_1 & u_2 \\ 0 & u_3 \end{array}\right]=\left[\begin{array}{ll} u_1 & u_2 \\ u_1 l_1 & u_2 l_1+u_3 \end{array}\right]$	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ l_1 & 1 \end{array}\right]\left[\begin{array}{cc} u_1 & u_2 \\ 0 & u_3 \end{array}\right]=\left[\begin{array}{cc} u_1 & u_2 \\ u_1 l_1 & u_2 l_1+u_3 \end{array}\right]$	$\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ l_1 & 1 \end{array}\right]\left[\begin{array}{cc} u_1 & u_2 \\ 0 & u_3 \end{array}\right]=\left[\begin{array}{cc} u_1 & u_2 \\ u_1 l_1 & u_2 l_1+u_3 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00110	058	■ 0.951 ● K +0	$\boldsymbol{A}_{\{1\}}=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \text{\quad} \boldsymbol{A}_{\{2\}}=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \text{\quad} \boldsymbol{A}_{\{3\}}=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right] \text{ .}$	$\boldsymbol{A}_1=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right] \text{ .}$	$\boldsymbol{A}_1=\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{rrr} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array}\right] \text{ .}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0278	117	■ 0.953 ● N -2	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad \text{(symmetry).}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad \text{(symmetry).}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle \quad \text{(symmetry).}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0570	211	■ 0.953 ● K +0	$\frac{ \hat{\boldsymbol{s}} - \boldsymbol{s} _{\infty}}{ \boldsymbol{s} _{\infty}} \quad \text{and} \quad \frac{ \hat{\boldsymbol{s}} - \boldsymbol{s} _1}{ \boldsymbol{s} _1}.$	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _{\infty}}{\ \boldsymbol{s}\ _{\infty}} \quad \text{and} \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$	$\frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _{\infty}}{\ \boldsymbol{s}\ _{\infty}} \quad \text{and} \quad \frac{\ \hat{\boldsymbol{s}} - \boldsymbol{s}\ _1}{\ \boldsymbol{s}\ _1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0620 (9.16)	227	■ 0.953 ● C -5	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2} = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{Ax}\ _2}{\ \boldsymbol{x}\ _2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0722	257	■ 0.954 ● C -4	$\boldsymbol{P}_4 = \left[\begin{array}{lll} \boldsymbol{e}_1 & \boldsymbol{e}_3 & \boldsymbol{e}_2 \\ \boldsymbol{e}_4 & & \end{array} \right] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{cc cc} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array} \right],$	$\boldsymbol{P}_4 = [\boldsymbol{e}_1 \ \boldsymbol{e}_3 \ \boldsymbol{e}_2 \ \boldsymbol{e}_4] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{cc cc} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array} \right],$	$\boldsymbol{P}_4 = [\boldsymbol{e}_1 \ \boldsymbol{e}_3 \ \boldsymbol{e}_2 \ \boldsymbol{e}_4] = \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \quad \boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{cc cc} 1 & 1 & 1 & 1 \\ 1 & -1 & -i & i \\ \hline 1 & 1 & -1 & -1 \\ 1 & -1 & i & -i \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0609 (9.12)	223	■ 0.955 ● N +1	$\begin{gathered} \boldsymbol{P}_{\mathcal{R}}(\boldsymbol{A}) = \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}, \\ \boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_2 \boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}. \end{gathered}$	$\boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})} = \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m},$ $\boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_2 \boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}.$	$\boldsymbol{P}_{\mathcal{R}(\boldsymbol{A})} = \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{U}_1^* = \sum_{j=1}^r \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m},$ $\boldsymbol{P}_{\mathcal{N}(\boldsymbol{A}^*)} = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_2 \boldsymbol{U}_2^* = \sum_{j=r+1}^m \boldsymbol{u}_j \boldsymbol{u}_j^* \in \mathbb{C}^{m \times m}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0753 (12.8)	270	■ 0.958 ● N +0	$\boldsymbol{M} \boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x} + \boldsymbol{b}.$	$\boldsymbol{M} \boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x} + \boldsymbol{b}.$	$\boldsymbol{M} \boldsymbol{x} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x} + \boldsymbol{b}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0654	235	■ 0.959 ● N +0	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \\ 2 \end{array}\right], \quad \epsilon \in \mathbb{R}.$	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \\ 2 \end{array}\right], \quad \epsilon \in \mathbb{R}.$	$\boldsymbol{A}=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{array}\right], \quad \boldsymbol{b}=\left[\begin{array}{c} 2 \\ 3 \\ 2 \end{array}\right], \quad \epsilon \in \mathbb{R}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0443	177	■ 0.959 ● N +0	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$	$\ \boldsymbol{A}\ _F \stackrel{3.}{=} \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{V}\ _F = \ \boldsymbol{\Sigma}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_n^2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0469 (8.5)	186	■ 0.960 ● N +0	$\ x\ _\infty \leq \ x\ _p \leq n^{1/p} \ x\ _\infty, \quad p \geq 1.$	$\ x\ _\infty \leq \ x\ _p \leq n^{1/p} \ x\ _\infty, \quad p \geq 1.$	$\ \boldsymbol{x}\ _\infty \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _\infty, \quad p \geq 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0606 (9.10)	222	■ 0.961 ● W +2	$\begin{aligned} & \boldsymbol{A}^{\dagger} \boldsymbol{A} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^* \\ & \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^* \end{aligned}$	$\boldsymbol{A}^{\dagger} \boldsymbol{A} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^* \\ \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$	$\boldsymbol{A}^{\dagger} \boldsymbol{A} = \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*, \\ \boldsymbol{A} \boldsymbol{A}^{\dagger} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^* \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0350	141	■ 0.962 ● N +0	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$	$\boldsymbol{P}_{1,2} \boldsymbol{P}_{2,3} \cdots \boldsymbol{P}_{m-2,m-1} \boldsymbol{P}_{m-1,m} \boldsymbol{w} = \begin{bmatrix} \alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0961 (13.71)	323	■ 0.962 ● N +0	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$	$\boldsymbol{B} \boldsymbol{r}_k = \alpha_k \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{r}_{k-1},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0332	135	■ 0.963 ● N +0	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$	$\boldsymbol{P} \boldsymbol{x} = \frac{1}{r} \begin{bmatrix} x_1 & x_2 \\ -x_2 & x_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} x_1^2 + x_2^2 \\ 0 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00057	040	■0.969 ●W +0	$\begin{array}{ll} \mathbf{P}_1=(\mathbf{A}+\mathbf{D})(\mathbf{E}+\mathbf{H}), & \mathbf{P}_5=(\mathbf{A}+\mathbf{B})\mathbf{H}, \\ \mathbf{P}_2=(\mathbf{C}+\mathbf{D})\mathbf{E}, & \mathbf{P}_6=(\mathbf{C}-\mathbf{A})(\mathbf{E}+\mathbf{F}), \\ \mathbf{P}_3=\mathbf{A}(\mathbf{F}-\mathbf{H}), & \mathbf{P}_7=(\mathbf{B}-\mathbf{D})(\mathbf{G}+\mathbf{H}), \\ \mathbf{P}_4=\mathbf{D}(\mathbf{G}-\mathbf{E}), & \mathbf{X}=\mathbf{P}_3+\mathbf{P}_5, \\ \mathbf{W}=\mathbf{P}_1+\mathbf{P}_4-\mathbf{P}_5+\mathbf{P}_7, & \mathbf{Z}=\mathbf{P}_1+\mathbf{P}_3-\mathbf{P}_2+\mathbf{P}_6. \\ \mathbf{Y}=\mathbf{P}_2+\mathbf{P}_4, & \end{array}$	$\begin{array}{ll} \mathbf{P}_1=(\mathbf{A}+\mathbf{D})(\mathbf{E}+\mathbf{H}), & \mathbf{P}_5=(\mathbf{A}+\mathbf{B})\mathbf{H}, \\ \mathbf{P}_2=(\mathbf{C}+\mathbf{D})\mathbf{E}, & \mathbf{P}_6=(\mathbf{C}-\mathbf{A})(\mathbf{E}+\mathbf{F}), \\ \mathbf{P}_3=\mathbf{A}(\mathbf{F}-\mathbf{H}), & \mathbf{P}_7=(\mathbf{B}-\mathbf{D})(\mathbf{G}+\mathbf{H}), \\ \mathbf{P}_4=\mathbf{D}(\mathbf{G}-\mathbf{E}), & \mathbf{X}=\mathbf{P}_3+\mathbf{P}_5, \\ \mathbf{W}=\mathbf{P}_1+\mathbf{P}_4-\mathbf{P}_5+\mathbf{P}_7, & \mathbf{Z}=\mathbf{P}_1+\mathbf{P}_3-\mathbf{P}_2+\mathbf{P}_6. \\ \mathbf{Y}=\mathbf{P}_2+\mathbf{P}_4, & \end{array}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0224 (3.16)	090	■0.969 ●W +0	$\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_k \\ \boldsymbol{C}_k & \boldsymbol{F}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \boldsymbol{0} \\ \boldsymbol{M}_k & \boldsymbol{N}_k \end{array}\right] \left[\begin{array}{cc} \boldsymbol{U}_{[k]} & \boldsymbol{S}_k \\ \boldsymbol{0} & \boldsymbol{T}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]} & \boldsymbol{L}_{[k]}\boldsymbol{S}_k \\ \boldsymbol{M}_k\boldsymbol{U}_{[k]} & \boldsymbol{M}_k\boldsymbol{S}_k + \boldsymbol{N}_k\boldsymbol{T}_k \end{array}\right],$	$\left[\begin{array}{cc} \boldsymbol{A}_{[k]} & \boldsymbol{B}_k \\ \boldsymbol{C}_k & \boldsymbol{F}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{[k]} & \boldsymbol{0} \\ \boldsymbol{M}_k & \boldsymbol{N}_k \end{array}\right] \left[\begin{array}{cc} \boldsymbol{U}_{[k]} & \boldsymbol{S}_k \\ \boldsymbol{0} & \boldsymbol{T}_k \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{[k]}\boldsymbol{U}_{[k]} & \boldsymbol{L}_{[k]}\boldsymbol{S}_k \\ \boldsymbol{M}_k\boldsymbol{U}_{[k]} & \boldsymbol{M}_k\boldsymbol{S}_k + \boldsymbol{N}_k\boldsymbol{T}_k \end{array}\right],$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0106	057	■0.970 ●W -1	$\begin{aligned} u^{\prime\prime}(x) &= \lim_{h \rightarrow 0} \frac{u^{\prime}\left(x+\frac{h}{2}\right)-u^{\prime}\left(x-\frac{h}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x+h)-2 u(x)+u(x-h)}{h^2} . \end{aligned}$	$u^{\prime\prime}(x)=\lim _{h \rightarrow 0} \frac{u^{\prime}\left(x+\frac{h}{2}\right)-u^{\prime}\left(x-\frac{h}{2}\right)}{h}=\lim _{h \rightarrow 0} \frac{\frac{u(x+h)-u(x)}{h}-\frac{u(x)-u(x-h)}{h}}{h}$ $=\lim _{h \rightarrow 0} \frac{u(x+h)-2 u(x)+u(x-h)}{h^2} .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00680	245	■ 0.970 ● W +0	$\begin{aligned} & (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F}) \\ & \Leftrightarrow \begin{bmatrix} \boldsymbol{A} \boldsymbol{b}_{11} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{s1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix} \\ & \Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s] \\ & \Leftrightarrow \boldsymbol{A} [\boldsymbol{V} \boldsymbol{b}_1, \dots, \boldsymbol{V} \boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}. \end{aligned}$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$ $\Leftrightarrow \begin{bmatrix} \boldsymbol{A} \boldsymbol{b}_{11} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{s1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 \\ \vdots \\ \boldsymbol{v}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_1 \\ \vdots \\ \boldsymbol{f}_s \end{bmatrix}$ $\Leftrightarrow \boldsymbol{A} \left[\sum_j b_{1j} \boldsymbol{v}_j, \dots, \sum_j b_{sj} \boldsymbol{v}_j \right] = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$ $\Leftrightarrow \boldsymbol{A} [\boldsymbol{V} \boldsymbol{b}_1, \dots, \boldsymbol{V} \boldsymbol{b}_s] = \boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00161	071	■ 0.971 ● N +0	$a_{\{j\}} = -1 - \frac{h}{2} r \left(x_{\{j\}} \right), \quad c_{\{j\}} = -1 + \frac{h}{2} r \left(x_{\{j\}} \right), \quad d_{\{j\}} = 2 + h^2 q \left(x_{\{j\}} \right),$	$a_j = -1 - \frac{h}{2} r(x_j), c_j = -1 + \frac{h}{2} r(x_j), d_j = 2 + h^2 q(x_j),$	$a_j = -1 - \frac{h}{2} r(x_j), \quad c_j = -1 + \frac{h}{2} r(x_j), \quad d_j = 2 + h^2 q(x_j),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0312 (5.14)	126	■ 0.971 ● N +0	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$	$\boldsymbol{u} := \frac{\boldsymbol{z} + \boldsymbol{e}_1}{\sqrt{1 + z_1}}, \quad \text{where } \boldsymbol{z} := \bar{\rho} \boldsymbol{x} / \ \boldsymbol{x}\ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01043	353	■ 0.971 ● C +78	$\begin{aligned} & \begin{bmatrix} x & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{23}^*} \begin{bmatrix} x & x & x & x \\ x & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}. \end{aligned}$	(not rendered)	$\boldsymbol{A} = \begin{bmatrix} x & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{12}^*} \begin{bmatrix} x & x & x & x \\ \mathbf{x} & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{23}^*} \begin{bmatrix} x & x & x & x \\ x & \mathbf{x} & x & x \\ 0 & 0 & 0 & x \\ 0 & 0 & 0 & x \end{bmatrix} \xrightarrow{P_{34}^*} \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & \mathbf{x} & x \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00407	162	■ 0.971 ● N +0	$\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_0 \\ 0 & 0 & -q_{\{0\}} \setminus 1 & 0 & \cdots & 0 & -q_{\{1\}} \setminus 0 & 1 & \cdots & 0 & -q_{\{2\}} \setminus \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & -q_{\{2\}} \setminus \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array}\right].$	$\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_0 \\ 1 & 0 & \cdots & 0 & -q_1 \\ 0 & 1 & \cdots & 0 & -q_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array}\right].$	$\boldsymbol{B}=\left[\begin{array}{ccccc} 0 & 0 & \cdots & 0 & -q_0 \\ 1 & 0 & \cdots & 0 & -q_1 \\ 0 & 1 & \cdots & 0 & -q_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -q_{n-1} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0289 (5.8)	120	■ 0.971 ● W +0	$\boldsymbol{v}_{\{1\}}:=\boldsymbol{s}_{\{1\}}, \quad \boldsymbol{v}_{\{j\}}:=\boldsymbol{s}_{\{j\}}-\sum_{i=1}^{j-1}\frac{\langle \boldsymbol{s}_{\{j\}}, \boldsymbol{v}_{\{i\}} \rangle}{\langle \boldsymbol{v}_{\{i\}}, \boldsymbol{v}_{\{i\}} \rangle} \boldsymbol{v}_{\{i\}}, \quad j=2, \dots, k.$	$\boldsymbol{v}_1 := \boldsymbol{s}_1, \quad \boldsymbol{v}_j := \boldsymbol{s}_j - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \quad j = 2, \dots, k.$	$\boldsymbol{v}_1 := \boldsymbol{s}_1, \quad \boldsymbol{v}_j := \boldsymbol{s}_j - \sum_{i=1}^{j-1} \frac{\langle \boldsymbol{s}_j, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} \boldsymbol{v}_i, \quad j = 2, \dots, k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0783	276	■ 0.973 ● K +0	$(\boldsymbol{A} \boldsymbol{B})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = (1 - \lambda) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} + \lambda (1 - \bar{\lambda}) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = (1 - \lambda ^2) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x},$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = (1 - \lambda) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} + \lambda (1 - \bar{\lambda}) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = (1 - \lambda ^2) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x},$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} + \boldsymbol{y}^* (\lambda \boldsymbol{A} \boldsymbol{x}) = (1 - \lambda) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} + \lambda (1 - \bar{\lambda}) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = (1 - \lambda ^2) \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0677 (10.14)	243	■ 0.974 ● K +0	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}).$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}).$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = (\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0527	202	■ 0.975 ● K +0	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 + \ \boldsymbol{x} - \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle + \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = 2\ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 + \ \boldsymbol{x} - \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle + \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = 2\ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 + \ \boldsymbol{x} - \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{x} + \boldsymbol{y} \rangle + \langle \boldsymbol{x} - \boldsymbol{y}, \boldsymbol{x} - \boldsymbol{y} \rangle = 2\ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{y}\ ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0994 (14.6)	336	■ 0.975 ● W -1	$\boldsymbol{A}=\left[\begin{array}{cccc} d_{\{1\}} & c_{\{1\}} & & \\ & d_{\{2\}} & c_{\{2\}} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cccc} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cccc} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0081 (2.4)	047	■0.975 ●W +0	$g(x) := \begin{cases} p_1(x), & \text{if } x_1 \leq x \leq x_2, \\ p_2(x), & \text{if } x_2 \leq x \leq x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1} \leq x \leq x_n, \\ p_n(x), & \text{if } x_n \leq x \leq x_{n+1}, \end{cases}$	$g(x) := \begin{cases} p_1(x), & \text{if } x_1 \leq x < x_2, \\ p_2(x), & \text{if } x_2 \leq x < x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1} \leq x < x_n, \\ p_n(x), & \text{if } x_n \leq x \leq x_{n+1}, \end{cases}$	$g(x) := \begin{cases} p_1(x), & \text{if } x_1 \leq x < x_2, \\ p_2(x), & \text{if } x_2 \leq x < x_3, \\ \vdots \\ p_{n-1}(x), & \text{if } x_{n-1} \leq x < x_n, \\ p_n(x), & \text{if } x_n \leq x \leq x_{n+1}, \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0998	336	■0.976 ●N +0	$p_3(x)=\left(x-d_3\right) p_2(x)-c_2^2 p_1(x)=\left(x-d_3\right)\left(x-y_1\right)\left(x-y_2\right)-c_2^2\left(x-d_1\right) .$	$p_3(x)=(x-d_3) p_2(x)-c_2^2 p_1(x)=(x-d_3)(x-y_1)(x-y_2)-c_2^2(x-d_1) .$	$p_3(x)=(x-d_3) p_2(x)-c_2^2 p_1(x)=(x-d_3)(x-y_1)(x-y_2)-c_2^2(x-d_1) .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0731	259	■0.976 ●W +0	$\begin{aligned} & \boldsymbol{F}_{2 m} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^T \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w} \\ & =\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^T \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w} \\ & =\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right]\left[\begin{array}{c} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \end{array}\right]=\left[\begin{array}{c} \boldsymbol{q}_1+\boldsymbol{q}_2 \\ \boldsymbol{q}_1-\boldsymbol{q}_2 \end{array}\right], \end{aligned}$	$\boldsymbol{F}_{2 m} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^T \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w}$ $=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right]\left[\begin{array}{c} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \end{array}\right]=\left[\begin{array}{c} \boldsymbol{q}_1+\boldsymbol{q}_2 \\ \boldsymbol{q}_1-\boldsymbol{q}_2 \end{array}\right],$	$\boldsymbol{F}_{2 m} \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{P}_{2 m}^T \boldsymbol{y}=\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m} \boldsymbol{w}$ $=\left[\begin{array}{c c} \boldsymbol{F}_m & \boldsymbol{D}_m \boldsymbol{F}_m \\ \hline \boldsymbol{F}_m & -\boldsymbol{D}_m \boldsymbol{F}_m \end{array}\right]\left[\begin{array}{c} \boldsymbol{w}_1 \\ \boldsymbol{w}_2 \end{array}\right]=\left[\begin{array}{c} \boldsymbol{q}_1+\boldsymbol{q}_2 \\ \boldsymbol{q}_1-\boldsymbol{q}_2 \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0046	037	■0.976 ●C -6	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array}\right]=\sum_k\left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array}\right]\left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array}\right],$	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array}\right]=\sum_k\left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array}\right]\left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array}\right],$	$\left[\begin{array}{ccc} c_{i_1, j_1} & \cdots & c_{i_1, j_r} \\ \vdots & & \vdots \\ c_{i_r, j_1} & \cdots & c_{i_r, j_r} \end{array}\right]=\sum_k\left[\begin{array}{ccc} a_{i_1, k_1} & \cdots & a_{i_1, k_r} \\ \vdots & & \vdots \\ a_{i_r, k_1} & \cdots & a_{i_r, k_r} \end{array}\right]\left[\begin{array}{ccc} b_{k_1, j_1} & \cdots & b_{k_1, j_r} \\ \vdots & & \vdots \\ b_{k_r, j_1} & \cdots & b_{k_r, j_r} \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0470 (8.6)	187	■0.977 ●N +0	$\ \boldsymbol{x}\ _{p'} \leq\ \boldsymbol{x}\ _p \leq n^{1 / p-1 / p'}\ \boldsymbol{x}\ _{p'}, \quad \boldsymbol{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty .$	$\ \boldsymbol{x}\ _{p'} \leq\ \boldsymbol{x}\ _p \leq n^{1 / p-1 / p'}\ \boldsymbol{x}\ _{p'}, \quad \boldsymbol{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty .$	$\ \boldsymbol{x}\ _{p'} \leq\ \boldsymbol{x}\ _p \leq n^{1 / p-1 / p'}\ \boldsymbol{x}\ _{p'}, \quad \boldsymbol{x} \in \mathbb{C}^n, \quad 1 \leq p \leq p' \leq \infty .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0969	327	0.977 ● N +1	$\begin{gathered} R_{\{i\}}=\left\{z\in\mathbb{C}:\right.\\\left. z-a_{\{i\}}\right \leq r_{\{i\}}\right\},\quad r_{\{i\}}:=\\\sum_{\substack{j=1\\j\neq i}}^n\left a_{ij}\right \\\left.\right\},\quad C_{\{j\}}=\left\{z\in\mathbb{C}:\right.\\\left. z-a_{\{j\}}\right \leq c_{\{j\}}\right\},\quad c_{\{j\}}:=\\\sum_{\substack{i=1\\i\neq j}}^n\left a_{ij}\right \\\left.\right\}.\end{gathered}$	$R_i=\left\{z\in\mathbb{C}: z-a_{ii} \leq r_i\right\},\quad r_i:=\sum_{\substack{j=1\\j\neq i}}^n a_{ij} ,$ $C_j=\left\{z\in\mathbb{C}: z-a_{jj} \leq c_j\right\},\quad c_j:=\sum_{\substack{i=1\\i\neq j}}^n a_{ij} .$	$R_i=\left\{z\in\mathbb{C}: z-a_{ii} \leq r_i\right\},\quad r_i:=\sum_{\substack{j=1\\j\neq i}}^n a_{ij} ,$ $C_j=\left\{z\in\mathbb{C}: z-a_{jj} \leq c_j\right\},\quad c_j:=\sum_{\substack{i=1\\i\neq j}}^n a_{ij} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0342	137	0.977 ● W +0	$\boldsymbol{A}=\left[\begin{array}{ll} 1&2\\ 1&2\\ 1&0\\ 1&0 \end{array}\right],\quad \boldsymbol{Q}=\frac{1}{2}\left[\begin{array}{llll} 1&1&1&1\\ 1&1&-1&-1\\ 1&-1&-1&1\\ 1&-1&1&-1 \end{array}\right],\quad \boldsymbol{R}=\left[\begin{array}{ll} 2&2\\ 0&2\\ 0&0\\ 0&0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ll} 1&2\\ 1&2\\ 1&0\\ 1&0 \end{array}\right],\quad \boldsymbol{Q}=\frac{1}{2}\left[\begin{array}{llll} 1&1&1&1\\ 1&1&-1&-1\\ 1&-1&-1&1\\ 1&-1&1&-1 \end{array}\right],\quad \boldsymbol{R}=\left[\begin{array}{ll} 2&2\\ 0&2\\ 0&0\\ 0&0 \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{ll} 1&2\\ 1&2\\ 1&0\\ 1&0 \end{array}\right],\quad \boldsymbol{Q}=\frac{1}{2}\left[\begin{array}{llll} 1&1&1&1\\ 1&1&-1&-1\\ 1&-1&-1&1\\ 1&-1&1&-1 \end{array}\right],\quad \boldsymbol{R}=\left[\begin{array}{ll} 2&2\\ 0&2\\ 0&0\\ 0&0 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0418 (7.2)	168	0.977 ● W +0	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14&2\\ 4&22\\ 16&13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1&2&2\\ 2&-2&1\\ 2&1&-2 \end{array}\right]\left[\begin{array}{cc} 2&0\\ 0&1\\ 0&0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3&4\\ 4&-3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14&2\\ 4&22\\ 16&13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1&2&2\\ 2&-2&1\\ 2&1&-2 \end{array}\right]\left[\begin{array}{cc} 2&0\\ 0&1\\ 0&0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3&4\\ 4&-3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14&2\\ 4&22\\ 16&13 \end{array}\right]=\frac{1}{3}\left[\begin{array}{ccc} 1&2&2\\ 2&-2&1\\ 2&1&-2 \end{array}\right]\left[\begin{array}{cc} 2&0\\ 0&1\\ 0&0 \end{array}\right]\frac{1}{5}\left[\begin{array}{cc} 3&4\\ 4&-3 \end{array}\right]=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0536	204	0.978 ● N +0	$(-a)\angle\boldsymbol{x},\boldsymbol{y}\rangle=\langle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ -\boldsymbol{ax}+\boldsymbol{y}\ ^2-\ -\boldsymbol{ax}-\boldsymbol{y}\ ^2\right)=-\langle\boldsymbol{ax},\boldsymbol{y}\rangle,$	$(-a)\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ -\boldsymbol{ax}+\boldsymbol{y}\ ^2-\ -\boldsymbol{ax}-\boldsymbol{y}\ ^2\right)=-\langle\boldsymbol{ax},\boldsymbol{y}\rangle,$	$(-a)\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle(-a)\boldsymbol{x},\boldsymbol{y}\rangle\stackrel{(8.36)}{=}\frac{1}{4}\left(\ -\boldsymbol{ax}+\boldsymbol{y}\ ^2-\ -\boldsymbol{ax}-\boldsymbol{y}\ ^2\right)=-\langle\boldsymbol{ax},\boldsymbol{y}\rangle,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0767	272	0.979 ● N +0	$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1)\\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0&1/2\\ 0&1/4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_k(1)\\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2\\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$	$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1)\\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0&1/2\\ 0&1/4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_k(1)\\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2\\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$	$\boldsymbol{x}_{k+1}=\left[\begin{array}{l} \boldsymbol{x}_{k+1}(1)\\ \boldsymbol{x}_{k+1}(2) \end{array}\right]=\left[\begin{array}{cc} 0&1/2\\ 0&1/4 \end{array}\right]\left[\begin{array}{l} \boldsymbol{x}_k(1)\\ \boldsymbol{x}_k(2) \end{array}\right]+\left[\begin{array}{c} 1/2\\ 3/4 \end{array}\right]=\boldsymbol{G}_1\boldsymbol{x}_k+\boldsymbol{c}.$
Conf. MathPix confidence: 0.9770.5-0.9<0.5no value Residual render vs scan (inkdrill): C componentW weakS stableN noiseK cleannot measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00125	061	■0.979 ●W +1	$\begin{aligned} & \boldsymbol{s}_j^T \boldsymbol{s}_j = \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h)) \\ & = \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2}, \end{aligned}$	$\boldsymbol{s}_j^T \boldsymbol{s}_j = \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h))$ $= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2},$	$\boldsymbol{s}_j^T \boldsymbol{s}_j = \sum_{k=1}^m \sin^2(kj\pi h) = \sum_{k=0}^m \sin^2(kj\pi h) = \frac{1}{2} \sum_{k=0}^m (1 - \cos(2kj\pi h))$ $= \frac{m+1}{2} - \frac{1}{2} \sum_{k=0}^m \cos(2kj\pi h) = \frac{m+1}{2},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00172	076	■0.980 ●W -2	$\begin{aligned} & a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I} \\ & a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}' \\ & a_{33}^{(3)} x_3 = b_3^{(3)}, \quad \text{III}'', \end{aligned}$	$a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \text{I}$ $a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \text{II}'$ $a_{33}^{(3)} x_3 = b_3^{(3)}, \text{III}'',$	$a_{11}^{(1)} x_1 + a_{12}^{(1)} x_2 + a_{13}^{(1)} x_3 = b_1^{(1)}, \quad \text{I}$ $a_{22}^{(2)} x_2 + a_{23}^{(2)} x_3 = b_2^{(2)}, \quad \text{II}'$ $a_{33}^{(3)} x_3 = b_3^{(3)}, \quad \text{III}'',$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00703	253	■0.981 ●W +0	$\begin{aligned} & \boldsymbol{T} \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j, \quad j = 1, \dots, m, \\ & \boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T, \\ & \lambda_j = 2 - 2 \cos(j\pi h) = 4 \sin^2(j\pi h/2), \quad h = 1/(m+1), \\ & \boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k. \end{aligned}$	$\boldsymbol{T} \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j, \quad j = 1, \dots, m,$ $\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T,$ $\lambda_j = 2 - 2 \cos(j\pi h) = 4 \sin^2(j\pi h/2), \quad h = 1/(m+1),$ $\boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k.$	$\boldsymbol{T} \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j, \quad j = 1, \dots, m,$ $\boldsymbol{s}_j = [\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T,$ $\lambda_j = 2 - 2 \cos(j\pi h) = 4 \sin^2(j\pi h/2), \quad h = 1/(m+1),$ $\boldsymbol{s}_j^T \boldsymbol{s}_k = \delta_{j,k}/(2h) \text{ for all } j, k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00674	243	■0.981 ●W +0	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	$\boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1 = \begin{bmatrix} \boldsymbol{T}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{T}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{T}_1 \end{bmatrix} + \begin{bmatrix} d\boldsymbol{I} & a\boldsymbol{I} & \mathbf{0} \\ a\boldsymbol{I} & d\boldsymbol{I} & a\boldsymbol{I} \\ \mathbf{0} & a\boldsymbol{I} & d\boldsymbol{I} \end{bmatrix} = \boldsymbol{T}_2$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00613 (9.13)	225	■0.981 ●N +0	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{e}_1\ }{\ \boldsymbol{b}_1\ }, \quad K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ .$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00127	062	■ 0.981 ● N +2	$\boldsymbol{\lambda} \boldsymbol{\boldsymbol{y}}^* = \boldsymbol{\boldsymbol{y}}^* \boldsymbol{A} = \left(\boldsymbol{\boldsymbol{y}}^* \boldsymbol{A} \right)^* = \left(\boldsymbol{\boldsymbol{y}}^* \boldsymbol{A} \right)^* = \left(\boldsymbol{\boldsymbol{y}}^* \boldsymbol{A} \right)^* = \boldsymbol{\mu} \boldsymbol{\boldsymbol{y}}^*,$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\boldsymbol{\mu} \boldsymbol{x}^* \boldsymbol{y})^* = \boldsymbol{\mu} \boldsymbol{y}^* \boldsymbol{x},$	$\lambda \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{y})^* = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{y})^* = (\boldsymbol{\mu} \boldsymbol{x}^* \boldsymbol{y})^* = \boldsymbol{\mu} \boldsymbol{y}^* \boldsymbol{x},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01020 (15.3)	345	■ 0.981 ● N +0	$\boldsymbol{z}_k = c_1 \boldsymbol{\lambda}_1^k + c_2 \boldsymbol{\lambda}_2^k + \cdots + c_n \boldsymbol{\lambda}_n^k, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \boldsymbol{\lambda}_1^k \boldsymbol{v}_1 + c_2 \boldsymbol{\lambda}_2^k \boldsymbol{v}_2 + \cdots + c_n \boldsymbol{\lambda}_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\boldsymbol{z}_k = c_1 \boldsymbol{\lambda}_1^k \boldsymbol{v}_1 + c_2 \boldsymbol{\lambda}_2^k \boldsymbol{v}_2 + \cdots + c_n \boldsymbol{\lambda}_n^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01021 (15.4)	345	■ 0.982 ● W +1	$\frac{\boldsymbol{z}_k}{\boldsymbol{\lambda}_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\boldsymbol{\lambda}_2}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\boldsymbol{\lambda}_n}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\frac{\boldsymbol{z}_k}{\boldsymbol{\lambda}_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\boldsymbol{\lambda}_2}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\boldsymbol{\lambda}_n}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$	$\frac{\boldsymbol{z}_k}{\boldsymbol{\lambda}_1^k} = c_1 \boldsymbol{v}_1 + c_2 \left(\frac{\boldsymbol{\lambda}_2}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_2 + \cdots + c_n \left(\frac{\boldsymbol{\lambda}_n}{\boldsymbol{\lambda}_1} \right)^k \boldsymbol{v}_n, \quad k = 0, 1, 2, \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00497	194	■ 0.982 ● N +0	$\left \boldsymbol{x}_i \right \leq \left \boldsymbol{y}_i \right , \quad i = 1, 2, \dots, n \implies \left\ \boldsymbol{x} \right\ \leq \left\ \boldsymbol{y} \right\ , \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\left \boldsymbol{x}_i \right \leq \left \boldsymbol{y}_i \right , \quad i = 1, 2, \dots, n \implies \left\ \boldsymbol{x} \right\ \leq \left\ \boldsymbol{y} \right\ , \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$	$\left \boldsymbol{x}_i \right \leq \left \boldsymbol{y}_i \right , \quad i = 1, 2, \dots, n \implies \left\ \boldsymbol{x} \right\ \leq \left\ \boldsymbol{y} \right\ , \quad \text{for all } \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00537	204	■ 0.983 ● N +0	$\left\ \boldsymbol{x} \right\ _A \leq \left\ \boldsymbol{y} \right\ _A \implies \left\ \boldsymbol{x} \right\ _2 \leq \sqrt{\frac{\boldsymbol{\lambda}_1}{\boldsymbol{\lambda}_n}} \left\ \boldsymbol{y} \right\ _2,$	$\left\ \boldsymbol{x} \right\ _A \leq \left\ \boldsymbol{y} \right\ _A \implies \left\ \boldsymbol{x} \right\ _2 \leq \sqrt{\frac{\boldsymbol{\lambda}_1}{\boldsymbol{\lambda}_n}} \left\ \boldsymbol{y} \right\ _2,$	$\left\ \boldsymbol{x} \right\ _A \leq \left\ \boldsymbol{y} \right\ _A \implies \left\ \boldsymbol{x} \right\ _2 \leq \sqrt{\frac{\boldsymbol{\lambda}_1}{\boldsymbol{\lambda}_n}} \left\ \boldsymbol{y} \right\ _2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00573	212	■ 0.984 ● N +0	$\left\ \boldsymbol{x} \right\ _p \leq \left\ \boldsymbol{x} \right\ _q \leq n^{1/q-1/p} \left\ \boldsymbol{x} \right\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$	$\left\ \boldsymbol{x} \right\ _p \leq \left\ \boldsymbol{x} \right\ _q \leq n^{1/q-1/p} \left\ \boldsymbol{x} \right\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$	$\left\ \boldsymbol{x} \right\ _p \leq \left\ \boldsymbol{x} \right\ _q \leq n^{1/q-1/p} \left\ \boldsymbol{x} \right\ _p, \quad \boldsymbol{x} \in \mathbb{C}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00184 (3.4)	079	■ 0.984 ● W +0	$\boldsymbol{L} = \begin{bmatrix} 1 & & \\ l_{21}^{(1)} & 1 & \\ \vdots & & \ddots \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{bmatrix},$	$\boldsymbol{L} = \begin{bmatrix} 1 & & \\ l_{21}^{(1)} & 1 & \\ \vdots & & \ddots \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{bmatrix},$	$\boldsymbol{L} = \begin{bmatrix} 1 & & \\ l_{21}^{(1)} & 1 & \\ \vdots & & \ddots \\ l_{n1}^{(1)} & l_{n2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} a_{11}^{(1)} & \cdots & a_{1n}^{(1)} \\ & \ddots & \vdots \\ & & a_{nn}^{(n)} \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00019 (1.7)	029	■ 0.984 ● N +0	$\dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T}).$	$\dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T}).$	$\dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T}).$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08582 (8.23)	196	0.984 N +0	$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$		$\frac{1}{K(\boldsymbol{A})} \frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ } \leq \frac{\ \boldsymbol{y} - \boldsymbol{x}\ }{\ \boldsymbol{x}\ } \leq K(\boldsymbol{A}) \frac{\ \boldsymbol{r}(\boldsymbol{y})\ }{\ \boldsymbol{b}\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08748	262	0.984 W +1	$x_{j,k} = \frac{h^4 g_{j,k}}{\sigma_j + \sigma_k - \frac{2}{3} \sigma_j \sigma_k}, \text{ where } \sigma_j = \sin^2((j\pi h)/2) \text{ for } j, k = 1, 2, \dots, m.$		$x_{j,k} = \frac{h^4 g_{j,k}}{\sigma_j + \sigma_k - \frac{2}{3} \sigma_j \sigma_k}, \text{ where } \sigma_j = \sin^2((j\pi h)/2) \text{ for } j, k = 1, 2, \dots, m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08114	059	0.985 N +0	$u^{(\prime \prime)}(jh) \approx \frac{u((j+1)h) - 2u(jh) + u((j-1)h)}{h^2}, \quad j = 1, \dots, m,$		$u''(jh) \approx \frac{u((j+1)h) - 2u(jh) + u((j-1)h)}{h^2}, \quad j = 1, \dots, m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08564	210	0.985 N +0	$A := \begin{bmatrix} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{bmatrix},$		$A := \begin{bmatrix} 2 & \mu_1 & 0 & \cdots & 0 & \lambda_1 \\ \lambda_2 & 2 & \mu_2 & \ddots & & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & & \ddots & \lambda_{n-1} & 2 & \mu_{n-1} \\ \mu_n & 0 & \cdots & 0 & \lambda_n & 2 \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08781 (11.3)	252	0.985 W +0	$\begin{aligned} \boldsymbol{y}_1 &= \boldsymbol{b}_1, \\ \boldsymbol{y}_k &= \boldsymbol{b}_k - \boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k = 2, 3, \dots, m, \\ \boldsymbol{x}_m &= \boldsymbol{U}_m^{-1} \boldsymbol{y}_m, \quad \boldsymbol{x}_k = \boldsymbol{U}_k^{-1} (\boldsymbol{y}_k - \boldsymbol{C}_k \boldsymbol{x}_{k+1}), \quad k = m-1, \dots, 2, 1. \end{aligned}$		$\begin{aligned} \boldsymbol{y}_1 &= \boldsymbol{b}_1, \quad \boldsymbol{y}_k = \boldsymbol{b}_k - \boldsymbol{L}_{k-1} \boldsymbol{y}_{k-1}, \quad k = 2, 3, \dots, m, \\ \boldsymbol{x}_m &= \boldsymbol{U}_m^{-1} \boldsymbol{y}_m, \quad \boldsymbol{x}_k = \boldsymbol{U}_k^{-1} (\boldsymbol{y}_k - \boldsymbol{C}_k \boldsymbol{x}_{k+1}), \quad k = m-1, \dots, 2, 1. \end{aligned}$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0848 (13.12)	294	■0.985 ●W +0	<pre>\begin{aligned} \boldsymbol{p}_{\textcolor{black}{j}} &\textcolor{black}{:=} \boldsymbol{r}_{\textcolor{black}{j}} - \sum_{i=0}^{\textcolor{black}{j}-1} \left(\frac{\boldsymbol{r}_{\textcolor{black}{j}}^T \boldsymbol{A} \boldsymbol{p}_{\textcolor{black}{i}}}{\boldsymbol{p}_{\textcolor{black}{i}}^T \boldsymbol{A} \boldsymbol{p}_{\textcolor{black}{i}}} \right) \boldsymbol{p}_{\textcolor{black}{i}}, \\ \boldsymbol{x}_{\textcolor{black}{j}+1} &\textcolor{black}{:=} \boldsymbol{x}_{\textcolor{black}{j}} + \alpha_{\textcolor{black}{j}} \boldsymbol{p}_{\textcolor{black}{j}} \quad \alpha_{\textcolor{black}{j}} \textcolor{black}{:=} \frac{\boldsymbol{r}_{\textcolor{black}{j}}^T \boldsymbol{r}_{\textcolor{black}{j}}}{\boldsymbol{p}_{\textcolor{black}{j}}^T \boldsymbol{A} \boldsymbol{p}_{\textcolor{black}{j}}}, \\ \boldsymbol{r}_{\textcolor{black}{j}+1} &= \boldsymbol{r}_{\textcolor{black}{j}} - \alpha_{\textcolor{black}{j}} \boldsymbol{A} \boldsymbol{p}_{\textcolor{black}{j}}. \end{aligned}</pre>	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$	$\boldsymbol{p}_j := \boldsymbol{r}_j - \sum_{i=0}^{j-1} \left(\frac{\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i} \right) \boldsymbol{p}_i,$ $\boldsymbol{x}_{j+1} := \boldsymbol{x}_j + \alpha_j \boldsymbol{p}_j \quad \alpha_j := \frac{\boldsymbol{r}_j^T \boldsymbol{r}_j}{\boldsymbol{p}_j^T \boldsymbol{A} \boldsymbol{p}_j},$ $\boldsymbol{r}_{j+1} = \boldsymbol{r}_j - \alpha_j \boldsymbol{A} \boldsymbol{p}_j.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0210	087	■0.985 ●N +0	<pre>\begin{array}{r} 10^{-4} \textcolor{black}{x}_1 + 2 \textcolor{black}{x}_2 = 4 \\ \textcolor{black}{x}_1 + \textcolor{black}{x}_2 = 3 \end{array}</pre>	$10^{-4}x_1 + 2x_2 = 4$ $x_1 + x_2 = 3$	$10^{-4}x_1 + 2x_2 = 4$ $x_1 + x_2 = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0429	172	■0.985 ●K +0	<pre>\boldsymbol{A} = \left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{array} \right].</pre>	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0145	065	■0.985 ●K +0	<pre>\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0},</pre>	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$	$\boldsymbol{B}_{11} \boldsymbol{A}_{11} = \boldsymbol{I}, \boldsymbol{B}_{21} \boldsymbol{A}_{11} = \mathbf{0}, \boldsymbol{B}_{21} \boldsymbol{A}_{12} + \boldsymbol{B}_{22} \boldsymbol{A}_{22} = \boldsymbol{I}, \boldsymbol{B}_{11} \boldsymbol{A}_{12} + \boldsymbol{B}_{12} \boldsymbol{A}_{22} = \mathbf{0}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0622 (9.17)	227	■0.985 ●W -1	<pre>\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.</pre>	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$	$\sigma_1 = \max_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}, \quad \sigma_n = \min_{\substack{\boldsymbol{x} \in \mathbb{C}^n \\ \boldsymbol{x} \neq \mathbf{0}}} \frac{\ \boldsymbol{A} \boldsymbol{x}\ _2}{\ \boldsymbol{x}\ _2}.$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 —no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00272	113	■ 0.986 ● K +0	$A=\left(\begin{array}{ccccc} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{array}\right).$	$A=\begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$	$A=\begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00268	110	■ 0.986 ● N +0	$\boldsymbol{L} \, \boldsymbol{L}^*=\left(\begin{array}{ll} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1 \end{array}\right)\left(\begin{array}{ll} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1^* \end{array}\right)=\left(\begin{array}{ll} 0 & \mathbf{0}^* \\ \mathbf{0} & B \end{array}\right)=A.$	$\boldsymbol{L} \boldsymbol{L}^*=\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1 \end{bmatrix}\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1^* \end{bmatrix}=\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & B \end{bmatrix}=A.$	$\boldsymbol{L} \boldsymbol{L}^*=\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1 \end{bmatrix}\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & L_1^* \end{bmatrix}=\begin{bmatrix} 0 & \mathbf{0}^* \\ \mathbf{0} & B \end{bmatrix}=A.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00205 (3.14)	086	■ 0.986 ● W -2	$\boldsymbol{L}=\left(\begin{array}{ccc} 1 & & \\ a_{p_2,1}^{(1)} & 1 & \\ \vdots & & \ddots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{array}\right), \quad \boldsymbol{U}=\left[\begin{array}{ccc} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{array}\right].$	$\boldsymbol{L}=\begin{bmatrix} 1 & & & \\ a_{p_2,1}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{U}=\begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{bmatrix}.$	$\boldsymbol{L}=\begin{bmatrix} 1 & & & \\ a_{p_2,1}^{(1)} & 1 & & \\ \vdots & & \ddots & \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots & 1 \end{bmatrix}, \quad \boldsymbol{U}=\begin{bmatrix} a_{p_1,1}^{(1)} & \cdots & a_{p_1,n}^{(1)} \\ & \ddots & \vdots \\ a_{p_n,1}^{(1)} & a_{p_n,2}^{(2)} & \cdots 1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00995	336	■ 0.986 ● N -1	$\boldsymbol{A}_1=\left(\begin{array}{cccc} d_1 & & & \\ c_1 & d_2 & & \\ & \ddots & \ddots & \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{array}\right) \text{ and } \boldsymbol{A}_2=\left(\begin{array}{cccc} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{array}\right).$	$\boldsymbol{A}_1=\begin{bmatrix} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{bmatrix} \text{ and } \boldsymbol{A}_2=\begin{bmatrix} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix}.$	$\boldsymbol{A}_1=\begin{bmatrix} d_1 & c_1 & & \\ c_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & c_{i-2} & d_{i-1} & c_{i-1} \\ & & & c_{i-1} & d_i \end{bmatrix} \text{ and } \boldsymbol{A}_2=\begin{bmatrix} d_{i+1} & c_{i+1} & & \\ c_{i+1} & d_{i+2} & c_{i+2} & \\ & \ddots & \ddots & \ddots \\ & & c_{n-2} & d_{n-1} & c_{n-1} \\ & & & c_{n-1} & d_n \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00300	124	■ 0.987 ● N +0	$\langle Ux,Uy\rangle=(Uy)^*(Ux)=y^*U^*Ux=y^*x=\langle x,y\rangle.$	$\langle Ux,Uy\rangle=(Uy)^*(Ux)=y^*U^*Ux=y^*x=\langle x,y\rangle.$	$\langle Ux,Uy\rangle=(Uy)^*(Ux)=y^*U^*Ux=y^*x=\langle x,y\rangle.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00426 (7.5)	172	■ 0.987 ● K +0	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*=\boldsymbol{U}_1\boldsymbol{\Sigma}_1\boldsymbol{V}_1^*.$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0397 (6.15)	159	0.987 N +1	$\sum_{j=1}^n \left \mu_{i_j} - \lambda_j - \lambda_{i_j} \right ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2$	$\sum_{j=1}^n \mu_{i_j} - \lambda_j ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2$	$\sum_{j=1}^n \mu_{i_j} - \lambda_j ^2 \leq \sum_{i=1}^n \sum_{j=1}^n a_{ij} - b_{ij} ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0511	199	0.987 N +1	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$	$\ \boldsymbol{x}\ _p := \left(\sum_{j=1}^n x_j ^p \right)^{1/p}, \quad \ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0139	064	0.987 W +0	$\boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{bmatrix}, \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{bmatrix}.$	$\boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{bmatrix}, \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{bmatrix}.$	$\boldsymbol{A} \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} \\ \boldsymbol{B}_{21} \end{bmatrix}, \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{12} \\ \boldsymbol{B}_{22} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0164 (2.38)	073	0.987 W +0	$\boldsymbol{L} = \begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{bmatrix}, \boldsymbol{U} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{bmatrix}.$	$\boldsymbol{L} = \begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{bmatrix}, \boldsymbol{U} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{bmatrix}.$	$\boldsymbol{L} = \begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ -\frac{1}{2} & 1 & \ddots & & \vdots \\ 0 & -\frac{2}{3} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{m-1}{m} & 1 \end{bmatrix}, \boldsymbol{U} = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ 0 & \frac{3}{2} & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \frac{m+1}{m} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0572	212	0.988 K +0	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$	$S_p = \{ \boldsymbol{x} \in \mathbb{R}^n : \ \boldsymbol{x}\ _p = 1 \}$
Conf. MathPix confidence: 0.987 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09949 (13.62)	321	0.988 W +0	$\begin{aligned} & \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \\ & \langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}. \end{aligned}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$ $\langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n,$ $\langle \boldsymbol{C}\boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{C}^T \boldsymbol{x} \rangle \stackrel{(13.61)}{=} \langle \boldsymbol{C}^T \boldsymbol{x}, \boldsymbol{y} \rangle, \quad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{C} \in \mathbb{R}^{n \times n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00166	073	0.988 N +0	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{a}_{:1} \boldsymbol{b}_{:1}^T + \boldsymbol{a}_{:2} \boldsymbol{b}_{:2}^T + \cdots + \boldsymbol{a}_{:n} \boldsymbol{b}_{:n}^T.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00589	217	0.989 C +5	$\boldsymbol{H}_3 = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$	$\boldsymbol{H}_3 = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$	$\boldsymbol{H}_3 = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00911 (13.44)	310	0.989 N +0	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b},$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b},$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b} \Leftrightarrow \boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{C}^{-T} \boldsymbol{x} = \boldsymbol{C}^T \boldsymbol{C} \boldsymbol{b} \Leftrightarrow \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00500	195	0.989 K +0	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ \text{ and } \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00137	064	0.989 W +1	$\begin{aligned} & (\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj} \\ & = (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}. \end{aligned}$	$(\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj}$ $= (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}.$	$(\boldsymbol{A} \boldsymbol{B})_{ij} = \sum_{k=1}^p a_{ik} b_{kj} = \sum_{k=1}^s a_{ik} b_{kj} + \sum_{k=s+1}^p a_{ik} b_{kj}$ $= (\boldsymbol{A}_1 \boldsymbol{B}_1)_{ij} + (\boldsymbol{A}_2 \boldsymbol{B}_2)_{ij} = (\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2)_{ij}.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0436 (7.7)	174	■ 0.989 ● W +0	$\begin{aligned} & \boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ & \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$	$\boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m.$	$\boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, \; i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, \; i = r + 1, \dots, n, \\ \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, \; i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, \; i = r + 1, \dots, m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0001	021	■ 0.990 ● K +0	$\boldsymbol{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right],$	$\boldsymbol{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right],$	$\boldsymbol{x} = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0060	041	■ 0.990 ● K +0	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$	$\left(\boldsymbol{A} + \boldsymbol{B} \boldsymbol{C}^T \right)^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1} \boldsymbol{B} \left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B} \right)^{-1} \boldsymbol{C}^T \boldsymbol{A}^{-1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0231 (3.21)	092	■ 0.990 ● W +1	$\begin{aligned} & \boldsymbol{A} \boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i, i = 1, \dots, r, \quad \boldsymbol{A} \boldsymbol{v}_i = 0, i = r + 1, \dots, n, \\ & \boldsymbol{A}^* \boldsymbol{u}_i = \sigma_i \boldsymbol{v}_i, i = 1, \dots, r, \quad \boldsymbol{A}^* \boldsymbol{u}_i = 0, i = r + 1, \dots, m. \end{aligned}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{bmatrix},$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_{\{m-1\}} & \boldsymbol{B} \\ \boldsymbol{C}^T & \boldsymbol{A}_{mm} \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}_{\{m-1\}} & \boldsymbol{0} \\ \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_{\{m-1\}} & \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \\ 0 & \boldsymbol{A}_{mm} - \boldsymbol{C}^T \boldsymbol{U}_{\{m-1\}}^{-1} \boldsymbol{L}_{\{m-1\}}^{-1} \boldsymbol{B} \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0293 (5.9)	122	■ 0.990 ● N +1	$\left\langle \boldsymbol{s}_0, \boldsymbol{s} \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{s} \right\rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \left\langle \boldsymbol{t}_0, \boldsymbol{t} \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{t} \right\rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle, \quad \text{for all } \boldsymbol{s} \in \mathcal{S}, \quad \langle \boldsymbol{t}_0, \boldsymbol{t} \rangle = \langle \boldsymbol{v}, \boldsymbol{t} \rangle, \quad \text{for all } \boldsymbol{t} \in \mathcal{T}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0543	206	■ 0.990 ● N +0	$\left\ \boldsymbol{A}^{-1} \right\ _2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{x} \right\ _2}{\left\ \boldsymbol{x} \right\ _2}.$	$\left\ \boldsymbol{A}^{-1} \right\ _2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{x} \right\ _2}{\left\ \boldsymbol{A} \boldsymbol{x} \right\ _2}.$	$\left\ \boldsymbol{A}^{-1} \right\ _2 = \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left\ \boldsymbol{x} \right\ _2}{\left\ \boldsymbol{A} \boldsymbol{x} \right\ _2}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0920 (13.52)	313	0.990 W +0	$\begin{aligned} &-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ &u(x,y)=0,\quad (x,y)\in\partial\Omega. \end{aligned}$	$-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ u(x,y)=0,\quad (x,y)\in\partial\Omega.$	$-\frac{\partial}{\partial x}\left(c(x,y)\frac{\partial u}{\partial x}\right)-\frac{\partial}{\partial y}\left(c(x,y)\frac{\partial u}{\partial y}\right)=f(x,y),\quad (x,y)\in\Omega=(0,1)^2, \\ u(x,y)=0,\quad (x,y)\in\partial\Omega.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0679	244	0.990 N +0	$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i,j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^t b_{i,k} d_{k,j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i,j}=((\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}))_{i,j}.$	$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i,j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^t b_{i,k} d_{k,j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i,j}=((\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}))_{i,j}.$	$((\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}))_{i,j}=\boldsymbol{A} \boldsymbol{C} \sum_{k=1}^t b_{i,k} d_{k,j}=(\boldsymbol{A} \boldsymbol{C})(\boldsymbol{B} \boldsymbol{D})_{i,j}=((\boldsymbol{A} \boldsymbol{C}) \otimes (\boldsymbol{B} \boldsymbol{D}))_{i,j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0042	036	0.991 N +1	$\begin{aligned} &\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ &=\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1). \end{aligned}$	$\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ =\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1).$	$\left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array}\right =\left \begin{array}{ccc} 1 & x & y \\ 0 & x_1-x & y_1-y \\ 0 & x_2-x_1 & y_2-y_1 \end{array}\right \\ =\left \begin{array}{cc} x_1-x & y_1-y \\ x_2-x_1 & y_2-y_1 \end{array}\right =(x_1-x)(y_2-y_1)-(y_1-y)(x_2-x_1).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0213	087	0.991 N +0	$\mathrm{fl}(x_2)=2,\quad \mathrm{fl}(x_1)=0.$	$\mathrm{fl}(x_2)=2,\quad \mathrm{fl}(x_1)=0.$	$\mathrm{fl}(x_2)=2,\quad \mathrm{fl}(x_1)=0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0234 (3.25)	094	0.991 W +0	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m(m+\frac{1}{2})(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m(m+\frac{1}{2})(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$	$\begin{aligned} &1+2+\cdots+m=\frac{1}{2}m(m+1), \\ &1^2+2^2+\cdots+m^2=\frac{1}{3}m(m+\frac{1}{2})(m+1), \\ &1+3+5+\cdots+2m-1=m^2, \\ &1*2+2*3+3*4+\cdots+(m-1)m=\frac{1}{3}(m-1)m(m+1). \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0468 (8.2)	186	0.991 N +0	$\ \boldsymbol{x}\ _p:=\left(\sum_{j=1}^n\left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty}\right)^p\right)^{1/p}.$	$\ \boldsymbol{x}\ _p:=\left(\sum_{j=1}^n\left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty}\right)^p\right)^{1/p}.$	$\ \boldsymbol{x}\ _p:=\ \boldsymbol{x}\ _\infty\left(\sum_{j=1}^n\left(\frac{ x_j }{\ \boldsymbol{x}\ _\infty}\right)^p\right)^{1/p}.$
Conf. MathPix confidence: 0.990 0.991 0.992 0.993 0.994 0.995 0.996 0.997 0.998 0.999 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09584 (9.1)	216	■0.991 ●W +0	$\boldsymbol{A} \, \boldsymbol{x} = \left[\begin{array}{c} p \left(t_1 \right) \\ \vdots \\ p \left(t_m \right) \end{array} \right] = \left[\begin{array}{ccc} \phi_1 \left(t_1 \right) & \cdots & \phi_n \left(t_1 \right) \\ \vdots & & \vdots \\ \phi_1 \left(t_m \right) & \cdots & \phi_n \left(t_m \right) \end{array} \right] \left[\begin{array}{c} x_1 \\ \vdots \\ x_n \end{array} \right] = \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] =: \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{c} p(t_1) \\ \vdots \\ p(t_m) \end{array} \right] = \left[\begin{array}{ccc} \phi_1(t_1) & \cdots & \phi_n(t_1) \\ \vdots & & \vdots \\ \phi_1(t_m) & \cdots & \phi_n(t_m) \end{array} \right] \left[\begin{array}{c} x_1 \\ \vdots \\ x_n \end{array} \right] = \left[\begin{array}{c} y_1 \\ \vdots \\ y_m \end{array} \right] =: \boldsymbol{b}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E0725	258	■0.991 ●K +0	$\boldsymbol{F}_{_4} \, \boldsymbol{P}_{_4} = \left[\begin{array}{c c} \boldsymbol{F}_{_2} & \boldsymbol{D}_2 \boldsymbol{F}_2 \\ \hline \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{array} \right].$	$\boldsymbol{F}_4 \boldsymbol{P}_4 = \left[\begin{array}{c c} \boldsymbol{F}_2 & \boldsymbol{D}_2 \boldsymbol{F}_2 \\ \hline \boldsymbol{F}_2 & -\boldsymbol{D}_2 \boldsymbol{F}_2 \end{array} \right].$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09169	074	■0.991 ●N +0	$\boldsymbol{C} \, \boldsymbol{A} \, \boldsymbol{B} = \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \boldsymbol{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{array} \right].$	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{B} = \left[\begin{array}{cc} \lambda & \boldsymbol{a}^T \boldsymbol{B}_1 \\ \boldsymbol{0} & \boldsymbol{C}_1 \boldsymbol{A}_1 \boldsymbol{B}_1 \end{array} \right].$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E09047 (1.15)	037	■0.991 ●K +0	$\operatorname{det}(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_{\boldsymbol{k}} \operatorname{det}(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \operatorname{det}(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$	$\det(\boldsymbol{C}(\boldsymbol{i}, \boldsymbol{j})) = \sum_k \det(\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{k})) \det(\boldsymbol{B}(\boldsymbol{k}, \boldsymbol{j})),$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E0422	170	■0.992 ●N +0	$\begin{aligned} & \boldsymbol{U} \, \boldsymbol{\Sigma} &= \boldsymbol{U} \left[\begin{array}{c} \boldsymbol{\Sigma}_1 \\ \vdots, \boldsymbol{\Sigma}_r \end{array} \right] \boldsymbol{e}_1, \dots, \boldsymbol{e}_r, 0, \dots, 0 \\ & &= \left[\begin{array}{c} \sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0 \end{array} \right] \\ & &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$	$\begin{aligned} \boldsymbol{U} \boldsymbol{\Sigma} &= \boldsymbol{U} [\sigma_1 \boldsymbol{e}_1, \dots, \sigma_r \boldsymbol{e}_r, 0, \dots, 0] \\ &= [\sigma_1 \boldsymbol{u}_1, \dots, \sigma_r \boldsymbol{u}_r, 0, \dots, 0] \\ &= [\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n]. \end{aligned}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E0792	252	■0.992 ●N -1	$\boldsymbol{T} \, \boldsymbol{V} + \boldsymbol{V} \, \boldsymbol{T} = h^2 \boldsymbol{F} \quad \text{with} \quad h = 1/(m+1),$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F} \quad \text{with} \quad h = 1/(m+1),$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E0857 (13.19)	299	■0.992 ●N +0	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k < e^{-\frac{2}{\kappa} k}, \quad , k > 0,$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _{\boldsymbol{A}}} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k < e^{-\frac{2}{\kappa} k}, \quad , k > 0,$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 —no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00687 (4h)	246	■ 0.992 ● N +0	$\left(\boldsymbol{s}_{\textcolor{teal}{j}} \otimes \boldsymbol{s}_{\textcolor{teal}{k}}\right)^T \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \otimes \boldsymbol{s}_{\textcolor{teal}{k}}^T\right) \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \boldsymbol{s}_{\textcolor{teal}{p}}\right) \otimes \left(\boldsymbol{s}_{\textcolor{teal}{k}}^T \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$	$\left(\boldsymbol{s}_{\textcolor{teal}{j}} \otimes \boldsymbol{s}_{\textcolor{teal}{k}}\right)^T \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \otimes \boldsymbol{s}_{\textcolor{teal}{k}}^T\right) \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \boldsymbol{s}_{\textcolor{teal}{p}}\right) \otimes \left(\boldsymbol{s}_{\textcolor{teal}{k}}^T \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$	$\left(\boldsymbol{s}_{\textcolor{teal}{j}} \otimes \boldsymbol{s}_{\textcolor{teal}{k}}\right)^T \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \otimes \boldsymbol{s}_{\textcolor{teal}{k}}^T\right) \left(\boldsymbol{s}_{\textcolor{teal}{p}} \otimes \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \left(\boldsymbol{s}_{\textcolor{teal}{j}}^T \boldsymbol{s}_{\textcolor{teal}{p}}\right) \otimes \left(\boldsymbol{s}_{\textcolor{teal}{k}}^T \boldsymbol{s}_{\textcolor{teal}{q}}\right) = \frac{1}{4h^2} \delta_{j,p} \delta_{k,q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00775 (12.20)	274	■ 0.992 ● N +0	$\left \left \boldsymbol{x}_{\textcolor{teal}{k}} - \boldsymbol{x}\right \right _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left \left \boldsymbol{x}_0 - \boldsymbol{x}\right \right _2, \quad k = 0, 1, 2, \dots$	$\left \left \boldsymbol{x}_{\textcolor{teal}{k}} - \boldsymbol{x}\right \right _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left \left \boldsymbol{x}_0 - \boldsymbol{x}\right \right _2, \quad k = 0, 1, 2, \dots$	$\left \left \boldsymbol{x}_{\textcolor{teal}{k}} - \boldsymbol{x}\right \right _2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \left \left \boldsymbol{x}_0 - \boldsymbol{x}\right \right _2, \quad k = 0, 1, 2, \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00904	309	■ 0.992 ● N +0	$\boldsymbol{\epsilon}_{\textcolor{teal}{j}} = \boldsymbol{x}_{m+1} - \boldsymbol{x}_{\textcolor{teal}{j}} = \boldsymbol{x}_m - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$	$\boldsymbol{\epsilon}_{\textcolor{teal}{j}} = \boldsymbol{x}_{m+1} - \boldsymbol{x}_{\textcolor{teal}{j}} = \boldsymbol{x}_m - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$	$\boldsymbol{\epsilon}_{\textcolor{teal}{j}} = \boldsymbol{x}_{m+1} - \boldsymbol{x}_{\textcolor{teal}{j}} = \boldsymbol{x}_m - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_m \boldsymbol{p}_m = \boldsymbol{x}_{m-1} - \boldsymbol{x}_{\textcolor{teal}{j}} + \alpha_{m-1} \boldsymbol{p}_{m-1} + \alpha_m \boldsymbol{p}_m = \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00462	183	■ 0.992 ● K +0	$\boldsymbol{A} = \left[\begin{array}{cc} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{array}\right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{array}\right].$	$\boldsymbol{A} = \left[\begin{array}{cc} 3 & 1 \\ 2 & 3 \\ -1 & 5 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00015 (1.3)	026	■ 0.993 ● N +0	$c_{\textcolor{teal}{1}} \boldsymbol{x}_{\textcolor{teal}{1}} + \dots + c_{\textcolor{teal}{n}} \boldsymbol{x}_{\textcolor{teal}{n}} = \mathbf{0} \quad \Longleftrightarrow \quad c_{\textcolor{teal}{1}} = \dots = c_{\textcolor{teal}{n}} = 0.$	$c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n = \mathbf{0} \quad \implies \quad c_1 = \dots = c_n = 0.$	$c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n = \mathbf{0} \quad \implies \quad c_1 = \dots = c_n = 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00240 (4.3)	101	■ 0.993 ● W +0	$\begin{aligned} \boldsymbol{A} &= \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{array}\right] \\ &= \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{array}\right] \end{aligned}$	$\boldsymbol{A} = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{array}\right]$ $= \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{array}\right]$	$\boldsymbol{A} = \left[\begin{array}{cc} \boldsymbol{L}_{n-1} & \mathbf{0} \\ \boldsymbol{l}_n^* & 1 \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{n-1} & \mathbf{0} \\ 0 & d_{nn} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n \\ \mathbf{0}^* & 1 \end{array}\right]$ $= \left[\begin{array}{cc} \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{l}_n \\ \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^* & \boldsymbol{l}_n^* \boldsymbol{D}_{n-1} \boldsymbol{l}_n + d_{nn} \end{array}\right]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00583	216	■ 0.993 ● N +0	$\boldsymbol{A} \, \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{c} \boldsymbol{u}_1^* \\ \boldsymbol{u}_2^* \\ \vdots \\ \boldsymbol{u}_m^* \end{array} \right] \boldsymbol{x} = \left[\begin{array}{c} y_1 \\ y_2 \\ \vdots \\ y_m \end{array} \right] = \boldsymbol{b}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00432	173	■ 0.993 ● K +0	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right], \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right].$	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right], \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right].$	$\boldsymbol{\Sigma} = \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array} \right], \quad \boldsymbol{\Sigma}_1 = [2], \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00769	273	■ 0.993 ● N +0	$\left \left \boldsymbol{x}_k - \boldsymbol{x} \right \right = \left \left \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right \right \leq \left \left \boldsymbol{G}^k \right \right \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \leq \left\ \boldsymbol{G} \right\ ^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \rightarrow \boldsymbol{0}, \quad k \rightarrow \infty.$	$\left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ = \left\ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right\ \leq \left\ \boldsymbol{G}^k \right\ \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \leq \left\ \boldsymbol{G} \right\ ^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \rightarrow \boldsymbol{0}, \quad k \rightarrow \infty.$	$\left\ \boldsymbol{x}_k - \boldsymbol{x} \right\ = \left\ \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}) \right\ \leq \left\ \boldsymbol{G}^k \right\ \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \leq \left\ \boldsymbol{G} \right\ ^k \left\ \boldsymbol{x}_0 - \boldsymbol{x} \right\ \rightarrow \boldsymbol{0}, \quad k \rightarrow \infty.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00863	300	■ 0.993 ● N -1	$\mathbb{W}_k = \text{span} \left(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0 \right), \quad k = 1, 2, 3, \dots.$	$\mathbb{W}_k = \text{span} \left(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0 \right), \quad k = 1, 2, 3, \dots.$	$\mathbb{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{A} \boldsymbol{r}_0, \boldsymbol{A}^2 \boldsymbol{r}_0, \dots, \boldsymbol{A}^{k-1} \boldsymbol{r}_0), \quad k = 1, 2, 3, \dots.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00279	117	■ 0.993 ● N +0	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00414	165	■ 0.994 ● N +1	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}.$	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}.$	$R(\boldsymbol{x}) := \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00802	281	■ 0.994 ● W -1	$\left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon.$	$\left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon.$	$\left\ \boldsymbol{A} \right\ = \left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{D}_t^{-1} \right\ _1 = \left\ \boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} \right\ _1 = \max_{1 \leq j \leq n} \sum_{i=1}^n \left (\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1})_{ij} \right $ $\leq \max_{1 \leq j \leq n} (\lambda_j + \epsilon) = \rho(\boldsymbol{A}) + \epsilon.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0175 (3.1)	077	■0.994 ●W +1	$\boldsymbol{A}^{(k)}=\left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n,k}^{(k)} & \cdots & a_{n,j}^{(k)} & \cdots & a_{n,n}^{(k)} \end{array}\right].$	$\boldsymbol{A}^{(k)}=\left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & \vdots & & \vdots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ \hline & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & \vdots & & \vdots & & \vdots \\ & & & a_{n,k}^{(k)} & \cdots & a_{n,j}^{(k)} & \cdots & a_{n,n}^{(k)} \end{array}\right].$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0833	288	■0.994 ●K +0	$A:=\left[\begin{array}{lll} 3&0&1\\ 0&7&2\\ 1&2&4 \end{array}\right]$	$A:=\left[\begin{array}{lll} 3&0&1\\ 0&7&2\\ 1&2&4 \end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{lll} 3&0&1\\ 0&7&2\\ 1&2&4 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0992	335	■0.994 ●K +0	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k\boldsymbol{Q}_{k+1} \text{ for } k=n-2,n-3,\ldots,1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k\boldsymbol{Q}_{k+1} \text{ for } k=n-2,n-3,\ldots,1.$	$\boldsymbol{Q}_{n-1}=\boldsymbol{I} \text{ and } \boldsymbol{Q}_k=\boldsymbol{H}_k\boldsymbol{Q}_{k+1} \text{ for } k=n-2,n-3,\ldots,1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1019 (15.2)	345	■0.994 ●W +0	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq\cdots\geq \lambda_n , \\ & \text{(ii) } z_0^T v_1\neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq\cdots\geq \lambda_n , \\ & \text{(ii) } z_0^T v_1\neq 0 \end{aligned}$	$\begin{aligned} & \text{(i) } \lambda_1 > \lambda_2 \geq \lambda_3 \geq\cdots\geq \lambda_n , \\ & \text{(ii) } z_0^T \boldsymbol{v}_1\neq 0 \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0124	061	■0.995 ●W +0	$\begin{aligned} & \left(\boldsymbol{T}_1\boldsymbol{s}_j\right)_k=\sum_{l=1}^m t_{k,l}\sin(lj\pi h) \\ & =a\big[\sin\big((k-1)j\pi h\big)+\sin\big((k+1)j\pi h\big)\big]+d\sin(kj\pi h) \\ & =2a\cos(j\pi h)\sin(kj\pi h)+d\sin(kj\pi h)=\lambda_js_{k,j}. \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{T}_1\boldsymbol{s}_j\right)_k=\sum_{l=1}^m t_{k,l}\sin(lj\pi h) \\ & =a\big[\sin\big((k-1)j\pi h\big)+\sin\big((k+1)j\pi h\big)\big]+d\sin(kj\pi h) \\ & =2a\cos(j\pi h)\sin(kj\pi h)+d\sin(kj\pi h)=\lambda_js_{k,j}. \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{T}_1\boldsymbol{s}_j\right)_k=\sum_{l=1}^m t_{k,l}\sin(lj\pi h) \\ & =a\big[\sin\big((k-1)j\pi h\big)+\sin\big((k+1)j\pi h\big)\big]+d\sin(kj\pi h) \\ & =2a\cos(j\pi h)\sin(kj\pi h)+d\sin(kj\pi h)=\lambda_js_{k,j}. \end{aligned}$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08384 (6.9)	155	■ 0.995 ● N +0	$R_{\{\mathbf{A}\}}(\mathbf{x})=\frac{\sum_{i=1}^n\lambda_i\left c_i\right ^2}{\sum_{j=1}^n\left c_j\right ^2},\quad \mathbf{x}\neq\mathbf{0},\quad \mathbf{x}=\sum_{j=1}^nc_j\mathbf{c}_j$	$R_A(\boldsymbol{x})=\frac{\sum_{i=1}^n\lambda_i c_i ^2}{\sum_{j=1}^n c_j ^2},\quad \boldsymbol{x}\neq\mathbf{0},\quad \boldsymbol{x}=\sum_{j=1}^nc_j\boldsymbol{u}_j.$	$R_A(\boldsymbol{x})=\frac{\sum_{i=1}^n\lambda_i c_i ^2}{\sum_{j=1}^n c_j ^2},\quad \boldsymbol{x}\neq\mathbf{0},\quad \boldsymbol{x}=\sum_{j=1}^nc_j\boldsymbol{u}_j.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08632	229	■ 0.995 ● W +0	$\begin{gathered} \lambda_1\geq\cdots\geq\lambda_{m+n}=\alpha_1\geq\cdots\geq\alpha_t\geq0=\cdots=0\geq-\alpha_t\geq\cdots\geq-\alpha_1, \\ \mu_1\geq\cdots\geq\mu_{m+n}=\beta_1\geq\cdots\geq\beta_t\geq0=\cdots=0\geq-\beta_t\geq\cdots\geq-\beta_1. \end{gathered}$	$\lambda_1\geq\cdots\geq\lambda_{m+n}=\alpha_1\geq\cdots\geq\alpha_t\geq0=\cdots=0\geq-\alpha_t\geq\cdots\geq-\alpha_1, \\ \mu_1\geq\cdots\geq\mu_{m+n}=\beta_1\geq\cdots\geq\beta_t\geq0=\cdots=0\geq-\beta_t\geq\cdots\geq-\beta_1.$	$\lambda_1\geq\cdots\geq\lambda_{m+n}=\alpha_1\geq\cdots\geq\alpha_t\geq0=\cdots=0\geq-\alpha_t\geq\cdots\geq-\alpha_1, \\ \mu_1\geq\cdots\geq\mu_{m+n}=\beta_1\geq\cdots\geq\beta_t\geq0=\cdots=0\geq-\beta_t\geq\cdots\geq-\beta_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08647	233	■ 0.995 ● K +0	$\mathbf{A}^{\dagger}=\frac{1}{\alpha}\mathbf{A}^*,\quad \alpha=\ \boldsymbol{u}\ _2^2\ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger}=\frac{1}{\alpha}\boldsymbol{A}^*,\quad \alpha=\ \boldsymbol{u}\ _2^2\ \boldsymbol{v}\ _2^2.$	$\boldsymbol{A}^{\dagger}=\frac{1}{\alpha}\boldsymbol{A}^*,\quad \alpha=\ \boldsymbol{u}\ _2^2\ \boldsymbol{v}\ _2^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E0881 (13.31)	304	■ 0.995 ● N +0	$x_{n+1}-2x_n+x_{n-1}=0\text{ for }n\geq 1,\text{ with }x_0=1,x_1=t.$	$x_{n+1}-2x_n+x_{n-1}=0\text{ for }n\geq 1,\text{ with }x_0=1,x_1=t.$	$x_{n+1}-2x_n+x_{n-1}=0\text{ for }n\geq 1,\text{ with }x_0=1,x_1=t.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08301	124	■ 0.995 ● N +0	$\left(\mathbf{U}^*\mathbf{U}\right)_{i,j}=\mathbf{e}_i^*\mathbf{U}^*\mathbf{U}\mathbf{e}_j=\left(\mathbf{U}\mathbf{e}_i\right)^*\left(\mathbf{U}\mathbf{e}_j\right)=\left\langle \mathbf{U}\mathbf{e}_j,\mathbf{U}\mathbf{e}_i\right\rangle =\left\langle \mathbf{e}_j,\mathbf{e}_i\right\rangle =\mathbf{e}_i^*\mathbf{e}_j,$	$\left(\boldsymbol{U}^*\boldsymbol{U}\right)_{i,j}=\boldsymbol{e}_i^*\boldsymbol{U}^*\boldsymbol{U}\boldsymbol{e}_j=\left(\boldsymbol{U}\boldsymbol{e}_i\right)^*\left(\boldsymbol{U}\boldsymbol{e}_j\right)=\left\langle \boldsymbol{U}\boldsymbol{e}_j,\boldsymbol{U}\boldsymbol{e}_i\right\rangle =\left\langle \boldsymbol{e}_j,\boldsymbol{e}_i\right\rangle =\boldsymbol{e}_i^*\boldsymbol{e}_j,$	$\left(\boldsymbol{U}^*\boldsymbol{U}\right)_{i,j}=\boldsymbol{e}_i^*\boldsymbol{U}^*\boldsymbol{U}\boldsymbol{e}_j=\left(\boldsymbol{U}\boldsymbol{e}_i\right)^*\left(\boldsymbol{U}\boldsymbol{e}_j\right)=\left\langle \boldsymbol{U}\boldsymbol{e}_j,\boldsymbol{U}\boldsymbol{e}_i\right\rangle =\left\langle \boldsymbol{e}_j,\boldsymbol{e}_i\right\rangle =\boldsymbol{e}_i^*\boldsymbol{e}_j,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08311	126	■ 0.995 ● W -1	$\mathbf{H}\mathbf{x}=\mathbf{a}\mathbf{e}_1,\quad \mathbf{a}=-\rho\ \mathbf{x}\ _2,\quad \rho:=\begin{cases}x_1/ x_1 ,&\text{if }x_1\neq 0,\\1,&\text{otherwise.}\end{cases}$	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{a}\boldsymbol{e}_1,\quad \boldsymbol{a}=-\rho\ \boldsymbol{x}\ _2,\quad \rho:=\begin{cases}x_1/ x_1 ,&\text{if }x_1\neq 0,\\1,&\text{otherwise.}\end{cases}$	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{a}\boldsymbol{e}_1,\quad \boldsymbol{a}=-\rho\ \boldsymbol{x}\ _2,\quad \rho:=\begin{cases}x_1/ x_1 ,&\text{if }x_1\neq 0,\\1,&\text{otherwise.}\end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08013	025	■ 0.995 ● N +0	$\left(\mathbf{f}+\mathbf{g}\right)(t):=\mathbf{f}(t)+\mathbf{g}(t),\quad \left(c\mathbf{f}\right)(t):=c\mathbf{f}(t),\quad t\in\mathcal{D},\quad c\in\mathbb{R}.$	$\left(\boldsymbol{f}+\boldsymbol{g}\right)(t):=\boldsymbol{f}(t)+\boldsymbol{g}(t),\quad \left(c\boldsymbol{f}\right)(t):=c\boldsymbol{f}(t),\quad t\in\mathcal{D},\quad c\in\mathbb{R}.$	$\left(\boldsymbol{f}+\boldsymbol{g}\right)(t):=\boldsymbol{f}(t)+\boldsymbol{g}(t),\quad \left(c\boldsymbol{f}\right)(t):=c\boldsymbol{f}(t),\quad t\in\mathcal{D},\quad c\in\mathbb{R}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E08061 (1.20)	041	■ 0.995 ● K +0	$\lambda:=\mathbf{a}^T\mathbf{C}\mathbf{e}_i\neq 0,$	$\lambda:=\boldsymbol{a}^T\boldsymbol{C}\boldsymbol{e}_i\neq 0,$	$\lambda:=\boldsymbol{a}^T\boldsymbol{C}\boldsymbol{e}_i\neq 0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00216	088	0.995 W +2	$x_2=\frac{3.9997}{1.9999}\approx 2, \quad \text{\quad}$ $x_1=3-x_2\approx 1 \text{ .}$	$x_2=\frac{3.9997}{1.9999}\approx 2, \quad x_1=3-x_2\approx 1.$	$x_2=\frac{3.9997}{1.9999}\approx 2, \quad x_1=3-x_2\approx 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00284	118	0.995 W +0	$\begin{aligned} & \ \boldsymbol{x}\ ^2 \text{ \& } = \langle \boldsymbol{x}, \boldsymbol{x} \rangle \\ & \ \boldsymbol{z} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{z} + a\boldsymbol{y}, \boldsymbol{z} + a\boldsymbol{y} \rangle \\ & = \langle \boldsymbol{z}, \boldsymbol{z} \rangle + 2\operatorname{Re}\langle \boldsymbol{z}, a\boldsymbol{y} \rangle + a ^2\ \boldsymbol{y}\ ^2 \\ & = \ \boldsymbol{z}\ ^2 + 2 a \operatorname{Re}\langle \boldsymbol{z}, \boldsymbol{y} \rangle + a ^2\ \boldsymbol{y}\ ^2 \\ & \geq a ^2\ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}. \end{aligned}$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + a\boldsymbol{y}, \boldsymbol{z} + a\boldsymbol{y} \rangle$ $\stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle a\boldsymbol{y}, a\boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + a ^2\ \boldsymbol{y}\ ^2$ $\geq a ^2\ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$	$\ \boldsymbol{x}\ ^2 = \langle \boldsymbol{x}, \boldsymbol{x} \rangle = \langle \boldsymbol{z} + a\boldsymbol{y}, \boldsymbol{z} + a\boldsymbol{y} \rangle$ $\stackrel{(5.4)}{=} \langle \boldsymbol{z}, \boldsymbol{z} \rangle + \langle a\boldsymbol{y}, a\boldsymbol{y} \rangle \stackrel{(5.1)}{=} \ \boldsymbol{z}\ ^2 + a ^2\ \boldsymbol{y}\ ^2$ $\geq a ^2\ \boldsymbol{y}\ ^2 = \frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle ^2}{\ \boldsymbol{y}\ ^2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00374	152	0.995 N -1	$\boldsymbol{W} = \left[\begin{array}{c} \boldsymbol{1} \\ \boldsymbol{0} \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V}\boldsymbol{W}.$	$\boldsymbol{W} = \left[\begin{array}{c} \boldsymbol{1} \\ \boldsymbol{0} \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V}\boldsymbol{W}.$	$\boldsymbol{W} = \left[\begin{array}{c} \boldsymbol{1} \\ \boldsymbol{0} \end{array} \right] \text{ and } \boldsymbol{U} = \boldsymbol{V}\boldsymbol{W}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00858 (13.20)	299	0.995 N +0	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k < 2e^{-\frac{2}{\sqrt{\kappa}}k}, \quad k \geq 0.$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k < 2e^{-\frac{2}{\sqrt{\kappa}}k}, \quad k \geq 0.$	$\frac{\ \boldsymbol{x} - \boldsymbol{x}_k\ _A}{\ \boldsymbol{x} - \boldsymbol{x}_0\ _A} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k < 2e^{-\frac{2}{\sqrt{\kappa}}k}, \quad k \geq 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00459	182	0.995 K +0	$\begin{aligned} & x_1 - x_3 = 4, \\ & x_1 + x_2 + x_3 = 12. \end{aligned}$	$x_1 - x_3 = 4,$ $x_1 + x_2 + x_3 = 12.$	$x_1 - x_3 = 4,$ $x_1 + x_2 + x_3 = 12.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00982	331	0.995 N -1	$\ \boldsymbol{r}\ _p = \left\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r} \right\ _p \leq \left\ \boldsymbol{D}_1^{-1} \right\ _p K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p = \frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min_j \lambda_j - \mu }.$	$1 = \ \boldsymbol{x}\ _p = \left\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r} \right\ _p \leq \left\ \boldsymbol{D}_1^{-1} \right\ _p K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p = \frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min_j \lambda_j - \mu }.$	$1 = \ \boldsymbol{x}\ _p = \left\ \boldsymbol{X}\boldsymbol{D}_1^{-1}\boldsymbol{X}^{-1}\boldsymbol{r} \right\ _p \leq \left\ \boldsymbol{D}_1^{-1} \right\ _p K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p = \frac{K_p(\boldsymbol{X})\ \boldsymbol{r}\ _p}{\min_j \lambda_j - \mu }.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0951 (13.63)	321	0.995 N -1	$\left \left \boldsymbol{A}^{-1}\right \right _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{x}\ _2}{\ \boldsymbol{A} \boldsymbol{x}\ _2},$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{x}\ _2}{\ \boldsymbol{A} \boldsymbol{x}\ _2},$	$\left\ \boldsymbol{A}^{-1}\right\ _2=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{x}\ _2}{\ \boldsymbol{A} \boldsymbol{x}\ _2},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0952 (8.39)	203	0.995 N +0	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle\boldsymbol{x}, \boldsymbol{y}\rangle.$	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle\boldsymbol{x}, \boldsymbol{y}\rangle.$	$\langle n \boldsymbol{x}, \boldsymbol{y}\rangle=\langle(n-1) \boldsymbol{x}+\boldsymbol{x}, \boldsymbol{y}\rangle \stackrel{(8.37)}{=} \langle(n-1) \boldsymbol{x}, \boldsymbol{y}\rangle+\langle\boldsymbol{x}, \boldsymbol{y}\rangle=n\langle\boldsymbol{x}, \boldsymbol{y}\rangle.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0648	233	0.995 K +0	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$	$\lambda_i^{\dagger}=\left\{\begin{array}{cc} 1 / \lambda_i, & \lambda_i \neq 0 \\ 0 & \lambda_i=0 . \end{array}\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0975	330	0.995 W +0	$\boldsymbol{A}_1:=\left(\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right), \quad \boldsymbol{E}:=\left(\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right)=\epsilon \boldsymbol{e}_n \boldsymbol{e}_1^T .$	$\boldsymbol{A}_1:=\left(\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right), \quad \boldsymbol{E}:=\left(\begin{array}{cccccc} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \epsilon & 0 & 0 & \cdots & 0 & 0 \end{array}\right)=\epsilon \boldsymbol{e}_n \boldsymbol{e}_1^T .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0926 (13.56)	314	0.996 W +1	$L_{\boldsymbol{h}} \boldsymbol{v}=\boldsymbol{F}, \quad L_{\boldsymbol{h}} \boldsymbol{v}:=\frac{1}{h^2}\left(\begin{array}{ccc} (d v)_{1,1} & \cdots & (d v)_{1, m} \\ \vdots & & \vdots \\ (d v)_{m, 1} & \cdots & (d v)_{m, m} \end{array}\right), \quad \boldsymbol{F}:=\left(\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right) .$	$L_{\boldsymbol{h}} \boldsymbol{v}=\boldsymbol{F}, \quad L_{\boldsymbol{h}} \boldsymbol{v}:=\frac{1}{h^2}\left(\begin{array}{ccc} (d v)_{1,1} & \cdots & (d v)_{1, m} \\ \vdots & & \vdots \\ (d v)_{m, 1} & \cdots & (d v)_{m, m} \end{array}\right), \quad \boldsymbol{F}:=\left(\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right) .$	$L_{\boldsymbol{h}} \boldsymbol{v}=\boldsymbol{F}, \quad L_{\boldsymbol{h}} \boldsymbol{v}:=\frac{1}{h^2}\left(\begin{array}{ccc} (d v)_{1,1} & \cdots & (d v)_{1, m} \\ \vdots & & \vdots \\ (d v)_{m, 1} & \cdots & (d v)_{m, m} \end{array}\right), \quad \boldsymbol{F}:=\left(\begin{array}{ccc} f_{1,1} & \cdots & f_{1, m} \\ \vdots & & \vdots \\ f_{m, 1} & \cdots & f_{m, m} \end{array}\right) .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1024	346	0.996 N +0	$\lim _k\left(\frac{\left \lambda_1\right }{\lambda_1}\right)^k \boldsymbol{x}_k=\frac{c_1}{\left c_1\right } \frac{\boldsymbol{v}_1}{\left\ \boldsymbol{v}_1\right\ } .$	$\lim _{k \rightarrow \infty}\left(\frac{\left \lambda_1\right }{\lambda_1}\right)^k \boldsymbol{x}_k=\frac{c_1}{\left c_1\right } \frac{\boldsymbol{v}_1}{\left\ \boldsymbol{v}_1\right\ } .$	$\lim _{k \rightarrow \infty}\left(\frac{\left \lambda_1\right }{\lambda_1}\right)^k \boldsymbol{x}_k=\frac{c_1}{\left c_1\right } \frac{\boldsymbol{v}_1}{\left\ \boldsymbol{v}_1\right\ } .$
Conf. MathPix confidence: — no value Residual render vs scan (inkdrill): <					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00031	032	■0.996 ●W +0	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array}\right]=\boldsymbol{b}.$	$\boldsymbol{A} \boldsymbol{x}=\left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array}\right]\left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array}\right]=\boldsymbol{b}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00207	086	■0.996 ●C +3	$\begin{aligned} & (\boldsymbol{L} \boldsymbol{U})_{ij}=\sum_{k=1}^n l_{i,k} u_{kj}=\sum_{k=1}^{i-1} a_{p_i,k}^{(k)} a_{p_k,j}^{(k)}+a_{p_i,j}^{(i)} \\ & \quad \quad \quad =\sum_{k=1}^{i-1}\left(a_{p_i,j}^{(k)}-a_{p_i,j}^{(k+1)}\right)+a_{p_i,j}^{(i)}=a_{p_i,j}^{(1)}=a_{p_i,j}=\left(\boldsymbol{P}^T \boldsymbol{A}\right)_{i j}, \end{aligned}$	$(\boldsymbol{L} \boldsymbol{U})_{ij}=\sum_{k=1}^n l_{i,k} u_{kj}=\sum_{k=1}^{i-1} a_{p_i,k}^{(k)} a_{p_k,j}^{(k)}+a_{p_i,j}^{(i)} \\ =\sum_{k=1}^{i-1}\left(a_{p_i,j}^{(k)}-a_{p_i,j}^{(k+1)}\right)+a_{p_i,j}^{(i)}=a_{p_i,j}^{(1)}=a_{p_i,j}=\left(\boldsymbol{P}^T \boldsymbol{A}\right)_{i j},$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00739	261	■0.996 ●N +0	$\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}-\frac{1}{6} \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}=4 h^4 \boldsymbol{G}, \text { where } \boldsymbol{G}=S \mu \boldsymbol{F} S,$	$\boldsymbol{D} \boldsymbol{X}+\boldsymbol{X} \boldsymbol{D}-\frac{1}{6} \boldsymbol{D} \boldsymbol{X} \boldsymbol{D}=4 h^4 \boldsymbol{G}, \text { where } \boldsymbol{G}=S \mu \boldsymbol{F} S,$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00787 (12.27)	277	■0.996 ●N +0	$\rho\left(\boldsymbol{G}_\omega\right)=\left\{\begin{array}{ll} \frac{1}{4}\left(\omega \beta+\sqrt{(\omega \beta)^2-4(\omega-1)}\right)^2, & \text { for } 0<\omega \leq \omega^*, \\ \omega-1, & \text { for } \omega^*<\omega<2, \end{array}\right.$	$\rho\left(\boldsymbol{G}_\omega\right)=\left\{\begin{array}{ll} \frac{1}{4}\left(\omega \beta+\sqrt{(\omega \beta)^2-4(\omega-1)}\right)^2, & \text { for } 0<\omega \leq \omega^*, \\ \omega-1, & \text { for } \omega^*<\omega<2, \end{array}\right.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00819	285	■0.996 ●K +0	$d(\omega):=(\omega \mu)^2-4(\omega-1) \text{ .}$	$d(\omega):=(\omega \mu)^2-4(\omega-1).$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00850 (13.16)	295	■0.996 ●N +0	$\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i \stackrel{(13.13)}{=} \boldsymbol{r}_j^T\left(\frac{\boldsymbol{r}_i-\boldsymbol{r}_{i+1}}{\alpha_i}\right)=0, \quad i=0,1, \ldots, j-2 .$	$\boldsymbol{r}_j^T \boldsymbol{A} \boldsymbol{p}_i \stackrel{(13.13)}{=} \boldsymbol{r}_j^T\left(\frac{\boldsymbol{r}_i-\boldsymbol{r}_{i+1}}{\alpha_i}\right)=0, \quad i=0,1, \ldots, j-2 .$	
Conf. MathPix confidence: ■ ■ ■ — ≥0.9 0.5-0.9 <0.5 no value Residual render vs scan (inkdrill): ● ● ● ● ● ● C component W weak S stable N noise K clean not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0991	334	0.996 W +0	<pre>\begin{aligned} \boldsymbol{A}_{k+1} &= \\ \boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k} &= \\ \left[\begin{array}{cc} \boldsymbol{I}_{k} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_{k} \end{array}\right] \begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \\ \boldsymbol{D}_{k} & \boldsymbol{E}_{k} \end{array} \begin{array}{cc} \boldsymbol{I}_{k} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_{k} \end{array} \\ &= \begin{array}{cc} \boldsymbol{B}_{k} & \boldsymbol{C}_{k} \boldsymbol{V}_{k} \\ \boldsymbol{V}_{k} \boldsymbol{D}_{k} & \boldsymbol{V}_{k} \boldsymbol{E}_{k} \boldsymbol{V}_{k} \end{array} . \end{aligned}</pre>	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{bmatrix} \\ = \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{bmatrix} .$	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k = \begin{bmatrix} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{D}_k & \boldsymbol{E}_k \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_k & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{V}_k \end{bmatrix} \\ = \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \boldsymbol{V}_k \\ \boldsymbol{V}_k \boldsymbol{D}_k & \boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k \end{bmatrix} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0454 (7.14)	181	0.997 K +0	<pre>\boldsymbol{X}_{k+1}=\frac{1}{2}\left(\boldsymbol{X}_{k}+\boldsymbol{X}_{k}^{-T}\right), k=0,1,2, \ldots \text { with } \\ \boldsymbol{X}_0=\boldsymbol{A},</pre>	$\boldsymbol{X}_{k+1} = \frac{1}{2} \left(\boldsymbol{X}_k + \boldsymbol{X}_k^{-T} \right), k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$	$\boldsymbol{X}_{k+1} = \frac{1}{2} \left(\boldsymbol{X}_k + \boldsymbol{X}_k^{-T} \right), \, k = 0, 1, 2, \dots \text{ with } \boldsymbol{X}_0 = \boldsymbol{A},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0135	064	0.997 K +0	<pre>\left[\begin{array}{l} A_1 \\ A_2 \end{array}\right] B=\left[\begin{array}{l} A_1 B \\ A_2 B \end{array}\right] .</pre>	$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} B = \begin{bmatrix} A_1 B \\ A_2 B \end{bmatrix} .$	$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} B = \begin{bmatrix} A_1 B \\ A_2 B \end{bmatrix} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0269 (4.11)	110	0.997 W +2	<pre>\begin{aligned} &\text { if } A(1,1)>0 \quad \& \quad A(1,1)= \\ &\sqrt{A(1,1)} \quad \& \quad A(2: n, 1)=A(2: n, 1) / A(1,1) \quad \\ &\text { for } j=2: n \quad \& \quad \quad A(j: n, j)=A(j: n, \\ &j)-A(j, 1) * A(j: n, 1) \end{aligned}</pre>	if $A(1,1) > 0$ $A(1,1) = \sqrt{A(1,1)}$ $A(2:n,1) = A(2:n,1)/A(1,1)$ for $j = 2:n$ $A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1)$	if $A(1,1) > 0$ $A(1,1) = \sqrt{A(1,1)}$ $A(2:n,1) = A(2:n,1)/A(1,1)$ for $j = 2:n$ $A(j:n,j) = A(j:n,j) - A(j,1) * A(j:n,1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0431	173	0.997 N +0	<pre>\boldsymbol{B}\left[\begin{array}{l} 1 \\ 1 \end{array}\right]=4\left[\begin{array}{l} 1 \\ 1 \end{array}\right], \quad \boldsymbol{B}\left[\begin{array}{l} 1 \\ -1 \end{array}\right]=0\left[\begin{array}{l} 1 \\ -1 \end{array}\right],</pre>	$\boldsymbol{B} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix},$	$\boldsymbol{B} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{B} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix},$
Conf. MathPix confidence: 0.996 0.997 0.999 1.000 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0656 (10.1)	238	0.997 N +0	$\begin{gathered} -\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega. \end{gathered}$	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega.$	$-\Delta u := -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f \text{ on } \Omega, \\ u := 0 \text{ on } \partial\Omega.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0666 (10.7)	240	0.997 N +0	$\mathbf{A} \mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{n \times n}, \quad \mathbf{b} \in \mathbb{R}^n, \quad n = m^2,$	$\mathbf{A} \mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{n \times n}, \quad \mathbf{b} \in \mathbb{R}^n, \quad n = m^2,$	$\mathbf{A} \mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{n \times n}, \quad \mathbf{b} \in \mathbb{R}^n, \quad n = m^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0671 (10.11)	242	0.997 W +2	$\mathbf{T}_2 = \left[\begin{array}{ccc ccc} 2d & a & 0 & a & 0 & 0 \\ a & 2d & a & 0 & a & 0 \\ 0 & a & 2d & 0 & 0 & a \\ \hline a & 0 & 0 & 2d & a & 0 \\ 0 & a & 0 & a & 2d & a \\ 0 & 0 & a & 0 & a & 2d \\ \hline 0 & 0 & 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 & a & 0 \\ 0 & 0 & 0 & 0 & 0 & a \end{array} \right]$	$\mathbf{T}_2 = \left[\begin{array}{ccc ccc} 2d & a & 0 & a & 0 & 0 \\ a & 2d & a & 0 & a & 0 \\ 0 & a & 2d & 0 & 0 & a \\ \hline a & 0 & 0 & 2d & a & 0 \\ 0 & a & 0 & a & 2d & a \\ 0 & 0 & a & 0 & a & 2d \\ \hline 0 & 0 & 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 & a & 0 \\ 0 & 0 & 0 & 0 & 0 & a \end{array} \right]$	$\mathbf{T}_2 = \left[\begin{array}{ccc ccc} 2d & a & 0 & a & 0 & 0 \\ a & 2d & a & 0 & a & 0 \\ 0 & a & 2d & 0 & 0 & a \\ \hline a & 0 & 0 & 2d & a & 0 \\ 0 & a & 0 & a & 2d & a \\ 0 & 0 & a & 0 & a & 2d \\ \hline 0 & 0 & 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 & a & 0 \\ 0 & 0 & 0 & 0 & 0 & a \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1010 (14.10)	340	0.997 W +1	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases}$	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases}$	$a_{kj} = \begin{cases} 1, & j = k + 1, \\ 0, & \text{otherwise} \end{cases}, \quad e_{kj} = \begin{cases} \epsilon, & k = n, j = 1, \\ 0, & \text{otherwise} \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0294 (5.10)	122	0.997 N +0	$\mathbf{s}_0 = \sum_{i=1}^k \frac{\langle \mathbf{v}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i.$	$\mathbf{s}_0 = \sum_{i=1}^k \frac{\langle \mathbf{v}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i.$	$\mathbf{s}_0 = \sum_{i=1}^k \frac{\langle \mathbf{v}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0416	166	0.997 N +0	$\mathbf{B}_1 := \{ \mathbf{x} \in \mathbb{R}^n \mid \ \mathbf{x}\ _2 = 1 \}.$	$\mathbf{B}_1 := \{ \mathbf{x} \in \mathbb{R}^n \mid \ \mathbf{x}\ _2 = 1 \}.$	$\mathbf{B}_1 := \{ \mathbf{x} \in \mathbb{R}^n \mid \ \mathbf{x}\ _2 = 1 \}.$

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0694 (10.24)	248	■0.997 ● W +1	$\begin{array}{lrl} -\Delta v(s,t) & = f(s,t) & (s,t) \in \Omega \\ -\Delta u(s,t) & = v(s,t) & (s,t) \in \Omega \\ u(s,t) & = v(s,t) = 0 & (s,t) \in \partial\Omega \end{array}$	$\begin{array}{lrl} -\Delta v(s,t) & = f(s,t) & (s,t) \in \Omega \\ -\Delta u(s,t) & = v(s,t) & (s,t) \in \Omega \\ u(s,t) & = v(s,t) = 0 & (s,t) \in \partial\Omega \end{array}$	$\begin{array}{lrl} -\Delta v(s,t) & = f(s,t) & (s,t) \in \Omega \\ -\Delta u(s,t) & = v(s,t) & (s,t) \in \Omega \\ u(s,t) & = v(s,t) = 0 & (s,t) \in \partial\Omega. \end{array}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0768 (12.16)	272	■0.997 ● K +0	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x}), \quad k = 0, 1, 2, \dots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0815	284	■0.997 ● N +0	$\boldsymbol{G}_J \boldsymbol{v} = (\boldsymbol{L} + \boldsymbol{R}) \boldsymbol{v} = \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}} \boldsymbol{v} = \mu \boldsymbol{v}.$	$\boldsymbol{G}_J \boldsymbol{v} = (\boldsymbol{L} + \boldsymbol{R}) \boldsymbol{v} = \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}} \boldsymbol{v} = \mu \boldsymbol{v}.$	$\boldsymbol{G}_J \boldsymbol{v} = (\boldsymbol{L} + \boldsymbol{R}) \boldsymbol{v} = \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}} \boldsymbol{v} = \mu \boldsymbol{v}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0847 (13.11)	294	■0.997 ● W +0	$\begin{aligned} & \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ & \boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}. \end{aligned}$	$\begin{aligned} & \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ & \boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}. \end{aligned}$	$\begin{aligned} & \boldsymbol{t}_{2k-2} = 3 * 4^{1-k} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \quad \boldsymbol{x}_{2k-1} = -4^{-k} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \boldsymbol{r}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ & \boldsymbol{t}_{2k-1} = 3 * 4^{-k} \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad \boldsymbol{x}_{2k} = -4^{-k} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad \boldsymbol{r}_{2k} = 3 * 4^{-k} \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0355 (5.22)	142	■0.998 ● N +0	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$	$P_{i_1, n+1} P_{i_2, n+1} \cdots P_{i_n, n+1} \begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}' \\ 0 \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0390 (6.11)	157	■0.998 ● K +0	$\lambda_k \leq \max_{\boldsymbol{x} \in \mathcal{S}} R(\boldsymbol{x}),$	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$	$\lambda_k \leq \max_{\substack{\boldsymbol{x} \in \mathcal{S} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0448	179	■0.998 ● W +0	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00765	272	0.998 N +1	$x_{k+1}(1)=\frac{1}{2} x_k(2)+\frac{1}{2}, \quad \quad x_{k+1}(2)=\frac{1}{2} x_{k+1}(1)+\frac{1}{2} \quad .$	$x_{k+1}(1)=\frac{1}{2}x_k(2)+\frac{1}{2}, \quad x_{k+1}(2)=\frac{1}{2}x_{k+1}(1)+\frac{1}{2}.$	$\mathbf{x}_{k+1}(1)=\frac{1}{2}\mathbf{x}_k(2)+\frac{1}{2}, \quad \mathbf{x}_{k+1}(2)=\frac{1}{2}\mathbf{x}_{k+1}(1)+\frac{1}{2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00228 (3.19)	092	0.998 N +0	$\mathbf{A}:=\left[\begin{array}{ccc}\mathbf{A}_{11} & \cdots & \mathbf{A}_{1m} \\ \vdots & & \vdots \\ \mathbf{A}_{m1} & \cdots & \mathbf{A}_{mm}\end{array}\right],$	$\mathbf{A}:=\left[\begin{array}{ccc}\mathbf{A}_{11} & \cdots & \mathbf{A}_{1m} \\ \vdots & & \vdots \\ \mathbf{A}_{m1} & \cdots & \mathbf{A}_{mm}\end{array}\right],$	$\mathbf{A}:=\left[\begin{array}{ccc}\mathbf{A}_{11} & \cdots & \mathbf{A}_{1m} \\ \vdots & & \vdots \\ \mathbf{A}_{m1} & \cdots & \mathbf{A}_{mm}\end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00603 (9.8)	222	0.998 N +0	$\mathbf{A} \mathbf{A}^{\dagger} \mathbf{A}=\mathbf{A}, \mathbf{A}^{\dagger} \mathbf{A} \mathbf{A}^{\dagger}=\mathbf{A}^{\dagger},\left(\mathbf{A}^{\dagger} \mathbf{A}\right)^{*}=\mathbf{A}^{\dagger} \mathbf{A},\left(\mathbf{A} \mathbf{A}^{\dagger}\right)^{*}=\mathbf{A} \mathbf{A}^{\dagger}.$	$\mathbf{A}\mathbf{A}^{\dagger}\mathbf{A}=\mathbf{A}, \mathbf{A}^{\dagger}\mathbf{A}\mathbf{A}^{\dagger}=\mathbf{A}^{\dagger}, \left(\mathbf{A}^{\dagger}\mathbf{A}\right)^{*}=\mathbf{A}^{\dagger}\mathbf{A}, \left(\mathbf{A}\mathbf{A}^{\dagger}\right)^{*}=\mathbf{A}\mathbf{A}^{\dagger}.$	$\mathbf{A}\mathbf{A}^{\dagger}\mathbf{A}=\mathbf{A}, \mathbf{A}^{\dagger}\mathbf{A}\mathbf{A}^{\dagger}=\mathbf{A}^{\dagger}, (\mathbf{A}^{\dagger}\mathbf{A})^*=\mathbf{A}^{\dagger}\mathbf{A}, (\mathbf{A}\mathbf{A}^{\dagger})^*=\mathbf{A}\mathbf{A}^{\dagger}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00691	247	0.998 N +0	$\mathbf{A}=\left[\begin{array}{rrrr}2 d & a & a & 0 \\ a & 2 d & 0 & a \\ a & 0 & 2 d & a \\ 0 & a & a & 2 d\end{array}\right] .$	$\mathbf{A}=\left[\begin{array}{rrrr}2d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d\end{array}\right].$	$\mathbf{A}=\left[\begin{array}{rrrr}2d & a & a & 0 \\ a & 2d & 0 & a \\ a & 0 & 2d & a \\ 0 & a & a & 2d\end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00981	331	0.998 K +0	$\mathbf{X} \mathbf{D}_1^{-1} \mathbf{X}^{-1} \mathbf{r}=\left(\mathbf{X}\left(\mathbf{D}-\mu \mathbf{I}\right) \mathbf{X}^{-1}\right)^{-1} \mathbf{r}=\left(\mathbf{A}-\mu \mathbf{I}\right)^{-1}(\mathbf{A}-\mu \mathbf{I}) \mathbf{x}=\mathbf{x} .$	$\mathbf{X} \mathbf{D}_1^{-1} \mathbf{X}^{-1} \mathbf{r}=\left(\mathbf{X}\left(\mathbf{D}-\mu \mathbf{I}\right) \mathbf{X}^{-1}\right)^{-1} \mathbf{r}=\left(\mathbf{A}-\mu \mathbf{I}\right)^{-1}(\mathbf{A}-\mu \mathbf{I}) \mathbf{x}=\mathbf{x} .$	$\mathbf{X} \mathbf{D}_1^{-1} \mathbf{X}^{-1} \mathbf{r}=\left(\mathbf{X}\left(\mathbf{D}-\mu \mathbf{I}\right) \mathbf{X}^{-1}\right)^{-1} \mathbf{r}=\left(\mathbf{A}-\mu \mathbf{I}\right)^{-1}(\mathbf{A}-\mu \mathbf{I}) \mathbf{x}=\mathbf{x} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E01035 (15.8)	351	0.998 W -1	$\begin{aligned} & \mathbf{A}_1=\mathbf{A} \\ & \text { for } k=1,2, \ldots \\ & \mathbf{Q}_k \mathbf{R}_k=\mathbf{A}_k \quad \text { (QR factorization of } \mathbf{A}_k) \\ & \mathbf{A}_{k+1}=\mathbf{R}_k \mathbf{Q}_k . \end{aligned}$	$\mathbf{A}_1=\mathbf{A}$ for $k=1,2,\dots$ $\mathbf{Q}_k \mathbf{R}_k=\mathbf{A}_k \quad \text { (QR factorization of } \mathbf{A}_k)$ $\mathbf{A}_{k+1}=\mathbf{R}_k \mathbf{Q}_k .$	$\mathbf{A}_1=\mathbf{A}$ for $k=1,2,\dots$ $\mathbf{Q}_k \mathbf{R}_k=\mathbf{A}_k \quad \text { (QR factorization of } \mathbf{A}_k)$ $\mathbf{A}_{k+1}=\mathbf{R}_k \mathbf{Q}_k .$
Conf. MathPix confidence: 0.998 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00012	024	■ 0.998 ● N +1	$\boldsymbol{A}\left(r_{1}: r_{2}, c_{1}: c_{2}\right):=\left[\begin{array}{cccc} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right].$	$\boldsymbol{A}\left(r_{1}: r_{2}, c_{1}: c_{2}\right):=\left[\begin{array}{cccc} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right].$	$\boldsymbol{A}\left(r_{1}: r_{2}, c_{1}: c_{2}\right):=\left[\begin{array}{cccc} a_{r_{1}, c_{1}} & a_{r_{1}, c_{1}+1} & \cdots & a_{r_{1}, c_{2}} \\ a_{r_{1}+1, c_{1}} & a_{r_{1}+1, c_{1}+1} & \cdots & a_{r_{1}+1, c_{2}} \\ \vdots & \vdots & \ddots & \vdots \\ a_{r_{2}, c_{1}} & a_{r_{2}, c_{1}+1} & \cdots & a_{r_{2}, c_{2}} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00247 (4.2)	101	■ 0.998 ● N +0	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left\{[k]\right\}} & \boldsymbol{B}_{\left\{k\right\}}^* \\ \boldsymbol{B}_{\left\{k\right\}} & \boldsymbol{F}_{\left\{k\right\}} \end{array}\right] \boldsymbol{B}_{\left\{k\right\}} & \boldsymbol{F}_{\left\{k\right\}} \\ \boldsymbol{L}_{\left\{[k]\right\}} & \boldsymbol{M}_{\left\{k\right\}} \\ \boldsymbol{N}_{\left\{k\right\}} & \boldsymbol{U}, \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{\left\{[k]\right\}} & \boldsymbol{E}_{\left\{k\right\}} \\ \boldsymbol{0} & \boldsymbol{I}_{\left\{k\right\}} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{\left\{[k]\right\}}^* & \boldsymbol{M}_{\left\{k\right\}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\left\{k\right\}}^* \end{array}\right] = \boldsymbol{L} \boldsymbol{D} \boldsymbol{U},$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left\{[k]\right\}} & \boldsymbol{B}_{\left\{k\right\}}^* \\ \boldsymbol{B}_{\left\{k\right\}} & \boldsymbol{F}_{\left\{k\right\}} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{\left\{[k]\right\}} & \boldsymbol{0} \\ \boldsymbol{M}_{\left\{k\right\}} & \boldsymbol{N}_{\left\{k\right\}} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{\left\{[k]\right\}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_{\left\{k\right\}} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{\left\{[k]\right\}}^* & \boldsymbol{M}_{\left\{k\right\}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\left\{k\right\}}^* \end{array}\right] = \boldsymbol{L} \boldsymbol{D} \boldsymbol{U},$	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\left\{[k]\right\}} & \boldsymbol{B}_{\left\{k\right\}}^* \\ \boldsymbol{B}_{\left\{k\right\}} & \boldsymbol{F}_{\left\{k\right\}} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{L}_{\left\{[k]\right\}} & \boldsymbol{0} \\ \boldsymbol{M}_{\left\{k\right\}} & \boldsymbol{N}_{\left\{k\right\}} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{D}_{\left\{[k]\right\}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{E}_{\left\{k\right\}} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{L}_{\left\{[k]\right\}}^* & \boldsymbol{M}_{\left\{k\right\}}^* \\ \boldsymbol{0} & \boldsymbol{N}_{\left\{k\right\}}^* \end{array}\right] = \boldsymbol{L} \boldsymbol{D} \boldsymbol{U},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00306 (5.13)	125	■ 0.998 ● N +0	$\boldsymbol{v}^* \boldsymbol{v} = (\boldsymbol{x} - \boldsymbol{y})^* (\boldsymbol{x} - \boldsymbol{y}) = 2 \boldsymbol{x}^* \boldsymbol{x} - 2 \operatorname{Re} (\boldsymbol{y}^* \boldsymbol{x}) = 2 \boldsymbol{v}^* \boldsymbol{x}.$	$\boldsymbol{v}^* \boldsymbol{v} = (\boldsymbol{x} - \boldsymbol{y})^* (\boldsymbol{x} - \boldsymbol{y}) = 2 \boldsymbol{x}^* \boldsymbol{x} - 2 \operatorname{Re} (\boldsymbol{y}^* \boldsymbol{x}) = 2 \boldsymbol{v}^* \boldsymbol{x}.$	$\boldsymbol{v}^* \boldsymbol{v} = (\boldsymbol{x} - \boldsymbol{y})^* (\boldsymbol{x} - \boldsymbol{y}) = 2 \boldsymbol{x}^* \boldsymbol{x} - 2 \operatorname{Re} (\boldsymbol{y}^* \boldsymbol{x}) = 2 \boldsymbol{v}^* \boldsymbol{x}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00320	129	■ 0.998 ● W +0	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$	$\boldsymbol{A}=\left[\begin{array}{lll} u_{11} & r_{12} & r_{13} \\ u_{21} & u_{22} & r_{23} \\ u_{31} & u_{32} & u_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right] \text { or } \boldsymbol{A}=\left[\begin{array}{lll} r_{11} & r_{12} & r_{13} \\ u_{21} & r_{22} & r_{23} \\ u_{31} & u_{32} & r_{33} \\ u_{41} & u_{42} & u_{43} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00438	175	■ 0.998 ● W +0	$1=\left\ z\right\ _2^2=\left\ \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\left\ z\right\ _2^2=\left\ \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$	$1=\left\ z\right\ _2^2=\left\ \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\left\ \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}\right\ _2^2=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00477 (8.10)	189	■ 0.998 ● N +0	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$	$\left\ \boldsymbol{A}^k\right\ \leq\left\ \boldsymbol{A}\right\ ^k \text { for } \boldsymbol{A} \in \mathbb{C}^{n \times n} \text { and } k \in \mathbb{N}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0487 (8.17)	191	■ 0.998 ● W -2	$\ \boldsymbol{A}\ _p := \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p} = \max_{\ \boldsymbol{y}\ _p=1} \ \boldsymbol{Ay}\ _p.$	$\ A\ _p := \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p} = \max_{\ \boldsymbol{y}\ _p=1} \ \boldsymbol{Ay}\ _p.$	$\ A\ _p := \max_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\ \boldsymbol{Ax}\ _p}{\ \boldsymbol{x}\ _p} = \max_{\ \boldsymbol{y}\ _p=1} \ \boldsymbol{Ay}\ _p.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0517 (8.32)	201	■ 0.998 ● N +0	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$	$a_1^{\lambda_1} a_2^{\lambda_2} \cdots a_n^{\lambda_n} \leq \sum_{j=1}^n \lambda_j a_j,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0544	206	■ 0.998 ● N +0	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0997	336	■ 0.998 ● N +0	$p_2(-M) > 0, \quad p_2(d_1) < 0, \quad p_2(+M) > 0.$	$p_2(-M) > 0, \quad p_2(d_1) < 0, \quad p_2(+M) > 0.$	$p_2(-M) > 0, \quad p_2(d_1) < 0, \quad p_2(+M) > 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ01049	358	■ 0.998 ● W +0	$D_i f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial x_i} := \lim_{h \rightarrow \boldsymbol{0}} \frac{f(\boldsymbol{x} + h \boldsymbol{e}_i) - f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$	$D_i f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial x_i} := \lim_{h \rightarrow \boldsymbol{0}} \frac{f(\boldsymbol{x} + h \boldsymbol{e}_i) - f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$	$D_i f(\boldsymbol{x}) := \frac{\partial f(\boldsymbol{x})}{\partial x_i} := \lim_{h \rightarrow \boldsymbol{0}} \frac{f(\boldsymbol{x} + h \boldsymbol{e}_i) - f(\boldsymbol{x})}{h}, \quad \boldsymbol{x} \in \mathbb{R}^n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0027 (1.10)	031	■ 0.998 ● N +0	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{x}^T \overline{\boldsymbol{y}} = \sum_{j=1}^n x_j \overline{y_j}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08318 (5.15)	128	0.998 W +1	$\begin{aligned} \boldsymbol{A}_{\{k\}} &= \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & & \vdots & & \vdots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}. \end{aligned}$	$\boldsymbol{A}_k = \left[\begin{array}{cccc cccc} a_{1,1}^{(1)} & \cdots & a_{1,k-1}^{(1)} & & a_{1,k}^{(1)} & \cdots & a_{1,j}^{(1)} & \cdots & a_{1,n}^{(1)} \\ & \ddots & \vdots & & \vdots & & \vdots & & \vdots \\ & & a_{k-1,k-1}^{(k-1)} & & a_{k-1,k}^{(k-1)} & \cdots & a_{k-1,j}^{(k-1)} & \cdots & a_{k-1,n}^{(k-1)} \\ & & & & a_{k,k}^{(k)} & \cdots & a_{k,j}^{(k)} & \cdots & a_{k,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{i,k}^{(k)} & \cdots & a_{i,j}^{(k)} & \cdots & a_{i,n}^{(k)} \\ & & & & \vdots & & \vdots & & \vdots \\ & & & & a_{m,k}^{(k)} & \cdots & a_{m,j}^{(k)} & \cdots & a_{m,n}^{(k)} \end{array} \right] \\ &= \begin{bmatrix} \boldsymbol{B}_k & \boldsymbol{C}_k \\ \boldsymbol{0} & \boldsymbol{D}_k \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08362 (6.1)	147	0.998 N +0	$\pi_{AC}(\lambda)=\lambda^{m-n}\pi_{CA}(\lambda), \quad \lambda \in \mathbb{C},$ $\quad \lambda \in \mathbb{C}.$	$\pi_{AC}(\lambda)=\lambda^{m-n}\pi_{CA}(\lambda), \quad \lambda \in \mathbb{C}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08611	224	0.998 K +0	$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1&1\end{bmatrix}, \boldsymbol{A}^\dagger=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1&1\end{bmatrix}=\frac{1}{4}\boldsymbol{A}, \boldsymbol{A}^\dagger\boldsymbol{b}=\begin{bmatrix}1/2\\1/2\end{bmatrix}.$	$\boldsymbol{A}:=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1&1\end{bmatrix}, \boldsymbol{A}^\dagger=\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1&1\end{bmatrix}=\frac{1}{4}\boldsymbol{A}, \boldsymbol{A}^\dagger\boldsymbol{b}=\begin{bmatrix}1/2\\1/2\end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E08872 (13.28)	303	0.998 N -1	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_k\\Q(0)=1}}\max_{a\leq x\leq b} Q(x) ,$	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_k\\Q(0)=1}}\max_{a\leq x\leq b} Q(x) ,$	$\frac{\ \boldsymbol{x}-\boldsymbol{x}_k\ _{\boldsymbol{A}}}{\ \boldsymbol{x}-\boldsymbol{x}_0\ _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_k\\Q(0)=1}}\max_{a\leq x\leq b} Q(x) ,$

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00016 (1.4)	028	■ 0.999 ● N -1	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}:\boldsymbol{s}\in\mathcal{S}\text{ and }\boldsymbol{t}\in\mathcal{T}\}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00073	044	■ 0.999 ● N +0	$b_{j,k}=(\alpha_k+\beta_j)A_k(-\beta_j)B_j(-\alpha_k),$		$b_{j,k}=(\alpha_k+\beta_j)A_k(-\beta_j)B_j(-\alpha_k),$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00101 (2.18)	055	■ 0.999 ● N +1	$\begin{aligned} & \left u_{k+1}\right \&=\\ & \left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }\\ & \geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right \end{aligned}$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }$	$\begin{aligned} \left u_{k+1}\right &= \left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right } \\ &\geq\left d_{k+1}\right -\left a_k\right =\sigma_{k+1}+\left c_{k+1}\right . \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00156	071	■ 0.999 ● N +0	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$	$\delta^2 f(x):=\frac{f(x+h)-2 f(x)+f(x-h)}{h^2}, \quad h>0, \quad f:[x-h, x+h] \rightarrow \mathbb{R} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00250	102	■ 0.999 ● N +1	$\left[\begin{array}{ll} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$		$\left[\begin{array}{cc} 3 & 1 \\ 1 & 3 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 1 / 3 & 1 \end{array}\right]\left[\begin{array}{cc} 3 & 0 \\ 0 & 8 / 3 \end{array}\right]\left[\begin{array}{cc} 1 & 1 / 3 \\ 0 & 1 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00296 (5.11)	123	■ 0.999 ● N +0	$\left \boldsymbol{v}-\boldsymbol{s}_0\right <\left \boldsymbol{v}-\boldsymbol{s}\right , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$	$\left\ \boldsymbol{v}-\boldsymbol{s}_0\right\ <\left\ \boldsymbol{v}-\boldsymbol{s}\right\ , \text { for all } \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s} \neq \boldsymbol{s}_0 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00361	147	■ 0.999 ● N +0	$\begin{aligned} & \pi_{\boldsymbol{B}}(\lambda)=\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\operatorname{det}\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ & =\operatorname{det}\left(\boldsymbol{S}^{-1}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) \operatorname{det}(\boldsymbol{S})=\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &= \operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\operatorname{det}\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ &= \operatorname{det}\left(\boldsymbol{S}^{-1}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) \operatorname{det}(\boldsymbol{S})=\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}$	$\begin{aligned} \pi_{\boldsymbol{B}}(\lambda) &= \operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}-\lambda \boldsymbol{I}\right)=\operatorname{det}\left(\boldsymbol{S}^{-1}(\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{S}\right) \\ &= \operatorname{det}\left(\boldsymbol{S}^{-1}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I}) \operatorname{det}(\boldsymbol{S})=\operatorname{det}\left(\boldsymbol{S}^{-1} \boldsymbol{S}\right) \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=\pi_{\boldsymbol{A}}(\lambda), \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0381	154	0.999 W -1	$e_{ii} = \sum_{k=1}^n \bar{b}_{ki} b_{ki} = \sum_{k=1}^i b_{ki} ^2 \text{ and } f_{ii} = \sum_{k=1}^n b_{ik} \bar{b}_{ik} = \sum_{k=i}^n b_{ik} ^2.$	$e_{ii} = \sum_{k=1}^n \bar{b}_{ki} b_{ki} = \sum_{k=1}^i b_{ki} ^2 \text{ and } f_{ii} = \sum_{k=1}^n b_{ik} \bar{b}_{ik} = \sum_{k=i}^n b_{ik} ^2.$	$e_{ii} = \sum_{k=1}^n \bar{b}_{ki} b_{ki} = \sum_{k=1}^i b_{ki} ^2 \text{ and } f_{ii} = \sum_{k=1}^n b_{ik} \bar{b}_{ik} = \sum_{k=i}^n b_{ik} ^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0406	162	0.999 N +0	$\mathbf{A} = \begin{bmatrix} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$	$\mathbf{A} = \begin{bmatrix} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$	$\mathbf{A} = \begin{bmatrix} -q_{n-1} & -q_{n-2} & \cdots & -q_1 & -q_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0446 (7.10)	178	0.999 N +0	$\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \mathbf{U} \mathbf{D} \mathbf{U}^T$	$\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \mathbf{U} \mathbf{D} \mathbf{U}^T$	$\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \mathbf{U} \mathbf{D} \mathbf{U}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0471 (8.7)	187	0.999 N +0	$m \ \mathbf{x}\ ' \leq \ \mathbf{x}\ \leq M \ \mathbf{x}\ '.$	$m \ \mathbf{x}\ ' \leq \ \mathbf{x}\ \leq M \ \mathbf{x}\ '.$	$m \ \mathbf{x}\ ' \leq \ \mathbf{x}\ \leq M \ \mathbf{x}\ '.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0481 (8.14)	190	0.999 N +0	$\mathcal{S} := \{x \in \mathbb{C}^n : \ \mathbf{x}\ = 1\}$	$\mathcal{S} := \{x \in \mathbb{C}^n : \ \mathbf{x}\ = 1\}$	$\mathcal{S} := \{x \in \mathbb{C}^n : \ \mathbf{x}\ = 1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0538	204	0.999 N +0	$\ \mathbf{x}\ _A := \sqrt{\mathbf{x}^T \mathbf{A} \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n.$	$\ \mathbf{x}\ _A := \sqrt{\mathbf{x}^T \mathbf{A} \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n.$	$\ \mathbf{x}\ _A := \sqrt{\mathbf{x}^T \mathbf{A} \mathbf{x}}, \quad \mathbf{x} \in \mathbb{R}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0634	230	0.999 N +0	$\ C - D\ _F^2 = \left\ \begin{bmatrix} \mathbf{0} & \mathbf{A} - \mathbf{B} \\ \mathbf{A}^* - \mathbf{B}^* & \mathbf{0} \end{bmatrix} \right\ _F^2 = \ \mathbf{B} - \mathbf{A}\ _F^2 + \ (\mathbf{B} - \mathbf{A})^*\ _F^2 = 2\ \mathbf{B} - \mathbf{A}\ _F^2.$	$\ C - D\ _F^2 = \left\ \begin{bmatrix} \mathbf{0} & \mathbf{A} - \mathbf{B} \\ \mathbf{A}^* - \mathbf{B}^* & \mathbf{0} \end{bmatrix} \right\ _F^2 = \ \mathbf{B} - \mathbf{A}\ _F^2 + \ (\mathbf{B} - \mathbf{A})^*\ _F^2 = 2\ \mathbf{B} - \mathbf{A}\ _F^2.$	$\ C - D\ _F^2 = \left\ \begin{bmatrix} \mathbf{0} & \mathbf{A} - \mathbf{B} \\ \mathbf{A}^* - \mathbf{B}^* & \mathbf{0} \end{bmatrix} \right\ _F^2 = \ \mathbf{B} - \mathbf{A}\ _F^2 + \ (\mathbf{B} - \mathbf{A})^*\ _F^2 = 2\ \mathbf{B} - \mathbf{A}\ _F^2.$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00816	284	■ 0.999 ● K +0	$\frac{1}{4}\left(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1}\right)=\mu w_{i,j}.$	$\frac{1}{4}\left(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1}\right)=\mu w_{i,j}.$	$\frac{1}{4}(\lambda^{1/2}w_{i-1,j}+\lambda^{1/2}w_{i,j-1}+\lambda^{-1/2}w_{i+1,j}+\lambda^{-1/2}w_{i,j+1})=\mu w_{i,j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00861	300	■ 0.999 ● K +0	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$	$\kappa_A=\frac{5+4\cos(\pi h)}{5-4\cos(\pi h)}\leq 9,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00954 (13.67)	322	■ 0.999 ● N -1	$\left \left \boldsymbol{x}^*\right -\left \boldsymbol{x}\right \right _{\mathcal{W}}\leq\left \left \boldsymbol{A}\right \right _2\min_{\boldsymbol{w}\in\mathcal{W}}\left\ \boldsymbol{x}^*-\boldsymbol{w}\right\ _2.$	$\left\ \boldsymbol{x}^*-\boldsymbol{x}\right\ _2\leq\left\ \boldsymbol{A}\right\ _2\min_{\boldsymbol{w}\in\mathcal{W}}\left\ \boldsymbol{x}^*-\boldsymbol{w}\right\ _2.$	$\left\ \boldsymbol{x}^*-\boldsymbol{x}\right\ _2\leq\left\ \boldsymbol{A}\right\ _2\min_{\boldsymbol{w}\in\mathcal{W}}\left\ \boldsymbol{x}^*-\boldsymbol{w}\right\ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00208	086	■ 0.999 ● W +0	$\begin{aligned} &(\boldsymbol{L}\boldsymbol{U})_{ij}=\sum_{k=1}^nl_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)} \\ &=\sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}. \end{aligned}$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^nl_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)}= \sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}.$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^nl_{ik}^{(k)}u_{kj}=\sum_{k=1}^{j-1}a_{pi,k}^{(k)}a_{pk,j}^{(k)}+a_{pi,j}^{(k)}a_{pj,j}^{(j)}= \sum_{k=1}^{j-1}\left(a_{pi,j}^{(k)}-a_{pi,j}^{(k+1)}\right)+a_{pi,j}^{(j)}=a_{pi,j}^{(1)}=a_{pi,j}=\left(\boldsymbol{P}^T\boldsymbol{A}\right)_{ij}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00265 (4.9)	109	■ 0.999 ● W +0	$\begin{aligned} &0\leq\left \left \boldsymbol{x}^*\right -\left \boldsymbol{x}\right \right _{\mathcal{W}}\leq\left \left \boldsymbol{A}\right \right _2\min_{\boldsymbol{w}\in\mathcal{W}}\left\ \boldsymbol{x}^*-\boldsymbol{w}\right\ _2. \\ &=\left \left \boldsymbol{A}\right \right _2\min_{\boldsymbol{w}\in\mathcal{W}}\left\ \boldsymbol{x}^*-\boldsymbol{w}\right\ _2. \end{aligned}$	$0\leq\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\left[-\boldsymbol{y}^*\boldsymbol{v}/\alpha,\boldsymbol{y}^*\right]\begin{bmatrix}\alpha&\boldsymbol{v}^*\\ \boldsymbol{v}&\boldsymbol{B}\end{bmatrix}\begin{bmatrix}-\boldsymbol{v}^*\boldsymbol{y}/\alpha\\ \boldsymbol{y}\end{bmatrix} \\ =\left[0,-\left(\boldsymbol{y}^*\boldsymbol{v}\right)\boldsymbol{v}^*/\alpha+\boldsymbol{y}^*\boldsymbol{B}\right]\begin{bmatrix}-\boldsymbol{v}^*\boldsymbol{y}/\alpha\\ \boldsymbol{y}\end{bmatrix} \\ =-\boldsymbol{y}^*\boldsymbol{v}\boldsymbol{v}^*\boldsymbol{y}/\alpha+\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y}=\boldsymbol{y}^*\boldsymbol{C}\boldsymbol{y}.$	$0\leq\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\left[-\boldsymbol{y}^*\boldsymbol{v}/\alpha,\boldsymbol{y}^*\right]\begin{bmatrix}\alpha&\boldsymbol{v}^*\\ \boldsymbol{v}&\boldsymbol{B}\end{bmatrix}\begin{bmatrix}-\boldsymbol{v}^*\boldsymbol{y}/\alpha\\ \boldsymbol{y}\end{bmatrix} \\ =\left[0,-\left(\boldsymbol{y}^*\boldsymbol{v}\right)\boldsymbol{v}^*/\alpha+\boldsymbol{y}^*\boldsymbol{B}\right]\begin{bmatrix}-\boldsymbol{v}^*\boldsymbol{y}/\alpha\\ \boldsymbol{y}\end{bmatrix} \\ =-\boldsymbol{y}^*\boldsymbol{v}\boldsymbol{v}^*\boldsymbol{y}/\alpha+\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y}=\boldsymbol{y}^*\boldsymbol{C}\boldsymbol{y}.$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00309	126	■ 0.999 ● C -12	$\begin{gathered} \boldsymbol{H}:=\boldsymbol{I}-\frac{2}{\boldsymbol{v}^T\boldsymbol{v}}\boldsymbol{v}\boldsymbol{v}^T \\ \boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} \end{gathered}$	$\boldsymbol{H}:=\boldsymbol{I}-\frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\frac{2}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\frac{1}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{H}:=\boldsymbol{I}-\frac{2\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\frac{2}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$ $\boldsymbol{P}:=\boldsymbol{I}-\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}-\frac{1}{4}\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \end{bmatrix}=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00645	232	■ 0.999 ● W +1	$\begin{aligned} B &= (BA)B = (A^*)B^*B = (A^*C^*)A^*B^*B = CA(A^*B^*)B \\ &= CA(BAB) = (C)AB = C(AC)AB = CC^*A^*(AB) \\ &= CC^*(A^*B^*A^*) = C(C^*A^*) = CAC = C. \end{aligned}$	$B = (BA)B = (A^*)B^*B = (A^*C^*)A^*B^*B = CA(A^*B^*)B$ $= CA(BAB) = (C)AB = C(AC)AB = CC^*A^*(AB)$ $= CC^*(A^*B^*A^*) = C(C^*A^*) = CAC = C.$	$\boldsymbol{B} = (\boldsymbol{B}\boldsymbol{A})\boldsymbol{B} = (\boldsymbol{A}^*)\boldsymbol{B}^*\boldsymbol{B} = (\boldsymbol{A}^*\boldsymbol{C}^*)\boldsymbol{A}^*\boldsymbol{B}^*\boldsymbol{B} = \boldsymbol{C}\boldsymbol{A}(\boldsymbol{A}^*\boldsymbol{B}^*)\boldsymbol{B}$ $= \boldsymbol{C}\boldsymbol{A}(\boldsymbol{B}\boldsymbol{A}\boldsymbol{B}) = (\boldsymbol{C})\boldsymbol{A}\boldsymbol{B} = \boldsymbol{C}(\boldsymbol{A}\boldsymbol{C})\boldsymbol{A}\boldsymbol{B} = \boldsymbol{C}\boldsymbol{C}^*\boldsymbol{A}^*(\boldsymbol{A}\boldsymbol{B})$ $= \boldsymbol{C}\boldsymbol{C}^*(\boldsymbol{A}^*\boldsymbol{B}^*\boldsymbol{A}^*) = \boldsymbol{C}(\boldsymbol{C}^*\boldsymbol{A}^*) = \boldsymbol{C}\boldsymbol{A}\boldsymbol{C} = \boldsymbol{C}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00827	287	■ 0.999 ● N +0	$\boldsymbol{x}:=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}+\begin{bmatrix} 1-a \\ 1-a \end{bmatrix}=: \boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}:=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} 0 & a \\ a & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}+\begin{bmatrix} 1-a \\ 1-a \end{bmatrix}=: \boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00942	319	■ 0.999 ● N +0	$\mathbb{W}_k:=\operatorname{span}\left(\boldsymbol{b},\boldsymbol{A}\boldsymbol{b},\dots,\boldsymbol{A}^{k-1}\boldsymbol{b}\right).$	$\mathbb{W}_k:=\operatorname{span}\left(\boldsymbol{b},\boldsymbol{A}\boldsymbol{b},\dots,\boldsymbol{A}^{k-1}\boldsymbol{b}\right).$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00062 (1.21)	041	■ 0.999 ● K +0	$\overline{\boldsymbol{C}}=\boldsymbol{C}\left(\boldsymbol{I}+\frac{1}{\lambda}\boldsymbol{e}_i\left(\boldsymbol{e}_i^T-\boldsymbol{a}^T\boldsymbol{C}\right)\right)$	$\overline{\boldsymbol{C}}=\boldsymbol{C}\left(\boldsymbol{I}+\frac{1}{\lambda}\boldsymbol{e}_i\left(\boldsymbol{e}_i^T-\boldsymbol{a}^T\boldsymbol{C}\right)\right)$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00202 (3.12)	085	■ 0.999 ● N +0	$\boldsymbol{P}^T=\boldsymbol{P}_{n-1}\cdots\boldsymbol{P}_1=\boldsymbol{I}(\boldsymbol{p},:),\quad \boldsymbol{P}=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n-1}=\boldsymbol{I}(:,\boldsymbol{p}).$	$\boldsymbol{P}^T=\boldsymbol{P}_{n-1}\cdots\boldsymbol{P}_1=\boldsymbol{I}(\boldsymbol{p},:),\quad \boldsymbol{P}=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n-1}=\boldsymbol{I}(:,\boldsymbol{p}).$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00254	104	■ 0.999 ● W +0	$\begin{aligned} & \boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i \\ & = \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2 \\ & = x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2. \end{aligned}$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i$ $= \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2$ $= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2.$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 2 \sum_{i=1}^n x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1} - \sum_{i=2}^n x_{i-1} x_i$ $= \sum_{i=1}^{n-1} x_i^2 - 2 \sum_{i=1}^{n-1} x_i x_{i+1} + \sum_{i=1}^{n-1} x_{i+1}^2 + x_1^2 + x_n^2$ $= x_1^2 + x_n^2 + \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00286	119	■ 0.999 ● N +0	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$	$\ \boldsymbol{x} + \boldsymbol{y}\ ^2 \leq \ \boldsymbol{x}\ ^2 + 2\ \boldsymbol{x}\ \ \boldsymbol{y}\ + \ \boldsymbol{y}\ ^2 = (\ \boldsymbol{x}\ + \ \boldsymbol{y}\)^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00310	126	■ 0.999 ● W +0	$\mathcal{M} := \{\boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$	$\mathcal{M} := \{\boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$	$\mathcal{M} := \{\boldsymbol{w} \in \mathbb{R}^3 : \boldsymbol{w}^T \boldsymbol{v} = 0\} = \left\{ \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} : 2w_1 = 0 \right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00348	141	■ 0.999 ● W +0	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix},$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix},$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00412 (6.20)	165	■ 0.999 ● N +0	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{A} \boldsymbol{x} = \sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \text{ where } c_j = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{A} \boldsymbol{x} = \sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \text{ where } c_j = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{u}_j, \quad \boldsymbol{A} \boldsymbol{x} = \sum_{j=1}^n c_j \lambda_j \boldsymbol{u}_j, \text{ where } c_j = \frac{\boldsymbol{u}_j^* \boldsymbol{x}}{\boldsymbol{u}_j^* \boldsymbol{u}_j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00440	176	■ 0.999 ● C -4	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$	$\frac{\left(\frac{3}{5}x_1 + \frac{4}{5}x_2\right)^2}{9} + \frac{\left(-\frac{4}{5}x_1 + \frac{3}{5}x_2\right)^2}{1} = 1,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00499	195	■ 0.999 ● K +0	$x_1 + (1 + 10^{-16})x_2 = 20 - 10^{-15},$	$x_1 + (1 + 10^{-16})x_2 = 20 - 10^{-15},$	$x_1 + (1 + 10^{-16})x_2 = 20 - 10^{-15},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00510	198	0.999 K +0	$\left \left \boldsymbol{B}^{-1}\right \right \leq \left \left \boldsymbol{A}^{-1}\right +\left \left \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right \right \leq \left \left \boldsymbol{A}^{-1}\right +r\left \left \boldsymbol{B}^{-1}\right \right $.	$\left\ \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +\left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ $.	$\left\ \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +\left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ $.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00559	209	0.999 W +0	$K\left(\boldsymbol{B}\right)^{-1} \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ } \leq \frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ }$.	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }$.	$K(\boldsymbol{B})^{-1} \frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ } \leq \frac{\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\ }{\ \boldsymbol{B}^{-1}\ }$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00617 (9.14)	226	0.999 N +0	$\frac{1}{4} 10^{-2} \leq \frac{\left\ \boldsymbol{y}-\boldsymbol{x}\right\ _{\infty}}{\left\ \boldsymbol{x}\right\ _{\infty}} \leq 4 \cdot 10^{-2}$.	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y}-\boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}$.	$\frac{1}{4} 10^{-2} \leq \frac{\ \boldsymbol{y}-\boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} \leq 4 \cdot 10^{-2}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00661 (10.5)	239	0.999 N +0	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F} \quad \text { with } \quad \boldsymbol{h}=1 /(m+1),$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F} \quad \text { with } \quad \boldsymbol{h}=1 /(m+1),$	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=\boldsymbol{h}^2 \boldsymbol{F} \quad \text { with } \quad \boldsymbol{h}=1 /(m+1),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00723	258	0.999 N +0	$\boldsymbol{D}_{2}=\operatorname{diag}\left(1,\omega_4\right)=\left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right]$.	$\boldsymbol{D}_2=\operatorname{diag}\left(1,\omega_4\right)=\left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right]$.	$\boldsymbol{D}_2=\operatorname{diag}\left(1,\omega_4\right)=\left[\begin{array}{cc} 1 & 0 \\ 0 & -i \end{array}\right]$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00846	293	0.999 N +0	$\begin{aligned} & \& \boldsymbol{t}_{\{0\}}=3\left[\begin{array}{c} 1 \\\ -1 / 2 \end{array}\right], \quad \alpha_{\{0\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{1\}}=-4^{\{-1\}}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_{\{1\}}=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ & \& \boldsymbol{r}_{\{1\}}=3 * 4^{\{-1\}}\left[\begin{array}{c} 0 \\ \boldsymbol{\theta} \end{array}\right] \& \boldsymbol{t}_{\{1\}}=3 * 4^{\{-1\}}\left[\begin{array}{c} \boldsymbol{\theta} \\\ 1 \end{array}\right] \& \boldsymbol{t}_{\{1\}}=3 * 4^{\{-1\}}\left[\begin{array}{c} -1 \\\ 2 \end{array}\right], \quad \alpha_{\{1\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{2\}}=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{\{2\}}=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \\ & \& \alpha_{\{1\}}=\frac{1}{2}, \quad \boldsymbol{x}_{\{2\}}=-4^{\{-1\}}\left[\begin{array}{c} 1 \\\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_{\{2\}}=3 * 4^{\{-1\}}\left[\begin{array}{c} 1 \\\ 2 \end{array}\right] \& \boldsymbol{\theta} \end{aligned}$	$\begin{aligned} \boldsymbol{t}_0 &= 3\left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_0=\frac{1}{2}, \quad \boldsymbol{x}_1=-4^{-1}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_1=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ \boldsymbol{t}_1 &= 3 * 4^{-1}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_1=\frac{1}{2}, \quad \boldsymbol{x}_2=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_2=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$	$\begin{aligned} \boldsymbol{t}_0 &= 3\left[\begin{array}{c} 1 \\ -1 / 2 \end{array}\right], \quad \alpha_0=\frac{1}{2}, \quad \boldsymbol{x}_1=-4^{-1}\left[\begin{array}{c} 1 \\ 2 \end{array}\right], \quad \boldsymbol{r}_1=3 * 4^{-1}\left[\begin{array}{c} 0 \\ 1 \end{array}\right] \\ \boldsymbol{t}_1 &= 3 * 4^{-1}\left[\begin{array}{c} -1 \\ 2 \end{array}\right], \quad \alpha_1=\frac{1}{2}, \quad \boldsymbol{x}_2=-4^{-1}\left[\begin{array}{c} 1 \\ 1 / 2 \end{array}\right], \quad \boldsymbol{r}_2=3 * 4^{-1}\left[\begin{array}{c} 1 / 2 \\ 0 \end{array}\right], \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00899 (13.40)	308	0.999 N +0	$\frac{\left \left \boldsymbol{\epsilon}_{k+1}\right \right _{\boldsymbol{A}}^2}{\left \left \boldsymbol{\epsilon}_k\right \right _{\boldsymbol{A}}^2} \leq\left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2, \ldots,$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} \leq\left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2, \ldots,$	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2} \leq\left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad k=0,1,2, \ldots,$
Conf. MathPix confidence: 0.999 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00937	317	0.999 K +0	$c_{\{0\}}\left(x^{\mathrm{T}}\,\,\boldsymbol{\mathrm{A}}_{\{p\}}\,x\right)\,\leq\,x^{\mathrm{T}}\,\,\boldsymbol{\mathrm{A}}\,\,\boldsymbol{\mathrm{x}}\,\,\leq\,c_{\{1\}}\left(x^{\mathrm{T}}\,\,\boldsymbol{\mathrm{A}}_{\{p\}}\,x\right)$	$c_0\left(x^T\boldsymbol{A}_px\right)\leq x^T\boldsymbol{A}\boldsymbol{x}\leq c_1\left(x^T\boldsymbol{A}_px\right)$	$c_0(x^T\boldsymbol{A}_px)\leq x^T\boldsymbol{A}\boldsymbol{x}\leq c_1(x^T\boldsymbol{A}_px)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00945	320	0.999 K +0	$\boldsymbol{\mathrm{A}}\,\,\boldsymbol{\mathrm{p}}_{\{k-1\}}\,\,\mathrm{in}\,\,\operatorname{span}\left(\boldsymbol{\mathrm{p}}_{\{k-2\}},\,\,\boldsymbol{\mathrm{p}}_{\{k-1\}},\,\,\boldsymbol{\mathrm{p}}_{\{k\}}\right)\,\,.$	$\boldsymbol{A}\boldsymbol{p}_{k-1}\in\operatorname{span}\left(\boldsymbol{p}_{k-2},\boldsymbol{p}_{k-1},\boldsymbol{p}_k\right).$	$\boldsymbol{A}\boldsymbol{p}_{k-1}\in\operatorname{span}(\boldsymbol{p}_{k-2},\boldsymbol{p}_{k-1},\boldsymbol{p}_k).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00960 (13.69)	323	0.999 W +0	$\begin{aligned} &\boldsymbol{\mathrm{x}}_{\{k+1\}}=\left(1-\omega_{\mathrm{k}}\right)\boldsymbol{\mathrm{x}}_{\{k-1\}}+\omega_{\mathrm{k}}\left(\boldsymbol{\mathrm{x}}_{\mathrm{k}}+\boldsymbol{\mathrm{r}}_{\mathrm{k}}\right),\\ &\boldsymbol{\mathrm{r}}_{\mathrm{k}+1}=\left(1-\omega_{\mathrm{k}}\right)\boldsymbol{\mathrm{r}}_{k-1}+\omega_{\mathrm{k}}\boldsymbol{B}\boldsymbol{r}_k,\\ &\boldsymbol{\mathrm{r}}_{\mathrm{k}+1}=\left(1-\omega_{\mathrm{k}}\right)\boldsymbol{\mathrm{r}}_{k-1}+\omega_{\mathrm{k}}\boldsymbol{B}\boldsymbol{r}_k, \end{aligned}$	$\boldsymbol{x}_{k+1}=\left(1-\omega_k\right)\boldsymbol{x}_{k-1}+\omega_k\left(\boldsymbol{x}_k+\boldsymbol{r}_k\right),$ $\boldsymbol{r}_{k+1}=\left(1-\omega_k\right)\boldsymbol{r}_{k-1}+\omega_k\boldsymbol{B}\boldsymbol{r}_k,$	$\boldsymbol{x}_{k+1}=\left(1-\omega_k\right)\boldsymbol{x}_{k-1}+\omega_k\left(\boldsymbol{x}_k+\boldsymbol{r}_k\right),$ $\boldsymbol{r}_{k+1}=\left(1-\omega_k\right)\boldsymbol{r}_{k-1}+\omega_k\boldsymbol{B}\boldsymbol{r}_k,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00986	332	0.999 N +0	$\frac{1}{\lambda}\leq\left\ \boldsymbol{\mathrm{A}}^{-1}\right\ _p=\frac{K_p(\boldsymbol{\mathrm{A}})}{\left\ \boldsymbol{\mathrm{A}}\right\ _p}$	$\frac{1}{\lambda}\leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\left\ \boldsymbol{A}\right\ _p}$	$\frac{1}{\lambda}\leq\left\ \boldsymbol{A}^{-1}\right\ _p=\frac{K_p(\boldsymbol{A})}{\left\ \boldsymbol{A}\right\ _p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01025	346	0.999 W +0	$\boldsymbol{\mathrm{x}}_{\mathrm{k}}=\frac{\boldsymbol{\mathrm{z}}_{\mathrm{k}}}{\left\ \boldsymbol{\mathrm{z}}_{\mathrm{k}}\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\,\,\,k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\,\,\,k=0,1,2,\ldots,$	$\boldsymbol{x}_k=\frac{\boldsymbol{z}_k}{\left\ \boldsymbol{z}_k\right\ }=\frac{c_1\lambda_1^k}{\left c_1\lambda_1^k\right }\frac{\boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n}{\left\ \boldsymbol{v}_1+\frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\boldsymbol{v}_2+\cdots+\frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\boldsymbol{v}_n\right\ },\,\,\,k=0,1,2,\ldots,$
Conf. MathPix confidence: 0.999 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ1034	349	■ 0.999 ● N +0	$\begin{aligned} & s_k = \frac{\boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k, \\ & \boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k. \end{aligned}$	$s_k = \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k,$ $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k.$	$s_k = \frac{\boldsymbol{y}_k^* \boldsymbol{A} \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* (\boldsymbol{A} - s_{k-1} \boldsymbol{I}) \boldsymbol{y}_k}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k} = s_{k-1} + \rho_k,$ $\boldsymbol{r}_k = \boldsymbol{A} \boldsymbol{x}_k - s_k \boldsymbol{x}_k = \frac{\boldsymbol{A} \boldsymbol{y}_k - (s_{k-1} + \rho_k) \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \frac{\boldsymbol{x}_{k-1} - \rho_k \boldsymbol{y}_k}{\ \boldsymbol{y}_k\ _2} = \boldsymbol{w}_k - \rho_k \boldsymbol{x}_k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0097	054	■ 0.999 ● W +0	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array} \right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array} \right], \quad \left[\begin{array}{lll} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right].$	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array} \right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array} \right], \quad \left[\begin{array}{lll} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right].$	$\left[\begin{array}{lll} 1 & 0 & 0 \\ l_1 & 1 & 0 \\ 0 & l_2 & 1 \end{array} \right] \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right] = \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array} \right], \quad \left[\begin{array}{lll} u_1 & c_1 & 0 \\ 0 & u_2 & c_2 \\ 0 & 0 & u_3 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{l} z_1 \\ z_2 \\ z_3 \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0300	154	■ 0.999 ● K +0	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{U}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{B}^* \boldsymbol{B}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0394 (6.13)	158	■ 0.999 ● W -2	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$	$\lambda_k = \min_{\dim(S)=n-k+1} \max_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) = \max_{\dim(S)=k} \min_{\substack{\boldsymbol{x} \in S \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}), \quad k = 1, \dots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0445	178	■ 0.999 ● W +1	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$	$\left\ \boldsymbol{A} - \boldsymbol{A}' \right\ _F = \min_{\substack{\boldsymbol{B} \in \mathbb{R}^{m \times n} \\ \text{rank}(\boldsymbol{B})=r}} \left\ \boldsymbol{A} - \boldsymbol{B} \right\ _F = \sqrt{\sigma_{r+1}^2 + \dots + \sigma_n^2}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00621	227	0.999 W +0	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$	$\frac{\ \boldsymbol{A} \boldsymbol{x}\ _2^2}{\ \boldsymbol{x}\ _2^2} = \frac{(\boldsymbol{A} \boldsymbol{x})^* (\boldsymbol{A} \boldsymbol{x})}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = R_{\boldsymbol{A}^* \boldsymbol{A}}(\boldsymbol{x}),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00903	309	0.999 N +0	$\left\ \boldsymbol{\epsilon}_k \right\ _2 - \left\ \boldsymbol{\epsilon}_{k+1} \right\ _2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} \left(\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2 \right)$	$\ \boldsymbol{\epsilon}_k\ _2^2 - \ \boldsymbol{\epsilon}_{k+1}\ _2^2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} \left(\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2 \right)$	$\ \boldsymbol{\epsilon}_k\ _2^2 - \ \boldsymbol{\epsilon}_{k+1}\ _2^2 = \frac{\ \boldsymbol{p}_k\ _2^2}{\ \boldsymbol{p}_k\ _A^2} (\ \boldsymbol{\epsilon}_k\ _A^2 + \ \boldsymbol{\epsilon}_{k+1}\ _A^2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00003	022	1.000 W +0	$\boldsymbol{A} = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array} \right].$	$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00008	023	1.000 K +0	$\boldsymbol{e}_1 := \left[\begin{array}{c} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_2 := \left[\begin{array}{c} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right], \quad \boldsymbol{e}_3 := \left[\begin{array}{c} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{array} \right], \quad \dots, \quad \boldsymbol{e}_n := \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{array} \right],$	$\boldsymbol{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_3 := \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \boldsymbol{e}_n := \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$	$\boldsymbol{e}_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_2 := \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_3 := \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad \boldsymbol{e}_n := \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00025	030	1.000 W +0	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$	$\sum_{i=1}^m b_i \boldsymbol{v}_i = \boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j \stackrel{(1.9)}{=} \sum_{j=1}^n c_j \left(\sum_{i=1}^m a_{ij} \boldsymbol{v}_i \right) = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} c_j \right) \boldsymbol{v}_i.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00053	039	1.000 N +0	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$	$\pi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_0,$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0078 (2.2)	046	■ 1.000 ● K +0	$g\left(x_i\right)=y_i, \text { for } i=1, \ldots, n+1$.	$g\left(x_i\right)=y_i, \text { for } i=1, \ldots, n+1.$	$g\left(x_i\right)=y_i, \text { for } i=1, \ldots, n+1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0087 (2.9)	051	■ 1.000 ● N +0	$\mu_{i-1}+4 \mu_i+\mu_{i+1}=\frac{6}{h^2}\left(y_{i+1}-2 y_i+y_{i-1}\right), \quad i=2, \ldots, n,$	$\mu_{i-1}+4 \mu_i+\mu_{i+1}=\frac{6}{h^2}\left(y_{i+1}-2 y_i+y_{i-1}\right), \quad i=2, \ldots, n,$	$\mu_{i-1}+4 \mu_i+\mu_{i+1}=\frac{6}{h^2}\left(y_{i+1}-2 y_i+y_{i-1}\right), \quad i=2, \ldots, n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0099 (2.16)	054	■ 1.000 ● N +0	$u_{\{1\}}=d_{\{1\}}, \quad \text {quad } l_{\{k\}}=a_{\{k\}} / u_{\{k\}}, \quad \text {quad } u_{\{k+1\}}=d_{\{k+1\}}-l_{\{k\}} c_{\{k\}}, \quad \text {quad } k=1,2, \ldots, n-1,$	$u_1=d_1, \quad l_k=a_k / u_k, \quad u_{k+1}=d_{k+1}-l_k c_k, \quad k=1,2, \ldots, n-1,$	$u_1=d_1, \quad l_k=a_k / u_k, \quad u_{k+1}=d_{k+1}-l_k c_k, \quad k=1,2, \ldots, n-1,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0102 (2.19)	056	■ 1.000 ● W +2	$\left l_{\{k\}}\right =\frac{\left a_{\{k\}}\right }{\left u_{\{k\}}\right } \leq \frac{\left a_{\{k\}}\right }{\left d_{\{k\}}\right -\left a_{\{k-1\}}\right },\left u_{\{k+1\}}\right \leq\left d_{\{k+1\}}\right +\frac{\left a_{\{k\}}\right \left c_{\{k\}}\right }{\left d_{\{k\}}\right -\left a_{\{k-1\}}\right }, k=1, \ldots, n-1.$	$\left l_k\right =\frac{\left a_k\right }{\left u_k\right } \leq \frac{\left a_k\right }{\left d_k\right -\left a_{k-1}\right },\left u_{k+1}\right \leq\left d_{k+1}\right +\frac{\left a_k\right \left c_k\right }{\left d_k\right -\left a_{k-1}\right }, k=1, \ldots, n-1.$	$\left l_k\right =\frac{\left a_k\right }{\left u_k\right } \leq \frac{\left a_k\right }{\left d_k\right -\left a_{k-1}\right },\left u_{k+1}\right \leq\left d_{k+1}\right +\frac{\left a_k\right \left c_k\right }{\left d_k\right -\left a_{k-1}\right }, k=1, \ldots, n-1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0103	056	■ 1.000 ● W +0	$\left l_{\{k\}}\right =\frac{\left a_{\{k\}}\right }{\left u_{\{k\}}\right } \leq \frac{\left a_{\{k\}}\right }{\left d_{\{k\}}\right -\left a_{\{k-1\}}\right },\left u_{\{k+1\}}\right \leq\left d_{\{k+1\}}\right +\left l_{\{k\}}\right \left c_{\{k\}}\right \leq\left d_{\{k+1\}}\right +\frac{\left a_{\{k\}}\right \left c_{\{k\}}\right }{\left d_{\{k\}}\right -\left a_{\{k-1\}}\right }.$	$\left l_k\right =\frac{\left a_k\right }{\left u_k\right } \leq \frac{\left a_k\right }{\left d_k\right -\left a_{k-1}\right },\left u_{k+1}\right \leq\left d_{k+1}\right +\left l_k\right \left c_k\right \leq\left d_{k+1}\right +\frac{\left a_k\right \left c_k\right }{\left d_k\right -\left a_{k-1}\right }.$	$\left l_k\right =\frac{\left a_k\right }{\left u_k\right } \leq \frac{\left a_k\right }{\left d_k\right -\left a_{k-1}\right },\left u_{k+1}\right \leq\left d_{k+1}\right +\left l_k\right \left c_k\right \leq\left d_{k+1}\right +\frac{\left a_k\right \left c_k\right }{\left d_k\right -\left a_{k-1}\right }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0121 (2.31)	061	■ 1.000 ● W +2	$\left.\boldsymbol{s}=\left[\boldsymbol{s}_{\{1\}}, \boldsymbol{s}_{\{2\}}, \boldsymbol{s}_{\{3\}}\right]=\left[\begin{array}{c} \sin \frac{\pi}{4} \sin \frac{2 \pi}{4} \sin \frac{3 \pi}{4} \\ \sin \frac{2 \pi}{4} \sin \frac{4 \pi}{4} \sin \frac{6 \pi}{4} \\ \sin \frac{3 \pi}{4} \sin \frac{6 \pi}{4} \sin \frac{9 \pi}{4} \end{array}\right]=\left[\begin{array}{ccc} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{array}\right], \quad t:=\frac{1}{\sqrt{2}} .\right.$	$\boldsymbol{S}=\left[s_1, s_2, s_3\right]=\left[\begin{array}{ccc} \sin \frac{\pi}{4} \sin \frac{2 \pi}{4} \sin \frac{3 \pi}{4} \\ \sin \frac{2 \pi}{4} \sin \frac{4 \pi}{4} \sin \frac{6 \pi}{4} \\ \sin \frac{3 \pi}{4} \sin \frac{6 \pi}{4} \sin \frac{9 \pi}{4} \end{array}\right]=\left[\begin{array}{ccc} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{array}\right], \quad t:=\frac{1}{\sqrt{2}} .$	$\boldsymbol{S}=\left[s_1, s_2, s_3\right]=\left[\begin{array}{ccc} \sin \frac{\pi}{4} \sin \frac{2 \pi}{4} \sin \frac{3 \pi}{4} \\ \sin \frac{2 \pi}{4} \sin \frac{4 \pi}{4} \sin \frac{6 \pi}{4} \\ \sin \frac{3 \pi}{4} \sin \frac{6 \pi}{4} \sin \frac{9 \pi}{4} \end{array}\right]=\left[\begin{array}{ccc} t & 1 & t \\ 1 & 0 & -1 \\ t & -1 & t \end{array}\right], \quad t:=\frac{1}{\sqrt{2}} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0123 (2.32)	061	■ 1.000 ● N +0	$\boldsymbol{s}_{\{j\}}^T \boldsymbol{s}_{\{k\}}=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$	$\boldsymbol{s}_j^T \boldsymbol{s}_k=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$	$\boldsymbol{s}_j^T \boldsymbol{s}_k=\frac{m+1}{2} \delta_{j, k}=\frac{1}{2 h} \delta_{j, k}, \quad j, k=1, \ldots, m.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00128	062	1.000 K +0	$\boldsymbol{A}=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0186 (3.5)	079	1.000 W +0	$(\boldsymbol{L} \ \boldsymbol{U})_{ij}=\sum_{k=1}^{i-1} l_{ik}^{(k)} a_{kj}^{(k)}+a_{ij}^{(i)}=\sum_{k=1}^{i-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(i)}=a_{ij}^{(1)}=a_{ij},$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^{i-1} l_{ik}^{(k)} a_{kj}^{(k)}+a_{ij}^{(i)}=\sum_{k=1}^{i-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(i)}=a_{ij}^{(1)}=a_{ij},$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^{i-1} l_{ik}^{(k)} a_{kj}^{(k)}+a_{ij}^{(i)}=\sum_{k=1}^{i-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(i)}=a_{ij}^{(1)}=a_{ij},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00193	082	1.000 C -8	$\left[\begin{array}{cccc} a_{k+1, k+1} & 0 & \cdots & 0 \\ a_{k+2, k+1} & a_{k+2, k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k+1} & & \cdots & a_{n, n} \end{array}\right] \left[\begin{array}{c} x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{c} b_{k+1} \\ \vdots \\ b_n \end{array}\right]-x_k\left[\begin{array}{c} a_{k+1, k} \\ \vdots \\ a_{n, k} \end{array}\right].$	$\left[\begin{array}{cccc} a_{k+1, k+1} & 0 & \cdots & 0 \\ a_{k+2, k+1} & a_{k+2, k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k+1} & & \cdots & a_{n, n} \end{array}\right] \left[\begin{array}{c} x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{c} b_{k+1} \\ \vdots \\ b_n \end{array}\right]-x_k\left[\begin{array}{c} a_{k+1, k} \\ \vdots \\ a_{n, k} \end{array}\right].$	$\left[\begin{array}{cccc} a_{k+1, k+1} & 0 & \cdots & 0 \\ a_{k+2, k+1} & a_{k+2, k+2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k+1} & & \cdots & a_{n, n} \end{array}\right] \left[\begin{array}{c} x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{c} b_{k+1} \\ \vdots \\ b_n \end{array}\right]-x_k\left[\begin{array}{c} a_{k+1, k} \\ \vdots \\ a_{n, k} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0240 (3.32)	097	1.000 N +0	$\boldsymbol{L}=\left[\begin{array}{cc} 1 & 0 \\ \ell_1 & \boldsymbol{L}_{2,2} \end{array}\right], \quad \boldsymbol{U}=\left[\begin{array}{cc} u_{1,1} & \boldsymbol{u}_1^T \\ 0 & \boldsymbol{U}_{2,2} \end{array}\right],$	$\boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ \ell_1 & \boldsymbol{L}_{2,2} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} u_{1,1} & \boldsymbol{u}_1^T \\ 0 & \boldsymbol{U}_{2,2} \end{bmatrix},$	$\boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ \ell_1 & \boldsymbol{L}_{2,2} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} u_{1,1} & \boldsymbol{u}_1^T \\ 0 & \boldsymbol{U}_{2,2} \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0270	113	1.000 N +1	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}$	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}$	$\boldsymbol{A}[a]:=\left[\begin{array}{cc} 2 & 2-a \\ a & 1 \end{array}\right], \quad a \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0274	114	1.000 K +0	$\boldsymbol{A}_{+}=\left[\begin{array}{cc} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^T & \alpha \end{array}\right],$	$\boldsymbol{A}_{+} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^T & \alpha \end{bmatrix},$	$\boldsymbol{A}_{+} = \begin{bmatrix} \boldsymbol{A} & \boldsymbol{a} \\ \boldsymbol{a}^T & \alpha \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0283 (5.4)	118	1.000 N +0	$\langle a \ \boldsymbol{y}, \boldsymbol{z} \rangle+\langle \boldsymbol{z}, a \ \boldsymbol{y} \rangle=a \overline{\langle \boldsymbol{z}, \boldsymbol{y} \rangle}+\bar{a}\langle \boldsymbol{z}, \boldsymbol{y} \rangle=0.$	$\langle a \boldsymbol{y}, \boldsymbol{z} \rangle+\langle \boldsymbol{z}, a \boldsymbol{y} \rangle=a \overline{\langle \boldsymbol{z}, \boldsymbol{y} \rangle}+\bar{a}\langle \boldsymbol{z}, \boldsymbol{y} \rangle=0.$	$\langle a \boldsymbol{y}, \boldsymbol{z} \rangle+\langle \boldsymbol{z}, a \boldsymbol{y} \rangle=a \overline{\langle \boldsymbol{z}, \boldsymbol{y} \rangle}+\bar{a}\langle \boldsymbol{z}, \boldsymbol{y} \rangle=0.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00328	133	■1.000 ● N +0	$\boldsymbol{R}_{\{1\}}=\boldsymbol{R}=\left[\begin{array}{cc} \left\ \boldsymbol{v}_{\{1\}}\right\ _2 & \boldsymbol{a}_2^T\boldsymbol{q}_{\{1\}} \\ 0 & \left\ \boldsymbol{v}_{\{2\}}\right\ _2 \end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$	$\boldsymbol{R}_1=\boldsymbol{R}=\left[\begin{array}{cc} \left\ \boldsymbol{v}_1\right\ _2 & \boldsymbol{a}_2^T\boldsymbol{q}_1 \\ 0 & \left\ \boldsymbol{v}_2\right\ _2 \end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$	$\boldsymbol{R}_1=\boldsymbol{R}=\left[\begin{array}{cc} \left\ \boldsymbol{v}_1\right\ _2 & \boldsymbol{a}_2^T\boldsymbol{q}_1 \\ 0 & \left\ \boldsymbol{v}_2\right\ _2 \end{array}\right]=\frac{1}{\sqrt{5}}\left[\begin{array}{cc} 5 & -4 \\ 0 & 3 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00341	137	■1.000 ● K +0	$\boldsymbol{B}:=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],$	$\boldsymbol{B}:=\left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],$	$\boldsymbol{B}:=\left[\begin{array}{cccc} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \epsilon & 0 & 0 & 0 \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00345	139	■1.000 ● N -1	$\boldsymbol{V}_{\{r\}}:=\boldsymbol{I}_{\{r\}}-2\frac{\boldsymbol{v}_{\{r\}}\boldsymbol{v}_{\{r\}}^*}{\boldsymbol{v}_{\{r\}}^*\boldsymbol{v}_{\{r\}}}=\boldsymbol{I}_{\{r\}}-\boldsymbol{u}_{\{r\}}\boldsymbol{u}_{\{r\}}^*, \text{ with } \boldsymbol{u}_{\{r\}}:=\sqrt{2}\frac{\boldsymbol{v}_{\{r\}}}{\left\ \boldsymbol{v}_{\{r\}}\right\ _2}.$	$\boldsymbol{V}_r:=\boldsymbol{I}_r-2\frac{\boldsymbol{v}_r\boldsymbol{v}_r^*}{\boldsymbol{v}_r^*\boldsymbol{v}_r}=\boldsymbol{I}_r-\boldsymbol{u}_r\boldsymbol{u}_r^*, \text{ with } \boldsymbol{u}_r:=\sqrt{2}\frac{\boldsymbol{v}_r}{\left\ \boldsymbol{v}_r\right\ _2}.$	$\boldsymbol{V}_r:=\boldsymbol{I}_r-2\frac{\boldsymbol{v}_r\boldsymbol{v}_r^*}{\boldsymbol{v}_r^*\boldsymbol{v}_r}=\boldsymbol{I}_r-\boldsymbol{u}_r\boldsymbol{u}_r^*, \text{ with } \boldsymbol{u}_r:=\sqrt{2}\frac{\boldsymbol{v}_r}{\left\ \boldsymbol{v}_r\right\ _2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00363	147	■1.000 ● K +0	$\boldsymbol{E}:=\left[\begin{array}{cc} A & C \\ C & A \end{array}\right], \quad \boldsymbol{F}:=\left[\begin{array}{cc} 0 & 0 \\ C & CA \end{array}\right], \quad \boldsymbol{S}:=\left[\begin{array}{cc} I & A \\ 0 & I \end{array}\right].$	$\boldsymbol{E}:=\left[\begin{array}{cc} AC & 0 \\ C & 0 \end{array}\right], \quad \boldsymbol{F}:=\left[\begin{array}{cc} 0 & 0 \\ C & CA \end{array}\right], \quad \boldsymbol{S}=\left[\begin{array}{cc} I & A \\ 0 & I \end{array}\right].$	$\boldsymbol{E}:=\left[\begin{array}{cc} AC & 0 \\ C & 0 \end{array}\right], \quad \boldsymbol{F}:=\left[\begin{array}{cc} 0 & 0 \\ C & CA \end{array}\right], \quad \boldsymbol{S}=\left[\begin{array}{cc} I & A \\ 0 & I \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00364 (6.2)	148	■1.000 ● N +0	$\pi_{\{\boldsymbol{A}\}}(\lambda):=\operatorname{det}(\boldsymbol{A}-\lambda\boldsymbol{I})=\left(\lambda_{\{1\}}-\lambda\right)^{a_{\{1\}}} \cdots\left(\lambda_{\{k\}}-\lambda\right)^{a_{\{k\}}}, \quad \lambda_{\{i\}} \neq \lambda_{\{j\}}, i \neq j, \sum_{i=1}^k a_{\{i\}}=n.$	$\pi_A(\lambda):=\det(\boldsymbol{A}-\lambda\boldsymbol{I})=(\lambda_1-\lambda)^{a_1}\cdots(\lambda_k-\lambda)^{a_k}, \quad \lambda_i\neq\lambda_j, i\neq j, \sum_{i=1}^ka_i=n.$	$\pi_A(\lambda):=\det(\boldsymbol{A}-\lambda\boldsymbol{I})=(\lambda_1-\lambda)^{a_1}\cdots(\lambda_k-\lambda)^{a_k}, \quad \lambda_i\neq\lambda_j, i\neq j, \sum_{i=1}^ka_i=n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00393 (6.12)	157	■1.000 ● N -1	$\lambda_{\{k\}}\geq\min_{\boldsymbol{x}\in S}\boldsymbol{R}(\boldsymbol{x}),$	$\lambda_k\geq\min_{\substack{\boldsymbol{x}\in S \\ \boldsymbol{x}\neq\boldsymbol{0}}}\boldsymbol{R}(\boldsymbol{x}),$	$\lambda_k\geq\min_{\substack{\boldsymbol{x}\in S \\ \boldsymbol{x}\neq\boldsymbol{0}}}\boldsymbol{R}(\boldsymbol{x}),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00413	165	■1.000 ● N -1	$\lambda_{\min}=\min_{\boldsymbol{x}\neq\boldsymbol{0}}\boldsymbol{R}(\boldsymbol{x})=\min_{\left\ \boldsymbol{x}\right\ _2=1}\boldsymbol{R}(\boldsymbol{x}),$	$\lambda_{\min}=\min_{\boldsymbol{x}\neq\boldsymbol{0}}\boldsymbol{R}(\boldsymbol{x})=\min_{\left\ \boldsymbol{x}\right\ _2=1}\boldsymbol{R}(\boldsymbol{x}),$	$\lambda_{\min}=\min_{\boldsymbol{x}\neq\boldsymbol{0}}\boldsymbol{R}(\boldsymbol{x})=\min_{\left\ \boldsymbol{x}\right\ _2=1}\boldsymbol{R}(\boldsymbol{x}),$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0427 (2)	172	■1.000 ● N +0	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{l} 2 \\ 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array}\right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array}\right] .$	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array}\right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array}\right] .$	$\left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{l} 1 \\ 1 \end{array}\right] [2] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \end{array}\right] .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0435	173	■1.000 ● N -2	$\boldsymbol{U} := \left[\begin{array}{ccc} 1 / \sqrt{2} & -1 / \sqrt{2} & 0 \\ 1 / \sqrt{2} & 1 / \sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma} := \left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V} := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .$	$\boldsymbol{U} := \left[\begin{array}{ccc} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma} := \left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V} := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .$	$\boldsymbol{U} := \left[\begin{array}{ccc} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{array}\right] \in \mathbb{R}^{3,3}, \quad \boldsymbol{\Sigma} := \left[\begin{array}{cc} 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{array}\right] \in \mathbb{R}^{3,2}, \quad \boldsymbol{V} := \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \in \mathbb{R}^{2,2} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0439	175	■1.000 ● K +0	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]$	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]$	$A := \frac{1}{25} \left[\begin{array}{ll} 11 & 48 \\ 48 & 39 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0442	177	■1.000 ● N +0	$\left \left \boldsymbol{A} \right \boldsymbol{B} \right _{\boldsymbol{F}}^2 = \sum_{i=1}^m \sum_{j=1}^k \left \boldsymbol{a}_{i:}^* \boldsymbol{b}_{:j} \right ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \left\ \boldsymbol{a}_{i:} \right\ _2^2 \left\ \boldsymbol{b}_{:j} \right\ _2^2 = \left\ \boldsymbol{A} \right\ _F^2 \left\ \boldsymbol{B} \right\ _F^2 .$	$\ \boldsymbol{A} \boldsymbol{B} \ _F^2 = \sum_{i=1}^m \sum_{j=1}^k \boldsymbol{a}_{i:}^* \boldsymbol{b}_{:j} ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \ \boldsymbol{a}_{i:} \ _2^2 \ \boldsymbol{b}_{:j} \ _2^2 = \ \boldsymbol{A} \ _F^2 \ \boldsymbol{B} \ _F^2 .$	$\ \boldsymbol{A} \boldsymbol{B} \ _F^2 = \sum_{i=1}^m \sum_{j=1}^k \boldsymbol{a}_{i:}^* \boldsymbol{b}_{:j} ^2 \leq \sum_{i=1}^m \sum_{j=1}^k \ \boldsymbol{a}_{i:} \ _2^2 \ \boldsymbol{b}_{:j} \ _2^2 = \ \boldsymbol{A} \ _F^2 \ \boldsymbol{B} \ _F^2 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0447 (7.11)	178	■1.000 ● N +1	$\boldsymbol{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} =: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array}\right] =: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{A} := \left[\begin{array}{ll} 1 & -1 \\ 1 & -1 \end{array}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & 1 \\ 1 & -1 \end{array}\right] \left[\begin{array}{ll} 2 & 0 \\ 0 & 0 \end{array}\right] \frac{1}{\sqrt{2}} \left[\begin{array}{ll} 1 & -1 \\ 1 & 1 \end{array}\right] =: \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0457	181	■1.000 ● K +0	$\boldsymbol{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .$	$\boldsymbol{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .$	$\boldsymbol{A} = \left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \\ -1 & 3 \end{array}\right] .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0463	183	■1.000 ● K +0	$\boldsymbol{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array}\right] .$	$\boldsymbol{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array}\right] .$	$\boldsymbol{T} = \left[\begin{array}{ll} 2 & -1 \\ -1 & 2 \end{array}\right] .$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0484 (8.16)	190	■ 1.000 ● K +0	$\ \boldsymbol{A}\ =\left\ \boldsymbol{A}\right\ \boldsymbol{x}^* \text { for some } \boldsymbol{x}^* \in \mathcal{S} .$	$\ \boldsymbol{A}\ =\left\ \boldsymbol{A} \boldsymbol{x}^*\right\ \text { for some } \boldsymbol{x}^* \in \mathcal{S} .$	$\ \boldsymbol{A}\ =\left\ \boldsymbol{A} \boldsymbol{x}^*\right\ \text { for some } \boldsymbol{x}^* \in \mathcal{S} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0490	192	■ 1.000 ● N +0	$\left\ \boldsymbol{A}\right\ _{\infty}=\max _{1 \leq k \leq m}\left \sum_{j=1}^n a_{k j} x_j\right \leq \max _{1 \leq k \leq m} \sum_{j=1}^n\left a_{k j}\right \left x_j\right \leq K_{\infty} .$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _{\infty}=\max _{1 \leq k \leq m}\left \sum_{j=1}^n a_{k j} x_j\right \leq \max _{1 \leq k \leq m} \sum_{j=1}^n\left a_{k j}\right \left x_j\right \leq K_{\infty} .$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _{\infty}=\max _{1 \leq k \leq m}\left \sum_{j=1}^n a_{k j} x_j\right \leq \max _{1 \leq k \leq m} \sum_{j=1}^n\left a_{k j}\right \left x_j\right \leq K_{\infty} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0493 (8.21)	193	■ 1.000 ● K +0	$\left\ \boldsymbol{A}\right\ _2^2\left\ \boldsymbol{v}\right\ _1=\sigma^2\left\ \boldsymbol{v}\right\ _1=\left\ \sigma^2 \boldsymbol{v}\right\ _1=\left\ \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}\right\ _1 \leq\left\ \boldsymbol{A}^*\right\ _1\left\ \boldsymbol{A}\right\ _1\left\ \boldsymbol{v}\right\ _1 .$	$\left\ \boldsymbol{A}\right\ _2^2\left\ \boldsymbol{v}\right\ _1=\sigma^2\left\ \boldsymbol{v}\right\ _1=\left\ \sigma^2 \boldsymbol{v}\right\ _1=\left\ \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}\right\ _1 \leq\left\ \boldsymbol{A}^*\right\ _1\left\ \boldsymbol{A}\right\ _1\left\ \boldsymbol{v}\right\ _1 .$	$\left\ \boldsymbol{A}\right\ _2^2\left\ \boldsymbol{v}\right\ _1=\sigma^2\left\ \boldsymbol{v}\right\ _1=\left\ \sigma^2 \boldsymbol{v}\right\ _1=\left\ \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}\right\ _1 \leq\left\ \boldsymbol{A}^*\right\ _1\left\ \boldsymbol{A}\right\ _1\left\ \boldsymbol{v}\right\ _1 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0505 (8.27)	198	■ 1.000 ● N +0	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ } \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ },$	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ } \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ },$	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{B}^{-1}\right\ } \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ },$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0509	198	■ 1.000 ● N +0	$\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \left\ \boldsymbol{B}^{-1}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\left\ \boldsymbol{B}^{-1}\right\ .$	$\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \left\ \boldsymbol{B}^{-1}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\left\ \boldsymbol{B}^{-1}\right\ .$	$\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ \left\ \boldsymbol{B}^{-1}\right\ \leq K(\boldsymbol{A}) \frac{\left\ \boldsymbol{E}\right\ }{\left\ \boldsymbol{A}\right\ }\left\ \boldsymbol{B}^{-1}\right\ .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0521 (8.34)	201	■ 1.000 ● N +1	$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+\frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .$	$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+\frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .$	$a^{\frac{1}{p}} b^{\frac{1}{q}} \leq \frac{1}{p} a+\frac{1}{q} b, \text { where } \frac{1}{p}+\frac{1}{q}=1 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0528 (8.36)	202	■ 1.000 ● N +0	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$	$\langle\boldsymbol{x}, \boldsymbol{y}\rangle:=\frac{1}{4}\left(\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2-\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2\right), \quad \boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0551	208	■ 1.000 ● N +0	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{c} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{c} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{c} e_1 \\ e_2 \end{array}\right], \quad\left\ \boldsymbol{e}\right\ _2=0.1 .$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{c} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{c} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{c} e_1 \\ e_2 \end{array}\right], \quad\left\ \boldsymbol{e}\right\ _2=0.1 .$	$\boldsymbol{A}:=\left[\begin{array}{cc} 1.1 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{b}:=\left[\begin{array}{c} b_1 \\ b_2 \end{array}\right]=\left[\begin{array}{c} 2.1 \\ 2.0 \end{array}\right], \quad \boldsymbol{e}:=\left[\begin{array}{c} e_1 \\ e_2 \end{array}\right], \quad\left\ \boldsymbol{e}\right\ _2=0.1 .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00598	220	■ 1.000 ● N +1	$\boldsymbol{A}^{\ast} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^{\ast} \boldsymbol{b} \Longrightarrow \boldsymbol{R}_1 \boldsymbol{x} = \boldsymbol{c}_1, \quad \boldsymbol{c}_1 := \boldsymbol{Q}_1^{\ast} \boldsymbol{b}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0608 (9.11)	223	■ 1.000 ● K +0	$\boldsymbol{A}^{\dagger} = \left(\boldsymbol{A}^{\ast} \boldsymbol{A} \right)^{-1} \boldsymbol{A}^{\ast}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0602 (10.15)	245	■ 1.000 ● N +0	$\lambda_j = d + 2 a \cos \left(j \pi h \right), \quad h := \frac{1}{m + 1},$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0684 (10.17)	246	■ 1.000 ● N +0	$\boldsymbol{s}_j^T \boldsymbol{s}_k = \frac{m + 1}{2} \delta_{j,k} = \frac{1}{2h} \delta_{j,k}, \quad j, k = 1, \dots, m.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0692 (10.22)	248	■ 1.000 ● N +1	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} - \frac{1}{6} \boldsymbol{T} \boldsymbol{V} \boldsymbol{T} = h^2 \mu \boldsymbol{F}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0695 (10.21)	248	■ 1.000 ● N +0	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}, \quad \boldsymbol{T} \boldsymbol{U} + \boldsymbol{U} \boldsymbol{T} = h^2 \boldsymbol{V}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0721	257	■ 1.000 ● N +0	$\boldsymbol{A} \boldsymbol{P}_N = [\boldsymbol{a}_1, \boldsymbol{a}_3, \dots, \boldsymbol{a}_{N-1}, \boldsymbol{a}_2, \boldsymbol{a}_4, \dots, \boldsymbol{a}_N],$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0754 (12.9)	270	■ 1.000 ● N +0	$\boldsymbol{M} \boldsymbol{x}_{k+1} = (\boldsymbol{M} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00789 (12.29)	277	■ 1.000 ● K +0	$\rho\left(\boldsymbol{G}_{\omega}\right)>\rho\left(\boldsymbol{G}_{\omega^*}\right)\text{ for }\omega\in\left(0,2\right)\backslash\left\{\omega^*\right\}.$	$\rho\left(\boldsymbol{G}_{\omega}\right)>\rho\left(\boldsymbol{G}_{\omega^*}\right)\text{ for }\omega\in\left(0,2\right)\backslash\left\{\omega^*\right\}.$	$\rho\left(\boldsymbol{G}_{\omega}\right)>\rho\left(\boldsymbol{G}_{\omega^*}\right)\text{ for }\omega\in\left(0,2\right)\backslash\left\{\omega^*\right\}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00791	278	■ 1.000 ● N +0	$\left \left \boldsymbol{x}_{\boldsymbol{k}}-\boldsymbol{x}\right \right \leq\left \left \boldsymbol{G}^{\boldsymbol{k}}\right \right \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx\rho\left(\boldsymbol{G}\right)^{\boldsymbol{k}}\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ \leq\left\ \boldsymbol{G}^k\right\ \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx\rho\left(\boldsymbol{G}\right)^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ \leq\left\ \boldsymbol{G}^k\right\ \left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ \approx\rho\left(\boldsymbol{G}\right)^k\left\ \boldsymbol{x}_0-\boldsymbol{x}\right\ .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00820	285	■ 1.000 ● K +0	$d^{\prime}(\omega)=2\left(\omega\mu^2-2\right)<2(\omega-2)<0.$	$d'(\omega)=2\left(\omega\mu^2-2\right)<2(\omega-2)<0.$	$d'(\omega)=2(\omega\mu^2-2)<2(\omega-2)<0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00840 (13.6)	292	■ 1.000 ● N +0	$\begin{aligned} &\frac{\partial}{\partial y_i}Q(\boldsymbol{y}) &:=\lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(Q(\boldsymbol{y}+\varepsilon\boldsymbol{e}_i)-Q(\boldsymbol{y})\right) \\ &= \lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(-\varepsilon\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y})\right)+\frac{1}{2}\varepsilon^2\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i\Big)=-\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y}),\quad i=1,\dots,n, \end{aligned}$	$\frac{\partial}{\partial y_i}Q(\boldsymbol{y}) := \lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(Q(\boldsymbol{y}+\varepsilon\boldsymbol{e}_i)-Q(\boldsymbol{y})\right) \\ = \lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(-\varepsilon\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y})\right)+\frac{1}{2}\varepsilon^2\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i\Big)=-\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y}),\quad i=1,\dots,n,$	$\frac{\partial}{\partial y_i}Q(\boldsymbol{y}) := \lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(Q(\boldsymbol{y}+\varepsilon\boldsymbol{e}_i)-Q(\boldsymbol{y})\right) \\ = \lim_{\varepsilon\rightarrow 0}\frac{1}{\varepsilon}\left(-\varepsilon\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y})\right)+\frac{1}{2}\varepsilon^2\boldsymbol{e}_i^T\boldsymbol{A}\boldsymbol{e}_i\Big)=-\boldsymbol{e}_i^T\boldsymbol{r}(\boldsymbol{y}),\quad i=1,\dots,n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00866 (13.23)	301	■ 1.000 ● N +0	$\boldsymbol{r}_{\boldsymbol{k}}^T\boldsymbol{w}=\boldsymbol{p}_{\boldsymbol{k}}^T\boldsymbol{A}\boldsymbol{w}=0,\quad \boldsymbol{w}\in\mathbb{W}_{\boldsymbol{k}}.$	$\boldsymbol{r}_k^T\boldsymbol{w}=\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{w}=0,\quad \boldsymbol{w}\in\mathbb{W}_k.$	$\boldsymbol{r}_k^T\boldsymbol{w}=\boldsymbol{p}_k^T\boldsymbol{A}\boldsymbol{w}=0,\quad \boldsymbol{w}\in\mathbb{W}_k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00874 (13.29)	303	■ 1.000 ● N -1	$\frac{\left \left \boldsymbol{x}-\boldsymbol{x}_{\boldsymbol{k}}\right \right _{\boldsymbol{A}}}{\left \left \boldsymbol{x}-\boldsymbol{x}_0\right \right _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_{\boldsymbol{k}}\\Q(0)=1}}\max_{\lambda_{\min}\leq x\leq\lambda_{\max}} Q(x) =\frac{2}{a^{-\boldsymbol{k}}+a^{\boldsymbol{k}}},\quad a:=\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$	$\frac{\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}}{\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_k\\Q(0)=1}}\max_{\lambda_{\min}\leq x\leq\lambda_{\max}} Q(x) =\frac{2}{a^{-k}+a^k},\quad a:=\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$	$\frac{\left\ \boldsymbol{x}-\boldsymbol{x}_k\right\ _{\boldsymbol{A}}}{\left\ \boldsymbol{x}-\boldsymbol{x}_0\right\ _{\boldsymbol{A}}}\leq\min_{\substack{Q\in\Pi_k\\Q(0)=1}}\max_{\lambda_{\min}\leq x\leq\lambda_{\max}} Q(x) =\frac{2}{a^{-k}+a^k},\quad a:=\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00875 (13.30)	303	■ 1.000 ● N +0	$\frac{x-1}{x+1}<e^{-2/x}\quad\text{for}\quad x>1$	$\frac{x-1}{x+1}<e^{-2/x}\quad\text{for}\quad x>1$	$\frac{x-1}{x+1}<e^{-2/x}\quad\text{for}\quad x>1$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0929 (13.58)	315	■ 1.000 ● N +0	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$	$\langle \boldsymbol{V}, \boldsymbol{W} \rangle := h^2 \sum_{j,k=1}^m v_{j,k} w_{j,k},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0930 (13.59)	315	■ 1.000 ● W -1	$\begin{aligned} \langle L_h v, \boldsymbol{W} \rangle &= \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k}) \\ &+ \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}). \end{aligned}$	$\langle L_h v, \boldsymbol{W} \rangle = \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k}) + \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}).$	$\langle L_h v, \boldsymbol{W} \rangle = \sum_{j=1}^m \sum_{k=0}^m c_{j,k+\frac{1}{2}} (v_{j,k+1} - v_{j,k}) (w_{j,k+1} - w_{j,k}) + \sum_{j=0}^m \sum_{k=1}^m c_{j+\frac{1}{2},k} (v_{j+1,k} - v_{j,k}) (w_{j+1,k} - w_{j,k}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0946	320	■ 1.000 ● N -1	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0957	322	■ 1.000 ● N +0	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \text{ and } \rho_k := \ \boldsymbol{r}_k\ _2^2.$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \text{ and } \rho_k := \ \boldsymbol{r}_k\ _2^2.$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k, \text{ and } \rho_k := \ \boldsymbol{r}_k\ _2^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0967	326	■ 1.000 ● W +0	$\boldsymbol{A} \boldsymbol{D} = \text{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \text{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \text{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) = \text{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{X} \boldsymbol{D}.$	$\boldsymbol{A} \boldsymbol{D} = \text{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \text{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \text{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) = \text{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{X} \boldsymbol{D}.$	$\boldsymbol{A} \boldsymbol{D} = \text{diag}(\boldsymbol{A}_1, \dots, \boldsymbol{A}_r) \text{diag}(\boldsymbol{X}_1, \dots, \boldsymbol{X}_r) = \text{diag}(\boldsymbol{A}_1 \boldsymbol{X}_1, \dots, \boldsymbol{A}_r \boldsymbol{X}_r) = \text{diag}(\boldsymbol{X}_1 \boldsymbol{D}_1, \dots, \boldsymbol{X}_r \boldsymbol{D}_r) = \boldsymbol{X} \boldsymbol{D}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1030	348	■ 1.000 ● N +0	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1}.$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1}.$	$\mu_1(s) = (\lambda_1 - s)^{-1}, \mu_2(s) = (\lambda_2 - s)^{-1}, \dots, \mu_n(s) = (\lambda_n - s)^{-1}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01041	352	■ 1.000 ● N +0	$A_{14}=\left[\begin{array}{rrrr} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{array}\right].$	$A_{14}=\left[\begin{array}{rrrr} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{array}\right].$	$A_{14}=\left[\begin{array}{rrrr} 2.323 & 0.047223 & -0.39232 & -0.65056 \\ -2.1e-10 & 0.13029 & 0.36125 & 0.15946 \\ -4.1e-10 & -0.58622 & 0.052576 & -0.25774 \\ 1.2e-14 & 3.3e-05 & -1.1e-05 & 0.22746 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01045	353	■ 1.000 ● N +0	$\tilde{\boldsymbol{Q}}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}(\boldsymbol{Q}_{\boldsymbol{k}}\boldsymbol{R}_{\boldsymbol{k}})\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\boldsymbol{A}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\left(\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*\right)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\boldsymbol{A}^{\boldsymbol{k}}.$	$\tilde{\boldsymbol{Q}}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}(\boldsymbol{Q}_{\boldsymbol{k}}\boldsymbol{R}_{\boldsymbol{k}})\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\boldsymbol{A}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\left(\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*\right)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\boldsymbol{A}^{\boldsymbol{k}}.$	$\tilde{\boldsymbol{Q}}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}(\boldsymbol{Q}_{\boldsymbol{k}}\boldsymbol{R}_{\boldsymbol{k}})\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\boldsymbol{A}_{\boldsymbol{k}}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\left(\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*\right)\boldsymbol{A}\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\tilde{\boldsymbol{R}}_{\boldsymbol{k}-1}=\boldsymbol{A}^{\boldsymbol{k}}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E01047	354	■ 1.000 ● N +1	$\hat{\boldsymbol{A}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\left(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right)\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*.$	$\hat{\boldsymbol{A}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}\left(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T\right)\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*.$	$\hat{\boldsymbol{A}}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*(\boldsymbol{A}+\boldsymbol{E})\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1},\quad \boldsymbol{E}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}(a_{i+1,i}^{(k)}\boldsymbol{e}_{i+1}\boldsymbol{e}_i^T)\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^*.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00002	022	■ 1.000 ● N +0	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_1+y_1 \\ \vdots \\ x_n+y_n \end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c} ax_1 \\ \vdots \\ ax_n \end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_1+y_1 \\ \vdots \\ x_n+y_n \end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c} ax_1 \\ \vdots \\ ax_n \end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$	$\boldsymbol{x}+\boldsymbol{y}:=\left[\begin{array}{c} x_1+y_1 \\ \vdots \\ x_n+y_n \end{array}\right],\quad \boldsymbol{a}\boldsymbol{x}:=\left[\begin{array}{c} ax_1 \\ \vdots \\ ax_n \end{array}\right],\quad \boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n,\quad \boldsymbol{a}\in\mathbb{R}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00004	022	■ 1.000 ● W +1	$\boldsymbol{a}_{\boldsymbol{j}}:=\left[\begin{array}{c} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{array}\right],\quad \boldsymbol{a}_{i:}^T:=[a_{i1},a_{i2},\ldots,a_{in}],\quad \boldsymbol{A}=[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\ldots,\boldsymbol{a}_{:n}]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{array}\right]$	$\boldsymbol{a}_{\boldsymbol{j}}:=\left[\begin{array}{c} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{array}\right],\quad \boldsymbol{a}_{i:}^T:=[a_{i1},a_{i2},\ldots,a_{in}],\quad \boldsymbol{A}=[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\ldots,\boldsymbol{a}_{:n}]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{array}\right]$	$\boldsymbol{a}_{\boldsymbol{j}}:=\left[\begin{array}{c} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{array}\right],\quad \boldsymbol{a}_{i:}^T:=[a_{i1},a_{i2},\ldots,a_{in}],\quad \boldsymbol{A}=[\boldsymbol{a}_{:1},\boldsymbol{a}_{:2},\ldots,\boldsymbol{a}_{:n}]=\left[\begin{array}{c} \boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \vdots \\ \boldsymbol{a}_{m:}^T \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00005	022	■ 1.000 ● N +0	$e^{\boldsymbol{x}}=e^{\boldsymbol{a}+\boldsymbol{i}\boldsymbol{b}}:=e^{\boldsymbol{a}}(\cos \boldsymbol{b}+\boldsymbol{i}\sin \boldsymbol{b}).$	$e^{\boldsymbol{x}}=e^{\boldsymbol{a}+\boldsymbol{i}\boldsymbol{b}}:=e^{\boldsymbol{a}}(\cos \boldsymbol{b}+\boldsymbol{i}\sin \boldsymbol{b}).$	$e^{\boldsymbol{x}}=e^{\boldsymbol{a}+\boldsymbol{i}\boldsymbol{b}}:=e^{\boldsymbol{a}}(\cos \boldsymbol{b}+\boldsymbol{i}\sin \boldsymbol{b}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00006	022	■ 1.000 ● N +0	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$	$e^{i\pi/2}=i,\quad e^{i\pi}=-1,\quad e^{2i\pi}=1.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00007	023	1.000 N +0	$x=a+ib=r e^{i\theta}, \quad r= x =\sqrt{a^2+b^2}, \quad \cos\theta=\frac{a}{r}, \quad \sin\theta=\frac{b}{r}.$	$x = a + ib = r e^{i\theta}, \quad r = x = \sqrt{a^2 + b^2}, \quad \cos\theta = \frac{a}{r}, \quad \sin\theta = \frac{b}{r}.$	$x = a + ib = r e^{i\theta}, \quad r = x = \sqrt{a^2 + b^2}, \quad \cos\theta = \frac{a}{r}, \quad \sin\theta = \frac{b}{r}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00014 (1.2)	025	1.000 N +1	$p(t):=a_{\{0\}}+a_{\{1\}}\,t+a_{\{2\}}\,t^2+\cdots+a_{\{n\}}\,t^n,$	$p(t) := a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n,$	$p(t) := a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00017 (1.5)	028	1.000 N +0	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}.$	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}.$	$\mathcal{S} \cap \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ and } \boldsymbol{x} \in \mathcal{T}\}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00018 (1.6)	028	1.000 N +1	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}.$	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}.$	$\mathcal{S} \cup \mathcal{T} := \{\boldsymbol{x} : \boldsymbol{x} \in \mathcal{S} \text{ or } \boldsymbol{x} \in \mathcal{T}\}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00020 (1.8)	029	1.000 N +1	$\operatorname{dim}(\mathcal{S} \oplus \mathcal{T}) = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{T}).$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$	$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00021	029	1.000 N +1	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00022	029	1.000 N +0	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00023	029	1.000 C -4	$\operatorname{dim}(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \operatorname{dim}(\mathcal{S}) + \operatorname{dim}(\mathcal{T}) - \operatorname{dim}(\mathcal{S} \cap \mathcal{T})$	$\dim(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$	$\dim(\mathcal{S} + \mathcal{T}) = p + q + r = (p + q) + (p + r) - p = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00026	030	1.000 N +0	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X}).$	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X}).$	$\mathcal{R}(\boldsymbol{X}) := \{\boldsymbol{X}\boldsymbol{c} : \boldsymbol{c} \in \mathbb{R}^n\} = \operatorname{span}(\mathcal{X}).$
Conf. MathPix confidence: 1.000 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00030	031	■ 1.000 ● N +0	$x_{\{1\}} \, \boldsymbol{\mathrm{a}}_{\{1\}} + x_{\{2\}} \, \boldsymbol{\mathrm{a}}_{\{2\}} + \cdots + x_{\{n\}} \, \boldsymbol{\mathrm{a}}_{\{n\}} = \boldsymbol{\mathrm{b}},$	$x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \cdots + x_n \boldsymbol{a}_n = \boldsymbol{b},$	$x_1 \boldsymbol{a}_1 + x_2 \boldsymbol{a}_2 + \cdots + x_n \boldsymbol{a}_n = \boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00035 (1.12)	035	■ 1.000 ● N +1	$\operatorname{det}(\boldsymbol{\mathrm{A}}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) \, \boldsymbol{\mathrm{a}}_{\sigma(1),1} \boldsymbol{\mathrm{a}}_{\sigma(2),2} \cdots \boldsymbol{\mathrm{a}}_{\sigma(n),n}$	$\det(\boldsymbol{A}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$	$\det(\boldsymbol{A}) = \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00036	035	■ 1.000 ● N +0	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right .$	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right .$	$\left \begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00037	035	■ 1.000 ● W +0	$\left \begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}.$	$\left \begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}.$	$\left \begin{array}{cc} a_{11} & a_{12} \\ a_{21} & a_{22} \end{array} \right = a_{11} a_{22} - a_{21} a_{12}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00039 (1.13)	036	■ 1.000 ● K +0	$\operatorname{det}(\boldsymbol{\mathrm{B}}) = \operatorname{det}(\boldsymbol{\mathrm{A}}).$	$\det(\boldsymbol{B}) = \det(\boldsymbol{A}).$	$\det(\boldsymbol{B}) = \det(\boldsymbol{A}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00041	036	■ 1.000 ● N +0	$\operatorname{det}(\boldsymbol{\mathrm{A}}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$	$\det(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$	$\det(\boldsymbol{A}) := \left \begin{array}{ccc} 1 & x & y \\ 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \end{array} \right = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00043	036	■ 1.000 ● N +0	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00044	037	■ 1.000 ● N +0	$x_{\{j\}} = \frac{\operatorname{det}(\boldsymbol{\mathrm{A}}_{\{j\}}(\boldsymbol{\mathrm{b}}))}{\operatorname{det}(\boldsymbol{\mathrm{A}})}, \quad j = 1, 2, \dots, n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$	$x_j = \frac{\det(\boldsymbol{A}_j(\boldsymbol{b}))}{\det(\boldsymbol{A})}, \quad j = 1, 2, \dots, n,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00045	037	■ 1.000 ● N +0	$\boldsymbol{A}^{-1}=\frac{1}{\operatorname{det}(\boldsymbol{A})}\operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$	$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \operatorname{adj}(\boldsymbol{A}),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00048	038	■ 1.000 ● W +0	$\operatorname{det}(\boldsymbol{A}-\boldsymbol{\lambda})=\left \begin{array}{ccc}a_{11}-\boldsymbol{\lambda}&a_{12}&a_{13}\\a_{21}&a_{22}-\boldsymbol{\lambda}&a_{23}\\a_{31}&a_{32}&a_{33}-\boldsymbol{\lambda}\end{array}\right .$	$\det(\boldsymbol{A}-\lambda\boldsymbol{I}) = \begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix}.$	$\det(\boldsymbol{A}-\lambda\boldsymbol{I}) = \begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00050 (1.16)	038	■ 1.000 ● N +0	$\operatorname{det}(\boldsymbol{A}-\boldsymbol{\lambda})=\left(a_{11}-\boldsymbol{\lambda}\right)\left(a_{22}-\boldsymbol{\lambda}\right)\cdots\left(a_{nn}-\boldsymbol{\lambda}\right)+r(\boldsymbol{\lambda}),$	$\det(\boldsymbol{A}-\lambda\boldsymbol{I}) = (a_{11}-\lambda)(a_{22}-\lambda)\cdots(a_{nn}-\lambda)+r(\lambda),$	$\det(\boldsymbol{A}-\lambda\boldsymbol{I}) = (a_{11}-\lambda)(a_{22}-\lambda)\cdots(a_{nn}-\lambda)+r(\lambda),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00051 (1.17)	039	■ 1.000 ● N +0	$\operatorname{trace}(\boldsymbol{A})=\boldsymbol{\lambda}_1+\boldsymbol{\lambda}_2+\cdots+\boldsymbol{\lambda}_n,\quad\operatorname{det}(\boldsymbol{A})=\boldsymbol{\lambda}_1\boldsymbol{\lambda}_2\cdots\boldsymbol{\lambda}_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$	$\operatorname{trace}(\boldsymbol{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \quad \det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00052 (1.18)	039	■ 1.000 ● N -1	$\operatorname{trace}(\boldsymbol{A}):=a_{11}+a_{22}+\cdots+a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$	$\operatorname{trace}(\boldsymbol{A}) := a_{11} + a_{22} + \cdots + a_{nn}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00054	039	■ 1.000 ● N +0	$\pi_{\boldsymbol{A}}(\boldsymbol{\lambda})=\left(\boldsymbol{\lambda}_1-\boldsymbol{\lambda}\right)\cdots\left(\boldsymbol{\lambda}_n-\boldsymbol{\lambda}\right)=(-1)^n\boldsymbol{\lambda}^n+d_{n-1}\boldsymbol{\lambda}^{n-1}+\cdots+d_0,$	$\pi_{\boldsymbol{A}}(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$	$\pi_{\boldsymbol{A}}(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda) = (-1)^n \lambda^n + d_{n-1} \lambda^{n-1} + \cdots + d_0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00055 (1.19)	039	■ 1.000 ● K +0	$\boldsymbol{\lambda}^2-\operatorname{trace}(\boldsymbol{A})\boldsymbol{\lambda}+\operatorname{det}(\boldsymbol{A})=0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$	$\lambda^2 - \operatorname{trace}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00056	040	■ 1.000 ● K +0	$\left[\begin{array}{ll}W&X\\Y&Z\end{array}\right]=\left[\begin{array}{ll}A&B\\C&D\end{array}\right]\left[\begin{array}{ll}E&F\\G&H\end{array}\right],$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$	$\begin{bmatrix} W & X \\ Y & Z \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00059	041	■ 1.000 ● W +0	$A=\left[\begin{array}{cc}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{array}\right].$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$	$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0063	042	1.000 K +0	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 1 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{l} 3 \\ 6 \end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0065	042	1.000 N +0	$\operatorname{adj}(\boldsymbol{A})=\left[\begin{array}{ccc} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{array}\right] .$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix} .$	$\operatorname{adj}(\boldsymbol{A}) = \begin{bmatrix} 14 & 21 & 42 \\ -42 & -14 & 21 \\ 21 & -42 & 14 \end{bmatrix} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0066	042	1.000 K +0	$\operatorname{adj}(\boldsymbol{A}) \boldsymbol{A}=\left[\begin{array}{ccc} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{array}\right] =\operatorname{det}(\boldsymbol{A}) \boldsymbol{I} .$	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A} = \begin{bmatrix} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{bmatrix} = \det(\boldsymbol{A})\boldsymbol{I}.$	$\operatorname{adj}(\boldsymbol{A})\boldsymbol{A} = \begin{bmatrix} 343 & 0 & 0 \\ 0 & 343 & 0 \\ 0 & 0 & 343 \end{bmatrix} = \det(\boldsymbol{A})\boldsymbol{I}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0067	042	1.000 W +1	$\left \begin{array}{llll} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{array}\right = 0$	$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$	$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0068	043	1.000 W +0	$A(T)=\frac{1}{2}\left \begin{array}{ccc} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{array}\right .$	$A(T) = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} .$	$A(T) = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0069	043	1.000 W +0	$\left \begin{array}{cccccc} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} & \cdots & x_n^{n-1} \end{array}\right = \prod_{i>j} (x_i - x_j) ,$	$\begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{vmatrix} = \prod_{i>j} (x_i - x_j) ,$	$\begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{vmatrix} = \prod_{i>j} (x_i - x_j) ,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0070	043	1.000 N +0	$\operatorname{det}(\boldsymbol{A})=Pg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$	$\det(\boldsymbol{A}) = Pg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$	$\det(\boldsymbol{A}) = Pg(\boldsymbol{\alpha})g(\boldsymbol{\beta})$
Conf. MathPix confidence: 1.000 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0071	044	■ 1.000 ● N +0	$g(\boldsymbol{\gamma})=\prod_{i=2}^n\left(\gamma_{\boldsymbol{i}}-\gamma_{\boldsymbol{1}}\right)\left(\gamma_{\boldsymbol{i}}-\gamma_{\boldsymbol{2}}\right)\cdots\left(\gamma_{\boldsymbol{i}}-\gamma_{\boldsymbol{i-1}}\right)$	$g(\gamma)=\prod_{i=2}^n(\gamma_i-\gamma_1)(\gamma_i-\gamma_2)\cdots(\gamma_i-\gamma_{i-1})$	$g(\boldsymbol{\gamma})=\prod_{i=2}^n(\gamma_i-\gamma_1)(\gamma_i-\gamma_2)\cdots(\gamma_i-\gamma_{i-1})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0072	044	■ 1.000 ● N +0	$q(\boldsymbol{\alpha},\boldsymbol{\beta})=k\,g(\boldsymbol{\alpha})\,g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha},\boldsymbol{\beta})=k\,g(\boldsymbol{\alpha})\,g(\boldsymbol{\beta})$	$q(\boldsymbol{\alpha},\boldsymbol{\beta})=k\,g(\boldsymbol{\alpha})\,g(\boldsymbol{\beta})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0074	044	■ 1.000 ● N +0	$A_{\boldsymbol{k}}(x)=\prod_{s\neq k}\left(\frac{\alpha_{\boldsymbol{s}}-x}{\alpha_{\boldsymbol{s}}-\alpha_{\boldsymbol{k}}}\right),\quad B_{\boldsymbol{k}}(x)=\prod_{s\neq k}\left(\frac{\beta_{\boldsymbol{s}}-x}{\beta_{\boldsymbol{s}}-\beta_{\boldsymbol{k}}}\right).$	$A_k(x)=\prod_{s\neq k}\left(\frac{\alpha_s-x}{\alpha_s-\alpha_k}\right),\quad B_k(x)=\prod_{s\neq k}\left(\frac{\beta_s-x}{\beta_s-\beta_k}\right).$	$A_k(x)=\prod_{s\neq k}\left(\frac{\alpha_s-x}{\alpha_s-\alpha_k}\right),\quad B_k(x)=\prod_{s\neq k}\left(\frac{\beta_s-x}{\beta_s-\beta_k}\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0075	044	■ 1.000 ● N +0	$t_{\boldsymbol{i},\boldsymbol{j}}^n=\frac{f(\boldsymbol{i})\,f(\boldsymbol{j})}{i+j-1},$	$t_{i,j}^n=\frac{f(i)f(j)}{i+j-1},$	$t_{i,j}^n=\frac{f(i)f(j)}{i+j-1},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0076	044	■ 1.000 ● N +0	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right)f(i),\quad i=1,2,\ldots,\quad f(1)=-n.$	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right)f(i),\quad i=1,2,\ldots,\quad f(1)=-n.$	$f(i+1)=\left(\frac{i^2-n^2}{i^2}\right)f(i),\quad i=1,2,\ldots,\quad f(1)=-n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Z-Library__EQ0077 (2.1)	046	■ 1.000 ● N +0	$x_{\boldsymbol{i}}=a+\frac{i-1}{n}(b-a),\quad i=1,2,\ldots,n+1$	$x_i=a+\frac{i-1}{n}(b-a),\quad i=1,2,\ldots,n+1$	$x_i=a+\frac{i-1}{n}(b-a),\quad i=1,2,\ldots,n+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0079 (2.3)	047	■ 1.000 ● N +0	$g(x)=y_{\boldsymbol{1}}+\frac{y_{\boldsymbol{2}}-y_{\boldsymbol{1}}}{x_{\boldsymbol{2}}-x_{\boldsymbol{1}}}\left(x-x_{\boldsymbol{1}}\right),$	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1),$	$g(x)=y_1+\frac{y_2-y_1}{x_2-x_1}(x-x_1),$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0080	047	■ 1.000 ● N -2	$f(x)=\arctan\left(10\,x\right)+\pi/2,\quad x\in[-1,1].$	$f(x)=\arctan(10x)+\pi/2,\quad x\in[-1,1].$	$f(x)=\arctan(10x)+\pi/2,\quad x\in[-1,1].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0082 (2.5)	049	■ 1.000 ● N +0	$g(x) := \begin{cases} p_1(x) = -\frac{1}{2}x + \frac{3}{2}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = 1 + 4(x-1) + \frac{9}{2}(x-1)^2 + \frac{13}{2}(x-1)^3, & \text{if } 1 \leq x \leq 2, \end{cases}$	$g(x) := \begin{cases} p_1(x) = -\frac{1}{2}x + \frac{3}{2}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = 1 + 4(x-1) + \frac{9}{2}(x-1)^2 + \frac{13}{2}(x-1)^3, & \text{if } 1 \leq x \leq 2, \end{cases}$	$g(x) := \begin{cases} p_1(x) = -\frac{1}{2}x + \frac{3}{2}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = 1 + 4(x-1) + \frac{9}{2}(x-1)^2 + \frac{13}{2}(x-1)^3, & \text{if } 1 \leq x \leq 2, \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0083	049	■ 1.000 ● N +0	$p_{i-1}^{(j)}(x_i) = p_i^{(j)}(x_i), \quad j = 0, 1, 2, \quad i = 2, \dots, n.$	$p_{i-1}^{(j)}(x_i) = p_i^{(j)}(x_i), \quad j = 0, 1, 2, \quad i = 2, \dots, n.$	$p_{i-1}^{(j)}(x_i) = p_i^{(j)}(x_i), \quad j = 0, 1, 2, \quad i = 2, \dots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0085 (2.7)	050	■ 1.000 ● N +0	$p_i(x) = c_{i,1} + c_{i,2}(x - x_i) + c_{i,3}(x - x_i)^2 + c_{i,4}(x - x_i)^3 \quad i = 1, \dots, n,$	$p_i(x) = c_{i,1} + c_{i,2}(x - x_i) + c_{i,3}(x - x_i)^2 + c_{i,4}(x - x_i)^3 \quad i = 1, \dots, n,$	$p_i(x) = c_{i,1} + c_{i,2}(x - x_i) + c_{i,3}(x - x_i)^2 + c_{i,4}(x - x_i)^3 \quad i = 1, \dots, n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0086 (2.8)	050	■ 1.000 ● W +2	$c_{i1} = y_i, c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}, c_{i3} = \frac{\mu_i}{2}, c_{i4} = \frac{\mu_{i+1} - \mu_i}{6h}.$	$c_{i1} = y_i, c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}, c_{i3} = \frac{\mu_i}{2}, c_{i4} = \frac{\mu_{i+1} - \mu_i}{6h}.$	$c_{i1} = y_i, c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}, c_{i3} = \frac{\mu_i}{2}, c_{i4} = \frac{\mu_{i+1} - \mu_i}{6h}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0088 (2.10)	051	■ 1.000 ● W +0	$\begin{aligned} & p_{i-1}^{(\prime)}(x_i) \\ & = c_{i-1,2} + 2hc_{i-1,3} + 3h^2c_{i-1,4} \\ & = \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ & = \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ & p_i'(x_i) = c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} p_{i-1}'(x_i) &= c_{i-1,2} + 2hc_{i-1,3} + 3h^2c_{i-1,4} \\ &= \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ &= \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ p_i'(x_i) &= c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$	$\begin{aligned} p_{i-1}'(x_i) &= c_{i-1,2} + 2hc_{i-1,3} + 3h^2c_{i-1,4} \\ &= \frac{y_i - y_{i-1}}{h} - \frac{h}{3}\mu_{i-1} - \frac{h}{6}\mu_i + 2h\frac{\mu_{i-1}}{2} + 3h^2\frac{\mu_i - \mu_{i-1}}{6h} \\ &= \frac{y_i - y_{i-1}}{h} + \frac{h}{6}\mu_{i-1} + \frac{h}{3}\mu_i \\ p_i'(x_i) &= c_{i2} = \frac{y_{i+1} - y_i}{h} - \frac{h}{3}\mu_i - \frac{h}{6}\mu_{i+1}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0090 (2.12)	052	■ 1.000 ● N +0	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i = 1, \dots, n.$	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i = 1, \dots, n.$	$ a_{ii} > \sum_{j \neq i} a_{ij} , \quad i = 1, \dots, n.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0091 (2.13)	052	■ 1.000 ● W +0	$\max_{1 \leq i \leq n} x_i \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right), \text{ where } \sigma_i := a_{ii} - \sum_{j \neq i} a_{ij} .$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0092	052	■ 1.000 ● W +0	$ b_k = \left a_{kk}x_k + \sum_{j \neq k} a_{kj}x_j \right \geq a_{kk} x_k - \sum_{j \neq k} a_{kj} x_j \geq x_k \left(a_{kk} - \sum_{j \neq k} a_{kj} \right),$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0093	052	■ 1.000 ● K +0	$\left[\begin{array}{lll} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{array} \right] \left[\begin{array}{l} \mu_2 \\ \mu_3 \\ \mu_4 \end{array} \right] = \left[\begin{array}{l} 2 \\ -6 \\ 2 \end{array} \right].$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0094 (2.14)	053	■ 1.000 ● W +0	$g(x) := \begin{cases} p_1(x) = \frac{1}{6}x^3, & \text{if } 0 \leq x < 1, \\ p_2(x) = \frac{1}{6} + \frac{1}{2}(x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & \text{if } 1 \leq x < 2, \\ p_3(x) = \frac{2}{3} - (x-2)^2 + \frac{1}{2}(x-2)^3, & \text{if } 2 \leq x < 3, \\ p_4(x) = \frac{1}{6} - \frac{1}{2}(x-3) + \frac{1}{2}(x-3)^2 - \frac{1}{6}(x-3)^3, & \text{if } 3 \leq x \leq 4, \end{cases}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0095 (2.15)	054	■ 1.000 ● W +2	$\left[\begin{array}{cccc} d_1 & c_1 & & \\ a_1 & d_2 & c_2 & \\ & \ddots & \ddots & \ddots \\ & & a_{n-2} & d_{n-1} & c_{n-1} \\ & & & a_{n-1} & d_n \end{array} \right] = \left[\begin{array}{cccc} 1 & & & \\ l_1 & 1 & & \\ & \ddots & \ddots & \\ & & l_{n-1} & 1 \end{array} \right] \left[\begin{array}{cccc} u_1 & c_1 & & \\ & \ddots & \ddots & \\ & & u_{n-1} & c_{n-1} \\ & & & u_n \end{array} \right].$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0100 (2.17)	055	■ 1.000 ● N +0	$ u_k \geq \sigma_k + c_k , \quad \text{where, } \sigma_k := d_k - a_{k-1} - c_k > 0, k = 1, \dots, n,$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00104 (2.20)	056	1.000 ● N +0	$-u^{\{\prime \prime\}}(x)=f(x), \quad x \in [0,1], \quad u(0)=0, \quad u(1)=0,$	$-u''(x)=f(x), \quad x \in [0,1], \quad u(0)=0, u(1)=0,$	$-u''(x)=f(x), \quad x \in [0,1], \quad u(0)=0, \quad u(1)=0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00105	056	1.000 ● W +0	$g^{\{\prime\}}(x)=\lim_{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$	$g'(x)=\lim_{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$	$g'(x)=\lim_{h \rightarrow 0} \frac{g\left(x+\frac{h}{2}\right)-g\left(x-\frac{h}{2}\right)}{h}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00107	057	1.000 ● N +0	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=f(jh), \quad j=1,\ldots,m, \quad v_0=v_{m+1}=0.$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=f(jh), \quad j=1,\ldots,m, \quad v_0=v_{m+1}=0.$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=f(jh), \quad j=1,\ldots,m, \quad v_0=v_{m+1}=0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00109 (2.22)	057	1.000 ● N +0	$\left a_{ii}\right \geq \sum_{j \neq i} \left a_{ij}\right , i=1,\ldots,n.$	$\left a_{ii}\right \geq \sum_{j \neq i} \left a_{ij}\right , i=1,\ldots,n.$	$\left a_{ii}\right \geq \sum_{j \neq i} \left a_{ij}\right , i=1,\ldots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00111 (2.23)	058	1.000 ● C +3	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$	$\left u_{k+1}\right =\left d_{k+1}-l_k c_k\right =\left d_{k+1}-\frac{a_k c_k}{u_k}\right \geq\left d_{k+1}\right -\frac{\left a_k\right \left c_k\right }{\left u_k\right }>\left d_{k+1}\right -\left a_k\right .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00112 (2.24)	059	1.000 ● N +0	$Ry^{\{\prime \prime\}}(x)=-Fy(x), \quad y(0)=y(L)=0,$	$Ry''(x)=-Fy(x), \quad y(0)=y(L)=0,$	$Ry''(x)=-Fy(x), \quad y(0)=y(L)=0,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00113 (2.25)	059	1.000 ● N +0	$u^{\{\prime \prime\}}(t)=-Ku(t), \quad u(0)=u(1)=0, \quad K:=\frac{FL^2}{R}.$	$u''(t)=-Ku(t), \quad u(0)=u(1)=0, \quad K:=\frac{FL^2}{R}.$	$u''(t)=-Ku(t), \quad u(0)=u(1)=0, \quad K:=\frac{FL^2}{R}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00115	059	1.000 ● N +0	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1,\ldots,m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1,\ldots,m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$	$\frac{-v_{j-1}+2v_j-v_{j+1}}{h^2}=Kv_j, \quad j=1,\ldots,m, h=\frac{1}{m+1}, \quad v_0=v_{m+1}=0,$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0116 (2.26)	060	■ 1.000 ● N -1	$\boldsymbol{T} \boldsymbol{v} = \lambda \boldsymbol{v}$, with $\boldsymbol{v} = [v_1, \dots, v_m]^T$,	$T\boldsymbol{v} = \lambda \boldsymbol{v}$, with $\boldsymbol{v} = [v_1, \dots, v_m]^T$,	$T\boldsymbol{v} = \lambda \boldsymbol{v}$, with $\boldsymbol{v} = [v_1, \dots, v_m]^T$,
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0118	060	■ 1.000 ● N +0	$F = \frac{4 \sin^2(\pi h / 2) R}{h^2 L^2}$.	$F = \frac{4 \sin^2(\pi h / 2) R}{h^2 L^2}$.	$F = \frac{4 \sin^2(\pi h / 2) R}{h^2 L^2}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0119 (2.28)	060	■ 1.000 ● N -1	$\boldsymbol{T}_1 := \text{tridiag}(a, d, a)$	$T_1 := \text{tridiag}(a, d, a)$	$T_1 := \text{tridiag}(a, d, a)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0120 (2.29)	060	■ 1.000 ● N +0	$\boldsymbol{S} = \left[\sin \frac{jk\pi}{m+1} \right]_{j,k=1}^m \in \mathbb{R}^{m \times m}$.	$S = \left[\sin \frac{jk\pi}{m+1} \right]_{j,k=1}^m \in \mathbb{R}^{m \times m}$.	$S = \left[\sin \frac{jk\pi}{m+1} \right]_{j,k=1}^m \in \mathbb{R}^{m \times m}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0126	062	■ 1.000 ● W +0	$\sum_{k=0}^m \cos(2kj\pi h) + i \sum_{k=0}^m \sin(2kj\pi h) = \sum_{k=0}^m e^{2ikj\pi h} = \frac{e^{2i(m+1)j\pi h} - 1}{e^{2ij\pi h} - 1} = 0$,	$\sum_{k=0}^m \cos(2kj\pi h) + i \sum_{k=0}^m \sin(2kj\pi h) = \sum_{k=0}^m e^{2ikj\pi h} = \frac{e^{2i(m+1)j\pi h} - 1}{e^{2ij\pi h} - 1} = 0$,	$\sum_{k=0}^m \cos(2kj\pi h) + i \sum_{k=0}^m \sin(2kj\pi h) = \sum_{k=0}^m e^{2ikj\pi h} = \frac{e^{2i(m+1)j\pi h} - 1}{e^{2ij\pi h} - 1} = 0$,
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0130	063	■ 1.000 ● N +1	$\boldsymbol{A} \boldsymbol{B} = \left[\boldsymbol{A} \boldsymbol{b}_{:1}, \boldsymbol{A} \boldsymbol{b}_{:2}, \dots, \boldsymbol{A} \boldsymbol{b}_{:n} \right]$.	$AB = [Ab_{:1}, Ab_{:2}, \dots, Ab_{:n}]$.	$AB = [Ab_{:1}, Ab_{:2}, \dots, Ab_{:n}]$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0131	063	■ 1.000 ● K +0	$\boldsymbol{A} = \boldsymbol{A} \boldsymbol{I} = \boldsymbol{A} [e_1, e_2, \dots, e_p] = [\boldsymbol{A} e_1, \boldsymbol{A} e_2, \dots, \boldsymbol{A} e_p]$	$A = AI = A[e_1, e_2, \dots, e_p] = [Ae_1, Ae_2, \dots, Ae_p]$	$A = AI = A[e_1, e_2, \dots, e_p] = [Ae_1, Ae_2, \dots, Ae_p]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0133	063	■ 1.000 ● N +0	$\boldsymbol{A} \boldsymbol{x} = x_1 \boldsymbol{a}_{:1} + x_2 \boldsymbol{a}_{:2} + \dots + x_p \boldsymbol{a}_{:p}$.	$A\boldsymbol{x} = x_1\boldsymbol{a}_{:1} + x_2\boldsymbol{a}_{:2} + \dots + x_p\boldsymbol{a}_{:p}$.	$A\boldsymbol{x} = x_1\boldsymbol{a}_{:1} + x_2\boldsymbol{a}_{:2} + \dots + x_p\boldsymbol{a}_{:p}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0134	064	■ 1.000 ● K +0	$\boldsymbol{A} [\boldsymbol{B}_1, \boldsymbol{B}_2] = [\boldsymbol{A} \boldsymbol{B}_1, \boldsymbol{A} \boldsymbol{B}_2]$.	$A[B_1, B_2] = [AB_1, AB_2]$.	$A[B_1, B_2] = [AB_1, AB_2]$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0136	064	■ 1.000 ● K +0	$\left[\boldsymbol{A}_{\{1\}}, \boldsymbol{A}_{\{2\}}\right]\left[\begin{array}{l} \boldsymbol{B}_{\{1\}} \\ \boldsymbol{B}_{\{2\}} \end{array}\right]=\left[\boldsymbol{A}_{\{1\}}\boldsymbol{B}_{\{1\}}+\boldsymbol{A}_{\{2\}}\boldsymbol{B}_{\{2\}}\right].$	$[\boldsymbol{A}_1, \boldsymbol{A}_2] \begin{bmatrix} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{bmatrix} = [\boldsymbol{A}_1 \boldsymbol{B}_1 + \boldsymbol{A}_2 \boldsymbol{B}_2].$	$[A_1, A_2] \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = [A_1 B_1 + A_2 B_2].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0140	065	■ 1.000 ● N +0	$\boldsymbol{C}_{\{i\ j\}}=\sum_{k=1}^s \boldsymbol{A}_{\{i\ k\}} \boldsymbol{B}_{\{k\ j\}}, \quad i=1, \ldots, p, \ j=1, \ldots, q$	$\boldsymbol{C}_{ij} = \sum_{k=1}^s \boldsymbol{A}_{ik} \boldsymbol{B}_{kj}, \quad i = 1, \dots, p, j = 1, \dots, q$	$\boldsymbol{C}_{ij} = \sum_{k=1}^s \boldsymbol{A}_{ik} \boldsymbol{B}_{kj}, \quad i = 1, \dots, p, \ j = 1, \dots, q$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0141	065	■ 1.000 ● N +0	$\left[\begin{array}{ccc} \boldsymbol{A}_{\{11\}} & \cdots & \boldsymbol{A}_{\{1s\}} \\ \vdots & & \vdots \\ \boldsymbol{A}_{\{p1\}} & \cdots & \boldsymbol{A}_{\{ps\}} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{B}_{\{11\}} & \cdots & \boldsymbol{B}_{\{1q\}} \\ \vdots & & \vdots \\ \boldsymbol{B}_{\{s1\}} & \cdots & \boldsymbol{B}_{\{sq\}} \end{array}\right]=\left[\begin{array}{ccc} \boldsymbol{C}_{\{11\}} & \cdots & \boldsymbol{C}_{\{1q\}} \\ \vdots & & \vdots \\ \boldsymbol{C}_{\{p1\}} & \cdots & \boldsymbol{C}_{\{pq\}} \end{array}\right].$	$\begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{bmatrix}.$	$\begin{bmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1s} \\ \vdots & & \vdots \\ \boldsymbol{A}_{p1} & \cdots & \boldsymbol{A}_{ps} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{11} & \cdots & \boldsymbol{B}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{B}_{s1} & \cdots & \boldsymbol{B}_{sq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{11} & \cdots & \boldsymbol{C}_{1q} \\ \vdots & & \vdots \\ \boldsymbol{C}_{p1} & \cdots & \boldsymbol{C}_{pq} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0143 (2.33)	065	■ 1.000 ● K +0	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{\{11\}}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{\{22\}}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1} = \begin{bmatrix} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{bmatrix},$	$\boldsymbol{A}^{-1} = \begin{bmatrix} \boldsymbol{A}_{11}^{-1} & \boldsymbol{C} \\ \boldsymbol{0} & \boldsymbol{A}_{22}^{-1} \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0146	066	■ 1.000 ● K +0	$\boldsymbol{B}_{\{12\}}=\boldsymbol{B}_{\{11\}}^{-1} \boldsymbol{A}_{\{12\}} \boldsymbol{A}_{\{22\}}^{-1}.$	$\boldsymbol{B}_{12} = \boldsymbol{C} = -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1}.$	$\boldsymbol{B}_{12} = \boldsymbol{C} = -\boldsymbol{A}_{11}^{-1} \boldsymbol{A}_{12} \boldsymbol{A}_{22}^{-1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0148	066	■ 1.000 ● N +0	$\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{\{k\}} & \boldsymbol{a}_{\{k+1, k+1\}} \\ \boldsymbol{0} & \boldsymbol{a}_{\{k+1, k+1\}} \end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & a_{k+1, k+1} \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{A}_k & \boldsymbol{a}_k \\ \boldsymbol{0} & a_{k+1, k+1} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0149	066	■ 1.000 ● N +0	$\boldsymbol{A}^{-1}=\left[\begin{array}{cc} \boldsymbol{A}_{\{k\}}^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & a_{\{k+1, k+1\}}^{-1} \end{array}\right],$	$\boldsymbol{A}^{-1} = \begin{bmatrix} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & a_{k+1, k+1}^{-1} \end{bmatrix},$	$\boldsymbol{A}^{-1} = \begin{bmatrix} \boldsymbol{A}_k^{-1} & \boldsymbol{c} \\ \boldsymbol{0} & a_{k+1, k+1}^{-1} \end{bmatrix},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00150	067	■ 1.000 ● W +0	$\max_{2 \leq j \leq n} \mu_j \leq \frac{3}{h^2} \max_{2 \leq i \leq n} y_{i+1} - 2y_i + y_{i-1} .$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00151 (2.34)	067	■ 1.000 ● C -8	$\left[\begin{array}{cccc} 2 & 1 & & \\ & 4 & 1 & \\ & & \ddots & \ddots \\ & & & 1 & 4 & 1 \\ & & & & \ddots & \ddots \end{array} \right] \left[\begin{array}{c} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \\ \mu_{n+1} \end{array} \right] = \frac{6}{h^2} \left[\begin{array}{c} y_2 - y_1 - h s_1 \\ \delta^2 y_2 \\ \delta^2 y_3 \\ \vdots \\ \delta^2 y_{n-1} \\ \delta^2 y_n \\ h s_{n+1} - y_{n+1} + y_n \end{array} \right],$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00152	068	■ 1.000 ● N +0	$\int_a^b (g''(x))^2 dx \leq \int_a^b (h''(x))^2 dx$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00153 (2.35)	068	■ 1.000 ● N +0	$\int_a^b g'' e'' = 0, \quad e := h - g.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00154	068	■ 1.000 ● N +0	$\int_a^b g''' e' = \sum_{i=1}^n \int_{x_i}^{x_{i+1}} g''' e' = \sum_{i=1}^n v_i \int_{x_i}^{x_{i+1}} e' = \sum_{i=1}^n v_i (e(x_{i+1}) - e(x_i)) = 0.$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0155	068	■ 1.000 ● W +0	$\begin{aligned} \int_a^b (h'')^2 &= \int_a^b (g'' + e'')^2 \\ &= \int_a^b (g'')^2 + \int_a^b (e'')^2 + 2 \int_a^b g'' e'' \\ &= \int_a^b (g'')^2 + \int_a^b (e'')^2 \geq \int_a^b (g'')^2 \end{aligned}$	$\int_a^b (h'')^2 = \int_a^b (g'' + e'')^2$ $= \int_a^b (g'')^2 + \int_a^b (e'')^2 + 2 \int_a^b g'' e''$ $= \int_a^b (g'')^2 + \int_a^b (e'')^2 \geq \int_a^b (g'')^2$	$\int_a^b (h'')^2 = \int_a^b (g'' + e'')^2$ $= \int_a^b (g'')^2 + \int_a^b (e'')^2 + 2 \int_a^b g'' e''$ $= \int_a^b (g'')^2 + \int_a^b (e'')^2 \geq \int_a^b (g'')^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0157	071	■ 1.000 ● N +0	$\delta^2 f(x) = f''(\eta_2), \quad x - h < \eta_2 < x + h.$	$\delta^2 f(x) = f''(\eta_2), \quad x - h < \eta_2 < x + h.$	$\delta^2 f(x) = f''(\eta_2), \quad x - h < \eta_2 < x + h.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0158	071	■ 1.000 ● N +0	$\delta^2 f(x) = f''(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$	$\delta^2 f(x) = f''(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$	$\delta^2 f(x) = f''(x) + \frac{h^2}{12} f^{(4)}(\eta_4), \quad x - h < \eta_4 < x + h.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0159 (2.36)	071	■ 1.000 ● W +0	$-u''(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$	$-u''(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$	$-u''(x) + r(x)u'(x) + q(x)u(x) = f(x), \text{ for } x \in [a, b],$ $u(a) = g_0, \quad u(b) = g_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0160 (2.37)	071	■ 1.000 ● N +0	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$	$\frac{-v_{j-1} + 2v_j - v_{j+1}}{h^2} + r(x_j) \frac{v_{j+1} - v_{j-1}}{2h} + q(x_j)v_j = f(x_j), \quad j = 1, \dots, m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0162	072	■ 1.000 ● W +0	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$	$b_j = \begin{cases} h^2 f(x_1) - a_1 g_0, & \text{if } j = 1, \\ h^2 f(x_j), & \text{if } 2 \leq j \leq m - 1, \\ h^2 f(x_m) - c_m g_1, & \text{if } j = m. \end{cases}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00163	072	■ 1.000 ● N +0	$F=\frac{4\sin^2(\pi h/2)R}{h^2L^2}=\frac{\pi^2R}{L^2}+O(h^2).$	$F=\frac{4\sin^2(\pi h/2)R}{h^2L^2}=\frac{\pi^2R}{L^2}+O(h^2).$	$F=\frac{4\sin^2(\pi h/2)R}{h^2L^2}=\frac{\pi^2R}{L^2}+O(h^2).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00165 (2.39)	073	■ 1.000 ● N +0	$s_{\{i, j\}}=s_{\{j, i\}}=\frac{1}{m+1}j(m+1-i), \quad 1\leq j\leq i\leq m.$	$s_{i,j}=s_{j,i}=\frac{1}{m+1}j(m+1-i), \quad 1\leq j\leq i\leq m.$	$s_{i,j}=s_{j,i}=\frac{1}{m+1}j(m+1-i), \quad 1\leq j\leq i\leq m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00167	073	■ 1.000 ● N +0	$\boldsymbol{A}\boldsymbol{X}=\boldsymbol{B}$ $\Longleftrightarrow \boldsymbol{x}_{:j}=\boldsymbol{b}_{:j}, j=1,\ldots,p.$	$\boldsymbol{A}\boldsymbol{X}=\boldsymbol{B} \iff \boldsymbol{A}\boldsymbol{x}_{:j}=\boldsymbol{b}_{:j}, j=1,\ldots,p.$	$\boldsymbol{A}\boldsymbol{X}=\boldsymbol{B} \iff \boldsymbol{A}\boldsymbol{x}_{:j}=\boldsymbol{b}_{:j}, j=1,\ldots,p.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00173	076	■ 1.000 ● W +0	$\begin{aligned} &x_3=b_3^{(3)}/a_{33}^{(3)} \\ &x_2=\left(b_2^{(2)}-a_{23}^{(2)}x_3\right)/a_{22}^{(2)} \\ &x_1=\left(b_1^{(1)}-a_{12}^{(1)}x_2-a_{13}^{(1)}x_3\right)/a_{11}^{(1)}. \end{aligned}$	$\begin{aligned} x_3 &= b_3^{(3)}/a_{33}^{(3)} \\ x_2 &= \left(b_2^{(2)}-a_{23}^{(2)}x_3\right)/a_{22}^{(2)} \\ x_1 &= \left(b_1^{(1)}-a_{12}^{(1)}x_2-a_{13}^{(1)}x_3\right)/a_{11}^{(1)}. \end{aligned}$	$\begin{aligned} x_3 &= b_3^{(3)}/a_{33}^{(3)} \\ x_2 &= (b_2^{(2)}-a_{23}^{(2)}x_3)/a_{22}^{(2)} \\ x_1 &= (b_1^{(1)}-a_{12}^{(1)}x_2-a_{13}^{(1)}x_3)/a_{11}^{(1)}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00177 (3.2)	077	■ 1.000 ● N +0	$\text{for } j=k:n$	$\text{for } j=k:n$	$\text{for } j=k:n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00178	077	■ 1.000 ● W +0	$a_{ij}^{(k+1)}=a_{ij}^{(k)}-l_{ik}^{(k)}a_{kj}^{(k)}$	$a_{ij}^{(k+1)}=a_{ij}^{(k)}-l_{ik}^{(k)}a_{kj}^{(k)}$	$a_{ij}^{(k+1)}=a_{ij}^{(k)}-l_{ik}^{(k)}a_{kj}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00179	078	■ 1.000 ● N +0	$\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1:k,1:k)=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1:k,1:k)=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{array}\right]$	$\boldsymbol{A}_{[k]}:=\boldsymbol{A}(1:k,1:k)=\left[\begin{array}{ccc} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00180	078	■ 1.000 ● K +0	$b_{\{i, j\}}=a_{\{r_{\{i\}}, r_{\{j\}}\}}, \quad i, j=1,\ldots,k.$	$b_{i,j}=a_{r_i,r_j}, \quad i,j=1,\ldots,k.$	$b_{i,j}=a_{r_i,r_j}, \quad i,j=1,\ldots,k.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0182	078	■ 1.000 ● K +0	$\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array}\right], \boldsymbol{A}.$	$[1], \left[\begin{array}{cc} 1 & 2 \\ 4 & 5 \end{array}\right], \boldsymbol{A}.$	$[1], \left[\begin{array}{cc} 1 & 2 \\ 4 & 5 \end{array}\right], \boldsymbol{A}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0183 (3.3)	078	■ 1.000 ● W +0	$\operatorname{det}\left(\boldsymbol{A}_{[k]}\right)=a_{11}^{\{1\}} a_{22}^{\{2\}} \cdots a_{kk}^{\{k\}}, \quad k=1, \ldots, n, \\ \quad k=1, \ldots, n.$	$\det \left(\boldsymbol{A}_{[k]}\right)=a_{11}^{(1)} a_{22}^{(2)} \cdots a_{kk}^{(k)}, \quad k=1, \ldots, n.$	$\det(\boldsymbol{A}_{[k]})=a_{11}^{(1)} a_{22}^{(2)} \cdots a_{kk}^{(k)}, \quad k=1, \ldots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0187 (3.6)	079	■ 1.000 ● W +0	$\left(\boldsymbol{L} \boldsymbol{U}\right)_{i j}=\sum_{k=1}^{j-1} l_{i k}^{(k)} a_{k j}^{(k)}+l_{i j} a_{j j}^{(j)}=\sum_{k=1}^{j-1}\left(a_{i j}^{(k)}-a_{i j}^{(k+1)}\right)+a_{i j}^{(j)}=a_{i j}.$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^{j-1} l_{ik}^{(k)} a_{kj}^{(k)}+l_{ij} a_{jj}^{(j)}=\sum_{k=1}^{j-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(j)}=a_{ij}.$	$(\boldsymbol{LU})_{ij}=\sum_{k=1}^{j-1} l_{ik}^{(k)} a_{kj}^{(k)}+l_{ij} a_{jj}^{(j)}=\sum_{k=1}^{j-1}\left(a_{ij}^{(k)}-a_{ij}^{(k+1)}\right)+a_{ij}^{(j)}=a_{ij}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0188	080	■ 1.000 ● W +0	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array}\right]\left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right]=\left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right].$	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array}\right]\left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right]=\left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right].$	$\left[\begin{array}{ccc} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{array}\right]\left[\begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array}\right]=\left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0191	080	■ 1.000 ● W +0	$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$	$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$	$\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & 0 \\ 0 & 0 & 0 & a_{54} & a_{55} \end{array}\right], \quad\left[\begin{array}{ccccc} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & a_{43} & a_{44} & 0 \\ 0 & 0 & a_{53} & a_{54} & a_{55} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0192	081	■ 1.000 ● W +0	$\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \\ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{l} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{l} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{array}\right].$	$\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \\ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{l} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{l} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{array}\right].$	$\left[\begin{array}{cccc} a_{k, k} & 0 & \cdots & 0 \\ a_{k+1, k} & a_{k+1, k+1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ a_{n, k} & & \cdots & a_{n \times n} \end{array}\right]\left[\begin{array}{l} x_k \\ x_{k+1} \\ \vdots \\ x_n \end{array}\right]=\left[\begin{array}{l} b_k \\ b_{k+1} \\ \vdots \\ b_n \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0194	082	■ 1.000 ● N +0	$b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k).$	$b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k).$	$b((k+1): n)=b((k+1): n)-x(k) * A((k+1): n, k).$

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00195 (3.9)	082	■ 1.000 ● C +10	$\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.3 cforwardsolve} \[N_{LU}:= \frac{2}{3} n^3 - \frac{1}{2} n^2 - \frac{1}{6} n \] \end{figure}$	(not rendered)	$N_{LU} := \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00196	083	■ 1.000 ● N +0	$M+D+A+S=\frac{2}{3} n(n-1)\left(n-\frac{1}{2}\right)+\frac{1}{2} n(n-1)=N_{LU}$	$M + D + A + S = \frac{2}{3}n(n-1)\left(n - \frac{1}{2}\right) + \frac{1}{2}n(n-1) = N_{LU}$	$M + D + A + S = \frac{2}{3}n(n-1)(n - \frac{1}{2}) + \frac{1}{2}n(n-1) = N_{LU}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00197	083	■ 1.000 ● N +0	$M+S=2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 d k \approx 2 \int_0^n(n-k)^2 d k=\frac{2}{3} n^3$	$M + S = 2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 dk \approx 2 \int_0^n(n-k)^2 dk = \frac{2}{3}n^3$	$M + S = 2 \sum_{k=1}^{n-1}(n-k)^2 \approx 2 \int_1^{n-1}(n-k)^2 dk \approx 2 \int_0^n(n-k)^2 dk = \frac{2}{3}n^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00198	083	■ 1.000 ● N +1	$N_{\{S\}}=2 \sum_{k=1}^n(2 k-1) \approx 2 \int_1^n(2 k-1) d k \approx 4 \int_0^n k d k=2 n^2$	$N_S = 2 \sum_{k=1}^n(2k-1) \approx 2 \int_1^n(2k-1) dk \approx 4 \int_0^n k dk = 2n^2$	$N_S = 2 \sum_{k=1}^n(2k-1) \approx 2 \int_1^n(2k-1) dk \approx 4 \int_0^n k dk = 2n^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00199	084	■ 1.000 ● N +0	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})=\left[\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \ldots, \boldsymbol{e}_{i_n}, \ldots, \boldsymbol{e}_{i_n}\right] \in \mathbb{R}^{n \times n},$	$\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p}) = [\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \dots, \boldsymbol{e}_{i_n}] \in \mathbb{R}^{n \times n},$	$\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p}) = [\boldsymbol{e}_{i_1}, \boldsymbol{e}_{i_2}, \dots, \boldsymbol{e}_{i_n}] \in \mathbb{R}^{n \times n},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00200 (3.10)	084	■ 1.000 ● K +0	$\boldsymbol{A} \boldsymbol{P}=\boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A}=\boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A} \boldsymbol{P} = \boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{A}(\boldsymbol{p},:).$	$\boldsymbol{A} \boldsymbol{P} = \boldsymbol{A}(:, \boldsymbol{p}), \quad \boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{A}(\boldsymbol{p},:).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00206	086	■ 1.000 ● N -1	$a_{p_i, k} a_{p_k, j}^{(k)}=a_{p_i, j}^{(k)}-a_{p_i, j}^{(k+1)} \text { for } k<\min (i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)}=a_{p_i, j}^{(j)} \text { for } i>j .$	$a_{p_i, k}^{(k)} a_{p_k, j}^{(k)} = a_{p_i, j}^{(k)} - a_{p_i, j}^{(k+1)} \text { for } k < \min(i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)} = a_{p_i, j}^{(j)} \text { for } i > j.$	$a_{p_i, k}^{(k)} a_{p_k, j}^{(k)} = a_{p_i, j}^{(k)} - a_{p_i, j}^{(k+1)} \text { for } k < \min(i, j), \text { and } a_{p_i, j}^{(k)} a_{p_j, j}^{(j)} = a_{p_i, j}^{(j)} \text { for } i > j.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00209	087	■ 1.000 ● N +0	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$	$\left a_{r_k, k}^{(k)}\right :=\max \left\{\left a_{i, k}^{(k)}\right : k \leq i \leq n\right\}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0211	087	■ 1.000 ● N +0	$\begin{aligned} 10^{-4} x_1 + 2 x_2 &= 4 \\ \times 10^4 \end{aligned} x_2 = 3 - 4 \times 10^4$	$10^{-4} x_1 + 2 x_2 = 4$ $(1 - 2 \times 10^4) x_2 = 3 - 4 \times 10^4$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0212	087	■ 1.000 ● N +1	$x_2 = \frac{-39997}{-19999} \approx 2, \quad \text{quad } x_1 = \frac{4 - 2 x_2}{10^{-4}} = \frac{20000}{19999} \approx 1.$	$x_2 = \frac{-39997}{-19999} \approx 2, \quad x_1 = \frac{4 - 2 x_2}{10^{-4}} = \frac{20000}{19999} \approx 1.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0214	087	■ 1.000 ● K +0	$\begin{aligned} x_1 + x_2 &= 3 \\ 10^{-4} x_1 + 2 x_2 &= 4 \end{aligned}$	$x_1 + x_2 = 3$ $10^{-4} x_1 + 2 x_2 = 4$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0215	087	■ 1.000 ● N +0	$\begin{aligned} x_1 + x_2 &= 3 \\ \left(2 - 10^{-4} \right) x_2 &= 4 - 3 \times 10^{-4} \end{aligned}$	$x_1 + x_2 = 3$ $(2 - 10^{-4}) x_2 = 4 - 3 \times 10^{-4}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0217	088	■ 1.000 ● W -1	$\frac{\left a_{r_k, k} \right }{s_k} := \max \left\{ \frac{\left a_{i, k} \right }{s_k} : k \leq i \leq n \right\}, \quad s_k := \max_{1 \leq j \leq n} a_{kj} .$	$\frac{ a_{r_k, k}^{(k)} }{s_k} := \max \left\{ \frac{ a_{i, k}^{(k)} }{s_k} : k \leq i \leq n \right\}, \quad s_k := \max_{1 \leq j \leq n} a_{kj} .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0218	088	■ 1.000 ● W -1	$a_{r_k, s_k}^{(k)} := \max \left\{ \left a_{i, j}^{(k)} \right : k \leq i, j \leq n \right\}$	$a_{r_k, s_k}^{(k)} := \max \{ a_{i, j}^{(k)} : k \leq i, j \leq n \}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0219	088	■ 1.000 ● W +0	$\boldsymbol{L} = \begin{bmatrix} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{bmatrix}.$	$\boldsymbol{L} = \begin{bmatrix} l_{1,1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n,1} & \cdots & l_{n,n} \end{bmatrix}, \quad \boldsymbol{U} = \begin{bmatrix} u_{1,1} & \cdots & u_{1,n} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & u_{n,n} \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0221 (3.15)	089	■ 1.000 ● N +0	$u_1 = a, \quad \text{quad } u_2 = b, \quad \text{quad } a l_1 = c, \quad \text{quad } b l_1 + u_3 = d.$	$u_1 = a, \quad u_2 = b, \quad a l_1 = c, \quad b l_1 + u_3 = d.$	

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0222	089	■1.000 ●N +0	$\backslash\boldsymbol{A}_{\{1\}}:=\left[\begin{array}{cc} 2 & -1 \\ \end{array}\right], \quad \backslash\boldsymbol{A}_{\{2\}}:=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \backslash\boldsymbol{A}_{\{3\}}:=\left[\begin{array}{lll} 0 & 1 & 1 \end{array}\right] \quad \backslash\boldsymbol{A}_{\{4\}}:=\left[\begin{array}{lll} 1 & 1 & 1 \end{array}\right]$.	$A_1 := \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad A_2 := \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_3 := \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \quad A_4 := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$	$A_1 := \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad A_2 := \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_3 := \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \quad A_4 := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0223	089	■1.000 ●K +0	$\backslash\boldsymbol{A}_{\{[k]\}}:=\left[\begin{array}{ccc} a_{11} & & \\ \cdots & & \\ a_{k1} & \cdots & a_{kk} \end{array}\right]$	$A_{[k]} := \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$	$A_{[k]} := \begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0225 (3.17)	090	■1.000 ●W +0	$\backslash\boldsymbol{A}=\left[\begin{array}{cc} \boldsymbol{A}_{[n-1]} & \boldsymbol{c}_n \\ \boldsymbol{r}_n^T & a_{nn} \end{array}\right]=\left[\begin{array}{cc} L_{n-1} & \mathbf{0} \\ l_n^T & 1 \end{array}\right]\left[\begin{array}{cc} U_{n-1} & u_n \\ 0 & u_{nn} \end{array}\right]=\left[\begin{array}{cc} L_{n-1}U_{n-1} & L_{n-1}u_n \\ l_n^TU_{n-1} & l_n^Tu_n + u_{nn} \end{array}\right],$	$A = \begin{bmatrix} A_{[n-1]} & c_n \\ r_n^T & a_{nn} \end{bmatrix} = \begin{bmatrix} L_{n-1} & \mathbf{0} \\ l_n^T & 1 \end{bmatrix} \begin{bmatrix} U_{n-1} & u_n \\ 0 & u_{nn} \end{bmatrix} = \begin{bmatrix} L_{n-1}U_{n-1} & L_{n-1}u_n \\ l_n^TU_{n-1} & l_n^Tu_n + u_{nn} \end{bmatrix},$	$A = \begin{bmatrix} A_{[n-1]} & c_n \\ r_n^T & a_{nn} \end{bmatrix} = \begin{bmatrix} L_{n-1} & \mathbf{0} \\ l_n^T & 1 \end{bmatrix} \begin{bmatrix} U_{n-1} & u_n \\ 0 & u_{nn} \end{bmatrix} = \begin{bmatrix} L_{n-1}U_{n-1} & L_{n-1}u_n \\ l_n^TU_{n-1} & l_n^Tu_n + u_{nn} \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0226 (3.18)	090	■1.000 ●W +0	$\backslash\boldsymbol{U}_{[n-1]^{\mathrm{T}}} \backslash\boldsymbol{l}_{\{n\}}= \backslash\boldsymbol{r}_{\{n\}}, \quad \backslash\boldsymbol{L}_{[n-1]} \backslash\boldsymbol{u}_{\{n\}}= \backslash\boldsymbol{c}_{\{n\}}, \quad \backslash u_{\{n\}}=a_{nn}-\backslash\boldsymbol{l}_{\{n\}}^{\mathrm{T}} \backslash\boldsymbol{u}_{\{n\}}.$	$U_{n-1}^T l_n = r_n, \quad L_{n-1} u_n = c_n, \quad u_{nn} = a_{nn} - l_n^T u_n.$	$U_{n-1}^T l_n = r_n, \quad L_{n-1} u_n = c_n, \quad u_{nn} = a_{nn} - l_n^T u_n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0227	091	■1.000 ●W +0	$\backslash\boldsymbol{U}_{[k-1]^{\mathrm{T}}} \backslash\boldsymbol{l}_{\{k\}}= \backslash\boldsymbol{r}_{\{k\}}, \quad \backslash\left[\begin{array}{cc} L_{k-1} & \mathbf{0} \\ l_k^T & 1 \end{array}\right] \backslash\left[\begin{array}{c} u_k \\ u_{kk} \end{array}\right]= \left[\begin{array}{c} c_k \\ a_{kk} \end{array}\right].$	$U_{k-1}^T l_k = r_k, \quad \begin{bmatrix} L_{k-1} & \mathbf{0} \\ l_k^T & 1 \end{bmatrix} \begin{bmatrix} u_k \\ u_{kk} \end{bmatrix} = \begin{bmatrix} c_k \\ a_{kk} \end{bmatrix}.$	$U_{k-1}^T l_k = r_k, \quad \begin{bmatrix} L_{k-1} & \mathbf{0} \\ l_k^T & 1 \end{bmatrix} \begin{bmatrix} u_k \\ u_{kk} \end{bmatrix} = \begin{bmatrix} c_k \\ a_{kk} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0230	092	■1.000 ●N +0	$\backslash\boldsymbol{A}_{\{\{k\}\}}:=\left[\begin{array}{ccc} \boldsymbol{A}_{\{11\}} & \cdots & \boldsymbol{A}_{\{1k\}} \\ \vdots & & \vdots \\ \boldsymbol{A}_{\{k1\}} & \cdots & \boldsymbol{A}_{\{kk\}} \end{array}\right]$	$A_{\{k\}} := \begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix}$	$A_{\{k\}} := \begin{bmatrix} A_{11} & \cdots & A_{1k} \\ \vdots & & \vdots \\ A_{k1} & \cdots & A_{kk} \end{bmatrix}$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0232 (3.22)	093	■ 1.000 ● C +54	\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 3.5 cbacksolve} \[\boldsymbol{A}((k+1):n,(k+1):n) \boldsymbol{b}_{_k}((k+1):n)=-\boldsymbol{A}((k+1):n,k) \boldsymbol{b}_{_k}(k), \quad k=1, \ldots, n-1 . \] \end{figure}	(not rendered)	$A((k+1):n,(k+1):n)\boldsymbol{b}_k((k+1):n)=-A((k+1):n,k)b_k(k), \quad k=1,\dots,n-1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0233 (3.23)	094	■ 1.000 ● K +0	\boldsymbol{A}(1:k,1:k) \boldsymbol{b}_{_k}(1:k)=\boldsymbol{I}(1:k,k), \quad k=n,n-1,\ldots,1,	$\boldsymbol{A}(1:k,1:k)\boldsymbol{b}_k(1:k)=\boldsymbol{I}(1:k,k), \quad k=n,n-1,\dots,1,$	$\boldsymbol{A}(1:k,1:k)\boldsymbol{b}_k(1:k)=\boldsymbol{I}(1:k,k), \quad k=n,n-1,\dots,1,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0235 (3.28)	095	■ 1.000 ● K +0	\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v} \boldsymbol{e}_{_1}^T,	$\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v}e_1^T,$	$\boldsymbol{C}:=\boldsymbol{U}+\boldsymbol{v}e_1^T,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0236 (3.29)	095	■ 1.000 ● N +0	\boldsymbol{P}:=\boldsymbol{I}_{_1,2} \boldsymbol{I}_{_2,3} \cdots \boldsymbol{I}_{_n-1,n}, \cdots \boldsymbol{I}_{_n-1,n},	$\boldsymbol{P}:=\boldsymbol{I}_{1,2}\boldsymbol{I}_{2,3}\cdots\boldsymbol{I}_{n-1,n},$	$\boldsymbol{P}:=\boldsymbol{I}_{1,2}\boldsymbol{I}_{2,3}\cdots\boldsymbol{I}_{n-1,n},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0237 (3.30)	096	■ 1.000 ● K +0	\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w} \boldsymbol{e}_{_1}^T.	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w}e_1^T.$	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{w}e_1^T.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0238	096	■ 1.000 ● N +0	\operatorname{det}\left(\boldsymbol{A}_{_k}\right)=u_{_11} u_{_22} \cdots u_{_k k} \text { for } k=1, \ldots, n .	$\det\left(\boldsymbol{A}_{[k]}\right)=u_{11}u_{22}\cdots u_{kk} \text { for } k=1,\dots,n.$	$\det(\boldsymbol{A}_{[k]})=u_{11}u_{22}\cdots u_{kk} \text { for } k=1,\dots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0239 (3.31)	097	■ 1.000 ● N +0	u_{_11}=a_{_11}, \quad u_{_k k}=\frac{\operatorname{det}\left(\boldsymbol{A}_{_k}\right)}{\operatorname{det}\left(\boldsymbol{A}_{_k-1}\right)}, \text { for } k=2, \ldots, n .	$u_{11}=a_{11}, \quad u_{kk}=\frac{\det\left(\boldsymbol{A}_{[k]}\right)}{\det\left(\boldsymbol{A}_{[k-1]}\right)}, \text { for } k=2,\dots,n.$	$u_{11}=a_{11}, \quad u_{kk}=\frac{\det(\boldsymbol{A}_{[k]})}{\det(\boldsymbol{A}_{[k-1]})}, \text { for } k=2,\dots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0242	098	■ 1.000 ● N +0	\boldsymbol{B}:=\left[\boldsymbol{a}_{_1}, \ldots, \boldsymbol{a}_{_p-1}, \boldsymbol{a}_{_p+1}, \ldots, \boldsymbol{a}_n, \boldsymbol{b}\right] \in \mathbb{R}^{n \times n} .	$\boldsymbol{B}:=[\boldsymbol{a}_1,\dots,\boldsymbol{a}_{p-1},\boldsymbol{a}_{p+1},\dots,\boldsymbol{a}_n,\boldsymbol{b}]\in\mathbb{R}^{n\times n}.$	$\boldsymbol{B}:=[\boldsymbol{a}_1,\dots,\boldsymbol{a}_{p-1},\boldsymbol{a}_{p+1},\dots,\boldsymbol{a}_n,\boldsymbol{b}]\in\mathbb{R}^{n\times n}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0243	098	■ 1.000 ● N +0	A:=\left[\begin{array}{rr} -3 & -2 \\ 4 & 2 \end{array}\right] .	$A:=\left[\begin{array}{cc} -3 & -2 \\ 4 & 2 \end{array}\right].$	$A:=\left[\begin{array}{cc} -3 & -2 \\ 4 & 2 \end{array}\right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00244	098	■ 1.000 ● N +0	$\boldsymbol{P}:=\left[\boldsymbol{e}_{\{n\}},\boldsymbol{e}_{\{n-1\}},\ldots,\boldsymbol{e}_{\{1\}}\right]$.	$\boldsymbol{P} := [e_n, e_{n-1}, \dots, e_1].$	$\boldsymbol{P} := [e_n, e_{n-1}, \dots, e_1].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00246 (4.1)	101	■ 1.000 ● N +0	$d_{\{1\}}=a$. $\quad a_{\{1\}}=b$, $\quad d_{\{2\}}=d-a\left l_{\{1\}}\right ^2$.	$d_1 = a.$ $\quad al_1 = b,$ $\quad d_2 = d - a\left l_1\right ^2.$	$d_1 = a.$ $\quad al_1 = b,$ $\quad d_2 = d - a\left l_1\right ^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00251	102	■ 1.000 ● N +0	$f(\boldsymbol{x})=\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\sum_{i=1}^n\sum_{j=1}^na_{ij}\bar{x}_ix_j$	$f(\boldsymbol{x}) = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j$	$f(\boldsymbol{x}) = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \bar{x}_i x_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00255 (4.5)	104	■ 1.000 ● N +0	$\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r})=\boldsymbol{X}^*\boldsymbol{A}\boldsymbol{X},\quad \boldsymbol{X}:=[\boldsymbol{e}_{r_1},\ldots,\boldsymbol{e}_{r_k}]\in\mathbb{C}^{n\times k}$.	$\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r}) = \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X} := [e_{r_1}, \dots, e_{r_k}] \in \mathbb{C}^{n \times k}.$	$\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r}) = \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X}, \quad \boldsymbol{X} := [e_{r_1}, \dots, e_{r_k}] \in \mathbb{C}^{n \times k}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00256 (4.6)	104	■ 1.000 ● W +1	$\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y}=\boldsymbol{y}^*\boldsymbol{X}^*\boldsymbol{A}\boldsymbol{X}\boldsymbol{y}=\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}\geq 0,$ $\quad \boldsymbol{y}\in\mathbb{C}^k,$ $\quad \boldsymbol{x}:=\boldsymbol{X}\boldsymbol{y}.$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0,$ $\quad \boldsymbol{y} \in \mathbb{C}^k,$ $\quad \boldsymbol{x} := \boldsymbol{X} \boldsymbol{y}.$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{X}^* \boldsymbol{A} \boldsymbol{X} \boldsymbol{y} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0,$ $\quad \boldsymbol{y} \in \mathbb{C}^k,$ $\quad \boldsymbol{x} := \boldsymbol{X} \boldsymbol{y}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00257 (0)	105	■ 1.000 ● W -1	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right]\left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right]=\left[\begin{array}{cc} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{array}\right]\left[\begin{array}{cc} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{array}\right].$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] = \left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array}\right] \left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right] \left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{array}\right] \left[\begin{array}{cc} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{array}\right].$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] = \left[\begin{array}{cc} 1 & 0 \\ -\frac{1}{2} & 1 \end{array}\right] \left[\begin{array}{cc} 2 & 0 \\ 0 & \frac{3}{2} \end{array}\right] \left[\begin{array}{cc} 1 & -\frac{1}{2} \\ 0 & 1 \end{array}\right] = \left[\begin{array}{cc} \sqrt{2} & 0 \\ -1/\sqrt{2} & \sqrt{3/2} \end{array}\right] \left[\begin{array}{cc} \sqrt{2} & -1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00258 (4.7)	106	■ 1.000 ● C +54	$\begin{figure} \captionsetup{labelformat=empty} \caption{Listing 4.2 bandcholesky} \[\theta<(\alpha\boldsymbol{e}_{\{i\}}+\beta\boldsymbol{e}_{\{j\}})^*\boldsymbol{A}(\alpha\boldsymbol{e}_{\{i\}}+\beta\boldsymbol{e}_{\{j\}})= \alpha ^2a_{ii}+ \beta ^2a_{jj}+2\operatorname{Re}(\overline{\alpha}\beta a_{ij}). \]$	<i>(not rendered)</i>	$0 < (\leq)(\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j)^* \boldsymbol{A} (\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j) = \alpha ^2 a_{ii} + \beta ^2 a_{jj} + 2 \operatorname{Re}(\overline{\alpha} \beta a_{ij}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00259	106	■ 1.000 ● N +0	$\theta<\left a_{\{i\}}\right ^2a_{\{i\}}+a_{\{i\}}^2a_{\{j\}}-2\left a_{\{i\}}\right \left a_{\{j\}}\right \left(a_{\{i\}}a_{\{j\}}-\left a_{\{i\}}\right \left a_{\{j\}}\right \right).$	$0 < \left a_{ij}\right ^2 a_{ii} + a_{ii}^2 a_{jj} - 2 \left a_{ij}\right ^2 a_{ii} = a_{ii} \left(a_{ii} a_{jj} - \left a_{ij}\right ^2\right).$	$0 < \left a_{ij}\right ^2 a_{ii} + a_{ii}^2 a_{jj} - 2 \left a_{ij}\right ^2 a_{ii} = a_{ii} (a_{ii} a_{jj} - \left a_{ij}\right ^2).$

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00260	106	■ 1.000 ● N +0	$\left a_{ij}\right =\left b_{ij}\right <\sqrt{b_{ii} b_{jj}}$ $b_{jj}=\sqrt{\left(a_{ii}+\varepsilon\right)\left(a_{jj}+\varepsilon\right)}, \quad i \neq j .$	$ a_{ij} = b_{ij} <\sqrt{b_{ii} b_{jj}}=\sqrt{\left(a_{ii}+\varepsilon\right)\left(a_{jj}+\varepsilon\right)}, \quad i \neq j .$	$ a_{ij} = b_{ij} <\sqrt{b_{ii} b_{jj}}=\sqrt{\left(a_{ii}+\varepsilon\right)\left(a_{jj}+\varepsilon\right)}, \quad i \neq j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00261	107	■ 1.000 ● N +0	$\boldsymbol{A}_{\{1\}}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_{\{2\}}=\left[\begin{array}{ll} 1 & 2 \\ 2 & 2 \end{array}\right],$ $\boldsymbol{A}_{\{3\}}=\left[\begin{array}{rr} -2 & 1 \\ 1 & 2 \end{array}\right] .$	$\boldsymbol{A}_1=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{cc} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{cc} -2 & 1 \\ 1 & 2 \end{array}\right] .$	$\boldsymbol{A}_1=\left[\begin{array}{cc} 0 & 1 \\ 1 & 1 \end{array}\right], \quad \boldsymbol{A}_2=\left[\begin{array}{cc} 1 & 2 \\ 2 & 2 \end{array}\right], \quad \boldsymbol{A}_3=\left[\begin{array}{cc} -2 & 1 \\ 1 & 2 \end{array}\right] .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00266 (4.10)	109	■ 1.000 ● K +0	$\boldsymbol{L}^*:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array}\right], \quad \beta:=\sqrt{\alpha},$ $\boldsymbol{L}^*:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array}\right], \quad \beta:=\sqrt{\alpha},$	$\boldsymbol{L}^*:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array}\right], \quad \beta:=\sqrt{\alpha},$	$\boldsymbol{L}^*:=\left[\begin{array}{cc} \beta & \boldsymbol{v}^* / \beta \\ \mathbf{0} & \boldsymbol{L}_1^* \end{array}\right], \quad \beta:=\sqrt{\alpha},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00271	113	■ 1.000 ● N +0	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}=2 x_1^2+(2-a) x_1 x_2+a x_2 x_1+x_2^2=x_1^2+\left(x_1+x_2\right)^2>0 .$ $x_{\{1\}}^2+(2-a) x_{\{1\}} x_{\{2\}}+a x_{\{2\}} x_{\{1\}}+x_{\{2\}}^2=x_{\{1\}}^2+\left(x_{\{1\}}+x_{\{2\}}\right)^2>0 .$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}=2 x_1^2+(2-a) x_1 x_2+a x_2 x_1+x_2^2=x_1^2+\left(x_1+x_2\right)^2>0 .$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}=2 x_1^2+(2-a) x_1 x_2+a x_2 x_1+x_2^2=x_1^2+\left(x_1+x_2\right)^2>0 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00273	114	■ 1.000 ● K +0	$\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+\boldsymbol{v} \boldsymbol{v}^T\right) \boldsymbol{L}^T,$ $\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+\boldsymbol{v} \boldsymbol{v}^T\right) \boldsymbol{L}^T,$	$\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+\boldsymbol{v} \boldsymbol{v}^T\right) \boldsymbol{L}^T,$	$\boldsymbol{A}=\boldsymbol{L}\left(\boldsymbol{I}+\boldsymbol{v} \boldsymbol{v}^T\right) \boldsymbol{L}^T,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00275	114	■ 1.000 ● N +0	$\boldsymbol{L}_{\{+\}}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^T & \lambda \end{array}\right],$	$\boldsymbol{L}_{+}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^T & \lambda \end{array}\right],$	$\boldsymbol{L}_{+}=\left[\begin{array}{cc} \boldsymbol{L} & 0 \\ \boldsymbol{y}^T & \lambda \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00276	117	■ 1.000 ● N +0	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\overline{\langle \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{\langle \boldsymbol{a}, \boldsymbol{y}\rangle}+\overline{\langle \boldsymbol{b}, \boldsymbol{z}\rangle}=\bar{a}\langle \boldsymbol{y}, \boldsymbol{x}\rangle+\bar{b}\langle \boldsymbol{z}, \boldsymbol{x}\rangle$ $\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\overline{\langle \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{\langle \boldsymbol{a}, \boldsymbol{y}\rangle}+\overline{\langle \boldsymbol{b}, \boldsymbol{z}\rangle}=\bar{a}\langle \boldsymbol{y}, \boldsymbol{x}\rangle+\bar{b}\langle \boldsymbol{z}, \boldsymbol{x}\rangle$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\overline{\langle \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{\langle \boldsymbol{a}, \boldsymbol{y}\rangle}+\overline{\langle \boldsymbol{b}, \boldsymbol{z}\rangle}=\bar{a}\langle \boldsymbol{y}, \boldsymbol{x}\rangle+\bar{b}\langle \boldsymbol{z}, \boldsymbol{x}\rangle$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\overline{\langle \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}, \boldsymbol{x}\rangle}=\overline{\langle \boldsymbol{a}, \boldsymbol{y}\rangle}+\overline{\langle \boldsymbol{b}, \boldsymbol{z}\rangle}=\bar{a}\langle \boldsymbol{y}, \boldsymbol{x}\rangle+\bar{b}\langle \boldsymbol{z}, \boldsymbol{x}\rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00277 (5.1)	117	■ 1.000 ● N +1	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\bar{a}\langle \boldsymbol{x}, \boldsymbol{y}\rangle+\bar{b}\langle \boldsymbol{x}, \boldsymbol{z}\rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}\rangle= a ^2\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$ $\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\bar{a}\langle \boldsymbol{x}, \boldsymbol{y}\rangle+\bar{b}\langle \boldsymbol{x}, \boldsymbol{z}\rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}\rangle= a ^2\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\bar{a}\langle \boldsymbol{x}, \boldsymbol{y}\rangle+\bar{b}\langle \boldsymbol{x}, \boldsymbol{z}\rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}\rangle= a ^2\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$	$\langle \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}+\boldsymbol{b} \boldsymbol{z}\rangle=\bar{a}\langle \boldsymbol{x}, \boldsymbol{y}\rangle+\bar{b}\langle \boldsymbol{x}, \boldsymbol{z}\rangle, \quad \langle \boldsymbol{a} \boldsymbol{x}, \boldsymbol{a} \boldsymbol{y}\rangle= a ^2\langle \boldsymbol{x}, \boldsymbol{y}\rangle .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0280 (5.2)	117	■1.000 ●N +0	$\backslash\!\!\!\backslash\!\!\!\cdot\!\!\!\backslash\!\!\!\!:\! \mathcal{V} \rightarrow \mathbb{R}, \quad \mathbf{x} \longmapsto \ \mathbf{x}\ := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \mathbf{x} \longmapsto \ \mathbf{x}\ := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}, \quad \mathbf{x} \longmapsto \ \mathbf{x}\ := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0281 (5.3)	118	■1.000 ●N +0	$ \langle \mathbf{x}, \mathbf{y} \rangle \leq \ \mathbf{x}\ \ \mathbf{y}\ ,$	$ \langle \mathbf{x}, \mathbf{y} \rangle \leq \ \mathbf{x}\ \ \mathbf{y}\ ,$	$ \langle \mathbf{x}, \mathbf{y} \rangle \leq \ \mathbf{x}\ \ \mathbf{y}\ ,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0285 (5.5)	119	■1.000 ●W +0	$\ \mathbf{x} + a\mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + a\langle \mathbf{y}, \mathbf{x} \rangle + \bar{a}\langle \mathbf{x}, \mathbf{y} \rangle + a ^2\ \mathbf{y}\ ^2, \quad a \in \mathbb{C}, \quad \mathbf{x}, \mathbf{y} \in \mathcal{V}.$	$\ \mathbf{x} + a\mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + a\langle \mathbf{y}, \mathbf{x} \rangle + \bar{a}\langle \mathbf{x}, \mathbf{y} \rangle + a ^2\ \mathbf{y}\ ^2, \quad a \in \mathbb{C}, \quad \mathbf{x}, \mathbf{y} \in \mathcal{V}.$	$\ \mathbf{x} + a\mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + a\langle \mathbf{y}, \mathbf{x} \rangle + \bar{a}\langle \mathbf{x}, \mathbf{y} \rangle + a ^2\ \mathbf{y}\ ^2, \quad a \in \mathbb{C}, \quad \mathbf{x}, \mathbf{y} \in \mathcal{V}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0287 (5.6)	119	■1.000 ●N +0	$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\ \mathbf{x}\ \ \mathbf{y}\ }.$	$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\ \mathbf{x}\ \ \mathbf{y}\ }.$	$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\ \mathbf{x}\ \ \mathbf{y}\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0288 (5.7)	119	■1.000 ●N +0	$\ \mathbf{x} + \mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + \ \mathbf{y}\ ^2, \quad \text{if } \mathbf{x} \perp \mathbf{y}.$	$\ \mathbf{x} + \mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + \ \mathbf{y}\ ^2, \quad \text{if } \mathbf{x} \perp \mathbf{y}.$	$\ \mathbf{x} + \mathbf{y}\ ^2 = \ \mathbf{x}\ ^2 + \ \mathbf{y}\ ^2, \quad \text{if } \mathbf{x} \perp \mathbf{y}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0290	120	■1.000 ●N +0	$\left\{ \mathbf{u}_1, \dots, \mathbf{u}_k \right\} := \left\{ \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }, \dots, \frac{\mathbf{v}_k}{\ \mathbf{v}_k\ } \right\}$	$\{\mathbf{u}_1, \dots, \mathbf{u}_k\} := \left\{ \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }, \dots, \frac{\mathbf{v}_k}{\ \mathbf{v}_k\ } \right\}$	$\{\mathbf{u}_1, \dots, \mathbf{u}_k\} := \left\{ \frac{\mathbf{v}_1}{\ \mathbf{v}_1\ }, \dots, \frac{\mathbf{v}_k}{\ \mathbf{v}_k\ } \right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0291	120	■1.000 ●C +3	$\left\langle \mathbf{v}_j, \mathbf{v}_l \right\rangle = \left\langle \mathbf{s}_j, \mathbf{v}_l \right\rangle - \sum_{i=1}^{j-1} \frac{\langle \mathbf{s}_j, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \langle \mathbf{v}_i, \mathbf{v}_l \rangle = \left\langle \mathbf{s}_j, \mathbf{v}_l \right\rangle - \frac{\langle \mathbf{s}_j, \mathbf{v}_l \rangle}{\langle \mathbf{v}_l, \mathbf{v}_l \rangle} \langle \mathbf{v}_l, \mathbf{v}_l \rangle = 0$	$\langle \mathbf{v}_j, \mathbf{v}_l \rangle = \langle \mathbf{s}_j, \mathbf{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \mathbf{s}_j, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \langle \mathbf{v}_i, \mathbf{v}_l \rangle = \langle \mathbf{s}_j, \mathbf{v}_l \rangle - \frac{\langle \mathbf{s}_j, \mathbf{v}_l \rangle}{\langle \mathbf{v}_l, \mathbf{v}_l \rangle} \langle \mathbf{v}_l, \mathbf{v}_l \rangle = 0$	$\langle \mathbf{v}_j, \mathbf{v}_l \rangle = \langle \mathbf{s}_j, \mathbf{v}_l \rangle - \sum_{i=1}^{j-1} \frac{\langle \mathbf{s}_j, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \langle \mathbf{v}_i, \mathbf{v}_l \rangle = \langle \mathbf{s}_j, \mathbf{v}_l \rangle - \frac{\langle \mathbf{s}_j, \mathbf{v}_l \rangle}{\langle \mathbf{v}_l, \mathbf{v}_l \rangle} \langle \mathbf{v}_l, \mathbf{v}_l \rangle = 0$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00292	121	1.000 ● N +0	$\operatorname{dim}(\operatorname{\mathcal{S}} \operatorname{\mathcal{O}} \operatorname{\mathcal{T}}) = \operatorname{dim}(\operatorname{\mathcal{S}}) + \operatorname{dim}(\operatorname{\mathcal{T}}) .$	$\dim(S \oplus \mathcal{T}) = \dim(S) + \dim(\mathcal{T}).$	$\dim(S \oplus \mathcal{T}) = \dim(S) + \dim(\mathcal{T}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00295	122	1.000 ● N +0	$\left\langle \boldsymbol{s}_0, \boldsymbol{v}_j \right\rangle = \left\langle \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \left\langle \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{v}_j \right\rangle .$	$\left\langle \boldsymbol{s}_0, \boldsymbol{v}_j \right\rangle = \left\langle \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \left\langle \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{v}_j \right\rangle .$	$\left\langle \boldsymbol{s}_0, \boldsymbol{v}_j \right\rangle = \left\langle \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \sum_{i=1}^k \frac{\left\langle \boldsymbol{v}, \boldsymbol{v}_i \right\rangle}{\left\langle \boldsymbol{v}_i, \boldsymbol{v}_i \right\rangle} \left\langle \boldsymbol{v}_i, \boldsymbol{v}_j \right\rangle = \left\langle \boldsymbol{v}, \boldsymbol{v}_j \right\rangle .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00297	123	1.000 ● N +0	$\left\ \boldsymbol{v} - \boldsymbol{s} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 + \left\ \boldsymbol{u} \right\ ^2 > \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 .$	$\left\ \boldsymbol{v} - \boldsymbol{s} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 + \left\ \boldsymbol{u} \right\ ^2 > \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 .$	$\left\ \boldsymbol{v} - \boldsymbol{s} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 + \boldsymbol{u} \right\ ^2 = \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 + \left\ \boldsymbol{u} \right\ ^2 > \left\ \boldsymbol{v} - \boldsymbol{s}_0 \right\ ^2 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00299	123	1.000 ● K +0	$\left\langle \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \right\rangle = \left\langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \right\rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j .$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \left\langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \right\rangle = \left\langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \right\rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j .$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j = \left\langle \boldsymbol{A} \boldsymbol{e}_j, \boldsymbol{e}_i \right\rangle = \left\langle \boldsymbol{e}_j, \overline{\boldsymbol{A} \boldsymbol{e}_i} \right\rangle = \boldsymbol{e}_i^T \boldsymbol{A}^T \boldsymbol{e}_j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00302	124	1.000 ● N -1	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^* \boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^* \boldsymbol{u} = 2$	$\boldsymbol{H} := \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*, \text{ where } \boldsymbol{u} \in \mathbb{C}^n \text{ and } \boldsymbol{u}^* \boldsymbol{u} = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00304	124	1.000 ● N +0	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I} .$	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I} .$	$\boldsymbol{H}^* \boldsymbol{H} = \boldsymbol{H}^2 = (\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*)(\boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^* + \boldsymbol{u} (\boldsymbol{u}^* \boldsymbol{u}) \boldsymbol{u}^* = \boldsymbol{I} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00305 (5.12)	124	1.000 ● K +0	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00307	125	1.000 ● K +0	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}, \text{ where } \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}},$	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}, \text{ where } \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}},$	$\boldsymbol{H} = \boldsymbol{I} - \frac{2 \boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}} = \boldsymbol{P} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}, \text{ where } \boldsymbol{P} := \boldsymbol{I} - \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}},$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00315	128	■1.000 ●K +0	$\left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{cccc} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{array}\right], \quad \left[\begin{array}{ccc} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{array}\right].$	$\begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix}, \quad \begin{bmatrix} x & x & x & x \\ 0 & x & x & x \\ 0 & 0 & x & x \\ 0 & 0 & 0 & x \end{bmatrix}, \quad \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{bmatrix}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00316	128	■1.000 ●N +0	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_{n+1}\boldsymbol{H}_n\cdots\boldsymbol{H}_1\boldsymbol{A}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}\end{bmatrix}=\boldsymbol{R},$	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_n\boldsymbol{H}_{n-1}\cdots\boldsymbol{H}_1\boldsymbol{A}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}\end{bmatrix}=\boldsymbol{R},$	$\boldsymbol{A}_{n+1}:=\boldsymbol{H}_n\boldsymbol{H}_{n-1}\cdots\boldsymbol{H}_1\boldsymbol{A}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}\end{bmatrix}=\boldsymbol{R},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00317	128	■1.000 ●N +0	$\boldsymbol{A}_1:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_k\boldsymbol{A}_k, \quad k=1,2,\ldots,n.$	$\boldsymbol{A}_1:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_k\boldsymbol{A}_k, \quad k=1,2,\ldots,n.$	$\boldsymbol{A}_1:=\boldsymbol{A}, \quad \boldsymbol{A}_{k+1}=\boldsymbol{H}_k\boldsymbol{A}_k, \quad k=1,2,\ldots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00319	129	■1.000 ●N +0	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_k\boldsymbol{A}_k=\begin{bmatrix}\boldsymbol{B}_k&\boldsymbol{C}_k\\\boldsymbol{0}&\hat{\boldsymbol{H}}_k\boldsymbol{D}_k\end{bmatrix}=\begin{bmatrix}\boldsymbol{B}_{k+1}&\boldsymbol{C}_{k+1}\\\boldsymbol{0}&\boldsymbol{D}_{k+1}\end{bmatrix},$	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_k\boldsymbol{A}_k=\begin{bmatrix}\boldsymbol{B}_k&\boldsymbol{C}_k\\\boldsymbol{0}&\hat{\boldsymbol{H}}_k\boldsymbol{D}_k\end{bmatrix}=\begin{bmatrix}\boldsymbol{B}_{k+1}&\boldsymbol{C}_{k+1}\\\boldsymbol{0}&\boldsymbol{D}_{k+1}\end{bmatrix},$	$\boldsymbol{A}_{k+1}:=\boldsymbol{H}_k\boldsymbol{A}_k=\begin{bmatrix}\boldsymbol{B}_k&\boldsymbol{C}_k\\\boldsymbol{0}&\hat{\boldsymbol{H}}_k\boldsymbol{D}_k\end{bmatrix}=\begin{bmatrix}\boldsymbol{B}_{k+1}&\boldsymbol{C}_{k+1}\\\boldsymbol{0}&\boldsymbol{D}_{k+1}\end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00322	131	■1.000 ●N +0	$\boldsymbol{R}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}_{m-n,n}\end{bmatrix}$	$\boldsymbol{R}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}_{m-n,n}\end{bmatrix}$	$\boldsymbol{R}=\begin{bmatrix}\boldsymbol{R}_1\\\boldsymbol{0}_{m-n,n}\end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00325 (5.17)	133	■1.000 ●N +0	$\boldsymbol{v}_1=\boldsymbol{a}_1, \quad \boldsymbol{v}_j=\boldsymbol{a}_j-\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i, \quad \text{for } j=2,\ldots,n.$	$\boldsymbol{v}_1=\boldsymbol{a}_1, \quad \boldsymbol{v}_j=\boldsymbol{a}_j-\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i, \quad \text{for } j=2,\ldots,n.$	$\boldsymbol{v}_1=\boldsymbol{a}_1, \quad \boldsymbol{v}_j=\boldsymbol{a}_j-\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i, \quad \text{for } j=2,\ldots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00327	133	■1.000 ●W +0	$\boldsymbol{a}_j=\boldsymbol{v}_j+\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i=r_{jj}\boldsymbol{q}_j+\sum_{i=1}^{j-1}\boldsymbol{q}_ir_{ij}=\boldsymbol{Q}_1\boldsymbol{R}_1\boldsymbol{e}_j, \quad j=1,\ldots,n.$	$\boldsymbol{a}_j=\boldsymbol{v}_j+\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i=r_{jj}\boldsymbol{q}_j+\sum_{i=1}^{j-1}\boldsymbol{q}_ir_{ij}=\boldsymbol{Q}_1\boldsymbol{R}_1\boldsymbol{e}_j, \quad j=1,\ldots,n.$	$\boldsymbol{a}_j=\boldsymbol{v}_j+\sum_{i=1}^{j-1}\frac{\boldsymbol{a}_j^T\boldsymbol{v}_i}{\boldsymbol{v}_i^T\boldsymbol{v}_i}\boldsymbol{v}_i=r_{jj}\boldsymbol{q}_j+\sum_{i=1}^{j-1}\boldsymbol{q}_ir_{ij}=\boldsymbol{Q}_1\boldsymbol{R}_1\boldsymbol{e}_j, \quad j=1,\ldots,n.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0333	135	■1.000 ● N +0	$\left[\begin{array}{cc} p_{i\ i} & p_{i\ j} \\ p_{j\ i} & p_{j\ j} \end{array}\right]=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text{ where } c^2+s^2=1 \ .$	$\left[\begin{array}{cc} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{array}\right]=\left[\begin{array}{cc} c & s \\ -s & c \end{array}\right], \text{ where } c^2+s^2=1.$	$\begin{bmatrix} p_{ii} & p_{ij} \\ p_{ji} & p_{jj} \end{bmatrix}=\begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \text{ where } c^2+s^2=1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0334	135	■1.000 ● C -6	$\boldsymbol{P}_{12}=\left[\begin{array}{cccc} c & s & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{13}=\left[\begin{array}{cccc} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{23}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$	$\boldsymbol{P}_{12}=\left[\begin{array}{cccc} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{13}=\left[\begin{array}{cccc} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \boldsymbol{P}_{23}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{array}\right].$	$\boldsymbol{P}_{12}=\begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{13}=\begin{bmatrix} c & 0 & s & 0 \\ 0 & 1 & 0 & 0 \\ -s & 0 & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{P}_{23}=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s & c & 0 \\ 0 & -s & c & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0337 (5.20)	136	■1.000 ● N +0	$\cos \theta=\frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \ .$	$\cos \theta=\frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$	$\cos \theta=\frac{ \langle \boldsymbol{x}, \boldsymbol{y} \rangle }{\ \boldsymbol{x}\ \ \boldsymbol{y}\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0339	137	■1.000 ● W +0	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right] \ .$	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right].$	$\boldsymbol{H}=\left[\begin{array}{cc} -\cos \phi & \sin \phi \\ \sin \phi & \cos \phi \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0340	137	■1.000 ● N +1	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text{where } \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text{where } \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{y}, \quad \text{where } \boldsymbol{H}:=\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0343	137	■1.000 ● K +0	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right] \ .$	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right].$	$\boldsymbol{A}:=\left[\begin{array}{ccc} 1 & 0 & 1 \\ -2 & -1 & 0 \\ 2 & 2 & 1 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0344 (5.21)	138	■1.000 ● N +0	$ \operatorname{det}(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2 \ .$	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2.$	$ \det(\boldsymbol{A}) \leq \prod_{j=1}^n\left\ \boldsymbol{a}_j\right\ _2.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00346	139	1.000 K +0	$A:=\left[\begin{array}{rr} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{array}\right]$	$A := \begin{bmatrix} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{bmatrix}$	$A := \begin{bmatrix} 2 & 1 \\ 2 & -3 \\ -2 & -1 \\ -2 & 3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00347	140	1.000 K +0	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H}=\boldsymbol{R}$	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H} = \boldsymbol{R}$	$\boldsymbol{G}_3\boldsymbol{G}_2\boldsymbol{G}_1\boldsymbol{H} = \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00349	141	1.000 W +0	$\boldsymbol{P}=\left[\begin{array}{cc} \cos \theta & \sin \theta \\ \sin \theta & -\sin \theta & \cos \theta \end{array}\right]$	$\boldsymbol{P} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$	$\boldsymbol{P} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00351	141	1.000 K +0	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n, n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n,n}\right)$	$\left(a_{1,1}, a_{2,2}, \ldots, a_{n,n}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00352	141	1.000 N +0	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n, n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}\right)$.	$\left(a_{2,1}, a_{3,2}, \ldots, a_{n,n-1}\right)$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00353	142	1.000 N +0	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$	$\boldsymbol{P}_{m-1, m} \boldsymbol{P}_{m-2, m-1} \cdots \boldsymbol{P}_{2,3} \boldsymbol{P}_{1,2} \boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00356	145	1.000 K +0	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\det(\boldsymbol{A}-\lambda \boldsymbol{I})$.	$\pi_{\boldsymbol{A}}(\lambda)=\det(\boldsymbol{A}-\lambda \boldsymbol{I})$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00357	146	1.000 N +0	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\mathbf{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j=\sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j=\mathbf{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j = \sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j = \mathbf{0}$.	$\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0} \Rightarrow \sum_{j=1}^m c_j \boldsymbol{A} \boldsymbol{x}_j = \sum_{j=1}^m c_j \lambda_j \boldsymbol{x}_j = \mathbf{0}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00358	146	1.000 K +0	$\boldsymbol{I}:=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \boldsymbol{J}:=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$\boldsymbol{I} := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.	$\boldsymbol{I} := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0359	146	1.000 N +0	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{x}_j$ for some scalars $c_1, \ldots, c_n$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{x}_j \text{ for some scalars } c_1, \ldots, c_n$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{x}_j \text{ for some scalars } c_1, \ldots, c_n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0360	146	1.000 K +0	$\boldsymbol{x}=\frac{x_1+x_2}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix}+\frac{x_1-x_2}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\boldsymbol{x} = \frac{x_1 + x_2}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x_1 - x_2}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$	$\boldsymbol{x} = \frac{x_1 + x_2}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x_1 - x_2}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0365 (6.3)	148	1.000 N +0	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: (\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}=\mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: (\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}=\mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{A}-\lambda \boldsymbol{I}):=\left\{\boldsymbol{x} \in \mathbb{C}^n: (\boldsymbol{A}-\lambda \boldsymbol{I}) \boldsymbol{x}=\mathbf{0}\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0368 (6.6)	150	1.000 W +0	$\boldsymbol{U}_i=\operatorname{diag}\left(\boldsymbol{J}_{m_i, 1}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_i, g_i}\left(\lambda_i\right)\right)$	$\boldsymbol{U}_i = \operatorname{diag}\left(\boldsymbol{J}_{m_{i,1}}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_{i,g_i}}\left(\lambda_i\right)\right).$	$\boldsymbol{U}_i = \operatorname{diag}\left(\boldsymbol{J}_{m_{i,1}}\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_{i,g_i}}\left(\lambda_i\right)\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0372	152	1.000 N +0	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1.$	$\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{V} \boldsymbol{e}_1=\boldsymbol{V}^* \boldsymbol{A} \boldsymbol{v}_1=\lambda_1 \boldsymbol{V}^* \boldsymbol{v}_1=\lambda_1 \boldsymbol{e}_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0376	153	1.000 N +0	$\boldsymbol{V}=\left[\begin{array}{ccc}-\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\end{array}\right]$	$\boldsymbol{V} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$	$\boldsymbol{V} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0377	153	1.000 N +0	$\boldsymbol{M}=\left[\begin{array}{cc} 2 & -\sqrt{2} \\ -\sqrt{2} & 2\end{array}\right]$	$\boldsymbol{M} = \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}.$	$\boldsymbol{M} = \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0378	153	1.000 N +0	$\boldsymbol{A}^* \boldsymbol{A}=\operatorname{diag}\left(\overline{d_1} d_1, \ldots, \overline{d_n} d_n\right)=\operatorname{diag}\left(\left d_1\right ^2, \ldots,\left d_n\right ^2\right)=\boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A}^* \boldsymbol{A} = \operatorname{diag}\left(\overline{d_1} d_1, \ldots, \overline{d_n} d_n\right) = \operatorname{diag}\left(\left d_1\right ^2, \ldots, \left d_n\right ^2\right) = \boldsymbol{A} \boldsymbol{A}^*,$	$\boldsymbol{A}^* \boldsymbol{A} = \operatorname{diag}\left(\overline{d_1} d_1, \ldots, \overline{d_n} d_n\right) = \operatorname{diag}\left(\left d_1\right ^2, \ldots, \left d_n\right ^2\right) = \boldsymbol{A} \boldsymbol{A}^*,$
Conf. MathPix confidence: 1.000 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0382	154	1.000 N +0	$e_{ii}=\sum_{k=1}^i\left b_{ki}\right ^2=\left b_{ii}\right ^2=\sum_{k=i}^n\left b_{ik}\right ^2=f_{ii}$	$e_{ii}=\sum_{k=1}^i b_{ki} ^2= b_{ii} ^2=\sum_{k=i}^n b_{ik} ^2=f_{ii}$	$e_{ii}=\sum_{k=1}^i b_{ki} ^2= b_{ii} ^2=\sum_{k=i}^n b_{ik} ^2=f_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0385 (6.10)	155	1.000 K +0	$\boldsymbol{M}=\left[\begin{array}{cc}\mu & v \\ -v & \mu\end{array}\right]$	$\boldsymbol{M}=\left[\begin{array}{cc}\mu & v \\ -v & \mu\end{array}\right]$	$\boldsymbol{M}=\left[\begin{array}{cc}\mu & v \\ -v & \mu\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0388	156	1.000 N +0	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$	$\boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0389	156	1.000 N +0	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\lambda_2,\ldots,\lambda_n\right)$	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\lambda_2,\ldots,\lambda_n\right)$	$\boldsymbol{U}^T\boldsymbol{A}\boldsymbol{U}=\operatorname{diag}\left(\lambda_1,\lambda_2,\ldots,\lambda_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0391	157	1.000 N +0	$\operatorname{dim}\left(\mathcal{S}\cap\mathcal{S}'\right)=\operatorname{dim}\left(\mathcal{S}\right)+\operatorname{dim}\left(\mathcal{S}'\right)-\operatorname{dim}\left(\mathcal{S}+\mathcal{S}'\right)\geq\left(n-k+1\right)+k-n=1$	$\operatorname{dim}\left(\mathcal{S}\cap\mathcal{S}'\right)=\operatorname{dim}\left(\mathcal{S}\right)+\operatorname{dim}\left(\mathcal{S}'\right)-\operatorname{dim}\left(\mathcal{S}+\mathcal{S}'\right)\geq\left(n-k+1\right)+k-n=1$	$\operatorname{dim}\left(\mathcal{S}\cap\mathcal{S}'\right)=\operatorname{dim}\left(\mathcal{S}\right)+\operatorname{dim}\left(\mathcal{S}'\right)-\operatorname{dim}\left(\mathcal{S}+\mathcal{S}'\right)\geq\left(n-k+1\right)+k-n=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0395 (6.14)	158	1.000 N +0	$\alpha_k+\varepsilon_n\leq\beta_k\leq\alpha_k+\varepsilon_1,\text{ for }k=1,\ldots,n,$	$\alpha_k+\varepsilon_n\leq\beta_k\leq\alpha_k+\varepsilon_1,\text{ for }k=1,\ldots,n,$	$\alpha_k+\varepsilon_n\leq\beta_k\leq\alpha_k+\varepsilon_1,\text{ for }k=1,\ldots,n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0398 (6.16)	159	1.000 W +1	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{y}_j^*\boldsymbol{v}}{\boldsymbol{y}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j=\sum_{k=1}^n\frac{\boldsymbol{x}_k^*\boldsymbol{v}}{\boldsymbol{y}_k^*\boldsymbol{x}_k}\boldsymbol{y}_k,\quad\boldsymbol{v}\in\mathbb{C}^n$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{y}_j^*\boldsymbol{v}}{\boldsymbol{y}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j=\sum_{k=1}^n\frac{\boldsymbol{x}_k^*\boldsymbol{v}}{\boldsymbol{y}_k^*\boldsymbol{x}_k}\boldsymbol{y}_k,\quad\boldsymbol{v}\in\mathbb{C}^n$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{y}_j^*\boldsymbol{v}}{\boldsymbol{y}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j=\sum_{k=1}^n\frac{\boldsymbol{x}_k^*\boldsymbol{v}}{\boldsymbol{y}_k^*\boldsymbol{x}_k}\boldsymbol{y}_k,\quad\boldsymbol{v}\in\mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0399	160	1.000 N -1	$\boldsymbol{X}:=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\right],\quad\boldsymbol{Y}:=\left[\boldsymbol{y}_1,\ldots,\boldsymbol{y}_n\right],\quad\boldsymbol{D}:=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$	$\boldsymbol{X}:=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\right],\quad\boldsymbol{Y}:=\left[\boldsymbol{y}_1,\ldots,\boldsymbol{y}_n\right],\quad\boldsymbol{D}:=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$	$\boldsymbol{X}:=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n\right],\quad\boldsymbol{Y}:=\left[\boldsymbol{y}_1,\ldots,\boldsymbol{y}_n\right],\quad\boldsymbol{D}:=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0400 (6.17)	160	■ 1.000 ● N +0	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j.$	$\boldsymbol{v}=\sum_{j=1}^n\frac{\boldsymbol{x}_j^*\boldsymbol{v}}{\boldsymbol{x}_j^*\boldsymbol{x}_j}\boldsymbol{x}_j.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0402	161	■ 1.000 ● N +0	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$	$\left(\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}\right)\boldsymbol{u}=\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{V}^*\boldsymbol{y}=\bar{\lambda}\boldsymbol{u},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0405 (6.18)	162	■ 1.000 ● N +0	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\dots,c_n$	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\dots,c_n$	$\boldsymbol{A}^k\boldsymbol{x}=\sum_{j=1}^nc_j\lambda_j^k\boldsymbol{x}_j$ for some scalars $c_1,\dots,c_n$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0411	164	■ 1.000 ● N +0	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$	$p(\boldsymbol{A}):=\sum_{j=0}^rb_j\boldsymbol{A}^j,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0417 (7.1)	168	■ 1.000 ● K +0	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^*,$ where $\boldsymbol{U}^*\boldsymbol{U}=\boldsymbol{I}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0419	169	■ 1.000 ● N +0	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$	$\lambda_1\geq\cdots\geq\lambda_r>0=\lambda_{r+1}=\cdots=\lambda_n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0421 (7.3)	170	■ 1.000 ● N +0	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq 0.$	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq 0.$	$\lambda=\frac{\boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}}{\boldsymbol{v}^*\boldsymbol{v}}=\frac{\ \boldsymbol{A}\boldsymbol{v}\ _2^2}{\ \boldsymbol{v}\ _2^2}\geq 0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0423	171	■ 1.000 ● K +0	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\begin{bmatrix}52&36\\36&73\end{bmatrix}$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\begin{bmatrix}52&36\\36&73\end{bmatrix}$	$\boldsymbol{B}:=\boldsymbol{A}^T\boldsymbol{A}=\frac{1}{25}\begin{bmatrix}52&36\\36&73\end{bmatrix}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0424	171	■ 1.000 ● N +0	$\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array}\right]=4\left[\begin{array}{l} 3 \\ 4 \end{array}\right], \quad \boldsymbol{B}\left[\begin{array}{l} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{l} 4 \\ -3 \end{array}\right].$	$\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array}\right]=4\left[\begin{array}{l} 3 \\ 4 \end{array}\right], \quad \boldsymbol{B}\left[\begin{array}{l} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{l} 4 \\ -3 \end{array}\right].$	$\boldsymbol{B}\left[\begin{array}{l} 3 \\ 4 \end{array}\right]=4\left[\begin{array}{l} 3 \\ 4 \end{array}\right], \quad \boldsymbol{B}\left[\begin{array}{l} 4 \\ -3 \end{array}\right]=1\left[\begin{array}{l} 4 \\ -3 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0428 (7.6)	172	■ 1.000 ● N +0	$\boldsymbol{A}=\sum_{j=1}^r \boldsymbol{\sigma}_j \boldsymbol{u}_j \boldsymbol{v}_j^*.$	$\boldsymbol{A}=\sum_{j=1}^r \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^*.$	$\boldsymbol{A}=\sum_{j=1}^r \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^*.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0430	172	■ 1.000 ● K +0	$\boldsymbol{B}:=\boldsymbol{A}^T \boldsymbol{A}=\left[\begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T \boldsymbol{A}=\left[\begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]$	$\boldsymbol{B}:=\boldsymbol{A}^T \boldsymbol{A}=\left[\begin{array}{ll} 2 & 2 \\ 2 & 2 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0433	173	■ 1.000 ● K +0	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right].$	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right].$	$\boldsymbol{A}=\frac{1}{\sqrt{2}}\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right][2] \frac{1}{\sqrt{2}}\left[\begin{array}{ll} 1 & 1 \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0434	173	■ 1.000 ● N +1	$\boldsymbol{s}_1=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_2=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_3=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$	$\boldsymbol{s}_1=\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_2=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_3=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$	$\boldsymbol{s}_1=\left[\begin{array}{l} 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_2=\left[\begin{array}{l} 0 \\ 1 \\ 0 \end{array}\right], \quad \boldsymbol{s}_3=\left[\begin{array}{l} 0 \\ 0 \\ 1 \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0437	174	■ 1.000 ● W +0	$\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}} \text { where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}=\left[y_1, \ldots, y_n\right]^T \in \mathbb{R}^n: \frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}=1\right\}.$	$\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}} \text { where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}=\left[y_1, \ldots, y_n\right]^T \in \mathbb{R}^n: \frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}=1\right\}.$	$\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}} \text { where } \tilde{\mathcal{E}}:=\left\{\boldsymbol{y}=\left[y_1, \ldots, y_n\right]^T \in \mathbb{R}^n: \frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}=1\right\}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0441 (7.9)	176	■ 1.000 ● K +0	$\left\ \boldsymbol{A}\right\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}.$	$\left\ \boldsymbol{A}\right\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}.$	$\left\ \boldsymbol{A}\right\ _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0444	177	■ 1.000 ● N +0	$\left\ \boldsymbol{A}-\boldsymbol{A}^{\prime}\right\ _F=\left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^*\right\ _F=\left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right]\right\ _F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}<\epsilon.$	$\left\ \boldsymbol{A}-\boldsymbol{A}^{\prime}\right\ _F=\left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^*\right\ _F=\left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right]\right\ _F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}<\epsilon.$	$\left\ \boldsymbol{A}-\boldsymbol{A}^{\prime}\right\ _F=\left\ \boldsymbol{U}\left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right] \boldsymbol{V}^*\right\ _F=\left\ \left[\begin{array}{c} \boldsymbol{D}-\boldsymbol{D}^{\prime} \\ \mathbf{0} \end{array}\right]\right\ _F=\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}<\epsilon.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0449	180	■ 1.000 ● N +0	$\boldsymbol{C}:=\left[\begin{array}{cc} \mathbf{0} & \boldsymbol{A} \\ \boldsymbol{A}^* & \mathbf{0} \end{array}\right] \in \mathbb{R}^{(m+n)\times(m+n)}$	$C := \begin{bmatrix} \mathbf{0} & A \\ A^* & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(m+n)\times(m+n)}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0450	180	■ 1.000 ● N +0	$\left(\left(\sigma_1, \boldsymbol{p}_1\right), \ldots, \left(\sigma_n, \boldsymbol{p}_n\right), \left(-\sigma_1, \boldsymbol{q}_1\right), \ldots, \left(-\sigma_n, \boldsymbol{q}_n\right), \left(0, \boldsymbol{r}_{n+1}\right), \ldots, \left(0, \boldsymbol{r}_m\right)\right),$	$\left\{\left(\sigma_1, \boldsymbol{p}_1\right), \ldots, \left(\sigma_n, \boldsymbol{p}_n\right), \left(-\sigma_1, \boldsymbol{q}_1\right), \ldots, \left(-\sigma_n, \boldsymbol{q}_n\right), \left(0, \boldsymbol{r}_{n+1}\right), \ldots, \left(0, \boldsymbol{r}_m\right)\right\},$	$\left\{\left(\sigma_1, \boldsymbol{p}_1\right), \ldots, \left(\sigma_n, \boldsymbol{p}_n\right), \left(-\sigma_1, \boldsymbol{q}_1\right), \ldots, \left(-\sigma_n, \boldsymbol{q}_n\right), \left(0, \boldsymbol{r}_{n+1}\right), \ldots, \left(0, \boldsymbol{r}_m\right)\right\},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0452 (7.12)	180	■ 1.000 ● N +0	$\boldsymbol{Q}:=\boldsymbol{U} \boldsymbol{V}^{\mathrm{T}}, \quad \quad \boldsymbol{P}:=\boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^{\mathrm{T}}$	$\boldsymbol{Q} := \boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P} := \boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$\boldsymbol{Q} := \boldsymbol{U} \boldsymbol{V}^T, \quad \boldsymbol{P} := \boldsymbol{V} \boldsymbol{\Sigma} \boldsymbol{V}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0453 (7.13)	180	■ 1.000 ● N +0	$A=Q \; P$	$A = \boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0455 (7.15)	181	■ 1.000 ● N +0	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2}\boldsymbol{X}_k^{-\mathrm{T}}\left(\boldsymbol{X}_k^{\mathrm{T}}-\boldsymbol{Q}^{\mathrm{T}}\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2}\boldsymbol{X}_k^{-T}\left(\boldsymbol{X}_k^T-\boldsymbol{Q}^T\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$	$\boldsymbol{X}_{k+1}-\boldsymbol{Q}=\frac{1}{2}\boldsymbol{X}_k^{-T}\left(\boldsymbol{X}_k^T-\boldsymbol{Q}^T\right)\left(\boldsymbol{X}_k-\boldsymbol{Q}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0456 (7.16)	181	■ 1.000 ● N +0	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\ _F \leq \frac{1}{2}\left\ \boldsymbol{X}_k^{-1}\right\ _F\left\ \boldsymbol{X}_k-\boldsymbol{Q}\right\ _F^2.$	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\ _F \leq \frac{1}{2}\left\ \boldsymbol{X}_k^{-1}\right\ _F\left\ \boldsymbol{X}_k-\boldsymbol{Q}\right\ _F^2.$	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{Q}\right\ _F \leq \frac{1}{2}\left\ \boldsymbol{X}_k^{-1}\right\ _F\left\ \boldsymbol{X}_k-\boldsymbol{Q}\right\ _F^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0458	181	■ 1.000 ● K +0	$\boldsymbol{B}=\left[\begin{array}{ccc} 1 & 0 & -1 \\ 1 & 1 & 1 \end{array}\right].$	$\boldsymbol{B}=\begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$	$\boldsymbol{B}=\begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0461	182	■ 1.000 ● N +0	$\boldsymbol{A}=\left[\begin{array}{rrrr} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{array}\right].$	$\boldsymbol{A}=\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$	$\boldsymbol{A}=\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0464	183	■ 1.000 ● N +0	$\left \left \boldsymbol{A}-\boldsymbol{A}^{\prime}\right _{F}^2=\sigma_{r+1}^2+\cdots+\sigma_n^2\right $.	$\left \left \boldsymbol{A}-\boldsymbol{A}^{\prime}\right _F^2=\sigma_{r+1}^2+\cdots+\sigma_n^2\right $.	$\left \left \boldsymbol{A}-\boldsymbol{A}^{\prime}\right _F^2=\sigma_{r+1}^2+\cdots+\sigma_n^2\right $.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0472	187	■ 1.000 ● N +0	$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}: \left \left \boldsymbol{y}\right ^{\prime}=1\right \right\}$.	$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}: \left \left \boldsymbol{y}\right ^{\prime}=1\right \right\}$.	$\mathcal{S}:=\left\{\boldsymbol{y} \in \mathcal{V}: \left \left \boldsymbol{y}\right ^{\prime}=1\right \right\}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0473 (8.8)	187	■ 1.000 ● N +0	$m \leq\left \left \boldsymbol{y}\right \right \leq M, \quad \boldsymbol{y} \in \mathcal{S}$.	$m \leq\left \left \boldsymbol{y}\right \right \leq M, \quad \boldsymbol{y} \in \mathcal{S}$.	$m \leq\left \left \boldsymbol{y}\right \right \leq M, \quad \boldsymbol{y} \in \mathcal{S}$.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0474	188	■ 1.000 ● K +0	$\left \left \boldsymbol{A}\right _{F}:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}\right $	$\left \left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}\right $	$\left \left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2}\right $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0475	188	■ 1.000 ● K +0	$\mu\left \left \boldsymbol{A}\right \right \leq\left \left \boldsymbol{A}\right ^{\prime}\right \leq M\left \left \boldsymbol{A}\right \right $	$\mu\left \left \boldsymbol{A}\right \right \leq\left \left \boldsymbol{A}\right ^{\prime}\right \leq M\left \left \boldsymbol{A}\right \right $	$\mu\left \left \boldsymbol{A}\right \right \leq\left \left \boldsymbol{A}\right ^{\prime}\right \leq M\left \left \boldsymbol{A}\right \right $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0476 (8.9)	188	■ 1.000 ● W -1	$\left \left \boldsymbol{A}\right _{S}:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ,\quad\left \left \boldsymbol{A}\right _{F}:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2},\quad\left \left \boldsymbol{A}\right _{M}:=\max _{i, j}\left a_{i j}\right \right $.	$\left \left \boldsymbol{A}\right _S:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ,\quad\left \left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2},\quad\left \left \boldsymbol{A}\right _M:=\max _{i, j}\left a_{i j}\right \right $.	$\left \left \boldsymbol{A}\right _S:=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ,\quad\left \left \boldsymbol{A}\right _F:=\left(\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2\right)^{1 / 2},\quad\left \left \boldsymbol{A}\right _M:=\max _{i, j}\left a_{i j}\right \right $.
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0478 (8.11)	189	■ 1.000 ● K +0	$\left \left \boldsymbol{A}\right \boldsymbol{A}\left \boldsymbol{x}\right \right \leq\left \left \boldsymbol{A}\right \right \left \boldsymbol{x}\right $,	$\left \left \boldsymbol{A}\boldsymbol{x}\right \right \leq\left \left \boldsymbol{A}\right \right \left \boldsymbol{x}\right $,	$\left \left \boldsymbol{A}\boldsymbol{x}\right \right \leq\left \left \boldsymbol{A}\right \right \left \boldsymbol{x}\right $,
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0479 (8.12)	190	■ 1.000 ● N +0	$\left \left \boldsymbol{A}\right \right :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left \left \boldsymbol{A}\boldsymbol{x}\right \right }{\left \left \boldsymbol{x}\right \right }$.	$\left \left \boldsymbol{A}\right \right :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left \left \boldsymbol{A}\boldsymbol{x}\right \right }{\left \left \boldsymbol{x}\right \right }$.	$\left \left \boldsymbol{A}\right \right :=\max _{\boldsymbol{x} \neq \boldsymbol{0}} \frac{\left \left \boldsymbol{A}\boldsymbol{x}\right \right }{\left \left \boldsymbol{x}\right \right }$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0480 (8.13)	190	■ 1.000 ● N +0	$\ \boldsymbol{A}\ =\max_{\ \boldsymbol{x}\ =1}\ \boldsymbol{A}\boldsymbol{x}\ $.	$\ \boldsymbol{A}\ = \max_{\ \boldsymbol{x}\ =1} \ \boldsymbol{A}\boldsymbol{x}\ .$	$\ \boldsymbol{A}\ = \max_{\ \boldsymbol{x}\ =1} \ \boldsymbol{A}\boldsymbol{x}\ .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0483 (8.15)	190	■ 1.000 ● N +0	$\ \boldsymbol{A}\boldsymbol{x}\ \leq\ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A}\in\mathbb{C}^{m\times n}$ and $\boldsymbol{x}\in\mathbb{C}^n$.	$\ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.	$\ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{x} \in \mathbb{C}^n$.
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0489	192	■ 1.000 ● N +0	$\left \sum_{j=1}^na_{kj}x_j\right \leq\sum_{k=1}^m\left a_{kj}\right x_j =\sum_{j=1}^n\left(\sum_{k=1}^m a_{kj} \right) x_j \leq K_1$.	$\ \boldsymbol{A}\boldsymbol{x}\ _1=\sum_{k=1}^m\left \sum_{j=1}^na_{kj}x_j\right \leq\sum_{k=1}^m\sum_{j=1}^n a_{kj} x_j =\sum_{j=1}^n\left(\sum_{k=1}^m a_{kj} \right) x_j \leq K_1.$	$\ \boldsymbol{A}\boldsymbol{x}\ _1=\sum_{k=1}^m\left \sum_{j=1}^na_{kj}x_j\right \leq\sum_{k=1}^m\sum_{j=1}^n a_{kj} x_j =\sum_{j=1}^n\left(\sum_{k=1}^m a_{kj} \right) x_j \leq K_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0491	192	■ 1.000 ● N +0	$\ \boldsymbol{A}\ _1=\frac{29}{15},\quad\ \boldsymbol{A}\ _2=2,\quad\ \boldsymbol{A}\ _\infty=\frac{37}{15},\quad\ \boldsymbol{A}\ _F=\sqrt{5}$.	$\ \boldsymbol{A}\ _1=\frac{29}{15},\quad\ \boldsymbol{A}\ _2=2,\quad\ \boldsymbol{A}\ _\infty=\frac{37}{15},\quad\ \boldsymbol{A}\ _F=\sqrt{5}.$	$\ \boldsymbol{A}\ _1=\frac{29}{15},\quad\ \boldsymbol{A}\ _2=2,\quad\ \boldsymbol{A}\ _\infty=\frac{37}{15},\quad\ \boldsymbol{A}\ _F=\sqrt{5}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0495	194	■ 1.000 ● N -1	$\ \boldsymbol{U}\boldsymbol{A}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{U}\boldsymbol{A}\boldsymbol{x}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{A}\boldsymbol{x}\ _2=\ \boldsymbol{A}\ _2$.	$\ \boldsymbol{U}\boldsymbol{A}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{U}\boldsymbol{A}\boldsymbol{x}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{A}\boldsymbol{x}\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{U}\boldsymbol{A}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{U}\boldsymbol{A}\boldsymbol{x}\ _2=\max_{\ \boldsymbol{x}\ _2=1}\ \boldsymbol{A}\boldsymbol{x}\ _2=\ \boldsymbol{A}\ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0496	194	■ 1.000 ● K +0	$\ \boldsymbol{A}\boldsymbol{V}\ _2=\ (\boldsymbol{A}\boldsymbol{V})^*\ _2=\ \boldsymbol{V}^*\boldsymbol{A}^*\ _2=\ \boldsymbol{A}^*\ _2=\ \boldsymbol{A}\ _2$.	$\ \boldsymbol{A}\boldsymbol{V}\ _2=\ (\boldsymbol{A}\boldsymbol{V})^*\ _2=\ \boldsymbol{V}^*\boldsymbol{A}^*\ _2=\ \boldsymbol{A}^*\ _2=\ \boldsymbol{A}\ _2.$	$\ \boldsymbol{A}\boldsymbol{V}\ _2=\ (\boldsymbol{A}\boldsymbol{V})^*\ _2=\ \boldsymbol{V}^*\boldsymbol{A}^*\ _2=\ \boldsymbol{A}^*\ _2=\ \boldsymbol{A}\ _2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0501 (8.22)	195	■ 1.000 ● N +0	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ },$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ },$	$\frac{1}{K(\boldsymbol{A})}\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ }\leq\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq K(\boldsymbol{A})\frac{\ \boldsymbol{e}\ }{\ \boldsymbol{b}\ },$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0503 (8.24)	196	■ 1.000 ● W +0	$\begin{aligned}&\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{y}\ }\leq r\leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },\\&\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq\frac{r}{1-r}\leq\frac{K(\boldsymbol{A})\ \boldsymbol{E}\ }{1-r\ \boldsymbol{A}\ }.\end{aligned}$	$\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{y}\ }\leq r\leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },$ $\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq\frac{r}{1-r}\leq\frac{K(\boldsymbol{A})\ \boldsymbol{E}\ }{1-r\ \boldsymbol{A}\ }.$	$\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{y}\ }\leq r\leq K(\boldsymbol{A})\frac{\ \boldsymbol{E}\ }{\ \boldsymbol{A}\ },$ $\frac{\ \boldsymbol{y}-\boldsymbol{x}\ }{\ \boldsymbol{x}\ }\leq\frac{r}{1-r}\leq\frac{K(\boldsymbol{A})\ \boldsymbol{E}\ }{1-r\ \boldsymbol{A}\ }.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0506 (8.28)	198	■ 1.000 ● N +0	$\frac{1}{1+r} \leq \frac{\left\ \boldsymbol{B}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{1}{1-r} .$	$\frac{1}{1+r} \leq \frac{\left\ \boldsymbol{B}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{1}{1-r} .$	$\frac{1}{1+r} \leq \frac{\left\ \boldsymbol{B}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{1}{1-r} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0507 (8.29)	198	■ 1.000 ● N +0	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A})\left\ \boldsymbol{E}\right\ }{1-r\left\ \boldsymbol{A}\right\ } .$	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A})\left\ \boldsymbol{E}\right\ }{1-r\left\ \boldsymbol{A}\right\ } .$	$\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ } \leq \frac{r}{1-r} \leq \frac{K(\boldsymbol{A})\left\ \boldsymbol{E}\right\ }{1-r\left\ \boldsymbol{A}\right\ } .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0508 (8.30)	198	■ 1.000 ● K +0	$\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}=-\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}=-\boldsymbol{B}^{-1} \boldsymbol{E} \boldsymbol{A}^{-1} .$	$\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}=-\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}=-\boldsymbol{B}^{-1} \boldsymbol{E} \boldsymbol{A}^{-1} .$	$\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}=-\boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{B}^{-1}=-\boldsymbol{B}^{-1} \boldsymbol{E} \boldsymbol{A}^{-1} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0512 (8.31)	199	■ 1.000 ● K +0	$f\left((1-\lambda) x_1+\lambda x_2\right) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$	$f\left((1-\lambda) x_1+\lambda x_2\right) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$	$f\left((1-\lambda) x_1+\lambda x_2\right) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0514	200	■ 1.000 ● K +0	$x=\frac{x_2-x}{x_2-x_1} x_1+\frac{x-x_1}{x_2-x_1} x_2=(1-\lambda) x_1+\lambda x_2$	$x=\frac{x_2-x}{x_2-x_1} x_1+\frac{x-x_1}{x_2-x_1} x_2=(1-\lambda) x_1+\lambda x_2$	$x=\frac{x_2-x}{x_2-x_1} x_1+\frac{x-x_1}{x_2-x_1} x_2=(1-\lambda) x_1+\lambda x_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0515	200	■ 1.000 ● N +0	$f\left(\sum_{j=1}^n \lambda_{j z_j}\right) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right) .$	$f\left(\sum_{j=1}^n \lambda_j z_j\right) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right) .$	$f\left(\sum_{j=1}^n \lambda_j z_j\right) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right) .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0516	200	■ 1.000 ● N +0	$f\left(\sum_{j=1}^n \lambda_{j z_j}\right)=f\left(\lambda_1 z_1+\left(1-\lambda_1\right) u\right) \leq \lambda_1 f\left(z_1\right)+\left(1-\lambda_1\right) f(u) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right)$	$f\left(\sum_{j=1}^n \lambda_j z_j\right)=f\left(\lambda_1 z_1+\left(1-\lambda_1\right) u\right) \leq \lambda_1 f\left(z_1\right)+\left(1-\lambda_1\right) f(u) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right)$	$f\left(\sum_{j=1}^n \lambda_j z_j\right)=f\left(\lambda_1 z_1+\left(1-\lambda_1\right) u\right) \leq \lambda_1 f\left(z_1\right)+\left(1-\lambda_1\right) f(u) \leq \sum_{j=1}^n \lambda_j f\left(z_j\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0518	201	■ 1.000 ● W +0	$-\log \left(\sum_{j=1}^n \lambda_{j a_j}\right) \leq-\sum_{j=1}^n \lambda_j \log \left(a_j\right)=-\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right)$	$-\log \left(\sum_{j=1}^n \lambda_{j a_j}\right) \leq-\sum_{j=1}^n \lambda_j \log \left(a_j\right)=-\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right)$	$-\log \left(\sum_{j=1}^n \lambda_{j a_j}\right) \leq-\sum_{j=1}^n \lambda_j \log \left(a_j\right)=-\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0519 (8.33)	201	■ 1.000 ● N +0	$\left(a_{\{1\}} a_{\{2\}} \cdots a_{\{n\}}\right)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_{\{j\}} .$	$(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_j .$	$(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} \sum_{j=1}^n a_j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0520	201	■ 1.000 ● N +1	$\sum_{j=1}^n\left x_{\{j\}} y_{\{j\}}\right \leq\left \boldsymbol{x}\right _{\{p\}}\left \boldsymbol{y}\right _{\{q\}}, \text { where }\frac{1}{p}+\frac{1}{q}=1$	$\sum_{j=1}^n\left x_j y_j\right \leq\left\ \boldsymbol{x}\right\ _p\left\ \boldsymbol{y}\right\ _q, \text { where }\frac{1}{p}+\frac{1}{q}=1$	$\sum_{j=1}^n\left x_j y_j\right \leq\left\ \boldsymbol{x}\right\ _p\left\ \boldsymbol{y}\right\ _q, \text { where }\frac{1}{p}+\frac{1}{q}=1 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0522	201	■ 1.000 ● W +0	$\frac{1}{\left \boldsymbol{x}\right _{\{p\}}\left \boldsymbol{y}\right _{\{q\}}}\sum_{j=1}^n\left x_{\{j\}} y_{\{j\}}\right =\sum_{j=1}^n\left(\frac{\left x_{\{j\}}\right ^p}{\left \boldsymbol{x}\right _{\{p\}}^p}\right)^{\frac{1}{p}}\left(\frac{\left y_{\{j\}}\right ^q}{\left \boldsymbol{y}\right _{\{q\}}^q}\right)^{\frac{1}{q}} \leq \sum_{j=1}^n\left(\frac{1}{p} \frac{\left x_{\{j\}}\right ^p}{\left \boldsymbol{x}\right _{\{p\}}^p}+\frac{1}{q} \frac{\left y_{\{j\}}\right ^q}{\left \boldsymbol{y}\right _{\{q\}}^q}\right)=1$	$\frac{1}{\left\ \boldsymbol{x}\right\ _p\left\ \boldsymbol{y}\right\ _q} \sum_{j=1}^n\left x_j y_j\right =\sum_{j=1}^n\left(\frac{\left x_j\right ^p}{\left\ \boldsymbol{x}\right\ _p^p}\right)^{\frac{1}{p}}\left(\frac{\left y_j\right ^q}{\left\ \boldsymbol{y}\right\ _q^q}\right)^{\frac{1}{q}} \leq \sum_{j=1}^n\left(\frac{1}{p} \frac{\left x_j\right ^p}{\left\ \boldsymbol{x}\right\ _p^p}+\frac{1}{q} \frac{\left y_j\right ^q}{\left\ \boldsymbol{y}\right\ _q^q}\right)=1$	$\frac{1}{\left\ \boldsymbol{x}\right\ _p\left\ \boldsymbol{y}\right\ _q} \sum_{j=1}^n\left x_j y_j\right =\sum_{j=1}^n\left(\frac{\left x_j\right ^p}{\left\ \boldsymbol{x}\right\ _p^p}\right)^{\frac{1}{p}}\left(\frac{\left y_j\right ^q}{\left\ \boldsymbol{y}\right\ _q^q}\right)^{\frac{1}{q}} \leq \sum_{j=1}^n\left(\frac{1}{p} \frac{\left x_j\right ^p}{\left\ \boldsymbol{x}\right\ _p^p}+\frac{1}{q} \frac{\left y_j\right ^q}{\left\ \boldsymbol{y}\right\ _q^q}\right)=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0523	202	■ 1.000 ● N +0	$\left \boldsymbol{x}+\boldsymbol{y}\right _{\{p\}} \leq\left \boldsymbol{x}\right _{\{p\}}+\left \boldsymbol{y}\right _{\{p\}} .$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p \leq\left\ \boldsymbol{x}\right\ _p+\left\ \boldsymbol{y}\right\ _p .$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p \leq\left\ \boldsymbol{x}\right\ _p+\left\ \boldsymbol{y}\right\ _p .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0524	202	■ 1.000 ● N +0	$\left \boldsymbol{x}+\boldsymbol{y}\right _{\{p\}}^p=\sum_{j=1}^n\left x_{\{j\}}+y_{\{j\}}\right ^p \leq \sum_{j=1}^n\left x_{\{j\}}\right ^p+\sum_{j=1}^n\left y_{\{j\}}\right ^p$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p^p=\sum_{j=1}^n\left x_j+y_j\right ^p \leq \sum_{j=1}^n\left x_j\right ^p+\sum_{j=1}^n\left y_j\right ^p$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p^p=\sum_{j=1}^n\left x_j+y_j\right ^p \leq \sum_{j=1}^n\left x_j\right ^p+\sum_{j=1}^n\left y_j\right ^p$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0525	202	■ 1.000 ● N +0	$\left \boldsymbol{x}+\boldsymbol{y}\right _{\{p\}}^p \leq\left \boldsymbol{x}\right _{\{p\}}^p+\left \boldsymbol{y}\right _{\{p\}}^p$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p^p \leq\left\ \boldsymbol{x}\right\ _p^p+\left\ \boldsymbol{y}\right\ _p^p$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ _p^p \leq\left\ \boldsymbol{x}\right\ _p^p+\left\ \boldsymbol{y}\right\ _p^p$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0526 (8.35)	202	■ 1.000 ● N +0	$\left \boldsymbol{x}+\boldsymbol{y}\right _{\{2\}}^2=\left \boldsymbol{x}\right _{\{2\}}^2+\left \boldsymbol{y}\right _{\{2\}}^2$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2+\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2=2\left\ \boldsymbol{x}\right\ ^2+2\left\ \boldsymbol{y}\right\ ^2,$	$\left\ \boldsymbol{x}+\boldsymbol{y}\right\ ^2+\left\ \boldsymbol{x}-\boldsymbol{y}\right\ ^2=2\left\ \boldsymbol{x}\right\ ^2+2\left\ \boldsymbol{y}\right\ ^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0534	203	■ 1.000 ● N +0	$\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\right\rangle=\frac{n}{m}\left\langle\boldsymbol{x}, \boldsymbol{y}\right\rangle$	$\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\right\rangle=\frac{n}{m}\left\langle\boldsymbol{x}, \boldsymbol{y}\right\rangle$	$\left\langle\frac{n}{m} \boldsymbol{x}, \boldsymbol{y}\right\rangle=\frac{n}{m}\left\langle\boldsymbol{x}, \boldsymbol{y}\right\rangle$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0539	204	1.000 N +0	$\tilde{\boldsymbol{b}}_k := \boldsymbol{b}_k - \boldsymbol{B}_{k-1} \boldsymbol{B}_{k-1}^T \boldsymbol{A} \boldsymbol{B}_{k-1} \boldsymbol{b}_k, \quad k = 2, \dots, n.$	$\tilde{\boldsymbol{b}}_k := \boldsymbol{b}_k - \boldsymbol{B}_{k-1} \left(\boldsymbol{B}_{k-1}^T \boldsymbol{A} \boldsymbol{B}_{k-1} \right)^{-1} \boldsymbol{B}_{k-1}^T \boldsymbol{A} \boldsymbol{b}_k, \quad k = 2, \dots, n.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0540	205	1.000 N +0	$\left\ \boldsymbol{b}_1 \right\ _A \left\ \boldsymbol{b}_2 \right\ _A \cdots \left\ \boldsymbol{b}_n \right\ _A \leq 2^{n(n-1)/4} \sqrt{\det(\boldsymbol{A})}.$	$\left\ \boldsymbol{b}_1 \right\ _A \left\ \boldsymbol{b}_2 \right\ _A \cdots \left\ \boldsymbol{b}_n \right\ _A \leq 2^{n(n-1)/4} \sqrt{\det(\boldsymbol{A})}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0541	205	1.000 N +0	$\left\ \boldsymbol{A} \right\ := \sqrt{mn} \left\ \boldsymbol{A} \right\ _M, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$\left\ \boldsymbol{A} \right\ := \sqrt{mn} \left\ \boldsymbol{A} \right\ _M, \quad \boldsymbol{A} \in \mathbb{C}^{m \times n}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0542	205	1.000 N -1	$\left\ \boldsymbol{A} \right\ _2 = \max_{\left\ \boldsymbol{x} \right\ _2 = 1} \left \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} \right .$	$\left\ \boldsymbol{A} \right\ _2 = \max_{\left\ \boldsymbol{x} \right\ _2 = \left\ \boldsymbol{y} \right\ _2 = 1} \left \boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} \right .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0545	207	1.000 N +0	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \frac{1}{\lambda_{k+1}} \boldsymbol{r}_k, \text{ where } \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k.$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \frac{1}{\lambda_{k+1}} \boldsymbol{r}_k, \text{ where } \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0546	207	1.000 N +0	$\boldsymbol{r}_k = \sum_{i=1}^n c_{ik} \boldsymbol{u}_i,$	$\boldsymbol{r}_k = \sum_{i=1}^n c_{ik} \boldsymbol{u}_i,$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0548	207	1.000 N +0	$\lambda_j = d + 2c \cos \left(\frac{j\pi}{n+1} \right), \quad j = 1, \dots, n.$	$\lambda_j = d + 2c \cos \left(\frac{j\pi}{n+1} \right), \quad j = 1, \dots, n.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0549 (8.40)	207	1.000 K +0	$\boldsymbol{T} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{B} \boldsymbol{x}_k.$	$\boldsymbol{T} \boldsymbol{x}_{k+1} = \boldsymbol{b} - \boldsymbol{B} \boldsymbol{x}_k.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0550	207	1.000 N -2	$\min \left\{ \max_i \sum_{j=1}^n b_{ij} , \max_j \sum_{i=1}^n b_{ij} \right\} < d - 2c.$	$\min \left\{ \max_i \sum_{j=1}^n b_{ij} , \max_j \sum_{i=1}^n b_{ij} \right\} < d - 2c.$	
Conf. MathPix confidence: 1.000 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0552 (8.41)	208	■ 1.000 ● N +0	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2,\quad p=1,2,\infty,$	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2,\quad p=1,2,\infty,$	$\frac{4}{\pi^2}(m+1)^2-2/3<\operatorname{cond}_p(\boldsymbol{T})\leq\frac{1}{2}(m+1)^2,\quad p=1,2,\infty,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0554	208	■ 1.000 ● N -1	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right).$	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right).$	$\operatorname{cond}_2(\boldsymbol{T})=\cot^2\left(\frac{\pi h}{2}\right)=1/\tan^2\left(\frac{\pi h}{2}\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0555 (8.43)	209	■ 1.000 ● N +0	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}.$	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}.$	$\frac{4}{\pi^2}h^{-2}-\frac{2}{3}<\operatorname{cond}_2(\boldsymbol{T})<\frac{4}{\pi^2}h^{-2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0556	209	■ 1.000 ● N +0	$\frac{\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{E}\right\ }{\left\ \boldsymbol{I}-\boldsymbol{E}\right\ }\leq\left\ \boldsymbol{E}\right\ $	$\frac{\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ }{\left\ \boldsymbol{I}-\boldsymbol{E}\right\ }\leq\left\ \boldsymbol{E}\right\ $	$\frac{\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ }{\left\ \boldsymbol{I}-\boldsymbol{E}\right\ }\leq\left\ \boldsymbol{E}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0557	209	■ 1.000 ● K +0	$\frac{1}{1+\left\ \boldsymbol{E}\right\ }\leq\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}\right\ \leq\frac{1}{1-\left\ \boldsymbol{E}\right\ }$	$\frac{1}{1+\left\ \boldsymbol{E}\right\ }\leq\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}\right\ \leq\frac{1}{1-\left\ \boldsymbol{E}\right\ }$	$\frac{1}{1+\left\ \boldsymbol{E}\right\ }\leq\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}\right\ \leq\frac{1}{1-\left\ \boldsymbol{E}\right\ }$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0558	209	■ 1.000 ● N +0	$\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$	$\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$	$\left\ \left(\boldsymbol{I}-\boldsymbol{E}\right)^{-1}-\boldsymbol{I}\right\ \leq\frac{\left\ \boldsymbol{E}\right\ }{1-\left\ \boldsymbol{E}\right\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0560	209	■ 1.000 ● N +0	$\frac{K(\boldsymbol{B})^{-1}\left\ \boldsymbol{E}\right\ }{1+r\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$	$\frac{K(\boldsymbol{B})^{-1}\left\ \boldsymbol{E}\right\ }{1+r\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$	$\frac{K(\boldsymbol{B})^{-1}\left\ \boldsymbol{E}\right\ }{1+r\left\ \boldsymbol{A}\right\ }\leq\frac{\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ }{\left\ \boldsymbol{A}^{-1}\right\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0561	209	■ 1.000 ● K +0	$a=x_0<x_1<\cdots<x_n=b,$	$a=x_0<x_1<\cdots<x_n=b,$	$a=x_0<x_1<\cdots<x_n=b,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0562	210	■ 1.000 ● W +0	$\begin{array}{ll} g\left(x_{\mathrm{i}}\right)=y_{\mathrm{i}}, & \text{\& i=0,1,} \\ \text{\&ldots, n,} & \text{\& g^{\prime}(a)=g^{\prime}(b), \& g^{\prime}(a)=g^{\prime}(b) .} \end{array}$	$g\left(x_{\mathrm{i}}\right)=y_{\mathrm{i}}, \quad i=0,1,\ldots,n, \\ g^{\prime}(a)=g^{\prime}(b), \quad g^{\prime \prime}(a)=g^{\prime \prime}(b).$	$g\left(x_{\mathrm{i}}\right)=y_{\mathrm{i}}, \quad i=0,1,\ldots,n, \\ g^{\prime}(a)=g^{\prime}(b), \quad g^{\prime \prime}(a)=g^{\prime \prime}(b).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0565	210	■ 1.000 ● W +0	$\lambda_{\mathrm{i}}:=\frac{h_{\mathrm{i}}}{h_{\mathrm{i}-1}+h_{\mathrm{i}}}, \quad \mu_{\mathrm{i}}:=\frac{h_{\mathrm{i}-1}}{h_{\mathrm{i}-1}+h_{\mathrm{i}}}, \quad i=1,\ldots,n,$	$\lambda_i := \frac{h_i}{h_{i-1} + h_i}, \quad \mu_i := \frac{h_{i-1}}{h_{i-1} + h_i}, \quad , i = 1, \ldots, n,$	$\lambda_i := \frac{h_i}{h_{i-1} + h_i}, \quad \mu_i := \frac{h_{i-1}}{h_{i-1} + h_i}, \quad , i = 1, \ldots, n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0566	210	■ 1.000 ● N -1	$h_{\mathrm{i}}=x_{\mathrm{i}+1}-x_{\mathrm{i}}, \quad \text{\&quad i=0, \&ldots, n-1, \&text { and } } h_{\mathrm{n}}=h_{\mathrm{0}} .$	$h_i = x_{i+1} - x_i, \quad i = 0, \ldots, n - 1, \text{ and } h_n = h_0.$	$h_i = x_{i+1} - x_i, \quad i = 0, \ldots, n - 1, \text{ and } h_n = h_0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0567	210	■ 1.000 ● N +0	$\frac{1}{2} \leq \frac{h_{\mathrm{i}}}{h_{\mathrm{i}-1}} \leq 2, \quad \text{\&ldots, n} .$	$\frac{1}{2} \leq \frac{h_i}{h_{i-1}} \leq 2, \quad i = 1, \ldots, n.$	$\frac{1}{2} \leq \frac{h_i}{h_{i-1}} \leq 2, \quad i = 1, \ldots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0568	211	■ 1.000 ● N -1	$\ \boldsymbol{A}\ _{\infty}=3 \quad \text{\&quad \&text { and that } } \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3} .$	$\ \boldsymbol{A}\ _{\infty} = 3 \quad \text{and that} \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3}.$	$\ \boldsymbol{A}\ _{\infty} = 3 \quad \text{and that} \quad \ \boldsymbol{A}\ _1 \leq \frac{10}{3}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0571 (8.45)	211	■ 1.000 ● N +0	$\angle \boldsymbol{x}, \boldsymbol{y}:=\frac{1}{4}\left(\left \boldsymbol{x}+\boldsymbol{y}\right ^2-\left \boldsymbol{x}-\boldsymbol{y}\right ^2+i\left \boldsymbol{x}+i\boldsymbol{y}\right ^2-i\left \boldsymbol{x}-i\boldsymbol{y}\right ^2\right),$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 + i \ \boldsymbol{x} + i \boldsymbol{y}\ ^2 - i \ \boldsymbol{x} - i \boldsymbol{y}\ ^2 \right),$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 + i \ \boldsymbol{x} + i \boldsymbol{y}\ ^2 - i \ \boldsymbol{x} - i \boldsymbol{y}\ ^2 \right),$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0574	213	■ 1.000 ● K +0	$E(\boldsymbol{x}):=\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\ _2^2,$	$E(\boldsymbol{x}) := \ \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\ _2^2,$	$E(\boldsymbol{x}) := \ \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\ _2^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0576	214	■ 1.000 ● N +1	$\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\ _2^2=\left(x_1-1\right)^2+\left(x_1-1\right)^2+\left(x_1-2\right)^2=3x_1^2-8x_1+6.$	$\ \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\ _2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2 = 3x_1^2 - 8x_1 + 6.$	$\ \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\ _2^2 = (x_1 - 1)^2 + (x_1 - 1)^2 + (x_1 - 2)^2 = 3x_1^2 - 8x_1 + 6.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0581	215	■ 1.000 ● N +0	$y=\boldsymbol{u}^*\boldsymbol{x}=\sum_{\mathrm{i}=1}^{\mathrm{n}}u_{\mathrm{i}}x_{\mathrm{i}},$	$y = \boldsymbol{u}^* \boldsymbol{x} = \sum_{i=1}^n u_i x_i,$	$y = \boldsymbol{u}^* \boldsymbol{x} = \sum_{i=1}^n u_i x_i,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0585 (9.2)	216	■ 1.000 ● N +0	$E(\boldsymbol{x}) := \ \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\ _2^2 = \sum_{k=1}^m \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0586	217	■ 1.000 ● N +0	$E(\boldsymbol{x}) := \sum_{k=1}^m w_k \left(\sum_{j=1}^n x_j \phi_j(t_k) - y_k \right)^2.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0587 (9.3)	217	■ 1.000 ● K +0	$p(\boldsymbol{t}) := \sum_{j=1}^n x_j \phi_j(t_k) = 0, \quad k = 1, \dots, m \Rightarrow x_1 = \dots = x_n = 0.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0595	219	■ 1.000 ● W +2	$\left(\boldsymbol{A}^* \boldsymbol{b} \right)_i = \sum_{k=1}^m \bar{a}_{k,i} b_k, \quad i = 1, \dots, n,$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0596 (9.6)	220	■ 1.000 ● N +1	$K_2(\boldsymbol{A}^* \boldsymbol{A}) := \ \boldsymbol{A}^* \boldsymbol{A}\ _2 \left\ (\boldsymbol{A}^* \boldsymbol{A})^{-1} \right\ _2 = \frac{\lambda_1}{\lambda_n} = \frac{\sigma_1^2}{\sigma_n^2} = K_2(\boldsymbol{A})^2,$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0597	220	■ 1.000 ● N +0	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{R}_1^* \boldsymbol{Q}_1^* \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{R}_1^* \boldsymbol{R}_1, \quad \boldsymbol{A}^* \boldsymbol{b} = \boldsymbol{R}_1^* \boldsymbol{Q}_1^* \boldsymbol{b}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0605	222	■ 1.000 ● N +0	$\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*, \quad \boldsymbol{A}^\dagger := \boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{U}_1^*$		
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0612	224	■ 1.000 ● K +0	$\boldsymbol{x} = \boldsymbol{A}^\dagger \boldsymbol{b} = \boldsymbol{A}^\dagger \boldsymbol{b}_1, \quad \boldsymbol{A}^\dagger = (\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^*,$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00616	226	■ 1.000 ● N -1	$\boldsymbol{b}_1=\boldsymbol{A} \boldsymbol{A}^{\dagger}\boldsymbol{b}=\left[10^{-4},0,0\right]^*, \quad \text{and} \quad \boldsymbol{e}_1=\boldsymbol{A} \boldsymbol{A}^{\dagger}\boldsymbol{e}=\left[10^{-6},0,0\right]^* .$	$\boldsymbol{b}_1 = \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = [10^{-4}, 0, 0]^*, \quad \text{and} \quad \boldsymbol{e}_1 = \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{e} = [10^{-6}, 0, 0]^* .$	$\boldsymbol{b}_1 = \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = [10^{-4}, 0, 0]^*, \quad \text{and} \quad \boldsymbol{e}_1 = \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{e} = [10^{-6}, 0, 0]^* .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00618	226	■ 1.000 ● N +1	$\frac{\left \left \boldsymbol{x}-\boldsymbol{y}\right \right _{\infty}}{\left \left \boldsymbol{x}\right \right _{\infty}}=\frac{10^{-6}}{10^{-4}}=10^{-2}$	$\frac{\ \boldsymbol{x}-\boldsymbol{y}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} = \frac{10^{-6}}{10^{-4}} = 10^{-2}$	$\frac{\ \boldsymbol{x}-\boldsymbol{y}\ _{\infty}}{\ \boldsymbol{x}\ _{\infty}} = \frac{10^{-6}}{10^{-4}} = 10^{-2},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00619 (9.15)	226	■ 1.000 ● W +2	$\rho=\frac{\left \left \boldsymbol{x}-\boldsymbol{y}\right \right _2}{\left \left \boldsymbol{x}\right \right _2} \leq \frac{1}{\alpha} K(1+\beta K) \frac{\left\ \boldsymbol{E}\right\ _2}{\left\ \boldsymbol{A}\right\ _2}, \quad \beta=\frac{\left\ \boldsymbol{b}_2\right\ _2}{\left\ \boldsymbol{b}_1\right\ _2}, \quad K=\left\ \boldsymbol{A}\right\ _2\left\ \boldsymbol{A}^{\dagger}\right\ _2 .$	$\rho = \frac{\ \boldsymbol{x}-\boldsymbol{y}\ _2}{\ \boldsymbol{x}\ _2} \leq \frac{1}{\alpha} K(1 + \beta K) \frac{\ \boldsymbol{E}\ _2}{\ \boldsymbol{A}\ _2}, \quad \beta = \frac{\ \boldsymbol{b}_2\ _2}{\ \boldsymbol{b}_1\ _2}, \quad K = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{\dagger}\ _2 .$	$\rho = \frac{\ \boldsymbol{x}-\boldsymbol{y}\ _2}{\ \boldsymbol{x}\ _2} \leq \frac{1}{\alpha} K(1 + \beta K) \frac{\ \boldsymbol{E}\ _2}{\ \boldsymbol{A}\ _2}, \quad \beta = \frac{\ \boldsymbol{b}_2\ _2}{\ \boldsymbol{b}_1\ _2}, \quad K = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{\dagger}\ _2 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00623 (9.18)	228	■ 1.000 ● N +0	$\left \alpha_j-\beta_j\right \leq\left \boldsymbol{A}-\boldsymbol{B}\right _2, \text { for } j=1,2, \ldots, n .$	$ \alpha_j - \beta_j \leq \ \boldsymbol{A} - \boldsymbol{B}\ _2, \text{ for } j = 1, 2, \dots, n.$	$ \alpha_j - \beta_j \leq \ \boldsymbol{A} - \boldsymbol{B}\ _2, \text{ for } j = 1, 2, \dots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00625	228	■ 1.000 ● W +0	$\left \left(\boldsymbol{A}+\boldsymbol{E}\right)^{\dagger}\right _2=\frac{1}{\beta_r} \leq \frac{1}{\alpha_r-\epsilon_1}=\frac{1 / \alpha_r}{1-\epsilon_1 / \alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2} .$	$\left\ (\boldsymbol{A}+\boldsymbol{E})^{\dagger}\right\ _2=\frac{1}{\beta_r} \leq \frac{1}{\alpha_r-\epsilon_1}=\frac{1 / \alpha_r}{1-\epsilon_1 / \alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2} .$	$\ (\boldsymbol{A}+\boldsymbol{E})^{\dagger}\ _2=\frac{1}{\beta_r} \leq \frac{1}{\alpha_r-\epsilon_1}=\frac{1 / \alpha_r}{1-\epsilon_1 / \alpha_r}=\frac{\left\ \boldsymbol{A}^{\dagger}\right\ _2}{1-\left\ \boldsymbol{A}^{\dagger}\right\ _2\left\ \boldsymbol{E}\right\ _2} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00626 (9.19)	228	■ 1.000 ● N +0	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n\left a_{i j}-b_{i j}\right ^2,$	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n \sum_{j=1}^n\left a_{i j}-b_{i j}\right ^2,$	$\sum_{j=1}^n\left \mu_j-\lambda_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2:=\sum_{i=1}^n \sum_{j=1}^n\left a_{i j}-b_{i j}\right ^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00627 (9.20)	229	■ 1.000 ● N +0	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2 .$	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2 .$	$\sum_{j=1}^n\left \beta_j-\alpha_j\right ^2 \leq\left\ \boldsymbol{A}-\boldsymbol{B}\right\ _F^2 .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0629 (9.21)	229	■ 1.000 ● N +0	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 \leq \ C - D\ _F^2$	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 \leq \ C - D\ _F^2$	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 \leq \ C - D\ _F^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0631	229	■ 1.000 ● N -1	$t := \max(r, s)$	$t := \max(r, s)$	$t := \max(r, s)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0633	230	■ 1.000 ● W +0	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 = \sum_{i=1}^t \alpha_i - \beta_i ^2 + \sum_{i=1}^t -\alpha_i + \beta_i ^2 = 2 \sum_{i=1}^t \alpha_i - \beta_i ^2$	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 = \sum_{i=1}^t \alpha_i - \beta_i ^2 + \sum_{i=1}^t -\alpha_i + \beta_i ^2 = 2 \sum_{i=1}^t \alpha_i - \beta_i ^2$	$\sum_{j=1}^{m+n} \lambda_j - \mu_j ^2 = \sum_{i=1}^t \alpha_i - \beta_i ^2 + \sum_{i=1}^t -\alpha_i + \beta_i ^2 = 2 \sum_{i=1}^t \alpha_i - \beta_i ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0635 (9.22)	230	■ 1.000 ● K +0	$\left(t_i - c_1\right)^2 + \left(y_i - c_2\right)^2 = r^2, i = 1, \dots, m,$	$\left(t_i - c_1\right)^2 + \left(y_i - c_2\right)^2 = r^2, i = 1, \dots, m,$	$\left(t_i - c_1\right)^2 + \left(y_i - c_2\right)^2 = r^2, i = 1, \dots, m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0636 (9.23)	230	■ 1.000 ● W +1	$t_i x_1 + y_i x_2 + x_3 = t_i^2 + y_i^2, i = 1, \dots, m,$	$t_i x_1 + y_i x_2 + x_3 = t_i^2 + y_i^2, i = 1, \dots, m,$	$t_i x_1 + y_i x_2 + x_3 = t_i^2 + y_i^2, i = 1, \dots, m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0637	231	■ 1.000 ● N +0	$\sum_{i=1}^3 \ p(x_i) - y_i\ ^2$	$\sum_{i=1}^3 \ p(x_i) - y_i\ ^2$	$\sum_{i=1}^3 \ p(x_i) - y_i\ ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0638	231	■ 1.000 ● N +0	$F(\mathbf{x}) := \ \mathbf{Ax} - \mathbf{b}\ _2^2 + \ \mathbf{x}\ _2^2$	$F(\mathbf{x}) := \ \mathbf{Ax} - \mathbf{b}\ _2^2 + \ \mathbf{x}\ _2^2$	$F(\mathbf{x}) := \ \mathbf{Ax} - \mathbf{b}\ _2^2 + \ \mathbf{x}\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0639	231	■ 1.000 ● N +1	$F(\mathbf{x}) = \mathbf{x}^T \mathbf{Bx} - 2\mathbf{c}^T \mathbf{x} + \mathbf{b}^T \mathbf{b}, \quad \text{where } \mathbf{c} = \mathbf{A}^T \mathbf{b}.$	$F(\mathbf{x}) = \mathbf{x}^T \mathbf{Bx} - 2\mathbf{c}^T \mathbf{x} + \mathbf{b}^T \mathbf{b}, \quad \text{where } \mathbf{c} = \mathbf{A}^T \mathbf{b}.$	$F(\mathbf{x}) = \mathbf{x}^T \mathbf{Bx} - 2\mathbf{c}^T \mathbf{x} + \mathbf{b}^T \mathbf{b}, \quad \text{where } \mathbf{c} = \mathbf{A}^T \mathbf{b}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00640 (9.24)	231	■ 1.000 ● N +1	$\ \boldsymbol{r}(\boldsymbol{x})\ _D^2 := \sum_{i=1}^m r_i(\boldsymbol{x})^2 d_i, \quad \boldsymbol{x} \in \mathbb{R}^n,$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00642	232	■ 1.000 ● K +0	$\boldsymbol{A}^T \boldsymbol{D} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{D} \boldsymbol{b}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00643	232	■ 1.000 ● N +0	$K_2(\boldsymbol{A}^T \boldsymbol{D} \boldsymbol{A}) \leq K_2(\boldsymbol{A}^T \boldsymbol{A}) K_2(\boldsymbol{D}),$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00646 (9.25)	233	■ 1.000 ● N +0	$\begin{aligned} & \boldsymbol{P}_{\mathcal{R}} = \boldsymbol{V}_1 \boldsymbol{V}_1^* = \sum_{j=1}^r \boldsymbol{v}_j \boldsymbol{v}_j^* \in \mathbb{C}^{n \times n} \\ & \boldsymbol{P}_{\mathcal{N}(\boldsymbol{A})} = \boldsymbol{I} - \boldsymbol{A}^\dagger \boldsymbol{A} = \boldsymbol{V}_2 \boldsymbol{V}_2^* = \sum_{j=r+1}^n \boldsymbol{v}_j \boldsymbol{v}_j^* \in \mathbb{C}^{n \times n}. \end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00652	234	■ 1.000 ● K +0	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{C} \end{bmatrix}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00653	235	■ 1.000 ● N +0	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E00655	235	■ 1.000 ● K +0	$ \sigma_i(\epsilon) - \sigma_i(0) \leq \epsilon , \quad i = 1, \dots, n,$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00657	238	■ 1.000 ● N +0	$\bar{\omega}_h := \{(j, h, k) : j, k = 0, 1, \dots, m+1\},$ $\quad \text{where } h = 1/(m+1).$	$\bar{\Omega}_h := \{(jh, kh) : j, k = 0, 1, \dots, m+1\}, \quad \text{where } h = 1/(m+1).$	$\overline{\mathcal{Q}}_h := \{(jh, kh) : j, k = 0, 1, \dots, m+1\}, \quad \text{where } h = 1/(m+1).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00658 (10.2)	239	■ 1.000 ● N +0	$\frac{\partial^2 u(j, h, k)}{\partial x^2} \approx \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2},$ $\frac{\partial^2 u(j, h, k)}{\partial y^2} \approx \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$	$\frac{\partial^2 u(jh, kh)}{\partial x^2} \approx \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2},$ $\frac{\partial^2 u(jh, kh)}{\partial y^2} \approx \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$	$\frac{\partial^2 u(jh, kh)}{\partial x^2} \approx \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2},$ $\frac{\partial^2 u(jh, kh)}{\partial y^2} \approx \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00660 (10.4)	239	■ 1.000 ● N +0	$-\Delta_h v_{j, k} := \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2} + \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$	$-\Delta_h v_{j, k} := \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2} + \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$	$-\Delta_h v_{j, k} := \frac{v_{j-1, k} - 2v_{j, k} + v_{j+1, k}}{h^2} + \frac{v_{j, k-1} - 2v_{j, k} + v_{j, k+1}}{h^2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00662 (10.6)	239	■ 1.000 ● C -4	$\mathbf{V} := \begin{bmatrix} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m},$ $\mathbf{F} := \begin{bmatrix} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m}.$	$\mathbf{V} := \begin{bmatrix} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m},$ $\mathbf{F} := \begin{bmatrix} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m}.$	$\mathbf{V} := \begin{bmatrix} v_{1,1} & \cdots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \cdots & v_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m},$ $\mathbf{F} := \begin{bmatrix} f_{1,1} & \cdots & f_{1,m} \\ \vdots & & \vdots \\ f_{m,1} & \cdots & f_{m,m} \end{bmatrix} \in \mathbb{R}^{m \times m}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00663	239	■ 1.000 ● W +0	$\sum_{i=1}^m \mathbf{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \mathbf{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$	$\sum_{i=1}^m \mathbf{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \mathbf{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$	$\sum_{i=1}^m \mathbf{T}_{j,i} v_{i,k} + \sum_{i=1}^m v_{j,i} \mathbf{T}_{i,k} = -v_{j-1,k} + 2v_{j,k} - v_{j+1,k} - v_{j,k-1} + 2v_{j,k} - v_{j,k+1},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00664	239	■ 1.000 ● W +0	$\operatorname{vec}(\mathbf{B}) := [b_{11}, \dots, b_{m1}, b_{12}, \dots, b_{m2}, \dots, b_{1n}, \dots, b_{mn}]^T \in \mathbb{R}^{mn}$	$\operatorname{vec}(\mathbf{B}) := [b_{11}, \dots, b_{m1}, b_{12}, \dots, b_{m2}, \dots, b_{1n}, \dots, b_{mn}]^T \in \mathbb{R}^{mn}$	$\operatorname{vec}(\mathbf{B}) := [b_{11}, \dots, b_{m1}, b_{12}, \dots, b_{m2}, \dots, b_{1n}, \dots, b_{mn}]^T \in \mathbb{R}^{mn}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00665	240	■ 1.000 ● N +0	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$	$4x_i - x_{i-1} - x_{i+1} - x_{i-m} - x_{i+m} = b_i,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00669 (10.9)	241	■ 1.000 ● N -2	$\mathbf{T}_1 := \operatorname{tridiag}(a, d, a).$	$\mathbf{T}_1 := \operatorname{tridiag}(a, d, a).$	$\mathbf{T}_1 := \operatorname{tridiag}(a, d, a).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00672	242	■ 1.000 ● N +0	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \boldsymbol{A} \boldsymbol{b}_{1,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,s} \\ \boldsymbol{A} \boldsymbol{b}_{2,1} & \boldsymbol{A} \boldsymbol{b}_{2,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \boldsymbol{A} \boldsymbol{b}_{r,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,s} \end{array}\right].$	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \boldsymbol{A} \boldsymbol{b}_{1,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,s} \\ \boldsymbol{A} \boldsymbol{b}_{2,1} & \boldsymbol{A} \boldsymbol{b}_{2,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \boldsymbol{A} \boldsymbol{b}_{r,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,s} \end{array}\right].$	$\boldsymbol{C}=\left[\begin{array}{cccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \boldsymbol{A} \boldsymbol{b}_{1,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,s} \\ \boldsymbol{A} \boldsymbol{b}_{2,1} & \boldsymbol{A} \boldsymbol{b}_{2,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{2,s} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \boldsymbol{A} \boldsymbol{b}_{r,2} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,s} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00673	242	■ 1.000 ● N +1	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$	$\boldsymbol{T}_1=\left[\begin{array}{ccc} d & a & 0 \\ a & d & a \\ 0 & a & d \end{array}\right], \quad \boldsymbol{I}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00675 (10.12)	243	■ 1.000 ● N +1	$\boldsymbol{T}_2=\boldsymbol{T}_1\otimes\boldsymbol{T}_1, \quad \boldsymbol{T}_1,\boldsymbol{I}\in\mathbb{R}^{m\times m}, \quad \boldsymbol{T}_2\in\mathbb{R}^{(m^2)\times(m^2)}.$	$\boldsymbol{T}_2=\boldsymbol{T}_1\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T}_1, \quad \boldsymbol{T}_1,\boldsymbol{I}\in\mathbb{R}^{m\times m}, \quad \boldsymbol{T}_2\in\mathbb{R}^{(m^2)\times(m^2)}.$	$\boldsymbol{T}_2=\boldsymbol{T}_1\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T}_1, \quad \boldsymbol{T}_1,\boldsymbol{I}\in\mathbb{R}^{m\times m}, \quad \boldsymbol{T}_2\in\mathbb{R}^{(m^2)\times(m^2)}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00678	243	■ 1.000 ● N +0	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,t} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{C} \boldsymbol{d}_{1,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} \boldsymbol{d}_{t,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{t,s} \end{array}\right].$	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,t} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{C} \boldsymbol{d}_{1,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} \boldsymbol{d}_{t,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{t,s} \end{array}\right].$	$(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})=\left[\begin{array}{ccc} \boldsymbol{A} \boldsymbol{b}_{1,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{1,t} \\ \vdots & & \vdots \\ \boldsymbol{A} \boldsymbol{b}_{r,1} & \cdots & \boldsymbol{A} \boldsymbol{b}_{r,t} \end{array}\right]\left[\begin{array}{ccc} \boldsymbol{C} \boldsymbol{d}_{1,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{1,s} \\ \vdots & & \vdots \\ \boldsymbol{C} \boldsymbol{d}_{t,1} & \cdots & \boldsymbol{C} \boldsymbol{d}_{t,s} \end{array}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00681	245	■ 1.000 ● W +0	$\begin{aligned} &\left(\boldsymbol{A}\otimes\boldsymbol{I}_s+\boldsymbol{I}_r\otimes\boldsymbol{B}\right)\text{vec}(\boldsymbol{V})=\text{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \left(\boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T+\boldsymbol{I}_r\boldsymbol{V}\boldsymbol{B}^T\right)=\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{B}^T=\boldsymbol{F}. \end{aligned}$	$\begin{aligned} &\left(\boldsymbol{A}\otimes\boldsymbol{I}_s+\boldsymbol{I}_r\otimes\boldsymbol{B}\right)\text{vec}(\boldsymbol{V})=\text{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \left(\boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T+\boldsymbol{I}_r\boldsymbol{V}\boldsymbol{B}^T\right)=\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{B}^T=\boldsymbol{F}. \end{aligned}$	$\begin{aligned} &\left(\boldsymbol{A}\otimes\boldsymbol{I}_s+\boldsymbol{I}_r\otimes\boldsymbol{B}\right)\text{vec}(\boldsymbol{V})=\text{vec}(\boldsymbol{F}) \\ &\Leftrightarrow \left(\boldsymbol{A}\boldsymbol{V}\boldsymbol{I}_s^T+\boldsymbol{I}_r\boldsymbol{V}\boldsymbol{B}^T\right)=\boldsymbol{F} \quad \Leftrightarrow \quad \boldsymbol{A}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{B}^T=\boldsymbol{F}. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00683 (10.16)	246	■ 1.000 ● N +0	$\boldsymbol{s}_j=[\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j=[\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$	$\boldsymbol{s}_j=[\sin(j\pi h), \sin(2j\pi h), \dots, \sin(mj\pi h)]^T.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0685 (10.18)	246	■ 1.000 ● N +0	$\boldsymbol{T}_2\boldsymbol{\left(s_j\right)}\boldsymbol{\otimes}\boldsymbol{s_k}=\boldsymbol{\left(\lambda_j+\lambda_k\right)s_j\otimes s_k}\quad j,\,k=1,\,\ldots,\,m,$	$\boldsymbol{T}_2\left(\boldsymbol{s}_j\otimes\boldsymbol{s}_k\right)=\left(\lambda_j+\lambda_k\right)\left(\boldsymbol{s}_j\otimes\boldsymbol{s}_k\right)\quad j,k=1,\ldots,m,$	$\boldsymbol{T}_2(\boldsymbol{s}_j\otimes\boldsymbol{s}_k)=(\lambda_j+\lambda_k)(\boldsymbol{s}_j\otimes\boldsymbol{s}_k)\quad j,k=1,\ldots,m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0688 (10.20)	246	■ 1.000 ● N +1	$\boldsymbol{A}^{-1}_2=\frac{\cos^2w}{\sin^2w},\quad w:=\frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2\ \boldsymbol{A}^{-1}\ _2=\frac{\cos^2w}{\sin^2w},\quad w:=\frac{\pi}{2(m+1)}$	$\ \boldsymbol{A}\ _2\ \boldsymbol{A}^{-1}\ _2=\frac{\cos^2w}{\sin^2w},\quad w:=\frac{\pi}{2(m+1)}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0689	246	■ 1.000 ● N +0	$\lambda_{j,k}=4-2\cos(2jw)-2\cos(2kw)=4\sin^2(jw)+4\sin^2(kw),\quad j,k=1,\ldots,m.$	$\lambda_{j,k}=4-2\cos(2jw)-2\cos(2kw)=4\sin^2(jw)+4\sin^2(kw),\quad j,k=1,\ldots,m.$	$\lambda_{j,k}=4-2\cos(2jw)-2\cos(2kw)=4\sin^2(jw)+4\sin^2(kw),\quad j,k=1,\ldots,m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0690	246	■ 1.000 ● N +0	$\lambda_{\max}=8\cos^2w,\quad \lambda_{\min}=8\sin^2w.$	$\lambda_{\max}=8\cos^2w,\quad \lambda_{\min}=8\sin^2w.$	$\lambda_{\max}=8\cos^2w,\quad \lambda_{\min}=8\sin^2w.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0696	249	■ 1.000 ● N +0	$\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{V}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{V}\boldsymbol{\otimes}\boldsymbol{F},\quad \boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{U}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{U}\boldsymbol{\otimes}\boldsymbol{F},\quad \boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{V}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{V}\boldsymbol{\otimes}\boldsymbol{F},\quad \boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{U}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{U}\boldsymbol{\otimes}\boldsymbol{F}.$	$\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{V}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{V}\boldsymbol{\otimes}\boldsymbol{F},\quad \boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{U}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{U}\boldsymbol{\otimes}\boldsymbol{F}.$	$\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{V}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{V}\boldsymbol{\otimes}\boldsymbol{F},\quad \boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{I}+\boldsymbol{I}\boldsymbol{\otimes}\boldsymbol{T}\boldsymbol{\otimes}\boldsymbol{U}=\boldsymbol{h}^2\boldsymbol{\otimes}\boldsymbol{U}\boldsymbol{\otimes}\boldsymbol{F}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0697 (10.25)	249	■ 1.000 ● N -1	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=\boldsymbol{h}^4\boldsymbol{F}\quad \text{and}\quad \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b},$	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=\boldsymbol{h}^4\boldsymbol{F}\quad \text{and}\quad \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b},$	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=\boldsymbol{h}^4\boldsymbol{F}\quad \text{and}\quad \boldsymbol{A}\boldsymbol{x}=\boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0700 (11.2)	252	■ 1.000 ● N +0	$\boldsymbol{U}_1=\boldsymbol{D}_1,\quad \boldsymbol{L}_k=\boldsymbol{A}_k\boldsymbol{U}_k^{-1},\quad \boldsymbol{U}_{k+1}=\boldsymbol{D}_{k+1}-\boldsymbol{L}_k\boldsymbol{C}_k,\quad k=1,2,\ldots,m-1.$	$\boldsymbol{U}_1=\boldsymbol{D}_1,\quad \boldsymbol{L}_k=\boldsymbol{A}_k\boldsymbol{U}_k^{-1},\quad \boldsymbol{U}_{k+1}=\boldsymbol{D}_{k+1}-\boldsymbol{L}_k\boldsymbol{C}_k,\quad k=1,2,\ldots,m-1.$	$\boldsymbol{U}_1=\boldsymbol{D}_1,\quad \boldsymbol{L}_k=\boldsymbol{A}_k\boldsymbol{U}_k^{-1},\quad \boldsymbol{U}_{k+1}=\boldsymbol{D}_{k+1}-\boldsymbol{L}_k\boldsymbol{C}_k,\quad k=1,2,\ldots,m-1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0704 (11.4)	253	■ 1.000 ● N +0	$\boldsymbol{S}:=\boldsymbol{\left[s_1,\ldots,s_m\right]}\boldsymbol{\in}\boldsymbol{\mathbb{R}}^{m\times m},\quad \boldsymbol{D}=\boldsymbol{\text{diag}}(\lambda_1,\ldots,\lambda_m).$	$\boldsymbol{S}:=\boldsymbol{\left[s_1,\ldots,s_m\right]}\boldsymbol{\in}\boldsymbol{\mathbb{R}}^{m\times m},\quad \boldsymbol{D}=\boldsymbol{\text{diag}}(\lambda_1,\ldots,\lambda_m).$	$\boldsymbol{S}:=\boldsymbol{\left[s_1,\ldots,s_m\right]}\boldsymbol{\in}\boldsymbol{\mathbb{R}}^{m\times m},\quad \boldsymbol{D}=\boldsymbol{\text{diag}}(\lambda_1,\ldots,\lambda_m).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0705 (2h)	253	■1.000 ● N +0	$\boldsymbol{T} \boldsymbol{S} = \left[\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m \right] = \left[\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m \right] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$	$\boldsymbol{T} \boldsymbol{S} = [\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$	$\boldsymbol{T} \boldsymbol{S} = [\boldsymbol{T} \boldsymbol{s}_1, \dots, \boldsymbol{T} \boldsymbol{s}_m] = [\lambda_1 \boldsymbol{s}_1, \dots, \lambda_m \boldsymbol{s}_m] = \boldsymbol{S} \boldsymbol{D}, \quad \boldsymbol{S}^2 = \boldsymbol{S}^T \boldsymbol{S} = \frac{1}{2h} \boldsymbol{I}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0708	253	■1.000 ● K +0	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$	$x_{j,k} = 4h^4 g_{j,k} / (\lambda_j + \lambda_k) = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad \sigma_j := \lambda_j / 4 = \sin^2(j\pi h / 2).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0710	255	■1.000 ● N +0	$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$	$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$	$w_j = \sum_{k=1}^m \sin\left(\frac{jk\pi}{m+1}\right) v_k, \quad j = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0711 (11.5)	255	■1.000 ● K +0	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$	$\omega_N = \exp^{-2\pi i / N} = \cos(2\pi / N) - i \sin(2\pi / N),$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0712	256	■1.000 ● N +0	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$	$z_{j+1} = \sum_{k=0}^{N-1} \omega_N^{jk} y_{k+1}, \quad j = 0, \dots, N-1,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0713	256	■1.000 ● K +0	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$	$\omega_4 = \exp^{-2\pi i / 4} = \cos(\pi / 2) - i \sin(\pi / 2) = -i$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0715	256	■1.000 ● K +0	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$	$\sin w = \frac{e^{iw} - e^{-iw}}{2i}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0716	256	■1.000 ● N +0	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$	$\boldsymbol{z}^T = [0, \boldsymbol{x}^T, 0, -\boldsymbol{x}_B^T] \in \mathbb{R}^{2m+2}, \quad \boldsymbol{x}_B^T := [x_m, \dots, x_2, x_1].$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0718	256	■1.000 ● N +0	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$	$\omega^{jk} = e^{-\pi i j k / (m+1)}, \quad \omega^{(2m+2-j)k} = e^{-2\pi i} e^{\pi i j k / (m+1)} = e^{\pi i j k / (m+1)}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00720	257	1.000 K +0	$\boldsymbol{P}_{\boldsymbol{N}}=\left[\boldsymbol{e}_{\boldsymbol{1}},\boldsymbol{e}_{\boldsymbol{3}},\ldots,\boldsymbol{e}_{\boldsymbol{N}-\boldsymbol{1}},\boldsymbol{e}_{\boldsymbol{2}},\boldsymbol{e}_{\boldsymbol{4}},\ldots,\boldsymbol{e}_{\boldsymbol{N}}\right],$	$\boldsymbol{P}_N=\left[e_1,e_3,\ldots,e_{N-1},e_2,e_4,\ldots,e_N\right],$	$\boldsymbol{P}_N=\left[e_1,e_3,\ldots,e_{N-1},e_2,e_4,\ldots,e_N\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00724	258	1.000 K +0	$\boldsymbol{F}_{\boldsymbol{2}}=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right],\quad \boldsymbol{D}_{\boldsymbol{2}}\boldsymbol{F}_{\boldsymbol{2}}=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$	$\boldsymbol{F}_2=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right],\quad \boldsymbol{D}_2\boldsymbol{F}_2=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$	$\boldsymbol{F}_2=\left[\begin{array}{rr} 1 & 1 \\ 1 & -1 \end{array}\right],\quad \boldsymbol{D}_2\boldsymbol{F}_2=\left[\begin{array}{rr} 1 & 1 \\ -i & i \end{array}\right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00727 (11.8)	258	1.000 N +1	$\boldsymbol{D}_{\boldsymbol{m}}=\operatorname{diag}\left(1,\omega_{\boldsymbol{N}},\omega_{\boldsymbol{N}}^{\boldsymbol{2}},\ldots,\omega_{\boldsymbol{N}}^{\boldsymbol{m}-\boldsymbol{1}}\right).$	$\boldsymbol{D}_m=\operatorname{diag}\left(1,\omega_N,\omega_N^2,\ldots,\omega_N^{m-1}\right).$	$\boldsymbol{D}_m=\operatorname{diag}(1,\omega_N,\omega_N^2,\ldots,\omega_N^{m-1}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00728	258	1.000 W +0	$\omega_{\boldsymbol{m}}^{\boldsymbol{m}}=1,\omega_{\boldsymbol{2}\boldsymbol{m}}^{\boldsymbol{2}\boldsymbol{k}}=\omega_{\boldsymbol{m}}^{\boldsymbol{k}},\omega_{\boldsymbol{2}\boldsymbol{m}}^{\boldsymbol{m}}=-1,\left(\boldsymbol{F}_{\boldsymbol{m}}\right)_{\boldsymbol{p},\boldsymbol{q}}=\omega_{\boldsymbol{m}}^{\boldsymbol{j}\boldsymbol{k}},\left(\boldsymbol{D}_{\boldsymbol{m}}\boldsymbol{F}_{\boldsymbol{m}}\right)_{\boldsymbol{p},\boldsymbol{q}}=\omega_{\boldsymbol{2}\boldsymbol{m}}^{\boldsymbol{j}}\omega_{\boldsymbol{m}}^{\boldsymbol{j}\boldsymbol{k}},$	$\omega_m^m=1,\omega_{2m}^{2k}=\omega_m^k,\omega_{2m}^m=-1,(\boldsymbol{F}_m)_{p,q}=\omega_m^{jk},(\boldsymbol{D}_m\boldsymbol{F}_m)_{p,q}=\omega_{2m}^j\omega_m^{jk},$	$\omega_m^m=1,\omega_{2m}^{2k}=\omega_m^k,\omega_{2m}^m=-1,(\boldsymbol{F}_m)_{p,q}=\omega_m^{jk},(\boldsymbol{D}_m\boldsymbol{F}_m)_{p,q}=\omega_{2m}^j\omega_m^{jk},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00730	258	1.000 N +0	$\boldsymbol{w}_{\boldsymbol{1}}^{\boldsymbol{T}}=\left[y_{\boldsymbol{1}},y_{\boldsymbol{3}},\ldots,y_{\boldsymbol{2}\boldsymbol{m}-\boldsymbol{1}}\right],\quad \boldsymbol{w}_{\boldsymbol{2}}^{\boldsymbol{T}}=\left[y_{\boldsymbol{2}},y_{\boldsymbol{4}},\ldots,y_{\boldsymbol{2}\boldsymbol{m}}\right].$	$\boldsymbol{w}_1^T=\left[y_1,y_3,\ldots,y_{2m-1}\right],\quad \boldsymbol{w}_2^T=\left[y_2,y_4,\ldots,y_{2m}\right].$	$\boldsymbol{w}_1^T=\left[y_1,y_3,\ldots,y_{2m-1}\right],\quad \boldsymbol{w}_2^T=\left[y_2,y_4,\ldots,y_{2m}\right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00732	259	1.000 N +0	$\boldsymbol{q}_{\boldsymbol{1}}=\boldsymbol{F}_{\boldsymbol{m}}\boldsymbol{w}_{\boldsymbol{1}},\quad \text{and}\quad \boldsymbol{q}_{\boldsymbol{2}}=\boldsymbol{D}_{\boldsymbol{m}}\left(\boldsymbol{F}_{\boldsymbol{m}}\boldsymbol{w}_{\boldsymbol{2}}\right).$	$\boldsymbol{q}_1=\boldsymbol{F}_m\boldsymbol{w}_1,\quad \text{and}\quad \boldsymbol{q}_2=\boldsymbol{D}_m\left(\boldsymbol{F}_m\boldsymbol{w}_2\right).$	$\boldsymbol{q}_1=\boldsymbol{F}_m\boldsymbol{w}_1,\quad \text{and}\quad \boldsymbol{q}_2=\boldsymbol{D}_m(\boldsymbol{F}_m\boldsymbol{w}_2).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00733	260	1.000 N +0	$\frac{8\boldsymbol{N}^{\boldsymbol{2}}}{5\boldsymbol{N}\log_{\boldsymbol{2}}\boldsymbol{N}}\approx 84000.$	$\frac{8N^2}{5N\log_2N}\approx 84000.$	$\frac{8N^2}{5N\log_2N}\approx 84000.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00734	260	1.000 W +0	$8\,\gamma\,m(m+1)\log_{\boldsymbol{2}}(2\,m+2)\approx 8\,\gamma\,m^{\boldsymbol{2}}\log_{\boldsymbol{2}}\boldsymbol{m}=4\gamma n\log_{\boldsymbol{2}}\boldsymbol{n},$	$8\gamma m(m+1)\log_2(2m+2)\approx 8\gamma m^2\log_2m=4\gamma n\log_2n,$	$8\gamma m(m+1)\log_2(2m+2)\approx 8\gamma m^2\log_2m=4\gamma n\log_2n,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00735	261	1.000 C +4	$\boldsymbol{v}_{\boldsymbol{i}\boldsymbol{j}}=\frac{1}{(\boldsymbol{m}+\boldsymbol{1})^{\boldsymbol{4}}}\sum_{\boldsymbol{p}=\boldsymbol{1}}^{\boldsymbol{m}}\sum_{\boldsymbol{r}=\boldsymbol{1}}^{\boldsymbol{m}}\frac{\sin\left(\frac{\boldsymbol{i}\boldsymbol{p}\pi}{\boldsymbol{m}+\boldsymbol{1}}\right)\sin\left(\frac{\boldsymbol{j}\boldsymbol{r}\pi}{\boldsymbol{m}+\boldsymbol{1}}\right)\sin\left(\frac{\boldsymbol{k}\boldsymbol{p}\pi}{\boldsymbol{m}+\boldsymbol{1}}\right)\sin\left(\frac{\boldsymbol{l}\boldsymbol{r}\pi}{\boldsymbol{m}+\boldsymbol{1}}\right)}{\left[\sin\left(\frac{\boldsymbol{p}\pi}{2(\boldsymbol{m}+\boldsymbol{1})}\right)\right]^{\boldsymbol{2}}+\left[\sin\left(\frac{\boldsymbol{r}\pi}{2(\boldsymbol{m}+\boldsymbol{1})}\right)\right]^{\boldsymbol{2}}}f_{\boldsymbol{k},\boldsymbol{l}}.$	$v_{ij}=\frac{1}{(m+1)^4}\sum_{p=1}^m\sum_{r=1}^m\sum_{k=1}^m\sum_{l=1}^m\frac{\sin\left(\frac{ip\pi}{m+1}\right)\sin\left(\frac{jr\pi}{m+1}\right)\sin\left(\frac{kp\pi}{m+1}\right)\sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2+\left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2}f_{k,l}.$	$v_{ij}=\frac{1}{(m+1)^4}\sum_{p=1}^m\sum_{r=1}^m\sum_{k=1}^m\sum_{l=1}^m\frac{\sin\left(\frac{ip\pi}{m+1}\right)\sin\left(\frac{jr\pi}{m+1}\right)\sin\left(\frac{kp\pi}{m+1}\right)\sin\left(\frac{lr\pi}{m+1}\right)}{\left[\sin\left(\frac{p\pi}{2(m+1)}\right)\right]^2+\left[\sin\left(\frac{r\pi}{2(m+1)}\right)\right]^2}f_{k,l}.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0737 (11.9)	261	■ 1.000 ● N +0	$\left(\boldsymbol{T}+\lambda_j\boldsymbol{I}\right)\boldsymbol{x}_j=\boldsymbol{c}_j\quad j=1,\ldots,m,$	$(\boldsymbol{T}+\lambda_j\boldsymbol{I})\boldsymbol{x}_j=\boldsymbol{c}_j\quad j=1,\ldots,m,$	$(\boldsymbol{T}+\lambda_j\boldsymbol{I})\boldsymbol{x}_j=\boldsymbol{c}_j\quad j=1,\ldots,m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0738	261	■ 1.000 ● N +0	$\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}-\frac{1}{6}\boldsymbol{T}\boldsymbol{V}\boldsymbol{T}=h^2\mu\boldsymbol{F},$	$\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}-\frac{1}{6}\boldsymbol{T}\boldsymbol{V}\boldsymbol{T}=h^2\mu\boldsymbol{F},$	$\boldsymbol{T}\boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}-\frac{1}{6}\boldsymbol{T}\boldsymbol{V}\boldsymbol{T}=h^2\mu\boldsymbol{F},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0741	262	■ 1.000 ● K +0	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=h^4\boldsymbol{F}.$	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=h^4\boldsymbol{F}.$	$\boldsymbol{T}^2\boldsymbol{U}+2\boldsymbol{T}\boldsymbol{U}\boldsymbol{T}+\boldsymbol{U}\boldsymbol{T}^2=h^4\boldsymbol{F}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0742	262	■ 1.000 ● N +0	$\boldsymbol{D}^2\boldsymbol{X}+2\boldsymbol{D}\boldsymbol{X}\boldsymbol{D}+\boldsymbol{X}\boldsymbol{D}^2=4h^6\boldsymbol{G},$ where $\boldsymbol{G}=\boldsymbol{S}\boldsymbol{F}\boldsymbol{S},$	$\boldsymbol{D}^2\boldsymbol{X}+2\boldsymbol{D}\boldsymbol{X}\boldsymbol{D}+\boldsymbol{X}\boldsymbol{D}^2=4h^6\boldsymbol{G},$ where $\boldsymbol{G}=\boldsymbol{S}\boldsymbol{F}\boldsymbol{S},$	$\boldsymbol{D}^2\boldsymbol{X}+2\boldsymbol{D}\boldsymbol{X}\boldsymbol{D}+\boldsymbol{X}\boldsymbol{D}^2=4h^6\boldsymbol{G},$ where $\boldsymbol{G}=\boldsymbol{S}\boldsymbol{F}\boldsymbol{S},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0743	262	■ 1.000 ● W +0	$x_{j,k}=\frac{h^6g_{j,k}}{4(\sigma_j+\sigma_k)^2},$ where $\sigma_j=\sin^2((j\pi h)/2)$ for $j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^6g_{j,k}}{4(\sigma_j+\sigma_k)^2},$ where $\sigma_j=\sin^2((j\pi h)/2)$ for $j,k=1,2,\ldots,m.$	$x_{j,k}=\frac{h^6g_{j,k}}{4(\sigma_j+\sigma_k)^2},$ where $\sigma_j=\sin^2((j\pi h)/2)$ for $j,k=1,2,\ldots,m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0744 (12.1)	266	■ 1.000 ● W +0	$\boldsymbol{x}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}(j)+b_i\right)/a_{ii},\quad i=1,2,\ldots,n.$	$\boldsymbol{x}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}(j)+b_i\right)/a_{ii},\quad i=1,2,\ldots,n.$	$\boldsymbol{x}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}(j)+b_i\right)/a_{ii},\quad i=1,2,\ldots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0745 (12.2)	266	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_k(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0746 (12.3)	266	■ 1.000 ● W +0	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$	$\boldsymbol{x}_{k+1}(i)=\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii},$ for $i=1,2,\ldots,n.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0747 (12.4)	266	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_k(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_k(i).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0748 (12.5)	267	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$	$\boldsymbol{x}_{k+1}(i)=\omega\left(-\sum_{j=1}^{i-1}a_{ij}\boldsymbol{x}_{k+1/2}(j)-\sum_{j=i+1}^na_{ij}\boldsymbol{x}_{k+1}(j)+b_i\right)/a_{ii}+(1-\omega)\boldsymbol{x}_{k+1/2}(i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0749	267	■ 1.000 ● N +0	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\dots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\dots,m,$	$4\boldsymbol{v}(i,j)-\boldsymbol{v}(i-1,j)-\boldsymbol{v}(i+1,j)-\boldsymbol{v}(i,j-1)-\boldsymbol{v}(i,j+1)=h^2f_{i,j},\quad i,j=1,\dots,m,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0750 (12.6)	267	■ 1.000 ● N +0	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$	$\boldsymbol{v}(i,j)=\left(\boldsymbol{v}(i-1,j)+\boldsymbol{v}(i+1,j)+\boldsymbol{v}(i,j-1)+\boldsymbol{v}(i,j+1)+e_{i,j}\right)/4,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0755 (12.10)	270	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c},\quad \boldsymbol{G}=\boldsymbol{I}-\boldsymbol{M}^{-1}\boldsymbol{A},\quad \boldsymbol{c}=\boldsymbol{M}^{-1}\boldsymbol{b}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0756	270	■ 1.000 ● W +0	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_k=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$	$\boldsymbol{x}=\lim_{k\rightarrow\infty}\boldsymbol{x}_{k+1}=\lim_{k\rightarrow\infty}(\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c})=\boldsymbol{G}\lim_{k\rightarrow\infty}\boldsymbol{x}_k+\boldsymbol{c}=\boldsymbol{G}\boldsymbol{x}+\boldsymbol{c}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0758 (12.12)	271	■ 1.000 ● N +0	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$	$\boldsymbol{M}_J=\boldsymbol{D},\quad \boldsymbol{M}_1=\boldsymbol{D}-\boldsymbol{A}_L,\quad \boldsymbol{M}_\omega=\omega^{-1}\boldsymbol{D}-\boldsymbol{A}_L.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0759 (12.13)	271	■ 1.000 ● N +0	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$	$\boldsymbol{D}\boldsymbol{x}_{k+1}=\omega(\boldsymbol{A}_L\boldsymbol{x}_{k+1}+\boldsymbol{A}_R\boldsymbol{x}_k+\boldsymbol{b})+(1-\omega)\boldsymbol{D}\boldsymbol{x}_k.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0760	271	■ 1.000 ● N +0	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] \left[\begin{array}{l} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{l} 1 \\ 1 \end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0761	271	■ 1.000 ● N +0	$\boldsymbol{A}_L = \left[\begin{array}{ll} 0 & 0 \\ 1 & 0 \end{array}\right], \quad \boldsymbol{D} = \left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{A}_R = \left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right],$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$	$\boldsymbol{A}_L = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{A}_R = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0762	271	■ 1.000 ● K +0	$\boldsymbol{M}_J = \boldsymbol{D} = \left[\begin{array}{ll} 2 & 0 \\ 0 & 2 \end{array}\right], \quad \boldsymbol{M}_\omega = \omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L = \left[\begin{array}{ll} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{array}\right].$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_\omega = \omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$	$\boldsymbol{M}_J = \boldsymbol{D} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{M}_\omega = \omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L = \begin{bmatrix} 2\omega^{-1} & 0 \\ -1 & 2\omega^{-1} \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0764 (12.15)	271	■ 1.000 ● K +0	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \left[\begin{array}{ll} 0 & 1/2 \\ 1/2 & 0 \end{array}\right], \quad \boldsymbol{G}_1 = \left[\begin{array}{ll} 0 & 1/2 \\ 0 & 1/4 \end{array}\right].$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 0 \end{bmatrix}, \quad \boldsymbol{G}_1 = \begin{bmatrix} 0 & 1/2 \\ 0 & 1/4 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0771 (12.18)	273	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha (\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha (\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k).$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha (\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0773	274	■ 1.000 ● N -1	$\rho_{\alpha} := \rho(\boldsymbol{I} - \alpha \boldsymbol{A}) := \max_j 1 - \alpha \lambda_j = \begin{cases} 1 - \alpha \lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha \lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha \lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$	$\rho_{\alpha} := \rho(\boldsymbol{I} - \alpha \boldsymbol{A}) := \max_j 1 - \alpha \lambda_j = \begin{cases} 1 - \alpha \lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha \lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha \lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$	$\rho_{\alpha} := \rho(\boldsymbol{I} - \alpha \boldsymbol{A}) := \max_j 1 - \alpha \lambda_j = \begin{cases} 1 - \alpha \lambda_1, & \text{if } \alpha \leq 0 \\ 1 - \alpha \lambda_n, & \text{if } 0 < \alpha \leq \alpha_o \\ \alpha \lambda_1 - 1, & \text{if } \alpha > \alpha_o, \end{cases}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0774	274	■ 1.000 ● N +1	$\rho_{\alpha_o} = \alpha_o \lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$	$\rho_{\alpha_o} = \alpha_o \lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$	$\rho_{\alpha_o} = \alpha_o \lambda_1 - 1 = \frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} = \frac{\kappa - 1}{\kappa + 1} < 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0776 (12.21)	275	■ 1.000 ● K +0	$\boldsymbol{G}_\omega = (\boldsymbol{I} - \omega \boldsymbol{L})^{-1} (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}).$	$\boldsymbol{G}_\omega = (\boldsymbol{I} - \omega \boldsymbol{L})^{-1} (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}).$	$\boldsymbol{G}_\omega = (\boldsymbol{I} - \omega \boldsymbol{L})^{-1} (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0777 (12.22)	275	■ 1.000 ● K +0	$\omega^{-1}(2-\omega) 1-\lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	$\omega^{-1}(2-\omega) 1-\lambda ^2\boldsymbol{x}^*\boldsymbol{D}\boldsymbol{x} = (1- \lambda ^2)\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x},$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0778	275	■ 1.000 ● N +0	$\boldsymbol{G}_{\omega}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_{\omega}^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$	$\boldsymbol{G}_{\omega}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{M}_{\omega}^{-1}\boldsymbol{A})\boldsymbol{x} = \boldsymbol{x} - (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)^{-1}\boldsymbol{A}\boldsymbol{x}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0779 (12.23)	276	■ 1.000 ● N +0	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1-\lambda)\boldsymbol{x}.$	$\boldsymbol{A}\boldsymbol{x} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L)\boldsymbol{y}, \quad \boldsymbol{y} := (1-\lambda)\boldsymbol{x}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0780 (12.24)	276	■ 1.000 ● N +0	$\boldsymbol{E}\boldsymbol{y} = \lambda\boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{E} := \omega^{-1}\boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}.$	$\boldsymbol{E}\boldsymbol{y} = \lambda\boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{E} := \omega^{-1}\boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D} = \omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0781	276	■ 1.000 ● K +0	$\boldsymbol{E}\boldsymbol{y} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A})\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{A}\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} - (1-\lambda)\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{A}\boldsymbol{x}.$	$\boldsymbol{E}\boldsymbol{y} = (\omega^{-1}\boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A})\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} - \boldsymbol{A}\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} - (1-\lambda)\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{A}\boldsymbol{x}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0784	276	■ 1.000 ● K +0	$\lambda_{j,k} = 4 - 2\cos(j\pi h) - 2\cos(k\pi h), \quad j, k = 1, \dots, m, h = 1/(m+1).$	$\lambda_{j,k} = 4 - 2\cos(j\pi h) - 2\cos(k\pi h), \quad j, k = 1, \dots, m, h = 1/(m+1).$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0785 (12.25)	276	■ 1.000 ● N +0	$\mu_{j,k} = 1 - \frac{1}{4}\lambda_{j,k} = \frac{1}{2}\cos(j\pi h) + \frac{1}{2}\cos(k\pi h), \quad j, k = 1, \dots, m.$	$\mu_{j,k} = 1 - \frac{1}{4}\lambda_{j,k} = \frac{1}{2}\cos(j\pi h) + \frac{1}{2}\cos(k\pi h), \quad j, k = 1, \dots, m.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0786 (12.26)	277	■ 1.000 ● N +0	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{G}_J\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2 = \cos^k(\pi h) \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2, \quad k = 0, 1, 2, \dots$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{G}_J\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2 = \cos^k(\pi h) \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2, \quad k = 0, 1, 2, \dots$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0788 (12.28)	277	■ 1.000 ● K +0	$\omega^* := \frac{2}{1 + \sqrt{1 - \beta^2}} > 1$.	$\omega^* := \frac{2}{1 + \sqrt{1 - \beta^2}} > 1.$	$\omega^* := \frac{2}{1 + \sqrt{1 - \beta^2}} > 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0790 (12.30)	277	■ 1.000 ● W +2	$\omega^* = \frac{2}{1 + \sin(\pi h)}$, $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1 = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)}$, $h = \frac{1}{m + 1}$.	$\omega^* = \frac{2}{1 + \sin(\pi h)}, \quad \rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1 = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)}, \quad h = \frac{1}{m + 1}.$	$\omega^* = \frac{2}{1 + \sin(\pi h)}, \quad \rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1 = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)}, \quad h = \frac{1}{m + 1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0792 (12.31)	278	■ 1.000 ● N +0	$\tilde{k} := \frac{s \log(10)}{\eta}$	$\tilde{k} := \frac{s \log(10)}{\eta}$	$\tilde{k} := \frac{s \log(10)}{\eta}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0794	279	■ 1.000 ● N +0	$\tilde{k}_n = \frac{2 \log(10)}{\pi^2} s n + O(n^{-1}) = O(n)$.	$\tilde{k}_n = \frac{2 \log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$	$\tilde{k}_n = \frac{2 \log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0795	279	■ 1.000 ● N +0	$\tilde{k}_n = \frac{\log(10)}{\pi^2} s n + O(n^{-1}) = O(n)$.	$\tilde{k}_n = \frac{\log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$	$\tilde{k}_n = \frac{\log(10)s}{\pi^2} n + O(n^{-1}) = O(n).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0796	279	■ 1.000 ● N +0	$\tilde{k}_n = \frac{\log(10)}{2\pi} s \sqrt{n} + O(n^{-1/2}) = O(\sqrt{n})$.	$\tilde{k}_n = \frac{\log(10)s}{2\pi} \sqrt{n} + O(n^{-1/2}) = O(\sqrt{n}).$	$\tilde{k}_n = \frac{\log(10)s}{2\pi} \sqrt{n} + O(n^{-1/2}) = O(\sqrt{n}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0797 (12.32)	280	■ 1.000 ● K +0	$\left \frac{\boldsymbol{x}_k - \boldsymbol{x}_{k-1}}{\boldsymbol{G}} \right \geq \frac{1 - \ \boldsymbol{G}\ }{\ \boldsymbol{G}\ } \ \boldsymbol{x}_k - \boldsymbol{x}\ $, $k \geq 1$.	$\ \boldsymbol{x}_k - \boldsymbol{x}_{k-1}\ \geq \frac{1 - \ \boldsymbol{G}\ }{\ \boldsymbol{G}\ } \ \boldsymbol{x}_k - \boldsymbol{x}\ , \quad k \geq 1.$	$\ \boldsymbol{x}_k - \boldsymbol{x}_{k-1}\ \geq \frac{1 - \ \boldsymbol{G}\ }{\ \boldsymbol{G}\ } \ \boldsymbol{x}_k - \boldsymbol{x}\ , \quad k \geq 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0799 (12.33)	280	■ 1.000 ● N +0	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{M}^{-1} \boldsymbol{r}_k$, $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$.	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{M}^{-1} \boldsymbol{r}_k, \quad \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k.$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{M}^{-1} \boldsymbol{r}_k, \quad \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k.$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00800	281	1.000 ● N +0	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0} \iff \rho(\boldsymbol{A}) < 1,$	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0} \iff \rho(\boldsymbol{A}) < 1,$	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0} \iff \rho(\boldsymbol{A}) < 1,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00801	281	1.000 ● N +0	$\boldsymbol{D}_{\{t\}} \boldsymbol{R} \boldsymbol{D}_{\{t\}}^{-1} = \left[\begin{array}{ccc} \lambda_1 & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_2 & t^{-1} r_{23} \\ 0 & 0 & \lambda_3 \end{array} \right].$	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} = \begin{bmatrix} \lambda_1 & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_2 & t^{-1} r_{23} \\ 0 & 0 & \lambda_3 \end{bmatrix}.$	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1} = \begin{bmatrix} \lambda_1 & t^{-1} r_{12} & t^{-2} r_{13} \\ 0 & \lambda_2 & t^{-1} r_{23} \\ 0 & 0 & \lambda_3 \end{bmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00803 (12.34)	282	1.000 ● N +0	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{A}^k \right\ ^{1/k} = \rho(\boldsymbol{A}).$	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{A}^k \right\ ^{1/k} = \rho(\boldsymbol{A}).$	$\lim_{k \rightarrow \infty} \ \boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00804	282	1.000 ● N +0	$\left\ \boldsymbol{A}^k \right\ = \left\ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k \right\ = (\rho(\boldsymbol{A}) + \epsilon)^k \left\ \boldsymbol{B}^k \right\ < (\rho(\boldsymbol{A}) + \epsilon)^k.$	$\left\ \boldsymbol{A}^k \right\ = \left\ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k \right\ = (\rho(\boldsymbol{A}) + \epsilon)^k \left\ \boldsymbol{B}^k \right\ < (\rho(\boldsymbol{A}) + \epsilon)^k.$	$\ \boldsymbol{A}^k\ = \ (\rho(\boldsymbol{A}) + \epsilon)^k \boldsymbol{B}^k\ = (\rho(\boldsymbol{A}) + \epsilon)^k \ \boldsymbol{B}^k\ < (\rho(\boldsymbol{A}) + \epsilon)^k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00805	282	1.000 ● N +0	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00807	283	1.000 ● W +0	$\left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \ \boldsymbol{B}\ ^k \leq \ \boldsymbol{B}\ ^{m+1} \sum_{k=0}^{\infty} \ \boldsymbol{B}\ ^k = \frac{\ \boldsymbol{B}\ ^{m+1}}{1 - \ \boldsymbol{B}\ }.$	$\ S_l - S_m\ = \left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \ \boldsymbol{B}\ ^k \leq \ \boldsymbol{B}\ ^{m+1} \sum_{k=0}^{\infty} \ \boldsymbol{B}\ ^k = \frac{\ \boldsymbol{B}\ ^{m+1}}{1 - \ \boldsymbol{B}\ }.$	$\ S_l - S_m\ = \left\ \sum_{k=m+1}^l \boldsymbol{B}^k \right\ \leq \sum_{k=m+1}^l \ \boldsymbol{B}\ ^k \leq \ \boldsymbol{B}\ ^{m+1} \sum_{k=0}^{\infty} \ \boldsymbol{B}\ ^k = \frac{\ \boldsymbol{B}\ ^{m+1}}{1 - \ \boldsymbol{B}\ }.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00808 (12.36)	283	1.000 ● N +0	$\left(\sum_{k=0}^m \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \dots + \boldsymbol{B}^m - (\boldsymbol{B} + \dots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}.$	$\left(\sum_{k=0}^m \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \dots + \boldsymbol{B}^m - (\boldsymbol{B} + \dots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}.$	$(\sum_{k=0}^m \boldsymbol{B}^k) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I} + \boldsymbol{B} + \dots + \boldsymbol{B}^m - (\boldsymbol{B} + \dots + \boldsymbol{B}^{m+1}) = \boldsymbol{I} - \boldsymbol{B}^{m+1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__E00809 (12.37)	283	1.000 ● N +0	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$	$\frac{1}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \mu v_{i,j}, \quad i, j = 1, \dots, m,$

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0810 (12.38)	283	■ 1.000 ● K +0	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}.$	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}.$	$(\omega \boldsymbol{R} + \lambda \omega \boldsymbol{L}) \boldsymbol{w} = (\lambda + \omega - 1) \boldsymbol{w}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0811 (12.39)	284	■ 1.000 ● N +1	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = (\lambda + \omega - 1) w_{i,j},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0812 (12.40)	284	■ 1.000 ● K +0	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$	$\mu := \frac{\lambda + \omega - 1}{\omega \lambda^{1/2}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0813 (12.41)	284	■ 1.000 ● K +0	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$	$\mu \omega \lambda^{1/2} = \lambda + \omega - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0814	284	■ 1.000 ● K +0	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}.$	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}.$	$\frac{\omega}{4} (v_{i-1,j} + v_{i,j-1} + v_{i+1,j} + v_{i,j+1}) = \lambda^{-1/2} (\lambda + \omega - 1) v_{i,j}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0817	284	■ 1.000 ● N +1	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j}$	$\frac{\omega}{4} (\lambda w_{i-1,j} + \lambda w_{i,j-1} + w_{i+1,j} + w_{i,j+1}) = \omega \mu \lambda^{1/2} w_{i,j},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0818 (12.42)	285	■ 1.000 ● K +0	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2.$	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2.$	$\lambda(\mu) := \frac{1}{4} \left(\omega \mu \pm \sqrt{(\omega \mu)^2 - 4(\omega - 1)} \right)^2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0821 (12.43)	285	■ 1.000 ● N +0	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}.$	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}.$	$\omega = \tilde{\omega}(\mu) := 2 \frac{1 - \sqrt{1 - \mu^2}}{\mu^2} = \frac{2}{1 + \sqrt{1 - \mu^2}}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0822	285	■ 1.000 ● K +0	$ \lambda(\mu) =\frac{1}{4}\left((\omega\mu)^2+4(\omega-1)-(\omega\mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$	$ \lambda(\mu) =\frac{1}{4}\left((\omega\mu)^2+4(\omega-1)-(\omega\mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$	$ \lambda(\mu) =\frac{1}{4}\left((\omega\mu)^2+4(\omega-1)-(\omega\mu)^2\right)=\omega-1, \quad \tilde{\omega}(\mu)<\omega<2.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0823 (12.44)	285	■ 1.000 ● K +0	$\lambda(\mu)=\frac{1}{4}\left((\omega\mu+\sqrt{(\omega\mu)^2-4(\omega-1)})^2-4(\omega-1)\right)^2, \quad \theta<\omega\leq\tilde{\omega}(\mu).$	$\lambda(\mu)=\frac{1}{4}\left((\omega\mu+\sqrt{(\omega\mu)^2-4(\omega-1)})^2\right), \quad 0<\omega\leq\tilde{\omega}(\mu).$	$\lambda(\mu)=\frac{1}{4}\left((\omega\mu+\sqrt{(\omega\mu)^2-4(\omega-1)})^2\right), \quad 0<\omega\leq\tilde{\omega}(\mu).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0824	285	■ 1.000 ● N +0	$\frac{d}{d\omega}\left(\omega\beta+\sqrt{(\omega\beta)^2-4(\omega-1)}\right)=\frac{\beta\sqrt{(\omega\beta)^2-4(\omega-1)}+\omega\beta^2-2}{\sqrt{(\omega\beta)^2-4(\omega-1)}}.$	$\frac{d}{d\omega}\left(\omega\beta+\sqrt{(\omega\beta)^2-4(\omega-1)}\right)=\frac{\beta\sqrt{(\omega\beta)^2-4(\omega-1)}+\omega\beta^2-2}{\sqrt{(\omega\beta)^2-4(\omega-1)}}.$	$\frac{d}{d\omega}\left(\omega\beta+\sqrt{(\omega\beta)^2-4(\omega-1)}\right)=\frac{\beta\sqrt{(\omega\beta)^2-4(\omega-1)}+\omega\beta^2-2}{\sqrt{(\omega\beta)^2-4(\omega-1)}}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0826	286	■ 1.000 ● N +0	$\rho\left(\boldsymbol{G}_1\right)=\left a_{21}a_{12}/a_{11}a_{22}\right .$	$\rho(\boldsymbol{G}_1)= a_{21}a_{12}/a_{11}a_{22} .$	$\rho(\boldsymbol{G}_1)= a_{21}a_{12}/a_{11}a_{22} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0829 (12.45)	287	■ 1.000 ● N +0	$\boldsymbol{M}\boldsymbol{x}_{k+1}=\boldsymbol{N}\boldsymbol{x}_k+\boldsymbol{b},$	$\boldsymbol{M}\boldsymbol{x}_{k+1}=\boldsymbol{N}\boldsymbol{x}_k+\boldsymbol{b},$	$\boldsymbol{M}\boldsymbol{x}_{k+1}=\boldsymbol{N}\boldsymbol{x}_k+\boldsymbol{b},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0832 (12.48)	288	■ 1.000 ● N +0	$x(i)=b(i)-\sum_{j\neq i}a(i,j)x(j)$	$x(i)=b(i)-\sum_{j\neq i}a(i,j)x(j)$	$x(i)=b(i)-\sum_{j\neq i}a(i,j)x(j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0838 (13.4)	291	■ 1.000 ● W +0	$Q(x,y):=\frac{1}{2}\left[\begin{array}{l}x\\y\end{array}\right]\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]\left[\begin{array}{l}x\\y\end{array}\right]=x^2-xy+y^2$	$Q(x,y):=\frac{1}{2}\left[\begin{array}{cc}x&y\end{array}\right]\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]\left[\begin{array}{c}x\\y\end{array}\right]=x^2-xy+y^2$	$Q(x,y):=\frac{1}{2}\left[\begin{array}{cc}x&y\end{array}\right]\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]\left[\begin{array}{c}x\\y\end{array}\right]=x^2-xy+y^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0839 (13.5)	291	■ 1.000 ● N +1	$Q(\boldsymbol{y}+\varepsilon\boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon\boldsymbol{h}^T\boldsymbol{r}(\boldsymbol{y})+\frac{1}{2}\varepsilon^2\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}, \text{ where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{y}.$	$Q(\boldsymbol{y}+\varepsilon\boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon\boldsymbol{h}^T\boldsymbol{r}(\boldsymbol{y})+\frac{1}{2}\varepsilon^2\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}, \text{ where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{y}.$	$Q(\boldsymbol{y}+\varepsilon\boldsymbol{h})=Q(\boldsymbol{y})-\varepsilon\boldsymbol{h}^T\boldsymbol{r}(\boldsymbol{y})+\frac{1}{2}\varepsilon^2\boldsymbol{h}^T\boldsymbol{A}\boldsymbol{h}, \text{ where } \boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{y}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00841	293	■ 1.000 ● N +0	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right) .$	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right) .$	$Q\left(\boldsymbol{x}_{k+1}\right)=Q\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\min _{\alpha \in \mathbb{R}} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right) .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00842 (13.7)	293	■ 1.000 ● W +0	$\alpha_k:=\frac{\boldsymbol{p}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} .$	$\alpha_k:=\frac{\boldsymbol{p}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} .$	$\alpha_k:=\frac{\boldsymbol{p}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00843 (13.8)	293	■ 1.000 ● W +0	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\left(\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}\right) \boldsymbol{r}_k .$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\left(\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}\right) \boldsymbol{r}_k .$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\left(\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}\right) \boldsymbol{r}_k .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00844 (13.9)	293	■ 1.000 ● W +0	$\begin{aligned} \boldsymbol{p}_k &= \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= \left(\boldsymbol{p}_k^T \boldsymbol{r}_k\right) / \left(\boldsymbol{p}_k^T \boldsymbol{t}_k\right), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{p}_k &= \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= \left(\boldsymbol{p}_k^T \boldsymbol{r}_k\right) / \left(\boldsymbol{p}_k^T \boldsymbol{t}_k\right), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k. \end{aligned}$	$\begin{aligned} \boldsymbol{p}_k &= \boldsymbol{r}_k, \boldsymbol{t}_k = \boldsymbol{A} \boldsymbol{p}_k, \\ \alpha_k &= \left(\boldsymbol{p}_k^T \boldsymbol{r}_k\right) / \left(\boldsymbol{p}_k^T \boldsymbol{t}_k\right), \\ \boldsymbol{x}_{k+1} &= \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ \boldsymbol{r}_{k+1} &= \boldsymbol{r}_k - \alpha_k \boldsymbol{t}_k. \end{aligned}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00845 (13.10)	293	■ 1.000 ● N +0	$\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A}\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\boldsymbol{r}_k-\alpha_k \boldsymbol{A} \boldsymbol{p}_k .$	$\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A}\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\boldsymbol{r}_k-\alpha_k \boldsymbol{A} \boldsymbol{p}_k .$	$\boldsymbol{r}_{k+1}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{k+1}=\boldsymbol{b}-\boldsymbol{A}\left(\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k\right)=\boldsymbol{r}_k-\alpha_k \boldsymbol{A} \boldsymbol{p}_k .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00853 (13.17)	296	■ 1.000 ● W +0	$\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+\beta_k \boldsymbol{p}_k, \quad \beta_k:=\frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k} .$	$\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+\beta_k \boldsymbol{p}_k, \quad \beta_k:=\frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k} .$	$\boldsymbol{p}_{k+1}:=\boldsymbol{r}_{k+1}+\beta_k \boldsymbol{p}_k, \quad \beta_k:=\frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00856	298	■ 1.000 ● W -2	$\boldsymbol{A}=\operatorname{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m+\boldsymbol{I}_m \otimes \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{\left(m^2\right) \times\left(m^2\right)}$	$\boldsymbol{A}=\operatorname{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m+\boldsymbol{I}_m \otimes \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{\left(m^2\right) \times\left(m^2\right)}$	$\boldsymbol{A}=\operatorname{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m+\boldsymbol{I}_m \otimes \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{\left(m^2\right) \times\left(m^2\right)} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00859	300	■ 1.000 ● C -3	$\boldsymbol{T}_2 := \operatorname{tridiag}_m(a, d, a) \otimes \boldsymbol{I}_m + \boldsymbol{I}_m \otimes \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$	$T_2 := \operatorname{tridiag}_m(a, d, a) \otimes I_m + I_m \otimes \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{(m^2) \times (m^2)}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00860 (13.21)	300	■ 1.000 ● N +0	$\lambda_{j,k} = 2d + 2a \cos(j\pi h) + 2a \cos(k\pi h), \quad j, k = 1, \dots, m.$	$\lambda_{j,k} = 2d + 2a \cos(j\pi h) + 2a \cos(k\pi h), \quad j, k = 1, \dots, m.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00864	301	■ 1.000 ● W +0	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$	$\mathbb{W}_0 \subset \mathbb{W}_1 \subset \mathbb{W}_2 \subset \dots \subset \mathbb{W}_n \subset \mathbb{R}^n$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00865 (13.22)	301	■ 1.000 ● N +0	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	$\boldsymbol{x}_k - \boldsymbol{x}_0 \in \mathbb{W}_k, \quad \boldsymbol{r}_k, \boldsymbol{p}_k \in \mathbb{W}_{k+1}, \quad k = 0, 1, \dots,$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00867 (13.24)	301	■ 1.000 ● N -1	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _A = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _A = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_k \right\ _A = \min_{\boldsymbol{w} \in \mathbb{W}_k} \left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00869 (13.25)	302	■ 1.000 ● N +0	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - \boldsymbol{w} \right\ _A^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j} Q(\lambda_j)^2, \quad Q(t) := 1 - tP(t).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00870 (13.26)	302	■ 1.000 ● K +0	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(A)\boldsymbol{r}_0 \right\ _A^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(A))\boldsymbol{r}_0 = Q(A)\boldsymbol{r}_0.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(A)\boldsymbol{r}_0 \right\ _A^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(A))\boldsymbol{r}_0 = Q(A)\boldsymbol{r}_0.$	$\left\ \boldsymbol{x} - \boldsymbol{x}_0 - P(A)\boldsymbol{r}_0 \right\ _A^2 = \boldsymbol{c}^T \boldsymbol{A}^{-1} \boldsymbol{c} \text{ where } \boldsymbol{c} = (\boldsymbol{I} - \boldsymbol{A}P(A))\boldsymbol{r}_0 = Q(A)\boldsymbol{r}_0.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00871 (13.27)	302	■ 1.000 ● N +0	$\boldsymbol{c} = \sum_{j=1}^n \sigma_j Q(\lambda_j) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c} = \sum_{i=1}^n \sigma_i \frac{Q(\lambda_i)}{\lambda_i} \boldsymbol{u}_i.$	$\boldsymbol{c} = \sum_{j=1}^n \sigma_j Q(\lambda_j) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c} = \sum_{i=1}^n \sigma_i \frac{Q(\lambda_i)}{\lambda_i} \boldsymbol{u}_i.$	$\boldsymbol{c} = \sum_{j=1}^n \sigma_j Q(\lambda_j) \boldsymbol{u}_j, \quad \boldsymbol{A}^{-1} \boldsymbol{c} = \sum_{i=1}^n \sigma_i \frac{Q(\lambda_i)}{\lambda_i} \boldsymbol{u}_i.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0876	303	■ 1.000 ● N +0	$e^{\{2 / x\}}=e^{\{2 y\}}=\sum_{k=0}^{\infty} \frac{\{2 y\}^k}{k!}<-1+2 \sum_{k=0}^{\infty} y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2/x}=e^{2y}=\sum_{k=0}^{\infty}\frac{(2y)^k}{k!}<-1+2\sum_{k=0}^{\infty}y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$	$e^{2/x}=e^{2y}=\sum_{k=0}^{\infty}\frac{(2y)^k}{k!}<-1+2\sum_{k=0}^{\infty}y^k=\frac{1+y}{1-y}=\frac{x+1}{x-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0877	304	■ 1.000 ● N -1	$\left f\right _{\infty}=\max_{a\leq x\leq b}\left f(x)\right .$	$\ f\ _{\infty}=\max_{a\leq x\leq b} f(x) .$	$\ f\ _{\infty}=\max_{a\leq x\leq b} f(x) .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0878	304	■ 1.000 ● W -1	$\left Q^*\right _{\infty}=\min_{Q\in S_k}\left Q\right _{\infty}.$	$\ Q^*\ _{\infty}=\min_{Q\in S_k}\ Q\ _{\infty}.$	$\ Q^*\ _{\infty}=\min_{Q\in S_k}\ Q\ _{\infty}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0879	304	■ 1.000 ● N +0	$T_{n+1}(t)=2\,t\,T_n(t)-T_{n-1}(t),\quad n\geq 1,\quad t\in\mathbb{R},$	$T_{n+1}(t)=2tT_n(t)-T_{n-1}(t),\quad n\geq 1,\quad t\in\mathbb{R},$	$T_{n+1}(t)=2tT_n(t)-T_{n-1}(t),\quad n\geq 1,\quad t\in\mathbb{R},$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0880	304	■ 1.000 ● N +0	$P_{n+1}(t)+P_{n-1}(t)=\cos{(n+1)\,\phi}+\cos{(n-1)\,\phi}=2\cos{\phi}\cos{n\,\phi}=2\,t\,P_n(t),$	$P_{n+1}(t)+P_{n-1}(t)=\cos(n+1)\phi+\cos(n-1)\phi=2\cos\phi\cos n\phi=2tP_n(t),$	$P_{n+1}(t)+P_{n-1}(t)=\cos(n+1)\phi+\cos(n-1)\phi=2\cos\phi\cos n\phi=2tP_n(t),$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0882	305	■ 1.000 ● K +0	$z_{\{1\}}=t+\sqrt{t^{\{2\}}-1},\quad z_{\{2\}}=t-\sqrt{t^{\{2\}}-1}=\left(t+\sqrt{t^{\{2\}}-1}\right)^{-1}.$	$z_1=t+\sqrt{t^2-1},\quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1}.$	$z_1=t+\sqrt{t^2-1},\quad z_2=t-\sqrt{t^2-1}=\left(t+\sqrt{t^2-1}\right)^{-1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Z-Library__EQ0883 (13.32)	305	■ 1.000 ● N +0	$Q^{\{*\}}(x)=\frac{T_{\{k\}}(u(x))}{T_{\{k\}}(u(c))},\quad u(x)=\frac{b+a-2\,x}{b-a}.$	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))},\quad u(x)=\frac{b+a-2x}{b-a}.$	$Q^*(x)=\frac{T_k(u(x))}{T_k(u(c))},\quad u(x)=\frac{b+a-2x}{b-a}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0884	305	■ 1.000 ● K +0	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right).$	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right).$	$\sigma_j=f\left(\mu_{j-1}\right)f\left(\mu_j\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0885	305	■ 1.000 ● N +0	$Q\left(\mu_j\right)\leq\left Q\right _{\infty}\leq\left Q^*\right _{\infty}=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=Q^*\left(\mu_j\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0886	305	■1.000 ● N +0	$-Q\left(\mu_{j-1}\right)\leq Q _{\infty}\leq Q^*\left(Q^*\right)^{-1}\left(\mu_{j-1}\right).$	$-Q\left(\mu_{j-1}\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=-Q^*\left(\mu_{j-1}\right).$	$-Q\left(\mu_{j-1}\right)\leq\ Q\ _{\infty}\leq\ Q^*\ _{\infty}=-Q^*\left(\mu_{j-1}\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0887 (13.33)	306	■1.000 ● K +0	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$	$Q^*(x)=T_k\left(\frac{b+a-2x}{b-a}\right)/T_k\left(\frac{b+a}{b-a}\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0888 (13.34)	306	■1.000 ● N -1	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1}\left T_k(t)\right =1.$	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1}\left T_k(t)\right =1.$	$\max_{a\leq x\leq b}\left T_k\left(\frac{b+a-2x}{b-a}\right)\right =\max_{-1\leq t\leq 1}\left T_k(t)\right =1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0889	306	■1.000 ● N +1	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad\kappa=b/a.$	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad\kappa=b/a.$	$t+\sqrt{t^2-1}=\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1},\quad\kappa=b/a.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0890 (13.35)	306	■1.000 ● W +0	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$	$T_k\left(\frac{b+a}{b-a}\right)=T_k\left(\frac{\kappa+1}{\kappa-1}\right)=\frac{1}{2}\left[\left(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}\right)^k+\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0893 (13.38)	307	■1.000 ● N +0	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$	$t_i=\frac{c_i^2}{\sum_{j=1}^nc_j^2}\geq 0,\quad i=1,\dots,n\text{ and }\sum_{i=1}^nt_i=1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0894	307	■1.000 ● N +0	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i(\lambda_ic+1/(\lambda_ic))=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i(\lambda_ic+1/(\lambda_ic))=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$	$\sqrt{ab}=\sqrt{(ac)(b/c)}\leq(ac+b/c)/2=\frac{1}{2}\sum_{i=1}^nt_i(\lambda_ic+1/(\lambda_ic))=\frac{1}{2}\sum_{i=1}^nt_if(\lambda_ic),$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0895	307	■1.000 ● N +0	$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$	$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$	$\sqrt{ab}\leq\frac{1}{2}\max_{mc\leq x\leq Mc}f(x).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0896 (13.39)	307	■ 1.000 ● N -1	$\sqrt{a\,b}\,\leq\,\frac{1}{2}\,\max\,\{\,f(m\,c)\,,\,f(M\,c)\,\}\,.$	$\sqrt{ab} \leq \frac{1}{2} \max\{f(mc), f(Mc)\}.$	$\sqrt{ab} \leq \frac{1}{2} \max\{f(mc), f(Mc)\}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0898	308	■ 1.000 ● W +0	$1\!=\!\left(\sum_{i=1}^n\,t_{\,i}\right)^{\!2}\!=\!\left(\sum_{i=1}^n\!\left(t_{\,i}\,\lambda_{\,i}\right)^{\!1/2}\!\left(t_{\,i}/\lambda_{\,i}\right)^{\!1/2}\right)^{\!2}\!\leq\!\left(\sum_{i=1}^n\,t_{\,i}\,\lambda_{\,i}\right)\!\left(\sum_{i=1}^n\,t_{\,i}/\lambda_{\,i}\right)\!=\!ab\,.$	$1 = \left(\sum_{i=1}^n t_i\right)^2 = \left(\sum_{i=1}^n (t_i \lambda_i)^{1/2} (t_i / \lambda_i)^{1/2}\right)^2 \leq \left(\sum_{i=1}^n t_i \lambda_i\right) \left(\sum_{i=1}^n t_i / \lambda_i\right) = ab.$	$1 = \left(\sum_{i=1}^n t_i\right)^2 = \left(\sum_{i=1}^n (t_i \lambda_i)^{1/2} (t_i / \lambda_i)^{1/2}\right)^2 \leq \left(\sum_{i=1}^n t_i \lambda_i\right) \left(\sum_{i=1}^n t_i / \lambda_i\right) = ab.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0900	308	■ 1.000 ● W +0	$\boldsymbol{\epsilon}_{k+1}\!=\!\boldsymbol{\epsilon}_k\!-\!\alpha_k\,\boldsymbol{r}_k,\quad\quad\alpha_k\!:=\!\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}.$	$\boldsymbol{\epsilon}_{k+1} = \boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}.$	$\boldsymbol{\epsilon}_{k+1} = \boldsymbol{\epsilon}_k - \alpha_k \boldsymbol{r}_k, \quad \alpha_k := \frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0902	308	■ 1.000 ● W +0	$\frac{\left\ \boldsymbol{\epsilon}_{k+1}\right\ _{\boldsymbol{A}}^2}{\left\ \boldsymbol{\epsilon}_k\right\ _{\boldsymbol{A}}^2}=1-\frac{\left(\boldsymbol{r}_k^T\boldsymbol{r}_k\right)^2}{\left(\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k\right)\left(\boldsymbol{r}_k^T\boldsymbol{A}^{-1}\boldsymbol{r}_k\right)}\leq1-\frac{4\lambda_{\min}\lambda_{\max}}{\left(\lambda_{\min}+\lambda_{\max}\right)^2}=\left(\frac{\kappa-1}{\kappa+1}\right)^2$	$\frac{\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2}{\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2} = 1 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{(\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k)(\boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k)} \leq 1 - \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min} + \lambda_{\max})^2} = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$	$\frac{\ \boldsymbol{\epsilon}_{k+1}\ _{\boldsymbol{A}}^2}{\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}^2} = 1 - \frac{(\boldsymbol{r}_k^T \boldsymbol{r}_k)^2}{(\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k)(\boldsymbol{r}_k^T \boldsymbol{A}^{-1} \boldsymbol{r}_k)} \leq 1 - \frac{4\lambda_{\min}\lambda_{\max}}{(\lambda_{\min} + \lambda_{\max})^2} = \left(\frac{\kappa - 1}{\kappa + 1}\right)^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0905 (13.41)	309	■ 1.000 ● N +0	$\boldsymbol{\epsilon}_j\!=\!\sum_{i=j}^m\,\alpha_i\,\boldsymbol{p}_i,\quad\quad\alpha_i\!=\!\frac{\boldsymbol{r}_i^T\boldsymbol{r}_i}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i \boldsymbol{p}_i, \quad \alpha_i = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$	$\boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i \boldsymbol{p}_i, \quad \alpha_i = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0906 (13.42)	309	■ 1.000 ● W +0	$\left\ \boldsymbol{\epsilon}_j\right\ _{\boldsymbol{A}}^2=\boldsymbol{\epsilon}_j\boldsymbol{A}\boldsymbol{\epsilon}_j=\sum_{i=j}^m\alpha_i^2\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i=\sum_{i=j}^m\frac{\left(\boldsymbol{r}_i^T\boldsymbol{r}_i\right)^2}{\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_i}.$	$\ \boldsymbol{\epsilon}_j\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_j \boldsymbol{A} \boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i^2 \boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i = \sum_{i=j}^m \frac{(\boldsymbol{r}_i^T \boldsymbol{r}_i)^2}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$	$\ \boldsymbol{\epsilon}_j\ _{\boldsymbol{A}}^2 = \boldsymbol{\epsilon}_j \boldsymbol{A} \boldsymbol{\epsilon}_j = \sum_{i=j}^m \alpha_i^2 \boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i = \sum_{i=j}^m \frac{(\boldsymbol{r}_i^T \boldsymbol{r}_i)^2}{\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_i}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0907	309	■ 1.000 ● W +0	$\boldsymbol{p}_i^T\boldsymbol{p}_k=\left(\boldsymbol{r}_i+\beta_{i-1}\boldsymbol{p}_{i-1}\right)^T\boldsymbol{p}_k=\beta_{i-1}\boldsymbol{p}_{i-1}^T\boldsymbol{p}_k=\cdots=\beta_{i-1}\cdots\beta_k\left(\boldsymbol{p}_k^T\boldsymbol{p}_k\right),$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \left(\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1}\right)^T \boldsymbol{p}_k = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k \left(\boldsymbol{p}_k^T \boldsymbol{p}_k\right),$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \left(\boldsymbol{r}_i + \beta_{i-1} \boldsymbol{p}_{i-1}\right)^T \boldsymbol{p}_k = \beta_{i-1} \boldsymbol{p}_{i-1}^T \boldsymbol{p}_k = \cdots = \beta_{i-1} \cdots \beta_k \left(\boldsymbol{p}_k^T \boldsymbol{p}_k\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0908 (13.43)	309	1.000 W +1	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k$	$\boldsymbol{p}_i^T \boldsymbol{p}_k = \frac{\boldsymbol{r}_i^T \boldsymbol{r}_i}{\boldsymbol{r}_k^T \boldsymbol{r}_k} \boldsymbol{p}_k^T \boldsymbol{p}_k, \quad i \geq k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0909	309	1.000 K +0	$\left \left \boldsymbol{\epsilon}_k \right \right _2^2 = \left \left \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k \right \right _2^2 = \left \left \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k \right \right _2^2,$	$\ \boldsymbol{\epsilon}_k\ _2^2 = \ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ _2^2 = \ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k\ _2^2,$	$\ \boldsymbol{\epsilon}_k\ _2^2 = \ \boldsymbol{\epsilon}_{k+1} + \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ _2^2 = \ \boldsymbol{\epsilon}_{k+1} + \alpha_k \boldsymbol{p}_k\ _2^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0913 (13.49)	311	1.000 K +0	$\boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$	$\boldsymbol{C}^T \left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$	$\boldsymbol{C}^T (\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T) \boldsymbol{C}^{-T} = \boldsymbol{B} \boldsymbol{A}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0916 (13.50)	311	1.000 W +0	$\begin{aligned} & \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \\ & \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \\ & \boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \\ & \boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k, \\ & \boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}. \end{aligned}$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}.$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k, \quad \alpha_k = \frac{\boldsymbol{s}_k^T \boldsymbol{r}_k}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k},$ $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{s}_{k+1} = \boldsymbol{s}_k - \alpha_k \boldsymbol{B} \boldsymbol{A} \boldsymbol{p}_k,$ $\boldsymbol{p}_{k+1} = \boldsymbol{s}_{k+1} + \beta_k \boldsymbol{p}_k, \quad \beta_k = \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{r}_k}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0921	313	1.000 W +0	$\begin{aligned} & I_m := \{(j, k) : 1 \leq j, k \leq m\}, \\ & \bar{I}_m := \{(j, k) : 0 \leq j, k \leq m+1\}, \\ & \partial I_m := \bar{I}_m \setminus I_m. \end{aligned}$	$I_m := \{(j, k) : 1 \leq j, k \leq m\},$ $\bar{I}_m := \{(j, k) : 0 \leq j, k \leq m+1\},$ $\partial I_m := \bar{I}_m \setminus I_m.$	$I_m := \{(j, k) : 1 \leq j, k \leq m\},$ $\bar{I}_m := \{(j, k) : 0 \leq j, k \leq m+1\},$ $\partial I_m := \bar{I}_m \setminus I_m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ0922	313	1.000 N +0	$\left\{ (x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m \right\}$	$\{(x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m\}$	$\{(x_j, y_k) = (jh, kh) : (j, k) \in \bar{I}_m\}$
Conf. MathPix confidence: 1.000 0.5-0.9 <0.5 no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0923 (13.54)	314	■1.000 ●W +2	$\begin{aligned} & \frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f\left(t + \frac{h}{2}\right) \frac{d}{dt} g\left(t + \frac{h}{2}\right) - f\left(t - \frac{h}{2}\right) \frac{d}{dt} g\left(t - \frac{h}{2}\right) \right) / h \\ & \approx \left(f\left(t + \frac{h}{2}\right) (g(t+h) - g(t)) - f\left(t - \frac{h}{2}\right) (g(t) - g(t-h)) \right) / h^2 \end{aligned}$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f\left(t + \frac{h}{2}\right) \frac{d}{dt} g\left(t + \frac{h}{2}\right) - f\left(t - \frac{h}{2}\right) \frac{d}{dt} g\left(t - \frac{h}{2}\right) \right) / h$ $\approx \left(f\left(t + \frac{h}{2}\right) (g(t+h) - g(t)) - f\left(t - \frac{h}{2}\right) (g(t) - g(t-h)) \right) / h^2$	$\frac{d}{dt} \left(f(t) \frac{d}{dt} g(t) \right) \approx \left(f\left(t + \frac{h}{2}\right) \frac{d}{dt} g\left(t + \frac{h}{2}\right) - f\left(t - \frac{h}{2}\right) \frac{d}{dt} g\left(t - \frac{h}{2}\right) \right) / h$ $\approx \left(f\left(t + \frac{h}{2}\right) (g(t+h) - g(t)) - f\left(t - \frac{h}{2}\right) (g(t) - g(t-h)) \right) / h^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0927	314	■1.000 ●N +0	$\boldsymbol{V} := \left(\begin{array}{ccc} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{array} \right)$	$\boldsymbol{V} := \begin{bmatrix} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{bmatrix}$	$\boldsymbol{V} := \begin{bmatrix} v_{1,1} & \dots & v_{1,m} \\ \vdots & & \vdots \\ v_{m,1} & \dots & v_{m,m} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0928 (13.57)	314	■1.000 ●N +0	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A} \operatorname{vec}(\boldsymbol{V}) := h^2 \operatorname{vec}(L_h \boldsymbol{v}).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0931 (13.60)	315	■1.000 ●W +0	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$	$\sum_{i=1}^m (a_{i-1} (b_i - b_{i-1}) - a_i (b_{i+1} - b_i)) c_i = \sum_{i=0}^m a_i (b_{i+1} - b_i) (c_{i+1} - c_i).$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0932	315	■1.000 ●W +0	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$	$\sum_{i=0}^m a_i (b_{i+1} - b_i) c_{i+1} - \sum_{i=0}^m a_i (b_{i+1} - b_i) c_i,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0933	316	■1.000 ●N +0	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$	$c(x, y) = e^{-x+y} \text{ and } f(x, y) = 1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ0934	317	■1.000 ●N +0	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$	$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \leq \frac{c_1}{c_0}.$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00936	317	1.000 N +0	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2 .$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2 .$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} = \sum_{i=1}^m \sum_{j=0}^m (v_{i,j+1} - v_{i,j})^2 + \sum_{j=1}^m \sum_{i=0}^m (v_{i+1,j} - v_{i,j})^2 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00938	318	1.000 N +0	$Q(\boldsymbol{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j .$	$Q(\boldsymbol{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j .$	$Q(\boldsymbol{y}) = \frac{1}{2} \sum_{j=1}^n \lambda_j v_j^2 - \sum_{j=1}^n c_j v_j .$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00939	318	1.000 K +0	$\left[\begin{array}{rr} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 0 \\ 3 \end{array} \right]$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 3 \end{array} \right]$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{c} 0 \\ 3 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00940	319	1.000 N -1	$\boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{A} \hat{\boldsymbol{x}} = \boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{b}, \quad \text{for all } \boldsymbol{w} \in \mathbb{W},$	$\boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{A} \hat{\boldsymbol{x}} = \boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{b}, \quad \text{for all } \boldsymbol{w} \in \mathbb{W},$	$\boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{A} \hat{\boldsymbol{x}} = \boldsymbol{w}^T \boldsymbol{A}^T \boldsymbol{b}, \quad \text{for all } \boldsymbol{w} \in \mathbb{W},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00941	319	1.000 N -1	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}.$	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}.$	$\ \boldsymbol{b} - \boldsymbol{A} \hat{\boldsymbol{x}}\ _2 \leq \ \boldsymbol{b} - \boldsymbol{A} \boldsymbol{w}\ _2, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00943	319	1.000 N +0	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$	$\langle \boldsymbol{v}, \boldsymbol{w} \rangle_{\boldsymbol{A}} := \boldsymbol{v}^T \boldsymbol{A}^T \boldsymbol{A} \boldsymbol{w}, \quad \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00944	320	1.000 K +0	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$	$\langle \boldsymbol{p}_k, \boldsymbol{w} \rangle_{\boldsymbol{A}} = 0, \quad \text{for all } \boldsymbol{w} \in \mathbb{W}_k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00952 (13.65)	321	1.000 K +0	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$	$\langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle \text{ for all } \boldsymbol{w} \in \mathcal{W},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_E00953 (13.66)	322	1.000 N +1	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k$	$\sum_{j=1}^k x_j \langle \boldsymbol{A} \boldsymbol{w}_j, \boldsymbol{w}_i \rangle = \langle \boldsymbol{b}, \boldsymbol{w}_i \rangle, \quad i = 1, \dots, k.$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean • not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0955	322	1.000 N -1	$\mathcal{W}_k := \operatorname{span}(\boldsymbol{b}, \boldsymbol{Bb}, \dots, \boldsymbol{B}^{k-1}\boldsymbol{b})$.	$\mathcal{W}_k := \operatorname{span}(\boldsymbol{b}, \boldsymbol{Bb}, \dots, \boldsymbol{B}^{k-1}\boldsymbol{b})$.	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0956	322	1.000 N +0	$\left\langle \boldsymbol{A} \boldsymbol{x}_k, \boldsymbol{w} \right\rangle = \left\langle \boldsymbol{b}, \boldsymbol{w} \right\rangle$, for all $\boldsymbol{w} \in \mathcal{W}_k$.	$\langle \boldsymbol{Ax}_k, \boldsymbol{w} \rangle = \langle \boldsymbol{b}, \boldsymbol{w} \rangle$, for all $\boldsymbol{w} \in \mathcal{W}_k$.	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0958	322	1.000 N +0	$\rho_k \neq 0$, $\quad k=0, \ldots, m$.	$\rho_k \neq 0, \quad k = 0, \dots, m.$	$\rho_k \neq 0, \quad k = 0, \dots, m.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0962 (13.72)	323	1.000 N +0	$\left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \right\rangle = \left\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \right\rangle$, $\quad j = 0, 1, \dots, k$.	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \rangle$, $\quad j = 0, 1, \dots, k.$	$\langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_{k+1} \rangle = \langle \boldsymbol{r}_{k+1}, \boldsymbol{A}^{-1} \boldsymbol{r}_j \rangle$, $\quad j = 0, 1, \dots, k.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0963 (13.70)	323	1.000 N -1	$T_n(t) = \cosh(n \operatorname{arccosh} t)$ for $t \geq 1$,	$T_n(t) = \cosh(n \operatorname{arccosh} t)$ for $t \geq 1$,	$T_n(t) = \cosh(n \operatorname{arccosh} t)$ for $t \geq 1$,
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0964	324	1.000 C -6	$\boldsymbol{A} \boldsymbol{x} = \left[\begin{array}{cccc} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{array} \right] \left[\begin{array}{c} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{array} \right] = \left[\begin{array}{c} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{array} \right],$	$\boldsymbol{Ax} = \left[\begin{array}{cccc} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{array} \right] \left[\begin{array}{c} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{array} \right] = \left[\begin{array}{c} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{array} \right],$	$\boldsymbol{Ax} = \left[\begin{array}{cccc} a_{1,1} & -c_{\frac{3}{2},1} & -c_{1,\frac{3}{2}} & 0 \\ -c_{\frac{3}{2},1} & a_{2,2} & 0 & -c_{2,\frac{3}{2}} \\ -c_{1,\frac{3}{2}} & 0 & a_{3,3} & -c_{\frac{3}{2},2} \\ 0 & -c_{2,\frac{3}{2}} & -c_{\frac{3}{2},2} & a_{4,4} \end{array} \right] \left[\begin{array}{c} v_{1,1} \\ v_{2,1} \\ v_{1,2} \\ v_{2,2} \end{array} \right] = \left[\begin{array}{c} (dv)_{1,1} \\ (dv)_{2,1} \\ (dv)_{1,2} \\ (dv)_{2,2} \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0965	324	1.000 C -8	$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array} \right] = \left[\begin{array}{l} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{array} \right]$.	$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array} \right] = \left[\begin{array}{l} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{array} \right]$.	$\left[\begin{array}{l} a_{1,1} \\ a_{2,2} \\ a_{3,3} \\ a_{4,4} \end{array} \right] = \left[\begin{array}{l} c_{\frac{1}{2},1} + c_{1,\frac{1}{2}} + c_{1,\frac{3}{2}} + c_{\frac{3}{2},1} \\ c_{\frac{3}{2},1} + c_{2,\frac{1}{2}} + c_{2,\frac{3}{2}} + c_{\frac{5}{2},1} \\ c_{\frac{1}{2},2} + c_{1,\frac{3}{2}} + c_{1,\frac{5}{2}} + c_{\frac{3}{2},2} \\ c_{\frac{3}{2},2} + c_{2,\frac{3}{2}} + c_{2,\frac{5}{2}} + c_{\frac{5}{2},2} \end{array} \right]$.
Conf. MathPix confidence: 1.000 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00966	326	■ 1.000 ● N -1	$\boldsymbol{A}=\operatorname{diag}\left(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}\right), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_i \times m_i} \in \mathbb{C}^{m_i \times m_i} \text { times } m_{i}\} .$	$\boldsymbol{A}=\operatorname{diag}\left(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}\right), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_i \times m_i} .$	$\boldsymbol{A}=\operatorname{diag}\left(\boldsymbol{A}_{1}, \boldsymbol{A}_{2}, \ldots, \boldsymbol{A}_{r}\right), \quad \boldsymbol{A}_{i} \in \mathbb{C}^{m_i \times m_i} .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00968	327	■ 1.000 ● N +0	$\operatorname{det}\left(\boldsymbol{A}-\lambda \boldsymbol{I}\right)=\prod_{i=1}^r \operatorname{det}\left(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I}\right)$	$\operatorname{det}\left(\boldsymbol{A}-\lambda \boldsymbol{I}\right)=\prod_{i=1}^r \operatorname{det}\left(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I}\right)$	$\operatorname{det}\left(\boldsymbol{A}-\lambda \boldsymbol{I}\right)=\prod_{i=1}^r \operatorname{det}\left(\boldsymbol{A}_{i i}-\lambda \boldsymbol{I}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00970	328	■ 1.000 ● N +0	$\left \lambda-a_{i i}\right =\left \sum_{j \neq i} a_{i j} x_j / x_i\right \leq \sum_{j \neq i}\left a_{i j}\right \left x_j / x_i\right \leq r_i$	$\left \lambda-a_{i i}\right =\left \sum_{j \neq i} a_{i j} x_j / x_i\right \leq \sum_{j \neq i}\left a_{i j}\right \left x_j / x_i\right \leq r_i$	$\left \lambda-a_{i i}\right =\left \sum_{j \neq i} a_{i j} x_j / x_i\right \leq \sum_{j \neq i}\left a_{i j}\right \left x_j / x_i\right \leq r_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00971	328	■ 1.000 ● N +0	$R_i=\left\{z \in \mathbb{R}: z-2 \leq 2\right\}, \quad i=2,3, \ldots, m-1 .$	$R_i=\left\{z \in \mathbb{R}: z-2 \leq 2\right\}, \quad i=2,3, \ldots, m-1 .$	$R_i=\left\{z \in \mathbb{R}: z-2 \leq 2\right\}, \quad i=2,3, \ldots, m-1 .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00972	328	■ 1.000 ● N +0	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$	$\lambda_j=4\left[\sin \frac{j \pi}{2(m+1)}\right]^2, \quad j=1,2, \ldots, m .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00973	329	■ 1.000 ● N +0	$\boldsymbol{A}(t):=\boldsymbol{D}+t\left(\boldsymbol{A}-\boldsymbol{D}\right), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in[0,1] .$	$\boldsymbol{A}(t):=\boldsymbol{D}+t\left(\boldsymbol{A}-\boldsymbol{D}\right), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in[0,1] .$	$\boldsymbol{A}(t):=\boldsymbol{D}+t\left(\boldsymbol{A}-\boldsymbol{D}\right), \quad \boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right), \quad t \in[0,1] .$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__E00974	329	■ 1.000 ● N +0	$R_i(t)=\left\{z \in \mathbb{C}: z-a_{i i} \leq t r_i\right\}, \quad r_i:=\sum_{\substack{j=1 \\ j \neq i}}^n\left a_{i j}\right .$	$R_i(t)=\left\{z \in \mathbb{C}: z-a_{i i} \leq t r_i\right\}, \quad r_i:=\sum_{\substack{j=1 \\ j \neq i}}^n\left a_{i j}\right .$	$R_i(t)=\left\{z \in \mathbb{C}: z-a_{i i} \leq t r_i\right\}, \quad r_i:=\sum_{\substack{j=1 \\ j \neq i}}^n\left a_{i j}\right .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09978	330	■ 1.000 ● N +1	$\begin{aligned} & \left \mu - \lambda_1 \right ^n \leq \left \prod_{j=1}^n \left \mu - \lambda_j \right \right = \left \operatorname{det}(\mu \boldsymbol{I} - \boldsymbol{A}) \right = \left \operatorname{det}(\mu \boldsymbol{I} - \boldsymbol{A}) \right \\ & \leq \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E} \ _2 (\ \boldsymbol{A} + \boldsymbol{E} \ _2 + \ \boldsymbol{A} \ _2)^{n-1}. \end{aligned}$	$ \mu - \lambda_1 ^n \leq \prod_{j=1}^n \mu - \lambda_j = \det(\mu \boldsymbol{I} - \boldsymbol{A}) = \det((\mu \boldsymbol{I} - \boldsymbol{A}) [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]) $ $\leq \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E} \ _2 (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A} \ _2)^{n-1}.$	$ \mu - \lambda_1 ^n \leq \prod_{j=1}^n \mu - \lambda_j = \det(\mu \boldsymbol{I} - \boldsymbol{A}) = \det((\mu \boldsymbol{I} - \boldsymbol{A}) [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]) $ $\leq \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_1\ _2 \prod_{j=2}^n \ (\mu \boldsymbol{I} - \boldsymbol{A}) \boldsymbol{u}_j\ _2 \leq \ \boldsymbol{E} \ _2 (\ (\boldsymbol{A} + \boldsymbol{E})\ _2 + \ \boldsymbol{A} \ _2)^{n-1}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09979 (14.2)	331	■ 1.000 ● N +0	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{r}\ _p, \quad 1 \leq p \leq \infty,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09980 (14.3)	331	■ 1.000 ● N +1	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$	$ \lambda - \mu \leq K_p(\boldsymbol{X}) \ \boldsymbol{E}\ _p, \quad 1 \leq p \leq \infty,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09983	332	■ 1.000 ● N +0	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$	$\ \boldsymbol{A} \boldsymbol{x}\ _p = \left\ [\lambda_1 x_1, \dots, \lambda_n x_n]^T \right\ _p = \left(\sum_{j=1}^n \lambda_j ^p x_j ^p \right)^{1/p} \leq \rho(\boldsymbol{A}) \ \boldsymbol{x}\ _p.$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09984 (14.4)	332	■ 1.000 ● N +0	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{r}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__E09985 (14.5)	332	■ 1.000 ● N +0	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$	$\frac{ \lambda - \mu }{ \lambda } \leq K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p \leq K_p(\boldsymbol{X}) K_p(\boldsymbol{A}) \frac{\ \boldsymbol{E}\ _p}{\ \boldsymbol{A}\ _p}, \quad 1 \leq p \leq \infty,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0987	333	■ 1.000 ● K +0	$\left(\boldsymbol{B}+\boldsymbol{F}-\boldsymbol{I}\right)\boldsymbol{x}=\left(\mu\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{E}-\boldsymbol{I}\right)\boldsymbol{x}=\boldsymbol{A}^{-1}\left(\mu\boldsymbol{I}-\left(\boldsymbol{E}+\boldsymbol{A}\right)\right)\boldsymbol{x}=\mathbf{0}.$	$\left(\boldsymbol{B}+\boldsymbol{F}-\boldsymbol{I}\right)\boldsymbol{x}=\left(\mu\boldsymbol{A}^{-1}-\boldsymbol{A}^{-1}\boldsymbol{E}-\boldsymbol{I}\right)\boldsymbol{x}=\boldsymbol{A}^{-1}\left(\mu\boldsymbol{I}-\left(\boldsymbol{E}+\boldsymbol{A}\right)\right)\boldsymbol{x}=\mathbf{0}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0988	333	■ 1.000 ● N +0	$\boldsymbol{A}_{k+1}^*=\left(\boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k\right)^*=\boldsymbol{H}_k\boldsymbol{A}_k^*\boldsymbol{H}_k=\boldsymbol{A}_{k+1}.$	$\boldsymbol{A}_{k+1}^*=\left(\boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k\right)^*=\boldsymbol{H}_k\boldsymbol{A}_k^*\boldsymbol{H}_k=\boldsymbol{A}_{k+1}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0990	334	■ 1.000 ● N +0	$\boldsymbol{H}_k=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{V}_k\end{array}\right]\in\mathbb{C}^{n\times n}.$	$\boldsymbol{H}_k=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{V}_k\end{array}\right]\in\mathbb{C}^{n\times n}.$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ0993	335	■ 1.000 ● N +0	$\boldsymbol{Q}_k=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{array}\right]^*\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{U}_k\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{U}_k-\boldsymbol{v}_k\left(\boldsymbol{v}_k^T\boldsymbol{U}_k\right)\end{array}\right].$	$\boldsymbol{Q}_k=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^T\end{array}\right]^*\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{U}_k\end{array}\right]=\left[\begin{array}{cc}\boldsymbol{I}_k&\mathbf{0}\\\mathbf{0}&\boldsymbol{U}_k-\boldsymbol{v}_k\left(\boldsymbol{v}_k^T\boldsymbol{U}_k\right)\end{array}\right].$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Z-Library__EQ0996 (14.7)	336	■ 1.000 ● N +0	$p_{k+1}(x)=\left(x-d_{k+1}\right)p_k(x)-c_k^2p_{k-1}(x).$	$p_{k+1}(x)=\left(x-d_{k+1}\right)p_k(x)-c_k^2p_{k-1}(x).$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ1001	337	■ 1.000 ● K +0	$p_{k+1}\left(y_j\right)=\left(-1\right)^{k+1-j}\left p_{k+1}\left(y_j\right)\right \neq 0, \text { for } j=0,1, \ldots, k+1 .$	$p_{k+1}(y_j)=\left(-1\right)^{k+1-j}\left p_{k+1}(y_j)\right \neq 0, \text { for } j=0,1, \ldots, k+1 .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ1002	337	■ 1.000 ● N +0	$p_{k+1}\left(y_j\right)=-c_k^2 p_{k-1}\left(y_j\right)=-c_k^2\left(y_j-z_1\right) \cdots\left(y_j-z_{k-1}\right) .$	$p_{k+1}(y_j)=-c_k^2 p_{k-1}(y_j)=-c_k^2\left(y_j-z_1\right) \cdots\left(y_j-z_{k-1}\right) .$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__EQ1003	337	■ 1.000 ● N +0	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y}=\sum_{i=1}^n \lambda_i\left y_i\right ^2=\sum_{i=1}^k \lambda_i\left x_i\right ^2>0 .$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{y}=\sum_{i=1}^n \lambda_i\left y_i\right ^2=\sum_{i=1}^k \lambda_i\left x_i\right ^2>0 .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1005 (14.8)	338	■ 1.000 ● N +0	$d_{\{1\}(\alpha)}=d_{\{1\}}-\alpha, \; d_{\{k\}(\alpha)}=d_{\{k\}}-\alpha-c_{\{k-1\}^{\{2\}}}/d_{\{k-1\}(\alpha)}, \; k=2,3, \; \ldots, \; n \; .$	$d_1(\alpha) = d_1 - \alpha, d_k(\alpha) = d_k - \alpha - c_{k-1}^2/d_{k-1}(\alpha), k = 2, 3, \dots, n.$	$d_1(\alpha) = d_1 - \alpha, \; d_k(\alpha) = d_k - \alpha - c_{k-1}^2/d_{k-1}(\alpha), \; k = 2, 3, \dots, n.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1006	338	■ 1.000 ● K +0	$\rho(x):=\#\left\{k: \; d_{\{k\}}(x)>0 \; \text{for } k=1, \; \ldots, \; n\right\}$	$\rho(x) := \# \{k : d_k(x) > 0 \text{ for } k = 1, \dots, n\}$	$\rho(x) := \#\{k : d_k(x) > 0 \text{ for } k = 1, \dots, n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1007	340	■ 1.000 ● K +0	$\boldsymbol{A}=\left(\begin{array}{llll} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right) \; .$	$\boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$	$\boldsymbol{A} = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 1 & 4 & 1 & 0 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1008	340	■ 1.000 ● N -1	$\boldsymbol{A}(t):=\boldsymbol{D}+t(\boldsymbol{A}-\boldsymbol{D}), \; \text{quad } \boldsymbol{D}:=\operatorname{diag}\left(a_{\{1\}}, \; \ldots, \; a_{\{n\}}\right), \; \text{quad } t \; \text{in } \mathbb{R} \; .$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn}), \quad t \in \mathbb{R}.$	$\boldsymbol{A}(t) := \boldsymbol{D} + t(\boldsymbol{A} - \boldsymbol{D}), \quad \boldsymbol{D} := \operatorname{diag}(a_{11}, \dots, a_{nn}), \quad t \in \mathbb{R}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1009	340	■ 1.000 ● N +0	$ \lambda-\mu \; \leq \; C \left(t_{\{2\}}-t_{\{1\}}\right)^{\{1/n\}}, \; \text{where } C \; \leq \; 2 \left(\ \boldsymbol{D}\ _2 + \ \boldsymbol{A}-\boldsymbol{D}\ _2\right) \; .$	$ \lambda - \mu \leq C (t_2 - t_1)^{1/n}, \text{ where } C \leq 2 (\ \boldsymbol{D}\ _2 + \ \boldsymbol{A} - \boldsymbol{D}\ _2).$	$ \lambda - \mu \leq C (t_2 - t_1)^{1/n}, \text{ where } C \leq 2 (\ \boldsymbol{D}\ _2 + \ \boldsymbol{A} - \boldsymbol{D}\ _2).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1013	342	■ 1.000 ● N +0	$\boldsymbol{D}^{\{-1/2\}}:=\operatorname{diag}\left(d_{\{1\}}^{\{-1/2\}}, \; \ldots, \; d_{\{n\}}^{\{-1/2\}}\right) \; .$	$\boldsymbol{D}^{-1/2} := \operatorname{diag}\left(d_1^{-1/2}, \dots, d_n^{-1/2}\right).$	$\boldsymbol{D}^{-1/2} := \operatorname{diag}\left(d_1^{-1/2}, \dots, d_n^{-1/2}\right).$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1027	347	■ 1.000 ● N +0	$\rho(\lambda)^{\{2\}}=\left \boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}-\lambda\right ^2+\left \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{u}\right _2^2,$	$\rho(\lambda)^2 = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u} - \lambda ^2 + \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{u}\ _2^2,$	$\rho(\lambda)^2 = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u} - \lambda ^2 + \ \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{u}\ _2^2,$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1028	347	■ 1.000 ● N +0	$\frac{ \lambda-\mu }{ \lambda } \leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\ \boldsymbol{A}\boldsymbol{u}-\mu\boldsymbol{u}\ _2}{\ \boldsymbol{A}\ _2}.$	$\frac{ \lambda - \mu }{ \lambda } \leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\ \boldsymbol{A} \boldsymbol{u} - \mu \boldsymbol{u}\ _2}{\ \boldsymbol{A}\ _2}.$	$\frac{ \lambda - \mu }{ \lambda } \leq K_2(\boldsymbol{X})K_2(\boldsymbol{A})\frac{\ \boldsymbol{A} \boldsymbol{u} - \mu \boldsymbol{u}\ _2}{\ \boldsymbol{A}\ _2}.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1033	349	■ 1.000 ● N +0	$\rho_{\{k\}}:=\frac{\boldsymbol{y}_{\{k\}}^*\boldsymbol{y}_{\{k-1\}}}{\boldsymbol{y}_{\{k\}}^*\boldsymbol{y}_{\{k\}}}, \; \text{quad } \boldsymbol{w}_{\{k\}}:=\frac{\boldsymbol{x}_{\{k-1\}}}{\ \boldsymbol{y}_{\{k\}}\ _2}.$	$\rho_k := \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}, \quad \boldsymbol{w}_k := \frac{\boldsymbol{x}_{k-1}}{\ \boldsymbol{y}_k\ _2}.$	$\rho_k := \frac{\boldsymbol{y}_k^* \boldsymbol{x}_{k-1}}{\boldsymbol{y}_k^* \boldsymbol{y}_k}, \quad \boldsymbol{w}_k := \frac{\boldsymbol{x}_{k-1}}{\ \boldsymbol{y}_k\ _2}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1036 (15.9)	351	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_{k+1} &= \boldsymbol{R}_k \boldsymbol{Q}_k \\ \boldsymbol{Q}_{k+1} &= \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k, \\ \boldsymbol{A}_{k+1} &= \boldsymbol{R}_k \boldsymbol{Q}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k,\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1037	351	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_1 &= \boldsymbol{A} = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right] = \left(\frac{1}{\sqrt{5}} \left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array} \right] \right) * \left(\frac{1}{\sqrt{5}} \left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array} \right] \right) = \boldsymbol{Q}_1 \boldsymbol{R}_1 \\ \boldsymbol{A}_1 &= \boldsymbol{A} = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right] = \left(\frac{1}{\sqrt{5}} \left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array} \right] \right) * \left(\frac{1}{\sqrt{5}} \left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array} \right] \right) = \boldsymbol{Q}_1 \boldsymbol{R}_1\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1038	351	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_2 &= \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array} \right] * \left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array} \right] = \frac{1}{5} \left[\begin{array}{cc} 14 & -3 \\ -3 & 6 \end{array} \right] = \left[\begin{array}{cc} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array} \right]. \\ \boldsymbol{A}_2 &= \boldsymbol{R}_1 \boldsymbol{Q}_1 = \frac{1}{5} \left[\begin{array}{cc} -5 & -4 \\ 0 & 3 \end{array} \right] * \left[\begin{array}{cc} -2 & -1 \\ -1 & 2 \end{array} \right] = \frac{1}{5} \left[\begin{array}{cc} 14 & -3 \\ -3 & 6 \end{array} \right] = \left[\begin{array}{cc} 2.8 & -0.6 \\ -0.6 & 1.2 \end{array} \right].\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1039	351	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_4 &\approx \left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array} \right], \quad \boldsymbol{A}_{10} \approx \left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array} \right] \\ \boldsymbol{A}_4 &\approx \left[\begin{array}{cc} 2.997 & -0.074 \\ -0.074 & 1.0027 \end{array} \right], \quad \boldsymbol{A}_{10} \approx \left[\begin{array}{cc} 3.0000 & -0.0001 \\ -0.0001 & 1.0000 \end{array} \right]\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1040	352	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_1 &= \boldsymbol{A} = \left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array} \right] \\ \boldsymbol{A}_1 &= \boldsymbol{A} = \left[\begin{array}{cccc} 0.9501 & 0.8913 & 0.8214 & 0.9218 \\ 0.2311 & 0.7621 & 0.4447 & 0.7382 \\ 0.6068 & 0.4565 & 0.6154 & 0.1763 \\ 0.4860 & 0.0185 & 0.7919 & 0.4057 \end{array} \right]\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1042 (15.10)	352	■ 1.000 ● K +0	$\begin{aligned}\tilde{\boldsymbol{Q}}_k &:= \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1, \\ \tilde{\boldsymbol{Q}}_k &:= \boldsymbol{Q}_1 \cdots \boldsymbol{Q}_k \text{ and } \tilde{\boldsymbol{R}}_k := \boldsymbol{R}_k \cdots \boldsymbol{R}_1,\end{aligned}$		
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__EQ1044 (15.11)	353	■ 1.000 ● N +0	$\begin{aligned}\boldsymbol{A}_k &= \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}. \\ \boldsymbol{A}_k &= \boldsymbol{Q}_{k-1}^* \boldsymbol{A}_{k-1} \boldsymbol{Q}_{k-1} = \boldsymbol{Q}_{k-1}^* \boldsymbol{Q}_{k-2}^* \boldsymbol{A}_{k-2} \boldsymbol{Q}_{k-2} \boldsymbol{Q}_{k-1} = \cdots = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}.\end{aligned}$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1046	353	■ 1.000 ● W +0	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{\boldsymbol{q}}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{\boldsymbol{q}}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$	$\boldsymbol{A}^k \boldsymbol{e}_1 = \tilde{\boldsymbol{Q}}_k \tilde{\boldsymbol{R}}_k \boldsymbol{e}_1 = \tilde{r}_{11}^{(k)} \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 \text{ or } \tilde{\boldsymbol{q}}_1^{(k)} := \tilde{\boldsymbol{Q}}_k \boldsymbol{e}_1 = \frac{1}{\tilde{r}_{11}^{(k)}} \boldsymbol{A}^k \boldsymbol{e}_1.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1048	355	■ 1.000 ● N +0	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{Q}_k^* (\boldsymbol{A}_k - s_k \boldsymbol{I}) \boldsymbol{Q}_k + s_k \boldsymbol{I} = \boldsymbol{Q}_k^* \boldsymbol{A}_k \boldsymbol{Q}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1050 (16.1)	358	■ 1.000 ● N +0	$\nabla f := \left[\begin{array}{c} D_1 f \\ \vdots \\ D_n f \end{array} \right], \quad \boldsymbol{H} f := \nabla \nabla^T f := \left[\begin{array}{ccc} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{array} \right],$	$\nabla f := \left[\begin{array}{c} D_1 f \\ \vdots \\ D_n f \end{array} \right], \quad \boldsymbol{H} f := \nabla \nabla^T f := \left[\begin{array}{ccc} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{array} \right],$	$\nabla f := \left[\begin{array}{c} D_1 f \\ \vdots \\ D_n f \end{array} \right], \quad \boldsymbol{H} f := \nabla \nabla^T f := \left[\begin{array}{ccc} D_1 D_1 f & \cdots & D_1 D_n f \\ \vdots & & \vdots \\ D_n D_1 & \cdots & D_n D_n f \end{array} \right],$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1051	359	■ 1.000 ● N +0	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{array} \right].$	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{array} \right].$	$\nabla^T \boldsymbol{f} := \left[\begin{array}{ccc} D_1 f_1 & \cdots & D_n f_1 \\ \vdots & & \vdots \\ D_1 f_m & \cdots & D_n f_m \end{array} \right].$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1053 (16.2)	359	■ 1.000 ● N +0	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^T \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L.$	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^T \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L.$	$f(\boldsymbol{x} + \boldsymbol{h}) = f(\boldsymbol{x}) + \boldsymbol{h}^T \nabla f(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \nabla \nabla^T f(\boldsymbol{c}) \boldsymbol{h}, \text{ for some } \boldsymbol{c} \in L.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1055	359	■ 1.000 ● N +0	$g(1) = g(0) + g'(\theta) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$	$g(1) = g(0) + g'(0) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$	$g(1) = g(0) + g'(0) + \frac{1}{2} g''(u), \text{ for some } u \in (0, 1),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_EQ1056	360	■ 1.000 ● W +1	<pre>\begin{aligned} D_{\{i\}}\left(\boldsymbol{x}^T\right. \\ \boldsymbol{B} \boldsymbol{x}\right) &=\lim _h \\ \rightarrow 0\} \frac{1}{h} \left(\left(\boldsymbol{x}+h\right. \\ \boldsymbol{e}_{\{i\}}\right)^T \boldsymbol{B} \left(\boldsymbol{x}+h\right. \\ \boldsymbol{e}_{\{i\}}\right)-\boldsymbol{B} \boldsymbol{x} \\ \right) \\ &=\lim _h \left(\boldsymbol{e}_{\{i\}}^T \boldsymbol{x}+\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{e}_{\{i\}}+h \boldsymbol{e}_{\{i\}}^T \boldsymbol{e}_{\{i\}}\right)= \\ \boldsymbol{e}_{\{i\}}^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right) \boldsymbol{x}, \end{aligned}</pre>	$D_i\left(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}\right)=\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)^T \boldsymbol{B}\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)-\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}\right)$ $=\lim _{h \rightarrow 0}\left(\boldsymbol{e}_i^T \boldsymbol{B} \boldsymbol{x}+\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{e}_i+h \boldsymbol{e}_i^T \boldsymbol{e}_i\right)=\boldsymbol{e}_i^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right) \boldsymbol{x},$	$D_i\left(\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}\right)=\lim _{h \rightarrow 0} \frac{1}{h}\left(\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)^T \boldsymbol{B}\left(\boldsymbol{x}+h \boldsymbol{e}_i\right)-\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{x}\right)$ $=\lim _{h \rightarrow 0}\left(\boldsymbol{e}_i^T \boldsymbol{B} \boldsymbol{x}+\boldsymbol{x}^T \boldsymbol{B} \boldsymbol{e}_i+h \boldsymbol{e}_i^T \boldsymbol{e}_i\right)=\boldsymbol{e}_i^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right) \boldsymbol{x},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00001		—	$\boldsymbol{A}$	$\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00002		—	$\boldsymbol{b}$	$\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00003		—	$\boldsymbol{x}$	$\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00004		—	$\boldsymbol{A} \ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{Ax} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00005		—	$\boldsymbol{b}-\boldsymbol{A} \ \boldsymbol{x}$	$\boldsymbol{b} - \boldsymbol{Ax}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00006		—	λ	λ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00007		—	$\boldsymbol{A} \ \boldsymbol{x}=\lambda \ \boldsymbol{x}$	$\boldsymbol{Ax} = \lambda \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00008		—	$\mathbb{R}^n$	$\mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00009		—	$\mathbb{C}^n$	$\mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00010		—	$\boldsymbol{A}^{*} \ \boldsymbol{A}, \ \boldsymbol{A} \ \boldsymbol{A}^{*}$	$\boldsymbol{A}^{*} \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00011		—	p	p
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00012		—	ω	ω
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00013		—	λ_m	λ_m
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00014		—	T	T
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00015		—	$P_{\{1\}}, P_{\{2\}}$	P_1, P_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00016		—	$P_{\{3\}}$	P_3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00017		—	$f(x)=\arctan (10\ x)+\pi / 2$	$f(x)=\arctan(10x)+\pi/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00018		—	$[-1,1]$	$[-1,1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00019		—	$n=14$	$n=14$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00020		—	$f(x)=x^{\{4\}}$	$f(x)=x^4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00021		—	5×5	5×5
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00022		—	$d=1$	$d=1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00023		—	d=2	$d = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00024		—	$\boldsymbol{v}_1$	$\boldsymbol{v}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00025		—	$\boldsymbol{v}_2$	$\boldsymbol{v}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00026		—	c	c
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00027		—	$c := \left\langle \boldsymbol{s}_2, \boldsymbol{v}_1 \right\rangle, \left\langle \boldsymbol{v}_1, \boldsymbol{v}_1 \right\rangle$	$c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00028		—	$\boldsymbol{s} + \boldsymbol{t}$	$\boldsymbol{s} + \boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00029		—	$\mathcal{S}$	$\mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00030		—	$\mathcal{T}$	$\mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00031		—	$y_1^2 / 9 + y_2^2 = 1$	$y_1^2/9 + y_2^2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00032		—	$\boldsymbol{A} \, \boldsymbol{\mathcal{S}}$	$\boldsymbol{A} \boldsymbol{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00033		—	n=100	$n = 100$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00034		—	$\alpha \rightarrow \left 1 - \alpha \lambda_{1} \right $	$\alpha \rightarrow \left 1 - \alpha \lambda_1 \right $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00035		—	$\alpha \rightarrow \left 1 - \alpha \lambda_{n} \right $	$\alpha \rightarrow \left 1 - \alpha \lambda_n \right $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00036		—	12.2 $\rho \left(\boldsymbol{G}_{\omega} \right)$	$12.2 \rho \left(G_{\omega} \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00037		—	$\omega \in [0, 2]$	$\omega \in [0, 2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00038		—	n=2500	$n = 2500$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00039		—	$Q(x, y)$	$Q(x, y)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00040		—	Q	Q
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00041		—	$\boldsymbol{x} - \boldsymbol{x}_{\{0\}}$	$\boldsymbol{x} - \boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00042		—	$\mathbb{W}_{\boldsymbol{k}}$	$\mathbb{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00043		—	$k=3 \text{ . f } \equiv Q - Q^{\ast}$	$k = 3.f \equiv Q - Q^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00044		—	$\mu_{\{1\}}$	μ_1
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00045		—	μ_{2}	μ_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00046		—	μ_{3}	μ_3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00047		—	R_{i}	R_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00048		—	k_{n}	k_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00049		—	n	n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00050		—	10^{-8}	10^{-8}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00051		—	$\boldsymbol{G}_J, \boldsymbol{G}_1, \boldsymbol{G}_{\omega^*}$	$\boldsymbol{G}_J, \boldsymbol{G}_1, \boldsymbol{G}_{\omega^*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00052		—	$\rho(\boldsymbol{G})^{k_n} \leq 10^{-8}$	$\rho(\boldsymbol{G})^{k_n} \leq 10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00053		—	K	K
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00054		—	$\sqrt{n} \times \sqrt{n}$	$\sqrt{n} \times \sqrt{n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00055		—	$K_{\text{p r e}}$	K_{pre}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00056		—	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00057		—	$v:=e$	$v := e$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00058		—	v	v
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00059		—	e	e
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00060		—	$\boldsymbol{x} \in \mathbb{R}^n$	$\boldsymbol{x} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00061		—	$x_{\mathrm{i}} \in \mathbb{R}$	$x_i \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00062		—	$i=1, \ldots, n$	$i = 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00063		—	$\boldsymbol{x}^{\mathrm{T}}$	$\boldsymbol{x}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00064		—	$\mathbb{R}^{m \times n}$	$\mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00065		—	m	m
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00066		—	i	i
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00067		—	j	j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00068		—	$a_{\{i, j\}}, a_{\{i\} j}$	$a_{i,j}, a_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00069		—	$\boldsymbol{A}(i, j)$	$\boldsymbol{A}(i, j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00070		—	$(\boldsymbol{A})_{i, j}$	$(\boldsymbol{A})_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00071		—	$\boldsymbol{a}_{\{ : j\}}$	$\boldsymbol{a}_{:j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00072		—	$\boldsymbol{a}_{\{i : \}^T}$	$\boldsymbol{a}_{i:}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00073		—	$\boldsymbol{a}_{\{j\}}$	$\boldsymbol{a}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00074		—	$\boldsymbol{a}_{\{i\}^T}$	$\boldsymbol{a}_i^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00075		—	$m=1$	$m = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00076		—	$n=1$	$n = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00077		—	$m=n$	$m = n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00078		—	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \cdots$	$A, B, C, \cdots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00079		—	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00080		—	$x = a + i \ b$	$x = a + ib$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00081		—	$a, \ b$	a, b
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00082		—	$i^2 = -1$	$i^2 = -1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00083		—	$\mathbb{C}$	$\mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00084		—	$a = \operatorname{Re} x$	$a = \operatorname{Re} x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00085		—	$b = \operatorname{Im} x$	$b = \operatorname{Im} x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00086		—	x	x
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00087		—	$\bar{x} := a - i \ b$	$\bar{x} := a - ib$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00088		—	$ x := \sqrt{\bar{x} x} = \sqrt{a^2 + b^2}$	$ x := \sqrt{\bar{x} x} = \sqrt{a^2 + b^2}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00089		—	$e^{x+y}=e^x e^y$	$e^{x+y} = e^x e^y$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00090		—	$x, y \in \mathbb{C}$	$x, y \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00091		—	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00092		—	$\boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{x} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00093		—	$\boldsymbol{x}^* := \overline{\boldsymbol{x}^T} = [\bar{x}_1, \dots, \bar{x}_n]$	$\boldsymbol{x}^* := \overline{\boldsymbol{x}}^T = [\bar{x}_1, \dots, \bar{x}_n]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00094		—	$\boldsymbol{x}^* = \boldsymbol{x}^T$	$\boldsymbol{x}^* = \boldsymbol{x}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00095		—	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00096		—	$a \in \mathbb{C}$	$a \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00097		—	$\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$	$\boldsymbol{C} := \boldsymbol{A} + \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00098		—	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00099		—	$c_{ij} := a_{ij} + b_{ij}$	$c_{ij} := a_{ij} + b_{ij}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00100		—	i, j	i, j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00101		—	$\boldsymbol{C}:=\alpha \boldsymbol{A}$	$\boldsymbol{C}:=\alpha \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00102		—	$c_{i j}:=\alpha a_{i j}$	$c_{i j}:=\alpha a_{i j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00103		—	$\boldsymbol{C}:=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{C}=\boldsymbol{A} \cdot \boldsymbol{B}$	$\boldsymbol{C}:=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{C}=\boldsymbol{A} \cdot \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00104		—	$\boldsymbol{C}=\boldsymbol{A} * \boldsymbol{B}$	$\boldsymbol{C}=\boldsymbol{A} * \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00105		—	$\boldsymbol{A} \in$	$\boldsymbol{A} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00106		—	$\mathbb{C}^{\{m \times p\}}, \boldsymbol{B} \in \mathbb{C}^{\{p \times n\}}, \boldsymbol{C} \in \mathbb{C}^{\{m \times n\}}$	$\mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}, \boldsymbol{C} \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00107		—	$c_{i j}:=\sum_{k=1}^p a_{i k} b_{k j}$	$c_{i j}:=\sum_{k=1}^p a_{i k} b_{k j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00108		—	$i=1, \ldots, m$	$i=1, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00109		—	$j=1, \ldots, n$	$j=1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00110		—	$\boldsymbol{C}:=\boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D}:=\boldsymbol{A} / \boldsymbol{B}$	$\boldsymbol{C}:=\boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D}:=\boldsymbol{A} / \boldsymbol{B}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00111		—	$\boldsymbol{E}:=$	$\boldsymbol{E}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00112		—	$\boldsymbol{A} \wedge r$	$\boldsymbol{A} \wedge r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00113		—	$c_{ij}:=a_{ij} \, b_{ij}, \, d_{ij}:=a_{ij} \, / \, b_{ij}$	$c_{ij}:=a_{ij}b_{ij}, d_{ij}:=a_{ij}/b_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00114		—	$e_{ij}:=a_{ij}^r$	$e_{ij}:=a_{ij}^r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00115		—	r	r
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00116		—	$\boldsymbol{A} \, / \, \boldsymbol{B}$	$\boldsymbol{A}/\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00117		—	$\boldsymbol{B}$	$\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00118		—	$\boldsymbol{C}=\boldsymbol{A} \, \times \, \boldsymbol{B}$	$\boldsymbol{C}=\boldsymbol{A} \times \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00119		—	$\boldsymbol{A} \, \text{in} \, \mathbb{R}^{m \, \times \, n}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00120		—	$\boldsymbol{A}^{\mathrm{T}}$	$\boldsymbol{A}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00121		—	$\boldsymbol{A}^*$	$\boldsymbol{A}^*$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00122		—	$n \times m$	$n \times m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00123		—	$a_{ij}^T := a_{ji}$	$a_{ij}^T := a_{ji}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00124		—	$a_{ij}^* := \bar{a}_{ji}$	$a_{ij}^* := \bar{a}_{ji}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00125		—	n, p	n, p
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00126		—	$(\boldsymbol{A} \boldsymbol{B})^T = \boldsymbol{B}^T \boldsymbol{A}^T$	$(\boldsymbol{A} \boldsymbol{B})^T = \boldsymbol{B}^T \boldsymbol{A}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00127		—	$(\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{B}^* \boldsymbol{A}^*$	$(\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{B}^* \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00128		—	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00129		—	$\boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A}^T = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00130		—	$\boldsymbol{A}^* = \boldsymbol{A}$	$\boldsymbol{A}^* = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00131		—	$\boldsymbol{I}_n = \boldsymbol{I} := [\delta_{ij}]_{i,j=1}^n$	$\boldsymbol{I}_n = \boldsymbol{I} := [\delta_{ij}]_{i,j=1}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00132		—	$\boldsymbol{I}$	$\boldsymbol{I}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00133		—	$\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$	$e_1, e_2, \dots, e_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00134		—	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00135		—	$a_{ij}=0$	$a_{ij} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00136		—	$i \neq j$	$i \neq j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00137		—	$i>j$	$i > j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00138		—	$i<j$	$i < j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00139		—	$i>j+1$	$i > j + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00140		—	$i<j+1$	$i < j + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00141		—	$ i-j >1$	$ i - j > 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00142		—	d	d
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00143		—	$ i-j >d$	$ i - j > d$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00144		—	$n \times n$	$n \times n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00145		—	$b_{ii} := d_i$	$b_{ii} := d_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00146		—	$i=1, \ldots, n, b_{i+1, i} := a_i, b_{i, i+1} := c_i$	$i = 1, \dots, n, b_{i+1, i} := a_i, b_{i, i+1} := c_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00147		—	$i=1, \ldots, n-1$	$i = 1, \dots, n - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00148		—	$b_{ij} := 0$	$b_{ij} := 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00149		—	$1 \leq i_1 < i_2 < \dots < i_r \leq m, 1 \leq j_1 < j_2 < \dots <$	$1 \leq i_1 < i_2 < \dots < i_r \leq m, 1 \leq j_1 < j_2 < \dots <$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00150		—	$j_c \leq n$	$j_c \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00151		—	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$	$\boldsymbol{A}(\boldsymbol{i}, \boldsymbol{j}) \in \mathbb{C}^{r \times c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00152		—	$\boldsymbol{i} := [i_1, \ldots, i_r]$	$\boldsymbol{i} := [i_1, \dots, i_r]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00153		—	$\boldsymbol{j} := [j_1, \ldots, j_c]$	$\boldsymbol{j} := [j_1, \dots, j_c]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00154		—	$\mathbb{R}^2$	$\mathbb{R}^2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00155		—	$\mathbb{R}^3$	$\mathbb{R}^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00156		—	$\mathcal{V}$	$\mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00157		—	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00158		—	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00159		—	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00160		—	c, d	c, d
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00161		—	$\mathbb{R}$	$\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00162		—	$\boldsymbol{u} + \boldsymbol{v}$	$\boldsymbol{u} + \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00163		—	$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$	$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00164		—	$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$	$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00165		—	$\mathbf{0}$	$\mathbf{0}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00166		—	$\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}$	$\boldsymbol{u}+\mathbf{0}=\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00167		—	$\boldsymbol{u}$	$\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00168		—	$-\boldsymbol{u}$	$-\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00169		—	$\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$	$\boldsymbol{u}+(-\boldsymbol{u})=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00170		—	$c\cdot\boldsymbol{u}$	$c\cdot\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00171		—	$c\cdot(\boldsymbol{u}+\boldsymbol{v})=c\cdot\boldsymbol{u}+c\cdot\boldsymbol{v}$	$c\cdot(\boldsymbol{u}+\boldsymbol{v})=c\cdot\boldsymbol{u}+c\cdot\boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00172		—	$(c+d)\cdot\boldsymbol{u}=c\cdot\boldsymbol{u}+d\cdot\boldsymbol{u}$	$(c+d)\cdot\boldsymbol{u}=c\cdot\boldsymbol{u}+d\cdot\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00173		—	$c\cdot(d\cdot\boldsymbol{u})=(cd)\cdot\boldsymbol{u}$	$c\cdot(d\cdot\boldsymbol{u})=(cd)\cdot\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00174		—	$1\cdot\boldsymbol{u}=\boldsymbol{u}$	$1\cdot\boldsymbol{u}=\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00175		—	$c\cdot\boldsymbol{v}$	$c\boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00176		—	$c\cdot\boldsymbol{v}$	$c\cdot\boldsymbol{v}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00177		—	$\boldsymbol{u}-\boldsymbol{v}:=\boldsymbol{u}+(-\boldsymbol{v})$	$\boldsymbol{u}-\boldsymbol{v}:=\boldsymbol{u}+(-\boldsymbol{v})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00178		—	$\boldsymbol{u} \in \mathcal{V}$	$\boldsymbol{u} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00179		—	$\boldsymbol{0} \, \boldsymbol{u}=\mathbf{0}, \, c \, \mathbf{0}=\mathbf{0}$	$\boldsymbol{0} \boldsymbol{u}=\mathbf{0}, c\mathbf{0}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00180		—	$-\boldsymbol{u}=(-1) \, \boldsymbol{u}$	$-\boldsymbol{u}=(-1)\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00181		—	$n \in \mathbb{N}$	$n \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00182		—	$\mathcal{D}$	$\mathcal{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00183		—	$d \in \mathbb{N}$	$d \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00184		—	$\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^d$	$\boldsymbol{f}, \boldsymbol{g}: \mathcal{D} \rightarrow \mathbb{R}^d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00185		—	$\boldsymbol{f}, \boldsymbol{g}$	$\boldsymbol{f}, \boldsymbol{g}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00186		—	$\boldsymbol{f}(t)=\boldsymbol{g}(t)$	$\boldsymbol{f}(t)=\boldsymbol{g}(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00187		—	$t \in \mathcal{D}$	$t \in \mathcal{D}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00188		—	$\boldsymbol{f}(t)=\mathbf{0}$	$f(t) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00189		—	$\boldsymbol{f}$	f
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00190		—	$-\boldsymbol{f}=(-1)\boldsymbol{f}$	$-f = (-1)f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00191		—	$d>1$	$d > 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00192		—	$n \geq 0$	$n \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00193		—	$\Pi_{\{n\}}$	Π_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00194		—	$p: \mathbb{R} \rightarrow \mathbb{R}, p: \mathbb{R} \rightarrow \mathbb{C}$	$p: \mathbb{R} \rightarrow \mathbb{R}, p: \mathbb{R} \rightarrow \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00195		—	$p: \mathbb{C} \rightarrow \mathbb{C}$	$p: \mathbb{C} \rightarrow \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00196		—	$a_{\{0\}}, \ldots, a_{\{n\}}$	$a_0, \ldots, a_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00197		—	$0 \leq k \leq n$	$0 \leq k \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00198		—	$p(t)=a_{\{0\}}+\cdots+a_{\{k\}}\ t^{\{k\}}$	$p(t) = a_0 + \cdots + a_k t^k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00199		—	$a_{\{k\}} \neq 0$	$a_k \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00200		—	$n \geq 1$	$n \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00201		—	$\mathcal{X} := \left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\}$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00202		—	$c_{\{1\}}, \ldots, c_{\{n\}}$	$c_1, \dots, c_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00203		—	$c_{\{1\}} \boldsymbol{x}_{\{1\}} + \cdots + c_{\{n\}} \boldsymbol{x}_{\{n\}}$	$c_1 \boldsymbol{x}_1 + \cdots + c_n \boldsymbol{x}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00204		—	$\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}}$	$\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00205		—	$c_{\{j\}} \boldsymbol{x}_{\{j\}} \neq \mathbf{0}$	$c_j \boldsymbol{x}_j \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00206		—	$\mathcal{X}$	$\mathcal{X}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00207		—	$\operatorname{span}(\mathcal{X})$	$\operatorname{span}(\mathcal{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00208		—	$\left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00209		—	$\mathcal{V} = \operatorname{span} \left(\left\{ \boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{n\}} \right\} \right)$	$\mathcal{V} = \operatorname{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00210		—	$\boldsymbol{x}=\left[x_{1}, \ldots, x_{m}\right]^T$	$\boldsymbol{x}=\left[x_1, \ldots, x_m\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00211		—	$\mathbb{C}^m$	$\mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00212		—	$\boldsymbol{x}=\boldsymbol{x}_1 \boldsymbol{e}_1+\boldsymbol{x}_2 \boldsymbol{e}_2+\cdots+\boldsymbol{x}_m \boldsymbol{e}_m$	$\boldsymbol{x}=x_1 \boldsymbol{e}_1+x_2 \boldsymbol{e}_2+\cdots+x_m \boldsymbol{e}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00213		—	$\mathbb{C}^m=\operatorname{span}\left(\left\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_m\right\}\right)$	$\mathbb{C}^m=\operatorname{span}\left(\left\{e_1, \ldots, e_m\right\}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00214		—	$\mathbb{R}^m$	$\mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00215		—	$\Pi=\cup_n \Pi_n$	$\Pi=\cup_n \Pi_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00216		—	Π	Π
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00217		—	$\Pi=$	$\Pi=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00218		—	$\operatorname{span}\left(\left\{p_1, \ldots, p_m\right\}\right)$	$\operatorname{span}\left(\left\{p_1, \ldots, p_m\right\}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00219		—	$p_1, \ldots, p_m$	$p_1, \ldots, p_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00220		—	$p_{\{j\}}$	p_j
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00221		—	$j=1, \ldots, m$	$j = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00222		—	$\mathcal{X}=\left\{\boldsymbol{x}_{_1}, \ldots, \boldsymbol{x}_{_n}\right\}$	$\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00223		—	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00224		—	$\boldsymbol{x}=c_{_1} \boldsymbol{x}_{_1}+\cdots+$	$\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \cdots +$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00225		—	$c_{_n} \boldsymbol{x}_{_n}$	$c_n \boldsymbol{x}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00226		—	$\boldsymbol{v}_{_1}, \ldots, \boldsymbol{v}_{_n}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00227		—	$\boldsymbol{w}_{_1}, \ldots, \boldsymbol{w}_{_k}$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00228		—	$k \leq n$	$k \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00229		—	$k>n$	$k > n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00230		—	$\boldsymbol{w}_{_1}$	$\boldsymbol{w}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00231		—	$\mathcal{X}_{_0}:=\left\{\boldsymbol{v}_{_1}, \ldots, \boldsymbol{v}_{_n}\right\}$	$\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00232		—	$\boldsymbol{w}_1 = \boldsymbol{c}_1 + \boldsymbol{c}_2 + \dots + \boldsymbol{c}_n$	$\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00233		—	$\boldsymbol{w}_1 \neq \mathbf{0}$	$\boldsymbol{w}_1 \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00234		—	c_{i_1}	c_{i_1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00235		—	$\boldsymbol{v}_{i_1}$	$\boldsymbol{v}_{i_1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00236		—	$\boldsymbol{v}$	$\boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00237		—	$\mathcal{X}_1 :=$	$\mathcal{X}_1 :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00238		—	$\left\{ \boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n \right\}$	$\{ \boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n \}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00239		—	$\boldsymbol{w}_2$	$\boldsymbol{w}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00240		—	$\mathcal{X}_1$	$\mathcal{X}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00241		—	$\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$	$\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00242		—	$d_{i_2} \neq 0$	$d_{i_2} \neq 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00243		—	$i_{\{2\}}$	i_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00244		—	$i_{\{2\}} \neq i_{\{1\}}$	$i_2 \neq i_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00245		—	$\boldsymbol{w}_{\{2\}} = d_{\{1\}} \boldsymbol{w}_{\{1\}}$	$\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00246		—	$\boldsymbol{w}$	$\boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00247		—	$\mathcal{X}_{\{2\}}$	$\mathcal{X}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00248		—	$\boldsymbol{v}_{\{i_{\{2\}}\}}$	$\boldsymbol{v}_{i_2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00249		—	$n-2$	$n - 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00250		—	$\mathcal{X}_{\{n\}}$	$\mathcal{X}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00251		—	$\boldsymbol{w}_{\{1\}}, \ldots, \boldsymbol{w}_{\{n\}}$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00252		—	$\boldsymbol{w}_{\{k\}}$	$\boldsymbol{w}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00253		—	$\square$	$\square$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00254		—	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}$	$\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00255		—	$\operatorname{span}\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}=\mathcal{V}$	$\operatorname{span}\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}=\mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00256		—	$\left\{\boldsymbol{v}_{i_1}, \ldots, \boldsymbol{v}_{i_k}\right\}$	$\left\{\boldsymbol{v}_{i_1}, \ldots, \boldsymbol{v}_{i_k}\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00257		—	$\mathcal{V}=\operatorname{span}\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}$	$\mathcal{V}=\operatorname{span}\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00258		—	$\operatorname{dim} \mathcal{V}$	$\operatorname{dim} \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00259		—	$\mathcal{X}=\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}$	$\mathcal{X}=\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00260		—	$\mathcal{Y}=\left\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\right\}$	$\mathcal{Y}=\left\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00261		—	$\mathcal{Y}$	$\mathcal{Y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00262		—	$n \leq k$	$n \leq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00263		—	$n=k$	$n=k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00264		—	$\left\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\right\}$	$\left\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\right\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00265		—	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\right\}$	$\{v_1, \dots, v_k\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00266		—	$\boldsymbol{v}_{k+1}$	v_{k+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00267		—	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	$u, v \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00268		—	$c \boldsymbol{u}$	cu
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00269		—	$\boldsymbol{u} \in \mathcal{S}$	$u \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00270		—	$V_1 - V_5$	$V1 - V5$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00271		—	$S_1 - S_5$	$S1 - S5$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00272		—	$\{\mathbf{0}\}$	$\{\mathbf{0}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00273		—	$\mathcal{X}=\left\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\right\} \subseteq \mathcal{V}$	$\mathcal{X}=\left\{x_1, \ldots, x_n\right\} \subseteq \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00274		—	$\mathcal{S} \oplus \mathcal{T}$	$\mathcal{S} \oplus \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00275		—	$\mathcal{S} \cap \mathcal{T}=\{\mathbf{0}\}$	$\mathcal{S} \cap \mathcal{T}=\{\mathbf{0}\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00276		—	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00277		—	$\mathcal{S} \cap \mathcal{T}$	$\mathcal{S} \cap \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00278		—	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}\right\}=\emptyset$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00279		—	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{s}_{1}, \ldots, \boldsymbol{s}_{q}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00280		—	$\left\{\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{p}, \boldsymbol{t}_{1}, \ldots, \boldsymbol{t}_{r}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00281		—	$\boldsymbol{x} \in \mathcal{S}+\mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00282		—	$\mathcal{S}+\mathcal{T}$	$\mathcal{S} + \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00283		—	$\boldsymbol{u}+\boldsymbol{s}+\boldsymbol{t}=\mathbf{0}$	$\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00284		—	$\boldsymbol{u}:=\sum_{j=1}^p \alpha_j \boldsymbol{u}_j, \boldsymbol{s}:=\sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\boldsymbol{u}:=\sum_{j=1}^p \alpha_j \boldsymbol{u}_j, \boldsymbol{s}:=\sum_{j=1}^q \rho_j \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00285		—	$\boldsymbol{t}:=\sum_{j=1}^r \sigma_j \boldsymbol{t}_j$	$\boldsymbol{t}:=\sum_{j=1}^r \sigma_j \boldsymbol{t}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00286		—	$\boldsymbol{s}=-\left(\boldsymbol{u}+\boldsymbol{t}\right)$	$\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00287		—	$\boldsymbol{s} \in$	$\boldsymbol{s} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00288		—	$\boldsymbol{s}$	$\boldsymbol{s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00289		—	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00290		—	$\boldsymbol{s} :=$	$\boldsymbol{s} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00291		—	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00292		—	$\boldsymbol{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\boldsymbol{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00293		—	$\beta_1 = \cdots = \beta_p = \rho_1 = \cdots = \rho_q = 0$	$\beta_1 = \cdots = \beta_p = \rho_1 = \cdots = \rho_q = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00294		—	$\boldsymbol{s} = \boldsymbol{0}$	$\boldsymbol{s} = \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00295		—	$\boldsymbol{u} + \boldsymbol{t} = \boldsymbol{0}$	$\boldsymbol{u} + \boldsymbol{t} = \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00296		—	$\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$	$\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00297		—	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00298		—	$\operatorname{dim}\{\mathbf{0}\}=0$	$\dim\{\mathbf{0}\}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00299		—	$\left\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\right\}$	$\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00300		—	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\right\}$	$\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00301		—	$\boldsymbol{s}_j$	$\boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00302		—	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00303		—	$\boldsymbol{x} \in \mathcal{S}$	$\boldsymbol{x} \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00304		—	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{s}_j=\sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{s}_j=\sum_{i=1}^m b_i \boldsymbol{v}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00305		—	$\boldsymbol{b}:=$	$\boldsymbol{b}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00306		—	$\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m\right]^T, \boldsymbol{c}:=\left[c_1, \ldots, c_n\right]^T$	$\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m\right]^T, \boldsymbol{c}:=\left[c_1, \ldots, c_n\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00307		—	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}$	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00308		—	$\boldsymbol{A}=\left[\boldsymbol{a}_{i j}\right] \in \mathbb{C}^{m \times n}$	$\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{m \times n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00309		—	$a_{i\ j}$	a_{ij}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00310		—	$\boldsymbol{s}_j \in \mathcal{V}$	$\boldsymbol{s}_j \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00311		—	$\left(c_j\right)$	(c_j)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00312		—	$\left(b_i\right)$	(b_i)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00313		—	$b_i=\sum_{j=1}^n a_{i\ j} \ c_j$	$b_i = \sum_{j=1}^n a_{ij} c_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00314		—	$i=$	$i =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00315		—	$1, \ldots, m$	$1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00316		—	$\boldsymbol{b}:=\boldsymbol{A} \ \boldsymbol{c}=\mathbf{0}$	$\boldsymbol{b} := \boldsymbol{A} \boldsymbol{c} = \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00317		—	$\boldsymbol{c}=\left[c_1, \ldots, c_n\right]^T$	$\boldsymbol{c} = [c_1, \dots, c_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00318		—	$\boldsymbol{x}:=\sum_{j=1}^n \ c_j \ \boldsymbol{s}_j$	$\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00319		—	$\boldsymbol{x}=\sum_{i=1}^m \ b_i \ \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00320		—	$\boldsymbol{b}=\mathbf{0}$	$b = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00321		—	$\boldsymbol{x}=\mathbf{0}$	$x = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00322		—	$\boldsymbol{c}=\mathbf{0}$	$c = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00323		—	$\mathcal{V}=\mathbf{R}^m$	$\mathcal{V} = \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00324		—	$m \times n$	$m \times n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00325		—	$\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right]$	$\boldsymbol{X} = \left[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00326		—	$\boldsymbol{X} \boldsymbol{c}$	$\boldsymbol{X} \boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00327		—	$\boldsymbol{c} =$	$\boldsymbol{c} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00328		—	$\left[c_1, \ldots, c_n\right]^T$	$\left[c_1, \ldots, c_n\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00329		—	$\boldsymbol{x}_j \in \mathcal{V}, j=1, \ldots, n$	$\boldsymbol{x}_j \in \mathcal{V}, j=1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00330		—	$\mathcal{R}(\boldsymbol{X})$	$\mathcal{R}(\boldsymbol{X})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00331		—	$\boldsymbol{X}$	$\boldsymbol{X}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00332		—	$\mathcal{R}(\boldsymbol{X}^T)$	$\mathcal{R}(\boldsymbol{X}^T)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00333		—	$\mathcal{N}(\boldsymbol{X}) := \left\{ \boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X}\boldsymbol{y} = \boldsymbol{0} \right\}$	$\mathcal{N}(\boldsymbol{X}) := \{ \boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X}\boldsymbol{y} = \boldsymbol{0} \}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00334		—	$\mathcal{N}(\boldsymbol{X})$	$\mathcal{N}(\boldsymbol{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00335		—	$\operatorname{null}(\boldsymbol{X})$	$\operatorname{null}(\boldsymbol{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00336		—	$\boldsymbol{X} \in$	$\boldsymbol{X} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00337		—	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00338		—	$\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$	$\operatorname{rank}(\boldsymbol{X}) = \operatorname{rank}(\boldsymbol{X}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00339		—	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}) = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00340		—	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$	$\operatorname{rank}(\boldsymbol{X}) + \operatorname{null}(\boldsymbol{X}^*) = m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00341		—	$x_{\{j\}}$	x_j
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00342		—	$\mathbf{b}_{\{i\}}$	b_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00343		—	$\boldsymbol{a}_{\{j\}}=\left[a_{\{1\ j\}}, \ldots, \boldsymbol{a}_{\{m\ j\}}\right]^T \in \mathbb{C}^m$	$\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00344		—	$\boldsymbol{b}=\left[b_{\{1\}}, \ldots, b_{\{m\}}\right]^T \in \mathbb{C}^m$	$\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00345		—	$m<n, \ m=n$	$m < n, m = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00346		—	$m>n$	$m > n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00347		—	$m<n$	$m < n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00348		—	$\boldsymbol{A\ x}=\mathbf{0}$	$\boldsymbol{Ax} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00349		—	$\boldsymbol{A}\ \boldsymbol{x}=\mathbf{0}$	$\boldsymbol{Ax} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00350		—	$\mathbb{R}^{\{n\ \times\ n\}}$	$\mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00351		—	$\boldsymbol{b} \in \mathbb{R}^{\{n\}}$	$\boldsymbol{b} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00352		—	$\boldsymbol{B}=[\boldsymbol{A}\ \boldsymbol{b}] \in \mathbb{R}^{\{n\ \times\ (n+1)\}}$	$\boldsymbol{B} = [\boldsymbol{Ab}] \in \mathbb{R}^{n \times (n+1)}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00353		—	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	$\boldsymbol{z} \in \mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00354		—	$\boldsymbol{B} \ \boldsymbol{z}=\mathbf{0}$	$\boldsymbol{B}\boldsymbol{z} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00355		—	$\boldsymbol{z}=\left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$	$\boldsymbol{z} = \left[\begin{array}{c}\tilde{\boldsymbol{z}} \\ z_{n+1}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00356		—	$\tilde{\boldsymbol{z}}=\left[z_1, \ldots, z_n\right]^T \in \mathbb{R}^n$	$\tilde{\boldsymbol{z}} = \left[z_1, \ldots, z_n\right]^T \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00357		—	$z_{n+1} \in \mathbb{R}$	$z_{n+1} \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00358		—	$z_{n+1}=0$	$z_{n+1} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00359		—	$\boldsymbol{A} \ \tilde{\boldsymbol{z}}=\mathbf{0}$	$\boldsymbol{A}\tilde{\boldsymbol{z}} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00360		—	$\tilde{\boldsymbol{z}}$	$\tilde{\boldsymbol{z}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00361		—	$\boldsymbol{x}:-\tilde{\boldsymbol{z}} / z_{n+1}$	$\boldsymbol{x} := -\tilde{\boldsymbol{z}} / z_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00362		—	$\boldsymbol{A} \ \boldsymbol{y}=\boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00363		—	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00364		—	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\mathbf{0}$	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00365		—	$\boldsymbol{x}-\boldsymbol{y}=\mathbf{0}$	$\boldsymbol{x}-\boldsymbol{y}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00366		—	$\boldsymbol{x}=\boldsymbol{y}$	$\boldsymbol{x}=\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00367		—	$m<$	$m<$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00368		—	$\boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{b} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00369		—	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00370		—	$\boldsymbol{A} \boldsymbol{B}=\boldsymbol{I}$	$\boldsymbol{A}\boldsymbol{B}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00371		—	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00372		—	$\boldsymbol{C} \boldsymbol{A}=\boldsymbol{I}$	$\boldsymbol{C}\boldsymbol{A}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00373		—	$\boldsymbol{C}$	$\boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00374		—	$\boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00375		—	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00376		—	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00377		—	$\boldsymbol{A} \boldsymbol{B} =$	$\boldsymbol{A} \boldsymbol{B} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00378		—	$\boldsymbol{B} \boldsymbol{A} = \boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{A} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00379		—	$\boldsymbol{A}^{-1} = \boldsymbol{B}$	$\boldsymbol{A}^{-1} = \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00380		—	$\boldsymbol{C} \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{C} \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00381		—	$\boldsymbol{A} \boldsymbol{B} \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{B} \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00382		—	$\boldsymbol{B} \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{B} \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00383		—	$\boldsymbol{C} \boldsymbol{x} = (\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x} = \boldsymbol{A} (\boldsymbol{B} \boldsymbol{x}) = \boldsymbol{A} \mathbf{0} = \mathbf{0}$	$\boldsymbol{C} \boldsymbol{x} = (\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x} = \boldsymbol{A} (\boldsymbol{B} \boldsymbol{x}) = \boldsymbol{A} \mathbf{0} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00384		—	$\tilde{\boldsymbol{x}}$	$\tilde{\boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00385		—	$\boldsymbol{A} \tilde{\boldsymbol{x}} = \mathbf{0}$	$\boldsymbol{A} \tilde{\boldsymbol{x}} = \mathbf{0}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00386		—	$\boldsymbol{B} \ \boldsymbol{x}=\tilde{\boldsymbol{x}}$	$\boldsymbol{B} \boldsymbol{x} = \tilde{\boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00387		—	$\boldsymbol{C} \ \boldsymbol{x}=(\boldsymbol{A} \ \boldsymbol{B}) \ \boldsymbol{x}=\boldsymbol{A}(\boldsymbol{B} \ \boldsymbol{x})=\boldsymbol{A} \tilde{\boldsymbol{x}}=\mathbf{0}$	$\boldsymbol{C} \boldsymbol{x} = (\boldsymbol{A} \boldsymbol{B}) \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{B} \boldsymbol{x}) = \boldsymbol{A} \tilde{\boldsymbol{x}} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00388		—	$\boldsymbol{A} \ \boldsymbol{b}_i=\boldsymbol{e}_i$	$\boldsymbol{A} \boldsymbol{b}_i = \boldsymbol{e}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00389		—	$\boldsymbol{b}_i$	$\boldsymbol{b}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00390		—	$\boldsymbol{B}=\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\right]$	$\boldsymbol{B} = [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00391		—	$\boldsymbol{A} \ \boldsymbol{B}=\left[\boldsymbol{A} \ \boldsymbol{b}_1, \ldots, \boldsymbol{A} \ \boldsymbol{b}_n\right]=\left[\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\right]=\boldsymbol{I}$	$\boldsymbol{A} \boldsymbol{B} = [\boldsymbol{A} \boldsymbol{b}_1, \ldots, \boldsymbol{A} \boldsymbol{b}_n] = [\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n] = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00392		—	1.8 $\boldsymbol{B}$	1.8 $\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00393		—	$\boldsymbol{B} \ \boldsymbol{C}=\boldsymbol{I}$	$\boldsymbol{B} \boldsymbol{C} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00394		—	$\boldsymbol{A} \ \boldsymbol{B}=\boldsymbol{B} \ \boldsymbol{C}=\boldsymbol{I}$	$\boldsymbol{A} \boldsymbol{B} = \boldsymbol{B} \boldsymbol{C} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00395		—	$\boldsymbol{A}=\boldsymbol{C}$	$\boldsymbol{A} = \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00396		—	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00397		—	$\left(\boldsymbol{A}^{-1}\right)^{-1}=\boldsymbol{A}$	$\left(\boldsymbol{A}^{-1}\right)^{-1}=\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00398		—	$\boldsymbol{C}=\boldsymbol{A} \ \boldsymbol{B}$	$\boldsymbol{C}=\boldsymbol{A}\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00399		—	$\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1} \ \boldsymbol{A}^{-1}$	$\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00400		—	$\left(\boldsymbol{A}^T\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^T=: \boldsymbol{A}^{-T}$	$\left(\boldsymbol{A}^T\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^T=: \boldsymbol{A}^{-T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00401		—	$\left(\boldsymbol{A}^*\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^*=:\boldsymbol{A}^{-*}$	$\left(\boldsymbol{A}^*\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^*=:\boldsymbol{A}^{-*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00402		—	$c \ \boldsymbol{A}$	$c\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00403		—	$(c \ \boldsymbol{A})^{-1}=\frac{1}{c} \ \boldsymbol{A}^{-1}$	$(c\boldsymbol{A})^{-1}=\frac{1}{c}\boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00404		—	$\boldsymbol{A}^{-1} \ \boldsymbol{A}=\boldsymbol{I}$	$\boldsymbol{A}^{-1}\boldsymbol{A}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00405		—	$\left(\boldsymbol{B}^{-1} \ \boldsymbol{A}^{-1}\right)\left(\boldsymbol{A} \ \boldsymbol{B}\right)=\boldsymbol{B}^{-1}\left(\boldsymbol{A}^{-1} \ \boldsymbol{A}\right) \boldsymbol{B}=\boldsymbol{B}^{-1} \boldsymbol{B}=\boldsymbol{I}$	$\left(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}\right)\left(\boldsymbol{A}\boldsymbol{B}\right)=\boldsymbol{B}^{-1}\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)\boldsymbol{B}=\boldsymbol{B}^{-1}\boldsymbol{B}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00406		—	$\boldsymbol{A} \ \boldsymbol{B}$	$\boldsymbol{A}\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00407		—	$\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1} \ \boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$	$\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1}\boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00408		—	$\left(\boldsymbol{A}^{-1}\right)^{T}$	$\left(\boldsymbol{A}^{-1}\right)^{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00409		—	$\frac{1}{c} \boldsymbol{A}^{-1}$	$\frac{1}{c} \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00410		—	$n!$	$n!$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00411		—	$\{1, 2, \ldots, n\}$	$\{1, 2, \ldots, n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00412		—	$\operatorname{sign}(\sigma)$	$\operatorname{sign}(\sigma)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00413		—	σ	σ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00414		—	ϵ	ϵ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00415		—	$\epsilon(i)=$	$\epsilon(i) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00416		—	$i, i=1, 2$	$i, i = 1, 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00417		—	$\sigma=\{2, 1\}$	$\sigma = \{2, 1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00418		—	$n=3$	$n = 3$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00419		—	$\{1,2,3\}$	$\{1,2,3\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00420		—	$\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1$	$\operatorname{sign}(\{1,2,3\})=\operatorname{sign}(\{2,3,1\})=\operatorname{sign}(\{3,1,2\})=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00421		—	$\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1$	$\operatorname{sign}(\{2,1,3\})=\operatorname{sign}(\{3,2,1\})=\operatorname{sign}(\{1,3,2\})=-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00422		—	$\operatorname{det}(\boldsymbol{A})=$	$\det(\boldsymbol{A})=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00423		—	$a_{11} \, a_{22} \, \cdots \, a_{nn}$	$a_{11}a_{22}\cdots a_{nn}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00424		—	$\operatorname{det}(\boldsymbol{I})=1$	$\det(\boldsymbol{I})=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00425		—	$\operatorname{det}(\boldsymbol{B})=-\operatorname{det}(\boldsymbol{A})$	$\det(\boldsymbol{B})=-\det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00426		—	$\alpha, \operatorname{det}(\boldsymbol{B})=\alpha \operatorname{det}(\boldsymbol{A})$	$\alpha, \det(\boldsymbol{B})=\alpha \det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00427		—	$\boldsymbol{A}_{i,j}$	$\boldsymbol{A}_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00428		—	$1 \leq i, j \leq n$	$1 \leq i, j \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00429		—	$\operatorname{det}\!\left(\boldsymbol{A}_{ij}\right)$	$\det(\boldsymbol{A}_{ij})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00430		—	$\left(x_{1}, y_{1}\right)$	(x_1, y_1)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00431		—	$\left(x_{2}, y_{2}\right)$	(x_2, y_2)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00432		—	$\operatorname{det}(\boldsymbol{A})=0$	$\det(\boldsymbol{A}) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00433		—	$\boldsymbol{A}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00434		—	$\operatorname{det}(\boldsymbol{A B})=\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{B})$	$\det(\boldsymbol{AB}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00435		—	$\operatorname{det}\left(\boldsymbol{A}^T\right)=\operatorname{det}(\boldsymbol{A})$	$\det\left(\boldsymbol{A}^T\right) = \det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00436		—	$\operatorname{det}\left(\boldsymbol{A}^*\right)=\overline{\operatorname{det}(\boldsymbol{A})}$	$\det\left(\boldsymbol{A}^*\right) = \overline{\det(\boldsymbol{A})}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00437		—	$\operatorname{det}(a \boldsymbol{A})=a^n \operatorname{det}(\boldsymbol{A})$	$\det(a \boldsymbol{A}) = a^n \det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00438		—	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{C} & \boldsymbol{D} \\ \boldsymbol{0} & \boldsymbol{E}\end{array}\right]$	$\boldsymbol{A} = \left[\begin{array}{cc} \boldsymbol{C} & \boldsymbol{D} \\ \boldsymbol{0} & \boldsymbol{E} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00439		—	$\boldsymbol{C}, \boldsymbol{E}$	$\boldsymbol{C}, \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00440		—	$\operatorname{det}(\boldsymbol{A})=\operatorname{det}(\boldsymbol{C}) \operatorname{det}(\boldsymbol{E})$	$\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00441		—	$\boldsymbol{x} =$	$\boldsymbol{x} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00442		—	$\left[x_1, x_2, \ldots, x_n\right]^T$	$[x_1, x_2, \ldots, x_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00443		—	$\boldsymbol{A}_j(\boldsymbol{b})$	$\boldsymbol{A}_j(\boldsymbol{b})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00444		—	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00445		—	$\operatorname{adj}(\boldsymbol{A})_{i,j} = (-1)^{i+j} \operatorname{det}\left(\boldsymbol{A}_{j,i}\right)$	$\operatorname{adj}(\boldsymbol{A})_{i,j} = (-1)^{i+j} \operatorname{det}(\boldsymbol{A}_{j,i})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00446		—	$\boldsymbol{A}_{j,i}$	$\boldsymbol{A}_{j,i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00447		—	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00448		—	$1 \leq r \leq \min \{m, n, p\}$	$1 \leq r \leq \min \{m, n, p\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00449		—	$\boldsymbol{i} = \left\{i_1, \ldots, i_r\right\}$	$\boldsymbol{i} = \{i_1, \ldots, i_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00450		—	$\boldsymbol{j} = \left\{j_1, \ldots, j_r\right\}$	$\boldsymbol{j} = \{j_1, \ldots, j_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00451		—	$1 \leq i_1 < i_2 < \cdots < i_r \leq m$	$1 \leq i_1 < i_2 < \cdots < i_r \leq m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00452		—	$1 \leq j_{\{1\}} < j_{\{2\}} < \cdots < j_{\{r\}} \leq n$	$1 \leq j_1 < j_2 < \cdots < j_r \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00453		—	$\boldsymbol{k} = \left\{ k_{\{1\}}, \ldots, k_{\{r\}} \right\}$	$\boldsymbol{k} = \{k_1, \dots, k_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00454		—	$1 \leq k_{\{1\}} < k_{\{2\}} < \cdots < k_{\{r\}} \leq p$	$1 \leq k_1 < k_2 < \cdots < k_r \leq p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00455		—	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00456		—	$\lambda, \boldsymbol{x}$	$\lambda, \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00457		—	$\sigma(\boldsymbol{A})$	$\sigma(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00458		—	$\sigma(\boldsymbol{I}) = \{1, \ldots, 1\} = \{1\}$	$\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00459		—	$\lambda \in$	$\lambda \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00460		—	$\sigma(\boldsymbol{A}) \xrightarrow{\operatorname{det}} \sigma(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$	$\sigma(\boldsymbol{A}) \iff \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00461		—	$(\lambda, \boldsymbol{x})$	$(\lambda, \boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00462		—	$(\boldsymbol{A} - \lambda \boldsymbol{I}) \boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} - \lambda \boldsymbol{I}) \boldsymbol{x} = \mathbf{0}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00463		—	$\boldsymbol{A}-\lambda \boldsymbol{I}$	$\boldsymbol{A}-\lambda \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00464		—	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=0$	$\det(\boldsymbol{A}-\lambda \boldsymbol{I})=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00465		—	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	$\det(\boldsymbol{A}-\lambda \boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00466		—	$r(\lambda)$	$r(\lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00467		—	$n-2, \operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	$n-2, \det(\boldsymbol{A}-\lambda \boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00468		—	$\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})=(-1)^n \operatorname{det}(\lambda \boldsymbol{I}-\boldsymbol{A})$	$\det(\boldsymbol{A}-\lambda \boldsymbol{I})=(-1)^n \det(\lambda \boldsymbol{I}-\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00469		—	$\operatorname{det}(\lambda \boldsymbol{I}-\boldsymbol{A})=0$	$\det(\lambda \boldsymbol{I}-\boldsymbol{A})=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00470		—	$\pi_{\boldsymbol{A}}: \mathbb{C} \rightarrow$	$\pi_{\boldsymbol{A}}: \mathbb{C} \rightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00471		—	$\pi_{\boldsymbol{A}}(\lambda)=\operatorname{det}(\boldsymbol{A}-\lambda \boldsymbol{I})$	$\pi_{\boldsymbol{A}}(\lambda)=\det(\boldsymbol{A}-\lambda \boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00472		—	$\lambda_1, \ldots, \lambda_n$	$\lambda_1, \ldots, \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00473		—	$\boldsymbol{A} \overline{x}=\bar{\lambda} \overline{x}$	$\boldsymbol{A} \overline{x}=\lambda \overline{x}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00474		—	$\pi_{\mathbf{A}}$	$\pi_{\mathbf{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00475		—	$c_{n-1}=(-1)^{n-1} \operatorname{trace}(\mathbf{A})$	$c_{n-1}=(-1)^{n-1} \operatorname{trace}(\mathbf{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00476		—	$c_0=\pi_{\mathbf{A}}(0)=\operatorname{det}(\mathbf{A})$	$c_0=\pi_{\mathbf{A}}(0)=\operatorname{det}(\mathbf{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00477		—	$d_{n-1}=(-1)^{n-1}\left(\lambda_1+\cdots+\lambda_n\right)$	$d_{n-1}=(-1)^{n-1}\left(\lambda_1+\cdots+\lambda_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00478		—	$d_0=\lambda_1 \cdots \lambda_n$	$d_0=\lambda_1 \cdots \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00479		—	$c_j=d_j$	$c_j=d_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00480		—	$\mathbf{A}=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right]$	$\mathbf{A}=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00481		—	$\operatorname{trace}(\mathbf{A})=4, \operatorname{det}(\mathbf{A})=3$	$\operatorname{trace}(\mathbf{A})=4, \operatorname{det}(\mathbf{A})=3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00482		—	$\pi_{\mathbf{A}}(\lambda)=\lambda^2-4 \lambda+3$	$\pi_{\mathbf{A}}(\lambda)=\lambda^2-4 \lambda+3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00483		—	$\operatorname{Leftrightarrow} \mathbf{A} \mathbf{x}=\mathbf{0}$	$\operatorname{Leftrightarrow} \mathbf{A} \mathbf{x}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00484		—	$\mathbf{x} \neq \mathbf{0} \operatorname{Leftrightarrow} \mathbf{A} \mathbf{x}=\mathbf{0} \mathbf{x}$	$\mathbf{x} \neq \mathbf{0} \operatorname{Leftrightarrow} \mathbf{A} \mathbf{x}=\mathbf{0} \mathbf{x}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00485		—	$\boldsymbol{x} \neq$	$x \neq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00486		—	$\Leftrightarrow$	$0 \Leftrightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00487		—	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{B} & \boldsymbol{D} \\ 0 & \boldsymbol{C}\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{B} & \boldsymbol{D} \\ 0 & \boldsymbol{C} \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00488		—	$\pi_{\boldsymbol{A}}=\pi_{\boldsymbol{B}}\cdot\pi_{\boldsymbol{C}}$	$\pi_{\boldsymbol{A}} = \pi_{\boldsymbol{B}} \cdot \pi_{\boldsymbol{C}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00489		—	$\boldsymbol{A},\boldsymbol{B}\in\mathbb{R}^{n\times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00490		—	$2n\times 2n$	$2n \times 2n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00491		—	$\boldsymbol{A},\ldots,\boldsymbol{Z}$	$\boldsymbol{A}, \ldots, \boldsymbol{Z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00492		—	$\boldsymbol{W},\boldsymbol{X},\boldsymbol{Y}$	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00493		—	$\boldsymbol{Z}$	$\boldsymbol{Z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00494		—	$m\times m$	$m \times m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00495		—	$m=2^k$	$m = 2^k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00496		—	$k \geq 0$	$k \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00497		—	$\mathcal{O}\left(m^{\log_2(7)}\right)$	$\mathcal{O}\left(m^{\log_2(7)}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00498		—	$\log_2(7) \approx 2.8<3$	$\log_2(7) \approx 2.8 < 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00499		—	2×2	2×2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00500		—	a, b, c, d	a, b, c, d
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00501		—	$a \, d - b \, c \neq 0$	$ad - bc \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00502		—	$\boldsymbol{B}, \boldsymbol{C} \in$	$\boldsymbol{B}, \boldsymbol{C} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00503		—	$\mathbb{R}^{n \times m}$	$\mathbb{R}^{n \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00504		—	$n, m \in \mathbb{N}$	$n, m \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00505		—	$\left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1}$	$\left(\boldsymbol{I} + \boldsymbol{C}^T \boldsymbol{A}^{-1} \boldsymbol{B}\right)^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00506		—	$\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$	$\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00507		—	$\boldsymbol{v}^T \boldsymbol{u} \neq 1$	$\boldsymbol{v}^T \boldsymbol{u} \neq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00508		—	$\boldsymbol{I} - \boldsymbol{u} \boldsymbol{v}^T$	$\boldsymbol{I} - \boldsymbol{u} \boldsymbol{v}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00509		—	$\boldsymbol{I} - \tau \boldsymbol{u} \boldsymbol{v}^T$	$\boldsymbol{I} - \tau \boldsymbol{u} \boldsymbol{v}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00510		—	$\tau = 1 / (\boldsymbol{v}^T \boldsymbol{u} - 1)$	$\tau = 1 / (\boldsymbol{v}^T \boldsymbol{u} - 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00511		—	$\boldsymbol{C} := \boldsymbol{A}^{-1}$	$\boldsymbol{C} := \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00512		—	$\boldsymbol{a} \in \mathbb{R}^n$	$\boldsymbol{a} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00513		—	$\overline{\boldsymbol{A}}$	$\overline{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00514		—	$\boldsymbol{a}^T$	$\boldsymbol{a}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00515		—	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i (\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$	$\overline{\boldsymbol{A}} = \boldsymbol{A} - \boldsymbol{e}_i (\boldsymbol{e}_i^T \boldsymbol{A} - \boldsymbol{a}^T)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00516		—	$\boldsymbol{e}_i$	$\boldsymbol{e}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00517		—	$\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$	$\overline{\boldsymbol{C}} = \overline{\boldsymbol{A}}^{-1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00518		—	$\boldsymbol{a}$	$\boldsymbol{a}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00519		—	$\overline{\boldsymbol{C}}$	$\overline{\boldsymbol{C}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00520		—	$\lambda \neq 0$	$\lambda \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00521		—	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00522		—	$\boldsymbol{B} \ \boldsymbol{C}$	$\boldsymbol{B} \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00523		—	$\boldsymbol{L} \in \mathbb{R}^{n \times n}$	$\boldsymbol{L} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00524		—	$\boldsymbol{A}=\boldsymbol{L}+\boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{L} + \boldsymbol{L}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00525		—	$\boldsymbol{E}^T \ \boldsymbol{A} \ \boldsymbol{E}=\boldsymbol{S}+\boldsymbol{S}^T$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \boldsymbol{S} + \boldsymbol{S}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00526		—	$\boldsymbol{S}=\boldsymbol{E}^T \ \boldsymbol{L} \ \boldsymbol{E}$	$\boldsymbol{S} = \boldsymbol{E}^T \boldsymbol{L} \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00527		—	$\boldsymbol{E}^T \ \boldsymbol{A} \ \boldsymbol{E}$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00528		—	$\left(x_1, y_1, z_1\right), \left(x_2, y_2, z_2\right), \left(x_3, y_3, z_3\right)$	$\left(x_1, y_1, z_1\right), \left(x_2, y_2, z_2\right), \left(x_3, y_3, z_3\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00529		—	$P_{\{i\}}=\left(x_{\{i\}}, y_{\{i\}}\right), i=1,2,3$	$P_i = \left(x_i, y_i\right), i = 1, 2, 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00530		—	$T \ \mathrm{is}^{\{2\}}$	$T\mathrm{i}s^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00531		—	$\prod_{i>j}\left(x_{\{i\}}-x_{\{j\}}\right)=\prod_{i=2}^n\left(x_{\{i\}}-x_{\{1\}}\right)\left(x_{\{i\}}-x_{\{2\}}\right)\cdots\left(x_{\{i\}}-x_{\{i-1\}}\right)$	$\prod_{i>j}\left(x_i-x_j\right)=\prod_{i=2}^n\left(x_i-x_1\right)\left(x_i-x_2\right)\cdots\left(x_i-x_{i-1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00532		—	$\boldsymbol{\alpha}=\left[\alpha_{\{1\}}, \ldots, \alpha_{\{n\}}\right]^{\{T\}}, \boldsymbol{\beta}=\left[\beta_{\{1\}}, \ldots, \beta_{\{n\}}\right]^{\{T\}}$	$\boldsymbol{\alpha}=\left[\alpha_1, \ldots, \alpha_n\right]^T, \boldsymbol{\beta}=\left[\beta_1, \ldots, \beta_n\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00533		—	$a_{\{i, j\}}=1 /\left(\alpha_{\{i\}}+\beta_{\{j\}}\right), i, j=1,2, \ldots, n$	$a_{i, j}=1 /\left(\alpha_i+\beta_j\right), i, j=1,2, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00534		—	$A(T)=A\left(A B P_{\{3\}} P_{\{1\}}\right)+A\left(P_{\{3\}} B C P_{\{2\}}\right)-A\left(P_{\{1\}} A C P_{\{2\}}\right)$	$A(T)=A\left(A B P_3 P_1\right)+A\left(P_3 B C P_2\right)-A\left(P_1 A C P_2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00535		—	$x_{\{n\}}^{\{k\}}$	x_n^k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00536		—	$k+1$	$k+1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00540		—	k=n-1, n-2, \ldots, 1	$k = n - 1, n - 2, \dots, 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00541		—	P=\prod_{i=1}^n \prod_{j=1}^n a_{i j}	$P = \prod_{i=1}^n \prod_{j=1}^n a_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00542		—	\boldsymbol{\gamma}=\left[\gamma_1, \ldots, \gamma_n\right]^T	$\boldsymbol{\gamma} = [\gamma_1, \dots, \gamma_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00543		—	\prod_{j=1}^n\left(\alpha_{i}+\beta_{j}\right)	$\prod_{j=1}^n (\alpha_i + \beta_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00544		—	i=1,2, \ldots, n	$i = 1, 2, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00545		—	n-1	$n - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00546		—	\alpha_{r}+\beta_{s}	$\alpha_r + \beta_s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00547		—	\operatorname{det}(\boldsymbol{C})	$\det(\boldsymbol{C})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00548		—	n(n-1)	$n(n - 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00549		—	\operatorname{det}(\boldsymbol{C})=q(\alpha, \beta)	$\det(\boldsymbol{C}) = q(\alpha, \beta)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00550		—	q	q
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00551		—	α_i	α_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00552		—	β_j	β_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00553		—	$\operatorname{det}(\boldsymbol{A})$	$\det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00554		—	$\alpha_i=\alpha_j$	$\alpha_i=\alpha_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00555		—	$\beta_r=\beta_s$	$\beta_r=\beta_s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00556		—	$r \neq s$	$r \neq s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00557		—	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00558		—	$g(\boldsymbol{\alpha})$	$g(\boldsymbol{\alpha})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00559		—	$g(\boldsymbol{\beta})$	$g(\boldsymbol{\beta})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00560		—	$\boldsymbol{\alpha}$	$\boldsymbol{\alpha}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00561		—	$\boldsymbol{\beta}$	$\boldsymbol{\beta}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00562		—	k=1	$k = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00563		—	$\beta_i + \alpha_i = 0, i = 1, 2, \ldots, n$	$\beta_i + \alpha_i = 0, i = 1, 2, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00564		—	$\boldsymbol{A}^{-1} = \left(b_{j, k}\right)$	$\boldsymbol{A}^{-1} = (b_{j,k})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00565		—	$\boldsymbol{H}_{\{n\}} = \left(h_{\{i, j\}}\right)$	$\boldsymbol{H}_n = (h_{i,j})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00566		—	$h_{\{i, j\}} = 1 / (i + j - 1)$	$h_{i,j} = 1 / (i + j - 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00567		—	$t_{\{i, j\}}^{\{n\}}$	$t_{i,j}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00568		—	$\boldsymbol{T}_{\{n\}} = \boldsymbol{H}_{\{n\}}^{-1}$	$\boldsymbol{T}_n = \boldsymbol{H}_n^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00569		—	$\boldsymbol{L}$	$\boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00570		—	$\boldsymbol{U}$	$\boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00571		—	$\boldsymbol{A} =$	$\boldsymbol{A} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00572		—	$\boldsymbol{L} \ \boldsymbol{U}$	$\boldsymbol{LU}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00573		—	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00574		—	$\boldsymbol{D}$	$\boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00575		—	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00576		—	$\boldsymbol{Q}$	$\boldsymbol{Q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00577		—	$\boldsymbol{Q}^{\ast} \boldsymbol{Q}=\boldsymbol{I}$	$\boldsymbol{Q}^{\ast} \boldsymbol{Q} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00578		—	$\boldsymbol{R}$	$\boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00579		—	$[a, b], n+1 \geq 2$	$[a, b], n + 1 \geq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00580		—	$[a, b]$	$[a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00581		—	y	y
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00582		—	$\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{y}:=\left[y_1, \ldots, y_{n+1}\right]^T \in \mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00583		—	$\mathbf{g}: [a, b] \rightarrow \mathbb{R}$	$g:[a, b] \rightarrow \mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00584		—	$a \leq x_1 < x_2 < \cdots < x_{n+1} \leq b$	$a \leq x_1 < x_2 < \cdots < x_{n+1} \leq b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00585		—	$n+1$	$n + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00586		—	g	g
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00587		—	$n=1, g$	$n = 1, g$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00588		—	f	f
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00589		—	$[a, b]=[-1,1]$	$[a, b] = [-1, 1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00590		—	$y_{\{i\}}=f\left(x_{\{i\}}\right), i=1, \ldots, 14$	$y_i = f\left(x_i\right), i = 1, \ldots, 14$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00591		—	$p_{\{i\}}$	p_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00592		—	≤ 1	≤ 1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00593		—	$p_{\{1\}}$	p_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00594		—	≤ 2	≤ 2
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00595		—	C^2	C^2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00596		—	D_2	D_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00597		—	$\boldsymbol{y} \in$	$\boldsymbol{y} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00598		—	$\mathbb{R}^{n+1}$	$\mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00599		—	$x_1, \ldots, x_{n+1}$	$x_1, \ldots, x_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00600		—	μ_1, μ_{n+1}	μ_1, μ_{n+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00601		—	$g \in C^2[a, b]$	$g \in C^2[a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00602		—	$g\text{\texttt{\textleftarrow}}(x_i)\text{\texttt{\textrightarrow}}y_i, \quad i=1,2, \ldots, n+1$	$g(x_i) = y_i, \quad i = 1, 2, \ldots, n+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00603		—	$\boldsymbol{D}_{\mathbf{2}}$	$\boldsymbol{D_2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00604		—	$g^{\prime \prime}(a)=\mu_1, \quad g^{\prime \prime}(b)=\mu_{n+1}$	$g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00605		—	N	N
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00606		—	$\mu_1=\mu_{n+1}=0$	$\mu_1 = \mu_{n+1} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00607		—	$n=2$	$n = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00608		—	$f:[0,2] \rightarrow \mathbb{R}$	$f : [0, 2] \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00609		—	$[a, \ b]=[0,2], \ \boldsymbol{y}:=[0,1,16]^T$	$[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00610		—	$\mu_1=g^{\{\text{prime } \text{prime}\}}(0)=0, \ \mu_3=g^{\{\text{prime } \text{prime}\}}(2)=$	$\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00611		—	$x_{\{1\}}=0, \ x_{\{2\}}=1$	$x_1 = 0, x_2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00612		—	$x_{\{3\}}=2$	$x_3 = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00613		—	$p_{\{2\}}$	p_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00614		—	$p_{\{1\}}(1)=p_{\{2\}}(1)=1, \ p_{\{1\}}^{\{\text{prime}\}}(1)=p_{\{2\}}^{\{\text{prime}\}}(1)=4, \ p_{\{1\}}^{\{\text{prime } \text{prime}\}}(1)=p_{\{2\}}^{\{\text{prime } \text{prime}\}}(1)=$	$p_1(1) = p_2(1) = 1, p_1'(1) = p_2'(1) = 4, p_1''(1) = p_2''(1) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00615		—	$g \in C^2[0,2]$	$g \in C^2[0, 2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00616		—	$g(0)=p_{\{1\}}(0)=0, \ g(1)=p_{\{2\}}(1)=1, \ g(2)=$	$g(0) = p_1(0) = 0, g(1) = p_2(1) = 1, g(2) =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00617		—	$p_{\{2\}}(2)=16, g^{\{\backslashprime \backslashprime\}}(\theta)=p_{\{1\}}^{\{\backslashprime \backslashprime\}}(\theta)=\theta$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00618		—	$g^{\{\backslashprime \backslashprime\}}(2)=p_{\{2\}}^{\{\backslashprime \backslashprime\}}(2)=48$	$g''(2) = p_2''(2) = 48$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00619		—	$p_{\{1\}}^{\{\backslashprime \backslashprime \backslashprime\}}(x)=9 \neq$	$p_1'''(x) = 9 \neq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00620		—	$39=p_{\{2\}}^{\{\backslashprime \backslashprime \backslashprime\}}(x)$	$39 = p_2'''(x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00621		—	$3(n-1) \ C^{\{2\}}$	$3(n - 1)C^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00622		—	$4 \ n$	$4n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00623		—	$p_{\{1\}}, \ \ldots, \ p_{\{n\}}$	$p_1, \dots, p_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00624		—	$[0,2]$	$[0, 2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00625		—	$(0,2)$	$(0, 2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00626		—	$a<b$	$a < b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00627		—	$h=(b-a) \ / \ n$	$h = (b - a)/n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00628		—	$n \geq 2, x_{\mathfrak{i}} = a + (i - 1) h$	$n \geq 2, x_i = a + (i - 1)h$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00629		—	$y_{\mathfrak{i}}, \mu_{\mathfrak{i}}$	y_i, μ_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00630		—	$i = 1, \ldots, n + 1$	$i = 1, \ldots, n + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00631		—	$1 \leq i \leq n$	$1 \leq i \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00632		—	$p_{\mathfrak{i}} \backslash \left(x_{\mathfrak{i}} \backslash \right) = c_{\mathfrak{i}, 1} = y_{\mathfrak{i}}$	$p_i \left(x_i\right) = c_{i,1} = y_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00633		—	$p_{\mathfrak{i}}^{\backslash \prime \prime} \backslash \left(x_{\mathfrak{i}} \backslash \right) = 2 c_{\mathfrak{i}, 3} + 6 c_{\mathfrak{i}, 4} \backslash \left(x - x_{\mathfrak{i}} \backslash \right)$	$p_i^{\prime \prime} (x) = 2 c_{i,3} + 6 c_{i,4} \left(x - x_i\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00634		—	$c_{\mathfrak{i}, 3}$	$c_{i,3}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00635		—	$p_{\mathfrak{i}}^{\backslash \prime \prime} \backslash \left(x_{\mathfrak{i}} \backslash \right) =$	$p_i^{\prime \prime} \left(x_i\right) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00636		—	$2 c_{\mathfrak{i}, 3} = \mu_{\mathfrak{i}}$	$2 c_{i,3} = \mu_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00637		—	$c_{\mathfrak{i}, 4}$	$c_{i,4}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00638		—	$p_{\mathfrak{i}}^{\backslash \prime \prime} \backslash \left(x_{\mathfrak{i} + 1} \backslash \right) = \mu_{\mathfrak{i}} + 6 h c_{\mathfrak{i}, 4} = \mu_{\mathfrak{i} + 1}$	$p_i^{\prime \prime} \left(x_{i + 1}\right) = \mu_i + 6 h c_{i,4} = \mu_{i + 1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00639		—	$c_{i, 2}$	$c_{i,2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00640		—	$p_{i}\backslash\left(x_{i+1}\right)=y_{i}+c_{i, 2} h+\frac{\mu_{i}}{2} h^2+\frac{\mu_{i+1}-\mu_{i}}{6} h^3=y_{i+1}$	$p_i\left(x_{i+1}\right)=y_i+c_{i,2} h+\frac{\mu_i}{2} h^2+\frac{\mu_{i+1}-\mu_i}{6 h} h^3=y_{i+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00641		—	$j=0,1,2,3$	$j=0,1,2,3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00642		—	$\backslash\left(x-x_{i}\right)^{j}$	$\left(x-x_i\right)^j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00643		—	$\mu_{1}, \ldots$	$\mu_1, \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00644		—	μ_{n+1}	μ_{n+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00645		—	$\mu_{1}, \ldots, \mu_{n+1}$	$\mu_1, \ldots, \mu_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00646		—	$p_{i-1}\backslash\left(x_{i}\right)=p_{i}\backslash\left(x_{i}\right)=y_{i}$	$p_{i-1}\left(x_i\right)=p_i\left(x_i\right)=y_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00647		—	$p_{i-1}^{\prime \prime} \backslash\left(x_{i}\right)=p_i^{\prime \prime} \backslash\left(x_{i}\right)=\mu_{i}$	$p_{i-1}^{\prime \prime}\left(x_i\right)=p_i^{\prime \prime}\left(x_i\right)=\mu_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00648		—	$i=2, \ldots, n$	$i=2, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00649		—	$g \in C^2$	$g \in C^2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00650		—	$p_{i-1}^{\prime }\left(x_{i}\right) =p_{i}^{\prime }\left(x_{i}\right)$	$p_{i-1}'(x_i)=p_i'(x_i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00651		—	$g\left(x_{i}\right) =y_{i},\text{ }i=1,\text{ }\ldots ,\text{ }n+1,\text{ }g^{\prime }\left(a\right) =p_{1}^{\prime }\left(x_{1}\right) =\mu _{1}$	$g(x_i)=y_i,i=1,\dots ,n+1,g''(a)=p_1''(x_1)=\mu _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00652		—	$g^{\prime }\left(b\right) =$	$g''(b)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00653		—	$p_{n}^{\prime }\left(x_{n+1}\right) =\mu _{n+1}$	$p_n''(x_{n+1})=\mu _{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00654		—	$\mu _{2},\text{ }\ldots ,\text{ }\mu _{n}$	$\mu _2,\dots ,\mu _n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00655		—	$n\geq 3$	$n\geq 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00656		—	$\boldsymbol{A}=\left[a_{ij}\right] \text{ in }\mathbb{C}^{n\times n}$	$\boldsymbol{A}=[a_{ij}]\in \mathbb{C}^{n\times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00657		—	$\left x_{k}\right =\max _{i}\left x_{i}\right $	$ x_k =\max_i x_i $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00658		—	$\max _{1\leq i\leq n}\left x_{i}\right =\left x_{k}\right \leq \frac{\left b_{k}\right }{\sigma _k}\leq \max _{1\leq i\leq n}\left(\frac{\left b_i\right }{\sigma _i}\right)$	$\max_{1\leq i\leq n} x_i = x_k \leq \frac{ b_k }{\sigma_k}\leq \max_{1\leq i\leq n}\left(\frac{ b_i }{\sigma_i}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00659		—	$\max _{1\leq i\leq n}\left x_{i}\right \leq 0$	$\max_{1\leq i\leq n} x_i \leq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00660		—	g_{1}	g_1
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00661		—	$g_{\{2\}}$	g_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00662		—	$g:=g_{\{1\}}\cdot g_{\{2\}}$	$g := g_1 - g_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00663		—	$\boldsymbol{y}=\mathbf{0}$	$\boldsymbol{y} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00664		—	$\left[\mu_{\{2\}}, \ldots, \mu_{\{n\}}\right]^T$	$[\mu_2, \ldots, \mu_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00665		—	$\mu_{\{1\}}=\mu_{\{n+1\}}$	$\mu_1 = \mu_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00666		—	$c_{\{i, j\}}$	$c_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00667		—	$g=0$	$g = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00668		—	$g_{\{1\}}=g_{\{2\}}$	$g_1 = g_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00669		—	B	B
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00670		—	$[a, b]=[0,4]$	$[a, b] = [0, 4]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00671		—	$\boldsymbol{y}=$	$\boldsymbol{y} =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00672		—	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]$	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00673		—	$\mu_2=\mu_4=1, \mu_3=-2$	$\mu_2 = \mu_4 = 1, \mu_3 = -2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00674		—	$\{0,1,2,3,4\}$	$\{0,1,2,3,4\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00675		—	$(0,4)$	$(0,4)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00676		—	$D^{\{2\}}$	D^2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00677		—	$\boldsymbol{A}=\operatorname{tridiag}\left(a_{i}, d_{i}, c_{i}\right) \in$	$\boldsymbol{A} = \operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00678		—	$\mathbb{C}^{n \times n}$	$\mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00679		—	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00680		—	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{L} \boldsymbol{U} \boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00681		—	$\boldsymbol{L} \boldsymbol{z}=\boldsymbol{b}$	$\boldsymbol{L} \boldsymbol{z} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00682		—	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{z}$	$\boldsymbol{U} \boldsymbol{x} = \boldsymbol{z}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00683		—	$u_{\{1\}}, u_{\{2\}}, \ldots, u_{\{n-1\}}$	$u_1, u_2, \dots, u_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00684		—	$u_{\{n\}} \neq 0$	$u_n \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00685		—	$\boldsymbol{z}$	$\boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00686		—	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00687		—	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00688		—	$\boldsymbol{l}, \in \mathbb{C}^{n-1}$	$\boldsymbol{l}, \in \mathbb{C}^{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00689		—	$\boldsymbol{u} \in \mathbb{C}^n$	$\boldsymbol{u} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00690		—	$\boldsymbol{b} \in \mathbb{C}^{n, r}$	$\boldsymbol{b} \in \mathbb{C}^{n, r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00691		—	$r \in \mathbb{N}$	$r \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00692		—	$u_{\{k\}}$	u_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00693		—	$k=1, \ldots, n$	$k = 1, \dots, n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00694		—	$a_{\{0\}}:=c_{\{n\}}:=0$	$a_0 := c_n := 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00695		—	$\left u_{\{1\}}\right =\left d_{\{1\}}\right =\sigma_{\{1\}}+\left c_{\{1\}}\right $	$ u_1 = d_1 = \sigma_1 + c_1 $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00696		—	$\left u_{\{k\}}\right \, \, \geq$	$ u_k \, \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00697		—	$\sigma_{\{k\}}+\left c_{\{k\}}\right $	$\sigma_k + c_k $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00698		—	$1 \, \leq k \, \leq n-1$	$1 \leq k \leq n-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00699		—	$\left c_{\{k\}}\right \, \, /\left u_{\{k\}}\right <1$	$ c_k / u_k < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00700		—	$u_{\{1\}}=d_{\{1\}}$	$u_1 = d_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00701		—	$l_{\{k\}}$	l_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00702		—	$3 \, n-3$	$3n-3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00703		—	$r(5 \, n-4)$	$r(5n-4)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00704		—	$0(n)$	$O(n)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00705		—	$8n-7$	$8n - 7$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00706		—	$r=1$	$r = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00707		—	$[0,1]$	$[0,1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00708		—	u	u
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00709		—	$u(0)=u(1)=$	$u(0) = u(1) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00710		—	$f(x)=1$	$f(x) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00711		—	$u(x)=x(x-1) \ / \ 2$	$u(x) = x(x-1)/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00712		—	$O\left(n^{\{3\}}\right)$	$O\left(n^3\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00713		—	$h:=1 \ / (m+1)$	$h := 1/(m+1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00714		—	$x_{\{j\}}:=j \ h$	$x_j := jh$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00715		—	$j=0,1, \ \ldots, \ m+1$	$j = 0,1,\ldots,m+1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00716		—	$v_{\{j\}}$	v_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00717		—	$u\left(x_{\{j\}}\right)$	$u\left(x_j\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00718		—	$h^{\{2\}}$	h^2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00719		—	$\boldsymbol{T}$	$\boldsymbol{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00720		—	$\boldsymbol{T}=\operatorname{tridiag}\left(a_{\{i\}}, d_{\{i\}}, c_{\{i\}}\right) \in$	$\boldsymbol{T}=\operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00721		—	$\mathbb{R}^{\{m \times m\}}$	$\mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00722		—	$a_{\{i\}}=c_{\{i\}}=-1$	$a_i=c_i=-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00723		—	$d_{\{i\}}=2$	$d_i=2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00724		—	$\boldsymbol{A}_{\{1\}}$	$\boldsymbol{A}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00725		—	$\boldsymbol{A}_{\{2\}}$	$\boldsymbol{A}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00726		—	$\boldsymbol{A}_{\{3\}}$	$\boldsymbol{A}_3$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00727		—	$\operatorname{det}(\boldsymbol{A}_3)=4 \neq 0$	$\det(A_3) = 4 \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00728		—	$ \left d_1\right >\left c_1\right $	$ d_1 > c_1 $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00729		—	$a_{i} \neq 0$	$a_i \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00730		—	$i=1, \ldots, n-2$	$i = 1, \dots, n - 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00731		—	$L \ U$	LU
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00732		—	$d_{\{n\}} \neq 0$	$d_n \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00733		—	$k=1, \ldots, n-1$	$k = 1, \dots, n - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00734		—	$ \left u_{\{k\}}\right >\left c_{\{k\}}\right $	$ u_k > c_k $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00735		—	$ \left u_{\{1\}}\right =\left d_{\{1\}}\right >\left c_{\{1\}}\right $	$ u_1 = d_1 > c_1 $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00736		—	$1 \leq k \leq n-2$	$1 \leq k \leq n - 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00737		—	$k=n-1$	$k = n - 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00738		—	$a_{n-1} \neq 0$	$a_{n-1} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00739		—	$\left u_{k+1}\right >\left d_{k+1}\right -\left a_k\right \geq\left c_{k+1}\right $	$\left u_{k+1}\right >\left d_{k+1}\right -\left a_k\right \geq\left c_{k+1}\right $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00740		—	$\left u_n\right >\left d_n\right -\left a_{n-1}\right \geq 0$	$\left u_n\right >\left d_n\right -\left a_{n-1}\right \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00741		—	$a_{n-1}=0$	$a_{n-1}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00742		—	$\left u_n\right \geq\left d_n\right >0$	$\left u_n\right \geq\left d_n\right >0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00743		—	$\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$	$\boldsymbol{T} \boldsymbol{v}=\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00744		—	$\boldsymbol{v}=\boldsymbol{T}^{-1} \boldsymbol{b}$	$\boldsymbol{v}=\boldsymbol{T}^{-1} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00745		—	$\boldsymbol{T}^{-1}$	$\boldsymbol{T}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00746		—	$\boldsymbol{T}^{-1} \boldsymbol{b}$	$\boldsymbol{T}^{-1} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00747		—	$O\left(n^2\right)$	$O\left(n^2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00748		—	$\boldsymbol{x} \in$	$\boldsymbol{x} \in$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00749		—	L	L
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00750		—	$x=0$	$x = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00751		—	$x=L$	$x = L$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00752		—	F	F
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00753		—	$(L, \ 0)$	$(L, 0)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00754		—	$y(x)$	$y(x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00755		—	R	R
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00756		—	$u:[0,1] \rightarrow \mathbb{R}$	$u : [0, 1] \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00757		—	$u(t):=y(t \ L)$	$u(t) := y(tL)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00758		—	$u^{\{\prime \prime\}}(t)=$	$u''(t) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00759		—	$L^2 \ y^{\{\prime \prime\}}(t \ L)$	$L^2 y''(tL)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00771		—	$\lambda=4 \sin^2(\pi h / 2)$	$\lambda = 4 \sin^2(\pi h/2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00772		—	$\lambda=$	$\lambda =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00773		—	$h^2 K=\frac{h^2 F L^2}{R}$	$h^2 K = \frac{h^2 FL^2}{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00774		—	h	h
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00775		—	$\frac{\pi^2}{L^2} R$	$\frac{\pi^2 R}{L^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00776		—	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1)$	$\boldsymbol{T} = \operatorname{tridiag}(-1,2,-1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00777		—	$a, d \in \mathbb{R}$	$a, d \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00778		—	$\mathbf{1}_D$	$\mathbf{1}_D$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00779		—	$ d >2 a $	$ d > 2 a $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00780		—	$m=3$	$m = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00781		—	$\boldsymbol{T}_1=\left(t_{kj}\right)_{k,j}=$	$\boldsymbol{T}_1 = (t_{kj})_{k,j} =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00782		—	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00783		—	$m \geq 2, a, d \in \mathbb{R}$	$m \geq 2, a, d \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00784		—	$h=1/(m+1)$	$h = 1/(m + 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00785		—	$\boldsymbol{T}_1 \boldsymbol{s}_j=\lambda_j \boldsymbol{s}_j$	$\boldsymbol{T}_1 \boldsymbol{s}_j = \lambda_j \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00786		—	$1<k<m$	$1 < k < m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00787		—	$k=1, m$	$k = 1, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00788		—	$j \pi h=j \pi /(m+1) \in(0, \pi)$	$j \pi h=j \pi /(m+1) \in(0, \pi)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00789		—	$(0, \pi)$	$(0, \pi)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00790		—	$\boldsymbol{T}_1$	$\boldsymbol{T}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00791		—	$i=\sqrt{-1}$	$i = \sqrt{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00792		—	$\boldsymbol{A}^{\ast}:=\overline{\boldsymbol{A}}^T$	$\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00793		—	$\boldsymbol{x} \neq 0$	$\boldsymbol{x} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00794		—	$\boldsymbol{x}^{\ast}$	$\boldsymbol{x}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00795		—	$\boldsymbol{x}^{\ast} \boldsymbol{x}$	$\boldsymbol{x}^{\ast} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00796		—	$\lambda=\frac{\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}}$	$\lambda = \frac{\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00797		—	$\bar{\lambda}=\lambda^{\ast}=\frac{\left(\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}\right)^{\ast}}{\left(\boldsymbol{x}^{\ast} \boldsymbol{x}\right)^{\ast}}=\frac{\boldsymbol{x}^{\ast} \boldsymbol{A}^{\ast} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}}=\frac{\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}}=\lambda$	$\bar{\lambda} = \lambda^{\ast} = \frac{(\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x})^{\ast}}{(\boldsymbol{x}^{\ast} \boldsymbol{x})^{\ast}} = \frac{\boldsymbol{x}^{\ast} \boldsymbol{A}^{\ast} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}} = \frac{\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^{\ast} \boldsymbol{x}} = \lambda$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00798		—	$(\mu, \boldsymbol{y})$	$(\mu, \boldsymbol{y})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00799		—	$\mu \neq \lambda$	$\mu \neq \lambda$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00800		—	$\boldsymbol{y}^{\ast}$	$\boldsymbol{y}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00801		—	μ	μ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00802		—	$\lambda \neq \mu$	$\lambda \neq \mu$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00803		—	$\boldsymbol{y}^{\ast} \boldsymbol{x}=0$	$\boldsymbol{y}^{\ast} \boldsymbol{x} = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00804		—	$\boldsymbol{y}$	$\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00805		—	$\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]=\left[\begin{array}{l ll}1 & 2 & 3 \\ 4 & 5 & 6 \\ \hline 7 & 8 & 9\end{array}\right]$	$\left[\begin{array}{cc c} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array}\right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ \hline 7 & 8 & 9 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00806		—	$\left[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3}\right]=\left[\begin{array}{c c c}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$	$[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3}] = \left[\begin{array}{c c c} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00807		—	$\left[\begin{array}{c}\boldsymbol{a}_{1:}^T \\ \boldsymbol{a}_{2:}^T \\ \boldsymbol{a}_{3:}^T\end{array}\right]=\left[\begin{array}{c}\frac{123}{45}6 \\ \hline \frac{789}{788}\end{array}\right]$	$\left[\begin{array}{c} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \boldsymbol{a}_3^T \end{array}\right] = \left[\begin{array}{c} \frac{123}{45}6 \\ \hline \frac{789}{788} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00808		—	$\left[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}\right]=\left[\begin{array}{c cc}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9\end{array}\right]$	$[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00809		—	$(i \ i)$	(ii)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00810		—	$(i \ i \ i) \ \boldsymbol{A}$	$(iii) \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00811		—	$\boldsymbol{A} \in \mathbb{C}^{m \times p}$	$\boldsymbol{A} \in \mathbb{C}^{m \times p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00812		—	$\boldsymbol{B} \in \mathbb{C}^{p \times n}$	$\boldsymbol{B} \in \mathbb{C}^{p \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00813		—	$\boldsymbol{B}=\left[\boldsymbol{b}_{:1}, \ldots, \boldsymbol{b}_{:n}\right]$	$\boldsymbol{B} = [\boldsymbol{b}_{:1}, \dots, \boldsymbol{b}_{:n}]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00814		—	$\boldsymbol{A} \, \boldsymbol{e}_{\boldsymbol{j}}$	$\boldsymbol{A} \boldsymbol{e}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00815		—	$\boldsymbol{j} \!=\! 1, \, \ldots, \, \boldsymbol{p}$	$\boldsymbol{j} = 1, \ldots, \boldsymbol{p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00816		—	$\boldsymbol{A} \!=\! \boldsymbol{I}$	$\boldsymbol{A} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00817		—	$\boldsymbol{e}_{\boldsymbol{i}}^{\boldsymbol{T}} \, \boldsymbol{B}$	$\boldsymbol{e}_i^T \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00818		—	$\boldsymbol{A} \, \boldsymbol{x}$	$\boldsymbol{A} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00819		—	$\boldsymbol{B} \!=\! \left[\boldsymbol{B}_{\boldsymbol{1}}, \, \boldsymbol{B}_{\boldsymbol{2}} \right]$	$\boldsymbol{B} = [\boldsymbol{B}_1, \boldsymbol{B}_2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00820		—	$\boldsymbol{B}_{\boldsymbol{1}} \, \text{in} \, \mathbb{C}^{\boldsymbol{p} \, \times \, \boldsymbol{r}}$	$\boldsymbol{B}_1 \in \mathbb{C}^{\boldsymbol{p} \times \boldsymbol{r}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00821		—	$\boldsymbol{B}_{\boldsymbol{2}} \, \text{in} \, \mathbb{C}^{\boldsymbol{p} \, \times \, (\boldsymbol{n} - \boldsymbol{r})}$	$\boldsymbol{B}_2 \in \mathbb{C}^{\boldsymbol{p} \times (\boldsymbol{n} - \boldsymbol{r})}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00822		—	$\boldsymbol{A} \!=\! \left[\begin{array}{l} \boldsymbol{A}_{\boldsymbol{1}} \\ \boldsymbol{A}_{\boldsymbol{2}} \end{array} \right]$	$\boldsymbol{A} = \left[\begin{array}{l} \boldsymbol{A}_1 \\ \boldsymbol{A}_2 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00823		—	$\boldsymbol{A}_{\boldsymbol{1}} \, \text{in} \, \mathbb{C}^{\boldsymbol{k} \, \times \, \boldsymbol{p}}$	$\boldsymbol{A}_1 \in \mathbb{C}^{\boldsymbol{k} \times \boldsymbol{p}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00824		—	$\boldsymbol{A}_{\boldsymbol{2}} \, \text{in} \, \mathbb{C}^{(\boldsymbol{m} - \boldsymbol{k}) \, \times \, \boldsymbol{p}}$	$\boldsymbol{A}_2 \in \mathbb{C}^{(\boldsymbol{m} - \boldsymbol{k}) \times \boldsymbol{p}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00825		—	$\boldsymbol{A}=\left[\boldsymbol{A}_{_1}, \boldsymbol{A}_{_2}\right]$	$\boldsymbol{A}=\left[\boldsymbol{A}_1, \boldsymbol{A}_2\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00826		—	$\boldsymbol{B}=\left[\begin{array}{l}\boldsymbol{B}_{_1} \\ \boldsymbol{B}_{_2}\end{array}\right]$	$\boldsymbol{B}=\left[\begin{array}{l}\boldsymbol{B}_1 \\ \boldsymbol{B}_2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00827		—	$\boldsymbol{A}_{_1} \in \mathbb{C}^{m \times s}, \boldsymbol{A}_{_2} \in \mathbb{C}^{m \times(p-s)}, \boldsymbol{B}_{_1} \in$ $\boldsymbol{B}_{_1} \in$	$\boldsymbol{A}_1 \in \mathbb{C}^{m \times s}, \boldsymbol{A}_2 \in \mathbb{C}^{m \times(p-s)}, \boldsymbol{B}_1 \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00828		—	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00829		—	$\boldsymbol{B}_{_2} \in \mathbb{C}^{(p-s) \times n}$	$\boldsymbol{B}_2 \in \mathbb{C}^{(p-s) \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00830		—	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{A}_{_11} & \boldsymbol{A}_{_12} \\ \boldsymbol{A}_{_21} & \boldsymbol{A}_{_22}\end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00831		—	$\boldsymbol{B}=\left[\begin{array}{ll}\boldsymbol{B}_{_11} & \boldsymbol{B}_{_12} \\ \boldsymbol{B}_{_21} & \boldsymbol{B}_{_22}\end{array}\right]$	$\boldsymbol{B}=\left[\begin{array}{ll}\boldsymbol{B}_{11} & \boldsymbol{B}_{12} \\ \boldsymbol{B}_{21} & \boldsymbol{B}_{22}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00832		—	$\boldsymbol{A}_{_11}$	$\boldsymbol{A}_{11}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00833		—	$\boldsymbol{A}_{_21}$	$\boldsymbol{A}_{21}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00834		—	$\boldsymbol{B}_{_11}$	$\boldsymbol{B}_{11}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00835		—	$\boldsymbol{B}_{_12}$	$\boldsymbol{B}_{12}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00836		—	$\boldsymbol{A}_{i\ k}\ \boldsymbol{B}_{k\ j}$	$\boldsymbol{A}_{ik}\boldsymbol{B}_{kj}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00837		—	$\boldsymbol{A},\ \boldsymbol{A}_{11}$	$\boldsymbol{A},\ \boldsymbol{A}_{11}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00838		—	$\boldsymbol{A}_{22}$	$\boldsymbol{A}_{22}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00839		—	$\boldsymbol{B}:=\boldsymbol{A}^{-1}$	$\boldsymbol{B}:=\boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00840		—	$\boldsymbol{B}_{21}=$	$\boldsymbol{B}_{21}=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00841		—	$\mathbf{0}\ \boldsymbol{A}_{11}^{-1}=\mathbf{0}$	$\mathbf{0}\boldsymbol{A}_{11}^{-1}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00842		—	$\boldsymbol{B}_{22}\ \boldsymbol{A}_{22}=\boldsymbol{I}$	$\boldsymbol{B}_{22}\boldsymbol{A}_{22}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00843		—	$a_{i\ i}$	a_{ii}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00844		—	$a_{i\ i}^{-1},\ i=1,\ \ldots,\ n$	$a_{ii}^{-1}, i=1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00845		—	a_{11}	a_{11}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00846		—	$a_{11}\ \neq\ \mathbf{0}$	$a_{11}\neq\mathbf{0}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00847		—	$\boldsymbol{A}^{-1}=\left[a_{11}^{-1}\right]$	$\boldsymbol{A}^{-1}=\left[a_{11}^{-1}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00848		—	$\boldsymbol{A} \in \mathbb{C}^{(k+1) \times (k+1)}$	$\boldsymbol{A} \in \mathbb{C}^{(k+1) \times (k+1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00849		—	$\boldsymbol{A}_{\boldsymbol{k}} \in \mathbb{C}^{k \times k}$	$\boldsymbol{A}_{\boldsymbol{k}} \in \mathbb{C}^{k \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00850		—	2.4 $\boldsymbol{A}$	2.4 $\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00851		—	$\boldsymbol{A}_{\boldsymbol{k}}$	$\boldsymbol{A}_{\boldsymbol{k}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00852		—	$\left(a_{\boldsymbol{k}+1, \boldsymbol{k}+1}\right)$	$\left(a_{\boldsymbol{k}+1, \boldsymbol{k}+1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00853		—	$\boldsymbol{c} \in \mathbb{C}^n$	$\boldsymbol{c} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00854		—	$a_{11}, \ldots, a_{\boldsymbol{k} \boldsymbol{k}}$	$a_{11}, \ldots, a_{\boldsymbol{k} \boldsymbol{k}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00855		—	$\boldsymbol{A}_{\boldsymbol{k}}^{-1}$	$\boldsymbol{A}_{\boldsymbol{k}}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00856		—	$a_{i i}^{-1}, i=1, \ldots, \boldsymbol{k}$	$a_{i i}^{-1}, i=1, \ldots, \boldsymbol{k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00857		—	$\boldsymbol{C}=\boldsymbol{A} \boldsymbol{B}=\left(c_{i j}\right)$	$\boldsymbol{C}=\boldsymbol{A} \boldsymbol{B}=\left(c_{i j}\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00858		—	$\boldsymbol{A}=\left(a_{i\ j}\right)$	$\boldsymbol{A} = (a_{ij})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00859		—	$\boldsymbol{B}=\left(b_{i\ j}\right)$	$\boldsymbol{B} = (b_{ij})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00860		—	$c_{i\ i}=a_{i\ i}\ b_{i\ i}$	$c_{ii} = a_{ii}b_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00861		—	$\left\{\left(x-x_{i}\right)^{j}\right\}_{0\leq j\leq n}$	$\left\{\left(x-x_i\right)^j\right\}_{0\leq j\leq n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00862		—	$n.^{\{3\}}$	$n.^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00863		—	$D_{\{1\}}$	D_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00864		—	$g^{\{\mathrm{prime}\}}(a)=$	$g'(a) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00865		—	$s_{\{1\}}$	s_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00866		—	$g^{\{\mathrm{prime}\}}(b)=s_{\{n+1\}}$	$g'(b) = s_{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00867		—	$s_{\{n+1\}}$	s_{n+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00868		—	$\delta^2 y_{\{i\}}:=y_{\{i+1\}}-2\ y_{\{i\}}+y_{\{i-1\}},\ i=2,\ \ldots,\ n$	$\delta^2 y_i := y_{i+1} - 2y_i + y_{i-1}, i = 2, \dots, n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00869		—	$g^{\{\backslashprime\}}\backslashleft(x_{\{1\}}\backslashright)$	$g'(x_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00870		—	$g^{\{\backslashprime\}}\backslashleft(x_{\{n+1\}}\backslashright)$	$g'(x_{n+1})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00871		—	$x_{\{i\}}$	x_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00872		—	$h\in C^2[a,b]$	$h\in C^2[a,b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00873		—	$h\backslashleft(x_{\{i\}}\backslashright)=g\backslashleft(x_{\{i\}}\backslashright),\ i=1,\ \backslashldots,\ n+1$	$h(x_i)=g(x_i),i=1,\ldots,n+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00874		—	$\backslashint_{\{a\}^{\{b\}}}g^{\{\backslashprime\}\backslashprime}e^{\{\backslashprime\}\backslashprime}=\backslashleft[g^{\{\backslashprime\}\backslashprime}e^{\{\backslashprime\}}\backslashright]_{\{a\}^{\{b\}}}-\backslashint_{\{a\}^{\{b\}}}g^{\{\backslashprime\}\backslashprime\backslashprime}e^{\{\backslashprime\}}$	$\int_a^bg''e''=[g''e']_a^b-\int_a^bg'''e'$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00875		—	$g^{\{\backslashprime\}\backslashprime}$	g''
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00876		—	$g^{\{\backslashprime\}\backslashprime}(b)=g^{\{\backslashprime\}\backslashprime}(a)=0$	$g''(b)=g''(a)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00877		—	$g^{\{\backslashprime\}\backslashprime\backslashprime}$	g'''
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00878		—	$v_{\{i\}}$	v_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00879		—	$x_{\{i\}},\ x_{\{i+1\}}$	x_i,x_{i+1}
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00880		—	$e\left(x_{i}\right)=0$	$e\left(x_{i}\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00881		—	$h=g+e$	$h=g+e$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00882		—	$\boldsymbol{y} \in \mathbb{R}^{n+1}$	$\boldsymbol{y} \in \mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00883		—	$\boldsymbol{x}=\left[x_{1}, \ldots, x_{n}\right]$	$\boldsymbol{x}=\left[x_{1}, \ldots, x_{n}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00884		—	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00885		—	$g\left(r_{j}\right)$	$g\left(r_{j}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00886		—	$\boldsymbol{r}=\left[r_{1}, \ldots, r_{m}\right] \in$	$\boldsymbol{r}=\left[r_{1}, \ldots, r_{m}\right] \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00887		—	i_{j}	i_{j}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00888		—	$g\left(r_{j}\right)=p_{i_{j}}\left(r_{j}\right)$	$g\left(r_{j}\right)=p_{i_{j}}\left(r_{j}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00889		—	$k \in \mathbb{N}, \boldsymbol{t}=\left[t_{1}, \ldots, t_{k}\right]$	$k \in \mathbb{N}, \boldsymbol{t}=\left[t_{1}, \ldots, t_{k}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00890		—	$i=1$	$i=1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00891		—	$x < t_2, i = k$	$x < t_2, i = k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00892		—	$x \geq t_k$	$x \geq t_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00893		—	$t_{\{i\}} \leq$	$t_i \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00894		—	$x < t_{i+1}$	$x < t_{i+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00895		—	$\boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{x} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00896		—	$\boldsymbol{i}$	$\boldsymbol{i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00897		—	$\boldsymbol{i} = \left[i_1, \dots, i_m \right]$	$\boldsymbol{i} = [i_1, \dots, i_m]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00898		—	$\boldsymbol{x}, \boldsymbol{C}$	$\boldsymbol{x}, \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00899		—	$\boldsymbol{G} = g(\boldsymbol{X})$	$\boldsymbol{G} = g(\boldsymbol{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00900		—	$f \in C^2[x-h, x+h]$	$f \in C^2[x - h, x + h]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00901		—	η_2	η_2
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00902		—	$f \in C^4[x-h, x+h]$	$f \in C^4[x-h, x+h]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00903		—	η_4	η_4
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00904		—	$\delta^2 f(x)$	$\delta^2 f(x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00905		—	f, q	f, q
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00906		—	$q(x) \geq 0$	$q(x) \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00907		—	$x \in [a, b]$	$x \in [a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00908		—	$m \in \mathbb{N}, h=(b-a)/(m+1), x_{\{j\}}=a+j \ h$	$m \in \mathbb{N}, h=(b-a)/(m+1), x_j=a+jh$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00909		—	$v_{\{0\}}=g_{\{0\}}$	$v_0=g_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00910		—	$v_{\{m+1\}}=g_{\{1\}}$	$v_{m+1}=g_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00911		—	$\boldsymbol{A} \ \boldsymbol{v}=\boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{v}=\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00912		—	$\operatorname{tridiag}\left(a_{\{j\}}, d_{\{j\}} \cdot c_{\{j\}}\right) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}\left(a_j, d_j.c_j\right) \in \mathbb{R}^{m \times m}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00913		—	$\frac{h}{2} r(x) < 1$	$\frac{h}{2} r(x) < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00914		—	$v_{\{1\}}, \ldots, v_{\{m\}}$	$v_1, \dots, v_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00915		—	$r=0, \text{ f=q=1}$	$r = 0, f = q = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00916		—	$u(0)=1, \text{ u(1)=0}$	$u(0) = 1, u(1) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00917		—	$u(x)=1-\sinh x / \sinh 1$	$u(x) = 1 - \sinh x / \sinh 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00918		—	$h=0.1,0.05,0.025,0.0125$	$h = 0.1, 0.05, 0.025, 0.0125$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00919		—	$\max_{\{1 \leq j \leq m\}} u\left(x_{\{j\}}\right)-v_{\{j\}} $	$\max_{1 \leq j \leq m} u(x_j) - v_j $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00920		—	$v_{\{j\}}, \text{ j=}$	$v_j, j =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00921		—	$0, \ldots, m+1$	$0, \dots, m + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00922		—	$h=0.1$	$h = 0.1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00923		—	$h^{\{p\}}$	h^p
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00924		—	$a_{i, i+1} a_{i+1, i} > 0$	$a_{i,i+1} a_{i+1,i} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00925		—	$\boldsymbol{D} = \operatorname{diag}(d_1, \ldots, d_n)$	$\boldsymbol{D} = \operatorname{diag}(d_1, \ldots, d_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00926		—	$d_{i} > 0$	$d_i > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00927		—	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00928		—	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00929		—	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00930		—	$s_{i \ j}$	s_{ij}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00931		—	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00932		—	$\boldsymbol{T}^{-1} = \boldsymbol{S}$	$\boldsymbol{T}^{-1} = \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00933		—	$a_{i \ j} = \boldsymbol{e}_{i}^{\boldsymbol{T}} \boldsymbol{A} \boldsymbol{e}_{j}$	$a_{ij} = \boldsymbol{e}_i^{\boldsymbol{T}} \boldsymbol{A} \boldsymbol{e}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00934		—	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_{i} \boldsymbol{e}_{j}^{\boldsymbol{T}}$	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^{\boldsymbol{T}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00935		—	$\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{A}^T \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00936		—	$\boldsymbol{B}=\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00937		—	$b_{i\ j}=\boldsymbol{a}_{\colon i}^T \boldsymbol{a}_{\colon j}$	$b_{ij} = \boldsymbol{a}_{\colon i}^T \boldsymbol{a}_{\colon j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00938		—	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00939		—	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00940		—	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00941		—	$\boldsymbol{B} =$	$\boldsymbol{B} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00942		—	$\left[\begin{array}{c}\boldsymbol{B}_{\scriptscriptstyle 1} \\ \boldsymbol{0}\end{array}\right]$	$\left[\begin{array}{c}\boldsymbol{B}_{\scriptscriptstyle 1} \\ \boldsymbol{0}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00943		—	$\boldsymbol{A} \ \boldsymbol{B}=\boldsymbol{A}_{\scriptscriptstyle 1} \ \boldsymbol{B}_{\scriptscriptstyle 1}$	$\boldsymbol{A} \boldsymbol{B} = \boldsymbol{A}_{\scriptscriptstyle 1} \boldsymbol{B}_{\scriptscriptstyle 1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00944		—	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00945		—	$\boldsymbol{A}_{\scriptscriptstyle 1}, \boldsymbol{B}_{\scriptscriptstyle 1}, \boldsymbol{C}_{\scriptscriptstyle 1} \in \mathbb{R}^{(n-1) \times (n-1)}$	$\boldsymbol{A}_{\scriptscriptstyle 1}, \boldsymbol{B}_{\scriptscriptstyle 1}, \boldsymbol{C}_{\scriptscriptstyle 1} \in \mathbb{R}^{(n-1) \times (n-1)}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00946		—	$\boldsymbol{T} \boldsymbol{x} = \boldsymbol{b}$?	$Tx = b?$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00947		—	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00948		—	$\boldsymbol{P}$	$\boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00949		—	3×3	3×3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00950		—	$a_{11}^{(1)} \neq 0$	$a_{11}^{(1)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00951		—	$l_{21}^{(1)} :=$	$l_{21}^{(1)} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00952		—	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00953		—	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00954		—	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_i^{(1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00955		—	$i = 2, 3$	$i = 2, 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00956		—	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i,1}^{(1)} a_{1j}^{(1)}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00957		—	$i, j=2,3$	$i, j = 2, 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00958		—	$a_{11}^{\wedge\{1\}}=0$	$a_{11}^{(1)} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00959		—	$a_{21}^{\wedge\{1\}} \neq 0$	$a_{21}^{(1)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00960		—	$a_{11}^{\wedge\{1\}}=$	$a_{11}^{(1)} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00961		—	$a_{21}^{\wedge\{1\}}=0$	$a_{21}^{(1)} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00962		—	$a_{22}^{\wedge\{2\}} \neq 0$	$a_{22}^{(2)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00963		—	$l_{32}^{\wedge\{2\}}:=a_{32}^{\wedge\{2\}} / a_{22}^{\wedge\{2\}}$	$l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00964		—	$\{ \}^{\wedge\{\mathrm{prime}\}}$	$'$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00965		—	$a_{33}^{\wedge\{3\}}=a_{33}^{\wedge\{2\}}-l_{32}^{\wedge\{2\}} a_{23}^{\wedge\{2\}}$	$a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00966		—	$b_{3}^{\wedge\{3\}}=b_{3}^{\wedge\{2\}}-l_{32}^{\wedge\{2\}} b_{2}^{\wedge\{2\}}$	$b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00967		—	$a_{22}^{\wedge\{2\}}=0$	$a_{22}^{(2)} = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00968		—	$a_{32}^{\{2\}} \neq 0$	$a_{32}^{(2)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00969		—	$a_{\{k\}k}^{\{(k)\}} \neq 0, \, k=1,2$	$a_{kk}^{(k)} \neq 0, \, k = 1, 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00970		—	$\boldsymbol{A}^{\{(1)\}}$	$\boldsymbol{A}^{(1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00971		—	$\boldsymbol{A}^{\{(k)\}} \boldsymbol{x}=\boldsymbol{b}^{\{(k)\}}$	$\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00972		—	$k=$	$k =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00973		—	$1, \, \ldots, \, n$	$1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00974		—	$\boldsymbol{A}^{\{(1)\}}=\boldsymbol{A}, \, \boldsymbol{b}^{\{(1)\}}=\boldsymbol{b}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00975		—	$\boldsymbol{A}^{\{(k)\}}$	$\boldsymbol{A}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00976		—	$k-1$	$k - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00977		—	$\boldsymbol{A}^{\{(n)\}}$	$\boldsymbol{A}^{(n)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00978		—	$\boldsymbol{A}^{\{(n)\}} \boldsymbol{x}=\boldsymbol{b}^{\{(n)\}}$	$\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00979		—	$\boldsymbol{A}^{\{(k+1)\}}$	$\boldsymbol{A}^{(k+1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00980		—	$j=k$	$j = k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00981		—	$a_{i\ k}^{\{(k+1)\}}=a_{i\ k}^{\{(k)\}}-\frac{a_{i\ k}^{\{(k)\}}}{a_{k\ k}^{\{(k)\}}}a_{k\ k}^{\{(k)\}}\ a_{k\ k}^{\{(k)\}}=0$	$a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}a_{kk}^{(k)} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00982		—	$k+1,\ \ldots,\ n$	$k + 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00983		—	$l_{i\ k}^{\{(k)\}}$	$l_{ik}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00984		—	$a_{k\ k}^{\{(k)\}}$	$a_{kk}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00985		—	$\boldsymbol{A}_{\{[k]\}}\ \text{in}\ \mathbb{C}^{k\ \times\ k}$	$\boldsymbol{A}_{[k]} \in \mathbb{C}^{k \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00986		—	$\boldsymbol{B}\ \text{in}\ \mathbb{C}^{k\ \times\ k}$	$\boldsymbol{B} \in \mathbb{C}^{k \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00987		—	$\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r},\ \boldsymbol{r})$	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00988		—	$\boldsymbol{r}=\left[r_1,\ \ldots,\ r_{k}\right]$	$\boldsymbol{r} = \left[r_1, \ldots, r_k\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F00989		—	$1\ \leq\ r_1<\cdots<r_{k}\ \leq\ n$	$1 \leq r_1 < \cdots < r_k \leq n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00990		—	$r_{\{j\}}=j$	$r_j = j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00991		—	$j=1, \, \ldots, \, k$	$j = 1, \ldots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00992		—	$\boldsymbol{A}=\left[\begin{array}{lll}1&2&3\\4&5&6\\7&8&9\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00993		—	$a_{\{k,\, k\}^{\{k\}}}\neq 0$	$a_{k,k}^{(k)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00994		—	$\boldsymbol{A}_{\{[k]\}}$	$\boldsymbol{A}_{[k]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00995		—	$\boldsymbol{B}_{\{k\}}=\boldsymbol{A}_{\{k-1\}^{\{k\}}}$	$\boldsymbol{B}_k = \boldsymbol{A}_{k-1}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00996		—	$\boldsymbol{B}_{\{k\}}$	$\boldsymbol{B}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00997		—	$\boldsymbol{A}_{\{[k-1]\}}$	$\boldsymbol{A}_{[k-1]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00998		—	$\boldsymbol{B}_{\{k+1\}}$	$\boldsymbol{B}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F00999		—	$a_{\{11\}^{\{1\}}}\,\cdots\, a_{\{k\, k\}^{\{k\}}}\neq 0$	$a_{11}^{(1)}\cdots a_{kk}^{(k)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01000		—	$1, \, \ldots, \, n-1$	$1, \ldots, n-1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01001		—	$\operatorname{det}\left(\boldsymbol{A}_{\left[k\right]}\right) \neq 0$	$\det\left(\boldsymbol{A}_{\left[k\right]}\right) \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01002		—	$\mathfrak{l}_{\left\{i\ j\right\}^{\left\{\left(j\right)\right\}}}$	$l_{ij}^{(j)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01003		—	$\mathfrak{a}_{\left\{i\ j\right\}^{\left\{\left(i\right)\right\}}}$	$a_{ij}^{(i)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01004		—	$i \leq j$	$i \leq j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01005		—	$\mathfrak{a}_{\left\{n\ n\right\}^{\left\{\left(n\right)\right\}}} = 0$	$a_{nn}^{(n)} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01006		—	$\boldsymbol{L}\ \boldsymbol{U}\ \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{L}\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01007		—	$\boldsymbol{L}\ \boldsymbol{y} = \boldsymbol{b}$	$\boldsymbol{L}\boldsymbol{y} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01008		—	$\boldsymbol{y} := \boldsymbol{U}\ \boldsymbol{x}$	$\boldsymbol{y} := \boldsymbol{U}\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01009		—	$\boldsymbol{U}\ \boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01010		—	$2.5\ \boldsymbol{A}$	$2.5\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01011		—	$x_{\left\{1\right\}} = b_{\left\{1\right\}}\ /\ \mathfrak{a}_{\left\{11\right\}}$	$x_1 = b_1/a_{11}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01012		—	$x_{\{2\}}$	x_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01013		—	$x_{\{2\}}=\left(b_{\{2\}}-a_{\{21\}}\ x_{\{1\}}\right)\ /\ a_{\{22\}}$	$x_2 = \left(b_2 - a_{21}x_1\right) / a_{22}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01014		—	$x_{\{3\}}=\left(b_{\{3\}}-a_{\{31\}}\ x_{\{1\}}-\right.$	$x_3 = \left(b_3 - a_{31}x_1 - \right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01015		—	$\left.a_{\{32\}}\ x_{\{2\}}\right)\ /\ a_{\{33\}}$	$a_{32}x_2) / a_{33}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01016		—	$a_{\{k,\ j\}}=0$	$a_{k,j} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01017		—	$j\notin\left\{l_{\{k\}},\ l_{\{k\}}+1,\ \ldots,\ k\right\}.$	$j\notin\{l_k,l_k+1,\ldots,k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01018		—	$k=1,2,\ \ldots,\ n$	$k = 1,2,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01019		—	$l_{\{k\}}:=\max\ \left(1,\ k-d\right)$	$l_k := \max(1,k-d)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01020		—	$d=n$	$d = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01021		—	$A\left(k,\ l_{\{k\}}:(k-1)\right)\ *\ x\left(l_{\{k\}}:(k-1)\right)$	$A\left(k,l_k:(k-1)\right)*x\left(l_k:(k-1)\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01022		—	$\sum_{j=l_{\{k\}}}^{k-1}\ a_{\{k\ j\}}\ x_{\{j\}}$	$\sum_{j=l_k}^{k-1} a_{kj}x_j$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01023		—	$x_{\{n\}}$	x_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01024		—	$\boldsymbol{A} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01025		—	$n-k+1$	$n - k + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01026		—	$x_{\{k\}}, \ldots, x_{\{n\}}$	$x_k, \ldots, x_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01027		—	$x_{\{k\}}=b_{\{k\}} / a_{\{k, k\}}$	$x_k = b_k/a_{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01028		—	$x_{\{k\}}$	x_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01029		—	$n-k$	$n - k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01030		—	$x_{\{k+1\}}, \ldots, x_{\{n\}}$	$x_{k+1}, \ldots, x_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01031		—	$k=1,2, \ldots n$	$k = 1, 2, \ldots n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01032		—	$x_{\{k\}}=b_{\{k\}} / A(k, k)$	$x_k = b_k/A(k,k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01033		—	b	b
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01034		—	M, D, A, S	M, D, A, S
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01035		—	$a_{i\ j}^{k+1}=a_{i\ j}^{(k)}-l_{i\ k}^{(k)}\ a_{k\ j}^{(k)}$	$a_{ij}^{k+1}=a_{ij}^{(k)}-l_{ik}^{(k)}a_{kj}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01036		—	$(n-k)^{2}$	$(n-k)^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01037		—	M+	$M+$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01038		—	$S=2\ \sum_{k=1}^{n-1}(n-k)^2=2\ \sum_{m=1}^{n-1}m^2=\frac{2}{3}\ n(n-1)\left(n-\frac{1}{2}\right)$	$S=2\sum_{k=1}^{n-1}(n-k)^2=2\sum_{m=1}^{n-1}m^2=\frac{2}{3}n(n-1)\left(n-\frac{1}{2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01039		—	$\sum_{k=1}^{n-1}(n-k)=\frac{1}{2}\ n(n-1)$	$\sum_{k=1}^{n-1}(n-k)=\frac{1}{2}n(n-1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01040		—	$N_{\{L\ U\}}$	N_{LU}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01041		—	$\frac{2}{3}\ n^3$	$\frac{2}{3}n^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01042		—	$G_{\{n\}}$	G_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01043		—	$3.2\left(G_{\{n\}}:=\frac{2}{3}\ n^3\right)$	$3.2\left(G_n:=\frac{2}{3}n^3\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01044		—	$G_{\{n\}}:=\frac{2}{3}\ n^3$	$G_n:=\frac{2}{3}n^3$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01045		—	$2\ n^{\{3\}}\ /\ 3$	$2n^3/3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01046		—	M+S	$M + S$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01047		—	$N_{\{S\}}$	N_S
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01048		—	$n{=}10^{\{6\}}$	$n = 10^6$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01049		—	$c{=}10^{\{-14\}}$	$c = 10^{-14}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01050		—	$c\ n^{\{3\}}{=}$	$cn^3 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01051		—	$10^{\{4\}}$	10^4
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01052		—	$\backslash approx\ 3$	≈ 3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01053		—	$c\ n^{\{2\}}{=}0.01$	$cn^2 = 0.01$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01054		—	$\backslash left[\backslash begin{array}{ll}0\ \&\ 1\ \backslash\ 1\ \&\ 1\backslash end{array}\backslash right]\backslash left[\backslash begin{array}{l}x_{\{1\}}\ \backslash\ x_{\{2\}}\backslash end{array}\backslash right]{=}\backslash left[\backslash begin{array}{l}1\ \backslash\ 1\backslash end{array}\backslash right]$	$\left[\begin{array}{cc}0 & 1 \\ 1 & 1\end{array}\right]\left[\begin{array}{c}x_1 \\ x_2\end{array}\right]=\left[\begin{array}{c}1 \\ 1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01055		—	(k, k)	(k,k)

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01056		—	$r_{\{k\}} \geq k$	$r_k \geq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01057		—	$a_{\{r_{\{k\}}, k\}^{\{k\}}} \neq 0$	$a_{r_k,k}^{(k)} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01058		—	$r_{\{k\}}$	r_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01059		—	$a_{\{i\ j\}^{\{k+1\}}}$	$a_{ij}^{(k+1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01060		—	$\boldsymbol{A}^{\{1\}}=\boldsymbol{A}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01061		—	$\boldsymbol{e}_{\{i_1\}}, \ldots, \boldsymbol{e}_{\{i_n\}}$	$\boldsymbol{e}_{i_1}, \ldots, \boldsymbol{e}_{i_n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01062		—	$\boldsymbol{e}_{\{1\}}, \ldots, \boldsymbol{e}_{\{n\}} \in \mathbb{R}^n$	$\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01063		—	$\boldsymbol{p}=\left[i_1, \ldots, i_n\right]^T$	$\boldsymbol{p} = [i_1, \ldots, i_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01064		—	$\boldsymbol{A} \ \boldsymbol{P}=\left(\boldsymbol{A} \ \boldsymbol{e}_{\{i_1\}}, \ldots, \boldsymbol{A} \ \boldsymbol{e}_{\{i_n\}}\right)=\boldsymbol{A}(:, \boldsymbol{p})$	$\boldsymbol{A} \boldsymbol{P} = (\boldsymbol{A} \boldsymbol{e}_{i_1}, \ldots, \boldsymbol{A} \boldsymbol{e}_{i_n}) = \boldsymbol{A}(:, \boldsymbol{p})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01065		—	$\boldsymbol{P}^T \ \boldsymbol{A}=\left(\boldsymbol{A}^T \ \boldsymbol{P}\right)^T=\left(\boldsymbol{A}^T \ \boldsymbol{A}\right)^T$	$\boldsymbol{P}^T \boldsymbol{A} = \left(\boldsymbol{A}^T \boldsymbol{P}\right)^T = \left(\boldsymbol{A}^T(\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01066		—	$, \boldsymbol{p}))^T=\boldsymbol{A}(\boldsymbol{p}, :)$	$, \boldsymbol{p}))^T = \boldsymbol{A}(\boldsymbol{p}, :)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01067		—	$\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$	$\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01068		—	$\boldsymbol{P}^{-1} = \boldsymbol{P}^T$	$\boldsymbol{P}^{-1} = \boldsymbol{P}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01069		—	$\boldsymbol{P} \boldsymbol{P}^T = \boldsymbol{I}$	$\boldsymbol{P} \boldsymbol{P}^T = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01070		—	(j, k)	(j, k)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01071		—	$\boldsymbol{I}_{j \ k}$	$\boldsymbol{I}_{jk}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01072		—	$\boldsymbol{I}_{j \ k} = \boldsymbol{I}_{k \ j}$	$\boldsymbol{I}_{jk} = \boldsymbol{I}_{kj}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01073		—	$\boldsymbol{I}_{j \ k}^2 = \boldsymbol{I}$	$\boldsymbol{I}_{jk}^2 = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01074		—	$\boldsymbol{p}_{k}$	$\boldsymbol{p}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01075		—	$\boldsymbol{p}_{k+1}$	$\boldsymbol{p}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01076		—	$\boldsymbol{P}_{k} := \boldsymbol{I}_{r_k, k}$	$\boldsymbol{P}_k := \boldsymbol{I}_{r_k, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01077		—	$\boldsymbol{P}_{k} \boldsymbol{P}_{k} \boldsymbol{I} \left(\boldsymbol{p}_{k}, : \right) = \boldsymbol{I} \left(\boldsymbol{P}_{k} \boldsymbol{p}_{k}, : \right) = \boldsymbol{I} \left(\boldsymbol{p}_{k+1}, : \right)$	$\boldsymbol{P}_k \boldsymbol{I}(\boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{P}_k \boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{p}_{k+1}, :)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01078		—	$a_{ij}^{(1)}=a_{ij}$	$a_{ij}^{(1)} = a_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01079		—	$\boldsymbol{A}=\boldsymbol{P} \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{L}\boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01080		—	$\boldsymbol{P}=\boldsymbol{I}(:, \boldsymbol{p})$	$\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01081		—	$\boldsymbol{p}=\boldsymbol{I}_{\{r_{n-1}, n-1\} \cdots \{r_1, 1\}}[1, \ldots, n]^T$	$\boldsymbol{p} = \boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_1, 1}[1, \ldots, n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01082		—	$\boldsymbol{P}^T \boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01083		—	$\mathrm{fl}(x_1)$	$\mathrm{fl}(x_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01084		—	$\mathrm{fl}(x_2)$	$\mathrm{fl}(x_2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01085		—	x_1	x_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01086		—	$\mathrm{fl}(x_1)=1$	$\mathrm{fl}(x_1) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01087		—	$\mathrm{fl}(x_2)=2$	$\mathrm{fl}(x_2) = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01088		—	s_k	s_k
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01089		—	$r_{\{k\}}, s_{\{k\}}$	r_k, s_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01090		—	$\mathbf{L\ U}$	LU
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01091		—	$\mathbf{L} \in \mathbb{C}^{n \times n}$	$\mathbf{L} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01092		—	$\mathbf{U} \in \mathbb{C}^{n \times n}$	$\mathbf{U} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01093		—	$n^{\{2\}}$	n^2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01094		—	$n^{\{2\}}+n$	$n^2 + n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01095		—	$\text{quad } l_{\{i\}}=1$	$l_{ii} = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01096		—	$\text{quad } u_{\{i\}}=1$	$u_{ii} = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01097		—	$\text{quad } \mathbf{A}=\mathbf{L} \mathbf{D} \mathbf{U}, l_{\{i\}}=u_{\{i\}}=1$	$\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{U}, l_{ii} = u_{ii} = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01098		—	$i, \mathbf{D}=\operatorname{diag}(d_{\{11\}}, \ldots, d_{\{n\ n\}})$	$i, \mathbf{D} = \operatorname{diag}(d_{11}, \ldots, d_{nn})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01099		—	$a, b, c, d \in \mathbb{C}$	$a, b, c, d \in \mathbb{C}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01100		—	$\boldsymbol{A}=\left[\begin{array}{ll}a & b \\ c & d\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01101		—	$\boldsymbol{l}_1$	l_1
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01102		—	u_1, u_2, u_3	u_1, u_2, u_3
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01103		—	$a \boldsymbol{l}_1=c$	$al_1 = c$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01104		—	$a \neq 0$	$a \neq 0$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01105		—	$a=c=0$	$a = c = 0$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01106		—	$a=0, c \neq 0$	$a = 0, c \neq 0$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01107		—	$\boldsymbol{A}_4$	$\boldsymbol{A}_4$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01108		—	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}$	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01109		—	$\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$	$\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01110		—	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$	$\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01111		—	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k, n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01112		—	$\boldsymbol{L}_{[k]}$	$\boldsymbol{L}_{[k]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01113		—	$\boldsymbol{U}_{[k]}$	$\boldsymbol{U}_{[k]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01114		—	1×1	1×1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01115		—	$\left[a_{11}\right]=\left[1\right]\left[a_{11}\right]$	$\left[a_{11}\right]=\left[1\right]\left[a_{11}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01116		—	$\boldsymbol{A}_{[n-1]}$	$\boldsymbol{A}_{[n-1]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01117		—	$\boldsymbol{A}_{[n-1]}=$	$\boldsymbol{A}_{[n-1]}=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01118		—	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01119		—	$\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{A}_{[1]}, \ldots, \boldsymbol{A}_{[n-1]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01120		—	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01121		—	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01122		—	$u_{\{n\ n\}} \in \mathbb{C}$	$u_{nn} \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01123		—	$\boldsymbol{l}_{\{n-1\}}$	$\boldsymbol{L}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01124		—	$\boldsymbol{U}_{\{n-1\}}$	$\boldsymbol{U}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01125		—	$\boldsymbol{l}_{\{n\}}, \boldsymbol{u}_{\{n\}}$	$\boldsymbol{l}_n, \boldsymbol{u}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01126		—	$u_{\{n\ n\}}$	u_{nn}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01127		—	$k \leq n-1$	$k \leq n-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01128		—	$\boldsymbol{A}_{\{[j]\}}$	$\boldsymbol{A}_{[j]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01129		—	$\boldsymbol{j} \leq k-1$	$\boldsymbol{j} \leq k-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01130		—	$\boldsymbol{U}_{\{[k]\}}^T \boldsymbol{M}_{\{k\}}^T = \boldsymbol{C}_{\{k\}}^T$	$\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01131		—	$\boldsymbol{M}_{\{k\}}^T$	$\boldsymbol{M}_k^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01132		—	$\boldsymbol{U}_{\{[k]\}}^T$	$\boldsymbol{U}_{[k]}^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01133		—	$\boldsymbol{M}_{\mathbf{k}}$	M_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01134		—	$k=n$	$k = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01135		—	$k-d$	$k - d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01136		—	$\boldsymbol{r}_{\mathbf{k}}$	$\boldsymbol{r}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01137		—	$\boldsymbol{c}_{\mathbf{k}}$	$\boldsymbol{c}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01138		—	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_{\mathbf{k}}=\boldsymbol{r}_{\mathbf{k}}$	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01139		—	$\boldsymbol{l}_{\mathbf{k}}^T$	$\boldsymbol{l}_k^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01140		—	$\left[\begin{array}{c}\boldsymbol{u}_{\mathbf{k}} \ \ u_{\mathbf{k}}\end{array}\right]$	$\left[\begin{array}{c}\boldsymbol{u}_k \\ u_k\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01141		—	$k=2, \ldots, n$	$k = 2, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01142		—	$d \geq 1$	$d \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01143		—	$O\left(d^2 n\right)$	$O\left(d^2 n\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01144		—	$k < n$	$k < n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01145		—	$\boldsymbol{A} = \boldsymbol{I} \boldsymbol{A}$	$\boldsymbol{A} = \boldsymbol{I} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01146		—	$a_{k \ k}$	a_{kk}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01147		—	$\boldsymbol{A}_{i \ i}$	$\boldsymbol{A}_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01148		—	$\mathbf{L}$	$\mathbf{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01149		—	$\boldsymbol{U}_{i \ i}$	$\boldsymbol{U}_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01150		—	$k=1, \ldots, m-1$	$k = 1, \dots, m - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01151		—	$\boldsymbol{A}_{\{k\}}$	$\boldsymbol{A}_{\{k\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01152		—	$\boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{m-1\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01153		—	$\boldsymbol{A}_{\{m-1\}} =$	$\boldsymbol{A}_{\{m-1\}} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01154		—	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01155		—	$\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01156		—	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01157		—	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{\{m-1\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01158		—	$\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$	$\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01159		—	$k=1, \ldots, m$	$k = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01160		—	$\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$	$\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01161		—	$\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$	$\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01162		—	$\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii})$	$\hat{\boldsymbol{L}} := \boldsymbol{L} \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01163		—	$\hat{\boldsymbol{U}} := \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1}) \boldsymbol{U}$	$\hat{\boldsymbol{U}} := \operatorname{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1}) \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01164		—	$\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\right]$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01165		—	$\boldsymbol{b}_k$	$\boldsymbol{b}_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01166		—	$\boldsymbol{A} \ \boldsymbol{b}_{\{k\}}=\boldsymbol{e}_{\{k\}}$	$\boldsymbol{A} \boldsymbol{b}_k = \boldsymbol{e}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01167		—	$\boldsymbol{b}_{\{k\}}(k)=1 \ / \ a(k, \ k)$	$\boldsymbol{b}_k(k) = 1/a(k, k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01168		—	$k=n, \ n-1, \ \ldots, \ 1$	$k = n, n - 1, \ldots, 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01169		—	$\frac{1}{3} \ n \left(2 \ n^2+1\right) \approx G_{\{n\}}$	$\frac{1}{3} n \left(2 n^2 + 1\right) \approx G_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01170		—	$\boldsymbol{B} \ \text{in} \ \mathbb{R}^{n \ \times \ n}$	$\boldsymbol{B} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01171		—	$\boldsymbol{B} \ \boldsymbol{A}$	$\boldsymbol{B} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01172		—	$\boldsymbol{P}, \ \boldsymbol{L}, \ \boldsymbol{U}$	$\boldsymbol{P}, \boldsymbol{L}, \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01173		—	$\boldsymbol{A}^{*} \ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{A}^{*} \boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01174		—	$2 \ G_{\{n\}}$	$2 G_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01175		—	$\boldsymbol{A}^{-1}=\boldsymbol{U}^{-1} \ \boldsymbol{L}^{-1} \ \boldsymbol{P}^T$	$\boldsymbol{A}^{-1} = \boldsymbol{U}^{-1} \boldsymbol{L}^{-1} \boldsymbol{P}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01176		—	$\boldsymbol{P}^T$	$\boldsymbol{P}^T$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01177		—	$k=1,2,\ldots,n-1$	$k = 1, 2, \dots, n - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01178		—	$a_{\{k, j\}}$	$a_{k,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01179		—	$a_{\{k+1, j\}}$	$a_{k+1,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01180		—	$j=k, k+1, \ldots, n$	$j = k, k + 1, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01181		—	$i=k+1, k+2, \ldots, n$	$i = k + 1, k + 2, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01182		—	$a_{\{i, k\}}=m_{\{i, k\}}=a_{\{i, k\}} / a_{\{k, k\}}$	$a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01183		—	$a_{\{i, j\}}=a_{\{i, j\}}-m_{\{i, k\}} a_{\{k, j\}}$	$a_{i,j} = a_{i,j} - m_{i,k}a_{k,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01184		—	$j=k+1, k+2, \ldots, n$	$j = k + 1, k + 2, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01185		—	$b_{\{k\}}$	b_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01186		—	$b_{\{k+1\}}$	b_{k+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01187		—	$b_{\{i\}}=b_{\{i\}}-a_{\{i, k\}} b_{\{k\}}$	$b_i = b_i - a_{i,k}b_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01188		—	$x_{\{n\}}=b_{\{n\}} / a_{\{n, n\}}$	$x_n = b_n/a_{n,n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01189		—	$=0$	$= 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01190		—	$+a_{\{k, j\}} x_{\{j\}}$	$+a_{k,j}x_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01191		—	$x_{\{k\}}=\left(b_{\{k\}}-\right.$	$x_k = (b_k-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01192		—	$\left. \right) / a_{\{k, k\}}$	$\left. \right)/a_{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01193		—	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01194		—	$h_{\{i, i-1\}} \neq 0$	$h_{i,i-1} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01195		—	$\boldsymbol{H} \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01196		—	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01197		—	$\boldsymbol{v} \in \mathbb{R}^n$	$\boldsymbol{v} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01198		—	$\boldsymbol{e}_{\{1\}}$	$\boldsymbol{e}_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01199		—	$\boldsymbol{I}_{\{i, j\}}$	$\boldsymbol{I}_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01200		—	$\boldsymbol{E}:=\boldsymbol{C} \ \boldsymbol{P}$	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01201		—	$\boldsymbol{W} \in \mathbb{R}^n$	$\boldsymbol{W} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01202		—	$\boldsymbol{B}=\boldsymbol{L} \ \boldsymbol{H} \ \boldsymbol{P}^{\mathrm{T}}$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01203		—	$\boldsymbol{H}$	$\boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01204		—	$\boldsymbol{B} \ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01205		—	$1 \leq$	$1 \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01206		—	$d \leq n$	$d \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01207		—	$A(k, k)=1$	$A(k,k) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01208		—	$N_{\{L \ U\}}(n, d):=\left(2 \ d^2+d\right) n-\left(d^2+d\right)(8 \ d+1) \ / \ 6=0\left(d^2 \ n\right)$	$N_{LU}(n, d) := \left(2 d^2 + d\right) n - \left(d^2 + d\right) (8 d + 1) / 6 = O\left(d^2 n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01209		—	$d=n-1$	$d = n - 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01210		—	$N_{\{L\ U\}}(n, n)=\frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \approx$	$N_{LU}(n,n)=\frac{2}{3}n^3-\frac{1}{2}n^2-\frac{1}{6}n\approx$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01211		—	$N_{\{L\ U\}}(n, 1)=3\ n-3=0(n)$	$N_{LU}(n,1)=3n-3=O(n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01212		—	$\boldsymbol{A}=\left[\left[\begin{array}{ll}1&1\\0&1\end{array}\right]\right]\right.$	$\boldsymbol{A}=\left[\left[\begin{array}{cc}1&1\\0&1\end{array}\right]\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01213		—	$2\leq k\leq d$	$2\leq k\leq d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01214		—	$d+1\leq k\leq n$	$d+1\leq k\leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01215		—	$u_{\{k\ k\}}$	u_{kk}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01216		—	$\boldsymbol{L}_{\{2,2\}},\ \boldsymbol{U}_{\{2,2\}}\in\mathbb{R}^{(n-1)\times(n-1)}$	$\boldsymbol{L}_{2,2},\boldsymbol{U}_{2,2}\in\mathbb{R}^{(n-1)\times(n-1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01217		—	$\boldsymbol{A}_{\{2,2\}}:=\boldsymbol{L}_{\{2,2\}}\boldsymbol{U}_{\{2,2\}}$	$\boldsymbol{A}_{2,2}:=\boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01218		—	$\boldsymbol{B}_{\{2,2\}}:=\boldsymbol{A}_{\{2,2\}}^{-1}$	$\boldsymbol{B}_{2,2}:=\boldsymbol{A}_{2,2}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01219		—	$l_{\{i,\ j\}},\ i>j$	$l_{i,j},i>j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01220		—	$u_{\{i,\ j\}},\ j\geq i$	$u_{i,j},j\geq i$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01221		—	$a_{i, j}$	$a_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01222		—	$\boldsymbol{s} \in \mathbb{R}^n$	$\boldsymbol{s} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01223		—	$\boldsymbol{T} \boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{THx} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01224		—	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01225		—	$\boldsymbol{T} = \left(t_{i j} \right)$	$\boldsymbol{T} = (t_{ij})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01226		—	$\boldsymbol{H} = \left(h_{i j} \right)$	$\boldsymbol{H} = (h_{ij})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01227		—	$\boldsymbol{S} := \boldsymbol{T} \boldsymbol{H}$	$\boldsymbol{S} := \boldsymbol{TH}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01228		—	$\boldsymbol{H} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{H} = \boldsymbol{LU}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01229		—	$\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$	$\ \boldsymbol{L}\ _{\infty} \ \boldsymbol{U}\ _{\infty} \leq K \ \boldsymbol{H}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01230		—	$\boldsymbol{S}$	$\boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01231		—	$\left(t_{i j}, i > j \right)$	$(t_{ij}, i > j)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01232		—	$\left(h_{i\ j},\ i>j+1\right)$	$(h_{ij}, i > j + 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01233		—	$\boldsymbol{T},\ \boldsymbol{H}$	$\boldsymbol{T}, \boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01234		—	$\boldsymbol{S}\ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{S} \boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01235		—	$\boldsymbol{S}=\boldsymbol{T}\ \boldsymbol{H}$	$\boldsymbol{S} = \boldsymbol{T} \boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01236		—	$\boldsymbol{T}\ \boldsymbol{z}=\boldsymbol{b}$	$\boldsymbol{T} \boldsymbol{z} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01237		—	$\boldsymbol{H}\ \boldsymbol{x}=\boldsymbol{z}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01238		—	$\boldsymbol{a}_1,\ \boldsymbol{a}_2,\ \ldots,\ \boldsymbol{a}_n$	$\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01239		—	$p\ \leq\ n$	$p \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01240		—	$\boldsymbol{H}:=\boldsymbol{L}^{-1}\ \boldsymbol{B}$	$\boldsymbol{H} := \boldsymbol{L}^{-1} \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01241		—	$\boldsymbol{H}=\boldsymbol{L}_{\boldsymbol{H}}\ \boldsymbol{U}_{\boldsymbol{H}}$	$\boldsymbol{H} = \boldsymbol{L}_H \boldsymbol{U}_H$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01242		—	$\boldsymbol{H}:=\boldsymbol{L}_{\boldsymbol{H}}\ \boldsymbol{U}_{\boldsymbol{H}}$	$\boldsymbol{H} := \boldsymbol{L}_H \boldsymbol{U}_H$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01243		—	$\boldsymbol{L}_H$	$\boldsymbol{L}_H$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01244		—	$\boldsymbol{U}_H$	$\boldsymbol{U}_H$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01245		—	$\boldsymbol{A}=\boldsymbol{U} \ \boldsymbol{L}$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01246		—	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01247		—	$\boldsymbol{P}^T=\boldsymbol{P}$	$\boldsymbol{P}^T=\boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01248		—	$\boldsymbol{P}^2=\boldsymbol{I}$	$\boldsymbol{P}^2=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01249		—	$\boldsymbol{P} \ \boldsymbol{A}$	$\boldsymbol{P}\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01250		—	$\boldsymbol{B}:=\boldsymbol{P} \ \boldsymbol{A} \ \boldsymbol{P}$	$\boldsymbol{B}:=\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01251		—	$r, \ s$	r, s
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01252		—	$i, \ j, \ n$	i, j, n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01253		—	$\mathbf{b}_{i, \ j}=\mathbf{a}_{r, \ s}$	$b_{i,j}=a_{r,s}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01254		—	$b_{\{i, j\}}$	$b_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01255		—	$w \in \mathbb{R}$	$w \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01256		—	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}=\boldsymbol{M} \boldsymbol{R}$	$PAP = MR$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01257		—	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}$	PAP
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01258		—	$\boldsymbol{M}$	M
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01259		—	$\boldsymbol{M}, \boldsymbol{R}$	M, R
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01260		—	$\hat{\boldsymbol{L}}$	$\hat{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01261		—	$\hat{\boldsymbol{U}}$	$\hat{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01262		—	$a_{\{j\ i\}}=\bar{a}_{\{i\ j\}}$	$a_{ji} = \bar{a}_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01263		—	$a_{\{i\ i\}}=\bar{a}_{\{i\ i\}}$	$a_{ii} = \bar{a}_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01264		—	$\boldsymbol{U}=\boldsymbol{L}^{\{*\}}$	$U = L^*$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01265		—	$\mathbb{I}_{ii} > 0$	$l_{ii} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01266		—	$\mathbf{A}^* = \left(\mathbf{L} \mathbf{D} \mathbf{L} \right)^* = \mathbf{L}^* \mathbf{D}^* \mathbf{L}^* = \mathbf{A}$	$\mathbf{A}^* = (\mathbf{L} \mathbf{D} \mathbf{L}^*)^* = \mathbf{L} \mathbf{D}^* \mathbf{L}^* = \mathbf{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01267		—	$b \in \mathbb{C}$	$b \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01268		—	d_1, d_2	d_1, d_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01269		—	d_1	d_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01270		—	d_2	d_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01271		—	$a=b=0$	$a = b = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01272		—	$a=0, \ b \neq 0$	$a = 0, b \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01273		—	$\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{L}^*$	$\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{L}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01274		—	$\mathbf{L} \mathbf{D} \mathbf{L}^*$	$\mathbf{L} \mathbf{D} \mathbf{L}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01275		—	$\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$	$\mathbf{A}_{[k]}, \mathbf{L}_{[k]}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01276		—	$\boldsymbol{D}_{\{k\}}$	$\boldsymbol{D}_{[k]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01277		—	$\boldsymbol{A}, \boldsymbol{L}$	$\boldsymbol{A}, \boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01278		—	$\boldsymbol{A}_{\{k\}}=$	$\boldsymbol{A}_{[k]}=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01279		—	$\boldsymbol{L}_{\{k\}} \boldsymbol{D}_{\{k\}} \boldsymbol{L}_{\{k\}}^*$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01280		—	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01281		—	$\boldsymbol{F}_{\{k\}}, \boldsymbol{N}_{\{k\}}, \boldsymbol{E}_{\{k\}} \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01282		—	$\boldsymbol{A}_{\{k\}}=\boldsymbol{L}_{\{k\}} \boldsymbol{D}_{\{k\}} \boldsymbol{L}_{\{k\}}^*$	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01283		—	$\boldsymbol{A}=\boldsymbol{A}^*$	$\boldsymbol{A}=\boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01284		—	$\boldsymbol{A}_{\{k\}}^*=\boldsymbol{A}_{\{k\}}$	$\boldsymbol{A}_{[k]}^*=\boldsymbol{A}_{[k]}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01285		—	$\left[a_{11}\right]=\left[1\right]\left[a_{11}\right]\left[1\right]$	$\left[a_{11}\right]=\left[1\right]\left[a_{11}\right]\left[1\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01286		—	$\boldsymbol{L}_{\{n-1\}} \boldsymbol{D}_{\{n-1\}} \boldsymbol{L}_{\{n-1\}}^*$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01287		—	$\boldsymbol{D}_{n-1}$	D_{n-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01288		—	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01289		—	$d_{n \ n}$	d_{nn}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01290		—	$a_{n \ n}$	a_{nn}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01291		—	$\frac{1}{2} \ G_n$	$\frac{1}{2} G_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01292		—	$\boldsymbol{m}$	$\boldsymbol{m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01293		—	$f: \mathbb{C}^n \rightarrow \mathbb{R}$	$f: \mathbb{C}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01294		—	$\overline{f(\boldsymbol{x})}=\overline{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}=\left(\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}\right)^*=\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x}=f(\boldsymbol{x})$	$\overline{f(\boldsymbol{x})} = \overline{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}} = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x} = f(\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01295		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}>0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01296		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01297		—	$-\boldsymbol{A}$	$-\boldsymbol{A}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01298		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \Rightarrow \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \Rightarrow \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01299		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01300		—	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01301		—	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01302		—	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01303		—	$\boldsymbol{z} \in \mathbb{C}^n$	$\boldsymbol{z} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01304		—	$\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01305		—	$\boldsymbol{x}, \boldsymbol{y}$	$\boldsymbol{x}, \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01306		—	∇f	∇f
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01307		—	$H f$	$H f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01308		—	$f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$	$f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01309		—	Ω	Ω
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01310		—	$\nabla f(\boldsymbol{x})=\mathbf{0}$	$\nabla f(\boldsymbol{x})=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01311		—	$H f(\boldsymbol{x})$	$H f(\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01312		—	$\boldsymbol{A}^*\boldsymbol{A}$	$\boldsymbol{A}^*\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01313		—	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01314		—	$\boldsymbol{z}:=\boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{z}:=\boldsymbol{A} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01315		—	$\boldsymbol{x}^*\boldsymbol{A}^*\boldsymbol{A} \boldsymbol{x}=\boldsymbol{z}^*\boldsymbol{z}=\ \boldsymbol{z}\ _2^2=\ \boldsymbol{A} \boldsymbol{x}\ _2^2\geq 0$	$\boldsymbol{x}^*\boldsymbol{A}^*\boldsymbol{A} \boldsymbol{x}=\boldsymbol{z}^*\boldsymbol{z}=\ \boldsymbol{z}\ _2^2=\ \boldsymbol{A} \boldsymbol{x}\ _2^2\geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01316		—	$\boldsymbol{T} =$	$\boldsymbol{T} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01317		—	$\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{n \times n}$	$\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01318		—	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01319		—	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01320		—	$x_{\{1\}}=x_{\{n\}}=0$	$x_1 = x_n = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01321		—	$x_{\{i\}}=x_{\{i+1\}}$	$x_i = x_{i+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01322		—	$\boldsymbol{x}=0$	$\boldsymbol{x} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01323		—	$\boldsymbol{B}=\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r})\in\mathbb{C}^{k\times k}$	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r},\boldsymbol{r}) \in \mathbb{C}^{k\times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01324		—	$b_{\{i,\,j\}}=a_{\{r_{\{i\}},\,r_{\{j\}}\}}$	$b_{i,j} = a_{r_i,r_j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01325		—	$i,\,j=1,\,\ldots,\,k$	$i,j = 1,\ldots,k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01326		—	$\boldsymbol{r}=[1,2,\,\ldots,\,k]^T$	$\boldsymbol{r} = [1,2,\ldots,k]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01327		—	$\boldsymbol{B}:=\boldsymbol{A}(\boldsymbol{r},\boldsymbol{r})$	$\boldsymbol{B} := \boldsymbol{A}(\boldsymbol{r},\boldsymbol{r})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01328		—	$\boldsymbol{y}^*\boldsymbol{B}\boldsymbol{y}=0$	$\boldsymbol{y}^* \boldsymbol{B} \boldsymbol{y} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01329		—	$\boldsymbol{A}^{\scriptscriptstyle{-}*}:=\left(\boldsymbol{A}^{\scriptscriptstyle{-1}}\right)^*=\left(\boldsymbol{A}^*\right)^{\scriptscriptstyle{-1}}$	$\boldsymbol{A}^{\scriptscriptstyle{-}*} := \left(\boldsymbol{A}^{\scriptscriptstyle{-1}}\right)^* = \left(\boldsymbol{A}^*\right)^{\scriptscriptstyle{-1}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01330		—	$1\,\Longrightarrow\,2\,\Longrightarrow\,3\,\Longrightarrow\,1$	$1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01331		—	1 \Longrightarrow	$1 \Rightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01332		—	$\boldsymbol{x}_i:=\boldsymbol{L}^{-*} \boldsymbol{e}_i$	$\boldsymbol{x}_i := \boldsymbol{L}^{-*} \boldsymbol{e}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01333		—	$\boldsymbol{L}^{-*}$	$\boldsymbol{L}^{-*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01334		—	$d_{\textit{i i}}=\boldsymbol{e}_{\textit{i}}^{*} \boldsymbol{D} \boldsymbol{e}_{\textit{i}}=\boldsymbol{e}_{\textit{i}}^{*} \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_{\textit{i}}=\boldsymbol{x}_{\textit{i}}^{*} \boldsymbol{A} \boldsymbol{x}_{\textit{i}}>0$	$d_{ii} = \boldsymbol{e}_i^* \boldsymbol{D} \boldsymbol{e}_i = \boldsymbol{e}_i^* \boldsymbol{L}^{-1} \boldsymbol{A} \boldsymbol{L}^{-*} \boldsymbol{e}_i = \boldsymbol{x}_i^* \boldsymbol{A} \boldsymbol{x}_i > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01335		—	2 \Longrightarrow	$2 \Rightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01336		—	LDL^{*}	LDL^*
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01337		—	$d_{\textit{i i}}$	d_{ii}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01338		—	$\boldsymbol{A}=\boldsymbol{S} \boldsymbol{S}^{*}$	$\boldsymbol{A} = \boldsymbol{S} \boldsymbol{S}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01339		—	$\boldsymbol{S}:=\boldsymbol{L} \boldsymbol{D}^{\frac{1}{2}}$	$\boldsymbol{S} := \boldsymbol{L} \boldsymbol{D}^{1/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01340		—	$\boldsymbol{D}^{\frac{1}{2}}:=$	$\boldsymbol{D}^{1/2} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01341		—	$\operatorname{diag}\left(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}}\right)$	$\operatorname{diag}\left(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}}\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01342		—	3 \Longrightarrow	$3 \Rightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01343		—	$\boldsymbol{A}=\boldsymbol{L} \ \boldsymbol{L}^{\ast}$	$\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01344		—	$\boldsymbol{L} \ \boldsymbol{L}^{\ast}=\boldsymbol{S} \ \boldsymbol{S}^{\ast}$	$\boldsymbol{L}\boldsymbol{L}^{\ast}=\boldsymbol{S}\boldsymbol{S}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01345		—	$\boldsymbol{S}^{-1} \ \boldsymbol{L}=\boldsymbol{S}^{\ast} \ \boldsymbol{L}^{-\ast}$	$\boldsymbol{S}^{-1}\boldsymbol{L}=\boldsymbol{S}^{\ast}\boldsymbol{L}^{-\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01346		—	2.5 $\boldsymbol{S}^{-1} \ \boldsymbol{L}$	$2.5\boldsymbol{S}^{-1}\boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01347		—	$\boldsymbol{S}^{\ast} \ \boldsymbol{L}^{-\ast}$	$\boldsymbol{S}^{\ast}\boldsymbol{L}^{-\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01348		—	$\ell_{ii} \ / \ s_{ii}$	ℓ_{ii}/s_{ii}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01349		—	$s_{ii} \ / \ \ell_{ii}$	s_{ii}/ℓ_{ii}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01350		—	$\ell_{ii}^2=s_{ii}^2$	$\ell_{ii}^2=s_{ii}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01351		—	$\ell_{ii}=s_{ii}$	$\ell_{ii}=s_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01352		—	$\boldsymbol{S}^{-1} \ \boldsymbol{L}=\boldsymbol{I}=\boldsymbol{S}^{\ast} \ \boldsymbol{L}^{-\ast}$	$\boldsymbol{S}^{-1}\boldsymbol{L}=\boldsymbol{I}=\boldsymbol{S}^{\ast}\boldsymbol{L}^{-\ast}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01353		—	$\boldsymbol{L}=\boldsymbol{S}$	$L=S$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01354		—	$\boldsymbol{A}=\boldsymbol{R}^*\boldsymbol{R}$	$A=R^*R$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01355		—	$\boldsymbol{R}=\boldsymbol{L}^*$	$R=L^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01356		—	$4.4(2\times 2)$	$4.4(2\times 2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01357		—	$\boldsymbol{A}=\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]$	$A=\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01358		—	$\frac{1}{2}G_n=n^3/3$	$\frac{1}{2}G_n=n^3/3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01359		—	$a_{ii}>0,\left(a_{ii}\geq 0\right)$	$a_{ii}>0,(a_{ii}\geq 0)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01360		—	$\left \operatorname{Re}\left(a_{ij}\right)\right <\left(a_{ii}+a_{jj}\right) / 2,\left(\left \operatorname{Re}\left(a_{ij}\right)\right \leq\left(a_{ii}+a_{jj}\right) / 2\right)$	$ \operatorname{Re}\left(a_{ij}\right) <\left(a_{ii}+a_{jj}\right) / 2,\left(\left \operatorname{Re}\left(a_{ij}\right)\right \leq\left(a_{ii}+a_{jj}\right) / 2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01361		—	$\left a_{ij}\right <\sqrt{a_{ii} a_{jj}},\left(\left a_{ij}\right \leq \sqrt{a_{ii} a_{jj}}\right)$	$ a_{ij} <\sqrt{a_{ii} a_{j j}},\left(\left a_{i j}\right \leq \sqrt{a_{i i} a_{j j}}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01362		—	$a_{ii}=0$	$a_{ii}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01363		—	$a_{ij}=a_{ji}=0$	$a_{ij}=a_{ji}=0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01364		—	$\mathbf{a}_i = \mathbf{A} \mathbf{e}_i \geq 0$	$a_{ii} = \mathbf{e}_i^T \mathbf{A} \mathbf{e}_i \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01365		—	$\alpha, \beta \in \mathbb{C}$	$\alpha, \beta \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01366		—	$\alpha \mathbf{e}_i + \beta \mathbf{e}_j \neq$	$\alpha \mathbf{e}_i + \beta \mathbf{e}_j \neq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01367		—	$\alpha = 1, \beta = \pm 1$	$\alpha = 1, \beta = \pm 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01368		—	$\mathbf{a}_i + \mathbf{a}_j \pm 2 \operatorname{Re} \mathbf{a}_{ij} > 0$	$a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01369		—	$\alpha = -\mathbf{a}_{ij}, \beta = \mathbf{a}_{ii}$	$\alpha = -a_{ij}, \beta = a_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01370		—	$\mathbf{a}_i > 0$	$a_{ii} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01371		—	$\varepsilon > 0$	$\varepsilon > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01372		—	$\mathbf{B} := \mathbf{A} + \varepsilon \mathbf{I}$	$\mathbf{B} := \mathbf{A} + \varepsilon \mathbf{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01373		—	$\mathbf{x}^* \mathbf{B} \mathbf{x} \geq \varepsilon \ \mathbf{x}\ _2^2 > 0$	$\mathbf{x}^* \mathbf{B} \mathbf{x} \geq \varepsilon \ \mathbf{x}\ _2^2 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01374		—	$\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array} \right]$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01375		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01376		—	$\lambda > 0 (\geq 0)$	$\lambda > 0 (\geq 0)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01377		—	$\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$	$\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01378		—	$\boldsymbol{U}^* \boldsymbol{U} =$	$\boldsymbol{U}^* \boldsymbol{U} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01379		—	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01380		—	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01381		—	$\boldsymbol{z} :=$	$\boldsymbol{z} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01382		—	$\boldsymbol{U}^* \boldsymbol{x} = \left[z_1, \ldots, z_n \right]^T \in \mathbb{C}^n$	$\boldsymbol{U}^* \boldsymbol{x} = \left[z_1, \ldots, z_n \right]^T \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01383		—	$\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$	$\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01384		—	$\boldsymbol{U}^*$	$\boldsymbol{U}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01385		—	$\boldsymbol{z} =$	$\boldsymbol{z} =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01386		—	$\boldsymbol{U}^* \boldsymbol{x}$	$\boldsymbol{U}^* \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01387		—	$\boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^*$	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01388		—	$\boldsymbol{B}=\boldsymbol{L}$	$\boldsymbol{B} = \boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01389		—	$4 \rightarrow$	$4 \Rightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01390		—	$\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 1 & 3\end{array}\right]$	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01391		—	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}=2 x_1^2+2 x_2^2+\left(x_1+x_2\right)^2>0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2 x_1^2 + 2 x_2^2 + \left(x_1 + x_2\right)^2 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01392		—	$\lambda_1=2$	$\lambda_1 = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01393		—	$\lambda_2=4$	$\lambda_2 = 4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01394		—	$\operatorname{det}\left(\boldsymbol{A}_{[1]}\right)=3$	$\det \left(\boldsymbol{A}_{[1]}\right) = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01395		—	$\operatorname{det}\left(\boldsymbol{A}_{[2]}\right)=8$	$\det \left(\boldsymbol{A}_{[2]}\right) = 8$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01396		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\left\ \boldsymbol{L}^* \boldsymbol{x}\right\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \left\ \boldsymbol{L}^* \boldsymbol{x}\right\ _2^2 \geq 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01397		—	$\alpha=\boldsymbol{e}_{1}^{*}\boldsymbol{A}\boldsymbol{e}_{1}>0$	$\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01398		—	$\boldsymbol{C}:=\boldsymbol{B}-$	$\boldsymbol{C} := \boldsymbol{B} -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01399		—	$\boldsymbol{v}\boldsymbol{v}^{*}/\alpha$	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01400		—	$\boldsymbol{y}\in\mathbb{C}^{n-1}$	$\boldsymbol{y} \in \mathbb{C}^{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01401		—	$\boldsymbol{x}^{*}:=\left[-\boldsymbol{y}^{*}\boldsymbol{v}/\alpha,\boldsymbol{y}^{*}\right]\in\mathbb{C}^n$	$\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01402		—	$\boldsymbol{C}\in\mathbb{C}^{\left(n-1\right)\times\left(n-1\right)}$	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01403		—	$\boldsymbol{C}=\boldsymbol{L}_{1}\boldsymbol{L}_{1}^{*}$	$\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01404		—	$\alpha=0$	$\alpha = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01405		—	$\boldsymbol{v}=\mathbf{0}$	$\boldsymbol{v} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01406		—	$\boldsymbol{B}\in$	$\boldsymbol{B} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01407		—	$\mathbb{C}^{\left(n-1\right)\times\left(n-1\right)}$	$\mathbb{C}^{(n-1) \times (n-1)}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01408		—	$\boldsymbol{B}=\boldsymbol{L}_{1}\boldsymbol{L}_{1}^{*}$	$\boldsymbol{B}=\boldsymbol{L}_{1}\boldsymbol{L}_{1}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01409		—	$\boldsymbol{L}^{*}$	$\boldsymbol{L}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01410		—	$\boldsymbol{L}=\left[\begin{array}{cc}0&\boldsymbol{0}^{*}\\\boldsymbol{0}&\boldsymbol{L}_{1}\end{array}\right]$	$\boldsymbol{L}=\left[\begin{array}{cc}0&\boldsymbol{0}^{*}\\\boldsymbol{0}&\boldsymbol{L}_{1}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01411		—	$\boldsymbol{v}^{*}=\left[\boldsymbol{u}^{*},\boldsymbol{0}^{*}\right]$	$\boldsymbol{v}^{*}=\left[\boldsymbol{u}^{*},\boldsymbol{0}^{*}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01412		—	$\boldsymbol{u}\in\mathbb{C}^d$	$\boldsymbol{u}\in\mathbb{C}^d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01413		—	$\boldsymbol{C}:=\boldsymbol{B}-\boldsymbol{v}\boldsymbol{v}^{*}/\alpha$	$\boldsymbol{C}:=\boldsymbol{B}-\boldsymbol{v}\boldsymbol{v}^{*}/\alpha$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01414		—	$d\times d$	$d\times d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01415		—	$n,\boldsymbol{C}=\boldsymbol{L}_{1}\boldsymbol{L}_{1}^{*}$	$n,\boldsymbol{C}=\boldsymbol{L}_{1}\boldsymbol{L}_{1}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01416		—	$\boldsymbol{L}_{1}^{*}$	$\boldsymbol{L}_{1}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01417		—	$\left[\beta,\boldsymbol{v}^{*}/\beta\right]^{*}$	$\left[\beta,\boldsymbol{v}^{*}/\beta\right]^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01418		—	$\alpha>0$	$\alpha>0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01419		—	$\boldsymbol{C}=\boldsymbol{B}-\boldsymbol{v}\boldsymbol{v}^* / \alpha$	$C = B - vv^* / \alpha$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01420		—	$A(2: n, 2: n)$	$A(2 : n, 2 : n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01421		—	$\min (i+d, n)$	$\min(i + d, n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01422		—	$\ell_{k\,k}=0$	$\ell_{kk} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01423		—	$1 \Longleftarrow 2$	$1 \Longleftrightarrow 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01424		—	$1 \Longleftarrow$	$1 \Longleftrightarrow$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01425		—	$1 \Longleftarrow 3$	$1 \Longleftrightarrow 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01426		—	$4 \longrightarrow 3$	$4 \Rightarrow 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01427		—	$1 \longrightarrow 3$	$1 \Rightarrow 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01428		—	$1 \Longleftarrow 4$	$1 \Longleftrightarrow 4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01429		—	$3 \longrightarrow 1$	$3 \Rightarrow 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01430		—	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\1&1\end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01431		—	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\boldsymbol{x}_1^2+\boldsymbol{x}_2^2+\boldsymbol{x}_1\boldsymbol{x}_2+\boldsymbol{x}_2\boldsymbol{x}_1=\left(\boldsymbol{x}_1+\boldsymbol{x}_2\right)^2\geq 0$	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=x_1^2+x_2^2+x_1x_2+x_2x_1=\left(x_1+x_2\right)^2\geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01432		—	$\boldsymbol{x}\in\mathbb{R}^2$	$\boldsymbol{x}\in\mathbb{R}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01433		—	$\lambda_2=0$	$\lambda_2=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01434		—	$\operatorname{det}\left(\left[a_{11}\right]\right)=$	$\det\left([a_{11}]\right)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01435		—	$\operatorname{det}\left(\left[a_{22}\right]\right)=1$	$\det\left([a_{22}]\right)=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01436		—	$\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$	$\boldsymbol{A}=\boldsymbol{B}\boldsymbol{B}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01437		—	$\boldsymbol{B}=\left[\begin{array}{ll}1&0\\1&0\end{array}\right]$	$\boldsymbol{B}=\left[\begin{array}{cc}1&0\\1&0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01438		—	$\boldsymbol{A}:=\left[\begin{array}{rr}0&0\\0&-1\end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{cc}0&0\\0&-1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01439		—	$\operatorname{det}\left(\boldsymbol{A}\right)>0$	$\det(\boldsymbol{A})>0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01440		—	$a_{i\ i}\ a_{j\ j}>a_{i\ j}\ a_{j\ i}$	$a_{ii}a_{jj}>a_{ij}a_{ji}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01441		—	$\lambda=\frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01442		—	$\left[\begin{array}{cc} a_{i \ i} & a_{i \ j} \\ a_{j \ i} & a_{j \ j} \end{array}\right]$	$\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01443		—	$a \in \mathbb{R}$	$a \in \mathbb{R}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01444		—	$\boldsymbol{A}[a]$	$\boldsymbol{A}[a]$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01445		—	$a \in \left[1-\frac{\sqrt{5}}{2}, 1+\frac{\sqrt{5}}{2}\right]$	$a \in \left[1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2}\right]$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01446		—	$\boldsymbol{a}$	$\boldsymbol{a}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01447		—	$\boldsymbol{A}:=\left[\begin{array}{cc} 1 & 0 \\ -2 & 1 \end{array}\right]$	$\boldsymbol{A} := \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01448		—	$\boldsymbol{E} \in \mathbb{R}^{n \times n}$	$\boldsymbol{E} \in \mathbb{R}^{n \times n}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01449		—	$\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u} \boldsymbol{u}^T$	$\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u} \boldsymbol{u}^T$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01450		—	$\boldsymbol{u} \in \mathbb{R}^n$	$\boldsymbol{u} \in \mathbb{R}^n$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F01451		—	$\boldsymbol{E}$	$\boldsymbol{E}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01452		—	$\boldsymbol{E}^{-1} .1$	$\boldsymbol{E}^{-1}.1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01453		—	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u} \ \boldsymbol{u}^{\mathrm{T}}$	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u}\boldsymbol{u}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01454		—	$\boldsymbol{A}=\boldsymbol{L} \ \boldsymbol{L}^{\mathrm{T}}$	$\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01455		—	$\boldsymbol{A}_{+}$	$\boldsymbol{A}_{+}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01456		—	$(n+1) \ \mathrm{times}(n+1)$	$(n+1)\times(n+1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01457		—	α	α
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01458		—	$\boldsymbol{A}_{+}=\boldsymbol{L}_{+} \ \boldsymbol{L}_{+}^{\mathrm{T}}$	$\boldsymbol{A}_{+}=\boldsymbol{L}_{+}\boldsymbol{L}_{+}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01459		—	$\boldsymbol{L}_{+}$	$\boldsymbol{L}_{+}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01460		—	$\alpha>\left \boldsymbol{L}^{-1} \ \boldsymbol{a}\right _{2}^2$	$\alpha>\left\ \boldsymbol{L}^{-1}\boldsymbol{a}\right\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01461		—	$\boldsymbol{E}^{-1}$	$\boldsymbol{E}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01462		—	$\boldsymbol{E}^{-1}=\boldsymbol{I}+\boldsymbol{a} \ \boldsymbol{u} \ \boldsymbol{u}^{\mathrm{T}}$	$\boldsymbol{E}^{-1}=\boldsymbol{I}+\boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01463		—	$\left[\begin{array}{lll}10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5\end{array}\right]$	$\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01464		—	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01465		—	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01466		—	$a, b \in \mathbb{C}$	$a, b \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01467		—	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle, \boldsymbol{x} \rangle \geq 0$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01468		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}, \boldsymbol{x} \rangle \rangle$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01469		—	$\langle a \boldsymbol{x} + b \boldsymbol{y}, \boldsymbol{z} \rangle = a \langle \boldsymbol{x}, \boldsymbol{z} \rangle + b \langle \boldsymbol{y}, \boldsymbol{z} \rangle$	$\langle a\boldsymbol{x} + b\boldsymbol{y}, \boldsymbol{z} \rangle = a\langle \boldsymbol{x}, \boldsymbol{z} \rangle + b\langle \boldsymbol{y}, \boldsymbol{z} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01470		—	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01471		—	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01472		—	$0 \boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$	$0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01473		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle, 0 \boldsymbol{y} \rangle = 0 \langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, 0\boldsymbol{y} \rangle = 0 \langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01474		—	$\ \boldsymbol{y}\ =0$	$\ \boldsymbol{y}\ =0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01475		—	$\boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{y} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01476		—	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle, \boldsymbol{y} \angle = \langle \boldsymbol{z}, \boldsymbol{y} \rangle - a \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - a \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01477		—	$\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{y}\ ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01478		—	$\boldsymbol{z} = \mathbf{0}$	$\boldsymbol{z} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01479		—	$\ \boldsymbol{x}\ \geq 0$	$\ \boldsymbol{x}\ \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01480		—	$ a \boldsymbol{x} = a \ \boldsymbol{x}\ $	$\ a\boldsymbol{x}\ = a \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01481		—	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01482		—	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01483		—	$\left\ \cdot \right\ : \mathbb{C}^n \rightarrow \mathbb{R}.$	$\ \cdot \ : \mathbb{C}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01484		—	$\ \boldsymbol{x} + a \boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a \boldsymbol{y}, \boldsymbol{x} + a \boldsymbol{y} \rangle$	$\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01485		—	$a=1$	$a = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01486		—	$-1 \leq \frac{\angle \boldsymbol{x}, \boldsymbol{y}}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$	$-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01487		—	θ	θ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01488		—	$[0, \pi]$	$[0, \pi]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01489		—	$\boldsymbol{x} \perp \boldsymbol{y}$	$\boldsymbol{x} \perp \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01490		—	$\angle \boldsymbol{x}, \boldsymbol{y} = 0$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01491		—	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01492		—	$\theta = \pi / 2$	$\theta = \pi / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01493		—	$\angle \boldsymbol{v}_i, \boldsymbol{v}_j = 0$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01494		—	$\angle \boldsymbol{v}_i, \boldsymbol{v}_j = \delta_{ij}$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01495		—	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01496		—	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01497		—	$S_j := \operatorname{span}\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j\}$	$S_j := \operatorname{span}\{s_1, \dots, s_j\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01498		—	$\boldsymbol{v}_1 = \boldsymbol{s}_1$	$\boldsymbol{v}_1 = \boldsymbol{s}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01499		—	S_1	S_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01500		—	$j \geq 2$	$j \geq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01501		—	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01502		—	S_{j-1}	S_{j-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01503		—	$\boldsymbol{v}_j$	$\boldsymbol{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01504		—	$\boldsymbol{v}_i$	$\boldsymbol{v}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01505		—	$\boldsymbol{s}_1, \ldots, \boldsymbol{s}_i$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01506		—	$\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$	$\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01507		—	$a_{\{0\}}, \ldots, a_{\{j\}}$	$a_0, \dots, a_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01508		—	$a_{\{j\}}=1$	$a_j = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01509		—	$\boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{s}_{\{j\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01510		—	$a_{\{j\}} \neq 0$	$a_j \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01511		—	$\boldsymbol{v}_{\{j\}} \neq 0$	$\boldsymbol{v}_j \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01512		—	$l=1, \ldots, j-1$	$l = 1, \dots, j - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01513		—	$\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{j\}}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01514		—	$S_{\{j\}}$	S_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01515		—	$\left\{\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{k\}}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01516		—	$\mathcal{S} \subset \mathcal{T}$	$\mathcal{S} \subset \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01517		—	$\operatorname{dim} \mathcal{S}:=k<\operatorname{dim} \mathcal{T}=n$	$\dim \mathcal{S} := k < \dim \mathcal{T} = n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01518		—	$\boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{s}_{\{k\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01519		—	$\boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{s}_{\{k\}}, \boldsymbol{s}_{\{k+1\}}, \ldots, \boldsymbol{s}_{\{n\}}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01520		—	$\boldsymbol{v}_{\{i\}}=\boldsymbol{s}_{\{i\}}$	$\boldsymbol{v}_i = \boldsymbol{s}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01521		—	$1, \ldots, k$	$1, \dots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01522		—	$2 \leq r < k$	$2 \leq r < k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01523		—	$\boldsymbol{v}_{\{j\}}=\boldsymbol{s}_{\{j\}}$	$\boldsymbol{v}_j = \boldsymbol{s}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01524		—	$j=1, \ldots, r-1$	$j = 1, \dots, r - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01525		—	$j=r$	$j = r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01526		—	$\left\langle \boldsymbol{s}_{\{r\}}, \boldsymbol{v}_{\{i\}} \right\rangle =$	$\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01527		—	$\left\langle \boldsymbol{s}_{\{r\}}, \boldsymbol{s}_{\{i\}} \right\rangle = 0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01528		—	$i < r$	$i < r$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01529		—	$\boldsymbol{v}_r=\boldsymbol{s}_r$	$\boldsymbol{v}_r = \boldsymbol{s}_r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01530		—	$\mathcal{S}=\operatorname{span}\left(\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k\right)$	$\mathcal{S} = \operatorname{span}\left(\boldsymbol{s}_1, \ldots, \boldsymbol{s}_k\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01531		—	$1 \leq k < n$	$1 \leq k < n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01532		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01533		—	$\mathcal{S}+\mathcal{T}:=\{\boldsymbol{s}+\boldsymbol{t}: \boldsymbol{s} \in \mathcal{S}$	$\mathcal{S} + \mathcal{T} := \{\boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01534		—	$\boldsymbol{t} \in \mathcal{T}\}$	$\boldsymbol{t} \in \mathcal{T}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01535		—	$\boldsymbol{\operatorname{sum}} \mathcal{S} \oplus \mathcal{T}$	$\boldsymbol{\operatorname{sum}} \mathcal{S} \oplus \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01536		—	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01537		—	$\langle \boldsymbol{s}, \boldsymbol{t} \rangle=0$	$\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01538		—	$\boldsymbol{s} \in \mathcal{S}$	$\boldsymbol{s} \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01539		—	$\boldsymbol{t} \in \mathcal{T}$	$\boldsymbol{t} \in \mathcal{T}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01540		—	$\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}$	$v \in \mathcal{S} \oplus \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01541		—	$\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$	$v = s + t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01542		—	$\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$	$v = s_1 + t_1 = s_2 + t_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01543		—	$\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$	$s_1, s_2 \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01544		—	$\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$	$t_1, t_2 \in \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01545		—	$\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$	$\mathbf{0} = s_1 - s_2 + t_1 - t_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01546		—	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$	$s_1 - s_2 = t_2 - t_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01547		—	$\boldsymbol{s}_1 - \boldsymbol{s}_2$	$s_1 - s_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01548		—	$\boldsymbol{t}_2 - \boldsymbol{t}_1$	$t_2 - t_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01549		—	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$	$s_1 - s_2 = t_2 - t_1 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01550		—	$\boldsymbol{s}_1 = \boldsymbol{s}_2$	$s_1 = s_2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01551		—	$\boldsymbol{t}_2=\boldsymbol{t}_1$	$t_2 = t_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01552		—	$\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$	$v \in \mathcal{S} \cap \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01553		—	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle, \boldsymbol{v} \rangle = 0$	$\langle v, v \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01554		—	$\boldsymbol{v}=0$	$v = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01555		—	$\mathcal{T}:=\mathcal{S}^{\perp}$	$\mathcal{T} := \mathcal{S}^\perp$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01556		—	$\boldsymbol{v}=\boldsymbol{s}_{\scriptscriptstyle 0}+\boldsymbol{t}_{\scriptscriptstyle 0} \in \mathcal{S} \oplus \mathcal{T}$	$v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01557		—	$\boldsymbol{s}_{\scriptscriptstyle 0} \in \mathcal{S}$	$s_0 \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01558		—	$\boldsymbol{t}_{\scriptscriptstyle 0} \in \mathcal{T}$	$t_0 \in \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01559		—	$\boldsymbol{s}_{\scriptscriptstyle 0}$	s_0
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01560		—	$\boldsymbol{t}_{\scriptscriptstyle 0}$	t_0
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01561		—	$\mathcal{T}=\mathcal{S}^{\perp}$	$\mathcal{T} = \mathcal{S}^\perp$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01562		—	$\langle \rangle \cdot, \cdot \rangle$	$\langle \cdot, \cdot \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01563		—	$\mathbf{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01564		—	$\mathbf{v} = \mathbf{s}_0 + \mathbf{t}_0$	$\boldsymbol{v} = \boldsymbol{s}_0 + \boldsymbol{t}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01565		—	$\langle \rangle \mathbf{s}_0, \mathbf{s} \rangle, \langle \rangle \mathbf{s} \rangle \mathbf{s}_0, \mathbf{s} \rangle = \langle \rangle \mathbf{v} - \mathbf{t}_0, \mathbf{s} \rangle = \langle \rangle \mathbf{v}, \mathbf{s} \rangle$	$\langle \boldsymbol{s}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v} - \boldsymbol{t}_0, \boldsymbol{s} \rangle = \langle \boldsymbol{v}, \boldsymbol{s} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01566		—	$\langle \rangle \mathbf{t}_0, \mathbf{s} \rangle = 0$	$\langle \boldsymbol{t}_0, \boldsymbol{s} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01567		—	$\ \mathbf{v} \ := \sqrt{\langle \rangle \mathbf{v}, \mathbf{v} \rangle}$	$\ \boldsymbol{v} \ := \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01568		—	$\mathbf{v} \in \mathcal{V}$	$\boldsymbol{v} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01569		—	$s_0 \neq s \in \mathcal{S}$	$s_0 \neq s \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01570		—	$0 \neq \mathbf{u} := \mathbf{s}_0 - \mathbf{s} \in \mathcal{S}$	$0 \neq \boldsymbol{u} := \boldsymbol{s}_0 - \boldsymbol{s} \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01571		—	$\langle \rangle \mathbf{v} - \mathbf{s}_0, \mathbf{u} \rangle = 0$	$\langle \boldsymbol{v} - \boldsymbol{s}_0, \boldsymbol{u} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01572		—	$\langle \rangle \mathbf{x}, \mathbf{y} \rangle := \mathbf{y}^* \mathbf{x} = \sum_{j=1}^n x_j \overline{y_j}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{x} = \sum_{j=1}^n x_j \overline{y_j}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01573		—	$\boldsymbol{A}^T=\boldsymbol{A} \quad \Longleftrightarrow \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \overline{\boldsymbol{A}} \boldsymbol{y}, \boldsymbol{x} \rangle$	$\boldsymbol{A}^T = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \overline{\boldsymbol{A}} \boldsymbol{y} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01574		—	$\boldsymbol{A}^*=\boldsymbol{A} \quad \Longleftrightarrow \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{A} \boldsymbol{y}, \boldsymbol{x} \rangle$	$\boldsymbol{A}^* = \boldsymbol{A} \iff \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01575		—	$\boldsymbol{x}=\boldsymbol{e}_j$	$\boldsymbol{x} = \boldsymbol{e}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01576		—	$\boldsymbol{y}=\boldsymbol{e}_i$	$\boldsymbol{y} = \boldsymbol{e}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01577		—	$\boldsymbol{A}=\boldsymbol{A}^T$	$\boldsymbol{A} = \boldsymbol{A}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01578		—	$\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01579		—	$\boldsymbol{U}^T \boldsymbol{U} =$	$\boldsymbol{U}^T \boldsymbol{U} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01580		—	$\boldsymbol{U}^{-1}=\boldsymbol{U}^*$	$\boldsymbol{U}^{-1} = \boldsymbol{U}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01581		—	$\boldsymbol{U} \boldsymbol{U}^* =$	$\boldsymbol{U} \boldsymbol{U}^* =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01582		—	$\boldsymbol{U} \boldsymbol{U}^{-1}=\boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^{-1} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01583		—	$\boldsymbol{U}_1^* \boldsymbol{U}_1=\boldsymbol{I}$	$\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01584		—	$\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01585		—	$\left(\boldsymbol{U}_1 \boldsymbol{U}_2\right)^* \left(\boldsymbol{U}_1 \boldsymbol{U}_2\right) =$	$(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01586		—	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01587		—	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle, \boldsymbol{U} \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01588		—	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01589		—	$\ \boldsymbol{x}\ _2$	$\ \boldsymbol{x}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01590		—	$i, j=1, \ldots, n$	$i, j = 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01591		—	$\left(\boldsymbol{U}^* \boldsymbol{U}\right)_{i, j} = \delta_{i, j}$	$(\boldsymbol{U}^* \boldsymbol{U})_{i, j} = \delta_{i, j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01592		—	$\boldsymbol{y} = \boldsymbol{x}$	$\boldsymbol{y} = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01593		—	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01594		—	$\boldsymbol{H}^* = (\boldsymbol{I} -$	$\boldsymbol{H}^* = (\boldsymbol{I} -$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01595		—	$\left.\boldsymbol{u} \ \boldsymbol{u}^*\right)^*=\boldsymbol{H}$	$\boldsymbol{u}\boldsymbol{u}^*)^* = \boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01596		—	$\boldsymbol{I}-2 \ \boldsymbol{u} \ \boldsymbol{u}^*$	$\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01597		—	$\boldsymbol{u}^* \ \boldsymbol{u}=1$	$\boldsymbol{u}^*\boldsymbol{u} = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01598		—	$\boldsymbol{v} \ \in \ \mathbb{C}^n$	$\boldsymbol{v} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01599		—	$\boldsymbol{H}=\boldsymbol{I}-\boldsymbol{u} \ \boldsymbol{u}^*$	$\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01600		—	$\boldsymbol{u}:=\sqrt{2} \ \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$	$\boldsymbol{u} := \sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01601		—	$\sqrt{2}$	$\sqrt{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01602		—	$\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2, \ \boldsymbol{y}^* \ \boldsymbol{x}$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2, \boldsymbol{y}^*\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01603		—	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \ \neq \ \mathbf{0}$	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01604		—	$\left(\boldsymbol{I}-2 \ \frac{\boldsymbol{v} \ \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{y}$	$\left(\boldsymbol{I} - 2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right) \boldsymbol{x} = \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01605		—	$\boldsymbol{x}^* \ \boldsymbol{x}=\boldsymbol{y}^* \ \boldsymbol{y}$	$\boldsymbol{x}^*\boldsymbol{x} = \boldsymbol{y}^*\boldsymbol{y}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01606		—	$\operatorname{Re}(\left(\boldsymbol{y}^* \boldsymbol{x}\right)=\boldsymbol{y}^* \boldsymbol{x}$	$\operatorname{Re}\left(\boldsymbol{y}^* \boldsymbol{x}\right)=\boldsymbol{y}^* \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01607		—	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{x}-\frac{2}{\boldsymbol{v}^* \boldsymbol{v}} \boldsymbol{v} \boldsymbol{v}^* \boldsymbol{x}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}\right) \boldsymbol{x}=\boldsymbol{x}-\frac{2 \boldsymbol{v}^* \boldsymbol{x}}{\boldsymbol{v}^* \boldsymbol{v}} \boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01608		—	$\boldsymbol{H} \boldsymbol{x}$	$\boldsymbol{H} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01609		—	$\mathcal{M}:=$	$\mathcal{M}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01610		—	$\left\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^* \boldsymbol{v}=0\right\}$	$\left\{\boldsymbol{w} \in \mathbb{R}^n: \boldsymbol{w}^* \boldsymbol{v}=0\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01611		—	$(\boldsymbol{x}+\boldsymbol{y}) / 2$	$(\boldsymbol{x}+\boldsymbol{y}) / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01612		—	$\boldsymbol{x}-\boldsymbol{y}$	$\boldsymbol{x}-\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01613		—	$\boldsymbol{x}:=\left[1,0,1\right]^T$	$\boldsymbol{x}:=\left[1,0,1\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01614		—	$\boldsymbol{y}:=\left[-1,0,1\right]^T$	$\boldsymbol{y}:=\left[-1,0,1\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01615		—	$\boldsymbol{v} =$	$\boldsymbol{v} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01616		—	$\boldsymbol{x}-\boldsymbol{y}=\left[2,0,0\right]^T$	$\boldsymbol{x}-\boldsymbol{y}=\left[2,0,0\right]^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01617		—	$y \ z$	$y z$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01618		—	$\boldsymbol{H} \ \boldsymbol{x} = [-1, 0, 1]^T = \boldsymbol{y}$	$\boldsymbol{H} \boldsymbol{x} = [-1, 0, 1]^T = \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01619		—	$\boldsymbol{P} \ \boldsymbol{x} = [0, 0, 1]^T =$	$\boldsymbol{P} \boldsymbol{x} = [0, 0, 1]^T =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01620		—	$(\boldsymbol{x} + \boldsymbol{y}) / 2 \in \mathcal{M}$	$(\boldsymbol{x} + \boldsymbol{y}) / 2 \in \mathcal{M}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01621		—	$\boldsymbol{H} \ \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{H} \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01622		—	$a = 0$	$a = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01623		—	$\ \boldsymbol{u}\ _2 = \sqrt{2}$	$\ \boldsymbol{u}\ _2 = \sqrt{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01624		—	$\boldsymbol{u} := \sqrt{2} \ \boldsymbol{e}_1$	$\boldsymbol{u} := \sqrt{2} \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01625		—	$\boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{x} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01626		—	$ \rho = 1$	$ \rho = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01627		—	$\rho \ \boldsymbol{x}\ _2 \boldsymbol{z} = \rho ^2 \boldsymbol{x} = \boldsymbol{x}$	$\rho \ \boldsymbol{x}\ _2 \boldsymbol{z} = \rho ^2 \boldsymbol{x} = \boldsymbol{x}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01628		—	$\ \boldsymbol{z}\ _2=1$	$\ \boldsymbol{z}\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01629		—	$\boldsymbol{z}_1=$	$\boldsymbol{z}_1 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01630		—	$\left x_1\right /\left\ \boldsymbol{x}\right\ _2$	$ x_1 /\ \boldsymbol{x}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01631		—	$\boldsymbol{u}^*\boldsymbol{u}=\frac{\left(z+\boldsymbol{e}_1\right)^*\left(z+\boldsymbol{e}_1\right)}{1+z_1}=\frac{2+2 z_1}{1+z_1}=2$	$\boldsymbol{u}^*\boldsymbol{u}=\frac{\left(z+\boldsymbol{e}_1\right)^*\left(z+\boldsymbol{e}_1\right)}{1+z_1}=\frac{2+2 z_1}{1+z_1}=2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01632		—	$\boldsymbol{u}^*\boldsymbol{u}=2$	$\boldsymbol{u}^*\boldsymbol{u}=2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01633		—	$\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=a \boldsymbol{e}_1$	$\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=a \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01634		—	$\boldsymbol{z}_1=\left x_1\right /\left\ \boldsymbol{x}\right\ _2 \geq 0$	$\boldsymbol{z}_1=\left x_1\right /\left\ \boldsymbol{x}\right\ _2 \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01635		—	$\ \boldsymbol{z}\ _2=$	$\ \boldsymbol{z}\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01636		—	$1 \leq 1+z_1 \leq 2$	$1 \leq 1+z_1 \leq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01637		—	$1+z_1$	$1+z_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01638		—	$\boldsymbol{H} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^* \boldsymbol{x}\right) \boldsymbol{u}$	$\boldsymbol{H} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^* \boldsymbol{x}\right) \boldsymbol{u}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01639		—	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01640		—	2 m	$2m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01641		—	$\boldsymbol{u}^T \boldsymbol{x}, m$	$\boldsymbol{u}^T \boldsymbol{x}, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01642		—	$\left(\boldsymbol{u}^T \boldsymbol{x}\right) \boldsymbol{u}$	$(\boldsymbol{u}^T \boldsymbol{x}) \boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01643		—	4 m	$4m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01644		—	4 m n	$4mn$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01645		—	$\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - \left(\boldsymbol{u}^T \boldsymbol{A}\right) \boldsymbol{u}$	$\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A}) \boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01646		—	$\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$	$\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01647		—	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01648		—	$1 \leq k <$	$1 \leq k <$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01649		—	$[\hat{\boldsymbol{u}}, a] := \operatorname{housegen}(\boldsymbol{z})$	$[\hat{\boldsymbol{u}}, a] := \operatorname{housegen}(\boldsymbol{z})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01650		—	$\hat{\boldsymbol{H}}=\boldsymbol{I}-\hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01651		—	$\hat{\boldsymbol{H}}\boldsymbol{z}=\boldsymbol{a}\boldsymbol{e}_1\in\mathbb{C}^{n-k}$	$\hat{\boldsymbol{H}}\boldsymbol{z} = \boldsymbol{a}\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01652		—	$\boldsymbol{u}:=\left[\begin{array}{c}\mathbf{0}\\\hat{\boldsymbol{u}}\end{array}\right]\in\mathbb{C}^n$	$\boldsymbol{u} := \left[\begin{array}{c}\mathbf{0}\\\hat{\boldsymbol{u}}\end{array}\right] \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01653		—	$\boldsymbol{u}^*\boldsymbol{u}=\boldsymbol{u}=\hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*\hat{\boldsymbol{u}}=2$	$\boldsymbol{u}^*\boldsymbol{u} = \hat{\boldsymbol{u}}^*\hat{\boldsymbol{u}} = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01654		—	$\boldsymbol{R}\in\mathbb{C}^{m\times n}$	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01655		—	$r_{i,j}=0$	$r_{i,j} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01656		—	$j<i$	$j < i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01657		—	$i=2,3\ldots,m$	$i = 2,3 \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01658		—	$m\leq n$	$m \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01659		—	$\boldsymbol{H}_1,\ldots,\boldsymbol{H}_n$	$\boldsymbol{H}_1,\ldots,\boldsymbol{H}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01660		—	$\boldsymbol{R}_1$	$\boldsymbol{R}_1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01661		—	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01662		—	$\left[a_{k,k}^{(k)}, \ldots, a_{m,k}^{(k)}\right]^T$	$\left[a_{k,k}^{(k)}, \ldots, a_{m,k}^{(k)}\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01663		—	$\boldsymbol{D}_k$	$\boldsymbol{D}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01664		—	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01665		—	$\left[\hat{\boldsymbol{u}}_k, a_k\right] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$	$[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01666		—	$\boldsymbol{H}_k := \left[\begin{array}{cc} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{array}\right]$	$\boldsymbol{H}_k := \left[\begin{array}{cc} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01667		—	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01668		—	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01669		—	$\boldsymbol{A}_{k+1}$	$\boldsymbol{A}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01670		—	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \mathbf{0} \end{array}\right]$	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \mathbf{0} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01671		—	$m > 1$	$m > 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01672		—	$m-1$	$m-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01673		—	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01674		—	$\hat{\boldsymbol{u}}_k = [\boldsymbol{u}_{k\ k}, \ldots, \boldsymbol{u}_{m\ k}]^T$	$\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01675		—	$\boldsymbol{u}_{k,\ k}$	$u_{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01676		—	$\boldsymbol{a}_k = \boldsymbol{r}_{k,\ k}$	$a_k = r_{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01677		—	$m=4$	$m=4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01678		—	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01679		—	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_s$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01680		—	$\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01681		—	$\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01682		—	$\boldsymbol{r}_{k,\ k}$	$r_{k,k}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01683		—	$\operatorname{side}(\mathrm{s})\ \boldsymbol{B}$	side(s) $\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01684		—	$\boldsymbol{B}=\boldsymbol{I}$	$\boldsymbol{B}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01685		—	$v=\hat{\boldsymbol{u}}_k$	$v=\hat{\boldsymbol{u}}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01686		—	$\hat{\boldsymbol{H}}_k\boldsymbol{C}=\left(\boldsymbol{I}-\boldsymbol{v}\boldsymbol{v}^*\right)\boldsymbol{C}=\boldsymbol{C}-\boldsymbol{v}\left(\boldsymbol{v}^*\boldsymbol{C}\right)$	$\hat{\boldsymbol{H}}_k\boldsymbol{C}=\left(\boldsymbol{I}-\boldsymbol{v}\boldsymbol{v}^*\right)\boldsymbol{C}=\boldsymbol{C}-\boldsymbol{v}\left(\boldsymbol{v}^*\boldsymbol{C}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01687		—	$n+1,\ldots,m$	$n+1,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01688		—	$\boldsymbol{C}-\boldsymbol{v}*\left(\boldsymbol{v}^*\boldsymbol{C}\right)$	$\boldsymbol{C}-\boldsymbol{v}*\left(\boldsymbol{v}^*\boldsymbol{C}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01689		—	$\boldsymbol{C}\in\mathbb{C}^{(m-k+1)\times(n+r-k)}$	$\boldsymbol{C}\in\mathbb{C}^{(m-k+1)\times(n+r-k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01690		—	$m\geq n$	$m\geq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01691		—	$\boldsymbol{C}-\boldsymbol{v}*\left(\boldsymbol{v}^T\boldsymbol{C}\right)$	$\boldsymbol{C}-\boldsymbol{v}*\left(\boldsymbol{v}^T\boldsymbol{C}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01692		—	$4(m-k+1)(n+r-k)$	$4(m-k+1)(n+r-k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01693		—	$r=0$	$r=0$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01694		—	$4n^3 / 3 = 2G_n$	$4n^3/3 = 2G_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01695		—	$\boldsymbol{R} \boldsymbol{x} = \boldsymbol{c}$	$\boldsymbol{R} \boldsymbol{x} = \boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01696		—	$4n^3 / 3 =$	$4n^3/3 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01697		—	$\boldsymbol{R} = \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{H}_{n-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01698		—	$\boldsymbol{Q} = \boldsymbol{H}_1 \cdots \boldsymbol{H}_{n-1}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \cdots \boldsymbol{H}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01699		—	$\mathbf{Q} \boldsymbol{R}$	$\boldsymbol{Q} \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01700		—	$\boldsymbol{Q} \in \mathbb{C}^{m, m}$	$\boldsymbol{Q} \in \mathbb{C}^{m, m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01701		—	$\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$	$\boldsymbol{R}_1 \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01702		—	$\mathbf{0}_{m-n, n}$	$\mathbf{0}_{m-n, n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01703		—	$m - n$	$m - n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche__Z-Library__F01704		—	$\boldsymbol{A} = \boldsymbol{Q}_1 \boldsymbol{R}_1$	$\boldsymbol{A} = \boldsymbol{Q}_1 \boldsymbol{R}_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01705		—	$\boldsymbol{Q}_1 \in \mathbb{C}^{m \times n}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01706		—	$\boldsymbol{Q} \boldsymbol{R}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01707		—	$\left[\boldsymbol{Q}_1, \boldsymbol{Q}_2\right]$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01708		—	$\boldsymbol{R}=\left[\begin{array}{c}\boldsymbol{R}_1 \\ \mathbf{0}\end{array}\right]$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01709		—	$\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01710		—	$\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01711		—	$\left\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\right\}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01712		—	$\boldsymbol{Q}_1$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01713		—	$\left\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \ldots, \boldsymbol{q}_m\right\}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01714		—	$\boldsymbol{Q}=\left[\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m\right]$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01715		—	$\boldsymbol{Q}^T \boldsymbol{Q}=\boldsymbol{I}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01716		—	$\mathbb{Q} \cap \mathbb{R}$	QR
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01717		—	$\mathbf{A} = \begin{bmatrix} 1 \end{bmatrix} \mathbf{A}$	$\mathbf{A} = \begin{bmatrix} 1 \end{bmatrix} \mathbf{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01718		—	$s := \min(m-1, n)$	$s := \min(m-1, n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01719		—	$\mathbf{R} = \mathbf{C} \mathbf{A}$	$\mathbf{R} = \mathbf{C} \mathbf{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01720		—	$\mathbf{C} =$	$\mathbf{C} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01721		—	$\mathbf{H}_s \cdots \mathbf{H}_2 \mathbf{H}_1$	$\mathbf{H}_s \cdots \mathbf{H}_2 \mathbf{H}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01722		—	$\mathbf{Q} := \mathbf{C}^* =$	$\mathbf{Q} := \mathbf{C}^* =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01723		—	$\mathbf{H}_1 \cdots \mathbf{H}_s$	$\mathbf{H}_1 \cdots \mathbf{H}_s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01724		—	$\mathbf{A}^* \mathbf{A} =$	$\mathbf{A}^* \mathbf{A} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01725		—	$\mathbf{R}_1^* \mathbf{Q}_1^* \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{R}_1^* \mathbf{R}_1$	$\mathbf{R}_1^* \mathbf{Q}_1^* \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{R}_1^* \mathbf{R}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01726		—	$\mathbf{R}_1^* \mathbf{R}_1$	$\mathbf{R}_1^* \mathbf{R}_1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01727		—	$\boldsymbol{Q}_{1}=\boldsymbol{A} \ \boldsymbol{R}_{1}^{-1}$	$\boldsymbol{Q}_1 = \boldsymbol{A} \boldsymbol{R}_1^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01728		—	$\boldsymbol{B}:=\boldsymbol{A}^{\mathrm{T}} \ \boldsymbol{A}=\left[\begin{array}{cc}5 & -4 \\ -4 & 5\end{array}\right]$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \left[\begin{array}{cc}5 & -4 \\ -4 & 5\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01729		—	$\boldsymbol{B}=\boldsymbol{R}^{\mathrm{T}} \ \boldsymbol{R}$	$\boldsymbol{B} = \boldsymbol{R}^T \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01730		—	$\boldsymbol{R}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}5 & -4 \\ 0 & 3\end{array}\right]$	$\boldsymbol{R} = \frac{1}{\sqrt{5}} \left[\begin{array}{cc}5 & -4 \\ 0 & 3\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01731		—	$\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\left[\begin{array}{cc}3 & 4 \\ 0 & 5\end{array}\right]$	$\boldsymbol{R}^{-1} = \frac{1}{3\sqrt{5}} \left[\begin{array}{cc}3 & 4 \\ 0 & 5\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01732		—	$\boldsymbol{Q}=\boldsymbol{A} \ \boldsymbol{R}^{-1}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}2 & 1 \\ -1 & 2\end{array}\right]$	$\boldsymbol{Q} = \boldsymbol{A} \boldsymbol{R}^{-1} = \frac{1}{\sqrt{5}} \left[\begin{array}{cc}2 & 1 \\ -1 & 2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01733		—	$\boldsymbol{a}_{1}, \ldots, \boldsymbol{a}_{n}$	$\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01734		—	$\mathcal{R}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01735		—	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]=\boldsymbol{a}_{1}, \boldsymbol{a}_{2}$	$\boldsymbol{A} = \left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right] = [\boldsymbol{a}_1, \boldsymbol{a}_2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01736		—	$\boldsymbol{v}_{1}=\boldsymbol{a}_{1}$	$\boldsymbol{v}_1 = \boldsymbol{a}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01737		—	$\boldsymbol{v}_{2}=\boldsymbol{a}_{2}-\frac{\boldsymbol{a}_{2}^{\mathrm{T}} \boldsymbol{v}_{1}}{\boldsymbol{v}_{1}^{\mathrm{T}} \boldsymbol{v}_{1}} \boldsymbol{v}_{1}=\frac{3}{5}\left[\begin{array}{l}1 \\ 2\end{array}\right]$	$\boldsymbol{v}_2 = \boldsymbol{a}_2 - \frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1 = \frac{3}{5} \left[\begin{array}{l}1 \\ 2\end{array}\right]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01738		—	$\boldsymbol{Q}=\left[\boldsymbol{q}_{1}, \boldsymbol{q}_{2}\right]$	$\boldsymbol{Q}=\left[\boldsymbol{q}_1, \boldsymbol{q}_2\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01739		—	$\boldsymbol{q}_{1}=\frac{1}{\sqrt{5}}\left[\begin{array}{c} 2 \\ -1 \end{array}\right]$	$\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\left[\begin{array}{c} 2 \\ -1 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01740		—	$\boldsymbol{q}_{2}=\frac{1}{\sqrt{5}}\left[\begin{array}{l} 1 \\ 2 \end{array}\right]$	$\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\left[\begin{array}{c} 1 \\ 2 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01741		—	n=4	$n=4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01742		—	$\boldsymbol{P} \in \mathbb{R}^{\{2,2\}}$	$\boldsymbol{P} \in \mathbb{R}^{2,2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01743		—	$\theta \in[0,2 \pi)$	$\theta \in[0,2 \pi)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01744		—	$c=\cos \theta$	$c=\cos \theta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01745		—	$s=\sin \theta$	$s=\sin \theta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01746		—	$\theta=0$	$\theta=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01747		—	$\boldsymbol{P}=\boldsymbol{R}$	$\boldsymbol{P}=\boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01748		—	$\boldsymbol{P}=\boldsymbol{I}$	$\boldsymbol{P}=\boldsymbol{I}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01749		—	$1 \leq i < j \leq n$	$1 \leq i < j \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01750		—	$\boldsymbol{P}_{ij} = \left(p_{kl}\right) \in \mathbb{R}^{n \times n}$	$\boldsymbol{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01751		—	$p_{kl} = \delta_{kl}$	$p_{kl} = \delta_{kl}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01752		—	$i\ i, \ j\ j, \ i\ j, \ j\ i$	ii, jj, ij, ji
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01753		—	$\boldsymbol{B} = \boldsymbol{P}_{ij} \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{P}_{ij} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01754		—	$\boldsymbol{C} = \boldsymbol{A} \boldsymbol{P}_{ij}$	$\boldsymbol{C} = \boldsymbol{A} \boldsymbol{P}_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01755		—	$\boldsymbol{B}(k,:) = \boldsymbol{A}(k,:)$	$\boldsymbol{B}(k,:) = \boldsymbol{A}(k,:)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01756		—	$\boldsymbol{C}(:, k) = \boldsymbol{A}(:, k)$	$\boldsymbol{C}(:, k) = \boldsymbol{A}(:, k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01757		—	$k \neq i, j$	$k \neq i, j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01758		—	$2\ n^{\{3\}}$	$2n^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01759		—	$\boldsymbol{P}_{i, i+1}$	$\boldsymbol{P}_{i, i+1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01760		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01761		—	$[0, \pi / 2]$	$[0, \pi / 2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01762		—	$\boldsymbol{x} = \boldsymbol{e}_1$	$\boldsymbol{x} = \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01763		—	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01764		—	$\left(\boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} \right) \boldsymbol{x} = \boldsymbol{y}$	$\left(\boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^T}{\boldsymbol{v}^T \boldsymbol{v}} \right) \boldsymbol{x} = \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01765		—	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01766		—	$\boldsymbol{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	$\boldsymbol{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01767		—	$\boldsymbol{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$	$\boldsymbol{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01768		—	$\boldsymbol{x} = [\cos \phi, \sin \phi]^T$	$\boldsymbol{x} = [\cos \phi, \sin \phi]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01769		—	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \boldsymbol{0}$	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01770		—	$\boldsymbol{B} \in \mathbb{R}^{4,4}$	$\boldsymbol{B} \in \mathbb{R}^{4,4}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01771		—	$\theta<\epsilon<1$	$0 < \epsilon < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01772		—	$\boldsymbol{B}_1$	$\boldsymbol{B}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01773		—	$\boldsymbol{B}_1:=\boldsymbol{H} \ \boldsymbol{B} \ \boldsymbol{H}$	$\boldsymbol{B}_1 := \boldsymbol{H} \boldsymbol{B} \boldsymbol{H}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01774		—	$\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$	$\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01775		—	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01776		—	$\boldsymbol{A}=\left[\boldsymbol{a}_{_1}, \ldots, \boldsymbol{a}_{_n}\right] \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01777		—	$ \operatorname{det}(\boldsymbol{Q}) =1$	$ \det(\boldsymbol{Q}) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01778		—	$\boldsymbol{R}=\left[\boldsymbol{r}_{_1}, \ldots, \boldsymbol{r}_{_n}\right]$	$\boldsymbol{R} = [\boldsymbol{r}_1, \dots, \boldsymbol{r}_n]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01779		—	$\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)_{j \ j}=\left \left \boldsymbol{a}_{_j}\right \right _{_2}^2=\left(\boldsymbol{R}^{*} \boldsymbol{R}\right)_{j \ j}=\left \left \boldsymbol{r}_{_j}\right \right _{_2}^2$	$(\boldsymbol{A}^* \boldsymbol{A})_{jj} = \ \boldsymbol{a}_j\ _2^2 = (\boldsymbol{R}^* \boldsymbol{R})_{jj} = \ \boldsymbol{r}_j\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01780		—	$ \operatorname{det}(\boldsymbol{A}) =\prod_{j=1}^n\left r_{_j \ j}\right \leq \prod_{j=1}^n\left \left \boldsymbol{a}_{_j}\right \right _{_2}$	$ \det(\boldsymbol{A}) = \prod_{j=1}^n r_{jj} \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01781		—	$\boldsymbol{B}=\boldsymbol{L}^{\mathrm{T}} \boldsymbol{L}$	$\boldsymbol{B} = \boldsymbol{L}^T \boldsymbol{L}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01782		—	$\boldsymbol{B}=\boldsymbol{L} \boldsymbol{L}^T$	$B = LL^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01783		—	$\mathfrak{l}_{\{i, j\}}$	$l_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01784		—	$i=n, n-1, \ldots, 1$	$i = n, n - 1, \ldots, 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01785		—	$j=i, 1,2 \ldots, i-1$	$j = i, 1, 2 \ldots, i - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01786		—	$\boldsymbol{L}^T \boldsymbol{L}$	$L^T L$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01787		—	$\boldsymbol{B} \boldsymbol{x}=\boldsymbol{c}$	$B\boldsymbol{x} = \boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01788		—	$\ \boldsymbol{L}\ _F$	$\ \boldsymbol{L}\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01789		—	$\boldsymbol{Q} \boldsymbol{L}$	QL
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01790		—	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01791		—	$\boldsymbol{a}:=\left[a_1, \ldots, a_n\right]^T \in \mathbb{R}^n$	$\boldsymbol{a} := [a_1, \ldots, a_n]^T \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01792		—	$\boldsymbol{e}_{\{n\}}:=[0, \ldots, 0,1]^T$	$\boldsymbol{e}_n := [0, \ldots, 0, 1]^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01793		—	$\boldsymbol{H}:=\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$	$\boldsymbol{H}:=\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01794		—	$\boldsymbol{H}\boldsymbol{a}=-s\boldsymbol{e}_n$	$\boldsymbol{H}\boldsymbol{a}=-se_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01795		—	$ s =\ \boldsymbol{a}\ _2$	$ s =\ \boldsymbol{a}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01796		—	s	s
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01797		—	$\Longleftarrow \boldsymbol{R}$	$\Longleftrightarrow \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01798		—	$\Longleftarrow \boldsymbol{A}^*\boldsymbol{A}$	$\Longleftrightarrow \boldsymbol{A}^*\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01799		—	$1\leq r\leq n,\boldsymbol{v}_r\in\mathbb{R}^r,\boldsymbol{v}_r\neq\mathbf{0}$	$1\leq r\leq n,\boldsymbol{v}_r\in\mathbb{R}^r,\boldsymbol{v}_r\neq\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01800		—	$\boldsymbol{H}:=\left[\begin{array}{cc}\boldsymbol{V}_r&\mathbf{0}\\\mathbf{0}&\boldsymbol{I}_{n-r}\end{array}\right]$	$\boldsymbol{H}:=\left[\begin{array}{cc}\boldsymbol{V}_r&\mathbf{0}\\\mathbf{0}&\boldsymbol{I}_{n-r}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01801		—	$a_{i,j}=0$	$a_{i,j}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01802		—	$i=1,\ldots,r$	$i=1,\ldots,r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01803		—	$j=r+1,\ldots,n$	$j=r+1,\ldots,n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01804		—	$\boldsymbol{H} \boldsymbol{A}$	$\boldsymbol{H} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01805		—	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_{n-1}$	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01806		—	$\boldsymbol{H}_{n-1}, \ldots, \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{n-1}, \ldots, \boldsymbol{H}_1 \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01807		—	$d \leq n-1$	$d \leq n-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01808		—	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01809		—	$\leq 2d$	$\leq 2d$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01810		—	$\mathrm{B} =$	$\mathrm{B} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01811		—	(A, d)	(A, d)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01812		—	$\mathcal{O}(cn^2)$	$\mathcal{O}(cn^2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01813		—	$\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$	$\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01814		—	$2d$	$2d$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01815		—	$\boldsymbol{x}=\left[\begin{array}{c}r\cos\alpha\\r\sin\alpha\end{array}\right]$	$\boldsymbol{x}=\left[\begin{array}{c}r\cos\alpha\\r\sin\alpha\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01816		—	$\boldsymbol{P}\boldsymbol{x} =$	$\boldsymbol{P}\boldsymbol{x} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01817		—	$\left[\begin{array}{l}r\cos\left(\alpha-\theta\right)\\r\sin\left(\alpha-\theta\right)\end{array}\right]$	$\left[\begin{array}{l}r\cos\left(\alpha-\theta\right)\\r\sin\left(\alpha-\theta\right)\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01818		—	$\boldsymbol{H}\in\mathbb{R}^{4,4}$	$\boldsymbol{H}\in\mathbb{R}^{4,4}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01819		—	$\boldsymbol{G}_1,\boldsymbol{G}_2,\boldsymbol{G}_3$	$\boldsymbol{G}_1,\boldsymbol{G}_2,\boldsymbol{G}_3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01820		—	$\boldsymbol{G}_{\boldsymbol{k}}=\boldsymbol{P}_{\boldsymbol{i},\boldsymbol{j}}$	$\boldsymbol{G}_k=\boldsymbol{P}_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01821		—	$\boldsymbol{i},\boldsymbol{j},\boldsymbol{c}$	$\boldsymbol{i},\boldsymbol{j},\boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01822		—	$\boldsymbol{P}_{\boldsymbol{k},\boldsymbol{k}+1}$	$\boldsymbol{P}_{k,k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01823		—	$\boldsymbol{G}:=\left[\begin{array}{cc}c&s\\-s&c\end{array}\right]\in\mathbb{R}^{2\times 2}$	$\boldsymbol{G}:=\left[\begin{array}{cc}c&s\\-s&c\end{array}\right]\in\mathbb{R}^{2\times 2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01824		—	$s^2+c^2=1$	$s^2+c^2=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01825		—	$\boldsymbol{G}$	$\boldsymbol{G}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01826		—	$x_{\{1\}}, x_{\{2\}} \in \mathbb{R}$	$x_1, x_2 \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01827		—	$r:=\sqrt{x_{\{1\}}^2+x_{\{2\}}^2}$	$r := \sqrt{x_1^2 + x_2^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01828		—	$y_{\{1\}}, y_{\{2\}}$	y_1, y_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01829		—	$y_{\{1\}}=y_{\{2\}}$	$y_1 = y_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01830		—	$\left[\begin{array}{l}y_{\{1\}} \ \ y_{\{2\}}\end{array}\right]=\boldsymbol{G}\left[\begin{array}{l}x_{\{1\}} \ \ x_{\{2\}}\end{array}\right]$	$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \boldsymbol{G} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01831		—	$\mathbf{i} \ j$	$i j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01832		—	$\boldsymbol{P}_{\{i, j\}}$	$\boldsymbol{P}_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01833		—	$\mathbf{i} \ \mathbf{i}, \ \mathbf{i} \ j, \ j \ \mathbf{i}, \ j \ j$	ii, ij, ji, jj
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01834		—	$\theta \in \mathbb{R}$	$\theta \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01835		—	$\theta \in (-\pi / 2, \pi / 2]$	$\theta \in (-\pi/2, \pi/2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01836		—	$\boldsymbol{P} \ \boldsymbol{x}= \pm\ \boldsymbol{x}\ _2 \boldsymbol{e}_{\{1\}}$	$\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01837		—	$\boldsymbol{e}_1=(1,0)^T$	$\boldsymbol{e}_1=(1,0)^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01838		—	$\boldsymbol{w} \in \mathbb{R}^m$	$\boldsymbol{w} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01839		—	$(m-1) \ m$	$(m-1)m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01840		—	$\alpha=\pm\ \boldsymbol{w}\ _2$	$\alpha=\pm\ \boldsymbol{w}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01841		—	$\boldsymbol{A}_{-}$	$\boldsymbol{A}_{-}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01842		—	$\boldsymbol{A} \cdot \boldsymbol{A}^2$	$\boldsymbol{A}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01843		—	$\boldsymbol{z}$	$\boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01844		—	$\boldsymbol{A}+\boldsymbol{z} \boldsymbol{z}^* \boldsymbol{z}$	$\boldsymbol{A}+\boldsymbol{z} \boldsymbol{z}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01845		—	$\boldsymbol{R}^* \boldsymbol{R}$	$\boldsymbol{R}^* \boldsymbol{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01846		—	$P_{i_1,n+1}, P_{i_2,n+1}, \ldots, P_{i_n,n+1}$	$P_{i_1,n+1}, P_{i_2,n+1}, \ldots, P_{i_n,n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01847		—	$\boldsymbol{R}^{\prime}$	$\boldsymbol{R}'$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01848		—	$\left[\begin{array}{c}\boldsymbol{L}^*\\ \boldsymbol{z}^*\end{array}\right]$	$\begin{bmatrix} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01849		—	$i_1, \ldots, i_n$	$i_1, \ldots, i_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01850		—	$\boldsymbol{Q}^T \boldsymbol{A}_-$	$\boldsymbol{Q}^T \boldsymbol{A}_-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01851		—	$\left(\lambda_k, \boldsymbol{x}_k\right), k=$	$(\lambda_k, \boldsymbol{x}_k), k=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01852		—	$\left\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\right\}$	$\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01853		—	$\sum_{j=1}^m c_j \boldsymbol{x}_j=\mathbf{0}$	$\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01854		—	c_j	c_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01855		—	$m \geq 2$	$m \geq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01856		—	$\sum_{j=1}^m c_j \lambda_m \boldsymbol{x}_j=\mathbf{0}$	$\sum_{j=1}^m c_j \lambda_m \boldsymbol{x}_j = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01857		—	$\sum_{j=1}^{m-1} c_j \left(\lambda_j -$	$\sum_{j=1}^{m-1} c_j (\lambda_j -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01858		—	$\left.\lambda_m\right) \boldsymbol{x}_j=\mathbf{0}$	$\lambda_m) \boldsymbol{x}_j = \mathbf{0}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01859		—	$\lambda_j - \lambda_m \neq 0$	$\lambda_j - \lambda_m \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01860		—	$j=1, \ldots, m-1$	$j = 1, \dots, m - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01861		—	$c_{\{j\}} \neq 0$	$c_j \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01862		—	$j<m$	$j < m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01863		—	$\left\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{m-1}\right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01864		—	$\boldsymbol{I} \, \boldsymbol{x} = \boldsymbol{x}$	$\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01865		—	$\lambda_1 = \lambda_2 = 1$	$\lambda_1 = \lambda_2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01866		—	$\boldsymbol{x} \in \mathbb{C}^2$	$\boldsymbol{x} \in \mathbb{C}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01867		—	$\boldsymbol{e}_{\{2\}}$	$\boldsymbol{e}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01868		—	$\mathbb{C}^2$	$\mathbb{C}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01869		—	$\boldsymbol{J}$	$\boldsymbol{J}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01870		—	$\boldsymbol{x}=\boldsymbol{x}$	$\boldsymbol{Jx} = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01871		—	$x_{2}=0$	$x_2 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01872		—	$\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]$	$\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01873		—	$\left(1,[1,1]^T\right)$	$(1,[1,1]^T)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01874		—	$\left(3,[1,-1]^T\right)$	$(3,[1,-1]^T)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01875		—	$\boldsymbol{x}=\left[x_1, x_2\right]^T \in \mathbb{C}^2$	$\boldsymbol{x}=\left[x_1, x_2\right]^T \in \mathbb{C}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01876		—	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01877		—	$\boldsymbol{B}=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\boldsymbol{B}=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01878		—	$\boldsymbol{A} \rightarrow \boldsymbol{B}$	$\boldsymbol{A} \rightarrow \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01879		—	$\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n$	$\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01880		—	$\operatorname{det}(\boldsymbol{A} \boldsymbol{C})=$	$\operatorname{det}(\boldsymbol{A} \boldsymbol{C})=$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01881		—	$\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{C})$	$\det(\boldsymbol{A}) \det(\boldsymbol{C})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01882		—	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01883		—	$\lambda, \boldsymbol{S} \boldsymbol{x}$	$\lambda, \boldsymbol{S} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01884		—	$\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}\right) \boldsymbol{x}=\lambda \boldsymbol{x}$	$(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}) \boldsymbol{x}=\lambda \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01885		—	$\boldsymbol{A}(\boldsymbol{S} \boldsymbol{x})=\lambda(\boldsymbol{S} \boldsymbol{x})$	$\boldsymbol{A}(\boldsymbol{S} \boldsymbol{x})=\lambda(\boldsymbol{S} \boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01886		—	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\boldsymbol{D}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\boldsymbol{D}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01887		—	$\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{D}$	$\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01888		—	$\left[\boldsymbol{A} \boldsymbol{s}_1,\ldots,\boldsymbol{A} \boldsymbol{s}_n\right]=\left[\lambda_1 \boldsymbol{s}_1,\ldots,\lambda_n \boldsymbol{s}_n\right]$	$\left[\boldsymbol{A} \boldsymbol{s}_1,\ldots,\boldsymbol{A} \boldsymbol{s}_n\right]=\left[\lambda_1 \boldsymbol{s}_1,\ldots,\lambda_n \boldsymbol{s}_n\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01889		—	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01890		—	$\boldsymbol{A} \boldsymbol{C}$	$\boldsymbol{A} \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01891		—	$\boldsymbol{C} \boldsymbol{A}$	$\boldsymbol{C} \boldsymbol{A}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01892		—	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01893		—	$n+m$	$n + m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01894		—	$\boldsymbol{S}^{-1}=\left[\begin{array}{cc}\boldsymbol{I} & -\boldsymbol{A} \\ \boldsymbol{0} & \boldsymbol{I}\end{array}\right]$	$\boldsymbol{S}^{-1}=\left[\begin{array}{cc}\boldsymbol{I} & -\boldsymbol{A} \\ \boldsymbol{0} & \boldsymbol{I}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01895		—	$\boldsymbol{E} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{F}$	$\boldsymbol{E} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01896		—	$\boldsymbol{F}$	$\boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01897		—	$\pi_{\boldsymbol{E}}(\lambda)=\lambda^n \pi_{\boldsymbol{A} \boldsymbol{C}}(\lambda)=\pi_{\boldsymbol{F}}(\lambda)=\lambda^m \pi_{\boldsymbol{C} \boldsymbol{A}}(\lambda)$	$\pi_{\boldsymbol{E}}(\lambda)=\lambda^n \pi_{\boldsymbol{A} \boldsymbol{C}}(\lambda)=\pi_{\boldsymbol{F}}(\lambda)=\lambda^m \pi_{\boldsymbol{C} \boldsymbol{A}}(\lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01898		—	$\lambda_1, \ldots, \lambda_k$	$\lambda_1, \ldots, \lambda_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01899		—	$a_1, \ldots, a_k$	$a_1, \ldots, a_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01900		—	$a_i=a\left(\lambda_i\right)=a_{\boldsymbol{A}}\left(\lambda_i\right)$	$a_i=a\left(\lambda_i\right)=a_{\boldsymbol{A}}\left(\lambda_i\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01901		—	λ_i	λ_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01902		—	a_i	a_i
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01903		—	$\lambda \in \sigma(\boldsymbol{A})$	$\lambda \in \sigma(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01904		—	$g(\lambda)=$	$g = g(\lambda) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01905		—	$g_{\boldsymbol{A}}(\lambda)$	$g\boldsymbol{A}(\lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01906		—	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I})$	$\mathcal{N}(\boldsymbol{A}-\lambda\boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01907		—	$\lambda=1$	$\lambda = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01908		—	$\pi_{\boldsymbol{I}}(\lambda)=(1-\lambda)^n$	$\pi_{\boldsymbol{I}}(\lambda) = (1-\lambda)^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01909		—	$\boldsymbol{I}-\lambda\boldsymbol{I}$	$\boldsymbol{I}-\lambda\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01910		—	$a=g=n$	$a = g = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01911		—	$\boldsymbol{J}:=\left[\begin{array}{ll}1&1\\0&1\end{array}\right]$	$\boldsymbol{J}:=\left[\begin{array}{cc}1&1\\0&1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01912		—	$a=2$	$a = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01913		—	$g=1$	$g = 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01914		—	$\operatorname{dim} \mathcal{N}(\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I}\right) \boldsymbol{I}) = k$	$\dim \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I}) = k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01915		—	$\operatorname{dim} \mathcal{N}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \ell$	$\dim \mathcal{N}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \ell$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01916		—	$k = \ell$	$k = \ell$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01917		—	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01918		—	$\mathcal{N}(\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I}\right) \boldsymbol{I})$	$\mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01919		—	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_i = \lambda \boldsymbol{v}_i$	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_i = \lambda \boldsymbol{v}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01920		—	$\boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_i = \lambda \boldsymbol{S} \boldsymbol{v}_i, i = 1, \ldots, k$	$\boldsymbol{A} \boldsymbol{S} \boldsymbol{v}_i = \lambda \boldsymbol{S} \boldsymbol{v}_i, i = 1, \ldots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01921		—	$\left\{\boldsymbol{S} \boldsymbol{v}_1, \ldots, \boldsymbol{S} \boldsymbol{v}_k\right\} \subset \mathcal{N}(\boldsymbol{A} - \lambda \boldsymbol{I})$	$\{\boldsymbol{S} \boldsymbol{v}_1, \ldots, \boldsymbol{S} \boldsymbol{v}_k\} \subset \mathcal{N}(\boldsymbol{A} - \lambda \boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01922		—	$k \leq \ell$	$k \leq \ell$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01923		—	$\boldsymbol{w}_1, \ldots, \boldsymbol{w}_\ell$	$\boldsymbol{w}_1, \ldots, \boldsymbol{w}_\ell$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F01924		—	$\left\{\boldsymbol{S}^{-1} \boldsymbol{w}_1, \ldots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell\right\} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I})$	$\{\boldsymbol{S}^{-1} \boldsymbol{w}_1, \ldots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell\} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda \boldsymbol{I})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01925		—	$k \geq \ell$	$k \geq \ell$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01926		—	$\boldsymbol{J}_m(\lambda)$	$\boldsymbol{J}_m(\lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01927		—	$\boldsymbol{J}_3(\lambda)=\left[\begin{array}{ccc}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{array}\right]$	$\boldsymbol{J}_3(\lambda)=\left[\begin{array}{ccc}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01928		—	$\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$	$\boldsymbol{J}_m(\lambda)\boldsymbol{v}=\lambda\boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01929		—	$\boldsymbol{v}=[v_1,\ldots,v_m]$	$\boldsymbol{v}=[v_1,\ldots,v_m]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01930		—	$\lambda v_{i-1}+v_i=\lambda v_{i-1}$	$\lambda v_{i-1}+v_i=\lambda v_{i-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01931		—	$i=2,\ldots,m$	$i=2,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01932		—	$v_2=\cdots=v_m=0$	$v_2=\cdots=v_m=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01933		—	$a=m$	$a=m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01934		—	$g_1,\ldots,g_k$	$g_1,\ldots,g_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01935		—	$\boldsymbol{U}_i$	$\boldsymbol{U}_i$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01936		—	$g_{\{i\}}$	g_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01937		—	$m_{\{i, 1\}, \ldots, m_{\{i, g_{\{i\}}\}}$	$m_{i,1}, \ldots, m_{i,g_i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01938		—	$m_{\{i, 1\}} \geq m_{\{i, 2\}} \geq \cdots \geq m_{\{i, g_{\{i\}}\}}$	$m_{i,1} \geq m_{i,2} \geq \cdots \geq m_{i,g_i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01939		—	$a_{\{i\}} = \sum_{j=1}^{g_{\{i\}}} m_{\{i, j\}}$	$a_i = \sum_{j=1}^{g_i} m_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01940		—	$\boldsymbol{A} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{J}$	$\boldsymbol{AS} = \boldsymbol{SJ}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01941		—	$\boldsymbol{U}_{\{1\}} = \operatorname{diag} \left(\boldsymbol{J}_{\{3\}}(2), \boldsymbol{J}_{\{2\}}(2), \boldsymbol{J}_{\{1\}}(2) \right)$	$\boldsymbol{U}_1 = \operatorname{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01942		—	$\boldsymbol{U}_{\{2\}} = \boldsymbol{J}_{\{2\}}(3)$	$\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01943		—	$\lambda = 2$	$\lambda = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01944		—	$\boldsymbol{S} = \left[\boldsymbol{s}_{\{1\}}, \ldots, \boldsymbol{s}_{\{8\}} \right]$	$\boldsymbol{S} = [\boldsymbol{s}_1, \ldots, \boldsymbol{s}_8]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01945		—	$\boldsymbol{J}_{\{2\}}(3), \boldsymbol{J}_{\{2\}}(2), \boldsymbol{J}_{\{1\}}(2), \boldsymbol{J}_{\{3\}}(2)$	$\boldsymbol{J}_2(3), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2), \boldsymbol{J}_3(2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01946		—	$m_{\{i, j\}} \geq 1$	$m_{i,j} \geq 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01947		—	$g_{\{i\}} \leq a_{\{i\}}$	$g_i \leq a_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01948		—	$\boldsymbol{e}_{\{1\}}, \boldsymbol{e}_{\{4\}}, \boldsymbol{e}_{\{6\}}, \boldsymbol{e}_{\{7\}}$	$\boldsymbol{e}_1, \boldsymbol{e}_4, \boldsymbol{e}_6, \boldsymbol{e}_7$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01949		—	$k=2$	$k = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01950		—	$g_{\{1\}}+g_{\{2\}}=$	$g_1 + g_2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01951		—	$3+1=4$	$3 + 1 = 4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01952		—	$\sum_{\{i\}} a_{\{i\}}=n$	$\sum_i a_i = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01953		—	$\sum_{\{i\}} g_{\{i\}}=n$	$\sum_i g_i = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01954		—	$a_{\{i\}}=g_{\{i\}}$	$a_i = g_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01955		—	$i=1, \ldots, k$	$i = 1, \ldots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01956		—	$\boldsymbol{S}=\boldsymbol{U}$	$\boldsymbol{S} = \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01957		—	$\boldsymbol{S}^{-1}=\boldsymbol{U}^{\{*\}}$	$\boldsymbol{S}^{-1} = \boldsymbol{U}^*$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01958		—	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01959		—	$\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01960		—	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01961		—	$k \times k$	$k \times k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01962		—	$n = k + 1$	$n := k + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01963		—	$(\lambda_1, \boldsymbol{v}_1)$	$(\lambda_1, \boldsymbol{v}_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01964		—	$\ \boldsymbol{v}_1\ _2 = 1$	$\ \boldsymbol{v}_1\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01965		—	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01966		—	$\boldsymbol{V} :=$	$\boldsymbol{V} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01967		—	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01968		—	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01969		—	$\boldsymbol{W}_1^{*} \boldsymbol{M} \boldsymbol{W}_1$	$\boldsymbol{W}_1^{*} \boldsymbol{M} \boldsymbol{W}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01970		—	$\boldsymbol{W}$	$\boldsymbol{W}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01971		—	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01972		—	λ_1	λ_1
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01973		—	$\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$	$\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01974		—	$\boldsymbol{V}$	$\boldsymbol{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01975		—	$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$	$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01976		—	$\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$	$\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01977		—	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01978		—	$n-1 \cdot \boldsymbol{M}$	$n-1 \cdot \boldsymbol{M}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F01979		—	$\boldsymbol{T} := \left[\begin{array}{ccc} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array} \right]$	$\boldsymbol{T} := \left[\begin{array}{ccc} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{array} \right]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01980		—	$\left(2, \boldsymbol{x}_1\right)$	$(2, \boldsymbol{x}_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01981		—	$\boldsymbol{x}_1 = [-1, 0, 1]^T$	$\boldsymbol{x}_1 = [-1, 0, 1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01982		—	$\boldsymbol{x}_1$	$\boldsymbol{x}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01983		—	$\left\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\right\}$	$\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01984		—	$\boldsymbol{x}_2 = [0, 1, 0]^T$	$\boldsymbol{x}_2 = [0, 1, 0]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01985		—	$\boldsymbol{x}_3 = [1, 0, 1]^T$	$\boldsymbol{x}_3 = [1, 0, 1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01986		—	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01987		—	$\boldsymbol{A}^* = -\boldsymbol{A}$	$\boldsymbol{A}^* = -\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01988		—	$\boldsymbol{A}^* = \boldsymbol{A}^{-1}$	$\boldsymbol{A}^* = \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01989		—	$\boldsymbol{A} = \operatorname{diag}\left(d_1, \ldots, d_n\right)$	$\boldsymbol{A} = \operatorname{diag}\left(d_1, \ldots, d_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01990		—	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01991		—	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$	$D=\operatorname{diag}\left(\lambda_{1}, \ldots, \lambda_{n}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01992		—	$\boldsymbol{U}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]$	$\boldsymbol{U}=\left[\boldsymbol{u}_{1}, \ldots, \boldsymbol{u}_{n}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01993		—	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=$	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01994		—	$\boldsymbol{B}=\boldsymbol{U}^{\ast} \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{B}=\boldsymbol{U}^{\ast} \boldsymbol{A} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01995		—	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{\ast}$	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01996		—	$\boldsymbol{B} \boldsymbol{B}^{\ast}=\boldsymbol{B}^{\ast} \boldsymbol{B}$	$\boldsymbol{B} \boldsymbol{B}^{\ast}=\boldsymbol{B}^{\ast} \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01997		—	$\boldsymbol{B}:=\boldsymbol{U}^{\ast} \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{B}:=\boldsymbol{U}^{\ast} \boldsymbol{A} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01998		—	$e_{i i}$	$e_{i i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F01999		—	$\boldsymbol{E}:=\boldsymbol{B}^{\ast} \boldsymbol{B}$	$\boldsymbol{E}:=\boldsymbol{B}^{\ast} \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02000		—	$f_{i i}$	$f_{i i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02001		—	$\boldsymbol{F}:=\boldsymbol{B} \boldsymbol{B}^{\ast}$	$\boldsymbol{F}:=\boldsymbol{B} \boldsymbol{B}^{\ast}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02002		—	$\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$	$ b_{11} ^2 = b_{11} ^2 + b_{12} ^2 + \cdots + b_{1n} ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02003		—	$b_{1 k}=0$	$b_{1k} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02004		—	$k=2,3, \ldots, n$	$k=2,3,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02005		—	$i-1$	$i-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02006		—	$b_{\{j k}=0$	$b_{jk} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02007		—	$j=1, \ldots, i-1$	$j=1,\ldots,i-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02008		—	$k=j+1, \ldots, n$	$k=j+1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02009		—	$b_{\{i k}=0, k=i+1, \ldots, n$	$b_{ik} = 0, k=i+1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02010		—	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U} =$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02011		—	$\operatorname{diag}(1,3)$	$\operatorname{diag}(1,3)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02012		—	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]$	$\boldsymbol{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02013		—	$R(\boldsymbol{x})=\frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}=\lambda$	$R(\boldsymbol{x})=\frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}=\lambda$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02014		—	$\lambda_j, \boldsymbol{u}_j$	$\lambda_j, \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02015		—	$j=1,2,\ldots,n$	$j=1,2,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02016		—	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\sum_{i=1}^n\sum_{j=1}^n\overline{\boldsymbol{c}_i}\boldsymbol{c}_j\boldsymbol{u}_j$	$\boldsymbol{x}^*\boldsymbol{x}=\sum_{i=1}^n\sum_{j=1}^n\bar{c}_i\bar{\boldsymbol{u}}_i c_j \boldsymbol{u}_j=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02017		—	$\sum_{j=1}^n\left c_j\right ^2$	$\sum_{j=1}^n\left c_j\right ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02018		—	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\sum_{i=1}^n\sum_{j=1}^n\overline{\boldsymbol{c}_i}\boldsymbol{c}_j\lambda_j\boldsymbol{u}_j$	$\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}=\sum_{i=1}^n\sum_{j=1}^n\bar{c}_i\bar{\boldsymbol{u}}_i c_j \lambda_j \boldsymbol{u}_j=\sum_{i=1}^n\lambda_i\left c_i\right ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02019		—	$\sum_{i=1}^n\left c_i\right ^2/\sum_{j=1}^n\left c_j\right ^2=1$	$\sum_{i=1}^n\left c_i\right ^2/\sum_{j=1}^n\left c_j\right ^2=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02020		—	$\lambda=\mu+i\nu, \bar{\lambda}=\mu-i\nu$	$\lambda=\mu+i\nu, \lambda=\mu-i\nu$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02021		—	μ, ν	μ, ν
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02022		—	$\lambda=\mu+i\nu$	$\lambda=\mu+i\nu$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02023		—	$\bar{\lambda}=\mu-i\nu$	$\lambda=\mu-i\nu$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02024		—	$\pi_{\boldsymbol{R}}=$	$\pi_{\boldsymbol{R}}=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02025		—	$\boldsymbol{\pi}_{\boldsymbol{D}_1} \boldsymbol{\pi}_{\boldsymbol{D}_2} \boldsymbol{\pi}_{\boldsymbol{D}_3}$	$\boldsymbol{\pi}_{D_1} \boldsymbol{\pi}_{D_2} \boldsymbol{\pi}_{D_3}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02026		—	$\boldsymbol{D}_1$	$\boldsymbol{D}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02027		—	$\boldsymbol{D}_3$	$\boldsymbol{D}_3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02028		—	$\lambda_1=2+i, \lambda_2=2-i$	$\lambda_1=2+i, \lambda_2=2-i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02029		—	$\boldsymbol{D}_2$	$\boldsymbol{D}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02030		—	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02031		—	$\boldsymbol{A} \boldsymbol{u}_j=\lambda_j \boldsymbol{u}_j$	$\boldsymbol{A} \boldsymbol{u}_j=\lambda_j \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02032		—	$\lambda_1, \lambda_2, \ldots, \lambda_n$	$\lambda_1, \lambda_2, \ldots, \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02033		—	$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02034		—	$\left(\lambda_j, \boldsymbol{u}_j\right), 1 \leq j \leq n$	$\left(\lambda_j, \boldsymbol{u}_j\right), 1 \leq j \leq n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02035		—	$\lambda_1 \geq \cdots \geq \lambda_n$	$\lambda_1 \geq \cdots \geq \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02036		—	$1 \leq k \leq n$	$1 \leq k \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02037		—	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02038		—	$\boldsymbol{x}=\boldsymbol{u}_k$	$\boldsymbol{x}=\boldsymbol{u}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02039		—	$\mathcal{S}^{\prime}:=$	$\mathcal{S}^{\prime}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02040		—	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02041		—	$\boldsymbol{y} \in \mathcal{S}$	$\boldsymbol{y} \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02042		—	$R(\boldsymbol{y}) \geq \lambda_k$	$R(\boldsymbol{y}) \geq \lambda_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02043		—	$\mathcal{S}+\mathcal{S}^{\prime}:=$	$\mathcal{S}+\mathcal{S}^{\prime}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02044		—	$\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}$	$\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02045		—	$\mathcal{S} \cap \mathcal{S}^{\prime}$	$\mathcal{S} \cap \mathcal{S}^{\prime}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02046		—	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}' = \sum_{j=1}^k c_j \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02047		—	$\sum_{j=1}^k \left c_j \right ^2 =$	$\sum_{j=1}^k \left c_j \right ^2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02048		—	$c_j=0$	$c_j=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02049		—	$k+1 \leq j \leq n$	$k+1 \leq j \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02050		—	$\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$	$\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02051		—	$\boldsymbol{z} = \sum_{j=k}^n d_j \boldsymbol{u}_j$	$\boldsymbol{z} = \sum_{j=k}^n d_j \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02052		—	$d_k, \ldots, d_n$	$d_k, \ldots, d_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02053		—	$\sum_{j=k}^n \left d_j \right ^2 = 1$	$\sum_{j=k}^n \left d_j \right ^2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02054		—	$6.8 \ R(z) = \sum_{j=k}^n \lambda_j \left d_j \right ^2 \leq$	$6.8 R(z) = \sum_{j=k}^n \lambda_j \left d_j \right ^2 \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02055		—	λ_k	λ_k
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02056		—	$\boldsymbol{z} \in \tilde{\mathcal{S}}$	$z \in \tilde{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02057		—	$\max_{\boldsymbol{x} \in \tilde{\mathcal{S}}} R(\boldsymbol{x}) \leq \lambda_k$	$\max_{\substack{\boldsymbol{x} \in \tilde{\mathcal{S}} \\ \boldsymbol{x} \neq \mathbf{0}}} R(\boldsymbol{x}) \leq \lambda_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02058		—	$\mathcal{S} = \tilde{\mathcal{S}}$	$\mathcal{S} = \tilde{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02059		—	$R(\boldsymbol{u}_k) = \lambda_k$	$R(\boldsymbol{u}_k) = \lambda_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02060		—	$\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$	$\mathcal{S} = \tilde{\mathcal{S}} := \operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02061		—	$(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq$	$(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02062		—	$\operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$	$\operatorname{span}(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02063		—	$R(\boldsymbol{y}) \leq \lambda_k$	$R(\boldsymbol{y}) \leq \lambda_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02064		—	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02065		—	$\boldsymbol{y} \in \tilde{\mathcal{S}}$	$\boldsymbol{y} \in \tilde{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02066		—	$\boldsymbol{A}, \boldsymbol{B} \in$	$\boldsymbol{A}, \boldsymbol{B} \in$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02067		—	$\alpha_{1} \geq \alpha_{2} \geq \cdots \geq \alpha_{n}$	$\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02068		—	$\beta_{1} \geq \beta_{2} \geq \cdots \geq \beta_{n}$	$\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02069		—	$\varepsilon_{1} \geq \varepsilon_{2} \geq \cdots \geq \varepsilon_{n}$	$\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02070		—	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02071		—	$\left(\alpha_{j}, \boldsymbol{u}_{j}\right), j=1, \ldots, n$	$(\alpha_j, \boldsymbol{u}_j), j = 1, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02072		—	$\mathcal{S} := \operatorname{span}\left\{\boldsymbol{u}_{k}, \ldots, \boldsymbol{u}_{n}\right\}$	$\mathcal{S} := \operatorname{span}\left\{\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02073		—	$\boldsymbol{D} := -\boldsymbol{E}$	$\boldsymbol{D} := -\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02074		—	$-\varepsilon_{n}$	$-\varepsilon_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02075		—	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02076		—	$\alpha_{k} \leq \beta_{k} - \varepsilon_{n}$	$\alpha_k \leq \beta_k - \varepsilon_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02077		—	$\mu_{1}, \ldots, \mu_{n}$	$\mu_1, \ldots, \mu_n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02078		—	$\boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{y} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02079		—	$\lambda, \boldsymbol{y}$	$\lambda, \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02080		—	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02081		—	$\boldsymbol{A}^* \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A}^* \boldsymbol{y} = \lambda \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02082		—	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02083		—	$(\lambda, \boldsymbol{y})$	$(\lambda, \boldsymbol{y})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02084		—	(λ, y)	(λ, y)
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02085		—	$\bar{\lambda}$	λ
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02086		—	$\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$	$\{\boldsymbol{y}_1, \dots, \boldsymbol{y}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02087		—	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02088		—	$(\lambda_1, \boldsymbol{x}_1), \dots, (\lambda_n, \boldsymbol{x}_n)$	$(\lambda_1, \boldsymbol{x}_1), \dots, (\lambda_n, \boldsymbol{x}_n)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02089		—	$\left(\lambda_1, \boldsymbol{y}_1\right), \ldots, \left(\lambda_n, \boldsymbol{y}_n\right)$	$(\lambda_1, \boldsymbol{y}_1), \dots, (\lambda_n, \boldsymbol{y}_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02090		—	$\boldsymbol{A} \ \boldsymbol{X}=\boldsymbol{X} \ \boldsymbol{D}, \ \boldsymbol{Y}^* \ \boldsymbol{A}=\boldsymbol{D} \ \boldsymbol{Y}^*$	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A}=\boldsymbol{D} \boldsymbol{Y}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02091		—	$\boldsymbol{A} \ \boldsymbol{X}=\boldsymbol{X} \ \boldsymbol{D} \ \Longrightarrow \boldsymbol{A}=\boldsymbol{X} \ \boldsymbol{D} \ \boldsymbol{X}^{-1} \ \Longrightarrow \boldsymbol{X}^{-1} \ \boldsymbol{A}=\boldsymbol{X} \ \boldsymbol{D} \ \boldsymbol{X}^{-1}$	$\boldsymbol{A} \boldsymbol{X}=\boldsymbol{X} \boldsymbol{D} \Rightarrow \boldsymbol{A}=\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \Rightarrow \boldsymbol{X}^{-1} \boldsymbol{A}=\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02092		—	$\boldsymbol{D} \ \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02093		—	$\boldsymbol{Y}^*:=\boldsymbol{X}^{-1}$	$\boldsymbol{Y}^*:=\boldsymbol{X}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02094		—	$\boldsymbol{Y}^* \ \boldsymbol{X}=\boldsymbol{I}$	$\boldsymbol{Y}^* \boldsymbol{X}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02095		—	$\boldsymbol{Y}$	$\boldsymbol{Y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02096		—	$\boldsymbol{A} \ \boldsymbol{Y}^{-*}=\boldsymbol{Y}^{-*} \ \boldsymbol{D}$	$\boldsymbol{A} \boldsymbol{Y}^{-*}=\boldsymbol{Y}^{-*} \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02097		—	$\boldsymbol{X}:=\boldsymbol{Y}^{-*}$	$\boldsymbol{X}:=\boldsymbol{Y}^{-*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02098		—	$\boldsymbol{v}=\sum_{j=1}^n \ c_j \ \boldsymbol{x}_j$	$\boldsymbol{v}=\sum_{j=1}^n c_j \boldsymbol{x}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02099		—	$\boldsymbol{y}_i^* \ \boldsymbol{v}=\sum_{j=1}^n \ c_j \ \boldsymbol{y}_i^* \ \boldsymbol{x}_j$ $\boldsymbol{x}_j=c_i \ \boldsymbol{y}_i^* \ \boldsymbol{x}_j$	$\boldsymbol{y}_i^* \boldsymbol{v}=\sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j=c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02100		—	$c_i=\boldsymbol{y}_i^*\boldsymbol{v}/\boldsymbol{y}_i^*\boldsymbol{x}_i$	$c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02101		—	$\mu, \boldsymbol{y}$	$\mu, \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02102		—	$\boldsymbol{y}^*\boldsymbol{A}\boldsymbol{x}=\lambda\boldsymbol{y}^*\boldsymbol{x}=\mu\boldsymbol{y}^*\boldsymbol{x}$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02103		—	$\boldsymbol{y}^*\boldsymbol{x}\neq 0$	$\boldsymbol{y}^* \boldsymbol{x} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02104		—	$\ \boldsymbol{x}\ _2=1$	$\ \boldsymbol{x}\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02105		—	$\boldsymbol{V}\boldsymbol{e}_1=\boldsymbol{x}$	$\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02106		—	$\boldsymbol{u}:=\boldsymbol{V}^*\boldsymbol{y}$	$\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02107		—	$(\bar{\lambda}, \boldsymbol{u})$	$(\lambda, \boldsymbol{u})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02108		—	$\boldsymbol{V}^*\boldsymbol{A}^*\boldsymbol{V}$	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02109		—	$\boldsymbol{y}^*\boldsymbol{x}=\boldsymbol{u}^*\boldsymbol{V}^*\boldsymbol{V}\boldsymbol{e}_1=\boldsymbol{u}^*\boldsymbol{e}_1$	$\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02110		—	$\boldsymbol{u}^*\boldsymbol{e}_1=0$	$\boldsymbol{u}^* \boldsymbol{e}_1 = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02111		—	$\boldsymbol{u}=\left[\begin{array}{l}0\\ \boldsymbol{v}\end{array}\right]$	$\boldsymbol{u}=\left[\begin{array}{l}0\\ \boldsymbol{v}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02112		—	$\boldsymbol{v}\in\mathbb{C}^{n-1}$	$\boldsymbol{v}\in\mathbb{C}^{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02113		—	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\ 0&1\end{array}\right]$	$\boldsymbol{A}:=\left[\begin{array}{ll}1&1\\ 0&1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02114		—	$\boldsymbol{y}=\boldsymbol{e}_2$	$\boldsymbol{y}=\boldsymbol{e}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02115		—	$\operatorname{det}\left(\boldsymbol{B}^T\right)=$	$\det\left(\boldsymbol{B}^T\right)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02116		—	$\operatorname{det}\left(\boldsymbol{B}\right)$	$\det(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02117		—	$\operatorname{det}\left(\overline{\boldsymbol{B}}\right)=\overline{\operatorname{det}\left(\boldsymbol{B}\right)}$	$\det(\overline{\boldsymbol{B}})=\overline{\det(\boldsymbol{B})}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02118		—	$\pi_{\boldsymbol{A}^T}=\pi_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}^T}=\pi_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02119		—	$\pi_{A^*}(\overline{\lambda})=\overline{\pi_A(\lambda)}$	$\pi_{A^*}(\overline{\lambda})=\overline{\pi_A(\lambda)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02120		—	$\left(\lambda^{-1},\boldsymbol{x}\right)$	$(\lambda^{-1},\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02121		—	$\left(\lambda^k,\boldsymbol{x}\right)$	$(\lambda^k,\boldsymbol{x})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02122		—	$\boldsymbol{A}^k$	$\boldsymbol{A}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02123		—	$k \in \mathbb{Z}$	$k \in \mathbb{Z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02124		—	$(\lambda_j, \boldsymbol{x}_j), j=1, \ldots, n$	$(\lambda_j, \boldsymbol{x}_j), j=1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02125		—	$k \in \mathbb{N}$	$k \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02126		—	$\boldsymbol{A}^2=$	$\boldsymbol{A}^2=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02127		—	$\lambda=0$	$\lambda=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02128		—	$\boldsymbol{A}^2=\boldsymbol{A}$	$\boldsymbol{A}^2=\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02129		—	$\boldsymbol{A}^k=0$	$\boldsymbol{A}^k=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02130		—	$\boldsymbol{A}^* \boldsymbol{A}=\boldsymbol{I}$	$\boldsymbol{A}^* \boldsymbol{A}=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02131		—	$ \lambda =1$	$ \lambda =1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02132		—	$\epsilon_0>0$	$\epsilon_0>0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02133		—	$\boldsymbol{A}+\epsilon \boldsymbol{I}$	$\boldsymbol{A}+\epsilon \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02134		—	$\epsilon \in \mathbb{C}$	$\epsilon \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02135		—	$ \epsilon <\epsilon_0$	$ \epsilon <\epsilon_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02136		—	$\operatorname{det}(\boldsymbol{A})=\lambda_1 \lambda_2 \cdots \lambda_n$	$\det(\boldsymbol{A})=\lambda_1 \lambda_2 \cdots \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02137		—	$q_{\{0\}}, \ldots, q_{\{n-1\}} \in \mathbb{C}$	$q_0, \ldots, q_{n-1} \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02138		—	$p(\lambda)=\lambda^n+$	$p(\lambda)=\lambda^n+$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02139		—	$q_{\{n-1\}} \lambda^{n-1}+\cdots+q_{\{0\}}$	$q_{n-1} \lambda^{n-1}+\cdots+q_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02140		—	$(-1)^n p$	$(-1)^n p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02141		—	$p=(-1)^n \pi_{\boldsymbol{A}}$	$p=(-1)^n \pi_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02142		—	$p=(-1)^n \pi_{\boldsymbol{B}}$	$p=(-1)^n \pi_{\boldsymbol{B}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02143		—	$\boldsymbol{A}=\left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{array}\right]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02144		—	$\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02145		—	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02146		—	$\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$	$\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02147		—	$\boldsymbol{A} \in \mathbb{R}^{n, n}$	$\boldsymbol{A} \in \mathbb{R}^{n, n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02148		—	$a_{i+1, i} a_{i, i+1} > 0$	$a_{i+1, i} a_{i, i+1} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02149		—	$\boldsymbol{A} = \left[\begin{array}{rrr} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{array} \right]$	$\boldsymbol{A} = \left[\begin{array}{rrr} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02150		—	$\boldsymbol{J} = \left[\begin{array}{lll} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$	$\boldsymbol{J} = \left[\begin{array}{lll} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02151		—	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I} \right)^r = \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0} \end{array} \right]$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^r = \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{I}_{m-r} \\ \mathbf{0} & \mathbf{0} \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02152		—	$1 \leq r \leq m-1$	$1 \leq r \leq m-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02153		—	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I} \right)^m = \mathbf{0}$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02154		—	$r=0, 1, 2, \ldots, m=2, 3, \ldots$	$r = 0, 1, 2, \dots, m = 2, 3, \dots$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02155		—	$\boldsymbol{A}^r=\boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$	$\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02156		—	$\boldsymbol{J}^r=\operatorname{diag}\!\left(\boldsymbol{U}_{1}^r, \ldots, \boldsymbol{U}_{k}^r\right)$	$\boldsymbol{J}^r = \operatorname{diag}\left(\boldsymbol{U}_1^r, \ldots, \boldsymbol{U}_k^r\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02157		—	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02158		—	$\boldsymbol{U}_{i}^r=\operatorname{diag}\!\left(\boldsymbol{J}_{m_{i,1}}\!\left(\lambda_i\right), \ldots, \boldsymbol{J}_{m_{i,g_i}}\!\left(\lambda_i\right)\right)$	$\boldsymbol{U}_i^r = \operatorname{diag}\left(\boldsymbol{J}_{m_{i,1}}\left(\lambda_i\right)^r, \ldots, \boldsymbol{J}_{m_{i,g_i}}\left(\lambda_i\right)^r\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02159		—	$\boldsymbol{J}_{m}(\lambda)^r=\left(\boldsymbol{E}_{m}+\lambda \boldsymbol{I}_{m}\right)^r=\sum_{k=0}^{\min \left\{r, m-1\right\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_{m}^k$	$\boldsymbol{J}_m(\lambda)^r = \left(\boldsymbol{E}_m + \lambda \boldsymbol{I}_m\right)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02160		—	$\boldsymbol{J}^{100}$	$\boldsymbol{J}^{100}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02161		—	$\boldsymbol{A}^{100}$	$\boldsymbol{A}^{100}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02162		—	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02163		—	$\lambda_i-\lambda$	$\lambda_i - \lambda$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02164		—	$\lambda_i \boldsymbol{I}-\boldsymbol{A}$	$\lambda_i \boldsymbol{I} - \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02165		—	$\pi_{\boldsymbol{A}}(\lambda)=\prod_{i=1}^k \prod_{j=1}^{g_i}\left(\lambda_i-\lambda\right)^{m_{i, j}}$	$\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02166		—	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02167		—	$\nu_{\boldsymbol{A}}$	$\nu_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02168		—	$\mu_{\boldsymbol{A}}(\boldsymbol{A})(\boldsymbol{A}) = \mathbf{0} \quad \Longleftrightarrow \quad \mu_{\boldsymbol{A}}(\boldsymbol{J})(\boldsymbol{J}) = \mathbf{0}$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0} \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02169		—	m_i	m_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02170		—	$\mu_{\boldsymbol{A}}(\boldsymbol{A})(\boldsymbol{A}) = 0$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02171		—	$p(\boldsymbol{A}) = \mathbf{0}$	$p(\boldsymbol{A}) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02172		—	$\pi_{\boldsymbol{A}}(\boldsymbol{A})(\boldsymbol{A}) = \mathbf{0}$	$\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02173		—	$p(t) := \sum_{j=0}^r b_j t^j$	$p(t) := \sum_{j=0}^r b_j t^j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02174		—	$b_j \in \mathbb{C}$	$b_j \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02175		—	$p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	$p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02176		—	$\boldsymbol{A}^0 := \boldsymbol{I}$	$\boldsymbol{A}^0 := \boldsymbol{I}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02177		—	$p(t) := \left(t - \alpha_1\right) \cdots \left(t - \alpha_r\right)$	$p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02178		—	$\alpha_0, \ldots, \alpha_r \in \mathbb{C}$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02179		—	$p(\boldsymbol{A}) = \left(\boldsymbol{A} - \alpha_1\right) \cdots \left(\boldsymbol{A} - \alpha_r\right)$	$p(\boldsymbol{A}) = (\boldsymbol{A} - \alpha_1) \cdots (\boldsymbol{A} - \alpha_r)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02180		—	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} = \boldsymbol{T}$	$\boldsymbol{U}^{*} \boldsymbol{A} \boldsymbol{U} = \boldsymbol{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02181		—	$\left[\begin{array}{rr} 2 & 1 \\ -1 & 4 \end{array}\right]$	$\left[\begin{array}{cc} 2 & 1 \\ -1 & 4 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02182		—	$\pi(\boldsymbol{A}) = \mathbf{0}$	$\pi(\boldsymbol{A}) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02183		—	$p(\boldsymbol{A}) =$	$p(\boldsymbol{A}) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02184		—	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^{*}$	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^{*}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02185		—	$n, k \in \mathbb{N}$	$n, k \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02186		—	$\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$	$\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02187		—	$c_{i, j} = 0$	$c_{i, j} = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02188		—	$i, j \leq k$	$i, j \leq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02189		—	$d_{k+1, k+1}=0$	$d_{k+1, k+1} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02190		—	$\boldsymbol{E}:=\boldsymbol{C} \ \boldsymbol{D}$	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02191		—	$e_{i, j}=0$	$e_{i, j} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02192		—	$i, j \leq k+1$	$i, j \leq k + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02193		—	$p:=\pi_{\boldsymbol{A}}$	$p := \pi_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02194		—	$p(\boldsymbol{T})=\mathbf{0}^{\,3}$	$p(\boldsymbol{T}) = \mathbf{0}^{\,3}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02195		—	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 2\end{array}\right]$	$\boldsymbol{A} = \left[\begin{array}{cc}1 & 2 \\ 3 & 2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02196		—	$\boldsymbol{U}^{\boldsymbol{T}} \ \boldsymbol{A} \ \boldsymbol{U}=\left[\begin{array}{cc}-1 & -1 \\ 0 & 4\end{array}\right]$	$\boldsymbol{U}^{\boldsymbol{T}} \boldsymbol{A} \boldsymbol{U} = \left[\begin{array}{cc}-1 & -1 \\ 0 & 4\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02197		—	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]$	$\boldsymbol{U} = \frac{1}{\sqrt{2}} \left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02198		—	$\boldsymbol{C}=\boldsymbol{A}+i \ \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A} + i \boldsymbol{B}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02199		—	$\boldsymbol{A}^T = -\boldsymbol{A}$	$\boldsymbol{A}^T = -\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02200		—	$\boldsymbol{B}^T = \boldsymbol{B}$	$\boldsymbol{B}^T = \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02201		—	$\left(\lambda_1, \boldsymbol{u}_1\right), \ldots, \left(\lambda_n, \boldsymbol{u}_n\right)$	$\left(\lambda_1, \boldsymbol{u}_1\right), \ldots, \left(\lambda_n, \boldsymbol{u}_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02202		—	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$	$\boldsymbol{u}_j^* \boldsymbol{u}_k = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02203		—	$j \neq k$	$j \neq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02204		—	$\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02205		—	λ_{min}	$\lambda_{\min}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02206		—	$R(\boldsymbol{x})$	$R(\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02207		—	$\boldsymbol{y} \neq 0$	$\boldsymbol{y} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02208		—	$t > 0$	$t > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02209		—	$\boldsymbol{x}^0 \in B_1$	$\boldsymbol{x}^0 \in B_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02210		—	$\boldsymbol{A} \, \boldsymbol{x}^{\{0\}} - R \left(\boldsymbol{x}^{\{0\}} \right) \, \boldsymbol{x}^{\{0\}} \neq 0$	$\boldsymbol{A} \boldsymbol{x}^0 - R \left(\boldsymbol{x}^0 \right) \boldsymbol{x}^0 \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02211		—	$\left(R \left(\boldsymbol{x}^{\{0\}} \right), \, \boldsymbol{x}^{\{0\}} \right)$	$\left(R \left(\boldsymbol{x}^0 \right), \boldsymbol{x}^0 \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02212		—	$\boldsymbol{x}^{\{1\}} \, \text{in} \, B_{\{1\}}$	$\boldsymbol{x}^1 \in B_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02213		—	$R \left(\boldsymbol{x}^{\{1\}} \right) <$	$R \left(\boldsymbol{x}^1 \right) <$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02214		—	$R \left(\boldsymbol{x}^{\{0\}} \right)$	$R \left(\boldsymbol{x}^0 \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02215		—	$\beta_{\{i\}} \, \text{geq} \, \alpha_{\{i\}}$	$\beta_i \geq \alpha_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02216		—	$\boldsymbol{A} := \left[\begin{array}{ll} 0 & 0 \\ 0 & 4 \end{array} \right]$	$\boldsymbol{A} := \left[\begin{array}{cc} 0 & 0 \\ 0 & 4 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02217		—	$\boldsymbol{B} := \left[\begin{array}{cc} -1 & -1 \\ 1 & 1 \end{array} \right]$	$\boldsymbol{B} := \left[\begin{array}{cc} -1 & -1 \\ 1 & 1 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02218		—	$\boldsymbol{A} := \left[\begin{array}{ll} 3 & 1 \\ 2 & 2 \end{array} \right]$	$\boldsymbol{A} := \left[\begin{array}{cc} 3 & 1 \\ 2 & 2 \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02219		—	$\boldsymbol{v} \, \text{in} \, \mathbb{R}^{\{2\}}$	$\boldsymbol{v} \in \mathbb{R}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02220		—	$\boldsymbol{y}, \, \boldsymbol{x} \, \text{in}$	$\boldsymbol{y}, \boldsymbol{x} \in$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02221		—	$R(\boldsymbol{y}, \boldsymbol{x}) = R_{\boldsymbol{A}}(\boldsymbol{y}, \boldsymbol{x}) := \frac{\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{y}^* \boldsymbol{x}}$	$R(\boldsymbol{y}, \boldsymbol{x}) = R_{\boldsymbol{A}}(\boldsymbol{y}, \boldsymbol{x}) := \frac{\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{y}^* \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02222		—	$R(\boldsymbol{y}, \boldsymbol{x}) = \lambda$	$R(\boldsymbol{y}, \boldsymbol{x}) = \lambda$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02223		—	$\boldsymbol{A}, \boldsymbol{A}^T$	$\boldsymbol{A}, \boldsymbol{A}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02224		—	$\boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A} \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02225		—	$f(t) = R(\boldsymbol{x} - t \boldsymbol{y})$	$f(t) = R(\boldsymbol{x} - t \boldsymbol{y})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02226		—	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n] \in \mathbb{C}^{n \times n}$	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n] \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02227		—	$\boldsymbol{D} :=$	$\boldsymbol{D} :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02228		—	$\operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$\operatorname{diag}(\lambda_1, \dots, \lambda_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02229		—	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02230		—	$\boldsymbol{\Sigma}$	$\boldsymbol{\Sigma}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02231		—	$\Sigma_{ij} = 0$	$\Sigma_{ij} = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02232		—	$\Sigma_{i\ i} \geq 0$	$\Sigma_{ii} \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02233		—	$\sigma_{i}:=\Sigma_{i\ i}$	$\sigma_i := \Sigma_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02234		—	$\sigma_{i} \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02235		—	$\sigma_{1}=2$	$\sigma_1 = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02236		—	$\sigma_{2}=1$	$\sigma_2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02237		—	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02238		—	$\mathcal{R}\!\left(\boldsymbol{A}^*\right)$	$\mathcal{R}(\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02239		—	$\mathcal{N}\!\left(\boldsymbol{A}^*\right)$	$\mathcal{N}(\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02240		—	$\left.\boldsymbol{A}^*\ \boldsymbol{A}, \boldsymbol{A}\boldsymbol{A}^*\right)$	$\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A}\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02241		—	$\boldsymbol{A}^*\ \boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}^* \boldsymbol{A} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02242		—	$\boldsymbol{A}\boldsymbol{A}^* \in \mathbb{C}^{m \times m}$	$\boldsymbol{A}\boldsymbol{A}^* \in \mathbb{C}^{m \times m}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02243		—	$\left(\lambda_j, \boldsymbol{v}_j\right)$	$(\lambda_j, \boldsymbol{v}_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02244		—	$\left\{\boldsymbol{A} \boldsymbol{v}_1, \ldots, \boldsymbol{A} \boldsymbol{v}_r\right\}$	$\{\boldsymbol{A} \boldsymbol{v}_1, \ldots, \boldsymbol{A} \boldsymbol{v}_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02245		—	$\mathcal{R}(\boldsymbol{A}) :=$	$\mathcal{R}(\boldsymbol{A}) :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02246		—	$\left\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m: \boldsymbol{y} \in \mathbb{C}^n\right\}$	$\{\boldsymbol{A} \boldsymbol{y} \in \mathbb{C}^m: \boldsymbol{y} \in \mathbb{C}^n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02247		—	$\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right\}$	$\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02248		—	$\mathcal{N}(\boldsymbol{A}) := \left\{\boldsymbol{y} \in \mathbb{C}^n: \boldsymbol{A} \boldsymbol{y} = \boldsymbol{0}\right\}$	$\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n: \boldsymbol{A} \boldsymbol{y} = \boldsymbol{0}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02249		—	$\left(\lambda_j, \boldsymbol{u}_j\right)$	$(\lambda_j, \boldsymbol{u}_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02250		—	$\lambda_j > 0, j=1, \ldots, r$	$\lambda_j > 0, j = 1, \ldots, r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02251		—	$\lambda_j = 0, j=r+1, \ldots, m$	$\lambda_j = 0, j = r + 1, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02252		—	$\left\{\boldsymbol{A}^* \boldsymbol{u}_1, \ldots, \boldsymbol{A}^* \boldsymbol{u}_r\right\}$	$\{\boldsymbol{A}^* \boldsymbol{u}_1, \ldots, \boldsymbol{A}^* \boldsymbol{u}_r\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02253		—	$\left\{\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\right\}$	$\{\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02254		—	$\pi_{\boldsymbol{A}^*} \boldsymbol{A}$	$\pi \boldsymbol{A}^* \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02255		—	$\pi_{\boldsymbol{A}} \boldsymbol{A}^*$	$\pi \boldsymbol{A} \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02256		—	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v} = \lambda \boldsymbol{v}$	$\boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v} = \lambda \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02257		—	$\boldsymbol{v} \neq \mathbf{0}$	$\boldsymbol{v} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02258		—	$\left(\boldsymbol{A} \boldsymbol{v}_{\{j\}}\right)^* \boldsymbol{A} \boldsymbol{v}_{\{k\}} = \boldsymbol{v}_{\{j\}}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}_{\{k\}} = \lambda_{\{k\}} \boldsymbol{v}_{\{j\}}^* \boldsymbol{v}_{\{k\}}$	$(\boldsymbol{A} \boldsymbol{v}_j)^* \boldsymbol{A} \boldsymbol{v}_k = \boldsymbol{v}_j^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{v}_k = \lambda_k \boldsymbol{v}_j^* \boldsymbol{v}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02259		—	$\boldsymbol{A} \boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{A} \boldsymbol{v}_{\{n\}}$	$\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02260		—	$\boldsymbol{A} \boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{A} \boldsymbol{v}_{\{r\}}$	$\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02261		—	$\boldsymbol{A} \boldsymbol{v}_{\{j\}} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{v}_j = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02262		—	$j = r +$	$j = r +$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02263		—	$\mathcal{N}(\boldsymbol{A})$	$\mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02264		—	$\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}\left(\boldsymbol{A} \boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{A} \boldsymbol{v}_{\{r\}}\right)$	$\mathcal{R}(\boldsymbol{A}) \subset \operatorname{span}(\boldsymbol{A} \boldsymbol{v}_1, \dots, \boldsymbol{A} \boldsymbol{v}_r)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02265		—	$\mathcal{N}(\mathbf{A}) \subset$	$\mathcal{N}(\mathbf{A}) \subset$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02266		—	$\operatorname{span}\left(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\right)$	$\operatorname{span}\left(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02267		—	$\mathbf{x} \in \mathcal{R}(\mathbf{A})$	$\mathbf{x} \in \mathcal{R}(\mathbf{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02268		—	$\mathbf{x}=\mathbf{A} \mathbf{y}$	$\mathbf{x}=\mathbf{A} \mathbf{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02269		—	$\mathbf{y}=\sum_{j=1}^n c_j \mathbf{v}_j$	$\mathbf{y}=\sum_{j=1}^n c_j \mathbf{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02270		—	$\mathbf{x}=\mathbf{A} \mathbf{y}=\sum_{j=1}^n c_j \mathbf{A} \mathbf{v}_j$ $\mathbf{v}_j=\sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j \in$	$\mathbf{x}=\mathbf{A} \mathbf{y}=\sum_{j=1}^n c_j \mathbf{A} \mathbf{v}_j=\sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02271		—	$\operatorname{span}\left(\mathbf{A} \mathbf{v}_1, \ldots, \mathbf{A} \mathbf{v}_r\right)$	$\operatorname{span}\left(\mathbf{A} \mathbf{v}_1, \ldots, \mathbf{A} \mathbf{v}_r\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02272		—	$\mathbf{y}=\sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$	$\mathbf{y}=\sum_{j=1}^n c_j \mathbf{v}_j \in \mathcal{N}(\mathbf{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02273		—	$\mathbf{A} \mathbf{y}=\sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j=\mathbf{0}$	$\mathbf{A} \mathbf{y}=\sum_{j=1}^r c_j \mathbf{A} \mathbf{v}_j=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02274		—	$c_1=\cdots=c_r=0$	$c_1=\cdots=c_r=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02275		—	$\mathbf{y}=\sum_{j=r+1}^n c_j \mathbf{v}_j \in \operatorname{span}\left(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\right)$	$\mathbf{y}=\sum_{j=r+1}^n c_j \mathbf{v}_j \in \operatorname{span}\left(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02276		—	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{B}^* \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02277		—	$\boldsymbol{B} := \boldsymbol{A}^*$	$\boldsymbol{B} := \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02278		—	$\boldsymbol{A} = \boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02279		—	$2 \boldsymbol{A}^* \boldsymbol{A}$	$2 \boldsymbol{A}^* \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02280		—	$4 \ r$	$4r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02281		—	$m, n, r \in \mathbb{N}$	$m, n, r \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02282		—	$\lambda_1 \geq \cdots \geq$	$\lambda_1 \geq \cdots \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02283		—	$\lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$	$\lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02284		—	$\boldsymbol{V} := [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$ $\times n$	$\boldsymbol{V} := [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02285		—	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02286		—	$\sigma_j := \sqrt{\lambda_j}$	$\sigma_j := \sqrt{\lambda_j}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02287		—	j=	$j =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02288		—	$1, \ldots, \min(m, n)$	$1, \dots, \min(m, n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02289		—	$\boldsymbol{U} := \left[\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{m\}} \right] \in \mathbb{C}^{m \times m} \times m$	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02290		—	$\boldsymbol{u}_{\{j\}} = \boldsymbol{\sigma}_{\{j\}}^{-1} \boldsymbol{A} \boldsymbol{v}_{\{j\}}$	$\boldsymbol{u}_j = \boldsymbol{\sigma}_j^{-1} \boldsymbol{A} \boldsymbol{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02291		—	j=1, \ldots, r	$j = 1, \dots, r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02292		—	$\boldsymbol{u}_{\{r+1\}}, \ldots, \boldsymbol{u}_{\{m\}}$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02293		—	$\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{r\}}$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02294		—	$\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{m\}}$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02295		—	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02296		—	$\boldsymbol{\sigma}_{\{j\}} = \left \left \boldsymbol{A} \boldsymbol{v}_{\{j\}} \right \right _2 > 0, j=1, \ldots, r$	$\sigma_j = \ \boldsymbol{A} \boldsymbol{v}_j\ _2 > 0, j = 1, \dots, r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02297		—	$\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma} = \boldsymbol{A} \boldsymbol{V}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02298		—	$\boldsymbol{U} \, \boldsymbol{\Sigma} \, \boldsymbol{V}^* = \boldsymbol{A} \, \boldsymbol{V} \, \boldsymbol{V}^* = \boldsymbol{A}$ $\boldsymbol{V}^* = \boldsymbol{A}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* = \boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02299		—	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02300		—	$\boldsymbol{A} = \frac{1}{15} \left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array} \right]$	$\boldsymbol{A} = \frac{1}{15} \begin{bmatrix} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02301		—	$\sigma_1 = 2, \, \sigma_2 = 1$	$\sigma_1 = 2, \sigma_2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02302		—	$\boldsymbol{V} = \frac{1}{5} \left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array} \right]$	$\boldsymbol{V} = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02303		—	$\boldsymbol{u}_1 = \boldsymbol{A} \, \boldsymbol{v}_1 \, / \, \sigma_1 = [1, 2, 2]^T / 3$	$\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = [1, 2, 2]^T / 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02304		—	$\boldsymbol{u}_2 = \boldsymbol{A} \, \boldsymbol{v}_2 \, / \, \sigma_2 = [2, -2, 1]^T / 3$	$\boldsymbol{u}_2 = \boldsymbol{A} \boldsymbol{v}_2 / \sigma_2 = [2, -2, 1]^T / 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02305		—	$\boldsymbol{u}_3$	$\boldsymbol{u}_3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02306		—	$\boldsymbol{u}_1$	$\boldsymbol{u}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02307		—	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3 = [2, 1, -2]^T / 3$	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3 = [2, 1, -2]^T / 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02308		—	Σ_1	Σ_1
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02309		—	$k, \, l \, \geq \, 0$	$k, l \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02310		—	$\mathbf{0}_{k, \, l} = []$	$\mathbf{0}_{k, l} = []$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02311		—	$k{=}0$	$k = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02312		—	$l{=}0$	$l = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02313		—	l	l
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02314		—	$\boldsymbol{A} = \boldsymbol{U}_{1} \, \boldsymbol{\Sigma}_{1} \, \boldsymbol{V}_{1}^{\ast}$	$\boldsymbol{A} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02315		—	$\boldsymbol{U}_{1} \, \text{in} \, \mathbb{C}^{m \, \times \, r}$	$\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02316		—	$\boldsymbol{V}_{1} \, \text{in} \, \mathbb{C}^{n \, \times \, r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02317		—	$\boldsymbol{\Sigma}_{1} \, \text{in} \, \mathbb{R}^{r \, \times \, r}$	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02318		—	$\sigma_{1} \, \geq \, \cdots \, \geq \, \sigma_{r} > 0$	$\sigma_1 \geq \cdots \geq \sigma_r > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02319		—	$\boldsymbol{U}_{1}, \, \boldsymbol{V}_{1}$	$\boldsymbol{U}_1, \boldsymbol{V}_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02320		—	$\boldsymbol{U}, \boldsymbol{V}$	U, V
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02321		—	$\Sigma_{\{1\}} \in \mathbb{R}^{r \times r}$	$\Sigma_1 \in \mathbb{R}^{r \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02322		—	$\boldsymbol{U}_{\{1\}}$	$\boldsymbol{U}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02323		—	$\boldsymbol{V}_{\{1\}}$	$\boldsymbol{V}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02324		—	$\boldsymbol{U} \in \mathbb{C}^{m \times m}$	$\boldsymbol{U} \in \mathbb{C}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02325		—	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$\boldsymbol{V} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02326		—	$\boldsymbol{A} = \left[\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{r\}} \right] \operatorname{diag} \left(\sigma_{\{1\}}, \ldots, \sigma_{\{r\}} \right) \left[\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{r\}} \right]^*$	$\boldsymbol{A} = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r) [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02327		—	$7.3(r < n < m)$	$7.3(r < n < m)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02328		—	$\sigma_{\{1\}}=2, \sigma_{\{2\}}=0$	$\sigma_1 = 2, \sigma_2 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02329		—	$r=1, m=3, n=2$	$r = 1, m = 3, n = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02330		—	$\boldsymbol{u}_{\{1\}} = \boldsymbol{A} \boldsymbol{v}_{\{1\}} / \sigma_{\{1\}} = \boldsymbol{s}_{\{1\}} / \sqrt{2}$	$\boldsymbol{u}_1 = \boldsymbol{A} \boldsymbol{v}_1 / \sigma_1 = \boldsymbol{s}_1 / \sqrt{2}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02331		—	$\boldsymbol{s}_1=[1,1,0]^T$	$\boldsymbol{s}_1 = [1,1,0]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02332		—	$\boldsymbol{s}_1$	$\boldsymbol{s}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02333		—	$\left\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\right\}$	$\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02334		—	$\boldsymbol{w}_1=\boldsymbol{s}_1, \boldsymbol{w}_2=\boldsymbol{s}_2-\frac{\boldsymbol{s}_2^T\boldsymbol{w}_1}{\boldsymbol{w}_1^T\boldsymbol{w}_1}\boldsymbol{w}_1$ $\boldsymbol{w}_1=\left[\begin{array}{c} -1/2 \\ 1/2 \\ 0 \end{array}\right], \boldsymbol{w}_3=\boldsymbol{s}_3-$	$\boldsymbol{w}_1 = \boldsymbol{s}_1, \boldsymbol{w}_2 = \boldsymbol{s}_2 - \frac{\boldsymbol{s}_2^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, \boldsymbol{w}_3 = \boldsymbol{s}_3-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02335		—	$\frac{\boldsymbol{s}_3^T\boldsymbol{w}_1}{\boldsymbol{w}_1^T\boldsymbol{w}_1}\boldsymbol{w}_1-\frac{\boldsymbol{s}_3^T\boldsymbol{w}_2}{\boldsymbol{w}_2^T\boldsymbol{w}_2}\boldsymbol{w}_2=\left[\begin{array}{c} 0 \\ 0 \\ 1 \end{array}\right]$	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02336		—	$\boldsymbol{w}_i$	$\boldsymbol{w}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02337		—	$\boldsymbol{u}_1=\boldsymbol{w}_1/\left\ \boldsymbol{w}_1\right\ _2=$	$\boldsymbol{u}_1 = \boldsymbol{w}_1/\left\ \boldsymbol{w}_1\right\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02338		—	$\left[1/\sqrt{2}, 1/\sqrt{2}, 0\right]^T, \boldsymbol{u}_2=\boldsymbol{w}_2/\left\ \boldsymbol{w}_2\right\ _2=\left[-1/\sqrt{2}, 1/\sqrt{2}, 0\right]^T$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, \boldsymbol{u}_2 = \boldsymbol{w}_2/\left\ \boldsymbol{w}_2\right\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02339		—	$\boldsymbol{u}_3=\boldsymbol{s}_3/\left\ \boldsymbol{s}_3\right\ _2=$	$\boldsymbol{u}_3 = \boldsymbol{s}_3/\left\ \boldsymbol{s}_3\right\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02340		—	$\left[0,0,1\right]^T$	$[0,0,1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02341		—	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$	$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02342		—	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^{\ast})$ $\text{\right)}$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^{\ast})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02343		—	m, n	m, n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02344		—	$\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\right] \boldsymbol{\Sigma} \left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right]^{\ast}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{\ast}$	$\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\right] \boldsymbol{\Sigma} \left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right]^{\ast}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02345		—	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right\}$	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02346		—	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right\}$	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02347		—	$\boldsymbol{A} \boldsymbol{V}=\boldsymbol{U} \boldsymbol{\Sigma}$	$\boldsymbol{A} \boldsymbol{V}=\boldsymbol{U} \boldsymbol{\Sigma}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02348		—	$\boldsymbol{A} \left[\boldsymbol{V}_1, \boldsymbol{V}_2\right]=\left[\boldsymbol{U}_1, \boldsymbol{U}_2\right] \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array}\right]$	$\boldsymbol{A} \left[\boldsymbol{V}_1, \boldsymbol{V}_2\right]=\left[\boldsymbol{U}_1, \boldsymbol{U}_2\right] \left[\begin{array}{cc} \boldsymbol{\Sigma}_1 & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02349		—	$\boldsymbol{A} \boldsymbol{V}_1=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1, \boldsymbol{A} \boldsymbol{V}_2=\mathbf{0}$	$\boldsymbol{A} \boldsymbol{V}_1=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1, \boldsymbol{A} \boldsymbol{V}_2=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02350		—	$\boldsymbol{A}^{\ast}=\boldsymbol{V} \boldsymbol{\Sigma}^{\ast} \boldsymbol{U}^{\ast}$	$\boldsymbol{A}^{\ast}=\boldsymbol{V} \boldsymbol{\Sigma}^{\ast} \boldsymbol{U}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02351		—	$\boldsymbol{A}^{\ast} \boldsymbol{U}=\boldsymbol{V} \boldsymbol{\Sigma}^{\ast}$	$\boldsymbol{A}^{\ast} \boldsymbol{U}=\boldsymbol{V} \boldsymbol{\Sigma}^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02352		—	$\mathcal{R}(\boldsymbol{A}), \left\{\boldsymbol{A}^{\ast} \boldsymbol{u}_1, \ldots, \boldsymbol{A}^{\ast} \boldsymbol{u}_r\right\}$	$\mathcal{R}(\boldsymbol{A}), \left\{\boldsymbol{A}^{\ast} \boldsymbol{u}_1, \ldots, \boldsymbol{A}^{\ast} \boldsymbol{u}_r\right\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02353		—	$\mathcal{R}\left(\boldsymbol{A}^*\right),\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_m\right. \right.$	$\mathcal{R}\left(\boldsymbol{A}^*\right),\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_m\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02354		—	$\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02355		—	$\boldsymbol{T} \ \boldsymbol{z}:=\boldsymbol{A} \ \boldsymbol{z}$	$\boldsymbol{T} \boldsymbol{z}:=\boldsymbol{A} \boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02356		—	$(\boldsymbol{A})=n$	$(\boldsymbol{A})=n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02357		—	$\mathcal{S}:=\left\{z \in \mathbb{R}^n\right\}.$	$\mathcal{S}:=\left\{z \in \mathbb{R}^n\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02358		—	$\left.\left.\boldsymbol{z}\right _2=1\right\}$	$\ \boldsymbol{z}\ _2=1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02359		—	$\mathcal{E}:=\boldsymbol{A} \ \mathcal{S}=\left\{\boldsymbol{A} \ \boldsymbol{z}: \boldsymbol{z} \in \mathcal{S} \in \mathcal{S}\right\}$	$\mathcal{E}:=\boldsymbol{A} \mathcal{S}=\left\{\boldsymbol{A} \boldsymbol{z}: \boldsymbol{z} \in \mathcal{S}\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02360		—	$r=n$	$r=n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02361		—	$\boldsymbol{U}_1 \Sigma_1 \boldsymbol{V}_1^T$	$\boldsymbol{U}_1 \Sigma_1 \boldsymbol{V}_1^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02362		—	$z \in \mathcal{S}$	$z \in \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02363		—	$\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \Sigma_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$	$\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \Sigma_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02364		—	$\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$	$\boldsymbol{y} := \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02365		—	$\operatorname{rank}(\boldsymbol{A}) = n$	$\operatorname{rank}(\boldsymbol{A}) = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02366		—	$\boldsymbol{V}_1 = \boldsymbol{V}$	$\boldsymbol{V}_1 = \boldsymbol{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02367		—	$\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$	$\boldsymbol{V}_1 \boldsymbol{V}_1^T = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02368		—	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y} = \boldsymbol{z}$	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y} = \boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02369		—	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	$\boldsymbol{y} \in \tilde{\mathcal{E}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02370		—	$\boldsymbol{x} = \boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$	$\boldsymbol{x} = \boldsymbol{A} \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z} = \boldsymbol{U}_1 \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02371		—	$\mathcal{E} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$	$\mathcal{E} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02372		—	$1 = \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2}$	$1 = \frac{y_1^2}{\sigma_1^2} + \cdots + \frac{y_n^2}{\sigma_n^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02373		—	σ_j	σ_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02374		—	$\boldsymbol{e}_j$	$\boldsymbol{e}_j$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02375		—	$\boldsymbol{U}_{\{1\}} \boldsymbol{y} \rightarrow \boldsymbol{x}$	$\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02376		—	$\mathcal{E}=\boldsymbol{A} \mathcal{S}$	$\mathcal{E} = \boldsymbol{A} \mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02377		—	$\boldsymbol{u}_{\{j\}}=\boldsymbol{U} \boldsymbol{e}_{\{j\}}$	$\boldsymbol{u}_j = \boldsymbol{U} \boldsymbol{e}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02378		—	$\sigma_{\{j\}}, j=1, \ldots, n$	$\sigma_j, j = 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02379		—	$\boldsymbol{A} \boldsymbol{v}_{\{j\}}=\sigma_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{A} \boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02380		—	$\mathcal{E}$	$\mathcal{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02381		—	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02382		—	$\sigma_{\{1\}}=3, \sigma_{\{2\}}=1, \boldsymbol{u}_{\{1\}}=[3,4]^T / 5$	$\sigma_1 = 3, \sigma_2 = 1, \boldsymbol{u}_1 = [3, 4]^T / 5$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02383		—	$\boldsymbol{u}_{\{2\}}=[-4,3]^T / 5$	$\boldsymbol{u}_2 = [-4, 3]^T / 5$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02384		—	$y_{\{1\}}^2 / \sigma_{\{1\}}^2+y_{\{2\}}^2 / \sigma_{\{2\}}^2=1$	$y_1^2 / \sigma_1^2 + y_2^2 / \sigma_2^2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02385		—	$\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_{\{1\}} \tilde{\mathcal{E}}$	$\mathcal{E} = \boldsymbol{A} \mathcal{S} = \boldsymbol{U}_1 \tilde{\mathcal{E}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02386		—	$\boldsymbol{A} \in \mathcal{S}$	$\mathcal{AS}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02387		—	$\boldsymbol{y}=\boldsymbol{U}_{\{1\}}^T \boldsymbol{x}=\left[3 / 5 x_{\{1\}}+4 / 5 x_{\{2\}},-4 / 5 x_{\{1\}}+3 / 5 x_{\{2\}}\right]^T$	$\boldsymbol{y}=\boldsymbol{U}_1^T \boldsymbol{x}=\left[3 / 5 x_1+4 / 5 x_2,-4 / 5 x_1+3 / 5 x_2\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02388		—	$\left\ \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}\right\ _F$	$\left\ \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}\right\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02389		—	$\left\ \boldsymbol{A}\right\ _F^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{\{j\}}\right\ _2^2$	$\left\ \boldsymbol{A}\right\ _F^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02390		—	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02391		—	$\boldsymbol{V} \in$	$\boldsymbol{V} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02392		—	$\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$	$\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02393		—	$\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$	$\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02394		—	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{x}\right\ _2$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{x}\right\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02395		—	$\left\ \boldsymbol{A}^*\right\ _F^2=\sum_{j=1}^n \sum_{i=1}^m\left \bar{a}_{i j}\right ^2=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2=\left\ \boldsymbol{A}\right\ _F^2$	$\left\ \boldsymbol{A}^*\right\ _F^2=\sum_{j=1}^n \sum_{i=1}^m\left \bar{a}_{i j}\right ^2=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2=\left\ \boldsymbol{A}\right\ _F^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02396		—	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02397		—	$\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{m \, n}$	$\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02398		—	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$	$\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02399		—	$\ \boldsymbol{U} \, \boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$	$\ \boldsymbol{U} \boldsymbol{x}\ _2 = \ \boldsymbol{x}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02400		—	$\ \boldsymbol{U} \, \boldsymbol{A}\ _F^2 \stackrel{2.}{=} \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 \stackrel{2.}{=} \ \boldsymbol{A}\ _F^2$	$\ \boldsymbol{U} \boldsymbol{A}\ _F^2 \stackrel{2.}{=} \sum_{j=1}^n \ \boldsymbol{U} \boldsymbol{a}_{:j}\ _2^2 = \sum_{j=1}^n \ \boldsymbol{a}_{:j}\ _2^2 \stackrel{2.}{=} \ \boldsymbol{A}\ _F^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02401		—	$\boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{I}$	$\boldsymbol{V} \boldsymbol{V}^* = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02402		—	$\ \boldsymbol{A} \boldsymbol{V}\ _F \stackrel{1.}{=} \ \boldsymbol{V}^* \boldsymbol{A}^*\ _F = \ \boldsymbol{A}^*\ _F \stackrel{1.}{=} \ \boldsymbol{A}\ _F$	$\ \boldsymbol{A} \boldsymbol{V}\ _F \stackrel{1.}{=} \ \boldsymbol{V}^* \boldsymbol{A}^*\ _F = \ \boldsymbol{A}^*\ _F \stackrel{1.}{=} \ \boldsymbol{A}\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02403		—	$\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$	$\ \boldsymbol{v}\ _F = \ \boldsymbol{v}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02404		—	$\boldsymbol{B} = \boldsymbol{x}$	$\boldsymbol{B} = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02405		—	$\ \boldsymbol{A}\ _F =$	$\ \boldsymbol{A}\ _F =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02406		—	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$	$\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02407		—	$\sigma_1, \ldots, \sigma_n$	$\sigma_1, \ldots, \sigma_n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02408		—	$m \geq n \geq 1$	$m \geq n \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02409		—	$\boldsymbol{U} \left[\begin{array}{c} \boldsymbol{D} \\ \boldsymbol{0} \end{array} \right] \boldsymbol{V}^*$	$\boldsymbol{U} \left[\begin{array}{c} \boldsymbol{D} \\ \boldsymbol{0} \end{array} \right] \boldsymbol{V}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02410		—	$\boldsymbol{D} = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$	$\boldsymbol{D} = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02411		—	$\epsilon > 0$	$\epsilon > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02412		—	$1 \leq r \leq n$	$1 \leq r \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02413		—	$\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$	$\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02414		—	$\boldsymbol{A}^{\prime} := \boldsymbol{U} \left[\begin{array}{c} \boldsymbol{D}^{\prime} \\ \boldsymbol{0} \end{array} \right] \boldsymbol{V}^*$	$\boldsymbol{A}^{\prime} := \boldsymbol{U} \left[\begin{array}{c} \boldsymbol{D}^{\prime} \\ \boldsymbol{0} \end{array} \right] \boldsymbol{V}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02415		—	$\boldsymbol{D}^{\prime} := \operatorname{diag}(\sigma_1, \ldots, \sigma_r, 0, \ldots, 0 \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}^{\prime} := \operatorname{diag}(\sigma_1, \ldots, \sigma_r, 0, \ldots, 0) \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02416		—	$\boldsymbol{A}^{\prime}$	$\boldsymbol{A}^{\prime}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02417		—	$\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$	$\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02418		—	$\operatorname{rank}(\boldsymbol{A}) = r$	$\operatorname{rank}(\boldsymbol{A}) = r$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02419		—	$r, \boldsymbol{A}^{\prime}$	$r, \boldsymbol{A}'$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02420		—	$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$	$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02421		—	$r \leq \operatorname{rank}(\boldsymbol{A})$	$r \leq \operatorname{rank}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02422		—	$\boldsymbol{A}=\left[\begin{array}{l}3\\4\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02423		—	$\boldsymbol{A}=\left[\begin{array}{ll}1&1\\2&2\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02424		—	$\boldsymbol{A}=\boldsymbol{e}_{1}$	$\boldsymbol{A} = \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02425		—	$\boldsymbol{A}=\boldsymbol{e}_{n}^T$	$\boldsymbol{A} = \boldsymbol{e}_n^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02426		—	$\boldsymbol{A}=\left[\begin{array}{rr}-1&0\\0&3\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02427		—	$\boldsymbol{U} \ \boldsymbol{\Sigma} \ \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02428		—	$\boldsymbol{A}^* \ \boldsymbol{A}=\boldsymbol{V} \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2) \boldsymbol{V}^*$	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{V} \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2) \boldsymbol{V}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02429		—	$\boldsymbol{A} \ \boldsymbol{A}^*=\boldsymbol{U} \operatorname{diag}(\sigma_1^2, \ldots, \sigma_m^2) \boldsymbol{U}^*$	$\boldsymbol{A} \boldsymbol{A}^* = \boldsymbol{U} \operatorname{diag}(\sigma_1^2, \ldots, \sigma_m^2) \boldsymbol{U}^*$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02430		—	$\left(\sigma_1, \ldots, \sigma_n\right)$	$(\sigma_1, \ldots, \sigma_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02431		—	$\boldsymbol{A}^2$	$\boldsymbol{A}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02432		—	$\left(\sigma_1^2, \ldots, \sigma_n^2\right)$	$(\sigma_1^2, \ldots, \sigma_n^2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02433		—	$\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}11 & 48 \\ 48 & 39\end{array}\right]$	$\boldsymbol{A}=\frac{1}{25}\left[\begin{array}{ll}11 & 48 \\ 48 & 39\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02434		—	$\boldsymbol{U} \ \boldsymbol{D} \ \boldsymbol{U}^T$	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02435		—	$\boldsymbol{A}=\frac{1}{5}\left[\begin{array}{cc}3 & -4 \\ 4 & 3\end{array}\right]\left[\begin{array}{cc}3 & 0 \\ 0 & 1\end{array}\right]\frac{1}{5}\left[\begin{array}{cc}3 & 4 \\ 4 & -3\end{array}\right]$	$\boldsymbol{A}=\frac{1}{5}\left[\begin{array}{cc}3 & -4 \\ 4 & 3\end{array}\right]\left[\begin{array}{cc}3 & 0 \\ 0 & 1\end{array}\right]\frac{1}{5}\left[\begin{array}{cc}3 & 4 \\ 4 & -3\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02436		—	$\operatorname{rank}(\boldsymbol{A})=\operatorname{rank}\left(\boldsymbol{A}^*\right)$	$\operatorname{rank}(\boldsymbol{A})=\operatorname{rank}\left(\boldsymbol{A}^*\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02437		—	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null}(\boldsymbol{A})=n$	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null}(\boldsymbol{A})=n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02438		—	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null}\left(\boldsymbol{A}^*\right)=m$	$\operatorname{rank}(\boldsymbol{A})+\operatorname{null}\left(\boldsymbol{A}^*\right)=m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02439		—	$\operatorname{null}(\boldsymbol{A})$	$\operatorname{null}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02440		—	$\operatorname{rank} \ \boldsymbol{A}=\operatorname{rank}\left(\boldsymbol{A}^* \ \boldsymbol{A} \right) \operatorname{rank}(\boldsymbol{A} \ \boldsymbol{A}^*)=\operatorname{rank}\left(\boldsymbol{A} \ \boldsymbol{A}^*\right)$	$\operatorname{rank} \boldsymbol{A}=\operatorname{rank}\left(\boldsymbol{A}^* \boldsymbol{A}\right)=\operatorname{rank}\left(\boldsymbol{A} \boldsymbol{A}^*\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02441		—	$\operatorname{null}\left(\boldsymbol{A}^* \boldsymbol{A}\right)=\operatorname{null}\boldsymbol{A}$	$\operatorname{null}\left(\boldsymbol{A}^* \boldsymbol{A}\right)=\operatorname{null}\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02442		—	$\operatorname{null}\left(\boldsymbol{A} \boldsymbol{A}^*\right)=\operatorname{null}\left(\boldsymbol{A}^*\right)$	$\operatorname{null}\left(\boldsymbol{A} \boldsymbol{A}^*\right)=\operatorname{null}\left(\boldsymbol{A}^*\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02443		—	$\mathcal{R}\left(\boldsymbol{B}\right)$	$\mathcal{R}\left(\boldsymbol{B}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02444		—	$\mathcal{N}\left(\boldsymbol{B}\right)$	$\mathcal{N}\left(\boldsymbol{B}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02445		—	$\mathcal{R}\left(\boldsymbol{A}^* \boldsymbol{A}\right)=$	$\mathcal{R}\left(\boldsymbol{A}^* \boldsymbol{A}\right)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02446		—	$\mathcal{R}\left(\boldsymbol{V}_1\right)=\mathcal{R}\left(\boldsymbol{A}^*\right)$	$\mathcal{R}\left(\boldsymbol{V}_1\right)=\mathcal{R}\left(\boldsymbol{A}^*\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02447		—	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m \in \mathbb{C}^m$	$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02448		—	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{C}^n$	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02449		—	$\boldsymbol{B}:=$	$\boldsymbol{B}:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02450		—	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02451		—	$\boldsymbol{P}=\boldsymbol{B}$	$\boldsymbol{P}=\boldsymbol{B}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02452		—	$\boldsymbol{X}_k=\boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$	$\boldsymbol{X}_k = \boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02453		—	Σ_k	Σ_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02454		—	$k=0,1,2, \ldots$	$k = 0, 1, 2, \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02455		—	$[\mathrm{Q}, \mathrm{P}, \mathrm{k}] =$	$[Q, P, k] =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02456		—	$(\mathrm{A}, \mathrm{tol}, \mathrm{K})$	(A, tol, K)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02457		—	$\boldsymbol{P}=\boldsymbol{Q}^T \boldsymbol{A}$	$\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02458		—	$\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02459		—	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{X}_k\right\ _F<$	$\left\ \boldsymbol{X}_{k+1}-\boldsymbol{X}_k\right\ _F <$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02460		—	$*\left\ \boldsymbol{X}_{k+1}\right\ _F$	$*\left\ \boldsymbol{X}_{k+1}\right\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02461		—	$k=K+1$	$k = K + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02462		—	$\left\ \boldsymbol{A}\right\ _1$	$\left\ \boldsymbol{A}\right\ _1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02463		—	$\ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02464		—	$\left(\boldsymbol{B}^T\right)$	$\left(\boldsymbol{B}^T\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02465		—	$\operatorname{ker}(\boldsymbol{B})$	$\ker(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02466		—	$\boldsymbol{x} \in \mathbb{R}^3$	$\boldsymbol{x} \in \mathbb{R}^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02467		—	$\boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{b} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02468		—	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02469		—	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^T\right), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}\left(\boldsymbol{A}^T\right), \mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02470		—	$\mathcal{N}\left(\boldsymbol{A}^T\right)$	$\mathcal{N}\left(\boldsymbol{A}^T\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02471		—	$\boldsymbol{B} \in \mathbb{R}^{4,3}$	$\boldsymbol{B} \in \mathbb{R}^{4,3}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02472		—	$\ \boldsymbol{A}-\boldsymbol{B}\ _F \geq 6$	$\ \boldsymbol{A}-\boldsymbol{B}\ _F \geq 6$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02473		—	$\left\ \boldsymbol{A}-\boldsymbol{A}_1\right\ _F=6$	$\ \boldsymbol{A}-\boldsymbol{A}_1\ _F=6$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02474		—	$(n, \ 1)$	$(n, 1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02475		—	$-2^{\{2-n\}}$	-2^{2-n}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02476		—	$\boldsymbol{x}:=\left[2^{\{n-2\}},\,2^{\{n-3\}},\,\ldots,\,2^{\{0\}},\,1\right]^T$	$\boldsymbol{x}:=\left[2^{n-2},2^{n-3},\ldots,2^0,1\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02477		—	$\operatorname{det}(\boldsymbol{A})=1$	$\det(\boldsymbol{A})=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02478		—	$\ \boldsymbol{A}-\boldsymbol{B}\ _F=2^{\{2-n\}}$	$\ \boldsymbol{A}-\boldsymbol{B}\ _F=2^{2-n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02479		—	$\boldsymbol{A}-\boldsymbol{B}$	$\boldsymbol{A}-\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02480		—	$\sigma_{\{n\}}$	σ_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02481		—	$2^{\{2-n\}}$	2^{2-n}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02482		—	$\ \boldsymbol{A}\ _1,\ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _1,\ \boldsymbol{A}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02483		—	$\ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02484		—	$\boldsymbol{T}=\boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{T}=\boldsymbol{L}\boldsymbol{L}^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02485		—	$\boldsymbol{A}^{\prime}=\sum_{i=1}^r \sigma_{i} \boldsymbol{u}_{i} \boldsymbol{v}_{i}^{\ast}$	$\boldsymbol{A}^{\prime}=\sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02486		—	$1 \leq r \leq n, \sigma_{i}$	$1 \leq r \leq n, \sigma_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02487		—	$\boldsymbol{u}_{i}, \boldsymbol{v}_{i}$	$\boldsymbol{u}_i, \boldsymbol{v}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02488		—	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02489		—	$(\mathcal{V}, \mathbb{R},\ \cdot\)$	$(\mathcal{V}, \mathbb{R}, \ \cdot\)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02490		—	$(\mathcal{V}, \mathbb{C},\ \cdot\)$	$(\mathcal{V}, \mathbb{C}, \ \cdot\)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02491		—	$\mathbb{R}^n, \mathbb{C}^n$	$\mathbb{R}^n, \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02492		—	$\mathcal{V}=\mathbb{C}^n$	$\mathcal{V}=\mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02493		—	$\mathcal{V}=\mathbb{R}^n$	$\mathcal{V}=\mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02494		—	$p \geq 1$	$p \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02495		—	$p=1,2, \infty$	$p=1,2, \infty$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02496		—	$\ \boldsymbol{x}\ _1:=\sum_{j=1}^n\left x_j\right $	$\ \boldsymbol{x}\ _1:=\sum_{j=1}^n x_j $
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02497		—	$\ \boldsymbol{x}\ _2:=\sqrt{\sum_{j=1}^n\left x_j\right ^2}$	$\ \boldsymbol{x}\ _2:=\sqrt{\sum_{j=1}^n x_j ^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02498		—	l_2	l_2
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02499		—	$\ \boldsymbol{x}\ _{\infty}:=\max_{1\leq j\leq n}\left x_j\right $	$\ \boldsymbol{x}\ _{\infty}:=\max_{1\leq j\leq n} x_j $
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02500		—	l_{∞}	l_{∞}
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02501		—	$1\leq p\leq\infty$	$1\leq p\leq\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02502		—	$\ \boldsymbol{x}+\boldsymbol{y}\ _p\leq\ \boldsymbol{x}\ _p+\ \boldsymbol{y}\ _p$	$\ \boldsymbol{x}+\boldsymbol{y}\ _p\leq\ \boldsymbol{x}\ _p+\ \boldsymbol{y}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02503		—	$\frac{1}{p}+\frac{1}{q}=1$	$\frac{1}{p}+\frac{1}{q}=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02504		—	$p=1$	$p=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02505		—	$q=\infty$	$q=\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02506		—	$p=2$	$p=2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02507		—	$\lim_{p \rightarrow \infty} n^{1/p} = 1$	$\lim_{p \rightarrow \infty} n^{1/p} = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02508		—	$\ \cdot\ $	$\ \cdot\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02509		—	$\ \cdot\ '$	$\ \cdot\ '$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02510		—	M	M
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02511		—	$\boldsymbol{x} \in \mathcal{V}$	$\boldsymbol{x} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02512		—	∞	∞
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02513		—	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02514		—	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n \quad$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02515		—	$\mathcal{V} \rightarrow \mathbb{R}$	$\mathcal{V} \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02516		—	$\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{x} - \boldsymbol{y} + \boldsymbol{y}\ \leq \ \boldsymbol{x} - \boldsymbol{y}\ + \ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02517		—	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $	$\ \boldsymbol{x} - \boldsymbol{y}\ \geq \ \boldsymbol{x}\ - \ \boldsymbol{y}\ $
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02518		—	$\ \boldsymbol{y}-\boldsymbol{x}\ =\ \boldsymbol{y}-\boldsymbol{x}\ \geq\ \boldsymbol{y}\ -\ \boldsymbol{x}\ $	$\ x-y\ =\ y-x\ \geq\ y\ -\ x\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02519		—	$f:\mathcal{S}\rightarrow\mathbb{R}$	$f:\mathcal{S}\rightarrow\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02520		—	$f(\boldsymbol{y})=\ \boldsymbol{y}\ $	$f(\boldsymbol{y})=\ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02521		—	$\boldsymbol{y}:=\boldsymbol{x}/\ \boldsymbol{x}\ ^{\prime}\in\mathcal{S}$	$\boldsymbol{y}:=\boldsymbol{x}/\ \boldsymbol{x}\ ^{\prime}\in\mathcal{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02522		—	$\left(\mathbb{C}^m\times n,\mathbb{C}\right)$	$\left(\mathbb{C}^{m\times n},\mathbb{C}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02523		—	$\left(\mathbb{R}^m\times n,\mathbb{R}\right)$	$\left(\mathbb{R}^{m\times n},\mathbb{R}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02524		—	$\left \right :\mathbb{C}^m\times n,\rightarrow\mathbb{R}.$	$\left \right :\mathbb{C}^{m\times n},\rightarrow\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02525		—	$m\ n$	mn
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02526		—	$\mathbb{C}^m\ n$	$\mathbb{C}^{mn}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02527		—	$\ \cdot\ _V$	$\ \cdot\ _V$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02528		—	$\ \boldsymbol{A}\ :=$	$\ \boldsymbol{A}\ :=$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02529		—	$\ \operatorname{vec}(\boldsymbol{A})\ _V$	$\ \operatorname{vec}(\boldsymbol{A})\ _V$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02530		—	$\ \boldsymbol{A} \boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ $	$\ \boldsymbol{A}\boldsymbol{B}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{B}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02531		—	$\ \ $	$\ \ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02532		—	$\left\ \right\ _{\beta}.$	$\ \ _{\beta}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02533		—	$\ \ \ $	$\ \ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02534		—	$\ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _{\beta}$	$\ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _{\beta}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02535		—	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02536		—	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02537		—	$n \in \mathbb{N}.^1$	$n \in \mathbb{N}.^1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02538		—	$\left\ \right\ .$	$\ \ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02539		—	$\ \ : \mathcal{S} \rightarrow \mathbb{R}$	$\ \ : \mathcal{S} \rightarrow \mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02540		—	$f(\ \boldsymbol{x}\)=\ \boldsymbol{A}\ \boldsymbol{x}\ $	$f(x)=\ \boldsymbol{A}x\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02541		—	$\ \ \ _{\beta}$	$\ \ _{\beta}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02542		—	$\ \boldsymbol{A}\ :=\max_{\boldsymbol{x}\neq 0}\frac{\ \boldsymbol{A}\ \boldsymbol{x}\ }{\ \boldsymbol{x}\ _{\beta}}$	$\ \boldsymbol{A}\ :=\max_{x\neq 0}\frac{\ \boldsymbol{A}x\ }{\ x\ _{\beta}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02543		—	$\ \boldsymbol{I}\ =1$	$\ \boldsymbol{I}\ =1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02544		—	$\ \boldsymbol{A}\ =0$	$\ \boldsymbol{A}\ =0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02545		—	$\ \boldsymbol{A}\ \boldsymbol{y}\ =0$	$\ \boldsymbol{A}y\ =0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02546		—	$\boldsymbol{A}=0$	$\boldsymbol{A}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02547		—	$\ c\ \boldsymbol{A}\ =\max_{\boldsymbol{x}}\ c\ \boldsymbol{A}\ \boldsymbol{x}\ =\max_{\boldsymbol{x}} c \ \boldsymbol{A}\ \boldsymbol{x}\ = c \ \boldsymbol{A}\ $	$\ c\boldsymbol{A}\ =\max_x\ c\boldsymbol{A}x\ =\max_x c \ \boldsymbol{A}x\ = c \ \boldsymbol{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02548		—	$\ \boldsymbol{A}+\boldsymbol{B}\ =\max_{\boldsymbol{x}}\ (\boldsymbol{A}+\boldsymbol{B})\ \boldsymbol{x}\ \leq\max_{\boldsymbol{x}}\ \boldsymbol{A}\ \boldsymbol{x}\ +\max_{\boldsymbol{x}}\ \boldsymbol{B}\ \boldsymbol{x}\ =\ \boldsymbol{A}\ +\ \boldsymbol{B}\ $	$\ \boldsymbol{A}+\boldsymbol{B}\ =\max_x\ (\boldsymbol{A}+\boldsymbol{B})x\ \leq\max_x\ \boldsymbol{A}x\ +\max_x\ \boldsymbol{B}x\ =\ \boldsymbol{A}\ +\ \boldsymbol{B}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02549		—	$\ \boldsymbol{I}\ _F=\sqrt{n}$	$\ \boldsymbol{I}\ _F=\sqrt{n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02550		—	$n>1$	$n>1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02551		—	$\boldsymbol{p}$	p
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02552		—	ℓ_p	ℓ_p
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02553		—	$\left \right _p$	$\left \right _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02554		—	$\left \boldsymbol{A} \right _2$	$\left \boldsymbol{A} \right _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02555		—	K_p	K_p
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02556		—	$\left \boldsymbol{A} \boldsymbol{x} \right _p \leq K_p$	$\left \boldsymbol{A} \boldsymbol{x} \right _p \leq K_p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02557		—	$\left \boldsymbol{x} \right _p = 1$	$\left \boldsymbol{x} \right _p = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02558		—	$\boldsymbol{y}_e \in \mathbb{C}^n$	$\boldsymbol{y}_e \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02559		—	$\left \boldsymbol{y}_e \right _p = 1$	$\left \boldsymbol{y}_e \right _p = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02560		—	$\left \boldsymbol{A} \boldsymbol{y}_e \right _p = K_p$	$\left \boldsymbol{A} \boldsymbol{y}_e \right _p = K_p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02561		—	$\left \boldsymbol{A} \right _p = \left \boldsymbol{A} \boldsymbol{y}_e \right _p = K_p$	$\left \boldsymbol{A} \right _p = \left \boldsymbol{A} \boldsymbol{y}_e \right _p = K_p$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02562		—	$K_{\{2\}}=$	$\bar K_2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02563		—	$\sigma_{\{1\}}, \boldsymbol{c}:=\boldsymbol{V}^*\boldsymbol{x}$	$\sigma_1, c := V^*x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02564		—	$\boldsymbol{y}_{\{e\}}=\boldsymbol{v}_{\{1\}}$	$y_e = v_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02565		—	$\sigma_{\{1\}}$	σ_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02566		—	$\boldsymbol{V}\boldsymbol{c}, \ \boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$	$Vc, \ c\ _2 = \ x\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02567		—	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*\boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$	$\ Ax\ _2^2 = \ U\Sigma V^*x\ _2^2 = \ \Sigma c\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02568		—	$\left\ \boldsymbol{A}\boldsymbol{v}_1\right\ _2 = \left\ \sigma_1\boldsymbol{u}_1\right\ _2 = \sigma_1$	$\ Av_1\ _2 = \ \sigma_1 u_1\ _2 = \sigma_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02569		—	$K_{\{1\}}, c$	K_1, c
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02570		—	$\boldsymbol{y}_{\{e\}}$	y_e
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02571		—	$K_{\{1\}}:=\left\ \boldsymbol{A}\boldsymbol{e}_{\{c\}}\right\ _1=\max _{1 \leq j \leq n}\left\ \boldsymbol{A}\boldsymbol{e}_{\{j\}}\right\ _1$	$K_1 := \ Ae_c\ _1 = \max_{1 \leq j \leq n} \ Ae_j\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02572		—	$\boldsymbol{y}_{\{e\}}:=$	$y_e :=$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02573		—	$\boldsymbol{e}_c$	$\boldsymbol{e}_c$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02574		—	$\ \boldsymbol{y}_e\ _1=1$	$\ \boldsymbol{y}_e\ _1=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02575		—	$\ \boldsymbol{A} \boldsymbol{y}_e\ _1=K_1$	$\ \boldsymbol{A}\boldsymbol{y}_e\ _1=K_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02576		—	K_{∞}, r	K_{∞}, r
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02577		—	$K_{\infty}:=\left\ \boldsymbol{e}_r^T\boldsymbol{A}\right\ _1=\max_{1\leq k\leq m}\left\ \boldsymbol{e}_k^T\boldsymbol{A}\right\ _1$	$K_{\infty}:=\left\ \boldsymbol{e}_r^T\boldsymbol{A}\right\ _1=\max_{1\leq k\leq m}\left\ \boldsymbol{e}_k^T\boldsymbol{A}\right\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02578		—	$\boldsymbol{y}_e:=\left[e^{-i\theta_1},\ldots,e^{-i\theta_n}\right]^T$	$\boldsymbol{y}_e:=\left[e^{-i\theta_1},\ldots,e^{-i\theta_n}\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02579		—	$a_{rj}=\left a_{rj}\right e^{i\theta_j}$	$a_{rj}=\left a_{rj}\right e^{i\theta_j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02580		—	$\left\ \boldsymbol{A} \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right =K_{\infty}$	$\left\ \boldsymbol{A}\boldsymbol{y}^*\right\ _{\infty}=\max_{1\leq k\leq m}\left \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right =K_{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02581		—	$\left \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right \leq\sum_{j=1}^n\left a_{kj}\right \leq K_{\infty}$	$\left \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right \leq\sum_{j=1}^n\left a_{kj}\right \leq K_{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02582		—	$k=r$	$k=r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02583		—	$\boldsymbol{A}:=\frac{1}{15}\left[\begin{array}{ccc}14&4&16\\2&22&13\end{array}\right]$	$\boldsymbol{A}:=\frac{1}{15}\left[\begin{array}{ccc}14&4&16\\2&22&13\end{array}\right]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02584		—	$\sigma_1 \geq \sigma_2 \geq$	$\sigma_1 \geq \sigma_2 \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02585		—	$\cdots \geq \sigma_n$	$\cdots \geq \sigma_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02586		—	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02587		—	$1 / \sigma_n$	$1/\sigma_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02588		—	$\ \boldsymbol{A}\ _2^2 \leq$	$\ \boldsymbol{A}\ _2^2 \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02589		—	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02590		—	$(\sigma^2, \boldsymbol{v})$	$(\sigma^2, \boldsymbol{v})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02591		—	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02592		—	$\ \boldsymbol{v}\ _1$	$\ \boldsymbol{v}\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02593		—	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02594		—	$\boldsymbol{U}(\boldsymbol{A}+$	$\boldsymbol{U}(\boldsymbol{A}+$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02595		—	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$	$\boldsymbol{E})\boldsymbol{V} = \boldsymbol{U}\boldsymbol{A}\boldsymbol{V} + \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02596		—	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02597		—	$\left\ \boldsymbol{A}^*\right\ _2 = \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02598		—	$\boldsymbol{V}^*$	$\boldsymbol{V}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02599		—	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02600		—	$ \boldsymbol{x} := [\left x_1\right , \ldots, \left x_n\right]^T$	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02601		—	$x_1 = x_2 = 10$	$x_1 = x_2 = 10$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02602		—	$x_1 = 30, x_2 = -10$	$x_1 = 30, x_2 = -10$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02603		—	$1 \cdot 10^{-16}$	$1 - 10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02604		—	$1 + 10^{-16}$	$1 + 10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02605		—	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}, \boldsymbol{b}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02606		—	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02607		—	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02608		—	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02609		—	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02610		—	$\boldsymbol{y} - \boldsymbol{x}$	$\boldsymbol{y} - \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02611		—	$\ \boldsymbol{y} - \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02612		—	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02613		—	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02614		—	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02615		—	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A} \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A} \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02616		—	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02617		—	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02618		—	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02619		—	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1} \boldsymbol{e}$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1} \boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02620		—	$ \boldsymbol{y} - \boldsymbol{x} = \left \boldsymbol{A}^{-1} \boldsymbol{e} \right \leq \left \boldsymbol{A}^{-1} \right \left \boldsymbol{e} \right $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1} \boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02621		—	$ \boldsymbol{b} = \boldsymbol{A} \boldsymbol{x} \leq \boldsymbol{A} \boldsymbol{x} $	$\ \boldsymbol{b}\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02622		—	$ \boldsymbol{e} \leq \boldsymbol{A} \boldsymbol{y} - \boldsymbol{x} $	$\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02623		—	$ \boldsymbol{x} \leq$	$\ \boldsymbol{x}\ \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02624		—	$\left \boldsymbol{A}^{-1} \right \boldsymbol{b} $	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02625		—	$ \boldsymbol{e} / \boldsymbol{b} $	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02626		—	$\boldsymbol{e}$	$\boldsymbol{e}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02627		—	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02628		—	$K(\boldsymbol{A}) \geq 1$	$K(\boldsymbol{A}) \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02629		—	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02630		—	$\ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{I}\ \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02631		—	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02632		—	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02633		—	ℓ_1, ℓ_∞	ℓ_1, ℓ_∞
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02634		—	$\boldsymbol{Ax} = \boldsymbol{b}$	$\boldsymbol{Ax} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02635		—	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{Ay} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{Ay} - \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02636		—	$\boldsymbol{r}$	$\boldsymbol{r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02637		—	$\boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02638		—	$\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{Ay} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{Ay} - \boldsymbol{b}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02639		—	$\boldsymbol{e}=\boldsymbol{r}(\boldsymbol{y})$	$\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02640		—	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02641		—	$\boldsymbol{A}, \boldsymbol{A}+\boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02642		—	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{y}=\boldsymbol{b}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02643		—	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02644		—	$r:=\left \boldsymbol{A}^{-1} \boldsymbol{E}\right <1$	$r := \left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02645		—	$\boldsymbol{A}+\boldsymbol{E}$	$\boldsymbol{A} + \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02646		—	$r \geq 1$	$r \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02647		—	$(\boldsymbol{A}+\boldsymbol{E}) \boldsymbol{x}=\mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02648		—	$\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \boldsymbol{E}\right) \boldsymbol{x}=\mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1} \boldsymbol{E}) \boldsymbol{x} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02649		—	$\left \boldsymbol{x}\right =\left \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{x}\right \leq r\left \boldsymbol{x}\right $	$\ \boldsymbol{x}\ = \left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{x}\right\ \leq r\ \boldsymbol{x}\ $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02650		—	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{E} \ \boldsymbol{y}$	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{E}\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02651		—	$\boldsymbol{x}-\boldsymbol{y}=\boldsymbol{A}^{-1} \ \boldsymbol{E} \ \boldsymbol{y}$	$\boldsymbol{x}-\boldsymbol{y}=\boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02652		—	$ \boldsymbol{y} $	$\ \boldsymbol{y}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02653		—	$\boldsymbol{y}=\left(\boldsymbol{I}+\boldsymbol{A}^{-1} \ \boldsymbol{E}\right)^{-1} \ \boldsymbol{x}$	$\boldsymbol{y}=\left(\boldsymbol{I}+\boldsymbol{A}^{-1}\boldsymbol{E}\right)^{-1}\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02654		—	$ \boldsymbol{y} \ \leq \left \left(\boldsymbol{I}+\boldsymbol{A}^{-1} \ \boldsymbol{E}\right)^{-1} \right \ \frac{\ \boldsymbol{x}\ }{1-r}$	$\ \boldsymbol{y}\ \leq \left\ \left(\boldsymbol{I}+\boldsymbol{A}^{-1}\boldsymbol{E}\right)^{-1}\right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02655		—	$ \boldsymbol{y}-\boldsymbol{x} \ \leq \ r \boldsymbol{y} \ \leq$	$\ \boldsymbol{y}-\boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02656		—	$\frac{r}{1-r} \boldsymbol{x} \ \leq \ \frac{K(\boldsymbol{A})}{1-r} \boldsymbol{E} \ \boldsymbol{A} $	$\frac{r}{1-r}\ \boldsymbol{x}\ \leq \frac{K(\boldsymbol{A})}{1-r}\ \boldsymbol{E}\ \ \boldsymbol{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02657		—	$ \boldsymbol{x} $	$\ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02658		—	$\boldsymbol{x}.\boldsymbol{E}/\ \boldsymbol{A} $	$\boldsymbol{x}.\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02659		—	$ \boldsymbol{E} /\ \boldsymbol{A} $	$\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02660		—	$\left \boldsymbol{A}^{-1} \ \boldsymbol{E} \ \boldsymbol{y}\right \ \leq \left \boldsymbol{A}^{-1} \ \boldsymbol{E} \ \boldsymbol{y} $	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ \leq \ \boldsymbol{A}^{-1}\ \ \ \boldsymbol{E}\ \ \boldsymbol{y}\ $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02661		—	$\left \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right $	$\left\ \boldsymbol{A}^{-1} \boldsymbol{E} \boldsymbol{y}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02662		—	$\sigma_{\{1\}} \geq \sigma_{\{2\}} \geq \cdots \geq \sigma_{\{n\}} > 0$	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02663		—	$\left \lambda_{\{1\}}\right \geq\left \lambda_{\{2\}}\right \geq \cdots \geq$	$\left \lambda_1\right \geq\left \lambda_2\right \geq \cdots \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02664		—	$\left \lambda_{\{n\}}\right > 0$	$\left \lambda_n\right > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02665		—	$\sigma_{\{1\}} / \sigma_{\{n\}}$	σ_1 / σ_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02666		—	$\lambda_{\{1\}} / \lambda_{\{n\}}$	λ_1 / λ_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02667		—	$\left \boldsymbol{y}-\boldsymbol{x}\right /\left \boldsymbol{x}\right \wedge\left \boldsymbol{x}\right \approx\left \boldsymbol{r}(\boldsymbol{y})\right /\left \boldsymbol{b}\right $	$\left\ \boldsymbol{y}-\boldsymbol{x}\right\ /\left\ \boldsymbol{x}\right\ \approx\left\ \boldsymbol{r}(\boldsymbol{y})\right\ /\left\ \boldsymbol{b}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02668		—	$\left \boldsymbol{b}\right \approx 1$	$\left\ \boldsymbol{b}\right\ \approx 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02669		—	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E}$	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02670		—	$\left \boldsymbol{B}^{-1}\right $	$\left\ \boldsymbol{B}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02671		—	$\left \boldsymbol{A}^{-1}\right $	$\left\ \boldsymbol{A}^{-1}\right\ $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02672		—	$\left \left \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right \right $	$\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02673		—	$\left \left \boldsymbol{B}^{-1}\right \right /\left \left \boldsymbol{A}^{-1}\right \right ,\left \left \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right \right /\left \left \boldsymbol{B}^{-1}\right \right $	$\left\ \boldsymbol{B}^{-1}\right\ /\left\ \boldsymbol{A}^{-1}\right\ ,\left\ \boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right\ /\left\ \boldsymbol{B}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02674		—	$\left \left \boldsymbol{B}^{-1}-\right.\right.$	$\left\ \boldsymbol{B}^{-1}-\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02675		—	$\boldsymbol{A}^{-1}\left \right /\left \left \boldsymbol{A}^{-1}\right \right $	$\boldsymbol{A}^{-1}\left\ \right /\left\ \boldsymbol{A}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02676		—	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{B}:=\boldsymbol{A}+\boldsymbol{E} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02677		—	$K(\boldsymbol{A}):=\left \left \boldsymbol{A}\right \right \left \left \boldsymbol{A}^{-1}\right \right $	$K(\boldsymbol{A}):=\left\ \boldsymbol{A}\right\ \left\ \boldsymbol{A}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02678		—	$\left \left \boldsymbol{A}^{-1} \boldsymbol{E}\right \right $	$\left\ \boldsymbol{A}^{-1} \boldsymbol{E}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02679		—	$\left \left \boldsymbol{E} \boldsymbol{A}^{-1}\right \right $	$\left\ \boldsymbol{E} \boldsymbol{A}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02680		—	$r<1$	$r<1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02681		—	$-\boldsymbol{E} =$	$-\boldsymbol{E} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02682		—	$\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{B}=\boldsymbol{B}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{A}$	$\boldsymbol{A}-\boldsymbol{B}=\boldsymbol{A}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{B}=\boldsymbol{B}\left(\boldsymbol{B}^{-1}-\boldsymbol{A}^{-1}\right) \boldsymbol{A}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02683		—	$\left\ \boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{B}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ $	$\left\ \boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{B}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02684		—	$\left\ \boldsymbol{B}^{-1}\right\ /\left\ \boldsymbol{A}^{-1}\right\ $	$\left\ \boldsymbol{B}^{-1}\right\ /\left\ \boldsymbol{A}^{-1}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02685		—	$1 \leq p \leq \infty,\left\ \boldsymbol{x}\right\ ,\left\ \boldsymbol{y}\right\ \in \mathbb{C}^n$	$1 \leq p \leq \infty,\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02686		—	$\left\ \boldsymbol{x}\right\ _p \geq 0$	$\left\ \boldsymbol{x}\right\ _p \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02687		—	$\left\ \boldsymbol{a}\right\ _p=\left\ \boldsymbol{a}\right\ _p$	$\left\ \boldsymbol{a x}\right\ _p=\left\ \boldsymbol{a}\right\ _p\left\ \boldsymbol{x}\right\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02688		—	$I \subset \mathbb{R}$	$I \subset \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02689		—	$f: I \rightarrow \mathbb{R}$	$f: I \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02690		—	$x_1, x_2 \in I$	$x_1, x_2 \in I$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02691		—	$x_1<x_2$	$x_1<x_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02692		—	$\lambda \in[0,1]$	$\lambda \in[0,1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02693		—	$\sum_{j=1}^n \lambda_j x_j$	$\sum_{j=1}^n \lambda_j x_j$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02694		—	$x_{\{1\}}, \ldots, x_{\{n\}}$	$x_1, \ldots, x_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02695		—	$\lambda_{\{j\}} \geq 0$	$\lambda_j \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02696		—	$\sum_{j=1}^n \lambda_{\{j\}}=1$	$\sum_{j=1}^n \lambda_j = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02697		—	$f \in C^2[a, b]$	$f \in C^2[a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02698		—	$f^{\{\prime \prime\prime\}}(x) \geq$	$f'''(x) \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02699		—	$a \leq x_{\{1\}} \leq x \leq x_{\{2\}} \leq b$	$a \leq x_1 \leq x \leq x_2 \leq b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02700		—	$c \in \left[x_{\{1\}}, x_{\{2\}}\right]$	$c \in [x_1, x_2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02701		—	$f(x) \leq (1-\lambda) f\left(x_{\{1\}}\right)+\lambda f\left(x_{\{2\}}\right)$	$f(x) \leq (1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02702		—	$I \in \mathbb{R}$	$I \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02703		—	$z_{\{1\}}, \ldots, z_{\{n\}} \in I$	$z_1, \ldots, z_n \in I$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02704		—	$n \geq 2$	$n \geq 2$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02705		—	λ_j, z_j	λ_j, z_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02706		—	$\lambda_i < 1$	$\lambda_i < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02707		—	$\lambda_1 < 1$	$\lambda_1 < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02708		—	$u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$	$u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02709		—	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02710		—	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} f(z_j)$	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} f(z_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02711		—	$\sum_{j=1}^n \lambda_j a_j$	$\sum_{j=1}^n \lambda_j a_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02712		—	$a_1, \ldots, a_n$	$a_1, \ldots, a_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02713		—	$0^{\{0\}} := 0$	$0^0 := 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02714		—	a_j	a_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02715		—	$a_j > 0$	$a_j > 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02716		—	$f:(0, \infty) \rightarrow \mathbb{R}$	$f : (0, \infty) \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02717		—	$f(x)=-\log x$	$f(x) = -\log x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02718		—	$f^{\{\prime \prime\}}(x)=1 / x^{\{2\}}>0$	$f''(x) = 1/x^2 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02719		—	$x \in (0, \infty)$	$x \in (0, \infty)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02720		—	$\log \left(a_{\{1\}}^{\{\lambda_{\{1\}\}} \cdots a_{\{n\}}^{\{\lambda_{\{n\}\}}}\right) \leq \log \left(\sum_{j=1}^{\{n\}} \lambda_{\{j\}} a_{\{j\}}\right)$	$\log \left(a_1^{\lambda_1} \cdots a_n^{\lambda_n}\right) \leq \log \left(\sum_{j=1}^n \lambda_j a_j\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02721		—	$\exp \left(\log x\right)$	$\exp(\log x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02722		—	$=x$	$= x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02723		—	$x>0$	$x > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02724		—	$\lambda_{\{j\}}=\frac{1}{\{n\}}$	$\lambda_j = \frac{1}{n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02725		—	$p=\infty$	$p = \infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02726		—	$1<p<$	$1 < p <$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02727		—	$a, b \geq 0$	$a, b \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02728		—	$(p-1) \, q=p$	$(p-1)q=p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02729		—	$a= \pm 1$	$a = \pm 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02730		—	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ ^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02731		—	$\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{y} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02732		—	$\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$	$\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02733		—	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \right\rangle$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \right\rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02734		—	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02735		—	$2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$	$2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02736		—	$a=n$	$a = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02737		—	$a>0$	$a > 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02738		—	$\left\{a_n\right\}$	$\{a_n\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02739		—	$a\angle\boldsymbol{x}, \boldsymbol{y}\angle=\angle a \boldsymbol{x}, \boldsymbol{y}\angle$	$a\langle\boldsymbol{x},\boldsymbol{y}\rangle=\langle a\boldsymbol{x},\boldsymbol{y}\rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02740		—	$a<0$	$a < 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02741		—	$(-a)>0$	$(-a) > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02742		—	$\mathcal{V}=\mathbb{C}^n, 1\leq p\leq\infty, n\geq 2$	$\mathcal{V}=\mathbb{C}^n, 1\leq p\leq\infty, n\geq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02743		—	$\angle\boldsymbol{x}, \boldsymbol{x}\angle= \boldsymbol{x} _p^2$	$\langle\boldsymbol{x},\boldsymbol{x}\rangle= \boldsymbol{x} _p^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02744		—	$p\neq 2$	$p\neq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02745		—	$\boldsymbol{x}:=\boldsymbol{e}_1$	$\boldsymbol{x}:=\boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02746		—	$\boldsymbol{y}:=\boldsymbol{e}_2$	$\boldsymbol{y}:=\boldsymbol{e}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02747		—	$0<\lambda_n\leq\cdots\leq\lambda_1$	$0<\lambda_n\leq\cdots\leq\lambda_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02748		—	$\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n$	$\boldsymbol{b}_1,\ldots,\boldsymbol{b}_n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02749		—	$\boldsymbol{B}_{\{k\}} := \left[\boldsymbol{b}_{\{1\}}, \ldots, \boldsymbol{b}_{\{k\}} \right] \in \mathbb{R}^{n \times k}$	$\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02750		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02751		—	$\ \boldsymbol{x} \ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$	$\ \boldsymbol{x} \ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02752		—	$\tilde{\boldsymbol{b}}_{\{1\}} := \boldsymbol{b}_{\{1\}}$	$\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02753		—	$\boldsymbol{B}_{\{k\}}^T \boldsymbol{A} \boldsymbol{B}_{\{k\}}$	$\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02754		—	$\langle \tilde{\boldsymbol{b}}_{\{k\}}, \boldsymbol{b}_{\{j\}} \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02755		—	$j=1, \ldots, k-1$	$j = 1, \dots, k-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02756		—	$\tilde{\boldsymbol{b}}_{\{k\}} - \boldsymbol{b}_{\{k\}} \in \operatorname{span}(\boldsymbol{b}_{\{1\}}, \ldots, \boldsymbol{b}_{\{k-1\}})$	$\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \operatorname{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02757		—	$\tilde{\boldsymbol{b}}_{\{1\}}, \ldots, \tilde{\boldsymbol{b}}_{\{n\}}$	$\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02758		—	$\langle \tilde{\boldsymbol{b}}_{\{i\}}, \tilde{\boldsymbol{b}}_{\{j\}} \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02759		—	$i, j \leq n, i \neq j$	$i, j \leq n, i \neq j$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02760		—	$\tilde{\boldsymbol{B}}_{\{n\}}:=\left[\tilde{\boldsymbol{b}}_{\{1\}},\,\ldots,\,\tilde{\boldsymbol{b}}_{\{n\}}\right]$	$\tilde{\boldsymbol{B}}_n := \left[\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02761		—	$\boldsymbol{T} \in$	$\boldsymbol{T} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02762		—	$\boldsymbol{B}_{\{n\}}=\tilde{\boldsymbol{B}}_{\{n\}}\,\boldsymbol{T}$	$\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02763		—	$\left t_{\{i\}\{j\}}\right \leq \frac{1}{2}$	$ t_{ij} \leq \frac{1}{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02764		—	$\left \left \tilde{\boldsymbol{b}}_{\{k\}}\right \right _{\boldsymbol{A}^{\{2\}}} \leq 2\left \left \tilde{\boldsymbol{b}}_{\{k+1\}}\right \right _{\boldsymbol{A}^{\{2\}}}$	$\left\ \tilde{\boldsymbol{b}}_k\right\ _{\boldsymbol{A}}^2 \leq 2\left\ \tilde{\boldsymbol{b}}_{k+1}\right\ _{\boldsymbol{A}}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02765		—	$\operatorname{det}\left(\boldsymbol{B}_{\{n\}}\right)=1$	$\det\left(\boldsymbol{B}_n\right)=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02766		—	$\left[\begin{array}{ll}1&1\\1&1\end{array}\right]$	$\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02767		—	$\sqrt{m\,n}$	$\sqrt{mn}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02768		—	$\left \left \boldsymbol{A}\,\boldsymbol{x}\right \right _{\infty} \leq$	$\left\ \boldsymbol{A}\boldsymbol{x}\right\ _{\infty} \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02769		—	$\left \left \boldsymbol{A}\right \right _{\boldsymbol{M}}\left \left \boldsymbol{x}\right \right _{\{1\}}$	$\left\ \boldsymbol{A}\right\ _M\left\ \boldsymbol{x}\right\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02770		—	$\left \left \boldsymbol{A}\right \right _{\boldsymbol{M}}=\left a_{\{k\}\{l\}}\right $	$\left\ \boldsymbol{A}\right\ _M = a_{kl} $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02771		—	$\left\ \boldsymbol{A} \boldsymbol{e}_{\ell}\right\ _{\infty}=\left\ \boldsymbol{A}\right\ _M\left\ e_{\ell}\right\ _1$	$\ \boldsymbol{A}e_l\ _{\infty} = \ \boldsymbol{A}\ _M \ e_l\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02772		—	$\left\ \boldsymbol{A}\right\ _M=\max _{\boldsymbol{x} \neq \mathbf{0}} \frac{\left\ \boldsymbol{A} \boldsymbol{x}\right\ _{\infty}}{\left\ \boldsymbol{x}\right\ _1}$	$\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _{\infty}}{\ \boldsymbol{x}\ _1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02773		—	$\left\ \tilde{\boldsymbol{b}}_1\right\ _{\boldsymbol{A}}^2 \cdots\left\ \tilde{\boldsymbol{b}}_n\right\ _{\boldsymbol{A}}^2=\operatorname{det}(\boldsymbol{A})$	$\left\ \tilde{\boldsymbol{b}}_1\right\ _{\boldsymbol{A}}^2 \cdots \left\ \tilde{\boldsymbol{b}}_n\right\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02774		—	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2 \geq \sigma_n$	$\ \boldsymbol{A}\boldsymbol{x}\ _2 \geq \sigma_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02775		—	$\left\ \boldsymbol{A}\right\ _p$	$\ \boldsymbol{A}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02776		—	$\left\ \boldsymbol{A}^{-1}\right\ _p$	$\ \boldsymbol{A}^{-1}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02777		—	$\left\ \boldsymbol{V} \boldsymbol{A}\right\ _2=$	$\ \boldsymbol{V}\boldsymbol{A}\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02778		—	$\boldsymbol{V}^* \boldsymbol{V}=\boldsymbol{I}$	$\boldsymbol{V}^*\boldsymbol{V} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02779		—	$8.12\left(\left\ \boldsymbol{A} \boldsymbol{U}\right\ _2\right)$	$8.12\left(\ \boldsymbol{A}\boldsymbol{U}\ _2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02780		—	$\boldsymbol{A})$	$\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02781		—	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02782		—	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02783		—	$\ \boldsymbol{A} \boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02784		—	$\ \boldsymbol{A} \boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02785		—	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A}) :=$	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A}) :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02786		—	$\max \lambda_i $	$\max \lambda_i $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02787		—	$\boldsymbol{a} \in \mathbb{C}^m$	$\boldsymbol{a} \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02788		—	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02789		—	$\ \boldsymbol{A}\ _p = \ \boldsymbol{a}\ _p$	$\ \boldsymbol{A}\ _p = \ \boldsymbol{a}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02790		—	$ \boldsymbol{A} \in \mathbb{R}^{m \times n}$	$ \boldsymbol{A} \in \mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02791		—	$ a_{ij} $	$ a_{ij} $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02792		—	$ \boldsymbol{A} $	$ \boldsymbol{A} $
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02793		—	$\boldsymbol{A}=\left[\begin{array}{cc}1+i & -2 \\ 1 & 1-i\end{array}\right], \quad i=\sqrt{-1}$	$\boldsymbol{A}=\left[\begin{array}{cc}1+i & -2 \\ 1 & 1-i\end{array}\right], \quad i=\sqrt{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02794		—	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$ $\boldsymbol{A} \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02795		—	$p=1, \infty$	$p=1, \infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02796		—	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _2$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02797		—	$\ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02798		—	$\left\{\lambda_1, \lambda_2, \ldots, \lambda_m\right\}$	$\{\lambda_1, \lambda_2, \ldots, \lambda_m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02799		—	$\boldsymbol{x}_{\emptyset} \in \mathbb{C}^n$	$\boldsymbol{x}_0 \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02800		—	$\left\{\boldsymbol{x}_k\right\}_{k=\emptyset}^{m-1}$	$\{\boldsymbol{x}_k\}_{k=0}^{m-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02801		—	$c_{i \ k}$	c_{ik}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02802		—	$\left\{\left(\sigma_i, \boldsymbol{u}_i\right)\right\}_{i=1}^n$	$\left\{\left(\sigma_i, \boldsymbol{u}_i\right)\right\}_{i=1}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02803		—	$l \leq m$	$l \leq m$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02804		—	$\boldsymbol{x}_{\{l\}}=\boldsymbol{x}_{\{l+1\}}=\cdots=\boldsymbol{x}_{\{m\}}=\boldsymbol{x}$	$\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \cdots = \boldsymbol{x}_m = \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02805		—	$\boldsymbol{T}=\operatorname{tridiag}(c, d, c)$	$\boldsymbol{T} = \operatorname{tridiag}(c, d, c)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02806		—	$d>2\ c$	$d > 2c$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02807		—	$\boldsymbol{T}\ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02808		—	$b_{\{i\ j\}}=0$	$b_{ij} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02809		—	$ i-j \ \leq 1$	$ i-j \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02810		—	$\boldsymbol{A}=\boldsymbol{T}+\boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02811		—	$\rho(\boldsymbol{E}):=\max_{\{i\}} \lambda_{\{i\}} $	$\rho(\boldsymbol{E}) := \max_i \lambda_i $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02812		—	$\rho\left(\boldsymbol{T}^{-1}\ \boldsymbol{B}\right)\leq \rho\left(\boldsymbol{T}^{-1}\right)\ \rho(\boldsymbol{B})$	$\rho\left(\boldsymbol{T}^{-1}\boldsymbol{B}\right)\leq \rho\left(\boldsymbol{T}^{-1}\right)\rho(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02813		—	$\delta:=\ \boldsymbol{x}-\boldsymbol{y}\ _{\{2\}}\ /\ \boldsymbol{x}\ _{\{2\}}$	$\delta := \ \boldsymbol{x} - \boldsymbol{y}\ _2 / \ \boldsymbol{x}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02814		—	$K_{\{2\}}(\boldsymbol{A})=\ \boldsymbol{A}\ _{\{2\}}\left\ \boldsymbol{A}^{-1}\right\ _{\{2\}}$	$K_2(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \left\ \boldsymbol{A}^{-1}\right\ _2$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02815		—	δ	δ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02816		—	$b_2=2.0$	$b_2 = 2.0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02817		—	$\ \boldsymbol{e}\ _2=0.1$	$\ \boldsymbol{e}\ _2 = 0.1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02818		—	$b_{\{1\}}$	b_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02819		—	$\boldsymbol{y}_{\boldsymbol{A}}$	$\boldsymbol{y}_A$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02820		—	$\boldsymbol{y}_{\boldsymbol{A}^{-1}}$	$\boldsymbol{y}_{\boldsymbol{A}^{-1}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02821		—	$\left \boldsymbol{y}_{\boldsymbol{A}}\right =\left \boldsymbol{y}_{\boldsymbol{A}^{-1}}\right $	$\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{\boldsymbol{A}^{-1}}\ = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02822		—	$\ \boldsymbol{A}\ =\left \boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}\right $	$\ \boldsymbol{A}\ = \ \boldsymbol{A}\boldsymbol{y}_A\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02823		—	$\left \boldsymbol{A}^{-1}\right =\left \boldsymbol{A}^{-1} \boldsymbol{y}_{\boldsymbol{A}^{-1}}\right $	$\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1}\boldsymbol{y}_{\boldsymbol{A}^{-1}}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02824		—	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_{\boldsymbol{A}}$	$\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02825		—	$\boldsymbol{e}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}$	$\boldsymbol{e} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02826		—	$\boldsymbol{b}=\boldsymbol{y}_{\boldsymbol{A}^{-1}}$	$\boldsymbol{b} = \boldsymbol{y}_{\boldsymbol{A}^{-1}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02827		—	$\boldsymbol{e}=\boldsymbol{A} \ \boldsymbol{y}_{\boldsymbol{A}}$	$\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02828		—	$m \geq 1$	$m \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02829		—	$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$	$\boldsymbol{T} := \operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02830		—	$\operatorname{cond}_{\mathfrak{p}}(\boldsymbol{T}):=\ \boldsymbol{T}\ _{\mathfrak{p}}\left\ \boldsymbol{T}^{-1}\right\ _{\mathfrak{p}}$	$\operatorname{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \left\ \boldsymbol{T}^{-1}\right\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02831		—	$m \geq 3$	$m \geq 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02832		—	$\operatorname{cond}_{\mathfrak{1}}(\boldsymbol{T})=\operatorname{cond}_{\infty}(\boldsymbol{T})=3$	$\operatorname{cond}_1(\boldsymbol{T}) = \operatorname{cond}_{\infty}(\boldsymbol{T}) = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02833		—	$m{=}2$	$m = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02834		—	$\tan x{>}x$	$\tan x > x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02835		—	$0{<}x{<}\pi \ / \ 2$	$0 < x < \pi/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02836		—	$\cot ^{2} x{>}\frac{1}{x^{2}}{-}\frac{2}{3}$	$\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02837		—	$\boldsymbol{I}-\boldsymbol{E}$	$\boldsymbol{I}-\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02838		—	$\ \boldsymbol{E}\ <1$	$\ \boldsymbol{E}\ <1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02839		—	$(\boldsymbol{I}-\boldsymbol{E})^{-1}$	$(\boldsymbol{I}-\boldsymbol{E})^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02840		—	$\boldsymbol{x}=\left[x_{0}, \ldots, x_{n}\right]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{x}=\left[x_{0}, \ldots, x_{n}\right]^T \in \mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02841		—	$\boldsymbol{y}:=\left[y_{0}, \ldots, y_{n}\right]^T \in \mathbb{R}^{n+1}$	$\boldsymbol{y}:=\left[y_{0}, \ldots, y_{n}\right]^T \in \mathbb{R}^{n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02842		—	$y_{\{0\}}=y_{\{n\}}$	$y_0=y_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02843		—	$\left(x_{i-1}, x_{\{i\}}\right)$	$\left(x_{i-1}, x_i\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02844		—	$C^2[a, b]$	$C^2[a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02845		—	$s_{\{i\}}:=g^{\{\prime\}}\left(x_{\{i\}}\right), i=0, \ldots, n$	$s_i:=g^{\prime}\left(x_i\right), i=0, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02846		—	$\left[s_{\{1\}}, \ldots, s_{\{n\}}\right]^T$	$\left[s_1, \ldots, s_n\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02847		—	$\left\ \boldsymbol{A}^{-1}\right\ _{\infty} \leq 1$	$\left\ \boldsymbol{A}^{-1}\right\ _{\infty} \leq 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02848		—	$\left\ \boldsymbol{A}^{-1}\right\ _{1} \leq \frac{3}{2}$	$\left\ \boldsymbol{A}^{-1}\right\ _{1} \leq \frac{3}{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02849		—	$\boldsymbol{e} \in \mathbb{R}^n$	$\boldsymbol{e} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02850		—	$\left \boldsymbol{e}\right _p/\left \boldsymbol{b}\right _p \leq 0.01$	$\left\ \boldsymbol{e}\right\ _p/\left\ \boldsymbol{b}\right\ _p \leq 0.01$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02851		—	$\hat{\boldsymbol{s}}$	$\hat{\boldsymbol{s}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02852		—	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02853		—	$\boldsymbol{U} \in \mathbb{R}^{m \times n}$	$\boldsymbol{U} \in \mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02854		—	$\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$	$\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02855		—	$[\mathrm{x}, \mathrm{K}]=l \mathrm{sq}(\mathrm{A}, \mathrm{b})$	$[\mathbf{x}, \mathbf{K}]=lsq(\mathbf{A}, \mathbf{b})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02856		—	$\boldsymbol{x}=\boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$	$\boldsymbol{x}=\boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02857		—	$\boldsymbol{A} \boldsymbol{x} =$	$\boldsymbol{A} \boldsymbol{x} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02858		—	$[\mathrm{U}, \mathrm{Sigma}, \mathrm{V}]=\operatorname{svd}(\mathrm{A}, \emptyset)$	$[\mathbf{U}, \mathbf{Sigma}, \mathbf{V}]=\operatorname{svd}(\mathbf{A}, 0)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02859		—	$\ \cdot\ _p$	$\ \cdot\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02860		—	$p=1, p=\infty$	$p = 1, p = \infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02861		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + i s(\boldsymbol{x}, i \boldsymbol{y})$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = s(\boldsymbol{x}, \boldsymbol{y}) + i s(\boldsymbol{x}, i \boldsymbol{y})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02862		—	$s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 \right)$	$s(\boldsymbol{x}, \boldsymbol{y}) := \frac{1}{4} \left(\ \boldsymbol{x} + \boldsymbol{y}\ ^2 - \ \boldsymbol{x} - \boldsymbol{y}\ ^2 \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02863		—	S_p	S_p
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02864		—	$\ \boldsymbol{x}\ _{-\infty} \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _{-\infty}$	$\ \boldsymbol{x}\ _{\infty} \leq \ \boldsymbol{x}\ _p \leq n^{1/p} \ \boldsymbol{x}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02865		—	$\boldsymbol{x}_l$	$\boldsymbol{x}_l$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02866		—	$\ \boldsymbol{x}_l\ _{\infty} = \ \boldsymbol{x}_l\ _p$	$\ \boldsymbol{x}_l\ _{\infty} = \ \boldsymbol{x}_l\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02867		—	$\boldsymbol{x}_u$	$\boldsymbol{x}_u$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02868		—	$\ \boldsymbol{x}_u\ _p = n^{1/p} \ \boldsymbol{x}_u\ _{\infty}$	$\ \boldsymbol{x}_u\ _p = n^{1/p} \ \boldsymbol{x}_u\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02869		—	$1 \leq q \leq p \leq \infty$	$1 \leq q \leq p \leq \infty$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02870		—	$f(z)=z^{\{p \ / \ q\}}$	$f(z) = z^{p/q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02871		—	$z_{\{i\}}=\left x_{\{i\}}\right ^{\{q\}}$	$z_i = x_i ^q$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02872		—	$y_{\{i\}}=x_{\{i\}} \ /\ \boldsymbol{x}\ _{\infty}$	$y_i = x_i/\ \boldsymbol{x}\ _\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02873		—	$\ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ _{\left\ _F\right\ } \ \boldsymbol{x}\ $	$\ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ _F \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02874		—	$\mathbb{C}^{\{n \times n\}}, \boldsymbol{x} \in \mathbb{C}^{\{n\}}, m, n \in \mathbb{N}$	$\mathbb{C}^{n \times n}, \boldsymbol{x} \in \mathbb{C}^n, m, n \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02875		—	$\ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _F$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02876		—	$\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b} \notin \mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02877		—	$\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\ $	$\ \boldsymbol{Ax}-\boldsymbol{b}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02878		—	$\boldsymbol{b} \in \mathbb{C}^{\{m\}}$	$\boldsymbol{b} \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02879		—	$E: \mathbb{C}^{\{n\}} \rightarrow \mathbb{R}$	$E: \mathbb{C}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02880		—	$E(\boldsymbol{x})$	$E(\boldsymbol{x})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02881		—	$\sqrt{E(\boldsymbol{x})}$	$\sqrt{E(x)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02882		—	$6\,x_1-8=0$	$6x_1 - 8 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02883		—	$x_1=4\ / \ 3$	$x_1 = 4/3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02884		—	$b_{\{1\}},\ b_{\{2\}},\ b_{\{3\}}$	b_1, b_2, b_3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02885		—	$\boldsymbol{A}^{\ast}\ \boldsymbol{A}\ \boldsymbol{x}=\boldsymbol{A}^{\ast}\ \boldsymbol{b}$	$\boldsymbol{A}^*\boldsymbol{A}x = \boldsymbol{A}^*\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02886		—	$p(t)=x_1+x_2\ t$	$p(t) = x_1 + x_2t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02887		—	$\left(t_{\{k\}},\ y_{\{k\}}\right)\in\mathbb{R}^{\{2\}},\ k=1,\ \ldots,\ m$	$(t_k, y_k) \in \mathbb{R}^2, k = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02888		—	$m>2$	$m > 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02889		—	$\left\{t_{\{1\}},\ \ldots,\ t_{\{m\}}\right\}$	$\{t_1, \dots, t_m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02890		—	$t_{\{i\}}\ \neq\ t_{\{j\}}$	$t_i \neq t_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02891		—	$t_{\{1\}}=\cdots=t_{\{m\}}$	$t_1 = \cdots = t_m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02892		—	$t_{\{1\}} \neq t_{\{2\}}$	$t_1 \neq t_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02893		—	$p\left(t_{\{k\}}\right)=y_{\{k\}},\; k=1,2$	$p\left(t_k\right)=y_k,k=1,2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02894		—	$\left[\begin{array}{cc}4&10\\10&30\end{array}\right]\left[\begin{array}{l}x_{\{1\}}\\x_{\{2\}}\end{array}\right]=\left[\begin{array}{c}6\\10.1\end{array}\right]$	$\left[\begin{array}{cc}4&10\\10&30\end{array}\right]\left[\begin{array}{l}x_1\\x_2\end{array}\right]=\left[\begin{array}{c}6\\10.1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02895		—	$p(t)=x_{\{1\}}+x_{\{2\}}\; t=3.95-0.98\; t$	$p(t)=x_1+x_2t=3.95-0.98t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02896		—	$y\in\mathbb{R}$	$y\in\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02897		—	$\min\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\ _{\{2\}}^2$	$\min\ \boldsymbol{Ax}-\boldsymbol{b}\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02898		—	$1\leq n\leq m$	$1\leq n\leq m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02899		—	$\mathcal{S}:=\left\{t_{\{1\}},\; t_{\{2\}},\; \ldots,\; t_{\{m\}}\right\}\subset[a,\; b]$	$\mathcal{S}:=\{t_1,t_2,\ldots,t_m\}\subset[a,b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02900		—	$\boldsymbol{y}=\left[y_{\{1\}},\; y_{\{2\}},\; \ldots,\; y_{\{m\}}\right]^*\in\mathbb{R}^{\{m\}}$	$\boldsymbol{y}=[y_1,y_2,\ldots,y_m]^*\in\mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02901		—	$\phi_{\{j\}}:[a,\; b]\rightarrow\mathbb{R},\; j=1,\; \ldots,\; n$	$\phi_j:[a,b]\rightarrow\mathbb{R},j=1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02902		—	$p:[a,\; b]\rightarrow\mathbb{R}$	$p:[a,b]\rightarrow\mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02903		—	$p:=\sum_{j=1}^n x_{\{j\}} \ \phi_{\{j\}}$	$p := \sum_{j=1}^n x_j \phi_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02904		—	$p\left(t_{\{k\}}\right) \ \approx \ y_{\{k\}}$	$p\left(t_k\right) \approx y_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02905		—	$\phi_{\{j\}}$	ϕ_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02906		—	$w_{\{k\}}>0$	$w_k > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02907		—	$y_{\{k\}}$	y_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02908		—	$w_{\{k\}}$	w_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02909		—	$p\left(t_{\{k\}}\right)-y_{\{k\}}$	$p\left(t_k\right)-y_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02910		—	$\delta y_{\{k\}}$	δy_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02911		—	$w_{\{k\}}=1 \ /\left(\delta y_{\{k\}}\right)^{\{2\}}, \ k=1,2, \ \ldots, \ m$	$w_k = 1/\left(\delta y_k\right)^2, k = 1,2,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02912		—	$w_{\{k\}}=1$	$w_k = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02913		—	4.2 $\boldsymbol{A}^{\{*\}}$ $\boldsymbol{A}$	4.2 $\boldsymbol{A}^*\boldsymbol{A}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02914		—	$(\boldsymbol{A} \, \boldsymbol{x})_k = \sum_{j=1}^n x_j \, \phi_j(t_k), \, k=1, \, \ldots, \, m$	$(\boldsymbol{Ax})_k = \sum_{j=1}^n x_j \phi_j(t_k), k = 1, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02915		—	$\phi_j(t) := t^{j-1}$	$\phi_j(t) := t^{j-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02916		—	$t_k = (k-1) \, /\, (m-1)$	$t_k = (k-1)/(m-1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02917		—	$\boldsymbol{B}_n \, \boldsymbol{x} = \boldsymbol{c}_n$	$\boldsymbol{B}_n \boldsymbol{x} = \boldsymbol{c}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02918		—	$\boldsymbol{B}_n$	$\boldsymbol{B}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02919		—	t	t
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02920		—	$\frac{1}{m} \, \boldsymbol{B}_n \, \approx \, \boldsymbol{H}_n$	$\frac{1}{m} \boldsymbol{B}_n \approx \boldsymbol{H}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02921		—	$\boldsymbol{H}_n \, \, \text{in} \, \, \mathbb{R}^{n \times n}$	$\boldsymbol{H}_n \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02922		—	$1 \, /\, (i+j-1)$	$1/(i+j-1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02923		—	$\frac{1}{m} \, \boldsymbol{B}_n$	$\frac{1}{m} \boldsymbol{B}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02924		—	$\boldsymbol{H}_n$	$\boldsymbol{H}_n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02925		—	$\boldsymbol{B}_n=\left[b_{i,j}\right]_{i,j=1}^n$	$\boldsymbol{B}_n=[b_{i,j}]_{i,j=1}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02926		—	$\boldsymbol{H}_n^{-1}$	$\boldsymbol{H}_n^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02927		—	$K_1\left(\boldsymbol{H}_6\right):=$	$K_1\left(\boldsymbol{H}_6\right):=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02928		—	$\left \boldsymbol{H}_6\right _1\left \boldsymbol{H}_6^{-1}\right _1\approx 3\cdot 10^7$	$\left\ \boldsymbol{H}_6\right\ _1\left\ \boldsymbol{H}_6^{-1}\right\ _1\approx 3\cdot 10^7$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02929		—	$\left(t-\tilde{t}\right)^{j-1},j=1,\ldots,n$	$\left(t-\tilde{t}\right)^{j-1},j=1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02930		—	$\tilde{t}$	$\tilde{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02931		—	$\langle\boldsymbol{x},\boldsymbol{y}\rangle=\boldsymbol{y}^*\boldsymbol{x}$	$\langle\boldsymbol{x},\boldsymbol{y}\rangle=\boldsymbol{y}^*\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02932		—	$\mathcal{S}:=\mathcal{R}\left(\boldsymbol{A}\right)$	$\mathcal{S}:=\mathcal{R}\left(\boldsymbol{A}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02933		—	$\mathcal{T}:=\mathcal{N}\left(\boldsymbol{A}^*\right)$	$\mathcal{T}:=\mathcal{N}\left(\boldsymbol{A}^*\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02934		—	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2$	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02935		—	$\boldsymbol{b}_1$	$\boldsymbol{b}_1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02936		—	$\boldsymbol{b}_2$	b_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02937		—	$\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02938		—	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02939		—	$\left\langle \boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \right\rangle = 0$	$\langle \boldsymbol{Ax} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02940		—	$\boldsymbol{A} \ \boldsymbol{x} = \boldsymbol{b}_1$	$\boldsymbol{Ax} = \boldsymbol{b}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02941		—	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02942		—	$\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02943		—	$\boldsymbol{A}^*(\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02944		—	$\boldsymbol{A}^* \ \boldsymbol{b}_2 = \mathbf{0}$	$\boldsymbol{A}^* \boldsymbol{b}_2 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02945		—	$\boldsymbol{A}^* \left(\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1 \right) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02946		—	$\boldsymbol{A} \ \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02947		—	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02948		—	$\boldsymbol{B} := \boldsymbol{A}^* \boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{A}^* \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02949		—	$\left(\boldsymbol{A}^* \boldsymbol{A}\right)_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \ldots, n$	$(\boldsymbol{A}^* \boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02950		—	$\boldsymbol{A}^* \boldsymbol{A} = \sum_{k=1}^m \left[\begin{array}{c} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{array} \right] \left[a_{k1} \cdots a_{kn} \right], \boldsymbol{A}^* \boldsymbol{b} = \sum_{k=1}^m \left[\begin{array}{c} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{array} \right] b_k$	$\boldsymbol{A}^* \boldsymbol{A} = \sum_{k=1}^m \left[\begin{array}{c} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{array} \right] \left[a_{k1} \cdots a_{kn} \right], \boldsymbol{A}^* \boldsymbol{b} = \sum_{k=1}^m \left[\begin{array}{c} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{array} \right] b_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02951		—	$n(n+1) / 2$	$n(n+1)/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02952		—	$m n^2$	$m n^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02953		—	$\boldsymbol{B}, 2 m n$	$\boldsymbol{B}, 2 m n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02954		—	$\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}, n^3 / 3$	$\boldsymbol{c} := \boldsymbol{A}^* \boldsymbol{b}, n^3 / 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02955		—	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^*, n^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02956		—	$\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$	$\boldsymbol{L} \boldsymbol{y} = \boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02957		—	$\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{L}^* \boldsymbol{x} = \boldsymbol{y}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02958		—	$m \approx n$	$m \approx n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02959		—	$\frac{4}{3} n^3 =$	$\frac{4}{3} n^3 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02960		—	$\lambda_1 \geq \cdots \geq \lambda_n > 0$	$\lambda_1 \geq \cdots \geq \lambda_n > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02961		—	$\sigma_1 \geq \cdots \geq \sigma_n > 0$	$\sigma_1 \geq \cdots \geq \sigma_n > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02962		—	$K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$	$K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02963		—	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02964		—	$\lambda_i = \sigma_i^2$	$\lambda_i = \sigma_i^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02965		—	$\boldsymbol{R}_1^*$	$\boldsymbol{R}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02966		—	$\boldsymbol{c}_1$	$\boldsymbol{c}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02967		—	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02968		—	$\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$	$\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02969		—	$\boldsymbol{c}$	$\boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02970		—	$[\mathrm{R}, \mathrm{c}] =$	$[\mathbf{R}, \mathbf{c}] =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02971		—	$\boldsymbol{x}=[1,0,0]^{*}$	$\boldsymbol{x} = [1, 0, 0]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02972		—	$\boldsymbol{R}_{1} \boldsymbol{x}=\boldsymbol{c}_{1}$	$\boldsymbol{R}_1 \boldsymbol{x} = \boldsymbol{c}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02973		—	$K_{2} \left(\boldsymbol{R}_{1}\right)=K_{2} \left(\boldsymbol{Q}_{1} \boldsymbol{R}_{1}\right)=K_{2}(\boldsymbol{A})$	$K_2(\boldsymbol{R}_1) = K_2(\boldsymbol{Q}_1 \boldsymbol{R}_1) = K_2(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02974		—	$K_{2} \left(\boldsymbol{A}^{*} \boldsymbol{A}\right)$	$K_2\left(\boldsymbol{A}^{*} \boldsymbol{A}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02975		—	$2 \mathrm{~m} \mathrm{~n}^2-$	$2 m n^2 -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02976		—	$\mathrm{x}=\mathrm{A} \backslash \mathrm{b}$	$\mathbf{x} = \mathbf{A} \backslash \mathbf{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02977		—	$\mathrm{x}=\operatorname{lscov}(\mathrm{A}, \mathrm{b})$	$\mathbf{x} = \operatorname{lscov}(\mathbf{A}, \mathbf{b})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02978		—	$\boldsymbol{U}_{2}$	$\boldsymbol{U}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F02979		—	$\boldsymbol{V}_{2}$	$\boldsymbol{V}_2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02980		—	$\boldsymbol{A}^{\dagger} \in \mathbb{C}^{n \times m}$	$\boldsymbol{A}^{\dagger} \in \mathbb{C}^{n \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02981		—	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02982		—	$\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$	$\left(\boldsymbol{A}^{\dagger} \boldsymbol{A}\right)^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02983		—	$\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$	$\left(\boldsymbol{A} \boldsymbol{A}^{\dagger}\right)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02984		—	$\boldsymbol{A}^{\dagger} \boldsymbol{A}$	$\boldsymbol{A}^{\dagger} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02985		—	$\boldsymbol{A} \boldsymbol{A}^{\dagger}$	$\boldsymbol{A} \boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02986		—	$\boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02987		—	$\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02988		—	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02989		—	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F02990		—	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02991		—	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02992		—	$\boldsymbol{v} \in \mathbb{C}^m$	$\boldsymbol{v} \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02993		—	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{P}_{\mathcal{S}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02994		—	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02995		—	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02996		—	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02997		—	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02998		—	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F02999		—	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03000		—	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^\dagger$	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^\dagger$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03001		—	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03002		—	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}$	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03003		—	$\boldsymbol{z}\in\mathcal{N}(\boldsymbol{A})$	$\boldsymbol{z}\in\mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03004		—	$\boldsymbol{b}_1:=\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}$	$\boldsymbol{b}_1:=\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03005		—	$\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger}\boldsymbol{b}$	$\boldsymbol{z}:=\boldsymbol{x}-\boldsymbol{A}^{\dagger}\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03006		—	$\boldsymbol{A}\boldsymbol{z} =$	$\boldsymbol{A}\boldsymbol{z} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03007		—	$\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}=\boldsymbol{b}_1-\boldsymbol{b}_1=\mathbf{0}$	$\boldsymbol{A}\boldsymbol{x}-\boldsymbol{A}\boldsymbol{A}^{\dagger}\boldsymbol{b}=\boldsymbol{b}_1-\boldsymbol{b}_1=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03008		—	$\boldsymbol{A}\boldsymbol{z}=\mathbf{0}$	$\boldsymbol{A}\boldsymbol{z}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03009		—	$\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}\left(\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}\right)=\boldsymbol{b}_1$	$\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}\left(\boldsymbol{A}^{\dagger}\boldsymbol{b}+\boldsymbol{z}\right)=\boldsymbol{b}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03010		—	$\boldsymbol{A}^{\dagger}\boldsymbol{b}$	$\boldsymbol{A}^{\dagger}\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03011		—	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}$	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03012		—	$\boldsymbol{z}=\boldsymbol{V}_2\boldsymbol{y}$	$\boldsymbol{z}=\boldsymbol{V}_2\boldsymbol{y}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03013		—	$\boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1}=\mathbf{0}$	$\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03014		—	$\boldsymbol{z}^{*} \boldsymbol{A}^{\dagger} \boldsymbol{b}=\boldsymbol{y}^{*} \boldsymbol{V}_{2}^{*} \boldsymbol{V}_{1} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_{1}^{*} \boldsymbol{b}=\mathbf{0}$	$\boldsymbol{z}^* \boldsymbol{A}^\dagger \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03015		—	$\left \boldsymbol{x}\right _{2}^{2}=\left \boldsymbol{A}^{\dagger} \boldsymbol{b}+\boldsymbol{z}\right _{2}^{2}=\left \boldsymbol{A}^{\dagger} \boldsymbol{b}\right _{2}^{2}+\ \boldsymbol{z}\ _{2}^{2} \geq\left\ \boldsymbol{A}^{\dagger} \boldsymbol{b}\right\ _{2}^{2}$	$\left\ \boldsymbol{x} \right\ _2^2 = \left\ \boldsymbol{A}^\dagger \boldsymbol{b} + \boldsymbol{z} \right\ _2^2 = \left\ \boldsymbol{A}^\dagger \boldsymbol{b} \right\ _2^2 + \left\ \boldsymbol{z} \right\ _2^2 \geq \left\ \boldsymbol{A}^\dagger \boldsymbol{b} \right\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03016		—	$\boldsymbol{A}=\left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03017		—	$\boldsymbol{b}=[1,1]^*$	$\boldsymbol{b} = [1, 1]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03018		—	$[1 / 2, 1 / 2]+[a,-a]$	$[1/2, 1/2] + [a, -a]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03019		—	$a=1 / 2$	$a = 1/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03020		—	$[1 / 2, 1 / 2]$	$[1/2, 1/2]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03021		—	$\min \left\ \boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\right\ _{2}$	$\min \left\ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b} \right\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03022		—	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03023		—	$\min \left\ \boldsymbol{A} \boldsymbol{y}-\boldsymbol{b}-\boldsymbol{e}\right\ _{2}$	$\min \left\ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e} \right\ _2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03024		—	$\boldsymbol{b}_1, \boldsymbol{e}_1$	$\boldsymbol{b}_1, \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03025		—	$\boldsymbol{b}_1 \neq \mathbf{0}$	$\boldsymbol{b}_1 \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03026		—	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03027		—	$\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	$\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03028		—	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03029		—	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03030		—	$\ \boldsymbol{b}_1\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{b}_1\ = \ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03031		—	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03032		—	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{e}_1$	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03033		—	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ $	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03034		—	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03035		—	$\left\ \boldsymbol{e}_1\right\ /\left\ \boldsymbol{b}_1\right\ $	$\left\ \boldsymbol{e}_1\right\ /\left\ \boldsymbol{b}_1\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03036		—	$\left\ \boldsymbol{b}\right\ /\left\ \boldsymbol{b}_1\right\ $	$\left\ \boldsymbol{b}\right\ /\left\ \boldsymbol{b}_1\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03037		—	$K_{\infty}(\boldsymbol{A})=\left\ \boldsymbol{A}\right\ _{\infty}\left\ \boldsymbol{A}^{\dagger}\right\ _{\infty}=2\cdot 2=4$	$K_{\infty}(\boldsymbol{A})=\left\ \boldsymbol{A}\right\ _{\infty}\left\ \boldsymbol{A}^{\dagger}\right\ _{\infty}=2\cdot 2=4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03038		—	$\boldsymbol{A}\boldsymbol{A}^{\dagger}=\left[\begin{array}{lll}1&0&0\\0&1&0\\0&0&0\end{array}\right]$	$\boldsymbol{A}\boldsymbol{A}^{\dagger}=\left[\begin{array}{lll}1&0&0\\0&1&0\\0&0&0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03039		—	$\left\ \boldsymbol{e}_1\right\ _{\infty}/\left\ \boldsymbol{b}_1\right\ _{\infty}=10^{-2}$	$\left\ \boldsymbol{e}_1\right\ _{\infty}/\left\ \boldsymbol{b}_1\right\ _{\infty}=10^{-2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03040		—	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}=\left[10^{-4},0\right]^*$	$\boldsymbol{x}=\boldsymbol{A}^{\dagger}\boldsymbol{b}=\left[10^{-4},0\right]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03041		—	$\boldsymbol{y}=\boldsymbol{A}^{\dagger}(\boldsymbol{b}+\boldsymbol{e})=\left[10^{-4}+10^{-6},0\right]^*$	$\boldsymbol{y}=\boldsymbol{A}^{\dagger}(\boldsymbol{b}+\boldsymbol{e})=\left[10^{-4}+10^{-6},0\right]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03042		—	$\boldsymbol{A},\boldsymbol{E}\in\mathbb{C}^{m\times n},m>n$	$\boldsymbol{A},\boldsymbol{E}\in\mathbb{C}^{m\times n},m>n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03043		—	$\alpha:=1-\left\ \boldsymbol{E}\right\ _2\left\ \boldsymbol{A}^{\dagger}\right\ _2>0$	$\alpha:=1-\left\ \boldsymbol{E}\right\ _2\left\ \boldsymbol{A}^{\dagger}\right\ _2>0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03044		—	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2\in\mathbb{C}^m$	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2\in\mathbb{C}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03045		—	$\min\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{y}-\boldsymbol{b}\right\ _2$	$\min\left\ \left(\boldsymbol{A}+\boldsymbol{E}\right)\boldsymbol{y}-\boldsymbol{b}\right\ _2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03046		—	$K(1+\beta K) / \alpha$	$K(1+\beta K)/\alpha$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03047		—	$\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03048		—	$\boldsymbol{A} . \beta$	$\boldsymbol{A} . \beta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03049		—	$\rho \leq \frac{1}{\alpha} K \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\rho \leq \frac{1}{\alpha} K \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03050		—	β	β
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03051		—	$\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$	$\frac{1}{\alpha} K^2 \beta \ \boldsymbol{E}\ _2 / \ \boldsymbol{A}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03052		—	$K(\boldsymbol{A})^2 \beta$	$K(\boldsymbol{A})^2 \beta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03053		—	$\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{b}_2 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03054		—	$\sigma_1, \sigma_2, \ldots, \sigma_n$	$\sigma_1, \sigma_2, \ldots, \sigma_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03055		—	$\sigma_1 \geq \cdots \geq \sigma_n$	$\sigma_1 \geq \cdots \geq \sigma_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03056		—	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03057		—	$n-j+1$	$n-j+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03058		—	$\beta_j \leq \alpha_j + \ \boldsymbol{A} - \boldsymbol{B}\ _2$	$\beta_j \leq \alpha_j + \ \boldsymbol{A} - \boldsymbol{B}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03059		—	$\boldsymbol{A}, \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{E} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03060		—	$\alpha_1 \geq \cdots \geq \alpha_n$	$\alpha_1 \geq \cdots \geq \alpha_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03061		—	$\epsilon_1 \geq \cdots \geq \epsilon_n$	$\epsilon_1 \geq \cdots \geq \epsilon_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03062		—	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) \leq$	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03063		—	$\left\ \boldsymbol{A}^{\dagger} \right\ _2 \left\ \boldsymbol{E} \right\ _2 < 1$	$\left\ \boldsymbol{A}^{\dagger} \right\ _2 \left\ \boldsymbol{E} \right\ _2 < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03064		—	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) = \operatorname{rank}(\boldsymbol{A})$	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) = \operatorname{rank}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03065		—	$\left\ (\boldsymbol{A} + \boldsymbol{E})^{\dagger} \right\ _2 \leq \frac{\left\ \boldsymbol{A}^{\dagger} \right\ _2}{1 - \left\ \boldsymbol{A}^{\dagger} \right\ _2 \left\ \boldsymbol{E} \right\ _2} = \frac{1}{\alpha_r - \epsilon_1}$	$\left\ (\boldsymbol{A} + \boldsymbol{E})^{\dagger} \right\ _2 \leq \frac{\left\ \boldsymbol{A}^{\dagger} \right\ _2}{1 - \left\ \boldsymbol{A}^{\dagger} \right\ _2 \left\ \boldsymbol{E} \right\ _2} = \frac{1}{\alpha_r - \epsilon_1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03066		—	$\beta_1 \geq$	$\beta_1 \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03067		—	$\cdots \geq \beta_n$	$\cdots \geq \beta_n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03068		—	$\epsilon_1 / \alpha_r < 1$	$\epsilon_1 / \alpha_r < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03069		—	$\alpha_r > \epsilon_1$	$\alpha_r > \epsilon_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03070		—	$\alpha_r - \beta_r \leq \ \boldsymbol{E}\ _2 = \epsilon_1$	$\alpha_r - \beta_r \leq \ \boldsymbol{E}\ _2 = \epsilon_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03071		—	$\beta_r \geq \alpha_r - \epsilon_1 > 0$	$\beta_r \geq \alpha_r - \epsilon_1 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03072		—	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) \geq r$	$\operatorname{rank}(\boldsymbol{A} + \boldsymbol{E}) \geq r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03073		—	$\beta_r \geq \alpha_r - \epsilon_1$	$\beta_r \geq \alpha_r - \epsilon_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03074		—	$\mu_1 \geq \cdots \geq \mu_n$	$\mu_1 \geq \cdots \geq \mu_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03075		—	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{m \times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03076		—	$\beta_1 \geq \cdots \geq \beta_n$	$\beta_1 \geq \cdots \geq \beta_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03077		—	$\lambda_1 \geq \cdots \geq \lambda_{m+n}$	$\lambda_1 \geq \cdots \geq \lambda_{m+n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03078		—	$\mu_1 \geq \cdots \geq \mu_{m+n}$	$\mu_1 \geq \cdots \geq \mu_{m+n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03079		—	$\operatorname{SVD}\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\right] \boldsymbol{\Sigma} \left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right]^*$	$\text{SVD} \left[u_1, \dots, u_m\right] \Sigma \left[v_1, \dots, v_n\right]^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03080		—	$2 \text{ } r$	$2r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03081		—	$\alpha_1, -\alpha_1, \ldots, \alpha_r, -\alpha_r$	$\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03082		—	$m+n-2 \text{ } r$	$m+n-2r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03083		—	$2 \text{ } s$	$2s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03084		—	$\beta_1, -\beta_1, \ldots, \beta_s, -\beta_s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03085		—	$m+n-2 \text{ } s$	$m+n-2s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03086		—	$\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2 \leq\left\ \boldsymbol{B}-\boldsymbol{A}\right\ _F^2$	$\sum_{i=1}^t\left \alpha_i-\beta_i\right ^2 \leq\left\ \boldsymbol{B}-\boldsymbol{A}\right\ _F^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03087		—	$t \leq n$	$t \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03088		—	$\alpha_i=\beta_i=0$	$\alpha_i=\beta_i=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03089		—	$i=t+1, \ldots, n$	$i=t+1, \dots, n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03090		—	$\left(t-c_{1}\right)^{2}+\left(y-c_{2}\right)^{2}=r^{2}$	$\left(t-c_{1}\right)^2+\left(y-c_{2}\right)^2=r^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03091		—	$\left(t_{i}, y_{i}\right)_{i=1}^m$	$\left(t_i, y_i\right)_{i=1}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03092		—	(t, y)	(t, y)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03093		—	c_{1}, c_{2}	c_1, c_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03094		—	$c_{1}=x_{1} / 2, c_{2}=x_{2} / 2$	$c_1=x_1/2, c_2=x_2/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03095		—	$r^{2}=c_{1}^{2}+c_{2}^{2}+x_{3}$	$r^2=c_1^2+c_2^2+x_3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03096		—	c_{1}, c_{2}, r	c_1, c_2, r
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03097		—	$\left[x_{1}, x_{2}, x_{3}\right]$	$\left[x_1, x_2, x_3\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03098		—	$(1,4),(3,2),(1,0)$	$(1,4),(3,2),(1,0)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03099		—	$\left[\begin{array}{cc}\sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3}\end{array}\right]$	$\left[\begin{array}{cc}\sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3}\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03100		—	$\boldsymbol{A}=\left[\begin{array}{cc}3 & \alpha \\ \alpha & 1\end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc}3 & \alpha \\ \alpha & 1\end{array}\right]$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03101		—	$\boldsymbol{p}_1=(0,1), \boldsymbol{p}_2=(1,0), \boldsymbol{p}_3=(2,1)$	$\boldsymbol{p}_1=(0,1), \boldsymbol{p}_2=(1,0), \boldsymbol{p}_3=(2,1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03102		—	$p(x)=m \ x+b$	$p(x)=mx+b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03103		—	$\boldsymbol{p}_i=\left(x_i, y_i\right)$	$\boldsymbol{p}_i=(x_i, y_i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03104		—	$F: \mathbb{R}^n \rightarrow \mathbb{R}$	$F: \mathbb{R}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03105		—	$\boldsymbol{B}:=\boldsymbol{I}+\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B}:=\boldsymbol{I}+\boldsymbol{A}^T \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03106		—	$\left(\boldsymbol{I}+\boldsymbol{A}^T \boldsymbol{A}\right) \boldsymbol{x}=\boldsymbol{A}^T \boldsymbol{b}$	$\left(\boldsymbol{I}+\boldsymbol{A}^T \boldsymbol{A}\right) \boldsymbol{x}=\boldsymbol{A}^T \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03107		—	$\operatorname{diag}\left(d_1, d_2, \ldots, d_m\right) \in \mathbb{R}^{m \times m}$	$\operatorname{diag}\left(d_1, d_2, \ldots, d_m\right) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03108		—	$d_i>0, i=1,2, \ldots, m$	$d_i>0, i=1,2, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03109		—	$r_i=r_i(\boldsymbol{x}), i=1,2, \ldots, m$	$r_i=r_i(\boldsymbol{x}), i=1,2, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03110		—	$\left \boldsymbol{r}(\boldsymbol{x})\right _D^2$	$\left\ \boldsymbol{r}(\boldsymbol{x})\right\ _D^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03111		—	$\boldsymbol{x}=\boldsymbol{x}_{\text {min }}$	$\boldsymbol{x}=\boldsymbol{x}_{\min }$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03112		—	$K_{\{2\}}(\boldsymbol{B}):=\ \boldsymbol{B}\ _{\{2\}}\left\ \boldsymbol{B}^{-1}\right\ _{\{2\}}$	$K_2(\boldsymbol{B}):=\ \boldsymbol{B}\ _2\left\ \boldsymbol{B}^{-1}\right\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03113		—	$\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03114		—	$\boldsymbol{B}=\boldsymbol{C}$	$\boldsymbol{B}=\boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03115		—	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 1 \\ 1 & 1 \\ 0 & 0\end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc}1 & 1 \\ 1 & 1 \\ 0 & 0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03116		—	$\boldsymbol{A}^{\dagger}=\frac{1}{4}\left[\begin{array}{lll}1 & 1 & 0 \\ 1 & 1 & 0\end{array}\right]$	$\boldsymbol{A}^{\dagger}=\frac{1}{4}\left[\begin{array}{ccc}1 & 1 & 0 \\ 1 & 1 & 0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03117		—	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^* \boldsymbol{A}\right)^{-1} \boldsymbol{A}^*$	$\boldsymbol{A}^{\dagger}=\left(\boldsymbol{A}^* \boldsymbol{A}\right)^{-1} \boldsymbol{A}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03118		—	$\boldsymbol{A}^{\dagger}=\boldsymbol{A}^*\left(\boldsymbol{A} \boldsymbol{A}^*\right)^{-1}$	$\boldsymbol{A}^{\dagger}=\boldsymbol{A}^*\left(\boldsymbol{A} \boldsymbol{A}^*\right)^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03119		—	$\mathcal{R}\left(\boldsymbol{A}^*\right), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}\left(\boldsymbol{A}^*\right), \mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03120		—	$\quad \mathbb{C}^n=\mathcal{R}\left(\boldsymbol{A}^*\right) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A})$	$\mathbb{C}^n=\mathcal{R}\left(\boldsymbol{A}^*\right) \stackrel{\perp}{\oplus} \mathcal{N}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03121		—	$\boldsymbol{u}^{\dagger}=\left(\boldsymbol{u}^* \boldsymbol{u}\right)^{-1} \boldsymbol{u}^*$	$\boldsymbol{u}^{\dagger}=\left(\boldsymbol{u}^* \boldsymbol{u}\right)^{-1} \boldsymbol{u}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03122		—	$\boldsymbol{u} \in \mathbb{C}^{n, 1}$	$\boldsymbol{u} \in \mathbb{C}^{n, 1}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03123		—	$\boldsymbol{A}=\boldsymbol{u} \, \boldsymbol{v}^*$	$\boldsymbol{A} = \boldsymbol{u} \boldsymbol{v}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03124		—	$\boldsymbol{u} \in \mathbb{C}^m, \boldsymbol{v} \in \mathbb{C}^n$	$\boldsymbol{u} \in \mathbb{C}^m, \boldsymbol{v} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03125		—	$\operatorname{diag}\!\left(\lambda_1, \ldots, \lambda_n\right)^{\dagger}=\operatorname{diag}\!\left(\lambda_1^{\dagger}, \ldots, \lambda_n^{\dagger}\right)$	$\operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)^{\dagger} = \operatorname{diag}\left(\lambda_1^{\dagger}, \ldots, \lambda_n^{\dagger}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03126		—	$\left(\boldsymbol{A}^*\right)^{\dagger}=\left(\boldsymbol{A}^{\dagger}\right)^*$	$\left(\boldsymbol{A}^*\right)^{\dagger} = \left(\boldsymbol{A}^{\dagger}\right)^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03127		—	$\left(\boldsymbol{A}^{\dagger}\right)^{\dagger}=\boldsymbol{A}$	$\left(\boldsymbol{A}^{\dagger}\right)^{\dagger} = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03128		—	$\left(\alpha \boldsymbol{A}\right)^{\dagger}=\frac{1}{\alpha} \boldsymbol{A}^{\dagger}, \quad \alpha \neq 0$	$\left(\alpha \boldsymbol{A}\right)^{\dagger} = \frac{1}{\alpha} \boldsymbol{A}^{\dagger}, \quad \alpha \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03129		—	$k, m, n \in \mathbb{N}, \boldsymbol{A} \in$	$k, m, n \in \mathbb{N}, \boldsymbol{A} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03130		—	$\mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$	$\mathbb{C}^{m \times n}, \boldsymbol{B} \in \mathbb{C}^{n \times k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03131		—	$(\boldsymbol{A} \, \boldsymbol{B})^{\dagger}=\boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$	$(\boldsymbol{A} \boldsymbol{B})^{\dagger} = \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03132		—	$\boldsymbol{E}=\boldsymbol{A} \, \boldsymbol{B}, \boldsymbol{F}=\boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$	$\boldsymbol{E} = \boldsymbol{A} \boldsymbol{B}, \boldsymbol{F} = \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03133		—	$\boldsymbol{A}^{\dagger} \boldsymbol{A}=\boldsymbol{B} \, \boldsymbol{B}^{\dagger}=\boldsymbol{I}$	$\boldsymbol{A}^{\dagger} \boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^{\dagger} = \boldsymbol{I}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03134		—	$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$	$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03135		—	$(\boldsymbol{A} \boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$	$(\boldsymbol{A}\boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger}\boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03136		—	$\boldsymbol{A}^* = \boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^* = \boldsymbol{A}^{\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03137		—	$\boldsymbol{A} \left(\boldsymbol{A}^* \boldsymbol{A} \right)^{-1} \boldsymbol{A}^* \boldsymbol{b}$	$\boldsymbol{A} \left(\boldsymbol{A}^* \boldsymbol{A} \right)^{-1} \boldsymbol{A}^* \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03138		—	$r > 0$	$r > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03139		—	$\boldsymbol{c} = \left[c_1, \ldots, c_n \right]^* = \boldsymbol{U}^* \boldsymbol{b}$	$\boldsymbol{c} = \left[c_1, \ldots, c_n \right]^* = \boldsymbol{U}^* \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03140		—	$\boldsymbol{y} = \left[y_1, \ldots, y_n \right]^* = \boldsymbol{V}^* \boldsymbol{x}$	$\boldsymbol{y} = \left[y_1, \ldots, y_n \right]^* = \boldsymbol{V}^* \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03141		—	$c_{r+1} = \cdots = c_n = 0$	$c_{r+1} = \cdots = c_n = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03142		—	$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03143		—	$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03144		—	$\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{y} \in \mathcal{N}(\boldsymbol{A}^*)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03145		—	$\boldsymbol{y}^* \boldsymbol{b} \neq 0$	$\boldsymbol{y}^* \boldsymbol{b} \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03146		—	$\mathbb{C}^{(m-n) \times n}$	$\mathbb{C}^{(m-n) \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03147		—	$\left\ \boldsymbol{A}^\dagger \right\ _2 \leq \left\ \boldsymbol{B}^{-1} \right\ _2$	$\left\ \boldsymbol{A}^\dagger \right\ _2 \leq \left\ \boldsymbol{B}^{-1} \right\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03148		—	$K(\boldsymbol{A}) = \left\ \boldsymbol{A} \right\ _2 \left\ \boldsymbol{A}^\dagger \right\ _2$	$K(\boldsymbol{A}) = \left\ \boldsymbol{A} \right\ _2 \left\ \boldsymbol{A}^\dagger \right\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03149		—	$\boldsymbol{y}_{\boldsymbol{A}^\dagger}$	$\boldsymbol{y}_{\boldsymbol{A}^\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03150		—	$\left\ \boldsymbol{y}_{\boldsymbol{A}} \right\ = \left\ \boldsymbol{y}_{\boldsymbol{A}^\dagger} \right\ = 1$	$\left\ \boldsymbol{y}_{\boldsymbol{A}} \right\ = \left\ \boldsymbol{y}_{\boldsymbol{A}^\dagger} \right\ = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03151		—	$\left\ \boldsymbol{A}^\dagger \right\ =$	$\left\ \boldsymbol{A}^\dagger \right\ =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03152		—	$\left\ \boldsymbol{A}^\dagger \boldsymbol{y}_{\boldsymbol{A}^\dagger} \right\ $	$\left\ \boldsymbol{A}^\dagger \boldsymbol{y}_{\boldsymbol{A}^\dagger} \right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03153		—	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_A, \boldsymbol{e}_1 = \boldsymbol{y}_{\boldsymbol{A}^\dagger}$	$\boldsymbol{b} = \boldsymbol{A} \boldsymbol{y}_A, \boldsymbol{e}_1 = \boldsymbol{y}_{\boldsymbol{A}^\dagger}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03154		—	$\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$	$\boldsymbol{A}^\dagger = \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03155		—	$\boldsymbol{e}_1 = \boldsymbol{e}$	$\boldsymbol{e}_1 = \boldsymbol{e}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03156		—	$(2,2)$	$(2,2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03157		—	$3+2\ \epsilon+\epsilon^2$	$3+2\epsilon+\epsilon^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03158		—	$3+2\ \epsilon$	$3+2\epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03159		—	$\epsilon<\sqrt{u},\ u$	$\epsilon<\sqrt{u},\ u$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03160		—	$u=10^{-16}$	$u=10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03161		—	$\sqrt{u}=10^{-8}$	$\sqrt{u}=10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03162		—	$\boldsymbol{A}(\epsilon)\in\mathbb{R}^{n\times n}$	$\boldsymbol{A}(\epsilon)\in\mathbb{R}^{n\times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03163		—	$j\neq i,\ i+1$	$j\neq i,\ i+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03164		—	$a_{\{k,\ k+1\}}=\epsilon\in\mathbb{R}$	$a_{k,k+1}=\epsilon\in\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03165		—	$\sigma_{\{i\}}(\epsilon),\ i=1,\ \ldots,\ n$	$\sigma_i(\epsilon),\ i=1,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03166		—	$\boldsymbol{A}(\epsilon)$	$\boldsymbol{A}(\epsilon)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03167		—	$\Omega := (0, 1)^2 = \{(x, y) : 0 < x, y < 1\}$	$\Omega := (0, 1)^2 = \{(x, y) : 0 < x, y < 1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03168		—	$\partial \Omega$	$\partial \Omega$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03169		—	$u =$	$u =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03170		—	$u(x, y)$	$u(x, y)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03171		—	$v_{j, k} \approx u(j h, k h)$	$v_{j, k} \approx u(j h, k h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03172		—	$\Omega_h := \{(j h, k h) : j, k = 1, \ldots, m\}$	$\Omega_h := \{(j h, k h) : j, k = 1, \ldots, m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03173		—	$\bar{\Omega}_h \backslash \Omega_h$	$\Omega_h \backslash \Omega_h$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03174		—	$f_{j, k} := f(j h, k h)$	$f_{j, k} := f(j h, k h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03175		—	$n = m^2$	$n := m^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03176		—	$v_{j, k}$	$v_{j, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03177		—	$\boldsymbol{T} = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$	$T = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03178		—	j, k	j, k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03179		—	$\boldsymbol{T} \, \boldsymbol{V} + \boldsymbol{V} \, \boldsymbol{T}$	$\boldsymbol{TV} + \boldsymbol{VT}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03180		—	$\boldsymbol{B} \in \mathbb{R}^{m \times n}$	$\boldsymbol{B} \in \mathbb{R}^{m \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03181		—	$\boldsymbol{x} := \operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^n$	$\boldsymbol{x} := \operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03182		—	$n = m^2$	$n = m^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03183		—	$x_i = v_{j, k}$	$x_i = v_{j, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03184		—	$b_i = h^2 f_{j, k}$	$b_i = h^2 f_{j, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03185		—	$\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{b} = h^2 \operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{b} = h^2 \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03186		—	$\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$	$\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03187		—	a, d	a, d
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03188		—	$\boldsymbol{T}_2 = [a_{ij}] \in$	$\boldsymbol{T}_2 = [a_{ij}] \in$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03189		—	$\boldsymbol{T}_2$	T_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03190		—	p, q, r, s	p, q, r, s
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03191		—	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03192		—	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03193		—	$\boldsymbol{C} \in \mathbb{R}^{pr \times qs}$	$\boldsymbol{C} \in \mathbb{R}^{pr \times qs}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03194		—	$\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03195		—	$\boldsymbol{B} \otimes \boldsymbol{A}$	$\boldsymbol{B} \otimes \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03196		—	$\boldsymbol{u} \otimes \boldsymbol{v} = \left[\boldsymbol{u}^T v_1, \dots, \boldsymbol{u}^T v_r \right]^T$	$\boldsymbol{u} \otimes \boldsymbol{v} = \left[\boldsymbol{u}^T v_1, \dots, \boldsymbol{u}^T v_r \right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03197		—	$\boldsymbol{u} \in \mathbb{R}^p$	$\boldsymbol{u} \in \mathbb{R}^p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03198		—	$\boldsymbol{v} \in \mathbb{R}^r$	$\boldsymbol{v} \in \mathbb{R}^r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03199		—	$p \cdot r$	$p \cdot r$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03200		—	r, s, k	r, s, k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03201		—	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03202		—	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s \times s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03203		—	$\boldsymbol{I}_k$	$\boldsymbol{I}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03204		—	$\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03205		—	λ, μ	λ, μ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03206		—	$\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$	$\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03207		—	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03208		—	$\boldsymbol{P}, \boldsymbol{Q}$	$\boldsymbol{P}, \boldsymbol{Q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03209		—	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03210		—	$\boldsymbol{B} \boldsymbol{D}$	$\boldsymbol{B} \boldsymbol{D}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03211		—	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03212		—	$\boldsymbol{B} \in \mathbb{R}^{r, t}$	$\boldsymbol{B} \in \mathbb{R}^{r, t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03213		—	$\boldsymbol{D} \in \mathbb{R}^{t, s}$	$\boldsymbol{D} \in \mathbb{R}^{t, s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03214		—	r, s, t	r, s, t
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03215		—	$r, s \in \mathbb{N}$	$r, s \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03216		—	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03217		—	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03218		—	$(\lambda_i, \boldsymbol{u}_i) \text{ } i=1, \ldots, r$	$(\lambda_i, \boldsymbol{u}_i) \text{ } i = 1, \ldots, r$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03219		—	$(\mu_j, \boldsymbol{v}_j) \text{ } j=1, \ldots, s$	$(\mu_j, \boldsymbol{v}_j) \text{ } j = 1, \ldots, s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03220		—	$\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$	$\boldsymbol{F}, \boldsymbol{V} \in \mathbb{R}^{r \times s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03221		—	$(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$	$(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03222		—	$\boldsymbol{A} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03223		—	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03224		—	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03225		—	$(\boldsymbol{A} \otimes \boldsymbol{B}) \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right) = \lambda_i \mu_j \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right), \quad i = 1, \dots, r, \quad j = 1, \dots, s$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j), \quad i = 1, \dots, r, \quad j = 1, \dots, s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03226		—	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B} \right) \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right) = (\lambda_i + \mu_j) \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right), \quad i = 1, \dots, r, \quad j = 1, \dots, s$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j), \quad i = 1, \dots, r, \quad j = 1, \dots, s$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03227		—	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03228		—	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03229		—	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03230		—	$\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$	$\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03231		—	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F}$	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03232		—	$\boldsymbol{A} \boldsymbol{W} =$	$\boldsymbol{A} \boldsymbol{W} =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03233		—	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03234		—	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03235		—	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03236		—	$(\boldsymbol{A} \otimes \boldsymbol{B}) \left(\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1} \right) = \left(\boldsymbol{A} \boldsymbol{A}^{-1} \right) \otimes \left(\boldsymbol{B} \boldsymbol{B}^{-1} \right) =$ $\left(\boldsymbol{I} \right) \otimes \left(\boldsymbol{I} \right) = \boldsymbol{I}$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$ $\boldsymbol{I} \otimes \boldsymbol{I} = \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03237		—	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03238		—	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$(\boldsymbol{A} \otimes \boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03239		—	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03240		—	$\boldsymbol{A} \otimes \boldsymbol{I}$	$\boldsymbol{A} \otimes \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03241		—	$\boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{I} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03242		—	$(\boldsymbol{A} \otimes \boldsymbol{B}) \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right) = \left(\boldsymbol{A} \boldsymbol{u}_i \right) \otimes \left(\boldsymbol{B} \boldsymbol{v}_j \right) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03243		—	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s \right) \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right) = \lambda_i \left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j \right),$ $\quad$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03244		—	$\quad\left(\boldsymbol{I}_r \otimes \boldsymbol{B}\right)\left(\boldsymbol{u}_{\{i\}} \otimes \boldsymbol{v}_{\{j\}}\right)=\mu_{\{j\}}\left(\boldsymbol{u}_{\{i\}} \otimes \boldsymbol{v}_{\{j\}}\right)$	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B}\right)\left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j\right)=\mu_j\left(\boldsymbol{u}_i \otimes \boldsymbol{v}_j\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03245		—	$1, \boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$	$1, \boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03246		—	$\lambda_{\{i\}}+\mu_{\{j\}}$	$\lambda_i+\mu_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03247		—	$\mu_{\{j\}}$	μ_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03248		—	$\boldsymbol{V}, \boldsymbol{F}$	$\boldsymbol{V}, \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03249		—	$\boldsymbol{B}^{\mathrm{T}}$	$\boldsymbol{B}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03250		—	$\boldsymbol{V}=\left[\boldsymbol{v}_{\{1\}}, \ldots, \boldsymbol{v}_{\{s\}}\right], \boldsymbol{F}=\left[\boldsymbol{f}_{\{1\}}, \ldots, \boldsymbol{f}_{\{s\}}\right]$	$\boldsymbol{V}=\left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_s\right], \boldsymbol{F}=\left[\boldsymbol{f}_1, \ldots, \boldsymbol{f}_s\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03251		—	$\boldsymbol{B}^{\mathrm{T}}=\left[\boldsymbol{b}_{\{1\}}, \ldots, \boldsymbol{b}_{\{s\}}\right]$	$\boldsymbol{B}^T=\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_s\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03252		—	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{\{s\}}\right) \operatorname{vec}(\boldsymbol{V})=\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_{\{s\}}^{\mathrm{T}}$	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s\right) \operatorname{vec}(\boldsymbol{V})=\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03253		—	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=$	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03254		—	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^{\mathrm{T}}$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03255		—	$\left(\lambda_j, \boldsymbol{s}_j\right)$	$(\lambda_j, \boldsymbol{s}_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03256		—	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03257		—	$d>0$	$d > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03258		—	$d \geq 2 a $	$d \geq 2 a $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03259		—	$j, k, p, q=1, \ldots, m$	$j, k, p, q = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03260		—	λ_j	λ_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03261		—	$\mathbb{R}^{m^2 \times m^2}$	$\mathbb{R}^{m^2 \times m^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03262		—	$d=2, a=-1$	$d = 2, a = -1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03263		—	$\lambda_{j, k}$	$\lambda_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03264		—	$\frac{\left\ \boldsymbol{A}\right\ _2}{\left\ \boldsymbol{A}\right\ _2}=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}\{$	$\left\ \boldsymbol{A}\right\ _2\left\ \boldsymbol{A}^{-1}\right\ _2=\frac{\lambda_{\max }}{\lambda_{\min }}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03265		—	$\lambda=2 a+2 d$	$\lambda = 2 a + 2 d$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03266		—	$[1,1,1,1]^T$	$[1,1,1,1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03267		—	$j=k=1$	$j = k = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03268		—	$-\Delta u=f, u=0$	$-\Delta u = f, u = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03269		—	$-\left(\square_h v\right)_{j, k}=(\mu f)_{j, k}$	$-\left(\square_h v\right)_{j, k}=(\mu f)_{j, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03270		—	$j, k=1, \ldots, m$	$j, k = 1, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03271		—	$\theta=v_{\{0, k\}}=v_{\{m+1, k\}}=v_{\{j, \theta\}}=v_{\{j, m+1\}}, j, k=0,1, \ldots, m+1$	$0=v_{0, k}=v_{m+1, k}=v_{j, 0}=v_{j, m+1}, j, k=0,1, \ldots, m+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03272		—	$-\left(\square_h v\right)_{j, k}=\left[20 v_{j, k}-4 v_{j-1, k}-4 v_{j, k-1}-4 v_{j+1, k}-4 v_{j, k+1}\right] \right.$	$-\left(\square_h v\right)_{j, k}=\left[20 v_{j, k}-4 v_{j-1, k}-4 v_{j, k-1}-4 v_{j+1, k}-4 v_{j, k+1}\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03273		—	$\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^2\right)$	$\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03274		—	$(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$	$(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03275		—	$j, k=1,2$	$j, k = 1, 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03276		—	$f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$	$f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03277		—	$v_{\{j, k\}=5 \pi^{\{2\}} / 66$	$v_{j,k} = 5\pi^2/66$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03278		—	$0\left(h^{\{4\}}\right)$	$O\left(h^4\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03279		—	$\mu \textbf{\boldsymbol{F}}$	$\mu \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03280		—	$(\mu f)_{\{j, k\}}$	$(\mu f)_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03281		—	$\textbf{\boldsymbol{A}}=\textbf{\boldsymbol{T}} \otimes \textbf{\boldsymbol{I}}+\textbf{\boldsymbol{I}} \otimes \textbf{\boldsymbol{T}}-\frac{1}{6} \textbf{\boldsymbol{T}} \otimes \textbf{\boldsymbol{T}}, \textbf{\boldsymbol{x}}=\operatorname{vec}(\textbf{\boldsymbol{V}})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{V})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03282		—	$\textbf{\boldsymbol{b}}=h^{\{2\}} \operatorname{vec}(\mu \textbf{\boldsymbol{F}})$	$\boldsymbol{b}=h^2 \operatorname{vec}(\mu \boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03283		—	$\Delta u=0$	$\Delta u=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03284		—	$\Delta^{\{2\}} u=u_{\{x \times x \times x\}}+2 u_{\{x \times y \ y\}}+u_{\{y \ y \ y\}}$	$\Delta^2 u=u_{xxxx}+2 u_{xxyy}+u_{yyyy}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03285		—	$v=-\Delta u$	$v=-\Delta u$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03286		—	$\textbf{\boldsymbol{T}}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03287		—	$\textbf{\boldsymbol{F}}=(f(j \ h, \ k \ h))_{\{j, k=1\}^{\{m\}}$	$\boldsymbol{F}=(f(jh,kh))_{j,k=1}^m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03288		—	$\boldsymbol{A}=(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^2$	$\boldsymbol{A} = (\boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T})^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03289		—	$\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T} \otimes \boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\operatorname{vec}(\boldsymbol{U})$	$\boldsymbol{A} = \boldsymbol{T}^2 \otimes \boldsymbol{I} + 2\boldsymbol{T} \otimes \boldsymbol{T} + \boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x} = \operatorname{vec}(\boldsymbol{U})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03290		—	$\boldsymbol{b}=h^4 \operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{b} = h^4 \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03291		—	$\mathbf{b}$	$\mathbf{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03292		—	$\mathbf{c}$	$\mathbf{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03293		—	$\boldsymbol{T}_1:=\operatorname{tridiagonal}(a, d, a)$	$\boldsymbol{T}_1 := \operatorname{tridiagonal}(a, d, a)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03294		—	$(\boldsymbol{A} \otimes$	$(\boldsymbol{A} \otimes$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03295		—	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03296		—	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_{\mathrm{s}}+\boldsymbol{I}_{\mathrm{r}} \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=\operatorname{vec}(\boldsymbol{F})$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03297		—	$n=9$	$n = 9$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03298		—	$d=\sqrt{n}$	$d = \sqrt{n}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03299		—	$0\left(n\ d^{2}\right)=0\left(n^{2}\right)$	$O\left(nd^2\right)=O\left(n^2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03300		—	$(n=100)$	$(n=100)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03301		—	$\boldsymbol{D}_{1},\ldots,\boldsymbol{D}_{m}$	$\boldsymbol{D}_1,\ldots,\boldsymbol{D}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03302		—	$\boldsymbol{U}_{1},\ldots,\boldsymbol{U}_{m}$	$\boldsymbol{U}_1,\ldots,\boldsymbol{U}_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03303		—	$\boldsymbol{A}_{1},\ldots,\boldsymbol{A}_{m-1},\boldsymbol{L}_{1}$	$\boldsymbol{A}_1,\ldots,\boldsymbol{A}_{m-1},\boldsymbol{L}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03304		—	$\ldots,\boldsymbol{L}_{m-1}$	$\ldots,\boldsymbol{L}_{m-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03305		—	$\boldsymbol{C}_{1},\ldots,\boldsymbol{C}_{m-1}$	$\boldsymbol{C}_1,\ldots,\boldsymbol{C}_{m-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03306		—	$\boldsymbol{b}^{\mathrm{T}}=$	$\boldsymbol{b}^T=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03307		—	$\left[\boldsymbol{b}_{1}^{\mathrm{T}},\ldots,\boldsymbol{b}_{m}^{\mathrm{T}}\right]$	$\left[\boldsymbol{b}_1^T,\ldots,\boldsymbol{b}_m^T\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03308		—	$\boldsymbol{x}^{\mathrm{T}}=\left[\boldsymbol{x}_{1}^{\mathrm{T}},\ldots,\boldsymbol{x}_{m}^{\mathrm{T}}\right]$	$\boldsymbol{x}^T=\left[\boldsymbol{x}_1^T,\ldots,\boldsymbol{x}_m^T\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03309		—	$\boldsymbol{L}_{k}$	$\boldsymbol{L}_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03310		—	$\boldsymbol{L}_{\boldsymbol{k}} \boldsymbol{U}_{\boldsymbol{k}} = \boldsymbol{A}_{\boldsymbol{k}}$	$\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03311		—	$\boldsymbol{x}_{\boldsymbol{k}}$	$\boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03312		—	$O\left(n^{\left\{3 \ / \ 2\right\}}\right)$	$O\left(n^{3/2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03313		—	$O(n \ \log \ n)$	$O(n \log n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03314		—	$\boldsymbol{V} =$	$\boldsymbol{V} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03315		—	$\left(v_{\boldsymbol{j}, \ \boldsymbol{k}}\right) \ \text{in} \ \mathbb{R}^{\left\{m \ \text{times} \ m\right\}}$	$\left(v_{j,k}\right) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03316		—	$\boldsymbol{F} = \left(f_{\boldsymbol{j}, \ \boldsymbol{k}}\right) = \left(f\left(j \ h, \ k \ h\right)\right) \ \text{in} \ \mathbb{R}^{\left\{m \ \text{times} \ m\right\}}$	$\boldsymbol{F} = \left(f_{j,k}\right) = \left(f(jh,kh)\right) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03317		—	$\boldsymbol{X} \ \text{in} \ \mathbb{R}^{\left\{m \ \text{times} \ m\right\}}$	$\boldsymbol{X} \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03318		—	$\boldsymbol{V} = \boldsymbol{S} \ \boldsymbol{X} \ \boldsymbol{S}$	$\boldsymbol{V} = \boldsymbol{S} \boldsymbol{X} \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03319		—	$\boldsymbol{T} \ \boldsymbol{V} + \boldsymbol{V} \ \boldsymbol{T} = h^{\left\{2\right\}} \ \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = h^2 \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03320		—	$\boldsymbol{D} \ \boldsymbol{X} + \boldsymbol{X} \ \boldsymbol{D} = 4 \ h^{\left\{4\right\}} \ \boldsymbol{G}$	$\boldsymbol{D} \boldsymbol{X} + \boldsymbol{X} \boldsymbol{D} = 4 h^4 \boldsymbol{G}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03321		—	$\boldsymbol{G}=\boldsymbol{S} \boldsymbol{F} \boldsymbol{S}$	$\boldsymbol{G} = \boldsymbol{S} \boldsymbol{F} \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03322		—	$x_{\{j, k\}=h^4} g_{\{j, k\}} /\left(\sigma_{\{j\}}+\sigma_{\{k\}}\right), \quad j, k=1, \ldots, m$	$x_{j,k} = h^4 g_{j,k} / (\sigma_j + \sigma_k), \quad j,k = 1, \dots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03323		—	$\boldsymbol{X}, \boldsymbol{S}$	$\boldsymbol{X}, \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03324		—	$-\Delta u=f$	$-\Delta u = f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03325		—	$\Omega=(0,1)^2$	$\Omega = (0,1)^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03326		—	$m \in$	$m \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03327		—	$\mathbb{N}, h=1 /(m+1)$	$\mathbb{N}, h=1 /(m+1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03328		—	$\boldsymbol{F}=(f(j h, k h)) \in \mathbb{R}^{m \times m}$	$\boldsymbol{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03329		—	$\boldsymbol{V} \in \mathbb{R}^{(m+2) \times(m+2)}$	$\boldsymbol{V} \in \mathbb{R}^{(m+2) \times(m+2)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03330		—	$O\left(m^3\right)$	$O\left(m^3\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03331		—	$O\left(4 \times 2 m^3\right)=$	$O\left(4 \times 2 m^3\right)=$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03332		—	$0\left(8\ n^{\{3\ / \ 2\}}\right) .{\ }^{\{1\}}$	$O\left(8n^{3/2}\right) .^1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03333		—	$10^{\{6\}}$	10^6
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03334		—	$1000\ \times\ 1000$	1000×1000
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03335		—	$0\left(4\ n^{\{3\ / \ 2\}}\right)$	$O\left(4n^{3/2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03336		—	$\boldsymbol{A}\ \in\ \mathbb{R}^{\{m\ \times\ m\}}$	$\boldsymbol{A}\in\mathbb{R}^{m\times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03337		—	$\boldsymbol{S}\ \boldsymbol{A}$	$\boldsymbol{S}\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03338		—	$\boldsymbol{A}\ \boldsymbol{S}$	$\boldsymbol{A}\boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03339		—	$0\left(m^{\{2\}\ \log_{\{2\}\ m}}\right)$	$O\left(m^2\log_2 m\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03340		—	$\boldsymbol{v}=\left[v_{\{1\}},\ \ldots,\ v_{\{m\}}\right]^{\{T\}}\ \in\ \mathbb{R}^{\{m\}}$	$\boldsymbol{v}=\left[v_1,\ldots,v_m\right]^T\in\mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03341		—	$\boldsymbol{w}=\left[w_{\{1\}},\ \ldots,\ w_{\{m\}}\right]^{\{T\}}$	$\boldsymbol{w}=\left[w_1,\ldots,w_m\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03342		—	$\boldsymbol{w}=\boldsymbol{S}\ \boldsymbol{v}$	$\boldsymbol{w}=\boldsymbol{S}\boldsymbol{v}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03343		—	$\boldsymbol{B}=\boldsymbol{S} \ \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{S} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03344		—	$\boldsymbol{A} \in \mathbb{R}^{m, n}$	$\boldsymbol{A} \in \mathbb{R}^{m,n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03345		—	$\boldsymbol{B}=\boldsymbol{A} \ \boldsymbol{S}$	$\boldsymbol{B} = \boldsymbol{A} \boldsymbol{S}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03346		—	$\boldsymbol{B}=\left(\boldsymbol{S} \ \boldsymbol{A}^T\right)^T$	$\boldsymbol{B} = \left(\boldsymbol{S} \boldsymbol{A}^T\right)^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03347		—	$N \in \mathbb{N}$	$N \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03348		—	$\boldsymbol{y}=\left[y_{\{1\}}, \ldots, y_{\{N\}}\right]^T \in \mathbb{R}^{\{N\}}$	$\boldsymbol{y} = \left[y_1, \dots, y_N\right]^T \in \mathbb{R}^N$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03349		—	$\boldsymbol{z}=\left[z_{\{1\}}, \ldots, z_{\{N\}}\right]^T$	$\boldsymbol{z} = \left[z_1, \dots, z_N\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03350		—	$\boldsymbol{z}=\boldsymbol{F}_{\{N\}} \ \boldsymbol{y}$	$\boldsymbol{z} = \boldsymbol{F}_N \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03351		—	$\boldsymbol{F}_{\{N\}} \in \mathbb{C}^{N \times N}$	$\boldsymbol{F}_N \in \mathbb{C}^{N \times N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03352		—	$\omega_{N}^{j \ k}, j, k=0,1, \ldots, N-1$	$\omega_N^{jk}, j, k=0,1, \dots, N-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03353		—	$\boldsymbol{B}=\boldsymbol{F}_{\{N\}} \ \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{F}_N \boldsymbol{A}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03354		—	$\backslash\omega_{4}^{\wedge\{2\}}=(-i)^{\wedge\{2\}}=-1$, $\backslash\omega_{4}^{\wedge\{3\}}=(-i)(-1)=i$, $\backslash\omega_{4}^{\wedge\{4\}}=(-1)^{\wedge\{2\}}=1$, $\backslash\omega_{4}^{\wedge\{6\}}=i^{\wedge\{2\}}=-1$	$\omega_4^2 = (-i)^2 = -1, \omega_4^3 = (-i)(-1) = i, \omega_4^4 = (-1)^2 = 1, \omega_4^6 = i^2 = -1$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03355		—	$\backslash\omega_{4}^{\wedge\{9\}}=i^{\wedge\{3\}}=-i$	$\omega_4^9 = i^3 = -i$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03356		—	$2\ m+2$	$2m + 2$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03357		—	w	w
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03358		—	$\backslash\boldsymbol{S}\ \backslash\boldsymbol{x}$	$\boldsymbol{Sx}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03359		—	$i\ /\ 2$	$i/2$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03360		—	$\backslash\boldsymbol{F}_{2\ m+2}\ \backslash\boldsymbol{z}$	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03361		—	$\backslash\omega=\backslash\omega_{2\ m+2}=\mathrm{e}^{\{-2\ \pi i\ i\ /\ (2\ m+2)\}}=\mathrm{e}^{\{-\pi i\ i\ /\ (m+1)\}}$	$\omega = \omega_{2m+2} = e^{-2\pi i/(2m+2)} = e^{-\pi i/(m+1)}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03362		—	$-2\ i$	$-2i$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03363		—	$-1\ /\ (2\ i)=-i\ /\left(2\ i^{\wedge\{2\}}\right)=i\ /\ 2$	$-1/(2i) = -i/\left(2i^2\right) = i/2$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03364		—	$N=2\ m+2$	$N = 2m + 2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03365		—	$\boldsymbol{F}_{\mathbf{N}} \boldsymbol{y}$	$\boldsymbol{F}_N \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03366		—	$\boldsymbol{F}_{\mathbf{N}}$	$\boldsymbol{F}_N$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03367		—	$\boldsymbol{F}_{\mathbf{N} / 2}$	$\boldsymbol{F}_{N/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03368		—	$\mathbf{N} / 2$	$N/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03369		—	$O\left(N^{\mathbf{2}}\right)$	$O\left(N^2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03370		—	$O\left(N \log _{\mathbf{2}} N\right)$	$O\left(N \log _2 N\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03371		—	$\boldsymbol{P}_{\mathbf{N}} \in \mathbb{R}^{\mathbf{N} \times \mathbf{N}}$	$\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03372		—	$\boldsymbol{e}_{\mathbf{k}}=\left(\delta_{\mathbf{j}, \mathbf{k}}\right)$	$\boldsymbol{e}_k=\left(\delta_{j, k}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03373		—	$\left[\boldsymbol{a}_{\mathbf{1}}, \ldots, \boldsymbol{a}_{\mathbf{N}}\right]$	$\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03374		—	$\boldsymbol{P}_{\mathbf{N}}$	$\boldsymbol{P}_N$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03375		—	$\boldsymbol{F}_{\mathbf{4}} \boldsymbol{P}_{\mathbf{4}}$	$\boldsymbol{F}_4 \boldsymbol{P}_4$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03376		—	$\boldsymbol{F}_2$	$\boldsymbol{F}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03377		—	$\omega_2=\exp ^{-2 \pi i \text { i } / 2}=-1$	$\omega_2=\exp ^{-2 \pi i / 2}=-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03378		—	$N=2 \text { m}$	$N=2 m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03379		—	$p, \text { q}$	p, q
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03380		—	$1 \leq p, \text { q } \leq m$	$1 \leq p, q \leq m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03381		—	$j:=p-1$	$j:=p-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03382		—	$k:=q-1$	$k:=q-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03383		—	$\boldsymbol{F}_{2 \text { m}} \boldsymbol{P}_{2 \text { m}}$	$\boldsymbol{F}_{2 m} \boldsymbol{P}_{2 m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03384		—	$\boldsymbol{y} \in \mathbb{R}^{2 \text { m}}$	$\boldsymbol{y} \in \mathbb{R}^{2 m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03385		—	$\boldsymbol{w}=\boldsymbol{P}_{2 \text { m}}^{\text {T}} \boldsymbol{y}=\left[\boldsymbol{w}_{1}^{\text {T}}\right]^{\text {T}}, \boldsymbol{w}_{2}^{\text {T}}\right]^{\text {T}}$	$\boldsymbol{w}=\boldsymbol{P}_{2 m}^T \boldsymbol{y}=\left[\boldsymbol{w}_1^T, \boldsymbol{w}_2^T\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03386		—	$\boldsymbol{F}_{2 \text { m}} \boldsymbol{y}$	$\boldsymbol{F}_{2 m} \boldsymbol{y}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03387		—	$\boldsymbol{F}_m \boldsymbol{w}_1$	$\boldsymbol{F}_m \boldsymbol{w}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03388		—	$\boldsymbol{F}_m \boldsymbol{w}_2$	$\boldsymbol{F}_m \boldsymbol{w}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03389		—	$n=2^{\{k\}}$	$n = 2^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03390		—	$\boldsymbol{y} \in \mathbb{R}^n$	$\boldsymbol{y} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03391		—	$\gamma N \log_2 N$	$\gamma N \log_2 N$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03392		—	γ	γ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03393		—	$N=2^{\{k\}}$	$N = 2^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03394		—	$N / 2=2^{\{k-1\}}$	$N/2 = 2^{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03395		—	$\boldsymbol{D}_{N / 2}$	$\boldsymbol{D}_{N/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03396		—	$x_{\{k\}}=2 x_{\{k-1\}}+\gamma 2^{\{k\}}$	$x_k = 2x_{k-1} + \gamma 2^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03397		—	$x_{\{0\}}=0$	$x_0 = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03398		—	$x_{\{k\}}=\gamma \text{ k } 2^{\{k\}}$	$x_k = \gamma k 2^k$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03399		—	$x_{\{k-1\}}=\gamma (k-1) \text{ } 2^{\{k-1\}}$	$x_{k-1} = \gamma (k-1) 2^{k-1}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03400		—	$x_{\{k\}}=2 \text{ } x_{\{k-1\}}+\gamma \text{ } 2^{\{k\}}=2 \text{ } \gamma (k-1) \text{ } 2^{\{k-1\}}+\gamma \text{ } 2^{\{k\}}=\gamma \text{ k } 2^{\{k\}}$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma (k-1) 2^{k-1} + \gamma 2^k = \gamma k 2^k$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03401		—	$\gamma \approx 5$	$\gamma \approx 5$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03402		—	$O\left(8 \text{ } n^{\{2\}}\right)$	$O\left(8n^2\right)$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03403		—	$5 \text{ } N \log _{\{2\}} N$	$5N \log _2 N$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03404		—	$N=2^{\{20\}} \approx 10^{\{6\}}$	$N = 2^{20} \approx 10^6$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03405		—	$\boldsymbol{H}=\boldsymbol{S} \text{ } \boldsymbol{F}$	$\boldsymbol{H} = \boldsymbol{S} \boldsymbol{F}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03406		—	$\boldsymbol{G}=\boldsymbol{H} \text{ } \boldsymbol{S}$	$\boldsymbol{G} = \boldsymbol{H} \boldsymbol{S}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03407		—	$\boldsymbol{Z}=\boldsymbol{S} \text{ } \boldsymbol{X}$	$\boldsymbol{Z} = \boldsymbol{S} \boldsymbol{X}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03408		—	$\boldsymbol{V}=\boldsymbol{Z} \text{ } \boldsymbol{S}$	$\boldsymbol{V} = \boldsymbol{Z} \boldsymbol{S}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03409		—	$0\left(\gamma(2\ m+2)\ \log_{_2}(2\ m+2)\right)$	$O\left(\gamma(2m+2)\log_2(2m+2)\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03410		—	$0\left(8\ n^{\{3\ /\ 2\}}\right)$	$O\left(8n^{3/2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03411		—	$\boldsymbol{F}_{_4}$	$\boldsymbol{F}_4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03412		—	$\left(v_{_i,\ j}\right)_{_i,\ j=1}^{\{m\}}$	$\left(v_{i,j}\right)_{i,j=1}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03413		—	$\boldsymbol{T}\ \boldsymbol{V}+\boldsymbol{V}\boldsymbol{T}=\mathsf{h}^{\{2\}}\ \boldsymbol{F}$	$\boldsymbol{TV}+\boldsymbol{VT}=\mathsf{h}^2\boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03414		—	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_{_1},\ \ldots,\ \lambda_{_m}\right)$	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_m\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03415		—	$\lambda_{_j}=4\ \sin^{\{2\}}(j\ \pi\ h\ /\ 2)$	$\lambda_j=4\sin^2(j\pi h/2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03416		—	$\boldsymbol{X}=\left[\boldsymbol{x}_{_1},\ \ldots,\ \boldsymbol{x}_{_m}\right]$	$\boldsymbol{X}=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_m\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03417		—	$\boldsymbol{C}=\left[\boldsymbol{c}_{_1},\ \ldots,\ \boldsymbol{c}_{_m}\right]$	$\boldsymbol{C}=\left[\boldsymbol{c}_1,\ldots,\boldsymbol{c}_m\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03418		—	$8\ m-7$	$8m-7$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03419		—	$0\left(\delta\ m^{\{2\}}\right)$	$O\left(\delta m^2\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03420		—	$0\left(4\ m^{\{3\}}\right)=0\left(4\ n^{\{3\}}\ / \ 2\right)$	$O\left(4m^3\right)=O\left(4n^{3/2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03421		—	$0\left(2\ \gamma n\ \log_{\{2\}}\ n\right)$	$O\left(2\gamma n\log_2n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03422		—	$\boldsymbol{V}=\boldsymbol{S}\ \boldsymbol{X}\ \boldsymbol{S}=\left(x_{\{j\},\ k}\right)$	$\boldsymbol{V}=\boldsymbol{SXS}=\left(x_{j,k}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03423		—	$\lambda_{\{j\}}=4\ \sin^{\{2\}}(j\ \pi\ h\ /\ 2),\ j=1,\ \ldots,\ m$	$\lambda_j=4\sin^2(j\pi h/2),j=1,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03424		—	$\sigma_{\{j\}}+\sigma_{\{k\}}-\frac{2}{3}\ \sigma_{\{j\}}\ \sigma_{\{k\}}>0$	$\sigma_j+\sigma_k-\frac{2}{3}\sigma_j\sigma_k>0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03425		—	$j,\ k=1,2,\ \ldots,\ m$	$j,k=1,2,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03426		—	$\boldsymbol{X}=\left(x_{\{j\},\ k}\right)$	$\boldsymbol{X}=\left(x_{j,k}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03427		—	$\boldsymbol{U}=\boldsymbol{S}\ \boldsymbol{X}\ \boldsymbol{S}$	$\boldsymbol{U}=\boldsymbol{SXS}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03428		—	$0\left(\delta\ n^{\{3\}}\ /\ 2\right)$	$O\left(\delta n^{3/2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03429		—	$\boldsymbol{F}=\$	$\boldsymbol{F}=\$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03430		—	$\boldsymbol{x}=\boldsymbol{A}\ \backslash\ \boldsymbol{b}$	$\boldsymbol{x}=\boldsymbol{A}\backslash\boldsymbol{b}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03431		—	$\boldsymbol{x} \in \mathbb{R}^{m^2}$	$\boldsymbol{x} \in \mathbb{R}^{m^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03432		—	U	U
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03433		—	$m=50$	$m = 50$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03434		—	$\boldsymbol{x}_{\{0\}}$	$\boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03435		—	$\left\{\boldsymbol{x}_{\{k\}}\right\}$	$\{\boldsymbol{x}_k\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03436		—	$\boldsymbol{x}_{\{k\}} \rightarrow \boldsymbol{x}$	$\boldsymbol{x}_k \rightarrow \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03437		—	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] \left[\begin{array}{l} y \\ z \end{array}\right] = \left[\begin{array}{l} 1 \\ 1 \end{array}\right]$	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right] \left[\begin{array}{l} y \\ z \end{array}\right] = \left[\begin{array}{l} 1 \\ 1 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03438		—	$y=(z+1) / 2$	$y = (z + 1)/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03439		—	$z=(y+1) / 2$	$z = (y + 1)/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03440		—	$y_{\{0\}}, z_{\{0\}}$	y_0, z_0
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03441		—	$\left\{y_{\{k\}}\right\}$	$\{y_k\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03442		—	$\left\{z_k\right\}$	$\{z_k\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03443		—	$y_{k+1}=\left(z_k+1\right) / 2$	$y_{k+1}=\left(z_k+1\right) / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03444		—	$z_{k+1}=\left(y_k+1\right) / 2$	$z_{k+1}=\left(y_k+1\right) / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03445		—	$y_{\{0\}}=z_{\{0\}}=0$	$y_0=z_0=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03446		—	$y_{\{1\}}=z_{\{1\}}=1 / 2$	$y_1=z_1=1 / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03447		—	$y_{\{k\}}=z_{\{k\}}=1-2^{\{-k\}}$	$y_k=z_k=1-2^{-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03448		—	$k=0,1,2,3, \ldots$	$k=0,1,2,3, \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03449		—	$[1,1]^{\{T\}}$	$[1,1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03450		—	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1}=\left(y_{k+1}+1\right) / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03451		—	$y_{\{1\}}=1 / 2, z_{\{1\}}=3 / 4, y_{\{2\}}=7 / 8$	$y_1=1 / 2, z_1=3 / 4, y_2=7 / 8$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03452		—	$z_{\{2\}}=15 / 16$	$z_2=15 / 16$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03453		—	$y_{\{k\}}=1\cdot 2 \cdot 4^{\{-k\}}$	$y_k = 1 - 2 \cdot 4^{-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03454		—	$z_{\{k\}}=1\cdot 4^{\{-k\}}$	$z_k = 1 - 4^{-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03455		—	$k=1,2,3, \ldots$	$k = 1,2,3,\ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03456		—	$\boldsymbol{x}_{\{k\}}=\left[\boldsymbol{x}_{\{k\}}(1), \ldots, \boldsymbol{x}_{\{k\}}(n)\right]^T$	$\boldsymbol{x}_k = [\boldsymbol{x}_k(1), \ldots, \boldsymbol{x}_k(n)]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03457		—	$\boldsymbol{x}(i)$	$\boldsymbol{x}(i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03458		—	$\boldsymbol{x}_{\{k+1\}}(i)$	$\boldsymbol{x}_{k+1}(i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03459		—	$0<\omega<2$	$0 < \omega < 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03460		—	$\boldsymbol{x}(i)=\omega \boldsymbol{x}(i)+(1-\omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i) = \omega \boldsymbol{x}(i) + (1 - \omega) \boldsymbol{x}(i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03461		—	$\omega=1$	$\omega = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03462		—	$\boldsymbol{x}_{\{k+1\}}^{\{g \ s\}}$	$\boldsymbol{x}_{k+1}^{gs}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03463		—	$\boldsymbol{x}_{\{k+1\}}=\omega \boldsymbol{x}_{\{k+1\}}^{\{g \ s\}}+(1-$	$\boldsymbol{x}_{k+1} = \omega \boldsymbol{x}_{k+1}^{gs} + (1-$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03464		—	$\omega) \, \boldsymbol{x}_{\{k\}}$	$\omega) \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03465		—	$\boldsymbol{x}_{\{k+1\}}$	$\boldsymbol{x}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03466		—	$1 \, \leq \, \omega < 2$	$1 \leq \omega < 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03467		—	$\boldsymbol{x}_{\{k+1 \, / \, 2\}}$	$\boldsymbol{x}_{k+1/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03468		—	$i=n, \, n-1, \, \ldots \, 1$	$i = n, n - 1, \ldots 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03469		—	$\boldsymbol{v}(i, \, j)$	$\boldsymbol{v}(i, j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03470		—	$\mathbf{e}_{\{i, \, j\}}:=\mathbf{f}_{\{i, \, j\}} \, /(\mathbf{m}+1)^{\{2\}}$	$e_{i,j} := f_{i,j}/(m+1)^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03471		—	$k_{\{100\}}$	k_{100}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03472		—	$k_{\{2500\}}$	k_{2500}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03473		—	$k_{\{10\}} \, 000$	$k_{10}000$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03474		—	$k_{\{40000\}}$	k_{40000}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03475		—	$k_{160} \ 000$	$k_{160}000$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03476		—	$\left(i_{1}, j_{1}\right)<$	$(i_1, j_1) <$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03477		—	i_{2}, j_{2}	i_2, j_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03478		—	$j_{1} \ \leq j_{2}$	$j_1 \leq j_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03479		—	$i_{1}<i_{2}$	$i_1 < i_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03480		—	$j_{1}=j_{2}$	$j_1 = j_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03481		—	$\boldsymbol{F}=\left(f_{i, j}\right) \in \mathbb{R}^{\{m \times m\}}$	$\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03482		—	$\boldsymbol{V}_{0}=\mathbf{0} \in$	$\boldsymbol{V}_0 = \boldsymbol{0} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03483		—	$\mathbb{R}^{\{ (m+2) \times (m+2) \}}$	$\mathbb{R}^{(m+2) \times (m+2)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03484		—	$\left \boldsymbol{V}^{\{k\}}-\boldsymbol{U}\right _{M}:=$	$\left\ \boldsymbol{V}^{(k)}-\boldsymbol{U}\right\ _M:=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03485		—	$\max _{i, j}\left v_{i j}-u_{i j}\right <$	$\max_{i,j} v_{ij} - u_{ij} <$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03486		—	$\left[u_{i\ j}\right] \in \mathbb{R}^{\left(m+2\right) \times \left(m+2\right)}$	$\left[u_{ij}\right] \in \mathbb{R}^{\left(m+2\right) \times \left(m+2\right)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03487		—	$k=k_{\left\{n\right\}}$	$k=k_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03488		—	$\boldsymbol{F}=\operatorname{ones}\left(m,\ m\right)$	$\boldsymbol{F}=\operatorname{ones}\left(m,m\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03489		—	$m=10,50,\ K=10^4$	$m=10,50,K=10^4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03490		—	$=10^{\left\{-8\right\}}$	$=10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03491		—	$\omega^{\ast}:=2\ /\left(1+\sin\ \left(\frac{\pi}{m+1}\right)\right)$	$\omega^{\ast}:=2/\left(1+\sin\left(\frac{\pi}{m+1}\right)\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03492		—	$0<w<2$	$0<w<2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03493		—	$w=1$	$w=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03494		—	$k_{\left\{n\right\}}=O\left(n\right)$	$k_n=O\left(n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03495		—	$k_{\left\{n\right\}}=O\left(\sqrt{n}\right)$	$k_n=O\left(\sqrt{n}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03496		—	$\sqrt{n}$	$\sqrt{n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03497		—	$0\left(k_n \times n\right)$	$O\left(k_n \times n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03498		—	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03499		—	$\boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$	$\boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03500		—	$\boldsymbol{I}-\boldsymbol{G}$	$\boldsymbol{I}-\boldsymbol{G}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03501		—	$\lim _k \rightarrow \infty \boldsymbol{x}_k=\boldsymbol{x}$	$\lim _{k \rightarrow \infty} \boldsymbol{x}_k=\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03502		—	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{M}^{-1} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03503		—	$\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}+\boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x}+\boldsymbol{M}^{-1} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03504		—	$\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}_{\boldsymbol{R}}$	$\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}_{\boldsymbol{R}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03505		—	$-\boldsymbol{A}_{\boldsymbol{L}}, \boldsymbol{D}$	$-\boldsymbol{A}_{\boldsymbol{L}}, \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03506		—	$-\boldsymbol{A}_{\boldsymbol{R}}$	$-\boldsymbol{A}_{\boldsymbol{R}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03507		—	$\boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right)$	$\boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{n n}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03508		—	$\boldsymbol{M}_J, \boldsymbol{M}_1$	M_J, M_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03509		—	$\boldsymbol{M}_{\omega}$	M_{ω}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03510		—	J, G S	J, GS
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03511		—	$\boldsymbol{D} \boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{D} \boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03512		—	$\boldsymbol{M}_J = \boldsymbol{D}$	$M_J = D$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03513		—	$\boldsymbol{A}_L \boldsymbol{x}_{k+1}$	$\boldsymbol{A}_L \boldsymbol{x}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03514		—	$(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{x}_k$	$(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + (\omega^{-1} - 1) \boldsymbol{D} \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03515		—	$\boldsymbol{M}_{\omega} =$	$M_{\omega} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03516		—	$\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L$	$\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03517		—	$\boldsymbol{M}_1$	M_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03518		—	$\boldsymbol{G}_{\omega} = \boldsymbol{I} - \boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}$	$G_{\omega} = I - M_{\omega}^{-1} A$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03519		—	$\boldsymbol{x}_{k+1}(1)$	$\boldsymbol{x}_{k+1}(1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03520		—	$\boldsymbol{x}_{k+1}:=\boldsymbol{G} \ \boldsymbol{x}_{k}+\boldsymbol{c}$	$\boldsymbol{x}_{k+1}:=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03521		—	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03522		—	$\lim _{k \rightarrow \infty} \boldsymbol{G}^k=\mathbf{0}$	$\lim_{k \rightarrow \infty} \boldsymbol{G}^k=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03523		—	$\boldsymbol{x}_{k+1}=\boldsymbol{G} \ \boldsymbol{x}_{k}+\boldsymbol{c}$	$\boldsymbol{x}_{k+1}=\boldsymbol{G}\boldsymbol{x}_k+\boldsymbol{c}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03524		—	$\boldsymbol{x}_{k+1}-\boldsymbol{x} =$	$\boldsymbol{x}_{k+1}-\boldsymbol{x} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03525		—	$\boldsymbol{G} \backslash \left(\boldsymbol{x}_{k}-\boldsymbol{x}\right)$	$\boldsymbol{G}\left(\boldsymbol{x}_k-\boldsymbol{x}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03526		—	$\boldsymbol{x}_{k}-\boldsymbol{x} \rightarrow \mathbf{0}$	$\boldsymbol{x}_k-\boldsymbol{x} \rightarrow \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03527		—	$\boldsymbol{G}^k \rightarrow \mathbf{0}$	$\boldsymbol{G}^k \rightarrow \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03528		—	$\boldsymbol{x}_{0}-\boldsymbol{x}=\boldsymbol{e}_{j}$	$\boldsymbol{x}_0-\boldsymbol{x}=\boldsymbol{e}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03529		—	$\boldsymbol{G}^k \boldsymbol{e}_{j} \rightarrow \mathbf{0}$	$\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03530		—	$\ \boldsymbol{G}\ <1$	$\ \boldsymbol{G}\ <1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03531		—	$\rho(\boldsymbol{G})<1$	$\rho(\boldsymbol{G})<1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03532		—	$\boldsymbol{r}_k:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{r}_k:=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03533		—	$\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_n>0$	$\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_n>0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03534		—	$\boldsymbol{x}_{k+1}=(\boldsymbol{I}-\alpha\boldsymbol{A})\boldsymbol{x}_k+\boldsymbol{b}$	$\boldsymbol{x}_{k+1}=(\boldsymbol{I}-\alpha\boldsymbol{A})\boldsymbol{x}_k+\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03535		—	$0<\alpha<2/\lambda_1$	$0<\alpha<2/\lambda_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03536		—	$\boldsymbol{I}-\alpha\boldsymbol{A}$	$\boldsymbol{I}-\alpha\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03537		—	$1-\alpha\lambda_j, \quad j=1, \ldots, n$	$1-\alpha\lambda_j, \quad j=1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03538		—	$1-\alpha\lambda_n=\alpha\lambda_1-1$	$1-\alpha\lambda_n=\alpha\lambda_1-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03539		—	$\alpha=\alpha_o$	$\alpha=\alpha_o$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03540		—	$\rho_\alpha<1$	$\rho_\alpha<1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03541		—	$\alpha \lambda_1 - 1 < 1$	$\alpha \lambda_1 - 1 < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03542		—	$\rho_{\alpha} > \rho_{\alpha_o}$	$\rho_{\alpha} > \rho_{\alpha_o}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03543		—	$\alpha \leq 0$	$\alpha \leq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03544		—	$\alpha \geq 2 / \lambda_1$	$\alpha \geq 2 / \lambda_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03545		—	$0 < \alpha < \alpha_o$	$0 < \alpha < \alpha_o$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03546		—	$\rho_{\alpha} = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$	$\rho_{\alpha} = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03547		—	$\alpha_o < \alpha < 2 / \lambda_1$	$\alpha_o < \alpha < 2 / \lambda_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03548		—	$\rho_{\alpha} = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$	$\rho_{\alpha} = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03549		—	$\mathbf{R}$	R
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03550		—	λ_{max}	λ_{max}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03551		—	$0 < \alpha < 2 / \lambda_{\text{max}}$	$0 < \alpha < 2 / \lambda_{\text{max}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03552		—	$\alpha=\alpha_{\text{o}}:=\frac{2}{\lambda_{\text{max}}+\lambda_{\text{min}}}$	$\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03553		—	$\kappa:=\lambda_{\text{max}} / \lambda_{\text{min}}$	$\kappa := \lambda_{\max} / \lambda_{\min}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03554		—	$\left\ \cdot \right\ _2$.	$\ \cdot \ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03555		—	$\ \boldsymbol{x}_{\text{k}} - \boldsymbol{x} \left\ _2 \leq \left\ \boldsymbol{I} -$	$\ \boldsymbol{x}_k - \boldsymbol{x} \ _2 \leq \ \boldsymbol{I} -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03556		—	$\alpha_{\text{o}} \boldsymbol{A} \left\ \cdot \right\ _2^k \boldsymbol{x}_0 - \boldsymbol{x} \left\ _2$	$\alpha_o \boldsymbol{A} \left\ \cdot \right\ _2^k \boldsymbol{x}_0 - \boldsymbol{x} \left\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03557		—	$\omega \in (0,2)$	$\omega \in (0,2)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03558		—	$\boldsymbol{D} \boldsymbol{x}_{\text{k+1}} = \omega \left(\boldsymbol{A}_{\text{L}} \boldsymbol{x}_{\text{k+1}} + \boldsymbol{A}_{\text{R}} \boldsymbol{x}_{\text{k}} + \boldsymbol{b} \right) + (1 - \omega) \boldsymbol{D} \boldsymbol{x}_{\text{k}}$	$\boldsymbol{D} \boldsymbol{x}_{k+1} = \omega \left(\boldsymbol{A}_L \boldsymbol{x}_{k+1} + \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} \right) + (1 - \omega) \boldsymbol{D} \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03559		—	$\boldsymbol{x}_{\text{k+1}} = \omega \left(\boldsymbol{L} \boldsymbol{x}_{\text{k+1}} + \boldsymbol{R} \boldsymbol{x}_{\text{k}} + \boldsymbol{D}^{-1} \boldsymbol{b} \right) + (1 - \omega) \boldsymbol{x}_{\text{k}}$	$\boldsymbol{x}_{k+1} = \omega \left(\boldsymbol{L} \boldsymbol{x}_{k+1} + \boldsymbol{R} \boldsymbol{x}_k + \boldsymbol{D}^{-1} \boldsymbol{b} \right) + (1 - \omega) \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03560		—	$\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_{\boldsymbol{L}}$	$\boldsymbol{L} := \boldsymbol{D}^{-1} \boldsymbol{A}_L$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03561		—	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_{\boldsymbol{R}}$	$\boldsymbol{R} := \boldsymbol{D}^{-1} \boldsymbol{A}_R$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03562		—	$(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{\text{k+1}} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_{\text{k}} + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$	$(\boldsymbol{I} - \omega \boldsymbol{L}) \boldsymbol{x}_{k+1} = (\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}) \boldsymbol{x}_k + \omega \boldsymbol{D}^{-1} \boldsymbol{b}$

Conf.
MathPix confidence:
■ ≥0.9
■ 0.5–0.9
■ <0.5
— no value

Residual
render vs scan (inkdrill):
● C component
● W weak
● S stable
● N noise
● K clean
● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03563		—	$\boldsymbol{G}_{\omega}$	$\boldsymbol{G}_{\omega}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03564		—	$\boldsymbol{I} - \omega \boldsymbol{L}$	$\boldsymbol{I} - \omega \boldsymbol{L}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03565		—	$\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}$	$\omega \boldsymbol{R} + (1 - \omega) \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03566		—	$1 - \omega$	$1 - \omega$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03567		—	$(1 - \omega)^n$	$(1 - \omega)^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03568		—	$\operatorname{det}(\boldsymbol{G}_{\omega}) = (1 - \omega)^n$	$\det(\boldsymbol{G}_{\omega}) = (1 - \omega)^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03569		—	$ \lambda \geq 1 - \omega $	$ \lambda \geq 1 - \omega $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03570		—	$\rho(\boldsymbol{G}_{\omega}) \geq \omega - 1 $	$\rho(\boldsymbol{G}_{\omega}) \geq \omega - 1 $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03571		—	$\rho(\boldsymbol{G}_{\omega}) \geq 1$	$\rho(\boldsymbol{G}_{\omega}) \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03572		—	$ \lambda < 1$	$ \lambda < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03573		—	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03574		—	$\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{D} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03575		—	$\boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{x} \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03576		—	$\boldsymbol{a}_{ii} =$	$\boldsymbol{a}_{ii} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03577		—	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > 0$	$\boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03578		—	$ \lambda \leq 1$	$ \lambda \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03579		—	$\boldsymbol{G}_\omega \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{G}_\omega \boldsymbol{x} = \lambda \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03580		—	$\boldsymbol{x} - (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{x} - (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03581		—	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03582		—	$\lambda \neq 1$	$\lambda \neq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03583		—	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03584		—	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03585		—	$\boldsymbol{y}^*(\lambda \boldsymbol{A} \boldsymbol{x}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03586		—	$(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03587		—	$\boldsymbol{y}^* = (1 - \lambda) \boldsymbol{x}^*$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03588		—	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A} / 4$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03589		—	$\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03590		—	$\boldsymbol{G}_J$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03591		—	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03592		—	$\alpha = 2 / (\lambda_{\max} + \lambda_{\min}) = 1 / 4$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03593		—	κ	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03594		—	$\rho(\boldsymbol{G}_\omega)$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03595		—	$\omega \in$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03596		—	$\backslash\mathrm{beta}:=\backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\mathrm{J}}\backslash\mathrm{right})=\backslash\cos\ (\backslash\mathrm{pi}\ h)$	$\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03597		—	$\backslash\mathrm{omega}^{\mathrm{*}}$	ω^*
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03598		—	$\backslash\mathrm{omega}=2$	$\omega = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03599		—	$\backslash\mathrm{beta}=\backslash\cos\ (\backslash\mathrm{pi}\ h)$	$\beta = \cos(\pi h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03600		—	$\backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\mathrm{1}}\backslash\mathrm{right})=\backslash\mathrm{beta}^{\mathrm{2}}=\backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\mathrm{J}}\backslash\mathrm{right})^{\mathrm{2}}=\backslash\cos\ ^{\mathrm{2}}(\backslash\mathrm{pi}\ h)$	$\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03601		—	$\backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\mathrm{J}}\backslash\mathrm{right}),\ \backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\mathrm{1}}\backslash\mathrm{right})$	$\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03602		—	$\backslash\rho\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{G}}_{\backslash\mathrm{omega}^{\mathrm{*}}}\backslash\mathrm{right})=\backslash\mathrm{omega}^{\mathrm{*}}-1$	$\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03603		—	$\backslash\rho(\backslash\mathrm{boldsymbol{G}})^{\mathrm{k}_{\mathrm{n}}}\ \backslash\mathrm{leq}$	$\rho(\boldsymbol{G})^{k_n} \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03604		—	$\backslash\mathrm{mathrm{n}}=100$	$\mathbf{n} = 100$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03605		—	$\backslash\mathrm{mathrm{n}}=2500$	$\mathbf{n} = 2500$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03606		—	$\backslash\mathrm{boldsymbol{x}}_{\mathrm{k}}-\backslash\mathrm{boldsymbol{x}}=\backslash\mathrm{boldsymbol{G}}^{\mathrm{k}}\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{x}}_{\mathrm{0}}-\backslash\mathrm{boldsymbol{x}}\backslash\mathrm{right})$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k(\boldsymbol{x}_0 - \boldsymbol{x})$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03607		—	$\lim_{k \rightarrow \infty} \left \boldsymbol{G}^k \right ^{\frac{1}{k}} =$	$\lim_{k \rightarrow \infty} \left\ \boldsymbol{G}^k \right\ ^{1/k} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03608		—	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03609		—	$\left \boldsymbol{G}_{\boldsymbol{J}}^k \right _2 = \rho(\boldsymbol{G}_{\boldsymbol{J}})^k$	$\left\ \boldsymbol{G}_J^k \right\ _2 = \rho(\boldsymbol{G}_J)^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03610		—	$\rho(\boldsymbol{G})=1-\eta$	$\rho(\boldsymbol{G}) = 1 - \eta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03611		—	$0<\eta<1$	$0 < \eta < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03612		—	$s \in \mathbb{N}$	$s \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03613		—	$\rho(\boldsymbol{G})^k \leq 10^{-s}$	$\rho(\boldsymbol{G})^k \leq 10^{-s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03614		—	$\tilde{k}$	$\tilde{k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03615		—	$\rho(\boldsymbol{G})^k=10^{-s}$	$\rho(\boldsymbol{G})^k = 10^{-s}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03616		—	$-\log(1-\eta) \approx \eta$	$-\log(1 - \eta) \approx \eta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03617		—	η	η
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03618		—	$\rho\left(\boldsymbol{G}_{\text{J}}\right)=\cos \left(\pi h\right)=1-\eta, \eta=1-\cos \left(\pi h\right)=\frac{1}{2} \pi^{2} h^{2}+O\left(h^{4}\right)=$	$\rho\left(\boldsymbol{G}_{\text{J}}\right)=\cos (\pi h)=1-\eta, \eta=1-\cos (\pi h)=\frac{1}{2} \pi^2 h^2+O\left(h^4\right)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03619		—	$\frac{\pi^2}{2} / n+O\left(n^{-2}\right)$	$\frac{\pi^2}{2} / n+O\left(n^{-2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03620		—	$\operatorname{GS}:\rho\left(\boldsymbol{G}_1\right)=\cos ^2(\pi h)=1-\eta, \eta=1-\cos ^2(\pi h)=\sin ^2 \pi h=\pi^2 h^2+$ $\pi h)=\sin ^2 \pi h=\pi^{2} h^2+$	$\operatorname{GS}:\rho\left(\boldsymbol{G}_1\right)=\cos ^2(\pi h)=1-\eta, \eta=1-\cos ^2(\pi h)=\sin ^2 \pi h=\pi^2 h^2+$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03621		—	$0\left(h^4\right)=\pi^{2} / n+O\left(n^{-2}\right)$	$O\left(h^4\right)=\pi^2 / n+O\left(n^{-2}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03622		—	$\rho\left(\boldsymbol{G}_{\omega^*}\right)=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}=1-2 \pi h+O\left(h^2\right)$	$\rho\left(\boldsymbol{G}_{\omega^*}\right)=\frac{1-\sin (\pi h)}{1+\sin (\pi h)}=1-2 \pi h+O\left(h^2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03623		—	$\boldsymbol{G}^k$	$\boldsymbol{G}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03624		—	$\lim _{k \rightarrow \infty}\left\ \boldsymbol{G}^k\right\ ^{1 / k}=\rho(\boldsymbol{G})$	$\lim _{k \rightarrow \infty}\left\ \boldsymbol{G}^k\right\ ^{1 / k}=\rho(\boldsymbol{G})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03625		—	$\left\ \boldsymbol{x}_{k+1}-\boldsymbol{x}_k\right\ $	$\left\ \boldsymbol{x}_{k+1}-\boldsymbol{x}_k\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03626		—	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ $	$\left\ \boldsymbol{x}_k-\boldsymbol{x}\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03627		—	$\ \boldsymbol{G}\ $	$\ \boldsymbol{G}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03628		—	$\boldsymbol{x}_k=\boldsymbol{G} \boldsymbol{x}_{k-1}+\boldsymbol{c}, \boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$	$\boldsymbol{x}_k=\boldsymbol{G} \boldsymbol{x}_{k-1}+\boldsymbol{c}, \boldsymbol{x}=\boldsymbol{G} \boldsymbol{x}+\boldsymbol{c}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03629		—	$(1-\ \boldsymbol{G}\)\ \boldsymbol{x}_{k+1}-\boldsymbol{x}_k\ \leq\ \boldsymbol{G}\ \ \boldsymbol{x}_k-\boldsymbol{x}\ $	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03630		—	$\boldsymbol{M}\boldsymbol{x}_{k+1}=\boldsymbol{M}\boldsymbol{x}_k-\boldsymbol{A}\boldsymbol{x}_k+\boldsymbol{b}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03631		—	$\ \boldsymbol{A}^k\ $	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03632		—	$\lim_{k\rightarrow\infty}\boldsymbol{A}^k=\mathbf{0}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03633		—	$\rho(\boldsymbol{A})$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03634		—	$\rho(\boldsymbol{A})<1$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03635		—	$ \lambda \geq 1$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03636		—	$\boldsymbol{A}^k\boldsymbol{x}=\lambda^k\boldsymbol{x}$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03637		—	$\ \boldsymbol{A}^k\ _2\geq\ \boldsymbol{A}^k\boldsymbol{x}\ _2=\ \lambda^k\boldsymbol{x}\ _2= \lambda ^k$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03638		—	$\ \boldsymbol{A}\ <1$	
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03639		—	$\rho(\boldsymbol{A})\leq$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03640		—	$\ \mathbf{A}\ $	$\ \mathbf{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03641		—	$\mathbf{A}, \ \cdot\ $	$\mathbf{A}, \ \cdot\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03642		—	$\mathbf{X} := [\mathbf{x}, \ldots, \mathbf{x}] \in \mathbb{C}^{n \times n}$	$\mathbf{X} := [\mathbf{x}, \ldots, \mathbf{x}] \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03643		—	$\lambda \mathbf{X} = \mathbf{A} \mathbf{X}$	$\lambda \mathbf{X} = \mathbf{A} \mathbf{X}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03644		—	$ \lambda \ \mathbf{X}\ =$	$ \lambda \ \mathbf{X}\ =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03645		—	$\ \lambda \mathbf{X}\ = \ \mathbf{A} \mathbf{X}\ \leq \ \mathbf{A}\ \ \mathbf{X}\ $	$\ \lambda \mathbf{X}\ = \ \mathbf{A} \mathbf{X}\ \leq \ \mathbf{A}\ \ \mathbf{X}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03646		—	$\ \mathbf{X}\ \neq 0$	$\ \mathbf{X}\ \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03647		—	$ \lambda \leq \ \mathbf{A}\ $	$ \lambda \leq \ \mathbf{A}\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03648		—	$\rho(\mathbf{A}) \leq \ \mathbf{A}\ \leq \rho(\mathbf{A}) + \epsilon$	$\rho(\mathbf{A}) \leq \ \mathbf{A}\ \leq \rho(\mathbf{A}) + \epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03649		—	$\mathbf{R} = [r_{ij}]$	$\mathbf{R} = [r_{ij}]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03650		—	$\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{R}$	$\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03651		—	$\operatorname{diag}\left(t, t^2, \ldots, t^n\right) \in \mathbb{R}^n \times n$	$D_t := \operatorname{diag}\left(t, t^2, \ldots, t^n\right) \in \mathbb{R}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03652		—	(i, j)	(i, j)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03653		—	$\boldsymbol{D}_{\boldsymbol{t}} \boldsymbol{R} \boldsymbol{D}_{\boldsymbol{t}}^{-1}$	$D_t R D_t^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03654		—	$t^{i-j} r_{ij}$	$t^{i-j} r_{ij}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03655		—	$\left \boldsymbol{B}\right _{\boldsymbol{t}}:=\left \boldsymbol{D}_{\boldsymbol{t}} \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{D}_{\boldsymbol{t}}^{-1}\right _1$	$\left\ \boldsymbol{B}\right\ _t:=\left\ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{D}_t^{-1}\right\ _1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03656		—	$\left \right _{\boldsymbol{t}}\right .$	$\left \right _{\boldsymbol{t}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03657		—	$\left \boldsymbol{B}\right :=\left \boldsymbol{B}\right _{\boldsymbol{t}}$	$\left\ \boldsymbol{B}\right\ :=\left\ \boldsymbol{B}\right\ _t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03658		—	λ^k	λ^k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03659		—	$\rho(\boldsymbol{A})^k=\rho\left(\boldsymbol{A}^k\right) \leq\left \boldsymbol{A}^k\right $	$\rho(\boldsymbol{A})^k=\rho\left(\boldsymbol{A}^k\right) \leq\left\ \boldsymbol{A}^k\right\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03660		—	$\rho(\boldsymbol{A}) \leq\left \boldsymbol{A}^k\right ^{1 / k}$	$\rho(\boldsymbol{A}) \leq\left\ \boldsymbol{A}^k\right\ ^{1 / k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03661		—	$\boldsymbol{B}:=\left(\rho(\boldsymbol{A})+\epsilon\right)^{-1} \boldsymbol{A}$	$\boldsymbol{B}:=\left(\rho(\boldsymbol{A})+\epsilon\right)^{-1} \boldsymbol{A}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03662		—	$\rho(\boldsymbol{B})=\rho(\boldsymbol{A})\ /\ (\rho(\boldsymbol{A})+\epsilon)<1$	$\rho(\boldsymbol{B})=\rho(\boldsymbol{A})/(\rho(\boldsymbol{A})+\epsilon)<1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03663		—	$\left \boldsymbol{B}^k\right \rightarrow 0$	$\left\ \boldsymbol{B}^k\right\ \rightarrow 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03664		—	$k\rightarrow \infty$	$k\rightarrow \infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03665		—	$\left \boldsymbol{B}^k\right <1$	$\left\ \boldsymbol{B}^k\right\ <1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03666		—	$k\geq N$	$k\geq N$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03667		—	$\rho(\boldsymbol{A})\ \leq\left \boldsymbol{A}^k\right ^{\frac{1}{k}}\leq \rho(\boldsymbol{A})+\epsilon$	$\rho(\boldsymbol{A})\leq\left\ \boldsymbol{A}^k\right\ ^{1/k}\leq\rho(\boldsymbol{A})+\epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03668		—	$\sum_{k=0}^{\infty}\boldsymbol{A}_k$	$\sum_{k=0}^{\infty}\boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03669		—	$\left\{\boldsymbol{S}_{_m}\right\}$	$\{\boldsymbol{S}_m\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03670		—	$\boldsymbol{S}_{_m}=\sum_{k=0}^m\boldsymbol{A}_{_k}$	$\boldsymbol{S}_m=\sum_{k=0}^m\boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03671		—	$\left \boldsymbol{S}_{_l}-\boldsymbol{S}_{_m}\right <\epsilon$	$\left\ \boldsymbol{S}_l-\boldsymbol{S}_m\right\ <\epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03672		—	$l>m\geq N$	$l>m\geq N$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03673		—	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03674		—	$\rho(\boldsymbol{B}) < 1$	$\rho(\boldsymbol{B}) < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03675		—	$(\boldsymbol{I} - \boldsymbol{B})$	$(\boldsymbol{I} - \boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03676		—	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03677		—	$\ \boldsymbol{B}\ < 1$	$\ \boldsymbol{B}\ < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03678		—	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03679		—	$l > m$	$l > m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03680		—	$\frac{\ \boldsymbol{B}\ ^{N+1}}{1 - \ \boldsymbol{B}\ } < \epsilon$	$\frac{\ \boldsymbol{B}\ ^{N+1}}{1 - \ \boldsymbol{B}\ } < \epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03681		—	$\boldsymbol{S}_m \boldsymbol{x} =$	$\boldsymbol{S}_m \boldsymbol{x} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03682		—	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03683		—	$\sum_{k=0}^{\infty} \lambda^k$	$\sum_{k=0}^{\infty} \lambda^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03684		—	$\left\{\boldsymbol{S}_{\mathfrak{m}} \boldsymbol{x}\right\}$	$\left\{\boldsymbol{S}_m \boldsymbol{x}\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03685		—	$\boldsymbol{B}^{m+1} \rightarrow 0$	$\boldsymbol{B}^{m+1} \rightarrow 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03686		—	$\left(\sum_{k=0}^{\infty} \boldsymbol{B}^k\right) (\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}$	$\left(\sum_{k=0}^{\infty} \boldsymbol{B}^k\right) (\boldsymbol{I}-\boldsymbol{B})=\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03687		—	$\left \left(\boldsymbol{I}-\boldsymbol{B}\right)^{-1}\right =\left \sum_{k=0}^{\infty} \boldsymbol{B}^k\right \leq \sum_{k=0}^{\infty}\left \boldsymbol{B}\right ^k=\frac{1}{1-\left \boldsymbol{B}\right }$	$\left\ \left(\boldsymbol{I}-\boldsymbol{B}\right)^{-1}\right\ =\left\ \sum_{k=0}^{\infty} \boldsymbol{B}^k\right\ \leq \sum_{k=0}^{\infty}\left\ \boldsymbol{B}\right\ ^k=\frac{1}{1-\left\ \boldsymbol{B}\right\ }$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03688		—	$\boldsymbol{\omega}$	$\boldsymbol{\omega}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03689		—	$\boldsymbol{G}_{\mathfrak{J}} \boldsymbol{v}=\mu \boldsymbol{v}$	$\boldsymbol{G}_J \boldsymbol{v}=\mu \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03690		—	$\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$	$\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03691		—	$v_{i, j}=0$	$v_{i, j}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03692		—	$i \in\{0, m+1\}$	$i \in\{0, m+1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03693		—	$j \in\{0, m+1\}$	$j \in\{0, m+1\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03694		—	$(\lambda, \boldsymbol{w})$	$(\lambda, \boldsymbol{w})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03695		—	$(12.21) (\boldsymbol{I} - \omega \boldsymbol{L})^{-1} (\omega \boldsymbol{R} + (1 -$	$(12.21) (\boldsymbol{I} - \omega \boldsymbol{L})^{-1} (\omega \boldsymbol{R} + (1 -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03696		—	$\omega) \boldsymbol{I}) \boldsymbol{w} = \lambda \boldsymbol{w}$	$\omega) \boldsymbol{I}) \boldsymbol{w} = \lambda \boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03697		—	$\boldsymbol{w} = \operatorname{vec}(\boldsymbol{W})$	$\boldsymbol{w} = \operatorname{vec}(\boldsymbol{W})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03698		—	$\boldsymbol{W} \in \mathbb{C}^{m \times m}$	$\boldsymbol{W} \in \mathbb{C}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03699		—	$w_{i,j} = 0$	$w_{i,j} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03700		—	$(\mu, \boldsymbol{v})$	$(\mu, \boldsymbol{v})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03701		—	$\boldsymbol{v} = (\boldsymbol{V})$	$\boldsymbol{v} = (\boldsymbol{V})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03702		—	$v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$	$v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03703		—	$w_{i,j}$	$w_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03704		—	$\lambda^{(i+j)/2} v_{i,j}$	$\lambda^{(i+j)/2} v_{i,j}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03705		—	$\lambda^{(i+j) / 2}$	$\lambda^{(i+j)/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03706		—	$\boldsymbol{v}:=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{w}=\operatorname{vec}(\boldsymbol{W})$	$\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03707		—	$w_{i, j}:=\lambda^{(i+j) / 2} v_{i, j}$	$w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03708		—	$\lambda^{-(i+j) / 2}$	$\lambda^{-(i+j)/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03709		—	$\omega \lambda^{1 / 2}$	$\omega \lambda^{1/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03710		—	$\omega \mu \lambda^{1 / 2}=\lambda+\omega-1$	$\omega \mu \lambda^{1/2} = \lambda + \omega - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03711		—	$\lambda, \boldsymbol{w}$	$\lambda, \boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03712		—	$\rho(\boldsymbol{G}_{\omega})= \lambda(\mu) $	$\rho(\boldsymbol{G}_{\omega}) = \lambda(\mu) $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03713		—	$\lambda(\mu)$	$\lambda(\mu)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03714		—	$\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h)$	$\frac{1}{2} \cos (j \pi h) + \frac{1}{2} \cos (k \pi h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03715		—	$-\mu$	$-\mu$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03716		—	$\backslash\lambda=\backslash\lambda(\mu)$	$\lambda=\lambda(\mu)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03717		—	$d(0)=4>0$	$d(0)=4>0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03718		—	$d(2)=4\ \mu^2-4<0$	$d(2)=4\mu^2-4<0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03719		—	$\omega,\ \backslash\lambda(\mu)$	$\omega,\lambda(\mu)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03720		—	$d(\omega)=0$	$d(\omega)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03721		—	$\tilde{\omega}(\mu)$	$\tilde{\omega}(\mu)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03722		—	$ \backslash\lambda(\mu) $	$ \lambda(\mu) $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03723		—	$\mu=\rho\left(\boldsymbol{G}_{J}\right)=:\beta$	$\mu=\rho(\boldsymbol{G}_J)=:\beta$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03724		—	$0<\omega\leq\tilde{\omega}(\beta)=\omega^*$	$0<\omega\leq\tilde{\omega}(\beta)=\omega^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03725		—	$\rho\left(\boldsymbol{G}_{\omega}\right)=\omega-1$	$\rho(\boldsymbol{G}_{\omega})=\omega-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03726		—	$\omega^*<\omega<2$	$\omega^*<\omega<2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03727		—	$0<\omega<\omega^{\ast}$	$0<\omega<\omega^{\ast}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03728		—	$\beta^2\left(\omega^2\beta^2-\beta^2-4\omega+4\right)<\left(2-\omega\beta^2\right)^2$	$\beta^2\left(\omega^2\beta^2-4\omega+4\right)<\left(2-\omega\beta^2\right)^2$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03729		—	$a_{ii}=d\neq 0$	$a_{ii}=d\neq 0$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03730		—	$\alpha:=1/d$	$\alpha:=1/d$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03731		—	$u_{\{j\}}$	u_j
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03732		—	$0<\alpha<\min_{\{j\}}\left(2u_{\{j\}}/\left \lambda_{\{j\}}\right ^2\right)$	$0<\alpha<\min_j\left(2u_j/ \lambda_j ^2\right)$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03733		—	$\boldsymbol{A}=\left[\begin{array}{ll}1&2\\3&4\end{array}\right]$	$\boldsymbol{A}=\left[\begin{array}{cc}1&2\\3&4\end{array}\right]$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03734		—	$\boldsymbol{A}:=\left[\begin{array}{l}a_{11}\text{ \& }a_{12}\\a_{21}\text{ \& }a_{22}\end{array}\right]\text{ \& }a_{21}\text{ \& }a_{22}\in\mathbb{R}^{2\times 2}$	$\boldsymbol{A}:=\left[\begin{array}{cc}a_{11}&a_{12}\\a_{21}&a_{22}\end{array}\right]\in\mathbb{R}^{2\times 2}$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03735		—	$\left[\begin{array}{ccc}1&a&a\\a&1&a\\a&a&1\end{array}\right]$	$\left[\begin{array}{ccc}1&a&a\\a&1&a\\a&a&1\end{array}\right]$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03736		—	$-1/2<a<1$	$-1/2<a<1$
Numerical_Linear_ Algebra_and_Matrix_ Factorizations_ _Tom_Lyche____ Z-Library__F03737		—	$1/2<a<1$	$1/2<a<1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03738		—	$\boldsymbol{G}_{1}$	$\boldsymbol{G}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03739		—	$\boldsymbol{A}:=\left[\begin{array}{ccc}1&0&1/2\\1&1&0\\-1&1&1\end{array}\right].1$	$\boldsymbol{A}:=\left[\begin{array}{ccc}1&0&1/2\\1&1&0\\-1&1&1\end{array}\right].1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03740		—	$\boldsymbol{G}_{1}:=\left[\begin{array}{ccc}0&0&-1/2\\0&0&1/2\\0&0&-1\end{array}\right]$	$\boldsymbol{G}_1:=\left[\begin{array}{ccc}0&0&-1/2\\0&0&1/2\\0&0&-1\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03741		—	$p(\lambda):=\operatorname{det}\left(\lambda\boldsymbol{I}-\boldsymbol{G}_J\right)=\lambda^3+\frac{1}{2}\lambda+\frac{1}{2}$	$p(\lambda):=\det(\lambda\boldsymbol{I}-\boldsymbol{G}_J)=\lambda^3+\frac{1}{2}\lambda+\frac{1}{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03742		—	$p(\lambda)\neq 0$	$p(\lambda)\neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03743		—	$\left a_{i\ i}\right >\sum_{j\neq i}\left a_{i\ j}\right $	$ a_{ii} >\sum_{j\neq i} a_{ij} $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03744		—	$r:=\max_{i}r_i<1$	$r:=\max_i r_i<1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03745		—	$r_i=\sum_{j\neq i}\frac{\left a_{i\ j}\right }{\left a_{i\ i}\right }$	$r_i=\sum_{j\neq i}\frac{ a_{ij} }{ a_{ii} }$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03746		—	$\left \boldsymbol{\epsilon}_{k+1}(j)\right \leq r\left \boldsymbol{\epsilon}_k\right _{\infty}$	$ \boldsymbol{\epsilon}_{k+1}(j) \leq r\ \boldsymbol{\epsilon}_k\ _\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03747		—	$j=1,\ldots,i$	$j=1,\dots,i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03748		—	$a\in$	$a\in$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03749		—	$\boldsymbol{x}_{\{0\}}=\mathbf{0}$	$\boldsymbol{x}_0 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03750		—	$\boldsymbol{x}=[1,1]^T$	$\boldsymbol{x} = [1,1]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03751		—	$ a <1$	$ a < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03752		—	$ a >1$	$ a > 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03753		—	$\eta=1- a $	$\eta = 1 - a $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03754		—	$a=0.9$	$a = 0.9$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03755		—	$s=16$	$s = 16$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03756		—	$ a ^{\{k\}} \leq 10^{\{-16\}}$	$ a ^k \leq 10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03757		—	$\rho\left(\boldsymbol{G}_{\{J\}}\right)=1 / 2$	$\rho\left(\boldsymbol{G}_J\right)=1 / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03758		—	$\boldsymbol{x}_{\{k\}}(1)=\boldsymbol{x}_{\{k\}}(2)=1-2^{\{-k\}}$	$\boldsymbol{x}_k(1)=\boldsymbol{x}_k(2)=1-2^{-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03759		—	$\boldsymbol{A}=\boldsymbol{I}-\boldsymbol{L}-\boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{I} - \boldsymbol{L} - \boldsymbol{L}^T$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03760		—	$\mathbb{I}_{i,j}=0$	$l_{i,j} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03761		—	$j \geq i$	$j \geq i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03762		—	$\mathbf{N}$	$\mathbf{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03763		—	$\mathbf{M}^{-1} \mathbf{N}$	$\mathbf{M}^{-1} \mathbf{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03764		—	$\mathbf{x}=[x(1), x(2), \ldots, x(n)]^T$	$\mathbf{x} = [x(1), x(2), \ldots, x(n)]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03765		—	$i=1,2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$	$i = 1, 2, \ldots, n-1, n, n, n-1, n-2, \ldots, 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03766		—	$\mathbf{b}:=[1,9,-2]^T$	$\mathbf{b} := [1, 9, -2]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03767		—	$\mathbf{x}_{\{0\}}=[1,1,1]^t$	$\mathbf{x}_0 = [1, 1, 1]^t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03768		—	$\mathbf{x}_{\{1\}} \in \mathbb{R}^3$	$\mathbf{x}_1 \in \mathbb{R}^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03769		—	$\mathbf{E} \in \mathbb{C}^{n \times n}$	$\mathbf{E} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03770		—	$\rho(\mathbf{A}^{-1} \mathbf{E}) < 1$	$\rho(\mathbf{A}^{-1} \mathbf{E}) < 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03771		—	$\lim_{k \rightarrow \infty} \parallel$	$\lim_{k \rightarrow \infty} \parallel$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03772		—	$\ \boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A})$	$\boldsymbol{A}^k\ ^{1/k} = \rho(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03773		—	$ \lambda = \rho(\boldsymbol{A}) < 1$	$ \lambda = \rho(\boldsymbol{A}) < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03774		—	$(1, n)$	$(1, n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03775		—	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03776		—	$k \geq n-1$	$k \geq n-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03777		—	$n=5$	$n = 5$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03778		—	$f(k)$	$f(k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03779		—	$\lambda=0.9, a=10$	$\lambda = 0.9, a = 10$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03780		—	$n-1 \leq k \leq 200$	$n-1 \leq k \leq 200$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03781		—	$\max_k f(k)$	$\max_k f(k)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03782		—	$f(k)<10^{-8}$	$f(k) < 10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03783		—	$\boldsymbol{E}:= (\boldsymbol{A}-\lambda \boldsymbol{I}) / a$	$\boldsymbol{E} := (\boldsymbol{A} - \lambda \boldsymbol{I}) / a$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03784		—	$\boldsymbol{E}^k=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{array}\right]$	$\boldsymbol{E}^k = \left[\begin{array}{cc} \mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0} \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03785		—	$\boldsymbol{E}^n=\mathbf{0}$	$\boldsymbol{E}^n = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03786		—	$\boldsymbol{A}^k=(a\boldsymbol{E}+\lambda \boldsymbol{I})^k=\sum_{j=0}^{\min\left\{k,n-1\right\}}\binom{k}{j}a^j\lambda^{k-j}\boldsymbol{E}^j$	$\boldsymbol{A}^k = (a\boldsymbol{E} + \lambda \boldsymbol{I})^k = \sum_{j=0}^{\min\{k,n-1\}} \binom{k}{j} a^j \lambda^{k-j} \boldsymbol{E}^j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03787		—	$\boldsymbol{A}^n\rightarrow \mathbf{0}$	$\boldsymbol{A}^n \rightarrow \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03788		—	$\ \boldsymbol{A}\ <\rho(\boldsymbol{A})$	$\ \boldsymbol{A}\ < \rho(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03789		—	$\kappa:=\frac{\lambda_{\text{max}}}{\lambda_{\text{min}}}$	$\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03790		—	$\langle \boldsymbol{x},\boldsymbol{y}\rangle,\boldsymbol{x}\rangle\boldsymbol{y}=\boldsymbol{x}^T\boldsymbol{y}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03791		—	$\ \boldsymbol{x}\ _2=\sqrt{\boldsymbol{x}^T\boldsymbol{x}}$	$\ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03792		—	$\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}\in$	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03793		—	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \geq 0$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03794		—	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03795		—	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} = (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y})^T = \boldsymbol{y}^T \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle_{\boldsymbol{A}}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} = (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y})^T = \boldsymbol{y}^T \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03796		—	$\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{z} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{z} = \langle \boldsymbol{x}, \boldsymbol{z} \rangle_{\boldsymbol{A}} + \langle \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}$	$\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{z} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{z} = \langle \boldsymbol{x}, \boldsymbol{z} \rangle_{\boldsymbol{A}} + \langle \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03797		—	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03798		—	$c \in \mathbb{R}$	$c \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03799		—	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03800		—	$\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$	$\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03801		—	$\varepsilon \in \mathbb{R}$	$\varepsilon \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03802		—	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{y}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations__Tom_Lyche____Z-Library__F03803		—	$\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$	$\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03804		—	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03805		—	$\boldsymbol{y} = \boldsymbol{x}, \varepsilon = 1$	$\boldsymbol{y} = \boldsymbol{x}, \varepsilon = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03806		—	$Q(\boldsymbol{x} + \boldsymbol{h}) =$	$Q(\boldsymbol{x} + \boldsymbol{h}) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03807		—	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03808		—	$Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$	$Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03809		—	$\boldsymbol{h} \in \mathbb{R}^n$	$\boldsymbol{h} \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03810		—	$\boldsymbol{A} \boldsymbol{x} \neq \boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{x} \neq \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03811		—	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03812		—	$Q(\boldsymbol{x}) + \varepsilon \boldsymbol{h} - Q(\boldsymbol{x}) = -\varepsilon \left(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h} \right) < 0$	$Q(\boldsymbol{x} + \varepsilon \boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon \left(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \varepsilon \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h} \right) < 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03813		—	$\boldsymbol{x}_0 \in \mathbb{R}^n$	$\boldsymbol{x}_0 \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03814		—	α_k	α_k
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03815		—	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\alpha_k \boldsymbol{p}_k$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03816		—	$Q\left(\boldsymbol{x}_{k+1}\right)$	$Q\left(\boldsymbol{x}_{k+1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03817		—	$Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=Q\left(\boldsymbol{x}_k\right)-\alpha \boldsymbol{p}_k^T \boldsymbol{r}_k+\frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$	$Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=Q\left(\boldsymbol{x}_k\right)-\alpha \boldsymbol{p}_k^T \boldsymbol{r}_k+\frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03818		—	$\boldsymbol{r}_k:=$	$\boldsymbol{r}_k :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03819		—	$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03820		—	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03821		—	$\frac{\partial}{\partial \alpha} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=0$	$\frac{\partial}{\partial \alpha} Q\left(\boldsymbol{x}_k+\alpha \boldsymbol{p}_k\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03822		—	$\boldsymbol{p}_k=\boldsymbol{r}_k$	$\boldsymbol{p}_k = \boldsymbol{r}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03823		—	$k=0,1,2 \ldots$	$k=0,1,2 \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03824		—	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k=\left(\boldsymbol{r}_k-\alpha_k \boldsymbol{A} \boldsymbol{r}_k\right)^T \boldsymbol{r}_k=0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k=\left(\boldsymbol{r}_k-\alpha_k \boldsymbol{A} \boldsymbol{r}_k\right)^T \boldsymbol{r}_k=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03825		—	$\alpha_k=$	$\alpha_k =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03826		—	$\frac{\boldsymbol{r}_{k}^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k} \boldsymbol{r}_k$	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03827		—	$\boldsymbol{x}_0 = [-1, -1 / 2]^T$	$\boldsymbol{x}_0 = [-1, -1/2]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03828		—	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3 / 2, 0]^T$	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03829		—	$k \geq 1$	$k \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03830		—	$\alpha_k = 1 / 2$	$\alpha_k = 1/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03831		—	$\frac{\ \boldsymbol{x}_{j+1}\ _2}{\ \boldsymbol{x}_j\ _2} = \frac{\ \boldsymbol{r}_{j+1}\ _2}{\ \boldsymbol{r}_j\ _2} = 1 / 2$	$\frac{\ \boldsymbol{x}_{j+1}\ _2}{\ \boldsymbol{x}_j\ _2} = \frac{\ \boldsymbol{r}_{j+1}\ _2}{\ \boldsymbol{r}_j\ _2} = 1/2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03832		—	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03833		—	$\boldsymbol{r}_0 =$	$\boldsymbol{r}_0 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03834		—	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03835		—	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03836		—	$\boldsymbol{p}_0 := \boldsymbol{r}_0$	$\boldsymbol{p}_0 := \boldsymbol{r}_0$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03837		—	$j \geq 0$	$j \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03838		—	$\boldsymbol{p}_j$	$\boldsymbol{p}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03839		—	$\boldsymbol{r}_0, \ldots, \boldsymbol{r}_j$	$\boldsymbol{r}_0, \ldots, \boldsymbol{r}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03840		—	α_j	α_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03841		—	$\boldsymbol{x}_j$	$\boldsymbol{x}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03842		—	$\boldsymbol{r}_j \neq 0$	$\boldsymbol{r}_j \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03843		—	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03844		—	$i, j = 0, 1, \ldots, k, i \neq j$	$i, j = 0, 1, \ldots, k, i \neq j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03845		—	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03846		—	$j = 0, 1, \ldots, k$	$j = 0, 1, \ldots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03847		—	$\boldsymbol{r}_j$	$\boldsymbol{r}_j$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03848		—	$j \leq k$	$j \leq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03849		—	$\boldsymbol{p}_{\boldsymbol{k}}^T \boldsymbol{A} \boldsymbol{p}_{\boldsymbol{i}} = 0$	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03850		—	$i < k$	$i < k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03851		—	$\boldsymbol{r}_{\boldsymbol{k}+1}$	$\boldsymbol{r}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03852		—	$j < k$	$j < k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03853		—	$\operatorname{dim} \mathbb{R}^n = n$	$\dim \mathbb{R}^n = n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03854		—	$\boldsymbol{r}_{\boldsymbol{0}}, \ldots, \boldsymbol{r}_{\boldsymbol{n}}$	$\boldsymbol{r}_0, \ldots, \boldsymbol{r}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03855		—	$\boldsymbol{r}_{\boldsymbol{k}} = 0$	$\boldsymbol{r}_k = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03856		—	$j = k + 1$	$j = k + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03857		—	$\boldsymbol{p}_{\boldsymbol{k}+1} = \boldsymbol{r}_{\boldsymbol{k}+1} + \beta_{\boldsymbol{k}} \boldsymbol{p}_{\boldsymbol{k}}$	$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03858		—	$\boldsymbol{x}_{\boldsymbol{0}}, \boldsymbol{p}_{\boldsymbol{0}} = \boldsymbol{r}_{\boldsymbol{0}} =$	$\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03859		—	$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$	$\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03860		—	$\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]\left[\begin{array}{l}x_1\\x_2\end{array}\right]=\left[\begin{array}{l}0\\0\end{array}\right]$	$\left[\begin{array}{cc}2&-1\\-1&2\end{array}\right]\left[\begin{array}{l}x_1\\x_2\end{array}\right]=\left[\begin{array}{l}0\\0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03861		—	$\boldsymbol{x}_0=\left[\begin{array}{c}-1\\-1/2\end{array}\right]$	$\boldsymbol{x}_0=\left[\begin{array}{c}-1\\-1/2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03862		—	$\boldsymbol{p}_0=\boldsymbol{r}_0=\left[\begin{array}{c}3/2\\0\end{array}\right]$	$\boldsymbol{p}_0=\boldsymbol{r}_0=\left[\begin{array}{c}3/2\\0\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03863		—	$\boldsymbol{x}_2$	$\boldsymbol{x}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03864		—	$\left \left \boldsymbol{r}_k\right \right _2/\left \left \boldsymbol{b}\right \right _2\leq$	$\left \left \boldsymbol{r}_k\right \right _2/\left \left \boldsymbol{b}\right \right _2\leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03865		—	$k>$	$k>$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03866		—	$(\boldsymbol{t}=\boldsymbol{A}\boldsymbol{p})$	$(\boldsymbol{t}=\boldsymbol{A}\boldsymbol{p})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03867		—	$\left(\boldsymbol{p}^T\boldsymbol{t}\right.$	$(\boldsymbol{p}^T\boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03868		—	$\left.\boldsymbol{r}^T\boldsymbol{r}\right)$	$\boldsymbol{r}^T\boldsymbol{r})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03869		—	$\boldsymbol{x}=\boldsymbol{x}+a\boldsymbol{p},\boldsymbol{r}=\boldsymbol{r}-a\boldsymbol{t}$	$\boldsymbol{x}=\boldsymbol{x}+a\boldsymbol{p},\boldsymbol{r}=\boldsymbol{r}-a\boldsymbol{t}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03870		—	$\boldsymbol{p}=\boldsymbol{r}+$	$\boldsymbol{p} = \boldsymbol{r} +$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03871		—	$\boldsymbol{t}=\boldsymbol{A} \ \boldsymbol{p}$	$\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03872		—	$\boldsymbol{T}_2:=\boldsymbol{T}_1 \ \otimes \boldsymbol{I}+\boldsymbol{I} \ \otimes \boldsymbol{T}_1$	$\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03873		—	$\boldsymbol{T}_1:=\operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$	$\boldsymbol{T}_1 := \operatorname{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03874		—	$\boldsymbol{f}=[1, \ldots, 1]^{\boldsymbol{T}} \in \mathbb{R}^n$	$\boldsymbol{f} = [1, \ldots, 1]^T \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03875		—	$a=1 \ / \ 9, d=5 \ / \ 18$	$a = 1/9, d = 5/18$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03876		—	$a=-1, d=2$	$a = -1, d = 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03877		—	$O(5 \ n)$	$O(5n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03878		—	$\boldsymbol{t}$	$\boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03879		—	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03880		—	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}), \boldsymbol{r}=\operatorname{vec}(\boldsymbol{R}), \boldsymbol{p}=\operatorname{vec}(\boldsymbol{P}), \boldsymbol{t}=\operatorname{vec}(\boldsymbol{T})$	$\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{r} = \operatorname{vec}(\boldsymbol{R}), \boldsymbol{p} = \operatorname{vec}(\boldsymbol{P}), \boldsymbol{t} = \operatorname{vec}(\boldsymbol{T})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03881		—	$h^{\{2\}} \ \boldsymbol{f}=\operatorname{vec}(\boldsymbol{B})$	$h^2 \boldsymbol{f} = \operatorname{vec}(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03882		—	$\boldsymbol{T}_{\{2\}} \ \boldsymbol{x}=h^{\{2\}} \ \boldsymbol{f} \ \Longleftarrow \boldsymbol{T}_{\{1\}} \ \boldsymbol{V}+\boldsymbol{V} \ \boldsymbol{T}_{\{1\}}=\boldsymbol{B}$	$\boldsymbol{T}_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff \boldsymbol{T}_1 \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T}_1 = \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03883		—	$\boldsymbol{t}=\boldsymbol{T}_{\{2\}} \ \boldsymbol{p} \ \Longleftarrow \boldsymbol{T}=\boldsymbol{T}_{\{1\}} \ \boldsymbol{P}+\boldsymbol{P} \ \boldsymbol{T}_{\{1\}}$	$\boldsymbol{t} = \boldsymbol{T}_2 \boldsymbol{p} \iff \boldsymbol{T} = \boldsymbol{T}_1 \boldsymbol{P} + \boldsymbol{P} \boldsymbol{T}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03884		—	$t \ o \ l=10^{\{-8\}}$	$tol = 10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03885		—	$K \ / \ \sqrt{n}$	$K/\sqrt{n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03886		—	$\ \boldsymbol{x}\ _{\boldsymbol{A}}:=\sqrt{\boldsymbol{x}^{\boldsymbol{T}} \ \boldsymbol{A} \ \boldsymbol{x}}$	$\ \boldsymbol{x}\ _{\boldsymbol{A}} := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03887		—	$\kappa=\operatorname{cond}_{\{2\}}(\boldsymbol{A}):=\lambda_{\text{max}} \ / \ \lambda_{\text{min}}$	$\kappa = \operatorname{cond}_2(\boldsymbol{A}) := \lambda_{\max} / \lambda_{\min}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03888		—	$\frac{\kappa-1}{\kappa+1}<1$	$\frac{\kappa-1}{\kappa+1} < 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03889		—	$\frac{\kappa-1}{\kappa+1}$	$\frac{\kappa-1}{\kappa+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03890		—	$d=5 \ / \ 18, \ a=1 \ / \ 9$	$d = 5/18, a = 1/9$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F03891		—	$\lambda_{\text{max}}=\frac{5}{9}+\frac{4}{9} \ \cos \ (\pi \ h)$	$\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03892		—	$\lambda_{\text{min}}=\frac{5}{9}-\frac{4}{9}\cos(\pi h)$	$\lambda_{\min}=\frac{5}{9}-\frac{4}{9}\cos(\pi h)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03893		—	$\mathbb{W}_{\emptyset}=\{\mathbf{0}\}$	$\mathbb{W}_0=\{\mathbf{0}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03894		—	$\operatorname{dim}\left(\mathbb{W}_{\mathfrak{k}}\right)\leq k$	$\dim(\mathbb{W}_k)\leq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03895		—	$\mathbf{v}\in\mathbb{W}_{\mathfrak{k}}$	$\boldsymbol{v}\in\mathbb{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03896		—	$\mathbf{A}\mathbf{v}\in\mathbb{W}_{\mathfrak{k}+1}$	$\boldsymbol{A}\boldsymbol{v}\in\mathbb{W}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03897		—	$\mathbf{p}_{\emptyset}=\mathbf{r}_{\emptyset}$	$\boldsymbol{p}_0=\boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03898		—	$\mathbf{r}_{\mathfrak{k}+1}=\mathbf{r}_{\mathfrak{k}}-\alpha_{\mathfrak{k}}\mathbf{A}\mathbf{p}_{\mathfrak{k}}\in\mathbb{W}_{\mathfrak{k}+2},\mathbf{p}_{\mathfrak{k}+1}=\mathbf{r}_{\mathfrak{k}+1}+\beta_{\mathfrak{k}}\mathbf{p}_{\mathfrak{k}}\in\mathbb{W}_{\mathfrak{k}+2},$	$\boldsymbol{r}_{k+1}=\boldsymbol{r}_k-\alpha_k\boldsymbol{A}\boldsymbol{p}_k\in\mathbb{W}_{k+2},\boldsymbol{p}_{k+1}=\boldsymbol{r}_{k+1}+\beta_k\boldsymbol{p}_k\in\mathbb{W}_{k+2},$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03899		—	$\mathbb{W}_{\mathfrak{k}+2}$	$\mathbb{W}_{k+2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03900		—	$\mathbf{x}_{\mathfrak{k}+1}-\mathbf{x}_{\emptyset}\stackrel{(13.12)}{=}\mathbf{x}_{\mathfrak{k}}-\mathbf{x}_{\emptyset}+\alpha_{\mathfrak{k}}\mathbf{p}_{\mathfrak{k}}\in\mathbb{W}_{\mathfrak{k}+1}$	$\boldsymbol{x}_{k+1}-\boldsymbol{x}_0\stackrel{(13.12)}{=}\boldsymbol{x}_k-\boldsymbol{x}_0+\alpha_k\boldsymbol{p}_k\in\mathbb{W}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03901		—	$\mathbf{w}\in\mathbb{W}_{\mathfrak{k}}$	$\boldsymbol{w}\in\mathbb{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03902		—	$\left\{\mathbf{r}_{\emptyset},\mathbf{r}_1,\ldots,\mathbf{r}_{\mathfrak{k}-1}\right\}$	$\{\boldsymbol{r}_0,\boldsymbol{r}_1,\ldots,\boldsymbol{r}_{k-1}\}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03903		—	$\left\{\boldsymbol{p}_{_0}, \boldsymbol{p}_{_1}, \ldots, \boldsymbol{p}_{_k-1}\right\}$	$\{\boldsymbol{p}_0, \boldsymbol{p}_1, \dots, \boldsymbol{p}_{k-1}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03904		—	$\boldsymbol{u}:=\boldsymbol{x}_{_k}-\boldsymbol{x}_{_0}-\boldsymbol{w}$	$\boldsymbol{u}:=\boldsymbol{x}_k-\boldsymbol{x}_0-\boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03905		—	$\boldsymbol{u} \in \mathbb{W}_{_k}$	$\boldsymbol{u} \in \mathbb{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03906		—	$\left\langle \boldsymbol{x}-\boldsymbol{x}_{_k}, \boldsymbol{u}\right\rangle =\boldsymbol{r}_{_k}^{\mathrm{T}} \boldsymbol{u}=0$ $\boldsymbol{u}=0$	$\langle \boldsymbol{x}-\boldsymbol{x}_k, \boldsymbol{u}\rangle=\boldsymbol{r}_k^T \boldsymbol{u}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03907		—	$\boldsymbol{u}=\mathbf{0}$	$\boldsymbol{u}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03908		—	$\boldsymbol{x}_{_0} \neq \mathbf{0}$	$\boldsymbol{x}_0 \neq \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03909		—	$\boldsymbol{x}-\boldsymbol{x}_{_k}=\left(\boldsymbol{x}-\boldsymbol{x}_{_0}\right)-$	$\boldsymbol{x}-\boldsymbol{x}_k=\left(\boldsymbol{x}-\boldsymbol{x}_0\right)-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03910		—	$\left(\boldsymbol{x}_{_k}-\boldsymbol{x}_{_0}\right)$	$\left(\boldsymbol{x}_k-\boldsymbol{x}_0\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03911		—	$\boldsymbol{x}_{_k}-\boldsymbol{x}_{_0}$	$\boldsymbol{x}_k-\boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03912		—	$p(t):=\sum_{j=0}^m a_{_j} t^m$	$p(t):=\sum_{j=0}^m a_j t^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03913		—	$p(\boldsymbol{A}):=\boldsymbol{a}_{_0} \boldsymbol{I}+\boldsymbol{a}_{_1} \boldsymbol{A}+\cdots+\boldsymbol{a}_{_m} \boldsymbol{A}^m$	$p(\boldsymbol{A}):=a_0 \boldsymbol{I}+a_1 \boldsymbol{A}+\dots+a_m \boldsymbol{A}^m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03914		—	$\left(p\left(\lambda_{j}\right), \boldsymbol{u}_{j}\right)$	$\left(p\left(\lambda_{j}\right), \boldsymbol{u}_{j}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03915		—	$p(\boldsymbol{A})$	$p(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03916		—	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=1,2, \ldots, n$	$\left(\lambda_{j}, \boldsymbol{u}_{j}\right), j=1,2, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03917		—	$\boldsymbol{r}_{0}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}$	$\boldsymbol{r}_{0}:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03918		—	$P(t):=\sum_{j=0}^{k-1} a_{j} t^{k-1}$	$P(t):=\sum_{j=0}^{k-1} a_{j} t^{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03919		—	$\boldsymbol{w}=P(\boldsymbol{A}) \boldsymbol{r}_{0}$	$\boldsymbol{w}=P(\boldsymbol{A}) \boldsymbol{r}_{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03920		—	$\sum_{j=1}^n \sigma_{j} \boldsymbol{u}_{j}$	$\sum_{j=1}^n \sigma_{j} \boldsymbol{u}_{j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03921		—	$\boldsymbol{w}=a_0 \boldsymbol{r}_0+\boldsymbol{A} a_1 \boldsymbol{r}_0+\cdots+\boldsymbol{A}^{k-1} a_{k-1} \boldsymbol{r}_0$	$\boldsymbol{w}=a_0 \boldsymbol{r}_0+\boldsymbol{A} a_1 \boldsymbol{r}_0+\cdots+\boldsymbol{A}^{k-1} a_{k-1} \boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03922		—	$a_0, \ldots, a_{k-1}$	$a_0, \ldots, a_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03923		—	$\boldsymbol{x}-\boldsymbol{x}_0-P(\boldsymbol{A}) \boldsymbol{r}_0=\boldsymbol{A}^{-1}\left(\boldsymbol{x}-\boldsymbol{A} \boldsymbol{x}_0-\boldsymbol{A} \boldsymbol{r}_0\right)$	$\boldsymbol{x}-\boldsymbol{x}_0-P(\boldsymbol{A}) \boldsymbol{r}_0=\boldsymbol{A}^{-1}\left(\boldsymbol{x}-\boldsymbol{A} \boldsymbol{x}_0-\boldsymbol{A} \boldsymbol{r}_0\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03924		—	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0=\boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0=\boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03925		—	$\boldsymbol{A}\left(\boldsymbol{x}-\boldsymbol{x}_{\{0\}}-P(\boldsymbol{A})\boldsymbol{r}_{\{0\}}\right)=Q(\boldsymbol{A})\boldsymbol{r}_{\{0\}}$	$\boldsymbol{A}\left(\boldsymbol{x}-\boldsymbol{x}_0-P(\boldsymbol{A})\boldsymbol{r}_0\right)=Q(\boldsymbol{A})\boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03926		—	$\boldsymbol{r}_{\{0\}}$	$\boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03927		—	$0<a<b$	$0<a<b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03928		—	$\boldsymbol{A}$	$\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03929		—	$\Pi_{\{k\}}$	Π_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03930		—	$\leq k$	$\leq k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03931		—	$Q(t)=1$	$Q(t)=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03932		—	$P(\boldsymbol{A})=\mathbf{0}$	$P(\boldsymbol{A})=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03933		—	$\ \boldsymbol{x}-$	$\ \boldsymbol{x}-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03934		—	$\boldsymbol{x}_{\{0\}}\ \boldsymbol{A}\ ^2=\sum_{j=1}^n\frac{\sigma_{\{j\}}^2}{\lambda_{\{j\}}}$	$\boldsymbol{x}_0\ _{\boldsymbol{A}}^2=\sum_{j=1}^n\frac{\sigma_j^2}{\lambda_j}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03935		—	$Q\in\Pi_{\{k\}}$	$Q\in\Pi_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03936		—	$Q(\emptyset)=1$	$Q(0) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03937		—	$\kappa=\lambda_{\text{max}} / \lambda_{\text{min}}$	$\kappa = \lambda_{\max} / \lambda_{\min}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03938		—	$y=1 \ / \ x$	$y = 1/x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03939		—	$2^{\{k\}} \ / \ k!=2, \ k=1,2$	$2^k/k! = 2, k = 1, 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03940		—	$2^{\{k\}} \ / \ k!<2$	$2^k/k! < 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03941		—	$k>2$	$k > 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03942		—	$c \ \text{notin}[a, \ b]$	$c \notin [a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03943		—	$\mathcal{S}_{\{k\}}$	$\mathcal{S}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03944		—	$Q(c)=1$	$Q(c) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03945		—	$Q^{\{*\}} \ \text{in} \ \mathcal{S}_{\{k\}}$	$Q^* \in \mathcal{S}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03946		—	$Q^{\{*\}}$	Q^*
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03947		—	$T_{\{n\}}$	T_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03948		—	$T_{\{0\}}(t)=1$	$T_0(t) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03949		—	$T_{\{1\}}(t)=t$	$T_1(t) = t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03950		—	$T_{\{2\}}(t)=2\ t^{\{2\}}-1, \ T_{\{3\}}(t)=4\ t^{\{3\}}-3\ t$	$T_2(t) = 2t^2 - 1, T_3(t) = 4t^3 - 3t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03951		—	$T_{\{n\}}(t)=\cos\ (n\ \arccos\ t)$	$T_n(t) = \cos(n\arccos t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03952		—	$t\ \mathrm{in}[-1,1]$	$t \in [-1,1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03953		—	$T_{\{n\}}(t)=\frac{1}{2}\left[\left(t+\sqrt{t^{\{2\}}-1}\right)^{\{n\}}+\left(t+\sqrt{t^{\{2\}}-1}\right)^{\{-n\}}\right]$	$T_n(t) = \frac{1}{2} \left[\left(t + \sqrt{t^2 - 1} \right)^n + \left(t + \sqrt{t^2 - 1} \right)^{-n} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03954		—	$ t \ \geq 1$	$ t \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03955		—	$P_{\{n\}}(t)=\cos\ (n\ \arccos\ t)$	$P_n(t) = \cos(n\arccos t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03956		—	$P_{\{n\}}(t)=\cos\ n\ \phi$	$P_n(t) = \cos n\phi$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03957		—	$t=\cos\ \phi$	$t = \cos \phi$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03958		—	$P_{\{n\}}$	P_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03959		—	$P_{\{0\}}=T_{\{0\}}$	$P_0 = T_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03960		—	$P_{\{1\}}=T_{\{1\}}$	$P_1 = T_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03961		—	$P_{\{n\}}=T_{\{n\}}$	$P_n = T_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03962		—	$x_{\{n\}}:=T_{\{n\}}(t)$	$x_n := T_n(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03963		—	$x_{\{n\}}=z^{\{n\}}$	$x_n = z^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03964		—	$z^{\{n+1\}}-$	$z^{n+1}-$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03965		—	$2\ t\ z^{\{n\}}+z^{\{n-1\}}=0$	$2tz^n + z^{n-1} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03966		—	$z^{\{2\}}-2\ t\ z+1=0$	$z^2 - 2tz + 1 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03967		—	$z_{\{1\}}^{\{n\}},\ z_{\{2\}}^{\{n\}}$	z_1^n, z_2^n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03968		—	$c_{\{1\}}\ z_{\{1\}}^{\{n\}}+c_{\{2\}}\ z_{\{2\}}^{\{n\}}$	$c_1z_1^n + c_2z_2^n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03969		—	$c_{\{1\}}$	c_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03970		—	$c_{\{2\}}$	c_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03971		—	$x_{\{0\}}=$	$x_0 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03972		—	$c_{\{1\}}+c_{\{2\}}=1$	$c_1 + c_2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03973		—	$x_{\{1\}}=c_{\{1\}} \ z_{\{1\}}+c_{\{2\}} \ z_{\{2\}}=t$	$x_1 = c_1 z_1 + c_2 z_2 = t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03974		—	$z_{\{1\}}+z_{\{2\}}=2 \ t$	$z_1 + z_2 = 2t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03975		—	$c_{\{1\}}=c_{\{2\}}=\frac{1}{2}$	$c_1 = c_2 = \frac{1}{2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03976		—	$a<b, \ c \ \notin [a, \ b]$	$a < b, c \notin [a, b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03977		—	$k \ \in$	$k \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03978		—	$\mathbb{N}$	$\mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03979		—	$Q \ \in \ \mathcal{S}_{\{k\}}$	$Q \in \mathcal{S}_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03980		—	$Q \neq Q^*$	$Q \neq Q^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03981		—	$\ Q\ _{\infty} > \ Q^*\ _{\infty}$	$\ Q\ _{\infty} > \ Q^*\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03982		—	$p(z)=p^{\{\prime\}}(z)=0$	$p(z)=p'(z)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03983		—	$\left Q^*\right $	$ Q^* $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03984		—	$1/\left T_k(u(c))\right $	$1/ T_k(u(c)) $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03985		—	$\mu_{\{0\}},\,\ldots,\,\mu_{\{k\}}$	$\mu_0,\ldots,\mu_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03986		—	$u\left(\mu_{\{i\}}\right)=\cos \left(i\,\pi /\,k\right)$	$u\left(\mu_i\right)=\cos (i\pi /k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03987		—	$i=0,1,\,\ldots,\,k$	$i=0,1,\ldots,k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03988		—	$Q\in S_{\{k\}}$	$Q\in S_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03989		—	$\ Q\ _{\infty}\leq\ Q^*\ _{\infty}$	$\ Q\ _{\infty}\leq\ Q^*\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03990		—	$Q\equiv Q^*$	$Q\equiv Q^*$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03991		—	$f \equiv Q - Q^{\ast}$	$f \equiv Q - Q^{\ast}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03992		—	$f(c)=0$	$f(c) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03993		—	$f \equiv 0$	$f \equiv 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03994		—	$I_{\{j\}}=\left[\mu_{\{j-1\}},\,\mu_{\{j\}}\right]$	$I_j = [\mu_{j-1}, \mu_j]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03995		—	$\sigma_{\{j\}} \leq 0$	$\sigma_j \leq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03996		—	$Q^{\ast}\left(\mu_{\{j\}}\right)>0$	$Q^{\ast}(\mu_j) > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03997		—	$f\left(\mu_{\{j\}}\right) \leq 0$	$f(\mu_j) \leq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03998		—	$f\left(\mu_{\{j-1\}}\right) \geq 0$	$f(\mu_{j-1}) \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F03999		—	$Q^{\ast}\left(\mu_{\{j\}}\right)<0$	$Q^{\ast}(\mu_j) < 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04000		—	$\sigma_{\{j\}}<0,\,f$	$\sigma_j < 0, f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04001		—	$I_{\{j\}}$	I_j
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04002		—	$\sigma_j=0$	$\sigma_j = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04003		—	$f\left(\mu_{j-1}\right)=0$	$f\left(\mu_{j-1}\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04004		—	$f\left(\mu_j\right)=0$	$f\left(\mu_j\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04005		—	$Q\left(\mu_j\right)=Q^*\left(\mu_j\right)$	$Q\left(\mu_j\right)=Q^*\left(\mu_j\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04006		—	$\mu_j \in (a, b)$	$\mu_j \in (a, b)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04007		—	$Q^{\prime}\left(\mu_j\right)=$	$Q^{\prime}\left(\mu_j\right)=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04008		—	$Q^{* \prime}\left(\mu_j\right)=0$	$Q^{* \prime}\left(\mu_j\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04009		—	$f\left(\mu_j\right)=f^{\prime}\left(\mu_j\right)=0$	$f\left(\mu_j\right)=f^{\prime}\left(\mu_j\right)=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04010		—	I_{j+1}	I_{j+1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04011		—	$\mu_j=b$	$\mu_j=b$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04012		—	μ_{j-1}	μ_{j-1}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04013		—	$\mu_{j-1} \in (a, b)$	$\mu_{j-1} \in (a, b)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04014		—	I_{j-1}	I_{j-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04015		—	$I_{\{j\}}, j=$	$I_j, j=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04016		—	$1, 2, \ldots, k$	$1, 2, \ldots, k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04017		—	$c=0$	$c = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04018		—	$t=(b+a) / (b-a)$	$t = (b + a)/(b - a)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04019		—	$M:=\lambda_{\text{max}}$	$M := \lambda_{\max}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04020		—	$m:=\lambda_{\text{min}}$	$m := \lambda_{\min}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04021		—	$\left(\lambda_j^{-1}, \boldsymbol{u}_{\{j\}}\right)$	$(\lambda_j^{-1}, \boldsymbol{u}_j)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04022		—	$\boldsymbol{A}^{-1}, j=1, \ldots, n$	$\boldsymbol{A}^{-1}, j = 1, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04023		—	$\boldsymbol{y}=\sum_{j=1}^n c_{\{j\}} \boldsymbol{u}_{\{j\}}$	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{u}_j$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04024		—	$f:[m\ c,\ M\ c]\rightarrow\mathbb{R}$	$f:[mc,Mc]\rightarrow\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04025		—	$f(x):=x+1\ /\ x$	$f(x):=x+1/x$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04026		—	$f\in C^2$	$f\in C^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04027		—	$f^{\prime\prime}$	f''
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04028		—	$c:=1\ /\ \sqrt{m\ M}$	$c:=1/\sqrt{mM}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04029		—	$f(m\ c)=f(M\ c)=\sqrt{\frac{M}{m}}+\sqrt{\frac{m}{M}}=\frac{M+m}{\sqrt{m\ M}}$	$f(mc)=f(Mc)=\sqrt{\frac{M}{m}}+\sqrt{\frac{m}{M}}=\frac{M+m}{\sqrt{mM}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04030		—	$\boldsymbol{\epsilon}_j:=\boldsymbol{x}-\boldsymbol{x}_j,\ j=0,1,\ldots$	$\boldsymbol{\epsilon}_j:=\boldsymbol{x}-\boldsymbol{x}_j,j=0,1,\ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04031		—	$\left \boldsymbol{\epsilon}_k\right _{\boldsymbol{A}}\leq\left(\frac{\kappa-1}{\kappa+1}\right)\left \boldsymbol{\epsilon}_{k-1}\right \leq\cdots\leq\left(\frac{\kappa-1}{\kappa+1}\right)^k\left \boldsymbol{\epsilon}_0\right $	$\ \boldsymbol{\epsilon}_k\ _{\boldsymbol{A}}\leq\left(\frac{\kappa-1}{\kappa+1}\right)\ \boldsymbol{\epsilon}_{k-1}\ \leq\cdots\leq\left(\frac{\kappa-1}{\kappa+1}\right)^k\ \boldsymbol{\epsilon}_0\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04032		—	$\boldsymbol{r}_{m+1}=\mathbf{0}$	$\boldsymbol{r}_{m+1}=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04033		—	$k\leq m$	$k\leq m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04034		—	$\left \boldsymbol{\epsilon}_{k+1}\right _2<\left \boldsymbol{\epsilon}_k\right _2$	$\ \boldsymbol{\epsilon}_{k+1}\ _2<\ \boldsymbol{\epsilon}_k\ _2$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04035		—	$\boldsymbol{\epsilon}_j=\boldsymbol{x}-\boldsymbol{x}_j$	$\boldsymbol{\epsilon}_j = \boldsymbol{x} - \boldsymbol{x}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04036		—	$j \leq m$	$j \leq m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04037		—	$\beta_{i-1} \cdots \beta_k=\left(\boldsymbol{r}_{i}^T \boldsymbol{r}_{i}\right) /\left(\boldsymbol{r}_{k}^T \boldsymbol{r}_{k}\right)$	$\beta_{i-1} \cdots \beta_k=\left(\boldsymbol{r}_i^T \boldsymbol{r}_i\right) /\left(\boldsymbol{r}_k^T \boldsymbol{r}_k\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04038		—	$\operatorname{cond}_2\left(\boldsymbol{A}\right)$	$\operatorname{cond}_2(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04039		—	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}=\boldsymbol{B} \boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{B} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04040		—	$\operatorname{cond}_2\left(\boldsymbol{B} \boldsymbol{A}\right)$	$\operatorname{cond}_2(\boldsymbol{B} \boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04041		—	$\boldsymbol{B}=\boldsymbol{M}^{-1}$	$\boldsymbol{B} = \boldsymbol{M}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04042		—	$\boldsymbol{B}=\boldsymbol{C}^T \boldsymbol{C}$	$\boldsymbol{B} = \boldsymbol{C}^T \boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04043		—	$\boldsymbol{y}:=\boldsymbol{C}^{-T} \boldsymbol{x}$	$\boldsymbol{y} := \boldsymbol{C}^{-T} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04044		—	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T$	$\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04045		—	$\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{y}=\boldsymbol{C} \boldsymbol{b}$	$\left(\boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T\right) \boldsymbol{y} = \boldsymbol{C} \boldsymbol{b}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04046		—	$\boldsymbol{q}_k$	$\boldsymbol{q}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04047		—	$\boldsymbol{z}_k := \boldsymbol{C} \boldsymbol{b} - \boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \boldsymbol{y}_k$	$\boldsymbol{z}_k := \boldsymbol{C} \boldsymbol{b} - \boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \boldsymbol{y}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04048		—	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04049		—	$\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T \left(\boldsymbol{C} - \boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y}_k = \boldsymbol{B} \boldsymbol{b} -$	$\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T \left(\boldsymbol{C} - \boldsymbol{C} \boldsymbol{A} \boldsymbol{C}^T \right) \boldsymbol{y}_k = \boldsymbol{B} \boldsymbol{b} -$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04050		—	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$	$\boldsymbol{B} \boldsymbol{A} \boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04051		—	$\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0, \boldsymbol{p}_0 =$	$\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0, \boldsymbol{p}_0 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04052		—	$\boldsymbol{s}_0 = \boldsymbol{B} \boldsymbol{r}_0$	$\boldsymbol{s}_0 = \boldsymbol{B} \boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04053		—	$\mathrm{x} \left(= \boldsymbol{x}_0 \right)$	$\mathbf{x} \left(= \boldsymbol{x}_0 \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04054		—	ρ	ρ
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04055		—	$w = B * t$	$w = B * t$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04056		—	$\boldsymbol{y}_k$	$\boldsymbol{y}_k$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04057		—	$\boldsymbol{B} \boldsymbol{x}$	Bx
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04058		—	$u=u(x, y)$	$u = u(x, y)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04059		—	$c(x, y)>0$	$c(x, y) > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04060		—	$(x, y) \in \Omega$	$(x, y) \in \Omega$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04061		—	$c(x, y)=1$	$c(x, y) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04062		—	$m \in \mathbb{N}$	$m \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04063		—	$v_{j, k} \approx u\left(x_{\{j\}}, y_{\{k\}}\right)$	$v_{j,k} \approx u\left(x_j, y_k\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04064		—	f, g	f, g
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04065		—	$f_{\{j, k\}}:=f\left(x_{\{j\}}, y_{\{k\}}\right),\left(d v\right)_{\{j, k\}}:=\left(d_1 v\right)_{\{j, k\}}+\left(d_2 v\right)_{\{j, k\}}$	$f_{j,k}:=f\left(x_j, y_k\right),\left(d v\right)_{j,k}:=\left(d_1 v\right)_{j,k}+\left(d_2 v\right)_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04066		—	$c_{\{p, q\}}=c(p h, q h)$	$c_{p,q} = c(ph, qh)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04067		—	$p, q \in \mathbb{R}$	$p, q \in \mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04068		—	$h^{\{2\}}L_{\{h\}}v=h^{\{2\}}\boldsymbol{F}$	$h^2L_hv=h^2\boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04069		—	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04070		—	$v_{\{j, k\}}=w_{\{j, k\}}=0$	$v_{j,k}=w_{j,k}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04071		—	$(j, k) \in \partial I_m$	$(j,k) \in \partial I_m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04072		—	$m \in \mathbb{N}, a_{\{i\}}, b_{\{i\}}, c_{\{i\}} \in \mathbb{R}$	$m \in \mathbb{N}, a_i, b_i, c_i \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04073		—	$i=0, \ldots, m$	$i=0, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04074		—	$b_{\{0\}}=c_{\{0\}}=b_{\{m+1\}}=c_{\{m+1\}}=0$	$b_0=c_0=b_{m+1}=c_{m+1}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04075		—	$\left(d_{\{1\}}v\right)_{\{j, k\}}w_{\{j, k\}}$	$(d_1v)_{j,k}w_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04076		—	$\left(d_{\{2\}}v\right)_{\{j, k\}}w_{\{j, k\}}$	$(d_2v)_{j,k}w_{j,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04077		—	$\left\langle L_{\{h\}}v, \boldsymbol{W} \right\rangle = \left\langle \boldsymbol{W}, L_{\{h\}}v \right\rangle$	$\langle L_hv, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_hv \rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04078		—	$\boldsymbol{W} =$	$\boldsymbol{W} =$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04079		—	$c_{j+\frac{1}{2}, k}$	$c_{j+\frac{1}{2}, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04080		—	$c_{j, k+\frac{1}{2}}$	$c_{j, k+\frac{1}{2}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04081		—	$\left\langle L_h v, \boldsymbol{V} \right\rangle \geq 0$	$\langle L_h v, \boldsymbol{V} \rangle \geq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04082		—	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04083		—	$\left\langle L_h v, \boldsymbol{V} \right\rangle = 0$	$\langle L_h v, \boldsymbol{V} \rangle = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04084		—	$v_{j, k+1} = v_{j, k}$	$v_{j, k+1} = v_{j, k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04085		—	$k=0, 1, \ldots, m$	$k = 0, 1, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04086		—	$v_{j, 0} = v_{j, m+1} = 0$	$v_{j, 0} = v_{j, m+1} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04087		—	$\boldsymbol{V} = \mathbf{0}$	$\boldsymbol{V} = \mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04088		—	$b \in \mathbb{R}^n$	$b \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04089		—	$x \in \mathbb{R}^n$	$x \in \mathbb{R}^n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04090		—	$m=\sqrt{n}$	$m = \sqrt{n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04091		—	$c(x, y) \equiv 1$	$c(x, y) \equiv 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04092		—	$\boldsymbol{A}_p$	$\boldsymbol{A}_p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04093		—	$\boldsymbol{B}=\boldsymbol{A}_p^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04094		—	$\boldsymbol{w}=\boldsymbol{B} \ \boldsymbol{t}$	$\boldsymbol{w} = \boldsymbol{B} \boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04095		—	$\boldsymbol{A}_p \ \boldsymbol{w}_k=\boldsymbol{t}_k$	$\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04096		—	$\boldsymbol{A}_p \ \boldsymbol{w}=\boldsymbol{t}$	$\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04097		—	$\boldsymbol{x}_0=0$	$\boldsymbol{x}_0 = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04098		—	$\epsilon=10^{-8}$	$\epsilon = 10^{-8}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04099		—	$O\left(n \log _{2} n\right)$	$O\left(n \log _2 n\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04100		—	$K_{\text {pre }}$	$\tilde{K}_{\text {pre }}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04101		—	$\theta < c_{\{0\}} \leq$	$0 < c_0 \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04102		—	$c(x, y) \leq c_{\{1\}}$	$c(x, y) \leq c_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04103		—	$(x, y) \in [0, 1]^2$	$(x, y) \in [0, 1]^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04104		—	$\boldsymbol{B} \boldsymbol{A} = \boldsymbol{A}_{\{p\}}^{-1} \boldsymbol{A}$	$\boldsymbol{B} \boldsymbol{A} = \boldsymbol{A}_p^{-1} \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04105		—	$\boldsymbol{A}_{\{p\}}^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$	$\boldsymbol{A}_p^{-1} \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04106		—	$\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$	$\boldsymbol{x} \in \mathbb{R}^n \setminus \{0\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04107		—	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_{\{p\}} \boldsymbol{x}$	$\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{A}_p \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04108		—	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04109		—	$\boldsymbol{x}^T \boldsymbol{A}_{\{p\}} \boldsymbol{x}$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04110		—	$\boldsymbol{x}^T \boldsymbol{A}_{\{p\}} \boldsymbol{x} > 0$	$\boldsymbol{x}^T \boldsymbol{A}_p \boldsymbol{x} > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04111		—	$c_{\{0\}}$	c_0
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04112		—	$c_{\{0\}} \leq \lambda \leq c_{\{1\}}$	$c_0 \leq \lambda \leq c_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04113		—	$c(x, y)=e^{\{-x+y\}}$	$c(x,y)=e^{-x+y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04114		—	$c_{\{0\}}=e^{\{-2\}}$	$c_0=e^{-2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04115		—	$c_{\{1\}}=1$	$c_1=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04116		—	$\kappa \leq e^{\{2\}} \approx$	$\kappa \leq e^2 \approx$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04117		—	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^{\mathrm{T}}$	$\boldsymbol{A}=\boldsymbol{U}\boldsymbol{D}\boldsymbol{U}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04118		—	$\boldsymbol{v}=\left[v_{\{1\}}, \ldots, v_{\{n\}}\right]^{\mathrm{T}}:=\boldsymbol{U}^{\mathrm{T}} \boldsymbol{y}$	$\boldsymbol{v}=\left[v_1, \ldots, v_n\right]^T:=\boldsymbol{U}^T\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04119		—	$\boldsymbol{c}:=\boldsymbol{U}^{\mathrm{T}} \boldsymbol{b}=\left[c_{\{1\}}, \ldots, c_{\{n\}}\right]^{\mathrm{T}}$	$\boldsymbol{c}:=\boldsymbol{U}^T\boldsymbol{b}=\left[c_1, \ldots, c_n\right]^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04120		—	$\boldsymbol{x}_{\{k+1\}}=\boldsymbol{x}_{\{k\}}+\alpha_{\{k\}} \boldsymbol{r}_{\{k\}}$	$\boldsymbol{x}_{k+1}=\boldsymbol{x}_k+\alpha_k\boldsymbol{r}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04121		—	$\boldsymbol{r}_{\{k\}}=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_{\{k\}}$	$\boldsymbol{r}_k=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04122		—	$\alpha_{\{k\}}=\frac{\boldsymbol{r}_{\{k\}}^{\mathrm{T}} \boldsymbol{r}_{\{k\}}}{\boldsymbol{r}_{\{k\}}^{\mathrm{T}} \boldsymbol{A} \boldsymbol{r}_{\{k\}}}$	$\alpha_k=\frac{\boldsymbol{r}_k^T\boldsymbol{r}_k}{\boldsymbol{r}_k^T\boldsymbol{A}\boldsymbol{r}_k}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04123		—	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right], \boldsymbol{b}=\left[\begin{array}{l}1 \\ 1\end{array}\right]^T$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04124		—	$\boldsymbol{e}_k=\boldsymbol{x}_k-\boldsymbol{x}$	$\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04125		—	$\boldsymbol{x}_1=\left(\frac{\boldsymbol{b}^T\boldsymbol{b}}{\boldsymbol{b}^T\boldsymbol{A}\boldsymbol{b}}\right)\boldsymbol{b}$	$\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T\boldsymbol{b}}{\boldsymbol{b}^T\boldsymbol{A}\boldsymbol{b}}\right)\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04126		—	$\boldsymbol{A} \boldsymbol{y}=\boldsymbol{r}_0:=\boldsymbol{b}-\boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{A}\boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04127		—	$\boldsymbol{y}_0=\mathbf{0}.^3$	$\boldsymbol{y}_0 = \mathbf{0}.^3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04128		—	$\boldsymbol{p}_k=\boldsymbol{r}_k+\sum_{j=0}^{k-1}a_{k,j}\boldsymbol{r}_j$	$\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1}a_{k,j}\boldsymbol{r}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04129		—	$\boldsymbol{A} \boldsymbol{y}=\boldsymbol{r}_0$	$\boldsymbol{A}\boldsymbol{y} = \boldsymbol{r}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04130		—	$\boldsymbol{y}_{k+1}:=\boldsymbol{y}_k+\gamma_k\boldsymbol{q}_k, \gamma_k:=$	$\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k\boldsymbol{q}_k, \gamma_k :=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04131		—	$\frac{\boldsymbol{s}_k^T\boldsymbol{s}_k}{\boldsymbol{q}_k^T\boldsymbol{A}\boldsymbol{q}_k}, \boldsymbol{s}_{k+1}:=\boldsymbol{s}_k-\gamma_k\boldsymbol{A}\boldsymbol{q}_k, \boldsymbol{q}_{k+1}:=\boldsymbol{s}_{k+1}+\delta_k\boldsymbol{q}_k, \delta_k:=\frac{\boldsymbol{s}_{k+1}^T\boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T\boldsymbol{s}_k}$	$\frac{\boldsymbol{s}_k^T\boldsymbol{s}_k}{\boldsymbol{q}_k^T\boldsymbol{A}\boldsymbol{q}_k}, \boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k\boldsymbol{A}\boldsymbol{q}_k, \boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k\boldsymbol{q}_k, \delta_k := \frac{\boldsymbol{s}_{k+1}^T\boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T\boldsymbol{s}_k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04132		—	$\boldsymbol{y}_k=\boldsymbol{x}_k-\boldsymbol{x}_0$	$\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04133		—	$\boldsymbol{s}_k=\boldsymbol{r}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04134		—	$\boldsymbol{q}_{\{k\}}=\boldsymbol{p}_{\{k\}}$	$\boldsymbol{q}_k = \boldsymbol{p}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04135		—	$n=100,400,1600,2500$	$n = 100,400,1600,2500$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04136		—	$K \text{ / } n$	K/n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04137		—	$\boldsymbol{A} \in \mathbb{R}^{m,n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m,n}, \boldsymbol{b} \in \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04138		—	$\mathrm{r}=\mathrm{A}^{\prime}\left(\mathrm{b}-\mathrm{A}^{\ast}\mathrm{x}\right) ; \mathrm{p}=\mathrm{r}$	$\boldsymbol{r} = \boldsymbol{A}' \left(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x}\right); \boldsymbol{p} = \boldsymbol{r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04139		—	$\mathrm{r}=\mathrm{r}-\mathrm{a}^{\ast}\mathrm{A}^{\ast}\mathrm{t}$	$\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04140		—	$\operatorname{cond}_2(\boldsymbol{A})^2$	$\operatorname{cond}_2(\boldsymbol{A})^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04141		—	$\left\{\boldsymbol{v}_{\{i\}}\right\}_{i=1}^k$	$\left\{\boldsymbol{v}_i\right\}_{i=1}^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04142		—	$n_{\{i \ j\}}=\left\langle\boldsymbol{v}_{\{i\}}, \boldsymbol{v}_{\{j\}}\right\rangle$	$n_{ij} = \left\langle\boldsymbol{v}_i, \boldsymbol{v}_j\right\rangle$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04143		—	$\mathbb{W} \subset \mathbb{R}^n$	$\mathbb{W} \subset \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04144		—	$\hat{\boldsymbol{x}} \in \mathbb{W}$	$\hat{\boldsymbol{x}} \in \mathbb{W}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04145		—	$\hat{\mathbf{x}}$	$\hat{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04146		—	$\mathbb{W}$	$\mathbb{W}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04147		—	$\mathbf{x}_k \in \mathbb{W}_k$	$x_k \in \mathbb{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04148		—	$\mathbf{p}_k \in \mathbb{W}_{k+1}$	$p_k \in \mathbb{W}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04149		—	$\ \mathbf{A}\mathbf{p}_k\ _2 =$	$\ \mathbf{A}p_k\ _2 =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04150		—	$\ \mathbf{p}_k\ _{\mathbf{A}} = 1$	$\ p_k\ _A = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04151		—	$\alpha_k \in \mathbb{R}$	$\alpha_k \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04152		—	$\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$	p_{k-2}, p_{k-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04153		—	$k \leq 3$	$k \leq 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04154		—	$\left[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}\right] = \left[\begin{array}{rrr} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{array}\right]$	$[b, \mathbf{A}b, \mathbf{A}^2b] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04155		—	$\dim(\mathbb{W}_k) = k$	$\dim(\mathbb{W}_k) = k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04156		—	$k=0,1,2,3$	$k = 0, 1, 2, 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04157		—	$\boldsymbol{x}_3$	$\boldsymbol{x}_3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04158		—	$\boldsymbol{r}_{\{0\}}, \ldots, \boldsymbol{r}_{\{k-1\}}$	$\boldsymbol{r}_0, \ldots, \boldsymbol{r}_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04159		—	$k=1,2,3$	$k = 1, 2, 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04160		—	$\boldsymbol{p}_{\{0\}}, \ldots, \boldsymbol{p}_{\{k-1\}}$	$\boldsymbol{p}_0, \ldots, \boldsymbol{p}_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04161		—	$\left\{\left\ \boldsymbol{r}_{\{k\}}\right\ \right\}$	$\{\ \boldsymbol{r}_k\ \}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04162		—	$\left\{\left\ \boldsymbol{x}_{\{k\}}-\boldsymbol{x}\right\ \right\}$	$\{\ \boldsymbol{x}_k-\boldsymbol{x}\ \}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04163		—	$\boldsymbol{B}^T=-\boldsymbol{B}$	$\boldsymbol{B}^T = -\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04164		—	$\boldsymbol{A}:=\boldsymbol{I}-\boldsymbol{B}$	$\boldsymbol{A} := \boldsymbol{I} - \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04165		—	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2=\ \boldsymbol{x}\ _2^2+\ \boldsymbol{B}\boldsymbol{x}\ _2^2$	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{x}\ _2^2 + \ \boldsymbol{B}\boldsymbol{x}\ _2^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04166		—	$\ \boldsymbol{A}\ _2=\sqrt{1+\ \boldsymbol{B}\ _2^2}$	$\ \boldsymbol{A}\ _2 = \sqrt{1 + \ \boldsymbol{B}\ _2^2}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04167		—	$ \boldsymbol{A} _2 \leq 1$	$\ \boldsymbol{A}\ _2 \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04168		—	$1 \leq k \leq n, \mathcal{W} = \operatorname{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k)$	$1 \leq k \leq n, \mathcal{W} = \operatorname{span}(\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04169		—	$\boldsymbol{x} \in \mathcal{W}$	$\boldsymbol{x} \in \mathcal{W}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04170		—	$ \boldsymbol{x} _2 \leq \boldsymbol{b} _2$	$\ \boldsymbol{x}\ _2 \leq \ \boldsymbol{b}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04171		—	$\boldsymbol{x} := \sum_{j=1}^k x_j \boldsymbol{w}_j$	$\boldsymbol{x} := \sum_{j=1}^k x_j \boldsymbol{w}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04172		—	$x_1, \ldots, x_k$	$x_1, \ldots, x_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04173		—	$\boldsymbol{x}^* := \boldsymbol{A}^{-1} \boldsymbol{b}$	$\boldsymbol{x}^* := \boldsymbol{A}^{-1} \boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04174		—	$\boldsymbol{x}_k \in \mathcal{W}_k$	$\boldsymbol{x}_k \in \mathcal{W}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04175		—	$\boldsymbol{x}^*$	$\boldsymbol{x}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04176		—	$k=0, \ldots, n$	$k = 0, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04177		—	$\omega_0, \omega_1, \ldots, \omega_m$	$\omega_0, \omega_1, \ldots, \omega_m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04178		—	$k=1,2,\ldots,m$	$k=1,2,\ldots,m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04179		—	$k=0,1,\ldots,m-1$	$k=0,1,\ldots,m-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04180		—	$\boldsymbol{x}_0=\boldsymbol{x}_{-1}=0$	$\boldsymbol{x}_0=\boldsymbol{x}_{-1}=0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04181		—	$\boldsymbol{r}_0=\boldsymbol{r}_{-1}=\boldsymbol{b}$	$\boldsymbol{r}_0=\boldsymbol{r}_{-1}=\boldsymbol{b}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04182		—	$0<\omega_k<1$	$0<\omega_k<1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04183		—	$\left\langle \boldsymbol{r}_k,\boldsymbol{r}_j\right\rangle =0$	$\left\langle \boldsymbol{r}_k,\boldsymbol{r}_j\right\rangle =0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04184		—	$j=0,1,\ldots,k-1$	$j=0,1,\ldots,k-1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04185		—	$k\leq m+1$	$k\leq m+1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04186		—	$\mathcal{W}_k=\operatorname{span}\left(\boldsymbol{r}_0,\boldsymbol{r}_1,\ldots,\boldsymbol{r}_{k-1}\right)$	$\mathcal{W}_k=\operatorname{span}\left(\boldsymbol{r}_0,\boldsymbol{r}_1,\ldots,\boldsymbol{r}_{k-1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04187		—	$\operatorname{dim}\mathcal{W}_k=k$	$\operatorname{dim}\mathcal{W}_k=k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04188		—	$1\leq k\leq m-1$	$1\leq k\leq m-1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04189		—	$\alpha_{k}:=\left\langle \boldsymbol{B\ r}_{k},\ \boldsymbol{r}_{k+1}\right\rangle /\rho_{k+1}$	$\alpha_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04190		—	$\beta_{k}:=\left\langle \boldsymbol{B\ r}_{k},\ \boldsymbol{r}_{k-1}\right\rangle /\rho_{k-1}$	$\beta_k := \langle \boldsymbol{B}\boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04191		—	$\alpha_{\{0\}}:=\left\langle \boldsymbol{B\ r}_{\{0\}},\ \boldsymbol{r}_{\{1\}}\right\rangle /\rho_{\{1\}}$	$\alpha_0 := \langle \boldsymbol{B}\boldsymbol{r}_0, \boldsymbol{r}_1 \rangle / \rho_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04192		—	$\alpha_{\{0\}}=1$	$\alpha_0 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04193		—	$\beta_{k}=-\alpha_{k-1}\rho_{k}/\rho_{k-1}$	$\beta_k = -\alpha_{k-1}\rho_k / \rho_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04194		—	$\alpha_{k}+\beta_{k}=1$	$\alpha_k + \beta_k = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04195		—	$k=1,2,\ldots,m-1$	$k = 1, 2, \dots, m - 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04196		—	$\alpha_{k}\geq 1$	$\alpha_k \geq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04197		—	$\boldsymbol{x}_{k},\ \boldsymbol{r}_{k}$	$\boldsymbol{x}_k, \boldsymbol{r}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04198		—	$\boldsymbol{\omega}_{k}$	$\boldsymbol{\omega}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04199		—	$\operatorname{arccosh}$	$\operatorname{arccosh}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04200		—	$\cosh x := \left(e^x + e^{-x} \right) / 2$	$\cosh x := \left(e^x + e^{-x} \right) / 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04201		—	$\boldsymbol{A}^{-1} \left(\boldsymbol{r}_{k+1} - \boldsymbol{r}_j \right) \in \mathcal{W}_{k+1}$	$\boldsymbol{A}^{-1} \left(\boldsymbol{r}_{k+1} - \boldsymbol{r}_j \right) \in \mathcal{W}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04202		—	$f:[a, b] \rightarrow \mathbb{R}$	$f:[a, b] \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04203		—	$\max_{\{a \leq x \leq b\}} f(x) \leq \max \{f(a), f(b)\}$	$\max_{a \leq x \leq b} f(x) \leq \max \{f(a), f(b)\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04204		—	$\boldsymbol{A} \boldsymbol{e}_i = a_{ii} \boldsymbol{e}_i$	$\boldsymbol{A} \boldsymbol{e}_i = a_{ii} \boldsymbol{e}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04205		—	$\left(\lambda_i, \boldsymbol{e}_i \right)$	$\left(\lambda_i, \boldsymbol{e}_i \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04206		—	$\lambda_i = a_{ii}$	$\lambda_i = a_{ii}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04207		—	$\operatorname{det}(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^n \left(a_{ii} - \lambda \right)$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04208		—	$\boldsymbol{A}_i \boldsymbol{X}_i = \boldsymbol{X}_i \boldsymbol{D}_i$	$\boldsymbol{A}_i \boldsymbol{X}_i = \boldsymbol{X}_i \boldsymbol{D}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04209		—	$\boldsymbol{A}_i$	$\boldsymbol{A}_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04210		—	$\boldsymbol{X} := \operatorname{diag} \left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r \right)$	$\boldsymbol{X} := \operatorname{diag} \left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_r \right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04211		—	$\boldsymbol{D}:=\operatorname{diag}\left(\boldsymbol{D}_{\{1\}},\,\ldots,\,\boldsymbol{D}_{\{r\}}\right)$	$D := \operatorname{diag}\left(D_1,\ldots,D_r\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04212		—	$\boldsymbol{A}_{\{11\}},\,\boldsymbol{A}_{\{22\}},\,\ldots,\,\boldsymbol{A}_{\{r\ r\}}$	$\boldsymbol{A}_{11},\boldsymbol{A}_{22},\ldots,\boldsymbol{A}_{rr}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04213		—	$\pi_{\boldsymbol{A}}(\lambda)$	$\pi_{\boldsymbol{A}}(\lambda)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04214		—	$\left.\pi_{\boldsymbol{A}}(\lambda)\right _{\lambda=\lambda^{16}}=\lambda^{16}$	$\pi_{\boldsymbol{A}}(\lambda):=\lambda^{16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04215		—	$q(\lambda)=\lambda^{16}-10^{-16}$	$q(\lambda)=\lambda^{16}-10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04216		—	$\lambda_j=10^{-1}\,e^{2\pi i\,j/16}$	$\lambda_j=10^{-1}e^{2\pi i j/16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04217		—	$j=1,\,\ldots,\,16$	$j=1,\ldots,16$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04218		—	10^{-16}	10^{-16}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04219		—	$R\cap C$	$R\cap C$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04220		—	$R=R_{\{1\}}\cup R_{\{2\}}\cup\cdots\cup R_{\{n\}}$	$R=R_1\cup R_2\cup\cdots\cup R_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04221		—	$C=C_{\{1\}}\cup C_{\{2\}}\cup\cdots\cup C_{\{n\}}$	$C=C_1\cup C_2\cup\cdots\cup C_n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04222		—	$\lambda \in R_i$	$\lambda \in R_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04223		—	$ \left x_i\right =\ \boldsymbol{x}\ _{\infty}$	$ x_i = \ \boldsymbol{x}\ _\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04224		—	$\sum_{j} a_{i j} x_j=\lambda x_i$	$\sum_j a_{ij} x_j = \lambda x_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04225		—	$\left(\lambda-a_{i i}\right) x_i=$	$(\lambda-a_{ii}) x_i =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04226		—	$\sum_{j \neq i} a_{i j} x_j$	$\sum_{j \neq i} a_{ij} x_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04227		—	$\left x_j\right / x_i \leq 1$	$ x_j / x_i \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04228		—	C_{j}	C_j
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04229		—	$\lambda \in C_j$	$\lambda \in C_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04230		—	r_{i}	r_i
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04231		—	$R_{\{1\}}, \ldots, R_{\{n\}}$	$R_1, \ldots, R_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04232		—	$C_{\{1\}}, \ldots, C_{\{n\}}$	$C_1, \ldots, C_n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04233		—	$R_{\{i\}}=C_{\{i\}}$	$R_i = C_i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04234		—	$R_{\{1\}}=R_{\{m\}}=\{z \in \mathbb{R}: z-2 \leq 1\}$	$R_1 = R_m = \{z \in \mathbb{R} : z - 2 \leq 1\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04235		—	$\lambda \in [0,4]$	$\lambda \in [0, 4]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04236		—	$4\left[\sin \frac{\pi}{2(m+1)}\right]^2$	$4 \left[\sin \frac{\pi}{2(m+1)} \right]^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04237		—	$4\left[\sin \frac{m \pi}{2(m+1)}\right]^2$	$4 \left[\sin \frac{m\pi}{2(m+1)} \right]^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04238		—	$\boldsymbol{A}(0)=\boldsymbol{D}$	$\boldsymbol{A}(0) = \boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04239		—	$\boldsymbol{A}(1)=\boldsymbol{A}$	$\boldsymbol{A}(1) = \boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04240		—	$\boldsymbol{A}(t)$	$\boldsymbol{A}(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04241		—	$R_{\{i\}}(t)$	$R_i(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04242		—	$0 \leq t_{\{1\}}<t_{\{2\}} \leq 1$	$0 \leq t_1 < t_2 \leq 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04243		—	$R_{\{i\}}\left(t_{\{1\}}\right) \subset R_{\{i\}}\left(t_{\{2\}}\right)$	$R_i(t_1) \subset R_i(t_2)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04244		—	$R_{\{i\}}(1)$	$R_i(1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04245		—	$\bigcup_{k=1}^p R_{\{i_k\}}(1)$	$\bigcup_{k=1}^p R_{i_k}(1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04246		—	$R^{\{p\}}(t) := \bigcup_{k=1}^p R_{\{i_k\}}(t)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04247		—	$t \in [0,1]$	$t \in [0,1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04248		—	$R^{\{p\}}(\emptyset)$	$R^p(0)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04249		—	$a_{\{i_1, i_1\}}, \ldots, a_{\{i_p, i_p\}}$	$a_{i_1, i_1}, \dots, a_{i_p, i_p}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04250		—	$R^{\{p\}}(t)$	$R^p(t)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04251		—	$R^{\{p\}}(1)$	$R^p(1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04252		—	$\boldsymbol{A} = \left[\begin{array}{ccc} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{array} \right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04253		—	$ \epsilon_i \leq 10^{-15}$	$ \epsilon_i \leq 10^{-15}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04254		—	$\lambda_1, \lambda_2, \lambda_3$	$\lambda_1, \lambda_2, \lambda_3$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04255		—	$\left \lambda_{\mathbf{j}}-\mathbf{j}\right \leq$	$ \lambda_j-j \leq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04256		—	2×10^{-15}	2×10^{-15}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04257		—	$\mathbf{j=1,2,3}$	$j=1,2,3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04258		—	$\boldsymbol{A}+$	$\boldsymbol{A}+$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04259		—	$\boldsymbol{A}_{\mathbf{0}}:=\mathbf{0}$	$\boldsymbol{A}_0:=\mathbf{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04260		—	$\lambda \in \sigma\left(\boldsymbol{A}_{\mathbf{0}}+\boldsymbol{E}\right)=\sigma\left(\boldsymbol{E}\right)$	$\lambda\in\sigma\left(\boldsymbol{A}_0+\boldsymbol{E}\right)=\sigma\left(\boldsymbol{E}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04261		—	$\left \lambda\right \leq\left \boldsymbol{E}\right _{\infty}$	$ \lambda \leq\ \boldsymbol{E}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04262		—	$\boldsymbol{A}_{\mathbf{0}}$	$\boldsymbol{A}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04263		—	$\left \boldsymbol{E}\right _{\infty}$	$\ \boldsymbol{E}\ _{\infty}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04264		—	$\boldsymbol{A}_{\mathbf{1}}+\boldsymbol{E}$	$\boldsymbol{A}_1+\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04265		—	$\pi(\lambda):=(-1)^n\left(\lambda^n-\epsilon\right)$	$\pi(\lambda):=(-1)^n\left(\lambda^n-\epsilon\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04266		—	$ \lambda = \ \boldsymbol{E}\ _{\infty}^{1/n}$	$ \lambda = \ \boldsymbol{E}\ _{\infty}^{1/n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04267		—	$n=16$	$n = 16$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04268		—	$\epsilon=10^{-16}$	$\epsilon = 10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04269		—	$\ \boldsymbol{E}\ _{\infty}^{1/n}$	$\ \boldsymbol{E}\ _{\infty}^{1/n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04270		—	$\mu \in$	$\mu \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04271		—	$\sigma(\boldsymbol{A}+\boldsymbol{E})$	$\sigma(\boldsymbol{A} + \boldsymbol{E})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04272		—	$\ \boldsymbol{u}_1\ _2=1$	$\ \boldsymbol{u}_1\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04273		—	$\ \boldsymbol{E}\ _2^{1/n}$	$\ \boldsymbol{E}\ _2^{1/n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04274		—	$\left[\begin{array}{cc} 1 & 1 \\ \epsilon & 1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1 \\ \epsilon & 1 \end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04275		—	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04276		—	$\epsilon=0$	$\epsilon = 0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04277		—	$1 / n$	$1/n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04278		—	$\ \boldsymbol{E}\ _2$	$\ \boldsymbol{E}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04279		—	$\mu \in \mathbb{C}$	$\mu \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04280		—	$\boldsymbol{r} := \boldsymbol{A} \boldsymbol{x} - \mu \boldsymbol{x}$	$\boldsymbol{r} := \boldsymbol{A} \boldsymbol{x} - \mu \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04281		—	$K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$	$K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04282		—	$(\mu, \boldsymbol{x})$	$(\mu, \boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04283		—	$\mu \in \sigma(\boldsymbol{A})$	$\mu \in \sigma(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04284		—	$\lambda = \mu$	$\lambda = \mu$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04285		—	$\mu \notin \sigma(\boldsymbol{A})$	$\mu \notin \sigma(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04286		—	$\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04287		—	$(\lambda_j, \boldsymbol{x}_j)$	$(\lambda_j, \boldsymbol{x}_j)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04288		—	$\boldsymbol{D}_{\{1\}} := \boldsymbol{D} - \mu \boldsymbol{I}$	$\boldsymbol{D}_1 := \boldsymbol{D} - \mu \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04289		—	$\boldsymbol{D}_{\{1\}}^{-1} = \operatorname{diag}\left(\left(\lambda_1 - \mu\right)^{-1}, \ldots, \left(\lambda_n - \mu\right)^{-1}\right)$	$\boldsymbol{D}_1^{-1} = \operatorname{diag}\left(\left(\lambda_1 - \mu\right)^{-1}, \ldots, \left(\lambda_n - \mu\right)^{-1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04290		—	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{x} = \mu \boldsymbol{x}$	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{x} = \mu \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04291		—	$\mathbf{0} = \boldsymbol{A} \boldsymbol{x} - \mu \boldsymbol{x} + \boldsymbol{E} \boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E} \boldsymbol{x}$	$\mathbf{0} = \boldsymbol{A} \boldsymbol{x} - \mu \boldsymbol{x} + \boldsymbol{E} \boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E} \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04292		—	$\ \boldsymbol{r}\ _p = \ \boldsymbol{E} \boldsymbol{x}\ _p \leq \ \boldsymbol{E}\ _p$	$\ \boldsymbol{r}\ _p = \ \boldsymbol{E} \boldsymbol{x}\ _p \leq \ \boldsymbol{E}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04293		—	$K_p(\boldsymbol{X})$	$K_p(\boldsymbol{X})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04294		—	$K_2(\boldsymbol{X}) = 1$	$K_2(\boldsymbol{X}) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04295		—	$ \lambda - \mu \leq \ \boldsymbol{E}\ _2$	$ \lambda - \mu \leq \ \boldsymbol{E}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04296		—	$\boldsymbol{A} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$	$\boldsymbol{A} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04297		—	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A})$	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04298		—	$p < \infty$	$p < \infty$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04299		—	$\ \boldsymbol{A}\ _p = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p} \leq \rho(\boldsymbol{A})$	$\ \boldsymbol{A}\ _p = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\ _p}{\ \boldsymbol{x}\ _p} \leq \rho(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04300		—	$\mu, \boldsymbol{x}$	$\mu, \boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04301		—	$\boldsymbol{B} := \mu \boldsymbol{A}^{-1}$	$\boldsymbol{B} := \mu \boldsymbol{A}^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04302		—	$\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$	$\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04303		—	$(\lambda_j, \boldsymbol{x})$	$(\lambda_j, \boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04304		—	$\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$	$\left(\frac{\mu}{\lambda_j}, \boldsymbol{x}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04305		—	$(1, \boldsymbol{x})$	$(1, \boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04306		—	$\boldsymbol{B} + \boldsymbol{F}$	$\boldsymbol{B} + \boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04307		—	$\left \frac{\mu}{\lambda} - 1\right \leq K_p(\boldsymbol{X}) \ \boldsymbol{F}\ _p = K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p$	$\left \frac{\mu}{\lambda} - 1\right \leq K_p(\boldsymbol{X}) \ \boldsymbol{F}\ _p = K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04308		—	$1, 2, \ldots, i-2, i=3, 4, \ldots, n$	$1, 2, \ldots, i-2, i=3, 4, \ldots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04309		—	$\boldsymbol{A}_1 = \boldsymbol{A}$	$\boldsymbol{A}_1 = \boldsymbol{A}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04310		—	$\boldsymbol{A}_{k+1}=\boldsymbol{H}_{k} \boldsymbol{A}_{k} \boldsymbol{H}_{k}$	$\boldsymbol{A}_{k+1}=\boldsymbol{H}_k\boldsymbol{A}_k\boldsymbol{H}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04311		—	$k=1,2,\ldots,n-2$	$k=1,2,\ldots,n-2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04312		—	$\boldsymbol{H}_{k}$	$\boldsymbol{H}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04313		—	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04314		—	$\boldsymbol{A}_{k}^{\ast}=\boldsymbol{A}_{k}$	$\boldsymbol{A}_k^{\ast}=\boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04315		—	$\boldsymbol{B}_{k} \in \mathbb{C}^{k,k}$	$\boldsymbol{B}_k \in \mathbb{C}^{k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04316		—	$\boldsymbol{D}_{k} \in$	$\boldsymbol{D}_k \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04317		—	$\mathbb{C}^{n-k,k}$	$\mathbb{C}^{n-k,k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04318		—	$\boldsymbol{D}_{k}=\left[\boldsymbol{0},\boldsymbol{0},\ldots,\boldsymbol{0},\boldsymbol{d}_{k}\right]$	$\boldsymbol{D}_k=[\boldsymbol{0},\boldsymbol{0},\ldots,\boldsymbol{0},\boldsymbol{d}_k]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04319		—	$\boldsymbol{V}_{k}=\boldsymbol{I}-\boldsymbol{v}_{k} \boldsymbol{v}_{k}^{\ast} \in \mathbb{C}^{n-k,n-k}$	$\boldsymbol{V}_k=\boldsymbol{I}-\boldsymbol{v}_k\boldsymbol{v}_k^{\ast} \in \mathbb{C}^{n-k,n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04320		—	$\boldsymbol{V}_{k} \boldsymbol{d}_{k}=\alpha_{k} \boldsymbol{e}_{1}$	$\boldsymbol{V}_k\boldsymbol{d}_k=\alpha_k\boldsymbol{e}_1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04321		—	$\boldsymbol{V}_{\{k\}} \boldsymbol{D}_{\{k\}} = \left[\boldsymbol{V}_{\{k\}} \mathbf{0}, \ldots, \boldsymbol{V}_{\{k\}} \mathbf{0}, \boldsymbol{V}_{\{k\}} \boldsymbol{d}_{\{k\}} \right] = \left(\mathbf{0}, \ldots, \mathbf{0}, \alpha_{\{k\}} \boldsymbol{e}_{\{1\}} \right)$	$\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \mathbf{0}, \dots, \boldsymbol{V}_k \mathbf{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\mathbf{0}, \dots, \mathbf{0}, \alpha_k \boldsymbol{e}_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04322		—	$(k+1) \times (k+1)$	$(k+1) \times (k+1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04323		—	$\boldsymbol{v}_{\{k\}}$	$\boldsymbol{v}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04324		—	$\boldsymbol{L}(k+1:n, k) = \boldsymbol{v}_{\{k\}}$	$\boldsymbol{L}(k+1:n, k) = \boldsymbol{v}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04325		—	$\boldsymbol{B} \cdot \boldsymbol{B}$	$\boldsymbol{B} \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04326		—	$\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{B} = \boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04327		—	$\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$	$\boldsymbol{Q}^* \boldsymbol{A} \boldsymbol{Q}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04328		—	$\boldsymbol{Q} = \boldsymbol{H}_{\{1\}} \boldsymbol{H}_{\{2\}} \cdots \boldsymbol{H}_{\{n-2\}}$	$\boldsymbol{Q} = \boldsymbol{H}_1 \boldsymbol{H}_2 \cdots \boldsymbol{H}_{n-2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04329		—	$\boldsymbol{H}_{\{k\}} = \left[\begin{array}{cc} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{array} \right]$	$\boldsymbol{H}_k = \left[\begin{array}{cc} \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^T \end{array} \right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04330		—	$\boldsymbol{v}_{\{k\}} \in \mathbb{R}^{n-k}$	$\boldsymbol{v}_k \in \mathbb{R}^{n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04331		—	$\boldsymbol{v}_{\{1\}} \in \mathbb{R}^{n-1}$	$\boldsymbol{v}_1 \in \mathbb{R}^{n-1}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04332		—	$\boldsymbol{v}_{n-2} \in \mathbb{R}^2$	$\boldsymbol{v}_{n-2} \in \mathbb{R}^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04333		—	$\boldsymbol{Q}_{k+1}$	$\boldsymbol{Q}_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04334		—	$\left[\begin{array}{cc}\boldsymbol{I}_{\{k\}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{\{k\}}\end{array}\right]$	$\left[\begin{array}{cc}\boldsymbol{I}_k & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_k\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04335		—	$\boldsymbol{U}_{\{k\}} \in \mathbb{R}^{n-k, n-k}$	$\boldsymbol{U}_k \in \mathbb{R}^{n-k, n-k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04336		—	$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$	$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04337		—	$1 \leq m \leq n$	$1 \leq m \leq n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04338		—	$c_{\{i\}}=0$	$c_i = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04339		—	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{\{1\}} & \mathbf{0} \\ \boldsymbol{A}_{\{2\}}\end{array}\right]$	$\boldsymbol{A} = \left[\begin{array}{cc}\boldsymbol{A}_1 & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_2\end{array}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04340		—	$c_{\{i\}} \neq 0$	$c_i \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04341		—	$x \in \mathbb{R}$	$x \in \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04342		—	$p_{\{k\}}(x):=\operatorname{det}\left(x \boldsymbol{I}_{\{k\}}-\boldsymbol{A}_{\{k\}}\right)$	$p_k(x) := \det(x\boldsymbol{I}_k - \boldsymbol{A}_k)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04343		—	$p_{\{n\}}$	p_n
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04344		—	$k \geq 2$	$k \geq 2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04345		—	$p_{\{k+1\}}(x)$	$p_{k+1}(x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04346		—	$p_{\{1\}}(x)=x-d_{\{1\}}$	$p_1(x) = x - d_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04347		—	$p_{\{2\}}(x)=\left(x-d_{\{2\}}\right)\left(x-d_{\{1\}}\right)-c_{\{1\}}^2$	$p_2(x) = (x - d_2)(x - d_1) - c_1^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04348		—	$k=0,1$	$k = 0, 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04349		—	$p_{\{-1\}}(x)=0$	$p_{-1}(x) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04350		—	$p_{\{0\}}(x)=1$	$p_0(x) = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04351		—	$y_{\{1\}} \in \left(-M, d_{\{1\}}\right)$	$y_1 \in (-M, d_1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04352		—	$y_{\{2\}} \in \left(d_{\{1\}}, M\right)$	$y_2 \in (d_1, M)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04353		—	$p_{\{2\}}\left(y_{\{1\}}\right)=p_{\{2\}}\left(y_{\{2\}}\right)=0$	$p_2\left(y_1\right)=p_2\left(y_2\right)=0$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04354		—	$y_{\{1\}}$	y_1
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04355		—	$y_{\{2\}}$	y_2
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04356		—	$y_{\{1\}}<d_{\{1\}}<y_{\{2\}}$	$y_1 < d_1 < y_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04357		—	$x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$	x_1, x_2, x_3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04358		—	$p_{\{3\}}$	p_3
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04359		—	$z_{\{1\}}, \ldots, z_{\{k-1\}}$	$z_1, \ldots, z_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04360		—	$p_{\{k-1\}}$	p_{k-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04361		—	$y_{\{1\}}, \ldots, y_{\{k\}}$	$y_1, \ldots, y_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04362		—	$p_{\{k\}}$	p_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04363		—	$y_{\{0\}}:=-M<y_{\{1\}}, y_{\{k+1\}}:=M>y_{\{k\}}$	$y_0 := -M < y_1, y_{k+1} := M > y_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04364		—	$j=0, k+1$	$j = 0, k + 1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04365		—	$x_{\{1\}}, \ldots, x_{\{k+1\}}$	$x_1, \dots, x_{k+1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04366		—	$\boldsymbol{A}=\boldsymbol{E}^*\boldsymbol{B}\boldsymbol{E}$	$\boldsymbol{A}=\boldsymbol{E}^*\boldsymbol{B}\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04367		—	$\boldsymbol{D}, \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\boldsymbol{D}$	$\boldsymbol{D}, \boldsymbol{U}^*\boldsymbol{A}\boldsymbol{U}=\boldsymbol{D}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04368		—	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04369		—	$v(\boldsymbol{A})$	$v(\boldsymbol{A})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04370		—	$\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+v(\boldsymbol{A})=n$	$\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+v(\boldsymbol{A})=n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04371		—	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B}), \zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B}), \zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04372		—	$v(\boldsymbol{A})=v(\boldsymbol{B})$	$v(\boldsymbol{A})=v(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04373		—	$\pi(\boldsymbol{A})=k$	$\pi(\boldsymbol{A})=\overline{k}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04374		—	$\pi(\boldsymbol{B})=m<k$	$\pi(\boldsymbol{B})=m<k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04375		—	$\boldsymbol{E}_{\{1\}}$	$\boldsymbol{E}_1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04376		—	$m \times k$	$m \times k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04377		—	$m < k$	$m < k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04378		—	$\boldsymbol{E}_1 \boldsymbol{x} = \boldsymbol{0}$	$\boldsymbol{E}_1 \boldsymbol{x} = \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04379		—	$\boldsymbol{y}^T =$	$\boldsymbol{y}^T =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04380		—	$\left[\boldsymbol{x}^T, \boldsymbol{0}^T\right] \in \mathbb{C}^n$	$\left[\boldsymbol{x}^T, \boldsymbol{0}^T\right] \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04381		—	$\boldsymbol{z} = \left[z_1, \ldots, z_n\right]^T = \boldsymbol{E} \boldsymbol{y}$	$\boldsymbol{z} = \left[z_1, \ldots, z_n\right]^T = \boldsymbol{E} \boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04382		—	$z_i = 0$	$z_i = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04383		—	$i = 1, 2, \ldots, m$	$i = 1, 2, \ldots, m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04384		—	$\mu_i \leq 0$	$\mu_i \leq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04385		—	$i \geq m + 1$	$i \geq m + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04386		—	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04387		—	$v(\boldsymbol{A})=$	$v(\boldsymbol{A})=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04388		—	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=v(\boldsymbol{B})$	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=v(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04389		—	$\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-v(\boldsymbol{A})=n-\pi(\boldsymbol{B})-v(\boldsymbol{B})=$	$\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-v(\boldsymbol{A})=n-\pi(\boldsymbol{B})-v(\boldsymbol{B})=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04390		—	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04391		—	$\boldsymbol{U}_1^*\boldsymbol{A}\boldsymbol{U}_1=\boldsymbol{D}_1$	$\boldsymbol{U}_1^*\boldsymbol{A}\boldsymbol{U}_1=\boldsymbol{D}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04392		—	$\boldsymbol{U}_2^*\boldsymbol{B}\boldsymbol{U}_2=\boldsymbol{D}_2$	$\boldsymbol{U}_2^*\boldsymbol{B}\boldsymbol{U}_2=\boldsymbol{D}_2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04393		—	$\boldsymbol{D}_1=\boldsymbol{F}^*\boldsymbol{D}_2\boldsymbol{F}$	$\boldsymbol{D}_1=\boldsymbol{F}^*\boldsymbol{D}_2\boldsymbol{F}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04394		—	$\boldsymbol{F}=\boldsymbol{U}_2^*\boldsymbol{E}\boldsymbol{U}_1$	$\boldsymbol{F}=\boldsymbol{U}_2^*\boldsymbol{E}\boldsymbol{U}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04395		—	$\boldsymbol{D}_1,\boldsymbol{B}$	$\boldsymbol{D}_1,\boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04396		—	$\pi(\boldsymbol{A})=\pi(\boldsymbol{D}_1)=\pi(\boldsymbol{D}_2)=\pi(\boldsymbol{B})$	$\pi(\boldsymbol{A})=\pi(\boldsymbol{D}_1)=\pi(\boldsymbol{D}_2)=\pi(\boldsymbol{B})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04397		—	ζ	ζ
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04398		—	$\operatorname{tridiag}\left(c_{\mathrm{i}},\,d_{\mathrm{i}},\,c_{\mathrm{i}}\right)\in\mathbb{R}^{\mathrm{n}\times\mathrm{n}}$	$\boldsymbol{A}=\operatorname{tridiag}\left(c_i,d_i,c_i\right)\in\mathbb{R}^{n\times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04399		—	$\alpha\in\mathbb{R}$	$\alpha\in\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04400		—	$\boldsymbol{A}-\alpha\,\boldsymbol{I}$	$\boldsymbol{A}-\alpha\boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04401		—	$\boldsymbol{A}-\alpha\,\boldsymbol{I}=\boldsymbol{L}\,\boldsymbol{D}\,\boldsymbol{L}^{\mathrm{T}}$	$\boldsymbol{A}-\alpha\boldsymbol{I}=\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04402		—	$d_1(\alpha),\,\ldots,\,d_n(\alpha)$	$d_1(\alpha),\ldots,d_n(\alpha)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04403		—	$\boldsymbol{L}\,\boldsymbol{D}\,\boldsymbol{L}^{\mathrm{T}}$	$\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04404		—	$v(\boldsymbol{A}-\alpha\,\boldsymbol{I})=v(\boldsymbol{D})$	$v(\boldsymbol{A}-\alpha\boldsymbol{I})=v(\boldsymbol{D})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04405		—	$(\boldsymbol{A}-\alpha\,\boldsymbol{I})\,\boldsymbol{x}=(\lambda-\alpha)\,\boldsymbol{x}$	$(\boldsymbol{A}-\alpha\boldsymbol{I})\boldsymbol{x}=(\lambda-\alpha)\boldsymbol{x}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04406		—	$\lambda-\alpha$	$\lambda-\alpha$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04407		—	$v(\boldsymbol{A}-\alpha\,\boldsymbol{I})$	$v(\boldsymbol{A}-\alpha\boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04408		—	$\lambda_1\geq\lambda_2\geq$	$\lambda_1\geq\lambda_2\geq$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04409		—	$\cdots \geq \lambda_n$	$\cdots \geq \lambda_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04410		—	(a, b)	(a, b)
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04411		—	$\rho(a)=$	$\rho(a) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04412		—	$n, \rho(b)=0$	$n, \rho(b) = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04413		—	$\left\{\left[a_j, b_j\right]\right\}$	$\{[a_j, b_j]\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04414		—	$b_j-a_j=$	$b_j - a_j =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04415		—	$2^{-j}(b-a)$	$2^{-j}(b - a)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04416		—	$d_k(\alpha)$	$d_k(\alpha)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04417		—	$\delta_k=$	$\delta_k =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04418		—	$c_k \ \epsilon_M$	$c_k \epsilon_M$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04419		—	ϵ_M	ϵ_M
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04420		—	2×10^{-16}	2×10^{-16}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04421		—	$ \left d_k(\alpha)\right <\left \delta_k\right $	$ d_k(\alpha) < \delta_k $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04422		—	$\boldsymbol{A}\left(\left a_{i, i}\right >\right).$	$\boldsymbol{A}\left(\left a_{i,i}\right >\right.$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04423		—	$\sum_{j \neq i} \left a_{i, j}\right $	$\sum_{j \neq i} a_{i,j} $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04424		—	$R\!:=\!R_{\{1\}} \cup \cdots \cup R_{\{n\}}$	$R := R_1 \cup \cdots \cup R_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04425		—	$C\!:=\!C_{\{1\}} \cup \cdots \cup C_{\{n\}}$	$C = C_1 \cup \cdots \cup C_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04426		—	$\boldsymbol{s}\!=\!\left[s_{\{1\}}, \ldots, s_{\{n\}}\right] \in \mathbb{R}^{\{n\}}$	$\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04427		—	$\boldsymbol{r}\!=\!\left[r_{\{1\}}, \ldots, r_{\{n\}}\right] \in \mathbb{R}^{\{n\}}$	$\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04428		—	$\boldsymbol{c}\!=\!\left[c_{\{1\}}, \ldots, c_{\{n\}}\right] \in \mathbb{R}^{\{n\}}$	$\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04429		—	$\boldsymbol{A}\left(t_{\{2\}}\right)$	$\boldsymbol{A}\left(t_2\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04430		—	$\boldsymbol{A}\!=\!\boldsymbol{A}\left(t_{\{1\}}\right)$	$\boldsymbol{A} = \boldsymbol{A}\left(t_1\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04431		—	$\boldsymbol{E}=\boldsymbol{A}\left(t_{2}\right)-\boldsymbol{A}\left(t_{1}\right)$	$\boldsymbol{E}=\boldsymbol{A}\left(t_{2}\right)-\boldsymbol{A}\left(t_{1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04432		—	$\boldsymbol{A}\left(t_{1}\right)$	$\boldsymbol{A}\left(t_{1}\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04433		—	$\left \boldsymbol{A}\right _{\infty} =$	$\left\ \boldsymbol{A}\right\ _{\infty} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04434		—	$\left[a_{k\text{ }j}\right], \boldsymbol{E}=\left[e_{k\text{ }j}\right]$	$\left[a_{kj}\right], \boldsymbol{E}=\left[e_{kj}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04435		—	$\boldsymbol{B}=\left[b_{k\text{ }j}\right]$	$\boldsymbol{B}=\left[b_{kj}\right]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04436		—	$\mathbb{R}^{\{n,\text{ }n\}}$	$\mathbb{R}^{n,n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04437		—	$\boldsymbol{B}=\boldsymbol{A}+\boldsymbol{E}$	$\boldsymbol{B}=\boldsymbol{A}+\boldsymbol{E}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04438		—	$\left \boldsymbol{A}\right _{2}=\left \boldsymbol{B}\right _{2}=1$	$\left\ \boldsymbol{A}\right\ _{2}=\left\ \boldsymbol{B}\right\ _{2}=1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04439		—	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04440		—	$\mu=\epsilon^{\{1\text{ } / \text{ } n\}}$	$\mu=\epsilon^{1/n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04441		—	$\frac{10}{3}n^{\{3\}}=5\text{ }G_n$	$\frac{10}{3}n^3=5G_n$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04442		—	$\frac{4}{3} n^3 = 2 G_n$	$\frac{4}{3} n^3 = 2 G_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04443		—	$\boldsymbol{V}_{\boldsymbol{k}} \boldsymbol{E}_{\boldsymbol{k}} \boldsymbol{V}_{\boldsymbol{k}}$	$\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04444		—	$\boldsymbol{E}_{\boldsymbol{k}}$	$\boldsymbol{E}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04445		—	$\boldsymbol{G} = \left(\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T \right) \boldsymbol{E} \left(\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T \right)$	$\boldsymbol{G} = \left(\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T \right) \boldsymbol{E} \left(\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04446		—	$\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}, \beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$	$\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}, \beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04447		—	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04448		—	$\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$	$\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04449		—	$O \left(4 n^3 / 3 \right)$	$O \left(4 n^3 / 3 \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04450		—	$d_{\boldsymbol{k}}$	d_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04451		—	$\boldsymbol{A}_{\boldsymbol{n}}$	$\boldsymbol{A}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04452		—	$5 + \sqrt{24} < d_{\boldsymbol{k}} \leq 10, \boldsymbol{k} = 1, 2, \ldots, \boldsymbol{n}$	$5 + \sqrt{24} < d_k \leq 10, k = 1, 2, \ldots, n$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5 - 0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04453		—	$D_{\{n\}}:=\operatorname{det}\left(\boldsymbol{A}_{\{n\}}\right)>(5+\sqrt{24})^{\{n\}}$	$D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04454		—	$n_{\{0\}} \in \mathbb{N}$	$n_0 \in \mathbb{N}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04455		—	$D_{\{n_{\{0\}}\}}$	D_{n_0}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04456		—	$\boldsymbol{B}=\boldsymbol{U}^{\mathrm{T}} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04457		—	$\hat{\boldsymbol{A}}=\boldsymbol{D}^{\{-1 / 2\}} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^{\mathrm{T}} \boldsymbol{D}^{\{-1 / 2\}}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04458		—	$\hat{\boldsymbol{A}}$	$\hat{\boldsymbol{A}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04459		—	$\hat{\boldsymbol{A}}=\hat{\boldsymbol{U}}^{\mathrm{T}} \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$	$\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04460		—	$\hat{\boldsymbol{D}}$	$\hat{\boldsymbol{D}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04461		—	$\boldsymbol{E} =$	$\boldsymbol{E} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04462		—	$\boldsymbol{U}^{\mathrm{T}} \boldsymbol{D}^{\{-1 / 2\}} \hat{\boldsymbol{U}}^{\mathrm{T}}$	$\boldsymbol{U}^T \boldsymbol{D}^{-1/2} \hat{\boldsymbol{U}}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04463		—	$\boldsymbol{E}^{\mathrm{T}} \boldsymbol{A} \boldsymbol{E}=\hat{\boldsymbol{D}}, \boldsymbol{E}^{\mathrm{T}} \boldsymbol{B} \boldsymbol{E}=\boldsymbol{I}$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \hat{\boldsymbol{D}}, \boldsymbol{E}^T \boldsymbol{B} \boldsymbol{E} = \boldsymbol{I}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04464		—	$\boldsymbol{A}=\operatorname{tridiag}(\boldsymbol{c},\boldsymbol{d},\boldsymbol{c})$	$\boldsymbol{A}=\operatorname{tridiag}(c,d,c)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04465		—	$d_{\{1\}},\ldots,d_{\{n\}}$	$d_1,\ldots,d_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04466		—	$c_{\{1\}},\ldots,c_{\{n-1\}}$	$c_1,\ldots,c_{n-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04467		—	$\mathrm{k}=\operatorname{counting}(\mathrm{c},\mathrm{d},\mathrm{x})$	$\mathbf{k}=\operatorname{counting}(c,d,x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04468		—	$d_{\{j\}}(x)$	$d_j(x)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04469		—	$(a,b]$	$(a,b]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04470		—	$\left\{\left[a_{\{j\}},b_{\{j\}}\right]\right\}$	$\{(a_j,b_j]\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04471		—	$b_{\{j\}}-a_{\{j\}}\leq(b-a)\epsilon_M$	$b_j-a_j\leq(b-a)\epsilon_M$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04472		—	$\epsilon_M\approx2.22\times10^{-16}$	$\epsilon_M\approx2.22\times10^{-16}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04473		—	$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1)$	$\boldsymbol{T}:=\operatorname{tridiag}(-1,2,-1)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04474		—	$\lambda_{\{5\}}$	λ_5
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04475		—	$x \in \mathbb{C}$	$x \in \mathbb{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04476		—	$f(x)=$	$f(x) =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04477		—	$\operatorname{det}(\boldsymbol{A}-x\boldsymbol{I})$	$\det(\boldsymbol{A}-x\boldsymbol{I})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04478		—	$\mu \in \sigma(\boldsymbol{A}+\boldsymbol{E})$	$\mu \in \sigma(\boldsymbol{A}+\boldsymbol{E})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04479		—	$2\,z_{\mathrm{k}}\,/\,3^{\mathrm{k}}$	$2z_k/3^k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04480		—	$[1,-1]$	$[1,-1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04481		—	$\lambda{=}3$	$\lambda = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04482		—	$\left\{z_{\mathrm{k}}^{\mathrm{T}}\boldsymbol{A}\,z_{\mathrm{k}}\,/\,z_{\mathrm{k}}^{\mathrm{T}}\,z_{\mathrm{k}}\right\}$	$\left\{z_k^T\boldsymbol{A}z_k/z_k^Tz_k\right\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04483		—	$\boldsymbol{z}_{\mathrm{0}}$	$\boldsymbol{z}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04484		—	$\left(\lambda_{\mathrm{j}}^{\mathrm{k}},\,\boldsymbol{v}_{\mathrm{j}}\right),\,\mathrm{j}=1,2$	$\left(\lambda_j^k,\boldsymbol{v}_j\right),j=1,2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04485		—	$3^{-\mathrm{k}}\,z_{\mathrm{k}}{=}c_{\mathrm{1}}\,\boldsymbol{v}_{\mathrm{1}}{+}3^{-\mathrm{k}}\,c_{\mathrm{2}}\,\boldsymbol{v}_{\mathrm{2}}\rightarrow c_{\mathrm{1}}\,\boldsymbol{v}_{\mathrm{1}}$	$3^{-k}z_k=c_1\boldsymbol{v}_1+3^{-k}c_2\boldsymbol{v}_2\rightarrow c_1\boldsymbol{v}_1$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04486		—	$c_{\{1\}} \neq 0$	$c_1 \neq 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04487		—	$\left(\lambda_{\{j\}}, \boldsymbol{v}_{\{j\}}\right), j=1, \ldots, n$	$(\lambda_j, \boldsymbol{v}_j), j = 1, \dots, n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04488		—	$\left \lambda_{\{1\}}\right >\left \lambda_{\{2\}}\right \geq \cdots \geq$	$ \lambda_1 > \lambda_2 \geq \cdots \geq$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04489		—	$\left \lambda_{\{n\}}\right $	$ \lambda_n $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04490		—	$z_{\{0\}} \in \mathbb{C}^{\{n\}}$	$z_0 \in \mathbb{C}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04491		—	$z_{\{0\}}$	z_0
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04492		—	$z_{\{0\}}=c_{\{1\}} \boldsymbol{v}_{\{1\}}+c_{\{2\}} \boldsymbol{v}_{\{2\}}+\cdots+c_{\{n\}} \boldsymbol{v}_{\{n\}}$	$z_0=c_1 \boldsymbol{v}_1+c_2 \boldsymbol{v}_2+\cdots+c_n \boldsymbol{v}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04493		—	$\boldsymbol{A}^{\{k\}} \boldsymbol{v}_{\{j\}}=\lambda_{\{j\}}^{\{k\}} \boldsymbol{v}_{\{j\}}$	$\boldsymbol{A}^k \boldsymbol{v}_j=\lambda_j^k \boldsymbol{v}_j$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04494		—	$\lambda_{\{1\}}^{\{k\}}$	λ_1^k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04495		—	$\left(\lambda_{\{j\}} / \lambda_{\{1\}}\right)^{\{k\}} \rightarrow 0$	$(\lambda_j / \lambda_1)^k \rightarrow 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04496		—	(i)	(i)
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04497		—	$z_{\{k\}}$	z_k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04498		—	$\boldsymbol{x}_{\{0\}}=z_{\{0\}} \ / \left z_{\{0\}} \right $	$\boldsymbol{x}_0 = z_0 / \ z_0\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04499		—	$k=1,2, \ \ldots$	$k = 1, 2, \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04500		—	$\lambda_{\{1\}}>0$	$\lambda_1 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04501		—	$c_{\{1\}}>0$	$c_1 > 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04502		—	$\boldsymbol{u}_{\{1\}}:=\boldsymbol{v}_{\{1\}} \ / \left \boldsymbol{v}_{\{1\}} \right $	$\boldsymbol{u}_1 := \boldsymbol{v}_1 / \ \boldsymbol{v}_1\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04503		—	$\boldsymbol{x}_{\{k\}}=\boldsymbol{z}_{\{k\}} \ / \left \boldsymbol{z}_{\{k\}} \right $	$\boldsymbol{x}_k = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04504		—	$\boldsymbol{z}_{\{k\}}=$	$\boldsymbol{z}_k =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04505		—	$\boldsymbol{A}^{\{k\}} \ \boldsymbol{z}_{\{0\}}$	$\boldsymbol{A}^k \boldsymbol{z}_0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04506		—	$\boldsymbol{y}_{\{k\}}=\boldsymbol{A} \ \boldsymbol{x}_{\{k-1\}}=$	$\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04507		—	$\boldsymbol{A} \ \boldsymbol{z}_{\{k-1\}} \ / \left \boldsymbol{z}_{\{k-1\}} \right = \boldsymbol{z}_{\{k\}} \ / \left \boldsymbol{z}_{\{k-1\}} \right $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04508		—	$\boldsymbol{x}_{k}=\left(\boldsymbol{z}_{k} \ / \left \boldsymbol{z}_{k-1}\right \right)\left(\left \boldsymbol{z}_{k-1}\right \ / \left \boldsymbol{z}_{k}\right \right)=\boldsymbol{z}_{k} \ / \left \boldsymbol{z}_{k}\right $	$\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\) (\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04509		—	$r(\lambda):=\boldsymbol{A} \ \boldsymbol{u}-\lambda \boldsymbol{u}$	$r(\lambda) := \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04510		—	$\boldsymbol{u} \in \mathbb{C}^n \backslash \{\boldsymbol{0}\}$	$\boldsymbol{u} \in \mathbb{C}^n \backslash \{\boldsymbol{0}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04511		—	$\rho: \mathbb{C} \rightarrow \mathbb{R}$	$\rho: \mathbb{C} \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04512		—	$\rho(\lambda)=\left \boldsymbol{A} \ \boldsymbol{u}-\lambda \boldsymbol{u}\right _2$	$\rho(\lambda) = \ \boldsymbol{A}\boldsymbol{u} - \lambda\boldsymbol{u}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04513		—	$\lambda:=\frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$	$\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04514		—	$\{\boldsymbol{u}, \boldsymbol{U}\}$	$\{\boldsymbol{u}, \boldsymbol{U}\}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04515		—	$\boldsymbol{U}^* \boldsymbol{u}=\boldsymbol{0}$	$\boldsymbol{U}^* \boldsymbol{u} = \boldsymbol{0}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04516		—	$\lambda=\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$	$\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04517		—	$(\mu, \boldsymbol{u})$	$(\mu, \boldsymbol{u})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04518		—	$\left \boldsymbol{u}\right _2=1$	$\ \boldsymbol{u}\ _2 = 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04519		—	$l, \boldsymbol{x}$	l, x
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04520		—	$\ \boldsymbol{A} \boldsymbol{x} - l \boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$	$\ \boldsymbol{A} \boldsymbol{x} - l \boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04521		—	$=K+1$	$= K + 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04522		—	$\boldsymbol{z} = [0.6602, 0.3420]$	$\boldsymbol{z} = [0.6602, 0.3420]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04523		—	$=10^{-6}$	$= 10^{-6}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04524		—	$\frac{ \lambda_2 }{ \lambda_1 }$	$\frac{ \lambda_2 }{ \lambda_1 }$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04525		—	$\lambda_1 = 5.3723$	$\lambda_1 = 5.3723$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04526		—	$\lambda_2 = -0.3723$	$\lambda_2 = -0.3723$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04527		—	$\lambda_2 = 1.9$	$\lambda_2 = 1.9$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04528		—	$\boldsymbol{A} - s \boldsymbol{I}$	$\boldsymbol{A} - s \boldsymbol{I}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04529		—	$\lambda - s$	$\lambda - s$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04530		—	$(\boldsymbol{A}-s\boldsymbol{I})^{-1}$	$(\boldsymbol{A}-s\boldsymbol{I})^{-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04531		—	$\lim_{s\rightarrow\lambda_1} \mu_1(s) =\infty$	$\lim_{s\rightarrow\lambda_1} \mu_1(s) =\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04532		—	$\lim_{s\rightarrow\lambda_1}\mu_j(s)=\left(\lambda_j-\lambda_1\right)^{-1}<\infty$	$\lim_{s\rightarrow\lambda_1}\mu_j(s)=\left(\lambda_j-\lambda_1\right)^{-1}<\infty$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04533		—	$j=2,\ldots,n$	$j=2,\ldots,n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04534		—	s_{k-1}	s_{k-1}
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04535		—	$\boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{A}\boldsymbol{x}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04536		—	$i\,i$	$i\dot{i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04537		—	$(s,\boldsymbol{x})$	$(s,\boldsymbol{x})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04538		—	$(\lambda,\boldsymbol{v})$	$(\lambda,\boldsymbol{v})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04539		—	$n\,r$	$n\boldsymbol{r}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04540		—	$\lambda_1=(5-\sqrt{33})\,/\,2\approx-0.37$	$\lambda_1=(5-\sqrt{33})/2\approx-0.37$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04541		—	$s=$	$s =$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04542		—	$\ \boldsymbol{r}\ _2$	$\ \boldsymbol{r}\ _2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04543		—	$1.0 \mathrm{e}+000$	$1.0\mathrm{e} + 000$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04544		—	$ \left s-\lambda_1\right $	$ s-\lambda_1 $
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04545		—	$\boldsymbol{Q} \in \mathbb{C}^{n \times n}$	$\boldsymbol{Q} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04546		—	$\boldsymbol{R} \in \mathbb{C}^{n \times n}$	$\boldsymbol{R} \in \mathbb{C}^{n \times n}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04547		—	$\boldsymbol{R}_{\boldsymbol{k}} \boldsymbol{Q}_{\boldsymbol{k}}$	$\boldsymbol{R}_k \boldsymbol{Q}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04548		—	$\boldsymbol{R}_{\boldsymbol{k}}=\boldsymbol{Q}_{\boldsymbol{k}}^{\ast} \boldsymbol{A}_{\boldsymbol{k}}$	$\boldsymbol{R}_k = \boldsymbol{Q}_k^* \boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04549		—	$\boldsymbol{A}_{10}$	$\boldsymbol{A}_{10}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04550		—	$\lambda_1=3$	$\lambda_1 = 3$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04551		—	$\lambda_2=1$	$\lambda_2 = 1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04552		—	$\lambda_1, \ldots, \lambda_4$	$\lambda_1, \ldots, \lambda_4$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04553		—	$\mathbf{A}_{14}$	$\mathbf{A}_{14}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04554		—	$\lambda_1 \approx 2.323$	$\lambda_1 \approx 2.323$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04555		—	$\lambda_4 \approx 0.2275$	$\lambda_4 \approx 0.2275$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04556		—	$\lambda_2 \approx 0.0914 + 0.4586 \text{ i}$	$\lambda_2 \approx 0.0914 + 0.4586i$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04557		—	$\lambda_3 \approx$	$\lambda_3 \approx$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04558		—	$\left(\mathbf{A}_k\right)_k$	$(\mathbf{A}_k)_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04559		—	$k=1,2,3 \ldots$	$k = 1, 2, 3 \ldots$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04560		—	$A^{\{k\}}$	A^k
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04561		—	$\mathbf{A}^{\{k\}}=\tilde{\mathbf{Q}}_{\{k\}} \tilde{\mathbf{R}}_{\{k\}}$	$\mathbf{A}^k = \tilde{\mathbf{Q}}_k \tilde{\mathbf{R}}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04562		—	$\mathbf{Q}_{\{1\}}, \ldots, \mathbf{Q}_{\{k\}}, \mathbf{R}_{\{1\}}, \ldots, \mathbf{R}_{\{k\}}$	$\mathbf{Q}_1, \ldots, \mathbf{Q}_k, \mathbf{R}_1, \ldots, \mathbf{R}_k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04563		—	$\tilde{\boldsymbol{Q}}_{\boldsymbol{1}} \tilde{\boldsymbol{R}}_{\boldsymbol{1}} = \boldsymbol{Q}_{\boldsymbol{1}} \boldsymbol{R}_{\boldsymbol{1}} = \boldsymbol{A}_{\boldsymbol{1}}$	$\tilde{\boldsymbol{Q}}_{\boldsymbol{1}} \tilde{\boldsymbol{R}}_{\boldsymbol{1}} = \boldsymbol{Q}_{\boldsymbol{1}} \boldsymbol{R}_{\boldsymbol{1}} = \boldsymbol{A}_{\boldsymbol{1}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04564		—	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$	$\tilde{\boldsymbol{Q}}_{k-1} \tilde{\boldsymbol{R}}_{k-1} = \boldsymbol{A}^{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04565		—	$\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$	$\boldsymbol{Q}_k \boldsymbol{R}_k = \boldsymbol{A}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04566		—	$\tilde{\boldsymbol{R}}_k$	$\tilde{\boldsymbol{R}}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04567		—	$\left \tilde{\boldsymbol{q}}_1^{(k)} \right _2 = 1$	$\left\ \tilde{\boldsymbol{q}}_1^{(k)} \right\ _2 = 1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04568		—	$\tilde{\boldsymbol{Q}}_k$	$\tilde{\boldsymbol{Q}}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04569		—	$\boldsymbol{x}_{\boldsymbol{0}} = \boldsymbol{e}_1$	$\boldsymbol{x}_0 = \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04570		—	$\lambda_1 \boldsymbol{e}_1$	$\lambda_1 \boldsymbol{e}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04571		—	$\left(\tilde{\boldsymbol{Q}}_k^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_k \right)_k$	$\left(\tilde{\boldsymbol{Q}}_k^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_k \right)_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04572		—	$\boldsymbol{H}_1$	$\boldsymbol{H}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library_F04573		—	$\boldsymbol{P}_{i,i+1}, i=1, \ldots, n-1$	$\boldsymbol{P}_{i,i+1}, i=1, \ldots, n-1$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04574		—	$\boldsymbol{P}_{\{n-1, n\}} \cdots \boldsymbol{P}_{\{1,2\}} \boldsymbol{H}_{\{1\}} = \boldsymbol{R}_{\{1\}}$	$\boldsymbol{P}_{n-1,n} \cdots \boldsymbol{P}_{1,2} \boldsymbol{H}_1 = \boldsymbol{R}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04575		—	$\boldsymbol{H}_{\{1\}} = \boldsymbol{Q}_{\{1\}} \boldsymbol{R}_{\{1\}}$	$\boldsymbol{H}_1 = \boldsymbol{Q}_1 \boldsymbol{R}_1$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04576		—	$\boldsymbol{Q}_{\{1\}} = \boldsymbol{P}_{\{1,2\}}^* \cdots \boldsymbol{P}_{\{n-1, n\}}^*$	$\boldsymbol{Q}_1 = \boldsymbol{P}_{1,2}^* \cdots \boldsymbol{P}_{n-1,n}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04577		—	$\boldsymbol{R}_{\{1\}} \boldsymbol{Q}_{\{1\}} = \boldsymbol{R}_{\{1\}} \boldsymbol{P}_{\{1,2\}}^* \cdots \boldsymbol{P}_{\{n-1, n\}}^*$	$\boldsymbol{R}_1 \boldsymbol{Q}_1 = \boldsymbol{R}_1 \boldsymbol{P}_{1,2}^* \cdots \boldsymbol{P}_{n-1,n}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04578		—	$(i+1, i)$	$(i+1, i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04579		—	$\boldsymbol{R} \boldsymbol{P}_{\{1,2\}}^* \cdots \boldsymbol{P}_{\{n-1, n\}}^*$	$\boldsymbol{R} \boldsymbol{P}_{1,2}^* \cdots \boldsymbol{P}_{n-1,n}^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04580		—	$\boldsymbol{a}_{\{i+1, i\}}$	$\boldsymbol{a}_{i+1,i}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04581		—	$\boldsymbol{A}(1: i, 1: i)$	$\boldsymbol{A}(1: i, 1: i)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04582		—	$\boldsymbol{A}(i+1: n, i+1: n)$	$\boldsymbol{A}(i+1: n, i+1: n)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04583		—	$\left \boldsymbol{a}_{\{i+1, i\}}^{\{k\}} \right \leq \epsilon$	$\left \boldsymbol{a}_{i+1,i}^{(k)} \right \leq \epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04584		—	$\hat{\boldsymbol{A}}_{\{k\}} := \boldsymbol{A}_{\{k\}} - \boldsymbol{a}_{\{i+1, i\}}^{\{k\}} \boldsymbol{e}_{\{i+1\}} \boldsymbol{e}_{\{i\}}^T$	$\hat{\boldsymbol{A}}_k := \boldsymbol{A}_k - \boldsymbol{a}_{i+1,i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04585		—	$\boldsymbol{A}_{\boldsymbol{k}}=\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}$	$\boldsymbol{A}_k = \tilde{\boldsymbol{Q}}_{k-1}^* \boldsymbol{A} \tilde{\boldsymbol{Q}}_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04586		—	$\tilde{\boldsymbol{Q}}_{\boldsymbol{k}-1}$	$\tilde{\boldsymbol{Q}}_{k-1}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04587		—	$\left \left \boldsymbol{E}\right \right _F=\left \left a_{i+1, i}^{\left(k\right)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T\right \right _F=\left a_{i+1, i}^{\left(k\right)}\right \leq \epsilon$	$\left\ \boldsymbol{E} \right\ _F = \left\ a_{i+1,i}^{(k)} \boldsymbol{e}_{i+1} \boldsymbol{e}_i^T \right\ _F = \left a_{i+1,i}^{(k)} \right \leq \epsilon$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04588		—	$a_{i+1, i}^{\left(k\right)}=0$	$a_{i+1,i}^{(k)} = 0$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04589		—	$\boldsymbol{R}_{\boldsymbol{k}}=\boldsymbol{Q}_{\boldsymbol{k}}^* \left(\boldsymbol{A}_{\boldsymbol{k}}-s_{\boldsymbol{k}} \boldsymbol{I}\right)$	$\boldsymbol{R}_k = \boldsymbol{Q}_k^* \left(\boldsymbol{A}_k - s_k \boldsymbol{I} \right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04590		—	$\boldsymbol{Q}_{\boldsymbol{k}}$	$\boldsymbol{Q}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04591		—	$\boldsymbol{A}-s_{\boldsymbol{k}} \boldsymbol{I}=\boldsymbol{Q}_{\boldsymbol{k}} \boldsymbol{R}_{\boldsymbol{k}}$	$\boldsymbol{A} - s_k \boldsymbol{I} = \boldsymbol{Q}_k \boldsymbol{R}_k$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04592		—	$\left(\boldsymbol{A}-s_{\boldsymbol{k}} \boldsymbol{I}\right)^*=\boldsymbol{R}_{\boldsymbol{k}}^* \boldsymbol{Q}_{\boldsymbol{k}}^*$	$\left(\boldsymbol{A} - s_k \boldsymbol{I} \right)^* = \boldsymbol{R}_k^* \boldsymbol{Q}_k^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04593		—	$\left(\boldsymbol{A}-s_{\boldsymbol{k}} \boldsymbol{I}\right)^* \boldsymbol{Q}_{\boldsymbol{k}}=\boldsymbol{R}_{\boldsymbol{k}}^*$	$\left(\boldsymbol{A} - s_k \boldsymbol{I} \right)^* \boldsymbol{Q}_k = \boldsymbol{R}_k^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04594		—	$\boldsymbol{R}_{\boldsymbol{k}}^*$	$\boldsymbol{R}_k^*$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04595		—	$n, \ n$	n, n

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04596		—	$\bar{r}_{nn}^{(k)}$	$\bar{r}_{nn}^{(k)}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04597		—	$(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n=$	$(\boldsymbol{A}-s_k\boldsymbol{I})^*\boldsymbol{Q}_k\boldsymbol{e}_n=$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04598		—	$\boldsymbol{R}_k^*\boldsymbol{e}_n=\bar{r}_{nn}^{(k)}\boldsymbol{e}_n$	$\boldsymbol{R}_k^*\boldsymbol{e}_n=\bar{r}_{nn}^{(k)}\boldsymbol{e}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04599		—	$s_k:=\boldsymbol{e}_n^T\boldsymbol{A}_k\boldsymbol{e}_n$	$s_k:=\boldsymbol{e}_n^T\boldsymbol{A}_k\boldsymbol{e}_n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04600		—	$\boldsymbol{A}:=\begin{bmatrix}0&1\\1&0\end{bmatrix}$	$\boldsymbol{A}:=\begin{bmatrix}0&1\\1&0\end{bmatrix}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04601		—	$\boldsymbol{A}_k=\boldsymbol{A}$	$\boldsymbol{A}_k=\boldsymbol{A}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04602		—	$\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}$	$\boldsymbol{A}\boldsymbol{u}-\lambda\boldsymbol{u}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04603		—	$\lambda=\frac{\boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}}{\boldsymbol{u}^*\boldsymbol{u}}$	$\lambda=\frac{\boldsymbol{u}^*\boldsymbol{A}\boldsymbol{u}}{\boldsymbol{u}^*\boldsymbol{u}}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04604		—	$f:\mathbb{R}^n\rightarrow\mathbb{R}$	$f:\mathbb{R}^n\rightarrow\mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04605		—	$\nabla f(\boldsymbol{x})\in\mathbb{R}^n$	$\nabla f(\boldsymbol{x})\in\mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04606		—	$\boldsymbol{H}f=\nabla\nabla^Tf(\boldsymbol{x})\in\mathbb{R}^{n\times n}$	$\boldsymbol{H}f=\nabla\nabla^Tf(\boldsymbol{x})\in\mathbb{R}^{n\times n}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04607		—	$\nabla^{\mathrm{T}} f := (\nabla f)^{\mathrm{T}}$	$\nabla^T f := (\nabla f)^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04608		—	$\nabla \nabla^{\mathrm{T}}$	$\nabla \nabla^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04609		—	$\nabla^{\mathrm{T}} \nabla$	$\nabla^T \nabla$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04610		—	$\nabla^{\mathrm{T}} \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$	$\nabla^T \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04611		—	$f, g: \mathbb{R}^n \rightarrow \mathbb{R}$	$f, g: \mathbb{R}^n \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04612		—	$\nabla(fg) = f \nabla g + g \nabla f, \quad \nabla^{\mathrm{T}}(fg) = f \nabla^{\mathrm{T}} g + g \nabla^{\mathrm{T}} f$	$\nabla(fg) = f \nabla g + g \nabla f, \quad \nabla^T(fg) = f \nabla^T g + g \nabla^T f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04613		—	$\nabla \nabla^{\mathrm{T}}(fg) = \nabla f \nabla^{\mathrm{T}} g + \nabla g \nabla^{\mathrm{T}} f + f \nabla \nabla^{\mathrm{T}} g + g \nabla \nabla^{\mathrm{T}} f$	$\nabla \nabla^T(fg) = \nabla f \nabla^T g + \nabla g \nabla^T f + f \nabla \nabla^T g + g \nabla \nabla^T f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04614		—	$\nabla^2(fg) = 2 \nabla^{\mathrm{T}} f \nabla g + f \nabla^2 g + g \nabla^2 f$	$\nabla^2(fg) = 2 \nabla^T f \nabla g + f \nabla^2 g + g \nabla^2 f$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04615		—	$\boldsymbol{f} = [f_1, \ldots, f_m]^{\mathrm{T}}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$\boldsymbol{f} = [f_1, \ldots, f_m]^T: \mathbb{R}^n \rightarrow \mathbb{R}^m$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04616		—	$f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$	$f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04617		—	$\boldsymbol{g}(x, y) := [f(x, y), x - y]^{\mathrm{T}}$	$\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04618		—	$f \in C^2(\Omega)$	$f \in C^2(\Omega)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04619		—	$\Omega \in \mathbb{R}^n$	$\Omega \in \mathbb{R}^n$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04620		—	$\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$	$\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04621		—	$L := \{\boldsymbol{x} + t \boldsymbol{h} :$	$L := \{\boldsymbol{x} + t \boldsymbol{h} :$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04622		—	$t \in (0, 1)\} \subset \Omega$	$t \in (0, 1)\} \subset \Omega$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04623		—	$g : [0, 1] \rightarrow \mathbb{R}$	$g : [0, 1] \rightarrow \mathbb{R}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04624		—	$g(t) := f(\boldsymbol{x} + t \boldsymbol{h})$	$g(t) := f(\boldsymbol{x} + t \boldsymbol{h})$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04625		—	$g \in C^2[0, 1]$	$g \in C^2[0, 1]$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04626		—	$\boldsymbol{c} = \boldsymbol{x} + u \boldsymbol{h}$	$\boldsymbol{c} = \boldsymbol{x} + u \boldsymbol{h}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04627		—	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in$	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04628		—	$\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$	$\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04629		—	$\nabla\left(\boldsymbol{y}^T\boldsymbol{C}\right)=\nabla^T\left(\boldsymbol{C}\boldsymbol{x}\right)=\boldsymbol{C}$	$\nabla\left(\boldsymbol{y}^T\boldsymbol{C}\right)=\nabla^T\left(\boldsymbol{C}\boldsymbol{x}\right)=\boldsymbol{C}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04630		—	$\nabla\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\left(\boldsymbol{B}+\boldsymbol{B}^T\right)\boldsymbol{x},\quad\nabla^T\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\boldsymbol{x}^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right)$	$\nabla\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\left(\boldsymbol{B}+\boldsymbol{B}^T\right)\boldsymbol{x},\quad\nabla^T\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\boldsymbol{x}^T\left(\boldsymbol{B}+\boldsymbol{B}^T\right)$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04631		—	$\nabla\nabla^T\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\boldsymbol{B}+\boldsymbol{B}^T$	$\nabla\nabla^T\left(\boldsymbol{x}^T\boldsymbol{B}\boldsymbol{x}\right)=\boldsymbol{B}+\boldsymbol{B}^T$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04632		—	$D_i\left(\boldsymbol{y}^T\boldsymbol{C}\right)=\lim_{h\rightarrow 0}\frac{1}{h}\left(\left(\boldsymbol{y}+h\boldsymbol{e}_i\right)^T\boldsymbol{C}-\boldsymbol{y}^T\boldsymbol{C}\right)=\boldsymbol{e}_i^T\boldsymbol{C}$	$D_i\left(\boldsymbol{y}^T\boldsymbol{C}\right)=\lim_{h\rightarrow 0}\frac{1}{h}\left(\left(\boldsymbol{y}+h\boldsymbol{e}_i\right)^T\boldsymbol{C}-\boldsymbol{y}^T\boldsymbol{C}\right)=\boldsymbol{e}_i^T\boldsymbol{C}$
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Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04635		—	$\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{y}$	$\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{y}$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04636		—	$\boldsymbol{T}\boldsymbol{H}\boldsymbol{x}=\boldsymbol{b},\,79$	$\boldsymbol{T}\boldsymbol{H}\boldsymbol{x}=\boldsymbol{b},\,79$
Numerical_Linear_Algebra_and_Matrix_Factorizations_Tom_Lyche_Z-Library__F04637		—	$\boldsymbol{A}^*\boldsymbol{A},\,86$	$\boldsymbol{A}^*\boldsymbol{A},\,86$
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Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

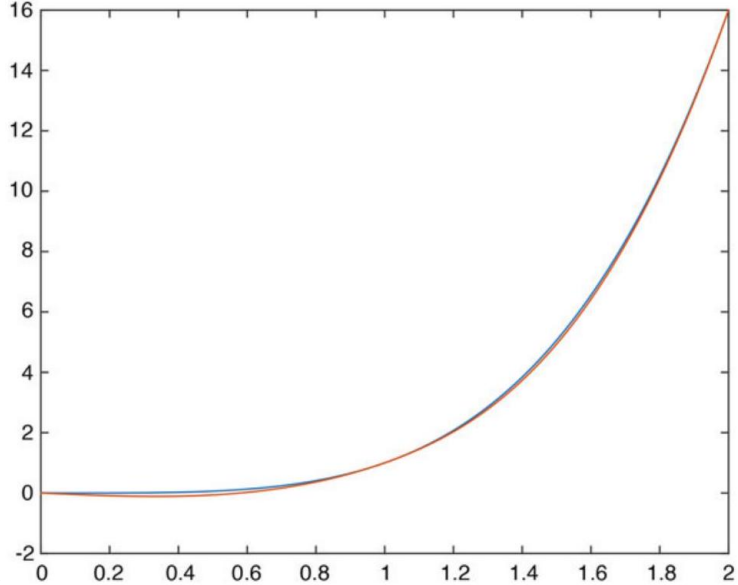
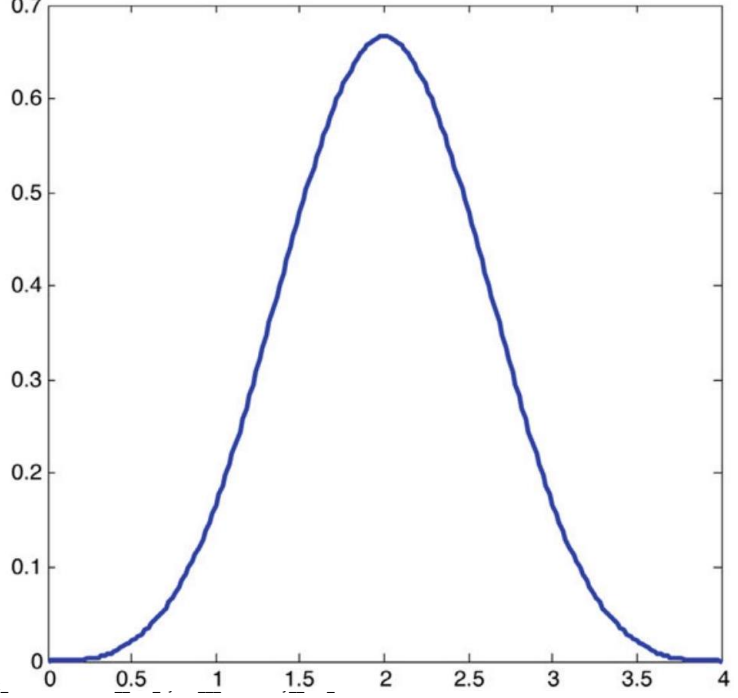
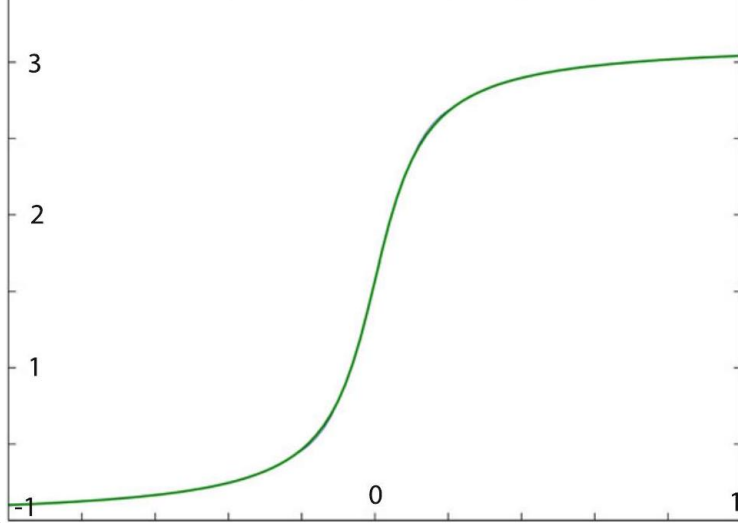
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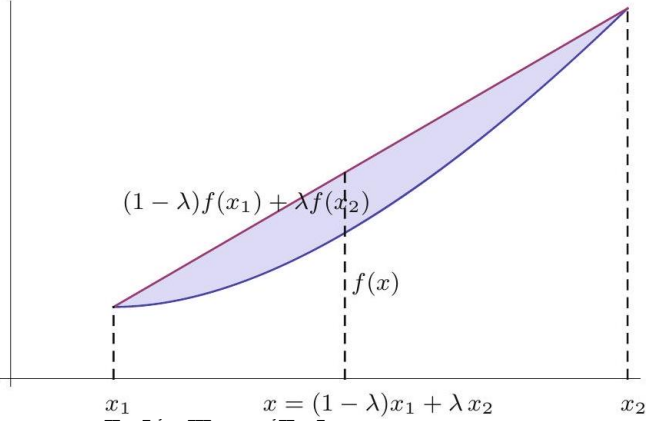
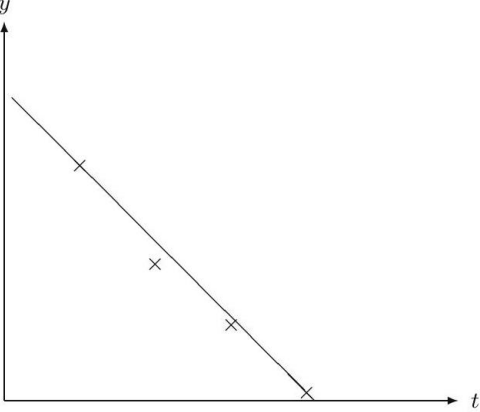
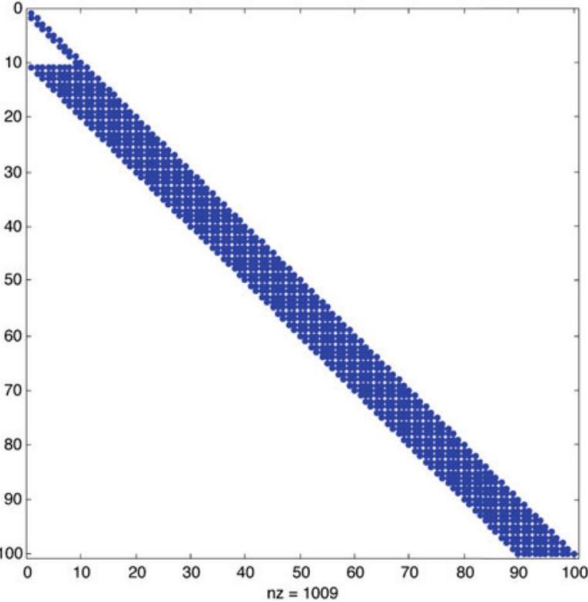
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Numerical_Linear_Algebra_and_Matrix_Factorization_Lecture_2	268	(no LaTeX source; 1161 × 175 px region — see tables.html)	—
Numerical_Linear_Algebra_and_Matrix_Factorization_Lecture_3	278	(no LaTeX source; 560 × 173 px region — see tables.html)	—
Numerical_Linear_Algebra_and_Matrix_Factorization_Lecture_4	298	(no LaTeX source; 1154 × 89 px region — see tables.html)	—
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Numerical_Linear_Algebra_and_Matrix_Factorization_Lecture_6	316	(no LaTeX source; 1152 × 173 px region — see tables.html)	—
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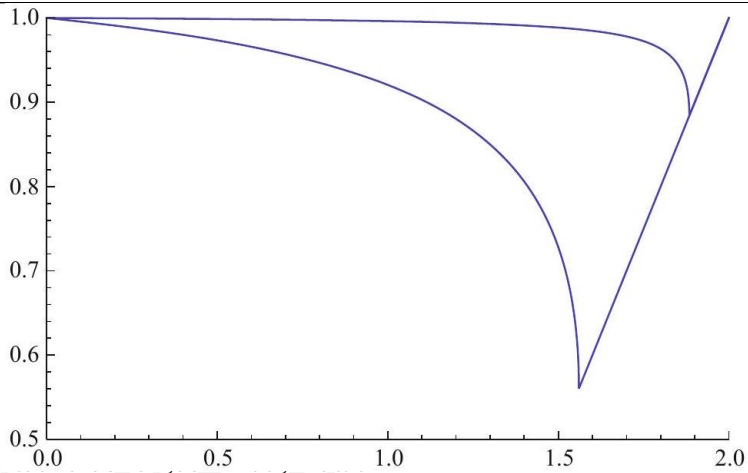
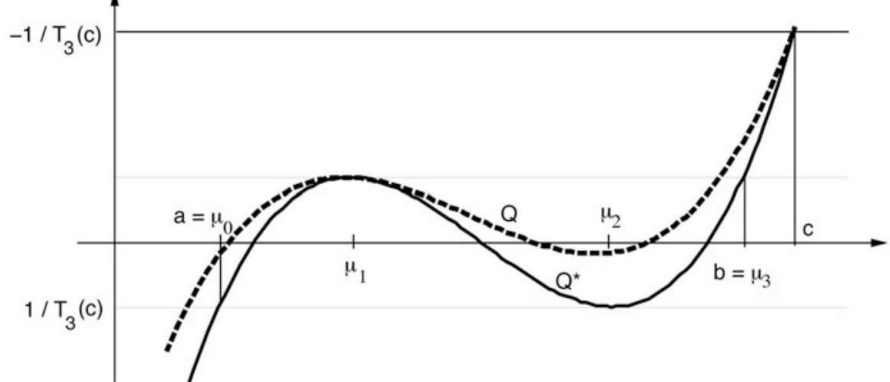
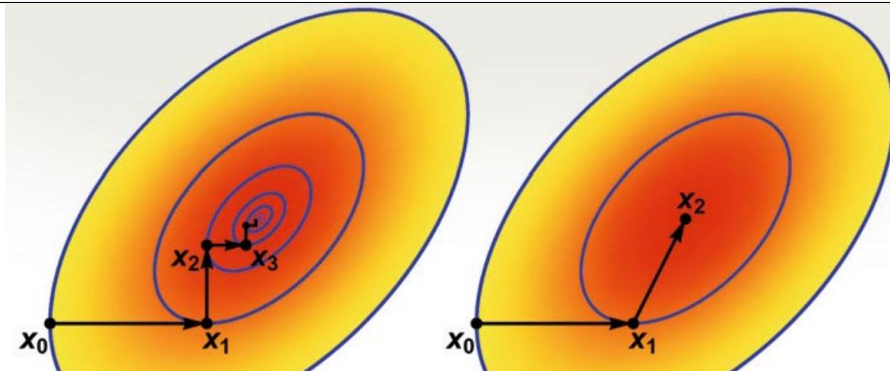
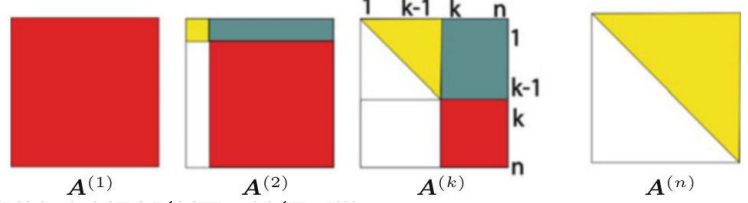
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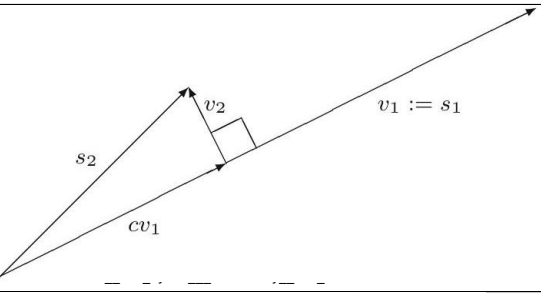
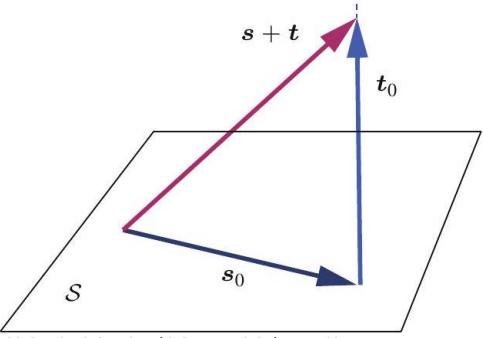
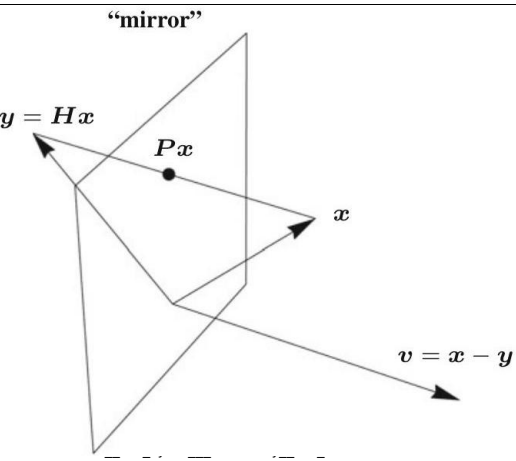
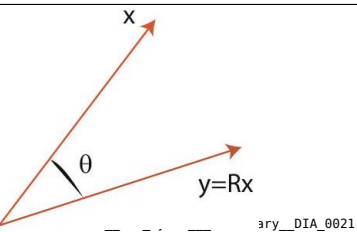
Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

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